

# niva

User Guide · Reference · Cookbook · Algorithm Appendix (v0.33.0)

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# About niva

## The idea

Automating geoprocessing in QGIS today means writing PyQGIS — initialize a `QgsApplication`, memorize algorithm IDs like `native:buffer`, build `ALL_CAPS` parameter dicts, juggle `TEMPORARY_OUTPUT`, and thread each tool's output into the next. That is a *programming* task, which puts everyday automation out of reach for the GUI-first analysts who most need it — and is tedious even for those who can.

niva's answer is a **short, readable text grammar** — a whole pipeline on one line that a non-programmer can write *and* read:

```
load roads.gpkg | buffer 100 dissolve | clip city.gpkg | save roads_local.gpkg
```

...running on QGIS's own Processing algorithms underneath.

## Goals

- **Readable over powerful.** Favor a grammar a non-programmer can pick up — not a new programming language.
- **Thin over clever.** A wrapper over PyQGIS / QGIS Processing (friendly alias names → `native:*` algorithms), not a reimplementation of GIS.
- **Meet people where they are.** A text-pipeline grammar, a Python API (`niva.flow(...)`), a CLI, and a QGIS plugin dock — pick the surface that fits.
- **One backend first.** In-process PyQGIS for v1; headless `qgis_process` later for batch.
- **Clean-room.** The grammar is derived from QGIS Processing's own model and plain readability — not from any proprietary GIS scripting language.
- **Provenance as a byproduct.** Every save records lineage; assess profiles data quality. You get a documented, reproducible pipeline without extra effort.

## Status & open questions

niva **runs** (v0.31.0): the full path — grammar → registry/binder → engine → PyQGIS backend — executes real geoprocessing, validated against real GIS data on QGIS 4.0.3. The surface is now broad: **45 alias verbs** plus **14 built-in verbs** (`load`, `save`, `sql`, `run`, `metadata`, `assess`, `catalog`, `show`, `info`, `project`, `style`, `split`, `notify`, `email`), with **every** QGIS algorithm reachable via `run`, PostGIS/SpatiaLite read-write-analyse, project & template files, a QGIS plugin, and a standalone CLI/API. It is covered by a QGIS-free unit/integration tier (**330+** tests) plus a live-QGIS tier.

The design is worked out in [planning/](#) (PRD, architecture, grammar spec, security model, the `Oscar_the_Grouch.md` failure register, and the [traceability matrix](#) of every verb ↔ algorithm).

Deliberately still-open questions:

- library vs CLI vs plugin as the *primary* surface for the target user;

- how far the friendly-error layer should go in translating cryptic QGIS/GDAL messages;
- whether the premise holds — that GUI-first analysts will adopt a text grammar at all (the risk `Oscar_the_Grouch.md` calls out first).

# niva User Guide

How to install and run niva — **inside QGIS** (the plugin) and **standalone** (the CLI and Python API) — plus configuration, logging, and troubleshooting. For the full verb/option catalogue see the [Reference](#); for worked examples see the [Cookbook](#).

- [1. What niva is](#)
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- 

## 1. What niva is

niva is a small text language for QGIS geoprocessing. You write a **flow** — a chain of verbs joined by | — and niva resolves each verb to a QGIS Processing algorithm and runs it:

```
load roads.gpkg | reproject EPSG:2262 | buffer 100m dissolve | save roads_buf.gpkg
```

It runs the same two ways: as a **QGIS plugin** (a dock panel, using the QGIS session's own Processing engine) and **standalone** (a niva CLI / Python API that boots QGIS headless). The language and results are identical; pick whichever fits the job. niva is pure Python with zero runtime dependencies beyond QGIS itself.

---

## 2. Inside QGIS — the plugin

### Install

1. Build the plugin zip from the repo: `bash plugin/build_plugin.sh` → produces `plugin/niva_qgis.zip`. (Each tagged release also attaches `niva_qgis.zip` — download it from the [Releases](#) page.)
2. In QGIS: **Plugins ► Manage and Install Plugins ► Install from ZIP**, choose `niva_qgis.zip`, install.
3. If it doesn't appear, enable **Show also experimental plugins** in the Plugin Manager settings (niva is marked experimental). It works on QGIS 3.22+ (Qt5) and QGIS 4 (Qt6), with no pip step — the package is vendored inside the zip.
4. Click the **niva** toolbar button (or the niva menu) to toggle the dock.

## The dock — three tabs

**Flow tab** — write and run flows.

- Type one or more flows in the editor (one flow per line; # comments). **Open...** loads a .niva file so call and relative paths resolve from it.
- **Run** executes for real in a background task (the UI stays responsive); progress streams to the output panel and the final output layer is added to the map.
- **Dry-run** validates without touching data — it parses, resolves each verb to its QGIS algorithm and parameters, and lists what it *would* run.
- **Stop** cancels a running flow; **Clear output** clears the panel.

**Convert tab** — bridge to PyQGIS (see §8). Export the current flow to a standalone PyQGIS script, save it, or import a niva-shaped script back into a flow.

**Setup tab** — configure the session:

- **Raster scratch** — the folder for large raster intermediates (sets NIVA\_TMPDIR). Defaults to a real-disk folder under your QGIS profile. Point it at a roomy drive for big raster jobs (see §6).
  - **Run log** — log each run to one file per QGIS session, in a folder you choose (§7).
  - **Email & notifications** — fields for the notify/email environment (ntfy topic/server, SMTP host/user/from, and the "Notify on errors / warnings" toggles), with **Send test notification / email** buttons. Non-secret values persist; secrets (passwords, tokens) are session-only unless you save them to the QGIS encrypted store.
  - **Environment report** — niva version, install path, the verbs and algorithms available, your saved database connections, and QGIS/GDAL/PROJ versions.
- 

## 3. Standalone — CLI

**A bare niva in a fresh shell usually does nothing — "command not found".** That's expected: there's no niva command until you put one on your PATH, *and* it must run on **QGIS's own Python** (niva imports qgis.core + the Processing providers — the system python3 typically can't). Fix it one of two ways:

### Make niva runnable

**Option A — no install (run it as a module).** Point QGIS's Python at the niva package and run the CLI module. **Two placeholders below must be replaced for your machine — copying the line verbatim will fail:**

- **/path/to/niva** — the directory holding the niva/ package (your repo root).
- **/path/to/qgis/python3** — the **full path to QGIS's Python interpreter**, *not* a bare python3. A bare python3 frequently resolves to a Homebrew / pyenv / conda Python that cannot import QGIS, giving No module named 'niva' or a PyQGIS ImportError. On Linux with a distro QGIS this is typically /usr/bin/python3.NN (match QGIS's exact minor version).

*# Substitute both placeholders:*

```
PYTHONPATH=/path/to/niva:/usr/share/qgis/python:/usr/lib/python3/dist-packages \  
QT_QPA_PLATFORM=offscreen /path/to/qgis/python3 -m niva.cli.main describe buffer
```

Wrap it in a shell alias so niva ... just works (put it in your ~/.bashrc):

*# Template:*

```
alias niva='PYTHONPATH=/path/to/niva:/usr/share/qgis/python:/usr/lib/python3/dist-packages  
↪ QT_QPA_PLATFORM=offscreen /path/to/qgis/python3 -m niva.cli.main'
```

*# Concrete example (Linux, distro QGIS 4.x, repo at ~/Github/niva):*

```
alias niva='PYTHONPATH=/home/jcz/Github/niva:/usr/share/qgis/python:/usr/lib/python3/dist-  
packages QT_QPA_PLATFORM=offscreen /usr/bin/python3.14 -m niva.cli.main'
```

Not sure of QGIS's Python or paths? In the QGIS **Python Console** run: `import sys, qgis; print(sys.executable); print([p for p in sys.path if 'qgis' in p.lower()])` — use that `sys.executable` as the interpreter. Then run `niva info`: if it lists your database connections, the alias is correctly wired to your real QGIS profile.

**Option B — install the niva command.** Install `niva` into QGIS's Python so it creates a `niva` console script, then make sure that interpreter's `bin/Scripts` is on your PATH:

```
<qgis-python> -m pip install git+https://github.com/johnzastrow/niva.git
```

On **Windows** run this from the **OSGeo4W shell**; on **macOS** use QGIS.app's bundled Python — both already have the bindings importable (no PYTHONPATH needed). On Linux, the QGIS bindings may still need to be on PYTHONPATH, so Option A's alias is often simplest.

--dry-run and --explain work on **any** Python (they don't import QGIS), so `python3 -m niva.cli.main "..."` --explain is a quick way to sanity-check your install.

## Commands

```
niva run <file.niva> [--dry-run | --explain] [--log <base>]  
niva "<flow>"          [--dry-run | --explain] [--log <base>]  
niva describe <verb-or-algorithm-id>  
niva export <file.niva> [-o <file.py>]  
niva import <file.py>   [-o <file.niva>]
```

- `niva run flow.niva / niva "load ... | save ..."` — execute a file or an inline flow.
- `--dry-run` — print the plan and validate it with no QGIS, no data touched.
- `--explain` — print the resolved algorithm and parameters for each stage.
- `--log <base>` — also write the journal (`<base>.jsonl` + `<base>.log`).
- `niva describe buffer / niva describe gdal:warpreproject` — introspect a verb or algorithm.

**Exit codes:** 0 ok · 1 runtime error · 2 usage/parse error · 3 QGIS not importable.

## Running headless (Linux example)

```
export QT_QPA_PLATFORM=offscreen          # no display needed  
export QGIS_PREFIX_PATH=/usr              # your QGIS install prefix  
export PYTHONPATH=/usr/share/qgis/python:/usr/lib/python3/dist-packages  
export NIVA_TMPDIR=$HOME/niva_scratch     # roomy disk for raster scratch (§6)
```

```
niva run myflow.niva  
# or, if the console script isn't on PATH:  
python3 -m niva.cli.main run myflow.niva
```

On Windows/macOS use the Python that ships with QGIS (the OSGeo4W shell on Windows, or QGIS.app's bundled Python on macOS); the same env vars apply with platform-appropriate paths. `--dry-run` and `--explain` work on **any** Python (they don't import QGIS).

---

## 4. Standalone — Python API

```
import niva  
  
# run an inline flow (needs QGIS on PYTHONPATH)
```



```

layer = niva.flow('load "data.gpkg|layername=roads" | buffer 100m dissolve | save
↳ out.gpkg')
print(layer.ref)          # '/abs/path/out.gpkg'

# run a file, with live progress
niva.run_file("myflow.niva")
niva.flow(open("myflow.niva").read(), file="myflow.niva",
          progress=lambda msg: print(msg))

# introspect a verb (no QGIS needed)
print(niva.describe("buffer"))

flow(text, *, backend=None, file=None, log=None, log_append=False, progress=None,
cancel=None) returns the final layer handle (or None for a terminal flow). Pass backend=niva.e
ngine.MockBackend() to dry-run without QGIS. Importing niva is always safe; QGIS is imported
lazily only when a real run executes. See the Reference for the full signatures.

```

## 5. Configuration & environment

niva is configured entirely through environment variables (the plugin Setup tab sets them for you). The full table is in the [Reference](#); the ones you are most likely to set:

| Variable                                            | Why                                                                 |
|-----------------------------------------------------|---------------------------------------------------------------------|
| NIVA_TMPDIR                                         | scratch dir for raster intermediates — set it to a roomy disk (\$6) |
| NIVA_LOG                                            | default journal base path (\$7)                                     |
| NIVA_TEMPLATES                                      | extra directory of named project templates                          |
| NIVA_NOTIFY_TOPIC (+ ..._SERVER, ..._TOKEN)         | the notify verb                                                     |
| NIVA_SMTP_HOST / ..._USER / ..._PASSWORD / ..._FROM | the email verb                                                      |
| QGIS_PREFIX_PATH, QT_QPA_PLATFORM                   | standalone QGIS bootstrap                                           |

Credentials for notify/email and for databases come **only** from the environment / QGIS — never from flow text.

## 6. Scratch space & large rasters

Raster steps (warp, clipraster, hillshade, resampling, ...) each write a full intermediate GeoTIFF — often gigabytes — before save re-encodes the final product. By default these land in the system temp dir, which on Linux is frequently a small RAM-backed /tmp (e.g. 16 GB). A multi-GB raster pipeline can exhaust it and abort with a "disk quota exceeded" error **even with terabytes of real disk free**.

**Fix:** set NIVA\_TMPDIR (or the plugin's *Raster scratch* field) to a roomy, disk-backed folder:

```
export NIVA_TMPDIR=$HOME/niva_scratch
```

niva routes its intermediates there (and points GDAL's own CPL\_TMPDIR at the same place). Scratch is **purged on every run, even a failed one**, so a crash never strands gigabytes; the run's final output is spared, and a niva-created scratch dir is removed when emptied. The plugin defaults this to a folder under your QGIS profile, always on real disk.

## 7. The run journal

Pass `--log <base>` (CLI), `log=<base>` (flow), or `NIVA_LOG`, or enable per-session logging in the plugin Setup tab, to record a run to two files:

- **<base>.log** — human-readable, one line per operation: timestamp, the stage text, the resolved algorithm, the output path, and the duration; warnings (⚠) and failures (✗ FAILED) are inlined.
- **<base>.jsonl** — machine-readable JSON Lines: a run-start record, one record per operation (including the exact `processing.run(...)` equivalent), and a run-finished summary.

The journal records the niva stage text and output paths — never raw parameter dicts, credentials, or SQL text. `log_append=True` (the plugin's per-session mode) keeps one growing log; the default truncates per invocation.

---

## 8. Export to / import from PyQGIS

niva flows convert to and from plain PyQGIS, so you can hand a script to someone without niva, or drop a flow into a larger Python tool:

- **Export** — `niva export flow.niva -o flow.py` (or the plugin Convert tab) writes a standalone PyQGIS script, one `processing.run(...)` per step.
- **Import** — `niva import flow.py -o flow.niva` recovers a flow from a **niva-shaped** script (a flat list of `processing.run(...)` calls, like Export produces). Arbitrary PyQGIS with loops/conditionals/functions can't round-trip and is reported, not guessed.

This is the audit trail: every niva run is reducible to the exact QGIS calls it made.

---

## 9. Databases & credentials

niva reads and writes Spatialite and PostGIS through **saved QGIS connections**, referenced as `@name` (see the [Reference](#)):

```
load @pg.public.roads | clip aoi.gpkg | save @pg.public.roads_clip mode=replace
sql @pg "SELECT id, ST_Buffer(geom,100) AS geom FROM homes WHERE has_cat" | save
↳ targets.gpkg
```

Set up the connection once in the **QGIS Browser** (Data Source Manager → PostgreSQL / Spatialite → New) or the plugin Setup tab. Only the connection *name* appears in a flow; host, database, user, and password stay in QGIS's own store. niva never logs or transmits credentials. See the SQL recipes in the [Cookbook](#).

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## 10. Troubleshooting

**"could not import QGIS — run niva with QGIS's own Python."** You ran the CLI/API on a plain Python. Use QGIS's interpreter or set `PYTHONPATH` to its bindings (\$3). `--dry-run` / `--explain` work without QGIS if you only need to validate.

**A raster run aborts with "disk quota exceeded" / "No space left" despite free disk.** Your scratch is on a small RAM-backed `/tmp`. Set `NIVA_TMPDIR` to a roomy disk folder (\$6).

**notify fails with "needs a topic."** Set NIVA\_NTFY\_TOPIC (or pass to=<topic>), or fill the ntfy fields in the plugin Setup tab. niva won't send to an unconfigured topic.

**load mydata.gpkg errors and lists layer names.** A multi-layer container needs an explicit layer: load "mydata.gpkg|layername=roads".

**save @pg.table fails because the table exists.** That's the fail-closed default (mode=create). Use mode=replace to overwrite or mode=append to add rows.

**A sql write piped into another stage errors.** A write statement (CREATE/UPDATE/...) is terminal — it returns no layer. Only SELECT/WITH/VALUES/TABLE/EXPLAIN/SHOW produce a pipeable layer.

**One file in an each batch failed but the run continued.** That's intended — a bad *data* item is skipped, logged, and counted so it can't abort the batch. A usage error (bad option, bad target) still stops everything. Check the output/journal for the skipped item.

**Headless QGIS exits with a segfault after the run finishes.** Known QGIS teardown race; the CLI hard-exits to avoid it, and results are already written. If you script the Python API directly, let the process exit rather than re-running in a loop.

# niva Reference

The complete reference for **niva v0.32.0** — every verb, alias, option, type, environment variable, CLI command, and Python entry point. For a task-oriented tour see the [Cookbook](#); for setup and day-to-day use see the [User Guide](#).

## Algorithm coverage at a glance

niva gives **45 alias verbs** friendly names; **every** QGIS Processing algorithm — **769** in QGIS 4.0.3 — is reachable through the [run](#) escape hatch. The complete, per-algorithm appendix (parameters, defaults, enum options, descriptions, and which verb aliases each) is in [docs/algorithms/](#).

| Provider     | Algorithms | niva alias verbs |
|--------------|------------|------------------|
| native:      | 339        | 40               |
| gdal:        | 59         | 5                |
| grass:       | 307        | 0                |
| qgis:        | 39         | 0                |
| pdal:        | 24         | 0                |
| 3d:          | 1          | 0                |
| <b>Total</b> | <b>769</b> | <b>45</b>        |

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## 1. The flow model

A niva program is one or more **flows**, one per line. A flow is a chain of **stages** separated by the pipe |. Each stage is a **verb** with optional arguments and options. A layer handle flows left-to-right: a stage receives the upstream layer, transforms it, and passes the result on.

```
load roads.gpkg | reproject EPSG:2262 | buffer 100m dissolve | save roads_buf.gpkg
```

Verbs fall into three behavioural classes:

| Class               | Meaning                                            | Examples                                             |
|---------------------|----------------------------------------------------|------------------------------------------------------|
| <b>Producing</b>    | returns a new layer to pipe onward                 | load, run, split, sql (SELECT), and every alias verb |
| <b>Pass-through</b> | returns the upstream layer unchanged, so it chains | save, assess, metadata, style, notify, email         |
| <b>Terminal</b>     | returns nothing; a following stage is an error     | catalog, show, info, project, sql (write)            |

A flow normally begins with `load` (or `run/sql`, which produce a layer), or with `each` to run the rest of the flow once per dataset in a directory/glob/container.

## 2. Syntax

### Stages, args, options, flags

A stage is verb [positional-args...] [key=value options...] [flags...], whitespace-separated.

- **Positional arguments** are bound in order (e.g. `buffer 100m` → the distance).
- **Options** are key=value (e.g. `segments=12, cap=flat`).
- **Flags** are bare words that switch a boolean on (e.g. `dissolve, percent`).

```
buffer 100m dissolve cap=flat segments=12
#      ^arg  ^flag  ^option ^option
```

### Quoting, comments, line joins

- **Quotes** keep a value with spaces or special characters atomic: `filter "landuse = 'R'", load "my data.gpkg|layername=roads"`. Single or double quotes; the outer pair is stripped.
- `#` starts a **comment** to end of line. A line that is only a comment is ignored.
- A trailing `\` **joins** the next line (for long flows).
- Blank lines are ignored. Each non-blank line is its own flow.

### Paths

A path argument has a leading `~` expanded to your home directory (`clip "~/aoi.gpkg"`). A layer inside a multi-layer container is addressed with the OGR suffix `|layername=<name>`: `load "data.gpkg|layername=roads"`.

### Per-item placeholder {name}

Inside an `each` batch, `{name}` in a save path or as clause is replaced with the current item's (sanitised) name — `save "out/{name}.tif"`. Outside a batch it is an error.

### Option keys

Option keys are identifier-like and may contain internal hyphens (`from-template=`, `to-template=`). Only tokens containing `=` are treated as options, so flags such as `-deep` are never mistaken for options.

### 3. Value types & units

Argument and option values are coerced by **type** (the binder in niva/registry/binder.py):

| Type           | Accepts                                             | Notes                                                                                       |
|----------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------|
| distance       | number + optional unit, e.g. 100, 100m, 1km, 0.5deg | unit resolved against the layer CRS at run time; <b>no unit = the layer's own CRS units</b> |
| number         | decimal, e.g. 2, 0.25, -9999                        |                                                                                             |
| int            | integer, e.g. 12                                    |                                                                                             |
| enum           | one word from a fixed vocab, e.g. cap=flat          | word → QGIS code; invalid word lists the valid set                                          |
| enumlist       | comma-separated enum words, e.g. stats=mean,min,max |                                                                                             |
| fields / list  | comma-separated list, e.g. fields=pop,income        |                                                                                             |
| layer / raster | a dataset path or @conn.table                       | ~ expanded; @conn passes through                                                            |
| crs            | a CRS string, e.g. EPSG:2262                        | verbatim                                                                                    |
| field          | an attribute field name                             | verbatim                                                                                    |
| string         | any text                                            | verbatim                                                                                    |
| expression     | a QGIS expression, e.g. "area > 2000"               | verbatim; quote it                                                                          |

**Supported distance units:** m, km, cm, mm, ft, yd, mi, nmi, deg. An unknown unit is a clear error. A distance with no unit is interpreted in the layer's CRS units (so on a geographic CRS, buffer 100 means 100 *degrees* — usually you want 100m).

**Enum-by-word:** options that map to a QGIS dropdown take a readable word, not a number — simplify 5m method=area, not method=2. Each alias below lists its vocab.

---

### 4. Built-in verbs

Built-in verbs are handled directly by the engine (not the alias registry).

#### load — read a dataset (*producing*)

```
load <path-or-uri>
load "<container.gpkg>|layername=<layer>"
load @conn[.schema].table
```

Exactly one argument; no options. ~ is expanded. A bare multi-layer container (load data.gpkg) errors and lists its layers — name one with |layername=. The @ form reads a table from a **saved database connection** (§7); a bare @conn with no table errors (use sql to query it), and an @ pointing at a file extension is rejected (use the path form).

```
load "parcels.gpkg|layername=residential"
load @pg.public.roads
```

#### save — write the current layer (*pass-through*)

```
save <path>
save <path> as <layer> # named layer inside a .gpkg/.sqlite/.db
save @conn[.schema].table [mode=create|replace|append]
```

- **File form:** extension chooses the format; parent directories are created. No options. as <layer> writes a named layer into a container (GeoPackage/SQLite only); writing into an existing container with a layer name **appends**.
- **Database form:** mode is the only option — create (default; **fails if the table exists** — never a silent overwrite), replace (drop + recreate), or append (insert, fields matched by name, CRS-transformed as needed). Rasters cannot be saved to a database.

- save chains: ... | save out.gpkg | style apply s.qml | notify "done".
- **In an each batch:** a file container gets **one layer per item** (named after the source); a {name} path gives **one file per item**; save @conn writes **one table per item** (a trailing qualifier is the schema, e.g. save @pg.niagara).

```
load roads.gpkg | clip aoi.gpkg | save out/roads_clip.gpkg
load roads.gpkg | save data.gpkg as roads
load roads.gpkg | save @pg.public.roads_clip mode=replace
```

## sql — query or mutate a database connection (*producing OR terminal*)

```
sql @conn "<statement>"
```

Exactly two arguments — a **bare** connection ref and a quoted statement — and no options. The leading keyword routes it:

- **SELECT / WITH / VALUES / TABLE / EXPLAIN / SHOW** → a **query layer** you can pipe onward (producing). CREATE TABLE ... AS SELECT ... reads as a **write** (leading CREATE); WITH ... SELECT ... reads as a query (leading WITH).
- **anything else** (CREATE/INSERT/UPDATE/DELETE/DROP/ALTER/...) → runs server-side and returns nothing (terminal).

Works against any saved connection — **Spatialite** or **PostGIS** — using that engine's spatial SQL (ST\_Buffer, ST\_Area, spatial joins, ...). Only the connection *name* leaves the flow; credentials never appear in flow text, errors, or logs (§7).

```
sql @cats_pg "SELECT id, ST_Buffer(geom, 100) AS geom FROM homes WHERE has_cat" | save
↳ targets.gpkg
sql @pg "CREATE TABLE roads_buf AS SELECT id, ST_Buffer(geom,100) AS geom FROM roads"
```

## run — call any QGIS algorithm directly (*producing*)

```
run <algorithm-id> [KEY=value ...]
```

The escape hatch: one positional (the algorithm id, e.g. native:slope), then QGIS parameters as KEY=value. Values are coerced (true/false → bool, numbers → int/float, else string); ~ is expanded; a ;-joined value becomes a list; a path with \*/?/[ is globbed. INPUT (from the pipe) and OUTPUT (a temp file) are supplied automatically when omitted. See §6.

```
load dem.tif | run native:slope Z_FACTOR=2 | save slope.tif
run gdal:merge INPUT="a.tif;b.tif;c.tif" DATA_TYPE=5 | save mosaic.tif
```

## split — keep one geometry type (*producing*)

```
split <point|line|polygon>
```

Extract a single geometry type from a mixed layer (native:filterbygeometry). Accepts singular or plural (points, lines, polygons). Pipe again to extract another type. Multipart features are preserved; GeometryCollection features match none of the sinks.

```
load mixed.gpkg | split line | save lines.gpkg
```

## metadata — stamp descriptive metadata (*pass-through*)

```
metadata set <field=value> [field=value ...]
```

Fields: title, abstract, keywords (comma-separated), identifier, license. The metadata persists on the next save.

```
load parcels.gpkg | metadata set title="Residential parcels" keywords="parcels,zoning" |
↳ save parcels.gpkg
```

## assess — data-quality report (*pass-through*)

```
assess [deep] to <report.md>
```

Profiles the current layer to a Markdown report: type, CRS, extent, fields, metadata, and — with **deep** — invalid/empty/duplicate-geometry counts and per-field null counts. Sits mid-pipe (returns the layer unchanged). **deep** may also be written **-deep**.

```
load raw_parcel.gpkg | assess deep to docs/raw_quality.md
```

## catalog — inventory a directory (*terminal*)

```
catalog <dir> [to=<out.md>]
```

Recursively profiles every geospatial dataset under **<dir>** (multi-layer containers expand per layer) to a Markdown inventory — type, CRS, geometry/bands, feature count, extent — listing unreadable files separately. Default output is **<dir>/catalog.md**.

```
catalog "data/" to=reports/inventory.md
```

## show — list available data at a location (*terminal*)

```
show <path|@conn[.schema[.table]]|service-url> [deep] [to=<out.md>]
```

Lists the loadable layers/tables at **one** location and what each is — a quick *"what can I load here, and what's its name?"* glance, the lighter cousin of **catalog** (which **deep**-profiles CRS/extent/fields/counts). Per entry: the **name**, **kind** (vector/raster/table), **type** (geometry like MultiPolygon, or a raster's N band · Float32), **format** (the file driver GPKG / GTiff / SQLite / ..., or the DB provider), and a copy-pasteable **source** you can hand straight to load (or ogrinfo for files). No feature counts — that keeps it instant even on big databases.

**<location>** may be:

- a **file** — show roads.shp, show dem.tif, or a multi-layer container show data.gpkg (one row per layer);
- a **directory** — show data/ lists the immediate children; add the **deep** flag (show data/deep) to recurse the whole tree. **Any QGIS-readable format is picked up** — not just a fixed extension list — so Spatialite (.sqlite/.db), FileGDB (.gdb), and any other GDAL/OGR driver appear; dataset sidecars (a shapefile's .dbf/.shx/...) and obviously non-geospatial files are skipped. Directory-based datasets like .gdb are listed as a container, not descended into;
- a **database connection** — show @conn (every table), show @conn.schema (one schema), show @conn.schema.table (one table). Connection names containing dots (e.g. @actual\_spatialite.sqlite) resolve correctly. Only the connection **name** is used — credentials stay in QGIS;
- a **remote service** — a **WFS** endpoint lists its feature types, a **WMS** endpoint its layers, an **ArcGIS REST** service (.../FeatureServer, .../MapServer, .../ImageServer) its layers and tables, and an **XYZ** {z}/{x}/{y} tile template is itself one layer. The kind is read from a service=WFS/service=WMS query parameter, the URL path/shape, or a WFS:/WMS: prefix (if it can't be told, show asks you to specify). Standard-library HTTP only — public services, no credentials sent; responses are fetched with a timeout and size cap, XML is parsed with DOCTYPE/entity expansion refused (no XXE), and ArcGIS REST JSON is read with json (no entity risk).

```
show "data/basemap.gpkg"          # layers in a GeoPackage
show data/                        # everything directly under data/
show data/ deep                   # recurse the whole tree
show @gisdb3                      # all PostGIS tables
show @gisdb3.public to=tables.md  # one schema, written to a file
show "https://demo.mapserver.org/cgi-bin/wfs?service=WFS"  # WFS feature types
show "https://ows.terrestris.de/osm/service?service=WMS"  # WMS layers
show "https://server/arcgis/rest/services/Foo/FeatureServer" # ArcGIS REST layers
```



```
show "https://tile.osm.org/{z}/{x}/{y}.png" # XYZ tile layer
```

The listing's footer includes **two runnable examples** built from the first row, so you can copy a real Source and pipe it onward (e.g. `load "data.gpkg|layername=roads" | buffer 100m | save out.gpkg`). `show` is terminal and can't itself be piped (neither it nor `catalog` produce or consume a layer); for a deep per-dataset report (CRS, extent, fields, feature counts) run `catalog` on the same location.

## info — inspect the local QGIS environment (*terminal*)

```
info [to=<report.md>]
```

Reports what a CLI user needs to know before writing a flow, especially when working **outside** QGIS where the Browser and connection dialogs aren't in front of you. Most useful: the **registered database connection names** — the valid @conn references for PostGIS and SpatiaLite — since a flow names them but you can't guess them. Because connections are **per QGIS profile**, `info` also lists **every profile and the connections in each**, and marks the active one (niva reads the active profile by default; `NIVA_QGIS_PROFILE=<name>` targets another). Also surfaces the Processing **providers** and reachable **algorithm count** (so you know run `grass:... / run pdal:... will work`), the **versions** (QGIS, Qt/PyQt, GDAL, PROJ, GEOS, SpatiaLite/SQLite, Python), niva's own build + import path, the verb list, and the **environment variables** niva honours (secrets masked — only *set / unset* is shown). With `to=`, writes the Markdown to a file; otherwise prints it to stdout. This is the CLI counterpart of the plugin's Setup-tab **Environment report**.

```
info # print the report
info to=env-report.md # save it
```

(Not to be confused with project `info <src.qgs>`, which inventories a *project file*; bare `info` inventories the *QGIS environment*.)

## project — manipulate QGIS project files (*terminal*)

Five forms; all use a standalone QgsProject (safe off the GUI thread) and all are terminal.

**Repoint / copy / convert** — copy a project, optionally repointing layer datasources:

```
project <src.qgs|qgz> to=<out.qgs|qgz> [repoint=<target>]
      [missing=fail|keep|drop] [rasters=<dir>] [paths=relative|absolute]
      [bookmark=<name> [at="x,y"] [scale=<N>] [width=<W>]]
```

`repoint=` is a `.gpkg` path **or** an `@conn[.schema]` connection; vector layers match by name, subset filters preserved. `missing=` (default **fail**) handles a layer absent from the target. `rasters=<dir>` repoints raster layers by basename. `paths=` rewrites datasource path storage. `bookmark=<name>` adds a spatial bookmark — the layers' **union** extent, or a centred box via `at="x,y"` with `width=` (exact) or `scale=` (approx). Omit `repoint=` to just copy/convert (`.qgs↔.qgz` by extension).

**project new** — build a fresh project from a folder of outputs:

```
project new from=<dir|glob> to=<out.qgs|qgz> [crs=<crs>] [title=<text>]
```

**project info** — inventory a project to Markdown:

```
project info <src.qgs|qgz> [to=<out.md>]
```

**project from-template=** — instantiate a template against your data (see [Template projects](#)):

```
project from-template=<name|path> to=<out.qgs|qgz> data=<dir|glob>
↪ [missing=keep|fail|drop]
```

The template is any existing project; each layer **slot** is repointed to the same-(display)- named dataset under `data=`, symbology and print layouts riding along. Resolves a bare name from `$NIVA_TEMPLATES → ~/.niva/templates → bundled (example)`. `missing=` default `keep`.

**project to-template=** — register an existing project as a reusable template:

```
project to-template=<name|path> from=<src.qgs|qgz> [paths=relative|absolute]
project "basemap.qgs" to="out/basemap.qgs" repoint="out/basemap_clip.gpkg" missing=keep
project from-template=example to="region_atlas.qgz" data="data/clips/"
```

## style — apply or export a layer style (*pass-through*)

```
style apply <file.qml|.qmd>
style save <file.qml|.qmd|.sld|.qlr>
```

apply reads a QGIS sidecar and **persists** it (a GeoPackage layer style goes into the container's layer\_styles table; a single-file layer gets a same-basename sidecar). save exports the current layer's style/metadata — .qml (symbolology), .qmd (metadata), and export-only .sld (OGC) and .qlr (portable layer definition). Chains after save.

```
load roads.gpkg | clip aoi.gpkg | save roads_clip.gpkg | style apply house.qml
```

## notify — push a message via ntfy (*pass-through*)

```
notify "<message>" [to=<topic>] [title=<t>] [priority=<p>] [server=<url>] [tags=<tags>]
```

Posts to an [ntfy](#) topic. to= falls back to NIVA\_NTFY\_TOPIC; server= to NIVA\_NTFY\_SERVER (default https://ntfy.sh); a bearer token comes only from NIVA\_NTFY\_TOKEN. The message interpolates {elapsed}, {last}, {now}, {started}, {ops}, {errors}. See also the auto-alert env vars in §8.

```
... | save out.gpkg | notify "done in {elapsed}, {errors} errors" to=geo-jobs priority=high
```

## email — send via SMTP (*pass-through*)

```
email to=<address> [subject=<s>] [body=<b>] [attach=<file>]
```

SMTP host/credentials come only from the environment (NIVA\_SMTP\_\*, §8); a @gmail.com sender auto-uses smtp.gmail.com:587 (needs a Gmail App Password). TLS is mandatory.

```
load r.gpkg | assess to q.md | email to=ops@example.com subject="Daily run" attach=q.md
```

## each — batch over many datasets (*flow prefix*)

```
each "<dir>" | <stages...> | save <target>
each "<glob>" | <stages...> | save out.gpkg
each <file.gpkg> | <stages...> | save out.gpkg
```

Must be the **first** stage. Resolves a directory (recursed), a glob, or a multi-layer container (expanded per layer) to a list of datasets, then runs the rest of the flow once per dataset. A failing item is **skipped** (logged, counted, warned) so one bad file can't abort the batch; a usage error (bad option/target) still stops everything. See save for per-item output behaviour.

```
each "in/*.shp" | reproject EPSG:6346 | save out.gpkg
```

## call — run another .niva file inline (*statement*)

```
call <other.niva>
```

A top-level statement (not a piped stage): runs another flow file's flows in place, resolved relative to the calling file, with cycle detection — for composing reusable flow libraries.

## 5. Alias verbs (the registry)

Each alias maps one verb to one QGIS algorithm, hiding its parameter names behind readable args/options/flags. The primary input is the piped layer; the output is the temp/saved destination. Source of truth: `niva/registry/definitions.py`. Run `niva describe <verb>` for the live signature.

**Reading the tables:** *Args* are positional (in order); a [name] is optional. *Options* are key= with the default in parentheses. *Flags* are bare switches (off by default). Enum options list their words.

### Vector — core overlay & selection

| Verb          | Algorithm                       | Args              | Options                                                                                                                                                                       | Flags    |
|---------------|---------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| clip          | native:clip                     | overlay (layer)   | —                                                                                                                                                                             | —        |
| intersect     | native:intersection             | overlay (layer)   | —                                                                                                                                                                             | —        |
| difference    | native:difference               | overlay (layer)   | —                                                                                                                                                                             | —        |
| union         | native:union                    | [overlay] (layer) | —                                                                                                                                                                             | —        |
| symdifference | native:symmetricaldifference    | overlay (layer)   | —                                                                                                                                                                             | —        |
| dissolve      | native:dissolve                 | [field]           | —                                                                                                                                                                             | separate |
| filter        | native:extractbyexpression      | expression        | —                                                                                                                                                                             | —        |
| selectloc     | native:extractbylocation        | against (layer)   | predicate= (intersect) (intersect, contain, disjoint, equal, touch, overlap, within, cross) — comma-list                                                                      | —        |
| spatialjoin   | native:joinattributesbylocation | —                 | with=(req, layer) · predicate=(intersect) (intersect, contain, equal, touch, overlap, within, cross) · method=(one-to-many) (one-to-many, first, largest) · fields= · prefix= | discard  |
| join          | native:joinattributetable       | —                 | with=(req) · field=(req) · field2=(req) · fields= · prefix= · method=(one-to-one) (one-to-many, one-to-one) · unmatched=                                                      | discard  |
| sample        | native:randomextract            | number (int)      | method=(count) (count, percent)                                                                                                                                               | —        |

## **Vector — buffering & geometry**

| Verb           | Algorithm                               | Args                         | Options                                                                                                  | Flags               |
|----------------|-----------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------|---------------------|
| buffer         | native:buffer                           | distance                     | segments=(5) · cap=(round) (round, flat, square) · join=(round) (round, miter, bevel) · miter=(2)        | dissolve · separate |
| offset         | native:offsetline                       | distance                     | segments=(8) · join=(round) (round, miter, bevel) · miter=(2)                                            | —                   |
| simplify       | native:simplifygeometries               | tolerance                    | method=(douglas) (douglas, grid, area)                                                                   | —                   |
| smooth         | native:smoothgeometry                   | —                            | iterations=(1) · offset=(0.25) · max_angle=(180)                                                         | —                   |
| densify        | native:densifygeometriesgivenaninterval | interval                     | —                                                                                                        | —                   |
| subdivide      | native:subdivide                        | —                            | max_nodes=(256)                                                                                          | —                   |
| centroid       | native:centroids                        | —                            | —                                                                                                        | —                   |
| pointonsurface | native:pointonsurface                   | —                            | —                                                                                                        | all_parts           |
| convexhull     | native:convexhull                       | —                            | —                                                                                                        | —                   |
| boundingbox    | native:boundingboxes                    | —                            | —                                                                                                        | —                   |
| minrect        | native:orientedminimumboundingbox       | —                            | —                                                                                                        | —                   |
| vertices       | native:extractvertices                  | —                            | —                                                                                                        | —                   |
| explode        | native:multiparttosingleparts           | —                            | —                                                                                                        | —                   |
| promote        | native:promotetomulti                   | —                            | —                                                                                                        | —                   |
| collect        | native:collect                          | [field]                      | —                                                                                                        | —                   |
| fixgeom        | native:fixgeometries                    | —                            | —                                                                                                        | —                   |
| swapxy         | native:swapxy                           | —                            | —                                                                                                        | —                   |
| forcerhr       | native:forcerhr                         | —                            | —                                                                                                        | —                   |
| snap           | native:snapgeometries                   | reference (layer), tolerance | behavior=(align) (align, closest, align-keep, closest-keep, ends-align, ends-closest, ends-only, anchor) | —                   |

## **Vector — CRS, attributes, counting**

| Verb        | Algorithm                   | Args          | Options                                                                                                                                                              | Flags                        |
|-------------|-----------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| reproject   | native:reprojectlayer       | target_crs    | operation=                                                                                                                                                           | convert_curved · transform_z |
| renamefield | native:renametablefield     | field, name   | —                                                                                                                                                                    | —                            |
| dropfields  | native:deletecolumn         | fields (list) | —                                                                                                                                                                    | —                            |
| keepfields  | native:retainfields         | fields (list) | —                                                                                                                                                                    | —                            |
| countpoints | native:countpointsinpolygon | —             | points=(req, layer) · field=(NUMPOINTS) · weight= · classfield=                                                                                                      | —                            |
| zonalstats  | native:zonalstatisticsfb    | —             | raster=(req) · band=(1) · stats=(count,sum,mean) (count, sum, mean, median, stdev, min, max, range, minority, majority, variety, variance) — comma-list · prefix=(_) | —                            |



## Vector — point creation

| Verb        | Algorithm                    | Args     | Options       | Flags |
|-------------|------------------------------|----------|---------------|-------|
| voronoi     | native:voronoipolygons       | —        | buffer=(0)    | —     |
| delaunay    | native:delaunaytriangulation | —        | —             | —     |
| pointsalong | native:pointsalonglines      | distance | start= · end= | —     |

## Raster

Raster verbs default CREATION\_OPTIONS to lossless COMPRESS=DEFLATE|TILED=YES so intermediates aren't left huge; the final product's compression is governed by save.

| Verb       | Algorithm                  | Args         | Options                                                                                                                                                 | Flags            |
|------------|----------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| warp       | gdal:warp                  | target_crs   | source_crs= · resampling=(nearest) (nearest, bilinear, cubic, cubicspline, lanczos, average, mode, max, min, median, q1, q3)<br>· nodata= · resolution= | —                |
| clipraster | gdal:cliprasterbymasklayer | mask (layer) | nodata=                                                                                                                                                 | —                |
| hillshade  | native:hillshade           | —            | z_factor=(1) · azimuth=(300) · altitude=(40)                                                                                                            | —                |
| slope      | gdal:slope                 | —            | band=(1) · scale=(1)                                                                                                                                    | percent          |
| aspect     | gdal:aspect                | —            | band=(1)                                                                                                                                                | trig · zero_flat |
| polygonize | gdal:polygonize            | —            | band=(1) · field=(DN)                                                                                                                                   | eight            |

Anything not aliased is still reachable with `run <algorithm-id> ... (§6)`. For the full catalogue of all 769 algorithms — parameters, defaults, enum options, and descriptions — see the [algorithm appendix](#).

---

## 6. The run escape hatch & describe

`run <id> KEY=value ...` calls any installed QGIS Processing algorithm with its native parameter names — the full catalogue (1000+ native:, gdal:, qgis:, grass:, pdal:, saga: algorithms), not just the aliased verbs. Use it for anything without an alias, or when you need a parameter an alias doesn't expose.

- INPUT is filled from the piped layer; OUTPUT from a temp file — supply either to override.
- Multi-input parameters take a ;-separated list: `INPUT="a.tif;b.tif"`.
- A value with \*/?/[ is globbed relative to the flow file: `INPUT="tiles/*.jp2"`.
- Values coerce: true/false → bool, integers → int, decimals → float, else string.

```
load dem.tif | run native:slope Z_FACTOR=2 NODATA=-9999 | save slope.tif
run gdal:translate INPUT=in.vrt OUTPUT=out.jp2 CREATION_OPTIONS="QUALITY=25"
```

`niva describe <name>` introspects either a niva verb (its alias mapping, args, options, flags — no QGIS needed) or a QGIS algorithm id (its live parameters and outputs — QGIS needed):

```
niva describe buffer
niva describe gdal:warpproject
```

The [algorithm appendix](#) lists all 769 algorithms by provider — each with its parameters (type, required, default, enum options), description, outputs, and niva alias (★) — so you can find and copy a `run <id> ...` for anything without an alias.

---

## 7. Database connections (@conn)

niva reads and writes databases through **saved QGIS connections**. A reference is:

```
@conn          # the connection itself (for sql)
@conn.table     # a table in the connection's default schema
@conn.schema.table # a schema-qualified table
```

@ always means a saved DB connection — never a file (a path is just a path). Used by `load @conn.table`, `save @conn.table`, `sql @conn "..."`, and `project ... reproject=@conn`.

**Default schema:** explicit wins; otherwise public for PostgreSQL/PostGIS, and the provider default (' ') for SpatiaLite/file databases.

**Credentials never enter a flow.** Only the connection *name* crosses from the flow into niva; host, database, user, and password are resolved from QGIS's own connection store (the single source of truth, configured in the QGIS Browser or the plugin Setup tab). niva never stores, logs, or transmits credentials — errors carry only the connection name and table, never the URI, password, or query text. SpatiaLite and PostGIS connections are used identically; the only difference is which engine's spatial SQL functions you write.

---

## 8. Environment variables

| Variable             | Default                  | Purpose                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NIVA_TMPDIR          | system temp              | <b>Disk-backed scratch dir for raster intermediates.</b> Point it at a roomy real-disk folder so a multi-GB raster pipeline doesn't exhaust a small RAM-backed /tmp. Strongly recommended for raster work.                                                                                                     |
| CPL_TMPDIR           | follows NIVA_TMPDIR      | GDAL's own scratch dir; niva points it at NIVA_TMPDIR unless you set it.                                                                                                                                                                                                                                       |
| NIVA_LOG             | unset                    | Default journal base path; writes <base>.jsonl + <base>.log. CLI --log overrides.                                                                                                                                                                                                                              |
| NIVA_TEMPLATES       | ~/.niva/templates        | Extra directory for named project templates; shadows the user library and bundled templates. Also where project to-template=<name> writes.                                                                                                                                                                     |
| NIVA_QGIS_PROFILE    | active desktop profile   | <b>Which QGIS user profile to read</b> for database connections (the @conn names) and other settings, when running standalone. By default niva uses the profile your QGIS desktop last used; set this to another profile's name to target its connections. Run info to see all profiles and their connections. |
| QGIS_PREFIX_PATH     | /usr                     | QGIS install prefix for standalone bootstrap.                                                                                                                                                                                                                                                                  |
| QT_QPA_PLATFORM      | offscreen (set if unset) | Headless Qt platform for standalone runs.                                                                                                                                                                                                                                                                      |
| NIVA_NTFY_TOPIC      | unset                    | Default ntfy topic for notify.                                                                                                                                                                                                                                                                                 |
| NIVA_NTFY_SERVER     | https://ntfy.sh          | ntfy server URL.                                                                                                                                                                                                                                                                                               |
| NIVA_NTFY_TOKEN      | unset                    | Bearer token for protected ntfy topics (never logged).                                                                                                                                                                                                                                                         |
| NIVA_NTFY_ON_ERROR   | unset                    | When truthy, auto-send a high-priority ntfy when a run fails.                                                                                                                                                                                                                                                  |
| NIVA_NTFY_ON_WARNING | unset                    | When truthy, auto-send an ntfy on warnings (mixed geometry, datum transforms, skipped batch items), de-duplicated per run.                                                                                                                                                                                     |
| NIVA_SMTP_HOST       | unset (auto for Gmail)   | SMTP server for email.                                                                                                                                                                                                                                                                                         |
| NIVA_SMTP_PORT       | 587                      | SMTP port; 465 = implicit TLS.                                                                                                                                                                                                                                                                                 |
| NIVA_SMTP_USER       | unset                    | SMTP username.                                                                                                                                                                                                                                                                                                 |
| NIVA_SMTP_PASSWORD   | unset                    | SMTP password (Gmail: an App Password).                                                                                                                                                                                                                                                                        |
| NIVA_SMTP_FROM       | = NIVA_SMTP_USER         | From address; a @gmail.com value auto-selects the Gmail host.                                                                                                                                                                                                                                                  |

Truthy = 1, true, yes, on (case-insensitive). Inside QGIS, the plugin's **Setup** tab sets these for you (and can store secrets in the QGIS encrypted store).

## 9. Command-line interface

The package installs a niva console script (niva.cli.main).

```
niva run <file.niva> [--dry-run | --explain] [--log <base>]
niva "<flow>"      [--dry-run | --explain] [--log <base>]
niva describe <verb-or-algorithm-id>
niva export <file.niva> [-o <file.py>]
niva import <file.py>  [-o <file.niva>]
```

| Command / flag                                 | Effect                                                                          |
|------------------------------------------------|---------------------------------------------------------------------------------|
| niva run flow.niva                             | execute a .niva file (real, via PyQGIS)                                         |
| niva "load a.gpkg   buffer 100m   save b.gpkg" | execute an inline flow                                                          |
| --dry-run                                      | print the plan and validate it over the mock backend (no QGIS, no data touched) |
| --explain                                      | parse + bind only; print the resolved algorithm + parameters per stage          |
| --log <base>                                   | also write the journal <base>.jsonl + <base>.log                                |
| niva describe <name>                           | introspect a verb or algorithm id                                               |
| niva export flow.niva [-o out.py]              | transpile a flow to a standalone PyQGIS script                                  |

| Command / flag                      | Effect                                          |
|-------------------------------------|-------------------------------------------------|
| niva import script.py [-o out.niva] | recover a flow from a niva-shaped PyQGIS script |

**Exit codes:** 0 ok · 1 runtime error (OpError) · 2 usage / parse error (FlowError) · 3 QGIS not importable. Progress and errors go to stderr; result markers to stdout. A real run needs QGIS's Python (see the [User Guide](#)).

## 10. Python API

The public API (import niva) is small and stable:

```
flow(text, *, backend=None, file=None, log=None, log_append=False, progress=None,
    ↪ cancel=None)
run_file(path, *, backend=None)
describe(name)
__version__
NivaError, FlowError, OpError          # exception types
```

### flow(text, ...)

Parse and execute a flow string; returns the final layer handle (its .ref is an on-disk path for a saved file, or a live QgsMapLayer), or None for a terminal flow.

| Parameter  | Default                         | Meaning                                                                     |
|------------|---------------------------------|-----------------------------------------------------------------------------|
| text       | —                               | the flow string                                                             |
| backend    | real PyqgisBackend (needs QGIS) | pass MockBackend() to dry-run without QGIS                                  |
| file       | None                            | path of the source .niva file — sets the base dir for relative paths / call |
| log        | NIVA_LOG                        | journal base path → <log>.jsonl + <log>.log                                 |
| log_append | False                           | True appends to the journal; False truncates                                |
| progress   | None                            | callable(str) — a ► <stage> line per stage plus 45% ticks                   |
| cancel     | None                            | callable() -> bool — return True to abort a long step                       |

```
import niva
layer = niva.flow('load "data.gpkg|layername=roads" | buffer 100m dissolve | save
    ↪ out.gpkg')
print(layer.ref)          # '/abs/path/out.gpkg'
```

### run\_file(path, \*, backend=None)

Read a .niva file and execute it (sets file=path so relative paths resolve).

```
import niva
niva.run_file("myflow.niva")
```

### describe(name)

Return a formatted description string for a verb or an algorithm id (see §6). Pure for a verb; needs QGIS for an algorithm id.

Importing niva is safe on any interpreter — QGIS is imported lazily only when a real run executes. To run for real outside QGIS you need QGIS's Python on PYTHONPATH; see the [User Guide](#) for the exact recipe.

# niva Cookbook

Fifty worked recipes, from a single transform to full pipelines — including a large block of spatial **SQL** for both **Spatialite** and **PostGIS** — plus a closing tour that reaches **every QGIS provider** (GDAL, GRASS, QGIS, PDAL, native, 3D) through the run escape hatch. Pair this with the [Reference](#) for exact signatures and the [User Guide](#) for setup.

## Conventions

- Replace file paths, layer names, CRS codes, and SQL table/column names with your own.
  - A multi-layer GeoPackage layer is addressed as "file.gpkg|layername=<layer>".
  - @name is a **saved QGIS database connection** (set up in the QGIS Browser or the plugin Setup tab) — Spatialite or PostGIS. Credentials live in QGIS, never in the flow.
  - SQL column names below (geom, geometry, id, ...) are illustrative — match your schema.
  - Run any recipe with niva "...", niva run file.niva, or the plugin Flow tab. Add --dry-run to validate without touching data.
- 

## A. First steps — one transform

### 1. Buffer a layer

```
load roads.gpkg | buffer 100m | save roads_buf.gpkg
```

Distances take a unit (m, km, ft, ...); with no unit they're in the layer's CRS units.

### 2. Buffer and merge overlaps

```
load wells.gpkg | buffer 1km dissolve | save well_zones.gpkg
```

dissolve merges the overlapping buffers into one feature.

### 3. Reproject to another CRS

```
load parcels.shp | reproject EPSG:2262 | save parcels_spcs.gpkg
```

### 4. Clip to an area of interest

```
load buildings.gpkg | clip aoi.gpkg | save buildings_aoi.gpkg
```

### 5. Keep only matching features (attribute filter)

```
load parcels.gpkg | filter "zoning = 'R1'" | save residential.gpkg
```

The expression is a QGIS expression — quote it.

### 6. Repair invalid geometries

```
load messy.shp | fixgeom | save clean.gpkg
```

### 7. Reduce a layer to centroids

```
load parcels.gpkg | centroid | save parcel_points.gpkg
```

---

## B. Multi-step chains

### 8. Filter → reproject → buffer

```
load roads.gpkg | filter "class = 'primary'" | reproject EPSG:2262 | buffer 100m dissolve |  
  ↪ save corridors.gpkg
```

### 9. Reproject → fixgeom → buffer with options

```
load buildings.gpkg | reproject EPSG:26918 | fixgeom | buffer 15m dissolve cap=flat  
  ↪ segments=12 | save footprints.gpkg
```

### 10. Clip → fixgeom → dissolve

```
load parcels.gpkg | clip township.gpkg | fixgeom | dissolve zoning | save  
  ↪ zoning_blocks.gpkg
```

### 11. Simplify then split to single parts

```
load streets.gpkg | reproject EPSG:26918 | simplify 5m method=area | explode | save  
  ↪ street_segments.gpkg
```

### 12. Stamp metadata and assess in one pass

```
load wells.gpkg | metadata set title="Monitoring wells" keywords="wells,gw" | assess deep  
  ↪ to docs/wells_quality.md | save wells.gpkg
```

metadata and assess are pass-through, so the chain still ends at save.

---

## C. Attributes, fields & joins

### 13. Keep only the fields you need

```
load parcels.gpkg | keepfields owner,zoning,area | save parcels_slim.gpkg
```

### 14. Drop fields / rename a field

```
load parcels.gpkg | dropfields temp1,temp2 | renamefield zoning zone_code | save  
  ↪ parcels.gpkg
```

### 15. Attribute join from a table

```
load homes.gpkg | join with=census.csv field=tract field2=GE0ID fields=pop,income  
  ↪ prefix=cen_ | save homes_enriched.gpkg
```

### 16. Join, keeping only matches

```
load homes.gpkg | join with=owners.gpkg field=parcel_id field2=pid discard | save  
  ↪ owned_homes.gpkg
```

discard drops input rows that found no match.

### 17. Count points inside each polygon

```
run native:countpointsinpolygon POLYGONS=parcels.gpkg POINTS=trees.gpkg FIELD=n_trees |  
  ↪ save parcels_treecount.gpkg
```

countpoints takes both layers as options, so call it without a piped input.

---

## D. Overlay & spatial selection

### 18. Intersection of two layers

```
load parcels.gpkg | intersect floodzone.gpkg | save parcels_in_flood.gpkg
```

### 19. Difference (erase)

```
load parcels.gpkg | fixgeom | difference buildings.gpkg | save open_land.gpkg
```

### 20. Select by location (spatial filter)

```
load buildings.gpkg | selectloc floodzone.gpkg predicate=intersect,within | save  
↪ at_risk.gpkg
```

### 21. Spatial join — attach attributes by location

```
load incidents.gpkg | spatialjoin with=districts.gpkg predicate=within method=first  
↪ fields=district_name | save incidents_tagged.gpkg
```

### 22. Zonal statistics (raster × polygons)

```
load watersheds.gpkg | zonalstats raster=dem.tif stats=mean,min,max prefix=elev_ | save  
↪ watersheds_elev.gpkg
```

---

## E. Batch processing with each

### 23. Reproject every file in a folder into one GeoPackage

```
each "in/*.shp" | reproject EPSG:6346 | save out.gpkg
```

Each input becomes its own layer inside out.gpkg (named after the file).

### 24. Process every layer of a multi-layer GeoPackage

```
each "basemap.gpkg" | fixgeom | clip aoi.gpkg | save basemap_clip.gpkg
```

### 25. Recurse a directory tree

```
each "data/" | reproject EPSG:6346 | save collected.gpkg
```

### 26. Per-item output files with {name}

```
each "rasters/*.tif" | warp EPSG:6346 | save "warped/{name}.tif"
```

### 27. Batch-load a folder into a PostGIS schema (one table per file)

```
each "deliverables/" | reproject EPSG:6346 | save @pg.staging
```

A trailing qualifier on a batch DB save is the **schema** (staging); each layer becomes a table there.

---

## F. Raster & terrain

### 28. Reproject (warp) a raster

```
load dem.tif | warp EPSG:6346 | save dem_utm.tif
```

### 29. Warp and resample to a target pixel size

```
load ortho.tif | warp EPSG:6346 resolution=5 resampling=average | save ortho_5m.tif
```

average is the right resampler when down-sampling imagery.

### 30. Clip a raster to a mask layer



```
load dem.tif | clipraster aoi.gpkg | save dem_aoi.tif
```

### 31. Hillshade from a DEM

```
load dem.tif | hillshade z_factor=2 azimuth=315 altitude=45 | save hillshade.tif
```

### 32. Slope (in percent) and aspect

```
load dem.tif | slope percent | save slope_pct.tif  
load dem.tif | aspect | save aspect.tif
```

### 33. Vectorise a classified raster

```
load landcover.tif | polygonize field=class | save landcover_zones.gpkg
```

---

## G. Spatial SQL in Spatialite

These use a saved **Spatialite** connection @sl. A SELECT/WITH returns a layer you can pipe; anything else runs in the database. Adjust table/column/geometry names to your schema (Spatialite geometry columns are often geometry or geom).

### 34. Read a filtered subset as a layer

```
sql @sl "SELECT * FROM parcels WHERE zoning = 'R1'" | save residential.gpkg
```

### 35. Filter by a spatial measure

```
sql @sl "SELECT id, zoning, geometry FROM parcels WHERE ST_Area(geometry) > 2000" | save  
↪ big_parcel.gpkg
```

### 36. Compute geometry server-side, then keep processing in niva

```
sql @sl "SELECT id, ST_Buffer(geometry, 50) AS geometry FROM wells" | reproject EPSG:6346 |  
↪ save well_buffers.gpkg
```

### 37. Spatial join — points within polygons

```
sql @sl "SELECT p.id, z.zone, p.geometry FROM points p JOIN zones z ON  
↪ ST_Within(p.geometry, z.geometry)" | save points_zoned.gpkg
```

### 38. Aggregate to an attribute table

```
sql @sl "SELECT zoning, COUNT(*) AS n, SUM(ST_Area(geometry)) AS total_area FROM parcels  
↪ GROUP BY zoning" | save zoning_summary.gpkg
```

### 39. Dissolve by attribute with ST\_Union

```
sql @sl "SELECT zoning, ST_Union(geometry) AS geometry FROM parcels GROUP BY zoning" | save  
↪ zoning_dissolved.gpkg
```

### 40. CTE, then hand off to a niva verb

```
sql @sl "WITH big AS (SELECT * FROM parcels WHERE ST_Area(geometry) > 5000) SELECT * FROM  
↪ big" | centroid | save big_centroids.gpkg
```

---

## H. Spatial SQL in PostGIS

These use a saved **PostGIS** connection @pg. Reads (SELECT/WITH) return a pipeable layer; writes (CREATE/UPDATE/INSERT/CREATE INDEX/...) run server-side and end the flow.

### 41. Read a table (schema-qualified)

```
load @pg.public.roads | save roads_local.gpkg
```

Equivalent read via SQL: `sql @pg "SELECT * FROM public.roads" | save roads_local.gpkg.`

#### 42. Filter in the database, finish in niva

```
sql @pg "SELECT * FROM roads WHERE class = 'primary'" | buffer 50m dissolve | save  
↪ primary_corridors.gpkg
```

#### 43. Do the spatial work in SQL (the server-side lever)

```
sql @pg "SELECT id, ST_Buffer(geom, 100) AS geom FROM homes WHERE has_cat AND NOT has_dog"  
↪ | save targets.gpkg
```

#### 44. Cross-table spatial join

```
sql @pg "SELECT a.id, a.geom FROM parcels a JOIN flood b ON ST_Intersects(a.geom, b.geom)"  
↪ | save parcels_at_risk.gpkg
```

#### 45. Write a niva result back into PostGIS

```
load parcels.gpkg | clip aoi.gpkg | save @pg.public.parcels_clip mode=replace  
mode=create (default) refuses to overwrite; replace drops + recreates; append inserts.
```

#### 46. Analyse-in-place: materialise a derived table

```
sql @pg "CREATE TABLE roads_buf AS SELECT id, ST_Buffer(geom, 100) AS geom FROM roads"
```

A leading CREATE routes server-side (terminal) — even CREATE TABLE ... AS SELECT ...

#### 47. Maintenance — update a column and add a spatial index

```
sql @pg "UPDATE parcels SET area_m2 = ST_Area(geom)"  
sql @pg "CREATE INDEX ON roads_buf USING GIST (geom)"
```

---

## I. Projects & templates

#### 48. Repoint a QGIS project to clipped data

```
project "city.qgs" to="out/city_aoi.qgs" repoint="out/clipped.gpkg" missing=keep
```

Copies the project and points each layer at the same-named layer in clipped.gpkg.

#### 49. Build a fresh project from a folder of outputs

```
project new from="out/" to="deliverable.qgz" crs=EPSG:6346 title="Deliverable"
```

#### 50. Instantiate a styled, laid-out template against your data

```
project from-template=example to="report.qgz" data="mydata/"
```

The bundled example template (boundary/roads/places slots + print layout) repoints to your same-named data, symbology and layout riding along. Register your own designed project with project to-template=<name> from="MyMap.qgz" paths=relative, then call it by name. See [Template projects](#).

---

## K. Reaching every provider with run

niva gives 45 algorithms friendly verbs; the other ~720 — across **every** Processing provider — are reachable with run <id> KEY=value .... Four per provider below. Find any algorithm's parameters with niva describe <id> or the [algorithm appendix](#).

Two things to know when using run directly:

- **Enum options take their integer index**, not the alias word — format=0, not format=degrees. (The index list is in each algorithm's appendix entry.)
- **Some providers (GRASS, PDAL) write *named outputs*** (output=, slope=, ...) instead of OUTPUT, so the step is terminal — give the output path directly rather than piping to save. GRASS and PDAL also need their backends installed.

## GDAL (gdal:) — raster/vector via GDAL/OGR

### 51. Contour lines from a DEM

```
load dem.tif | run gdal:contour INTERVAL=10 FIELD_NAME=ELEV | save contours.gpkg
```

### 52. Rasterize a vector field (10-unit pixels)

```
load parcels.gpkg | run gdal:rasterize FIELD=value UNITS=1 WIDTH=10 HEIGHT=10 | save  
↳ parcels_value.tif
```

### 53. Proximity (distance-to-target) raster

```
load targets.tif | run gdal:proximity UNITS=0 MAX_DISTANCE=500 | save distance.tif
```

### 54. Fill small NoData gaps by interpolation

```
load gappy.tif | run gdal:fillnodata DISTANCE=20 | save filled.tif
```

## GRASS (grass:) — named outputs (terminal), enums as integers

### 55. Slope and aspect from a DEM

```
run grass:r.slope.aspect elevation=dem.tif format=0 slope=slope.tif aspect=aspect.tif
```

### 56. Watershed flow accumulation and basins

```
run grass:r.watershed elevation=dem.tif threshold=1000 accumulation=flowacc.tif  
↳ basin=basins.tif
```

### 57. Least-cost surface from a start point

```
run grass:r.cost input=cost_surface.tif start_coordinates=600100,4800100  
↳ output=cumulative_cost.tif
```

### 58. Viewshed from an observer

```
run grass:r.viewshed input=dem.tif coordinates=600100,4800100 observer_elevation=1.75  
↳ max_distance=5000 output=visible.tif
```

## QGIS (qgis:) — the QGIS-Python toolbox

### 59. Virtual-layer SQL across files (input1, input2, ...)

```
run qgis:executesql INPUT_DATASOURCES="roads.gpkg;parcels.gpkg" INPUT_QUERY="SELECT * FROM  
↳ input1 WHERE class = 'primary'" | save primary.gpkg
```

This is the file-backed SQL form (the sql verb itself only queries @conn databases).

### 60. Concave hull (k-nearest neighbour)

```
load points.gpkg | run qgis:knearestconcavehull KNEIGHBORS=5 | save hull.gpkg
```

### 61. Random sample points inside polygons

```
load tracts.gpkg | run qgis:randompointsinsidepolygons STRATEGY=0 VALUE=100  
↳ MIN_DISTANCE=50 | save sample.gpkg
```

### 62. Spread overlapping (stacked) points apart

```
load stacked.gpkg | run qgis:pointsdisplacement PROXIMITY=5 DISTANCE=10 HORIZONTAL=false |  
↪ save displaced.gpkg
```

## **PDAL (pdal:) — point clouds (use run; load/save are vector/raster)**

### **63. Export a DEM from a LiDAR cloud, then hillshade it**

```
run pdal:exportraster INPUT=lidar.copc.laz ATTRIBUTE=Z RESOLUTION=1 | hillshade z_factor=2  
↪ | save lidar_hillshade.tif
```

exportraster outputs a raster (OUTPUT), so it pipes into niva's raster verbs.

### **64. Classify ground returns**

```
run pdal:classifyground INPUT=lidar.laz OUTPUT=ground_classified.laz
```

### **65. Extract the cloud's data-boundary polygon**

```
run pdal:boundary INPUT=lidar.laz RESOLUTION=10 THRESHOLD=10 OUTPUT=cloud_extent.gpkg
```

### **66. Thin a dense cloud by sampling radius**

```
run pdal:thinbyradius INPUT=lidar.laz SAMPLING_RADIUS=2 OUTPUT=thinned.laz
```

## **Native (native:) — beyond the alias verbs**

### **67. Join the nearest feature from another layer**

```
load schools.gpkg | run native:joinbynearest INPUT_2=parcels.gpkg NEIGHBORS=1  
↪ MAX_DISTANCE=500 | save schools_parcel.gpkg
```

### **68. DBSCAN point clustering**

```
load incidents.gpkg | run native:dbscanclustering MIN_SIZE=5 EPS=250 | save  
↪ incident_clusters.gpkg
```

### **69. Shortest path over a road network**

```
load roads.gpkg | run native:shortestpathpointtopoint STRATEGY=0  
↪ START_POINT="600100,4800100" END_POINT="601000,4801000" | save route.gpkg
```

### **70. Hub-and-spoke (spider) lines**

```
run native:hublines HUBS=depots.gpkg HUB_FIELD=id SPOKES=stops.gpkg SPOKE_FIELD=depot_id |  
↪ save spider.gpkg
```

## **3D (3d:)**

### **71. Tessellate polygons into 3D geometry**

```
load buildings.gpkg | run 3d:tessellate | save buildings_3d.gpkg
```

The 3d: provider ships a single algorithm in QGIS 4.0.3 — tessellate — so this is the whole provider.

---

## **Capstone — full pipelines**

Two complete, runnable flows ship in [examples/](#):

- **analyst\_plan.niva** — catalog data sources → reproject/warp everything to a target CRS → clip to a study-area bounding box → repoint QGIS projects to the clips. A whole regional compilation, batched end to end.

- **youngstown\_cat\_canvassing.niva** — a multi-step analysis: assess → reproject → clip → geocode → server-side SQL selection → terrain routing → atlas export.

Run one with `niva run examples/analyst_plan.niva` (set `NIVA_TMPDIR` to a roomy disk first for the raster steps — see the [User Guide](#)).

# Template projects

A **niva template is just a saved QGIS project**. There is no special format: you design a project in QGIS — print layout, styled layers, bookmarks, the lot — and niva reuses it against fresh data. `project from-template=` copies the template and **repoints each layer to your same-named data**, so everything you designed rides along.

```
# instantiate a template against your data
project from-template="my_basemap.qgz" to="acme.qgz" data="acme/clips/"
```

```
# ...or use the bundled example, by name
project from-template=example to="acme.qgz" data="acme/clips/"
```

This page explains **what a template can carry**, **how slots are matched**, and **how to author a good one** — including the bundled example you can open in QGIS as a starting point.

---

## How instantiation works

```
project from-template=<name|path> to=<out> data=<dir|glob> [missing=keep|fail|drop]
```

1. niva **copies the whole template project** — so every project-level thing you designed (layouts, bookmarks, themes, layer tree, title, CRS) is preserved verbatim.
2. For each layer (a **slot**), niva **repoints its datasource** to the dataset in `data=` whose name matches the slot, **preserving the layer's symbology and subset filter**.
3. The result is written to `to=` (`.qgs` or `.qgz` by extension). Nothing else changes.

`data=` is resolved like each `/` project new: a **directory** (recursed), a **glob**, or a **multi-layer container** (a GeoPackage, expanded per layer). Each dataset's **name** is its file stem (`parcels.gpkg` → `parcels`) or its `|layername=` inside a container.

## Slot matching — by display name

A slot's identity is the **layer's display name** — the label in the QGIS *Layers* panel — falling back to the datasource name (`|layername=`, else file stem) when the display name isn't found. So a layer shown as **parcels** is filled by **parcels.gpkg** in `data=`, *regardless of the placeholder data the template currently points at*.

**Author tip:** name your template's layers exactly what you'll name the matching data files. Rename a layer in the *Layers* panel (double-click) to set its slot name.

Slots with no match in `data=` follow `missing=`:

| missing=                | Unmatched slot behaviour                                                                |
|-------------------------|-----------------------------------------------------------------------------------------|
| keep ( <i>default</i> ) | left pointing at the template's example data — <b>preserves layout/legend structure</b> |
| drop                    | removed from the project                                                                |

|          |                                                  |
|----------|--------------------------------------------------|
| missing= | <b>Unmatched slot behaviour</b>                  |
| fail     | error (never silently produce a half-filled map) |

## What a template can carry

Because from-template copies the project and only swaps datasources, **everything in the project file rides along**. Author it once; it all comes through.

### Project-level — carried verbatim

| Element                                    | Carried? | Notes                                                                                                                                                            |
|--------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Print layouts</b>                       | ✓        | All items: map frames, titles/labels, <b>legend</b> , <b>scale bar</b> , pictures / north arrows, attribute tables, HTML frames, and <b>atlas</b> configuration. |
| <b>Spatial bookmarks</b>                   | ✓        | Jump-to extents (see project ... bookmark=).                                                                                                                     |
| <b>Map themes</b>                          | ✓        | Visibility / style presets.                                                                                                                                      |
| <b>Layer tree</b>                          | ✓        | Group structure, layer order, visibility, expanded state.                                                                                                        |
| <b>Project title &amp; CRS</b>             | ✓        |                                                                                                                                                                  |
| <b>Project metadata, colors, variables</b> | ✓        | Project color scheme, custom project variables.                                                                                                                  |

### Per-layer (per-slot) — carried, datasource repointed

| Element                                            | Carried?           | Notes                                                                                                                        |
|----------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Symbology / renderer</b>                        | ✓                  | The <i>style</i> — single/categorized/graduated/rule-based, including data-defined overrides. Rides along onto the new data. |
| <b>Display name</b>                                | ✓                  | This is the <b>slot key</b> .                                                                                                |
| <b>Opacity &amp; blend mode</b>                    | ✓                  |                                                                                                                              |
| <b>Labels</b>                                      | ✓                  | Label placement / expressions.                                                                                               |
| <b>Subset / filter string</b>                      | ✓                  | Re-applied to the <b>new</b> data (so the filter's fields must exist there — see caveats).                                   |
| <b>Layer metadata</b>                              | ✓                  |                                                                                                                              |
| <b>Joins, diagrams, actions, form/field config</b> | ✓                  | Carried as layer properties; schema-dependent ones assume matching fields.                                                   |
| <b>Datasource</b>                                  | ↻ <b>repointed</b> | To the same-named dataset in data= (vector → ogr, raster → gdal).                                                            |

### Not carried / unchanged

- **Layers that aren't vector or raster** are left untouched.
- **Slots with no matching data** follow missing= (above).
- from-template does **not** rewrite datasource **path storage** — that's a to-template option (paths=relative).

## Caveats when authoring

- **Schema-dependent styling.** A categorized/graduated/rule-based renderer, a subset filter, a join, or a label expression references **field names / values**. The style rides along, but it only *renders* correctly if your data has those fields. Keep slot schemas consistent, or use simple symbology in the template.

- **Layout map extent.** A layout map frame stores a fixed extent (the template's example area). After instantiation it still shows that extent — which may not cover your data. Pan the layout map to your area, or add a **bookmark** at your AOI and use it. (A future option may auto-fit the layout map to the new data.)
  - **Geometry type.** Symbology assumes a geometry type — fill the boundary (polygon) slot with polygon data, roads (line) with lines, etc. Mismatched geometry still repoints but the symbol won't apply meaningfully.
- 

## Authoring a template, end to end

1. **Design in QGIS.** Add your layers, style them, build a print layout, set bookmarks / themes / title / CRS. Save as .qgz.
2. **Name the slots.** Rename each layer in the *Layers* panel to the name you'll give the matching data file (e.g. parcels, roads, boundary).
3. *(Optional)* **Make it portable.** Register it into your library with relative paths:  
`project to-template=parcel_map from="MyParcelMap.qgz" paths=relative`  
`to-template=<name>` copies the project into the **template library** — \$NIVA\_TEMPLATES if set, else ~/.niva/templates — so `from-template=<name>` finds it. A **path** value (`to-template="out/tmpl.qgz"`) writes anywhere instead.
4. **Instantiate** against any dataset whose layers are named for your slots:  
`project from-template=parcel_map to="acme.qgz" data="acme/parcels/"`

## Where named templates resolve

`from-template=<name>` (a bare name, no path) is looked up in order:

1. \$NIVA\_TEMPLATES (if set)
2. ~/.niva/templates (your personal library)
3. the **bundled** templates that ship inside niva (e.g. example)

An earlier hit shadows a later one, so a personal `example.qgz` overrides the bundled one. An unknown name errors with the available names listed.

---

## The bundled example template

niva ships a fully-populated example template (under `niva/templates/`, regenerated by `templates/build_example.py`) so you can see exactly what a good template contains. It has:

- **Three styled slots:** boundary (polygon outline), roads (line), places (point) — each with distinct symbology.
- **A print layout** (Map): title, map frame, legend, scale bar.
- **A spatial bookmark** (Study Area) and a project **title + CRS**.

Open `niva/templates/example.qgz` in QGIS to inspect it, or instantiate it directly:

```
project from-template=example to="my_map.qgz" data="my_data/"
```

where `my_data/` holds `boundary.*`, `roads.*`, and `places.*` (any OGR vector format). The slots repoint to your data; the layout, legend, scale bar, bookmark, title, and CRS all ride along. Copy it into `~/.niva/templates/` and edit it to make your own.



# FAQ

Short answers to common questions. See the [User Guide](#) for setup, the [Reference](#) for the full surface, and the [Cookbook](#) for recipes.

## What software libraries are needed to use niva?

Almost nothing beyond **QGIS** itself. niva is **pure Python with zero third-party runtime dependencies** — it runs on QGIS's own bundled Python.

- **Required: QGIS 3.22+ (or 4.x)** with its Processing framework. QGIS already ships everything niva relies on — GDAL/OGR, PROJ, GEOS, and the native:, qgis:, and gdal: algorithm providers. There is **no pip install** for the plugin (it bundles niva), and even the standalone package installs into QGIS's Python with no extra dependencies.
- **Optional providers** (only if you call those algorithms via run): **GRASS GIS** for the grass: algorithms and **PDAL** for the pdal: point-cloud algorithms. Install these the usual way for your platform; niva just calls them through QGIS Processing.
- **Databases:** handled entirely by QGIS's own providers — **PostgreSQL/PostGIS** (libpq) and **Spatialite** (mod\_spatialite), both bundled with QGIS. You configure a connection in QGIS; niva references it as @conn.
- **notify / email:** Python standard library only (urllib, smtplib) — no extra libs.

The only thing that needs extra tools is **building the guide as a PDF** (scripts/build\_guide\_pdf.py): that needs **pandoc** + a LaTeX engine (e.g. xelatex). Those are *not* needed to use niva.

## Do I need to know Python or PyQGIS?

No. That's the point — you write a readable one-line flow instead of PyQGIS code:

```
load roads.gpkg | buffer 100m dissolve | clip city.gpkg | save roads_local.gpkg
```

If you *want* the Python, niva can **export any flow to a standalone PyQGIS script** (niva export, or the plugin's Convert tab).

## How do I run niva — in QGIS or from the command line?

Both, identically:

- **In QGIS** — install the plugin, open its dock, type a flow, hit **Run**. No pip step.
- **Standalone** — the niva CLI / Python API, run with QGIS's own Python (it boots QGIS headless). See the [User Guide](#).

## I typed niva in a terminal and nothing happened (or "command not found")

Two things have to be true for the niva command to work, and a fresh shell usually has neither:

1. **There's a niva command on your PATH.** niva isn't installed as a command until you either `pip install` it into a Python (which creates the niva script) or alias it (below).
2. **It runs on QGIS's Python.** niva needs the PyQGIS bindings (`qgis.core`, `Processing`) — your system `python3` usually can't import them, only the Python that QGIS ships with.

The quickest fix — **no install, just an alias. Two parts must point at *your* system — don't paste the line as-is:**

- replace `/path/to/niva` with the directory that contains the niva/ package (the repo root), and
- use the **full path to QGIS's Python**, not a bare `python3`. A bare `python3` often resolves to a Homebrew / `pyenv` / `conda` Python that *can't* import QGIS — you'll get `No module named 'niva'` or a `PyQGIS ImportError`.

*# Template – substitute both placeholders:*

```
alias niva='PYTHONPATH=/path/to/niva:/usr/share/qgis/python:/usr/lib/python3/dist-packages
↳ QT_QPA_PLATFORM=offscreen /path/to/qgis/python3 -m niva.cli.main'
```

*# Concrete example (Linux, distro QGIS, repo at ~/Github/niva):*

```
alias niva='PYTHONPATH=/home/jcz/Github/niva:/usr/share/qgis/python:/usr/lib/python3/dist-
↳ packages QT_QPA_PLATFORM=offscreen /usr/bin/python3.14 -m niva.cli.main'
niva describe buffer # now it works
```

To find QGIS's Python + paths, open the QGIS **Python Console** and run: `import sys, qgis; print(sys.executable); print([p for p in sys.path if 'qgis' in p.lower()])` — use that `sys.executable` as the interpreter above. Then `niva info` confirms it's wired to your real QGIS profile (it lists your `@conn` connections). Or install the command into QGIS's Python — full instructions in the [User Guide](#). Sanity-check the parser without QGIS using `--explain: <qgis-python> -m niva.cli.main "load a.gpkg | buffer 100m | save b.gpkg" --explain`.

## Is niva on PyPI? How do I install it?

Not yet on PyPI. Get it as the **plugin zip** (`niva_qgis.zip`, from the [latest release](#) or `plugin/build_plugin.sh`) and *Install from ZIP* in QGIS, or install the package from source into QGIS's Python:

```
<qgis-python> -m pip install git+https://github.com/johnzastrow/niva.git
```

## Which QGIS versions are supported?

QGIS **3.22+** (Qt5) and **QGIS 4.x** (Qt6). The plugin declares `qgisMinimumVersion=3.22`; development targets QGIS 4.0.3.

## A raster job fails with "disk quota exceeded" or "No space left" even though I have free disk — why?

Raster steps write multi-gigabyte intermediates, and by default they land in the system temp dir — often a small RAM-backed `/tmp`. Point niva's scratch at a roomy disk-backed folder:

```
export NIVA_TMPDIR=$HOME/niva_scratch
```

(or set the **Raster scratch** field in the plugin's Setup tab). Scratch is purged on every run, even a failed one. See the [User Guide](#).

## How do I read and write a database?

Configure a PostGIS or Spatialite connection in QGIS, then reference it by name as @conn:

```
load @pg.public.roads | clip aoi.gpkg | save @pg.public.roads_clip mode=replace
sql @pg "SELECT id, ST_Buffer(geom,100) AS geom FROM homes WHERE has_cat" | save
↳ targets.gpkg
```

Only the connection **name** appears in a flow — credentials stay in QGIS. See the [Reference](#).

## How do I see what layers/tables are in a file or database?

Use show — it lists the loadable layers at one location with each one's kind, geometry/raster type, format, and a copy-pasteable load source:

```
show data.gpkg           # layers in a GeoPackage (or any container/shapefile/GeoTIFF)
show data/               # everything directly under a directory
show data/ deep          # recurse the whole tree
show @gisdb3             # all tables in a PostGIS connection
show @gisdb3.public      # just one schema
show "https://host/geoserver/wfs?service=WFS" # remote WFS feature types
show "https://host/geoserver/ows?service=WMS" # remote WMS layers
show "https://host/arcgis/rest/services/X/FeatureServer" # ArcGIS REST layers
show "https://tile.osm.org/{z}/{x}/{y}.png"   # XYZ tile layer
```

The listing's footer shows two runnable examples built from the first row (e.g. load "...|layername=roads" | buffer 100m | save out.gpkg). Heads-up for the shell: the Source cells are wrapped in Markdown backticks — copy the value *inside* them, and quote the whole flow so your shell doesn't run the backticks or split on the |: niva 'load "...|layername=roads"'.

Directory listings are **shallow by default** — add the deep flag to recurse. Any QGIS-readable format is picked up (Spatialite, FileGDB, ... — not a fixed extension list); dataset sidecars and non-geospatial files are skipped. It's the quick "*what can I load here?*" glance; for a deep per-dataset inventory (CRS, extent, fields, feature counts) use catalog instead (the two can't be piped together — both are terminal). For files you can take show's source column straight to ogrinfo <file> <layer>.

## How do I find my database connection names (the @conn values)?

Run info. From a shell — where the QGIS Browser isn't in front of you — this is the quickest way to see the registered PostGIS/Spatialite connections, plus your QGIS/GDAL/PROJ versions, the reachable algorithm count, and the niva environment variables (secrets masked):

```
niva info                # prints the report; the "Database connections" section lists
↳ each @conn
niva info to=env.md      # or save it to a file
```

It's the CLI counterpart of the plugin's Setup-tab **Environment report**. See the [Reference](#).

## Standalone niva shows different connections than my QGIS — why?

Connections are **per QGIS user profile**. A standalone niva reads the **same profile your QGIS desktop last used**, so the @conn names match the GUI. (Earlier builds accidentally read a generic, empty Qt settings store and saw none of your connections — fixed in 0.28.0.) If you still see a mismatch, you're probably looking at a *different* profile than you think — run `niva info` and check the **QGIS profiles** section, which lists every profile and its connections.

## Can niva use a connection from another QGIS profile?

Yes — niva uses **one** profile at a time. `niva info` lists all profiles and their connections; to use a connection that lives in another profile, point niva at it:

```
NIVA_QGIS_PROFILE=staging niva load @prod_db.public.roads | save roads.gpkg
```

(Inside QGIS, niva always uses the profile you have open.)

## Can I use an algorithm that doesn't have a niva verb?

Yes. The ~45 [alias verbs](#) are conveniences; **every** QGIS algorithm (769 in QGIS 4.0.3) is reachable with `run <id> KEY=value ...`. Discover one with `niva describe <id>`, or browse the [algorithm appendix](#) — each entry has a worked example.

## How do I turn a niva flow into a normal PyQGIS script?

`niva export flow.niva -o flow.py` (or the plugin's Convert tab) writes a standalone PyQGIS script — one `processing.run(...)` per step. `niva import` reverses it for niva-shaped scripts.

## Does niva send my data anywhere, or need an internet connection?

No. niva runs locally on QGIS Processing; it never transmits your data or credentials. The only outbound features are **opt-in**: notify (ntfy), email (SMTP), and any algorithm that is itself online (e.g. a geocoder you call via `run`). Their credentials come **only** from the environment, never from flow text.

## How do I get the whole guide as one document?

Build a single PDF of this guide (and the full algorithm appendix) with:

```
python3 scripts/build_guide_pdf.py # -> docs/guide/niva-guide.pdf
```

It's also attached to each [release](#) as `niva-guide.pdf`.

# QGIS algorithm appendix

Every QGIS Processing algorithm reachable from niva via `run <id> KEY=value ...` — **769 algorithms** across 6 providers (QGIS 4.0.3). For each: its parameters (with types, defaults, and enum options), description, outputs, a worked **Example usage**, and which niva **alias verb** (if any) wraps it (★).

This is auto-generated — regenerate with `scripts/gen_algorithms.py` after a QGIS upgrade. Most users only need the 45 [alias verbs](#); this appendix is for reaching everything else through `run`. Discover one live with `niva describe <id>`.

| Provider     | Algorithms | niva alias verbs | Reference                 |
|--------------|------------|------------------|---------------------------|
| 3d:          | 1          | 0                | <a href="#">3d.md</a>     |
| gdal:        | 59         | 5                | <a href="#">gdal.md</a>   |
| grass:       | 307        | 0                | <a href="#">grass.md</a>  |
| native:      | 339        | 40               | <a href="#">native.md</a> |
| pdal:        | 24         | 0                | <a href="#">pdal.md</a>   |
| qgis:        | 39         | 0                | <a href="#">qgis.md</a>   |
| <b>Total</b> | <b>769</b> | <b>45</b>        |                           |

# native: algorithms

native — QGIS native (C++) algorithms — 339 algorithms (QGIS 4.0.3). Auto-generated by scripts/gen\_algorithms.py; see [the index](#).

Call any of these with run <id> KEY=value ... (the **Name** column is the KEY); each entry includes a worked **Example usage**. A ★ marks an algorithm with a friendly niva alias verb.

## native:addautoincrementalfield — Add autoincremental field

**group:** Vector table

This algorithm adds a new integer field to a vector layer, with a sequential value for each feature.

This field can be used as a unique ID for features in the layer. The new attribute is not added to the input layer but a new layer is generated instead.

The initial starting value for the incremental series can be specified.

Specifying an optional modulus value will restart the count to START whenever the field value reaches the modulus value.

Optionally, grouping fields can be specified. If group fields are present, then the field value will be reset for each combination of these group field values.

The sort order for features may be specified, if so, then the incremental field will respect this sort order.

| Parameter                | Type       | Required | Default | Description      |
|--------------------------|------------|----------|---------|------------------|
| INPUT                    | source     | yes      | —       | Input layer      |
| FIELD_NAME               | string     | no       | AUTO    | Field name       |
| START                    | number     | no       | 0       | Start values at  |
| MODULUS                  | number     | no       | 0       | Modulus value    |
| GROUP_FIELDS             | field      | no       | —       | Group values by  |
| SORT_EXPRESSION *(adv)*  | expression | no       | —       | Sort expression  |
| SORT_ASCENDING *(adv)*   | boolean    | no       | True    | Sort ascending   |
| SORT_NULLS_FIRST *(adv)* | boolean    | no       | False   | Sort nulls first |
| OUTPUT                   | sink       | yes      | —       | Incremented      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:addautoincrementalfield INPUT=input.gpkg FIELD_NAME=AUTO START=0 MODULUS=0  
↳ GROUP_FIELDS=field1 OUTPUT=output.gpkg
```

This calls **Add autoincremental field**: INPUT=input.gpkg — Input layer; FIELD\_NAME=AUTO — Field name; START=0 — Start values at; MODULUS=0 — Modulus value; GROUP\_FIELDS=field1 — Group values by; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:addfieldtoattributetable — Add field to attributes table

**group:** Vector table

This algorithm adds a new attribute to a vector layer.

The name and characteristics of the attribute are defined as parameters. The new attribute is not added to the input layer but a new layer is generated instead.

| Parameter       | Type   | Required | Default | Description                                                                                                                                                                                              |
|-----------------|--------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT           | source | yes      | —       | Input layer                                                                                                                                                                                              |
| FIELD_NAME      | string | yes      | —       | Field name                                                                                                                                                                                               |
| FIELD_TYPE      | enum   | no       | 0       | Field type — options: 0=Integer (32 bit), 1=Decimal (double), 2=Text (string), 3=Boolean, 4=Date, 5=Time, 6=Date & Time, 7=Binary Object (BLOB), 8=String List, 9=Integer List, 10=Decimal (double) List |
| FIELD_LENGTH    | number | no       | 10      | Field length                                                                                                                                                                                             |
| FIELD_PRECISION | number | no       | 0       | Field precision                                                                                                                                                                                          |
| FIELD_ALIAS     | string | no       | —       | Field alias                                                                                                                                                                                              |
| FIELD_COMMENT   | string | no       | —       | Field comment                                                                                                                                                                                            |
| OUTPUT          | sink   | yes      | —       | Added                                                                                                                                                                                                    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:addfieldtoattributetable INPUT=input.gpkg FIELD_NAME=value FIELD_TYPE=0  
↳ FIELD_LENGTH=10 FIELD_PRECISION=0 FIELD_ALIAS=value FIELD_COMMENT=value  
↳ OUTPUT=output.gpkg
```

This calls **Add field to attributes table**: INPUT=input.gpkg — Input layer; FIELD\_NAME=value — Field name; FIELD\_TYPE=0 selects *Integer (32 bit)*; FIELD\_LENGTH=10 — Field length; FIELD\_PRECISION=0 — Field precision; FIELD\_ALIAS=value — Field alias; FIELD\_COMMENT=value — Field comment; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:addhistorymetadata — Add history metadata

**group:** Metadata tools

This algorithm adds a new history entry to the layer's metadata.

| Parameter | Type   | Required | Default | Description   |
|-----------|--------|----------|---------|---------------|
| INPUT     | layer  | yes      | —       | Layer         |
| HISTORY   | string | yes      | —       | History entry |

**Outputs:** OUTPUT (outputLayer)

### Example usage

```
run native:addhistorymetadata INPUT=input.gpkg HISTORY=value
```

This calls **Add history metadata**: INPUT=input.gpkg — Layer; HISTORY=value — History entry. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:adduniquevalueindexfield — Add unique value index field

**group:** Vector table

This algorithm takes a vector layer and an attribute and adds a new numeric field. Values in this field correspond to values in the specified attribute, so features with the same value for the

attribute will have the same value in the new numeric field. This creates a numeric equivalent of the specified attribute, which defines the same classes.

The new attribute is not added to the input layer but a new layer is generated instead.

Optionally, a separate table can be output which contains a summary of the class field values mapped to the new unique numeric value.

| Parameter      | Type   | Required | Default   | Description            |
|----------------|--------|----------|-----------|------------------------|
| INPUT          | source | yes      | —         | Input layer            |
| FIELD          | field  | yes      | —         | Class field            |
| FIELD_NAME     | string | no       | NUM_FIELD | Output field name      |
| OUTPUT         | sink   | no       | —         | Layer with index field |
| SUMMARY_OUTPUT | sink   | no       | —         | Class summary          |

**Outputs:** OUTPUT (outputVector), SUMMARY\_OUTPUT (outputVector)

### Example usage

```
run native:adduniquevalueindexfield INPUT=input.gpkg FIELD=field1 FIELD_NAME=NUM_FIELD  
↪ OUTPUT=output.gpkg
```

This calls **Add unique value index field**: INPUT=input.gpkg — Input layer; FIELD=field1 — Class field; FIELD\_NAME=NUM\_FIELD — Output field name; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:addxyfields — Add X/Y fields to layer

**group:** Vector table

This algorithm adds X and Y (or latitude/longitude) fields to a point layer. The X/Y fields can be calculated in a different CRS to the layer (e.g. creating latitude/longitude fields for a layer in a project CRS).

| Parameter | Type   | Required | Default   | Description       |
|-----------|--------|----------|-----------|-------------------|
| INPUT     | source | yes      | —         | Input layer       |
| CRS       | crs    | no       | EPSG:4326 | Coordinate system |
| PREFIX    | string | no       | —         | Field prefix      |
| OUTPUT    | sink   | yes      | —         | Added fields      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:addxyfields INPUT=input.gpkg CRS=EPSG:6346 PREFIX=value OUTPUT=output.gpkg
```

This calls **Add X/Y fields to layer**: INPUT=input.gpkg — Input layer; CRS=EPSG:6346 — Coordinate system; PREFIX=value — Field prefix; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:affinetransform — Affine transform

**group:** Vector geometry

This algorithm applies an affine transformation to the geometries from a layer. Affine transformations can include translation, scaling and rotation. The operations are performed in a scale, rotation, translation order.

Z and M values present in the geometry can also be translated and scaled independently.



| Parameter  | Type     | Required | Default | Description                                        |
|------------|----------|----------|---------|----------------------------------------------------|
| INPUT      | source   | yes      | —       | Input layer                                        |
| DELTA_X    | distance | no       | 0.0     | Translation (x-axis)                               |
| DELTA_Y    | distance | no       | 0.0     | Translation (y-axis)                               |
| DELTA_Z    | number   | no       | 0.0     | Translation (z-axis)                               |
| DELTA_M    | number   | no       | 0.0     | Translation (m values)                             |
| SCALE_X    | number   | no       | 1.0     | Scale factor (x-axis)                              |
| SCALE_Y    | number   | no       | 1.0     | Scale factor (y-axis)                              |
| SCALE_Z    | number   | no       | 1.0     | Scale factor (z-axis)                              |
| SCALE_M    | number   | no       | 1.0     | Scale factor (m values)                            |
| ROTATION_Z | number   | no       | 0.0     | Rotation around z-axis (degrees counter-clockwise) |
| OUTPUT     | sink     | yes      | —       | Transformed                                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:affinetransform INPUT=input.gpkg DELTA_X=0 DELTA_Y=0 DELTA_Z=0 DELTA_M=0
↳ SCALE_X=1 SCALE_Y=1 OUTPUT=output.gpkg
```

This calls **Affine transform**: INPUT=input.gpkg — Input layer; DELTA\_X=0 — Translation (x-axis); DELTA\_Y=0 — Translation (y-axis); DELTA\_Z=0 — Translation (z-axis); DELTA\_M=0 — Translation (m values); SCALE\_X=1 — Scale factor (x-axis); SCALE\_Y=1 — Scale factor (y-axis); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:aggregate — Aggregate

**group:** Vector geometry

This algorithm takes a vector or table layer and aggregates features based on a group by expression. Features for which group by expression return the same value are grouped together.

It is possible to group all source features together using constant value in group by parameter, example: NULL.

It is also possible to group features using multiple fields using Array function, example: Array("Field1", "Field2").

Geometries (if present) are combined into one multipart geometry for each group.

Output attributes are computed depending on each given aggregate definition.

| Parameter  | Type       | Required | Default | Description                                      |
|------------|------------|----------|---------|--------------------------------------------------|
| INPUT      | source     | yes      | —       | Input layer                                      |
| GROUP_BY   | expression | no       | NULL    | Group by expression (NULL to group all features) |
| AGGREGATES | aggregates | yes      | —       | Aggregates                                       |
| OUTPUT     | sink       | yes      | —       | Aggregated                                       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:aggregate INPUT=input.gpkg AGGREGATES=... GROUP_BY="value > 0" OUTPUT=output.gpkg
```

This calls **Aggregate**: INPUT=input.gpkg — Input layer; AGGREGATES=... — Aggregates; GROUP\_BY="value > 0" — Group by expression (NULL to group all features); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:alignrasters — Align rasters

**group:** Raster tools

This algorithm aligns rasters by resampling them to the same cell size and reprojecting to the same CRS.

| Parameter       | Type              | Required | Default | Description                      |
|-----------------|-------------------|----------|---------|----------------------------------|
| LAYERS          | alignrasterlayers | yes      | —       | Input layers                     |
| REFERENCE_LAYER | raster            | yes      | —       | Reference layer                  |
| CRS             | crs               | no       | —       | Override reference CRS           |
| CELL_SIZE_X     | number            | no       | —       | Override reference cell size X   |
| CELL_SIZE_Y     | number            | no       | —       | Override reference cell size Y   |
| GRID_OFFSET_X   | number            | no       | —       | Override reference grid offset X |
| GRID_OFFSET_Y   | number            | no       | —       | Override reference grid offset Y |
| EXTENT          | extent            | no       | —       | Clip to extent                   |

**Outputs:** OUTPUT\_LAYERS (outputMultilayer)

### Example usage

```
run native:alignrasters LAYERS=... REFERENCE_LAYER=dem.tif CRS=EPSG:6346 CELL_SIZE_X=10  
↪ CELL_SIZE_Y=10 GRID_OFFSET_X=10 GRID_OFFSET_Y=10
```

This calls **Align rasters**: LAYERS=... — Input layers; REFERENCE\_LAYER=dem.tif — Reference layer; CRS=EPSG:6346 — Override reference CRS; CELL\_SIZE\_X=10 — Override reference cell size X; CELL\_SIZE\_Y=10 — Override reference cell size Y; GRID\_OFFSET\_X=10 — Override reference grid offset X; GRID\_OFFSET\_Y=10 — Override reference grid offset Y. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:alignsingleraster — Align raster

**group:** Raster tools

This algorithm aligns raster by resampling it to the same cell size and reprojecting to the same CRS as a reference raster.

| Parameter         | Type              | Required | Default | Description                                                                                                                                                                                                                                               |
|-------------------|-------------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT             | raster            | yes      | —       | Input layer                                                                                                                                                                                                                                               |
| RESAMPLING_METHOD | enum              | no       | 0       | Resampling method — options: 0=Nearest Neighbour, 1=Bilinear (2x2 Kernel), 2=Cubic (4x4 Kernel), 3=Cubic B-Spline (4x4 Kernel), 4=Lanczos (6x6 Kernel), 5=Average, 6=Mode, 7=Maximum, 8=Minimum, 9=Median, 10=First Quartile (Q1), 11=Third Quartile (Q3) |
| RESCALE           | boolean           | no       | False   | Rescale values according to the cell size                                                                                                                                                                                                                 |
| REFERENCE_LAYER   | raster            | yes      | —       | Reference layer                                                                                                                                                                                                                                           |
| CRS               | crs               | no       | —       | Override reference CRS                                                                                                                                                                                                                                    |
| CELL_SIZE_X       | number            | no       | —       | Override reference cell size X                                                                                                                                                                                                                            |
| CELL_SIZE_Y       | number            | no       | —       | Override reference cell size Y                                                                                                                                                                                                                            |
| GRID_OFFSET_X     | number            | no       | —       | Override reference grid offset X                                                                                                                                                                                                                          |
| GRID_OFFSET_Y     | number            | no       | —       | Override reference grid offset Y                                                                                                                                                                                                                          |
| EXTENT            | extent            | no       | —       | Clip to extent                                                                                                                                                                                                                                            |
| OUTPUT            | rasterDestination | yes      | —       | Aligned raster                                                                                                                                                                                                                                            |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:alignsingleraster INPUT=dem.tif REFERENCE_LAYER=dem.tif RESAMPLING_METHOD=0  
↪ RESCALE=false CRS=EPSG:6346 CELL_SIZE_X=10 CELL_SIZE_Y=10 OUTPUT=output.tif
```

This calls **Align raster**: INPUT=dem.tif — Input layer; REFERENCE\_LAYER=dem.tif — Reference layer; RESAMPLING\_METHOD=0 selects *Nearest Neighbour*; RESCALE=false — Rescale values according to the cell size; CRS=EPSG:6346 — Override reference CRS; CELL\_SIZE\_X=10 — Override reference cell size X; CELL\_SIZE\_Y=10 — Override reference cell size Y; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:angletonearest — Align points to features

**group:** Cartography

This algorithm calculates the rotation required to align point features with their nearest feature from another reference layer. A new field is added to the output layer which is filled with the angle (in degrees, clockwise) to the nearest reference feature.

Optionally, the output layer's symbology can be set to automatically use the calculated rotation field to rotate marker symbols.

If desired, a maximum distance to use when aligning points can be set, to avoid aligning isolated points to distant features.

| Parameter        | Type     | Required | Default  | Description                   |
|------------------|----------|----------|----------|-------------------------------|
| INPUT            | source   | yes      | —        | Input layer                   |
| REFERENCE_LAYER  | source   | yes      | —        | Reference layer               |
| MAX_DISTANCE     | distance | no       | —        | Maximum distance to consider  |
| FIELD_NAME       | string   | no       | rotation | Angle field name              |
| APPLY_SYMBOLLOGY | boolean  | no       | True     | Automatically apply symbology |
| OUTPUT           | sink     | yes      | —        | Aligned layer                 |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:angletonearest INPUT=input.gpkg REFERENCE_LAYER=input.gpkg MAX_DISTANCE=10
↳ FIELD_NAME=rotation APPLY_SYMBOLLOGY=true OUTPUT=output.gpkg
```

This calls **Align points to features**: INPUT=input.gpkg — Input layer; REFERENCE\_LAYER=input.gpkg — Reference layer; MAX\_DISTANCE=10 — Maximum distance to consider; FIELD\_NAME=rotation — Angle field name; APPLY\_SYMBOLLOGY=true — Automatically apply symbology; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:antimeridiansplit — Geodesic line split at antimeridian

**group:** Vector geometry

This algorithm splits a line into multiple geodesic segments, whenever the line crosses the antimeridian ( $\pm 180$  degrees longitude).

Splitting at the antimeridian helps the visual display of the lines in some projections. The returned geometry will always be a multi-part geometry.

Whenever line segments in the input geometry cross the antimeridian, they will be split into two segments, with the latitude of the breakpoint being determined using a geodesic line connecting the points either side of this segment. The current project ellipsoid setting will be used when calculating this breakpoint.

If the input geometry contains M or Z values, these will be linearly interpolated for the new vertices created at the antimeridian.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Split       |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:antimeridiansplit INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Geodesic line split at antimeridian**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:approximatemedialaxis — Approximate medial axis

**group:** Vector geometry

The Approximate Medial Axis algorithm generates a simplified skeleton of a shape by approximating its medial axis.

The output is a collection of lines that follow the central structure of the shape. The result is a thin, stable set of curves that capture the main topology while ignoring noise.

This algorithm ignores the Z dimensions. If the geometry is 3D, the approximate medial axis will be calculated from its 2D projection.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Medial axis |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:approximatemedialaxis INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Approximate medial axis**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:arrayoffsetlines — Array of offset (parallel) lines

**group:** Vector creation

This algorithm creates copies of line features in a layer, by creating multiple offset versions of each feature. Each copy is offset by a preset distance.

| Parameter           | Type     | Required | Default | Description                                     |
|---------------------|----------|----------|---------|-------------------------------------------------|
| INPUT               | source   | yes      | —       | Input layer                                     |
| COUNT               | number   | no       | 10      | Number of features to create                    |
| OFFSET              | distance | no       | 1.0     | Offset step distance                            |
| SEGMENTS *(adv)*    | number   | no       | 8       | Segments                                        |
| JOIN_STYLE *(adv)*  | enum     | no       | 0       | Join style — options: 0=Round, 1=Miter, 2=Bevel |
| MITER_LIMIT *(adv)* | number   | no       | 2       | Miter limit                                     |
| OUTPUT              | sink     | yes      | —       | Offset lines                                    |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:arrayoffsetlines INPUT=input.gpkg COUNT=10 OFFSET=1 OUTPUT=output.gpkg
```

This calls **Array of offset (parallel) lines**: INPUT=input.gpkg — Input layer; COUNT=10 — Number of features to create; OFFSET=1 — Offset step distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:arraytranslatedfeatures — Array of translated features

**group:** Vector creation

This algorithm creates copies of features in a layer, by creating multiple translated versions of each feature. Each copy is incrementally displaced by a preset amount in the x/y/z/m axis.

| Parameter | Type     | Required | Default | Description                  |
|-----------|----------|----------|---------|------------------------------|
| INPUT     | source   | yes      | —       | Input layer                  |
| COUNT     | number   | no       | 10      | Number of features to create |
| DELTA_X   | distance | no       | 0.0     | Step distance (x-axis)       |
| DELTA_Y   | distance | no       | 0.0     | Step distance (y-axis)       |
| DELTA_Z   | number   | no       | 0.0     | Step distance (z-axis)       |
| DELTA_M   | number   | no       | 0.0     | Step distance (m values)     |
| OUTPUT    | sink     | yes      | —       | Translated                   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:arraytranslatedfeatures INPUT=input.gpkg COUNT=10 DELTA_X=0 DELTA_Y=0 DELTA_Z=0  
↪ DELTA_M=0 OUTPUT=output.gpkg
```

This calls **Array of translated features**: INPUT=input.gpkg — Input layer; COUNT=10 — Number of features to create; DELTA\_X=0 — Step distance (x-axis); DELTA\_Y=0 — Step distance (y-axis); DELTA\_Z=0 — Step distance (z-axis); DELTA\_M=0 — Step distance (m values); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:aspect — Aspect

**group:** Raster terrain analysis

This algorithm calculates the aspect of the Digital Terrain Model in input.

The final aspect raster layer contains values from 0 to 360 that express the slope direction: starting from North (0°) and continuing clockwise.

| Parameter                | Type              | Required | Default | Description         |
|--------------------------|-------------------|----------|---------|---------------------|
| INPUT                    | raster            | yes      | —       | Elevation layer     |
| Z_FACTOR                 | number            | no       | 1.0     | Z factor            |
| NODATA *(adv)*           | number            | no       | -9999.0 | Output NoData value |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options    |
| OUTPUT                   | rasterDestination | yes      | —       | Aspect              |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:aspect INPUT=dem.tif Z_FACTOR=1 OUTPUT=output.tif
```

This calls **Aspect**: INPUT=dem.tif — Elevation layer; Z\_FACTOR=1 — Z factor; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:assignprojection — Assign projection

**group:** Vector general

This algorithm assigns a new projection to a vector layer. It creates a new layer with the exact same features and geometries as the input one, but assigned to a new CRS. E.g. the geometries are not reprojected, they are just assigned to a different CRS. This algorithm can be used to repair layers which have been assigned an incorrect projection.

Attributes are not modified by this algorithm.

| Parameter | Type   | Required | Default   | Description  |
|-----------|--------|----------|-----------|--------------|
| INPUT     | source | yes      | —         | Input layer  |
| CRS       | crs    | no       | EPSG:4326 | Assigned CRS |
| OUTPUT    | sink   | yes      | —         | Assigned CRS |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:assignprojection INPUT=input.gpkg CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Assign projection**: INPUT=input.gpkg — Input layer; CRS=EPSG:6346 — Assigned CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:atlaslayouttoimage — Export atlas layout as image

**group:** Cartography

This algorithm outputs an atlas layout to a set of image files (e.g. PNG or JPEG images).

If a coverage layer is set, the selected layout's atlas settings exposed in this algorithm will be overwritten. In this case, an empty filter or sort by expression will turn those settings off.

| Parameter                | Type       | Required | Default                         | Description                                                                                                                                                                                                                                                                                                             |
|--------------------------|------------|----------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LAYOUT                   | layout     | yes      | —                               | Atlas layout                                                                                                                                                                                                                                                                                                            |
| COVERAGE_LAYER           | vector     | no       | —                               | Coverage layer                                                                                                                                                                                                                                                                                                          |
| FILTER_EXPRESSION        | expression | no       | —                               | Filter expression                                                                                                                                                                                                                                                                                                       |
| SORTBY_EXPRESSION        | expression | no       | —                               | Sort expression                                                                                                                                                                                                                                                                                                         |
| SORTBY_REVERSE           | boolean    | no       | False                           | Reverse sort order (used when a sort expression is provided)                                                                                                                                                                                                                                                            |
| FILENAME_EXPRESSION      | expression | no       | 'output_'  @atlas_featurenumber | Output filename expression                                                                                                                                                                                                                                                                                              |
| FOLDER                   | file       | yes      | —                               | Output folder                                                                                                                                                                                                                                                                                                           |
| LAYERS *(adv)*           | multilayer | no       | —                               | Map layers to assign to unlocked map item(s)                                                                                                                                                                                                                                                                            |
| EXTENSION *(adv)*        | enum       | no       | 23                              | Image format — options: 0=avif, 1=bmp, 2=bw, 3=cur, 4=dds, 5=eps, 6=epsf, 7=epsi, 8=exr, 9=heic, 10=heif, 11=icns, 12=ico, 13=j2k, 14=jfif, 15=jp2, 16=jpeg, 17=jpg, 18=jxl, 19=pbm, 20=pcx, 21=pgm, 22=pic, 23=png, 24=ppm, 25=qoi, 26=rgb, 27=rgba, 28=sgi, 29=tga, 30=tif, 31=tiff, 32=wbmp, 33=webp, 34=xbm, 35=xpm |
| DPI *(adv)*              | number     | no       | —                               | DPI (leave blank for default layout DPI)                                                                                                                                                                                                                                                                                |
| GEOREFERENCE *(adv)*     | boolean    | no       | True                            | Generate world file                                                                                                                                                                                                                                                                                                     |
| INCLUDE_METADATA *(adv)* | boolean    | no       | True                            | Export RDF metadata (title, author, etc.)                                                                                                                                                                                                                                                                               |
| ANTIALIAS *(adv)*        | boolean    | no       | True                            | Enable antialiasing                                                                                                                                                                                                                                                                                                     |

## Example usage

```
run native:atlaslayouttoimage LAYOUT=... FOLDER=input.dat COVERAGE_LAYER=input.gpkg
↳ FILTER_EXPRESSION="value > 0" SORTBY_EXPRESSION="value > 0" SORTBY_REVERSE=false
↳ FILENAME_EXPRESSION="value > 0"
```

This calls **Export atlas layout as image**: LAYOUT=... — Atlas layout; FOLDER=input.dat — Output folder; COVERAGE\_LAYER=input.gpkg — Coverage layer; FILTER\_EXPRESSION="value > 0" — Filter expression; SORTBY\_EXPRESSION="value > 0" — Sort expression; SORTBY\_REVERSE=false — Reverse sort order (used when a sort expression is provided); FILENAME\_EXPRESSION="value > 0" — Output filename expression. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:atlaslayouttomultiplepdf — Export atlas layout as PDF (multiple files)

**group:** Cartography

This algorithm outputs an atlas layout to multiple PDF files.

If a coverage layer is set, the selected layout's atlas settings exposed in this algorithm will be overwritten. In this case, an empty filter or sort by expression will turn those settings off.

| Parameter                 | Type       | Required | Default | Description                                                                                                                                       |
|---------------------------|------------|----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| LAYOUT                    | layout     | yes      | —       | Atlas layout                                                                                                                                      |
| COVERAGE_LAYER            | vector     | no       | —       | Coverage layer                                                                                                                                    |
| FILTER_EXPRESSION         | expression | no       | —       | Filter expression                                                                                                                                 |
| SORTBY_EXPRESSION         | expression | no       | —       | Sort expression                                                                                                                                   |
| SORTBY_REVERSE            | boolean    | no       | False   | Reverse sort order (used when a sort expression is provided)                                                                                      |
| LAYERS *(adv)*            | multilayer | no       | —       | Map layers to assign to unlocked map item(s)                                                                                                      |
| DPI *(adv)*               | number     | no       | —       | DPI (leave blank for default layout DPI)                                                                                                          |
| FORCE_VECTOR *(adv)*      | boolean    | no       | False   | Always export as vectors                                                                                                                          |
| FORCE_RASTER *(adv)*      | boolean    | no       | False   | Always export as raster                                                                                                                           |
| GEOREFERENCE *(adv)*      | boolean    | no       | True    | Append georeference information                                                                                                                   |
| INCLUDE_METADATA *(adv)*  | boolean    | no       | True    | Export RDF metadata (title, author, etc.)                                                                                                         |
| DISABLE_TILED *(adv)*     | boolean    | no       | False   | Disable tiled raster layer exports                                                                                                                |
| SIMPLIFY *(adv)*          | boolean    | no       | True    | Simplify geometries to reduce output file size                                                                                                    |
| TEXT_FORMAT *(adv)*       | enum       | no       | 0       | Text export — options: 0=Always Export Text as Paths (Recommended), 1=Always Export Text as Text Objects, 2=Prefer Exporting Text as Text Objects |
| IMAGE_COMPRESSION *(adv)* | enum       | no       | 0       | Image compression — options: 0=Lossy (JPEG), 1=Lossless                                                                                           |
| OUTPUT_FILENAME           | expression | no       | —       | Output filename                                                                                                                                   |
| OUTPUT_FOLDER             | file       | yes      | —       | Output folder                                                                                                                                     |

## Example usage

```
run native:atlaslayouttomultiplepdf LAYOUT=... OUTPUT_FOLDER=input.dat
↳ COVERAGE_LAYER=input.gpkg FILTER_EXPRESSION="value > 0" SORTBY_EXPRESSION="value > 0"
↳ SORTBY_REVERSE=false OUTPUT_FILENAME="value > 0"
```

This calls **Export atlas layout as PDF (multiple files)**: LAYOUT=... — Atlas layout; OUTPUT\_FOLDER=input.dat — Output folder; COVERAGE\_LAYER=input.gpkg — Coverage layer; FILTER\_EXPRESSION="value > 0" — Filter expression; SORTBY\_EXPRESSION="value > 0" — Sort expression; SORTBY\_REVERSE=false — Reverse sort order (used when a sort expression is provided); OUTPUT\_FILENAME="value > 0" — Output filename. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:atlaslayouttopdf — Export atlas layout as PDF (single file)

**group:** Cartography

This algorithm outputs an atlas layout as a single PDF file.

If a coverage layer is set, the selected layout's atlas settings exposed in this algorithm will be overwritten. In this case, an empty filter or sort by expression will turn those settings off.

| Parameter                 | Type            | Required | Default | Description                                                                                                                                       |
|---------------------------|-----------------|----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| LAYOUT                    | layout          | yes      | —       | Atlas layout                                                                                                                                      |
| COVERAGE_LAYER            | vector          | no       | —       | Coverage layer                                                                                                                                    |
| FILTER_EXPRESSION         | expression      | no       | —       | Filter expression                                                                                                                                 |
| SORTBY_EXPRESSION         | expression      | no       | —       | Sort expression                                                                                                                                   |
| SORTBY_REVERSE            | boolean         | no       | False   | Reverse sort order (used when a sort expression is provided)                                                                                      |
| LAYERS *(adv)*            | multilayer      | no       | —       | Map layers to assign to unlocked map item(s)                                                                                                      |
| DPI *(adv)*               | number          | no       | —       | DPI (leave blank for default layout DPI)                                                                                                          |
| FORCE_VECTOR *(adv)*      | boolean         | no       | False   | Always export as vectors                                                                                                                          |
| FORCE_RASTER *(adv)*      | boolean         | no       | False   | Always export as raster                                                                                                                           |
| GEOREFERENCE *(adv)*      | boolean         | no       | True    | Append georeference information                                                                                                                   |
| INCLUDE_METADATA *(adv)*  | boolean         | no       | True    | Export RDF metadata (title, author, etc.)                                                                                                         |
| DISABLE_TILED *(adv)*     | boolean         | no       | False   | Disable tiled raster layer exports                                                                                                                |
| SIMPLIFY *(adv)*          | boolean         | no       | True    | Simplify geometries to reduce output file size                                                                                                    |
| TEXT_FORMAT *(adv)*       | enum            | no       | 0       | Text export — options: 0=Always Export Text as Paths (Recommended), 1=Always Export Text as Text Objects, 2=Prefer Exporting Text as Text Objects |
| IMAGE_COMPRESSION *(adv)* | enum            | no       | 0       | Image compression — options: 0=Lossy (JPEG), 1=Lossless                                                                                           |
| OUTPUT                    | fileDestination | yes      | —       | PDF file                                                                                                                                          |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:atlaslayouttopdf LAYOUT=... COVERAGE_LAYER=input.gpkg FILTER_EXPRESSION="value > 0" SORTBY_EXPRESSION="value > 0" SORTBY_REVERSE=false OUTPUT=output.txt
```

This calls **Export atlas layout as PDF (single file)**: LAYOUT=... — Atlas layout; COVERAGE\_LAYER=input.gpkg — Coverage layer; FILTER\_EXPRESSION="value > 0" — Filter expression; SORTBY\_EXPRESSION="value > 0" — Sort expression; SORTBY\_REVERSE=false — Reverse sort order (used when a sort expression is provided); OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:b3dmtogltf — Convert B3DM to GLTF

**group:** 3D Tiles

This algorithm converts files from the legacy B3DM format to GLTF.

| Parameter | Type            | Required | Default | Description |
|-----------|-----------------|----------|---------|-------------|
| INPUT     | file            | yes      | —       | Input B3DM  |
| OUTPUT    | fileDestination | yes      | —       | Output file |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:b3dmtogltf INPUT=input.dat OUTPUT=output.txt
```



This calls **Convert B3DM to GLTF**: INPUT=input.dat — Input B3DM; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:basicstatisticsforfields — Basic statistics for fields

**group:** Vector analysis

This algorithm generates basic statistics from the analysis of values in a field in the attribute table of a vector layer. Numeric, date, time and string fields are supported. The statistics returned will depend on the field type.

| Parameter        | Type            | Required | Default | Description                      |
|------------------|-----------------|----------|---------|----------------------------------|
| INPUT_LAYER      | source          | yes      | —       | Input layer                      |
| FIELD_NAME       | field           | yes      | —       | Field to calculate statistics on |
| OUTPUT           | sink            | no       | —       | Statistics                       |
| OUTPUT_HTML_FILE | fileDestination | no       | —       | Statistics report                |

**Outputs:** OUTPUT (outputVector), OUTPUT\_HTML\_FILE (outputHtml), COUNT (outputNumber), UNIQUE (outputNumber), EMPTY (outputNumber), FILLED (outputNumber), MIN (outputNumber), MAX (outputNumber), MIN\_LENGTH (outputNumber), MAX\_LENGTH (outputNumber), MEAN\_LENGTH (outputNumber), CV (outputNumber), SUM (outputNumber), MEAN (outputNumber), STD\_DEV (outputNumber), RANGE (outputNumber), MEDIAN (outputNumber), MINORITY (outputNumber), MAJORITY (outputNumber), FIRSTQUARTILE (outputNumber), THIRDQUARTILE (outputNumber), IQR (outputNumber)

### Example usage

```
run native:basicstatisticsforfields INPUT_LAYER=input.gpkg FIELD_NAME=field1
↳ OUTPUT=output.gpkg
```

This calls **Basic statistics for fields**: INPUT\_LAYER=input.gpkg — Input layer; FIELD\_NAME=field1 — Field to calculate statistics on; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:batchnominatimgeocoder — Batch Nominatim geocoder

**group:** Vector general

This algorithm performs batch geocoding using the Nominatim service against an input layer string field.

The output layer will have a point geometry reflecting the geocoded location as well as a number of attributes associated to the geocoded location.

| Parameter | Type   | Required | Default | Description   |
|-----------|--------|----------|---------|---------------|
| INPUT     | source | yes      | —       | Input layer   |
| FIELD     | field  | yes      | —       | Address field |
| OUTPUT    | sink   | yes      | —       | Geocoded      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:batchnominatimgeocoder INPUT=input.gpkg FIELD=field1 OUTPUT=output.gpkg
```

This calls **Batch Nominatim geocoder**: INPUT=input.gpkg — Input layer; FIELD=field1 — Address field; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:bookmarkstolayer — Convert spatial bookmarks to layer

**group:** Vector general

This algorithm creates a new layer containing polygon features for stored spatial bookmarks.

The export can be filtered to only bookmarks belonging to the current project, to all user bookmarks, or a combination of both.

| Parameter | Type | Required | Default                                   | Description                                                      |
|-----------|------|----------|-------------------------------------------|------------------------------------------------------------------|
| SOURCE    | enum | no       | [0, 1]                                    | Bookmark source — options: 0=Project bookmarks, 1=User bookmarks |
| CRS       | crs  | no       | <QgsCoordinateReferenceSystem: EPSG:4326> | Output CRS                                                       |
| OUTPUT    | sink | yes      | —                                         | Output                                                           |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:bookmarkstolayer SOURCE=0 CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Convert spatial bookmarks to layer**: SOURCE=0 selects *Project bookmarks*; CRS=EPSG:6346 — Output CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:boundary — Boundary

**group:** Vector geometry

This algorithm returns the closure of the combinatorial boundary of the input geometries (ie the topological boundary of the geometry). For instance, a polygon geometry will have a boundary consisting of the linestrings for each ring in the polygon. Only valid for polygon or line layers.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Boundary    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:boundary INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Boundary**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:boundingboxes — Bounding boxes ★

**group:** Vector geometry · **niva verb:** boundingbox

This algorithm calculates the bounding box (envelope) for each feature in an input layer.

See the 'Minimum bounding geometry' algorithm for a bounding box calculation which covers the whole layer or grouped subsets of features.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Bounds      |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:boundingboxes INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Bounding boxes**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**native:buffer — Buffer ★**

**group:** Vector geometry · **niva verb:** buffer

This algorithm computes a buffer area for all the features in an input layer, using a fixed or dynamic distance.

The segments parameter controls the number of line segments to use to approximate a quarter circle when creating rounded offsets.

The end cap style parameter controls how line endings are handled in the buffer.

The join style parameter specifies whether round, miter or beveled joins should be used when offsetting corners in a line.

The miter limit parameter is only applicable for miter join styles, and controls the maximum distance from the offset curve to use when creating a mitered join.

| Parameter                 | Type     | Required | Default | Description                                        |
|---------------------------|----------|----------|---------|----------------------------------------------------|
| INPUT                     | source   | yes      | —       | Input layer                                        |
| DISTANCE                  | distance | no       | 10      | Distance                                           |
| SEGMENTS                  | number   | no       | 5       | Segments                                           |
| END_CAP_STYLE             | enum     | no       | 0       | End cap style — options: 0=Round, 1=Flat, 2=Square |
| JOIN_STYLE                | enum     | no       | 0       | Join style — options: 0=Round, 1=Miter, 2=Bevel    |
| MITER_LIMIT               | number   | no       | 2       | Miter limit                                        |
| DISSOLVE                  | boolean  | no       | False   | Dissolve result                                    |
| SEPARATE_DISJOINT *(adv)* | boolean  | no       | False   | Keep disjoint results separate                     |
| OUTPUT                    | sink     | yes      | —       | Buffered                                           |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:buffer INPUT=input.gpkg DISTANCE=10 SEGMENTS=5 END_CAP_STYLE=0 JOIN_STYLE=0  
↪ MITER_LIMIT=2 DISSOLVE=false OUTPUT=output.gpkg
```

This calls **Buffer**: INPUT=input.gpkg — Input layer; DISTANCE=10 — Distance; SEGMENTS=5 — Segments; END\_CAP\_STYLE=0 selects *Round*; JOIN\_STYLE=0 selects *Round*; MITER\_LIMIT=2 — Miter limit; DISSOLVE=false — Dissolve result; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**native:bufferbym — Variable width buffer (by M value)**

**group:** Vector geometry

This algorithm creates variable width buffers along lines, using the M value of the line geometries as the diameter of the buffer at each vertex.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| SEGMENTS  | number | no       | 16      | Segments    |
| OUTPUT    | sink   | yes      | —       | Buffered    |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:bufferbym INPUT=input.gpkg SEGMENTS=16 OUTPUT=output.gpkg
```

This calls **Variable width buffer (by M value)**: INPUT=input.gpkg — Input layer; SEGMENTS=16 — Segments; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:calculateexpression — Calculate expression

**group:** Modeler tools

This algorithm calculates the result of a QGIS expression and makes it available for use in other parts of the model.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | string | yes      | —       | Input       |

**Outputs:** OUTPUT (outputVariant)

#### Example usage

```
run native:calculateexpression INPUT=value
```

This calls **Calculate expression**: INPUT=value — Input. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:calculatevectoroverlaps — Overlap analysis

**group:** Vector analysis

This algorithm calculates the area and percentage cover by which features from an input layer are overlapped by features from a selection of overlay layers.

New attributes are added to the output layer reporting the total area of overlap and percentage of the input feature overlapped by each of the selected overlay layers.

| Parameter         | Type       | Required | Default | Description    |
|-------------------|------------|----------|---------|----------------|
| INPUT             | source     | yes      | —       | Input layer    |
| LAYERS            | multilayer | yes      | —       | Overlay layers |
| OUTPUT            | sink       | yes      | —       | Overlap        |
| GRID_SIZE *(adv)* | number     | no       | —       | Grid size      |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:calculatevectoroverlaps INPUT=input.gpkg LAYERS="a.gpkg;b.gpkg"
↪ OUTPUT=output.gpkg
```

This calls **Overlap analysis**: INPUT=input.gpkg — Input layer; LAYERS="a.gpkg;b.gpkg" — Overlay layers; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:categorizeusingstyle — Create categorized renderer from styles

**group:** Cartography

This algorithm sets a vector layer's renderer to a categorized renderer using matching symbols from a style database. If no style file is specified, symbols from the user's current style library are used instead.

The specified expression (or field name) is used to create categories for the renderer. A category will be created for each unique value within the layer.

Each category is individually matched to the symbols which exist within the specified QGIS XML style database. Whenever a matching symbol name is found, the category's symbol will be set to this matched symbol.

The matching is case-insensitive by default, but can be made case-sensitive if required.

Optionally, non-alphanumeric characters in both the category value and symbol name can be ignored while performing the match. This allows for greater tolerance when matching categories to symbols.

If desired, tables can also be output containing lists of the categories which could not be matched to symbols, and symbols which were not matched to categories.

| Parameter               | Type       | Required | Default | Description                                       |
|-------------------------|------------|----------|---------|---------------------------------------------------|
| INPUT                   | vector     | yes      | —       | Input layer                                       |
| FIELD                   | expression | yes      | —       | Categorize using expression                       |
| STYLE                   | file       | no       | —       | Style database (leave blank to use saved symbols) |
| CASE_SENSITIVE          | boolean    | no       | False   | Use case-sensitive match to symbol names          |
| TOLERANT                | boolean    | no       | False   | Ignore non-alphanumeric characters while matching |
| NON_MATCHING_CATEGORIES | sink       | no       | —       | Non-matching categories                           |
| NON_MATCHING_SYMBOLS    | sink       | no       | —       | Non-matching symbol names                         |

**Outputs:** OUTPUT (outputVector), NON\_MATCHING\_CATEGORIES (outputVector), NON\_MATCHING\_SYMBOLS (outputVector)

### Example usage

```
run native:categorizeusingstyle INPUT=input.gpkg FIELD="value > 0" STYLE=input.dat
↳ CASE_SENSITIVE=false TOLERANT=false
↳ NON_MATCHING_CATEGORIES=non_matching_categories.gpkg
```

This calls **Create categorized renderer from styles**: INPUT=input.gpkg — Input layer; FIELD="value > 0" — Categorize using expression; STYLE=input.dat — Style database (leave blank to use saved symbols); CASE\_SENSITIVE=false — Use case-sensitive match to symbol names; TOLERANT=false — Ignore non-alphanumeric characters while matching; NON\_MATCHING\_CATEGORIES=non\_matching\_categories.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:cellstackpercentile — Cell stack percentile

**group:** Raster analysis

This algorithm generates a raster containing the cell-wise percentile value of a stack of input rasters. The percentile to return is determined by the percentile input value (ranges between 0 and 1). At each cell location, the specified percentile is obtained using the respective value from the stack of all overlaid and sorted cell values of the input rasters.

There are three methods for percentile calculation: Nearest rank Inclusive linear interpolation (PERCENTILE.INC) Exclusive linear interpolation (PERCENTILE.EXC) While the output value can stay the same for the nearest rank method (obtains the value that is nearest to the specified percentile),

the linear interpolation method return unique values for different percentiles. Both interpolation methods follow their counterpart methods implemented by LibreOffice or Microsoft Excel.

The output raster's extent and resolution is defined by a reference raster. If the input raster layers that do not match the cell size of the reference raster layer will be resampled using nearest neighbor resampling. NoData values in any of the input layers will result in a NoData cell output if the Ignore NoData parameter is not set. The output raster data type will be set to the most complex data type present in the input datasets.

| Parameter                   | Type              | Required | Default | Description                                                                                                                            |
|-----------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                       | multilayer        | yes      | —       | Input layers                                                                                                                           |
| METHOD                      | enum              | no       | 0       | Method — options: 0=Nearest rank, 1=Inclusive linear interpolation (PERCENTILE.INC), 2=Exclusive linear interpolation (PERCENTILE.EXC) |
| PERCENTILE                  | number            | no       | 0.25    | Percentile                                                                                                                             |
| IGNORE_NODATA               | boolean           | no       | True    | Ignore NoData values                                                                                                                   |
| REFERENCE_LAYER             | raster            | yes      | —       | Reference layer                                                                                                                        |
| OUTPUT_NODATA_VALUE *(adv)* | number            | no       | -9999   | Output NoData value                                                                                                                    |
| CREATE_OPTIONS              | string            | no       | —       | Creation options                                                                                                                       |
| CREATION_OPTIONS *(adv)*    | string            | no       | —       | Creation options                                                                                                                       |
| OUTPUT                      | rasterDestination | yes      | —       | Output layer                                                                                                                           |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:cellstackpercentile INPUT="a.gpkg;b.gpkg" REFERENCE_LAYER=dem.tif METHOD=0
↳ PERCENTILE=0.25 IGNORE_NODATA=true CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Cell stack percentile**: INPUT="a.gpkg;b.gpkg" — Input layers; REFERENCE\_LAYER=dem.tif — Reference layer; METHOD=0 selects *Nearest rank*; PERCENTILE=0.25 — Percentile; IGNORE\_NODATA=true — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:cellstackpercentrankfromrasterlayer — Cell stack percentrank from raster layer

**group:** Raster analysis

This algorithm generates a raster containing the cell-wise percent rank value of a stack of input rasters based on an input value raster.

At each cell location, the current value of the value raster is used ranked among the respective values in the stack of all overlaid and sorted cell values of the input rasters. For values outside of the the stack value distribution, the algorithm returns NoData because the value cannot be ranked among the cell values.

There are two methods for percentile calculation: Inclusive linearly interpolated percent rank (PERCENTRANK.INC) Exclusive linearly interpolated percent rank (PERCENTRANK.EXC) The linear interpolation method return the unique percent rank for different values. Both interpolation methods follow their counterpart methods implemented by LibreOffice or Microsoft Excel.

The output raster's extent and resolution is defined by a reference raster. If the input raster layers that do not match the cell size of the reference raster layer will be resampled using nearest neighbor resampling. NoData values in any of the input layers will result in a NoData cell output if the Ignore NoData parameter is not set. The output raster data type will always be Float32.

| Parameter                      | Type              | Required | Default | Description                                                                                                              |
|--------------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------|
| INPUT                          | multilayer        | yes      | —       | Input layers                                                                                                             |
| INPUT_VALUE_RASTER             | raster            | yes      | —       | Value raster layer                                                                                                       |
| VALUE_RASTER_BAND              | band              | no       | 1       | Value raster band                                                                                                        |
| METHOD                         | enum              | no       | 0       | Method — options: 0=Inclusive linear interpolation (PERCENTRANK.INC), 1=Exclusive linear interpolation (PERCENTRANK.EXC) |
| IGNORE_NODATA                  | boolean           | no       | True    | Ignore NoData values                                                                                                     |
| REFERENCE_LAYER                | raster            | yes      | —       | Reference layer                                                                                                          |
| OUTPUT_NODATA_VALUE<br>*(adv)* | number            | no       | -9999   | Output NoData value                                                                                                      |
| CREATE_OPTIONS                 | string            | no       | —       | Creation options                                                                                                         |
| CREATION_OPTIONS *(adv)*       | string            | no       | —       | Creation options                                                                                                         |
| OUTPUT                         | rasterDestination | yes      | —       | Output layer                                                                                                             |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:cellstackpercentrankfromrasterlayer INPUT="a.gpkg;b.gpkg"
↳ INPUT_VALUE_RASTER=dem.tif REFERENCE_LAYER=dem.tif VALUE_RASTER_BAND=1 METHOD=0
↳ IGNORE_NODATA=true CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Cell stack percentrank from raster layer**: INPUT="a.gpkg;b.gpkg" — Input layers; INPUT\_VALUE\_RASTER=dem.tif — Value raster layer; REFERENCE\_LAYER=dem.tif — Reference layer; VALUE\_RASTER\_BAND=1 — Value raster band; METHOD=0 selects *Inclusive linear interpolation* (PERCENTRANK.INC); IGNORE\_NODATA=true — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:cellstackpercentrankfromvalue — Cell stack percent rank from value

**group:** Raster analysis

This algorithm generates a raster containing the cell-wise percent rank value of a stack of input rasters based on a single input value.

At each cell location, the specified value is ranked among the respective values in the stack of all overlaid and sorted cell values from the input rasters. For values outside of the stack value distribution, the algorithm returns NoData because the value cannot be ranked among the cell values.

There are two methods for percentile calculation: Inclusive linearly interpolated percent rank (PERCENTRANK.INC) Exclusive linearly interpolated percent rank (PERCENTRANK.EXC) The linear interpolation method return the unique percent rank for different values. Both interpolation methods follow their counterpart methods implemented by LibreOffice or Microsoft Excel.

The output raster's extent and resolution is defined by a reference raster. If the input raster layers that do not match the cell size of the reference raster layer will be resampled using nearest neighbor resampling. NoData values in any of the input layers will result in a NoData cell output if the Ignore NoData parameter is not set. The output raster data type will always be Float32.

| Parameter | Type       | Required | Default | Description                                                                                                              |
|-----------|------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------|
| INPUT     | multilayer | yes      | —       | Input layers                                                                                                             |
| METHOD    | enum       | no       | 0       | Method — options: 0=Inclusive linear interpolation (PERCENTRANK.INC), 1=Exclusive linear interpolation (PERCENTRANK.EXC) |

| Parameter                      | Type              | Required | Default | Description          |
|--------------------------------|-------------------|----------|---------|----------------------|
| VALUE                          | number            | no       | 10      | Value                |
| IGNORE_NODATA                  | boolean           | no       | True    | Ignore NoData values |
| REFERENCE_LAYER                | raster            | yes      | —       | Reference layer      |
| OUTPUT_NODATA_VALUE<br>*(adv)* | number            | no       | -9999   | Output NoData value  |
| CREATE_OPTIONS                 | string            | no       | —       | Creation options     |
| CREATION_OPTIONS *(adv)*       | string            | no       | —       | Creation options     |
| OUTPUT                         | rasterDestination | yes      | —       | Output layer         |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:cellstackpercentrankfromvalue INPUT="a.gpkg;b.gpkg" REFERENCE_LAYER=dem.tif
↳ METHOD=0 VALUE=10 IGNORE_NODATA=true CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Cell stack percent rank from value**: INPUT="a.gpkg;b.gpkg" — Input layers; REFERENCE\_LAYER=dem.tif — Reference layer; METHOD=0 selects *Inclusive linear interpolation (PERCENTRANK.INC)*; VALUE=10 — Value; IGNORE\_NODATA=true — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:cellstatistics — Cell statistics

**group:** Raster analysis

The Cell statistics algorithm computes a value for each cell of the output raster. At each cell location, the output value is defined as a function of all overlaid cell values of the input rasters.

The output raster's extent and resolution is defined by a reference raster. The following functions can be applied on the input raster cells per output raster cell location: Sum Count Mean Median Standard deviation Variance Minimum Maximum Minority (least frequent value) Majority (most frequent value) Range (max-min) Variety (count of unique values) Input raster layers that do not match the cell size of the reference raster layer will be resampled using nearest neighbor resampling. The output raster data type will be set to the most complex data type present in the input datasets except when using the functions Mean, Standard deviation and Variance (data type is always Float32/Float64 depending on input float type) or Count and Variety (data type is always Int32). Calculation details - general: NoData values in any of the input layers will result in a NoData cell output if the Ignore NoData parameter is not set. Calculation details - Count: Count will always result in the number of cells without NoData values at the current cell location. Calculation details - Median: If the number of input layers is even, the median will be calculated as the arithmetic mean of the two middle values of the ordered cell input values. In this case the output data type is Float32. Calculation details - Minority/Majority: If no unique minority or majority could be found, the result is NoData, except all input cell values are equal.

| Parameter                      | Type       | Required | Default | Description                                                                                                                                                 |
|--------------------------------|------------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                          | multilayer | yes      | —       | Input layers                                                                                                                                                |
| STATISTIC                      | enum       | no       | 0       | Statistic — options: 0=Sum, 1=Count, 2=Mean, 3=Median, 4=Standard deviation, 5=Variance, 6=Minimum, 7=Maximum, 8=Minority, 9=Majority, 10=Range, 11=Variety |
| IGNORE_NODATA                  | boolean    | no       | True    | Ignore NoData values                                                                                                                                        |
| REFERENCE_LAYER                | raster     | yes      | —       | Reference layer                                                                                                                                             |
| OUTPUT_NODATA_VALUE<br>*(adv)* | number     | no       | -9999   | Output NoData value                                                                                                                                         |
| CREATE_OPTIONS                 | string     | no       | —       | Creation options                                                                                                                                            |



| Parameter                | Type              | Required | Default | Description      |
|--------------------------|-------------------|----------|---------|------------------|
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options |
| OUTPUT                   | rasterDestination | yes      | —       | Output layer     |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:cellstatistics INPUT="a.gpkg;b.gpkg" REFERENCE_LAYER=dem.tif STATISTIC=0
↳ IGNORE_NODATA=true CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Cell statistics**: INPUT="a.gpkg;b.gpkg" — Input layers; REFERENCE\_LAYER=dem.tif — Reference layer; STATISTIC=0 selects *Sum*; IGNORE\_NODATA=true — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:centroids — Centroids ★

**group:** Vector geometry · **niva verb:** centroid

This algorithm creates a new point layer with points representing the centroid of the geometries in an input layer.

The attributes associated to each point in the output layer are the same ones associated to the original features.

| Parameter | Type    | Required | Default | Description                   |
|-----------|---------|----------|---------|-------------------------------|
| INPUT     | source  | yes      | —       | Input layer                   |
| ALL_PARTS | boolean | no       | False   | Create centroid for each part |
| OUTPUT    | sink    | yes      | —       | Centroids                     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:centroids INPUT=input.gpkg ALL_PARTS=false OUTPUT=output.gpkg
```

This calls **Centroids**: INPUT=input.gpkg — Input layer; ALL\_PARTS=false — Create centroid for each part; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryangle — Small angles

**group:** Check geometry

This algorithm checks the angles of line or polygon geometries. Angles below the minimum angle are errors.

| Parameter         | Type   | Required | Default | Description                |
|-------------------|--------|----------|---------|----------------------------|
| INPUT             | source | yes      | —       | Input layer                |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier  |
| MIN_ANGLE         | number | no       | 0       | Minimum angle (in degrees) |
| ERRORS            | sink   | yes      | —       | Small angle errors         |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                  |

**Outputs:** ERRORS (outputVector)

### Example usage

```
run native:checkgeometryangle INPUT=input.gpkg UNIQUE_ID=field1 MIN_ANGLE=0
↳ ERRORS=errors.gpkg
```

This calls **Small angles**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; MIN\_ANGLE=0 — Minimum angle (in degrees); ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryarea — Small polygons

**group:** Check geometry

This algorithm checks the areas of polygon geometries. Areas below the area threshold are errors.

| Parameter         | Type   | Required | Default | Description               |
|-------------------|--------|----------|---------|---------------------------|
| INPUT             | source | yes      | —       | Input layer               |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier |
| AREATHRESHOLD     | area   | no       | 0       | Area threshold            |
| ERRORS            | sink   | yes      | —       | Small polygons errors     |
| OUTPUT            | sink   | no       | —       | Small polygons features   |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                 |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometryarea INPUT=input.gpkg UNIQUE_ID=field1 AREATHRESHOLD=0
↳ ERRORS=errors.gpkg
```

This calls **Small polygons**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; AREATHRESHOLD=0 — Area threshold; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrycontained — Features inside polygon

**group:** Check geometry

This algorithm checks the input geometries contained in the polygons from the polygon layers list. A polygon layer can be checked against itself. Input features contained in the polygon layers features are errors.

| Parameter         | Type       | Required | Default | Description                    |
|-------------------|------------|----------|---------|--------------------------------|
| INPUT             | source     | yes      | —       | Input layer                    |
| UNIQUE_ID         | field      | yes      | —       | Unique feature identifier      |
| POLYGONS          | multilayer | yes      | —       | Polygon layers                 |
| ERRORS            | sink       | yes      | —       | Errors from contained features |
| OUTPUT            | sink       | no       | —       | Contained features             |
| TOLERANCE *(adv)* | number     | no       | 8       | Tolerance                      |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrycontained INPUT=input.gpkg UNIQUE_ID=field1
↳ POLYGONS="a.gpkg;b.gpkg" ERRORS=errors.gpkg
```

This calls **Features inside polygon**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; POLYGONS="a.gpkg;b.gpkg" — Polygon layers; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrydangle — Dangle-end lines

**group:** Check geometry

This algorithm checks lines with a dangle end. Lines with a dangle end are errors.

| Parameter         | Type   | Required | Default | Description               |
|-------------------|--------|----------|---------|---------------------------|
| INPUT             | source | yes      | —       | Input layer               |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier |
| ERRORS            | sink   | yes      | —       | Dangle-end errors         |
| OUTPUT            | sink   | no       | —       | Dangle-end features       |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                 |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrydangle INPUT=input.gpkg UNIQUE_ID=field1 ERRORS=errors.gpkg
```

This calls **Dangle-end lines**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrydegeneratepolygon — Degenerate polygons

**group:** Check geometry

This algorithm checks the polygons with less than 3 points, which are degenerate polygons. Degenerate polygons are errors.

| Parameter         | Type   | Required | Default | Description                  |
|-------------------|--------|----------|---------|------------------------------|
| INPUT             | source | yes      | —       | Input layer                  |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier    |
| ERRORS            | sink   | yes      | —       | Degenerate polygons errors   |
| OUTPUT            | sink   | no       | —       | Degenerate polygons features |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                    |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrydegeneratepolygon INPUT=input.gpkg UNIQUE_ID=field1  
↪ ERRORS=errors.gpkg
```

This calls **Degenerate polygons**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryduplicate — Duplicated geometries

**group:** Check geometry

This algorithm checks the duplicate geometries. Duplicate geometries are errors.

| Parameter | Type   | Required | Default | Description                 |
|-----------|--------|----------|---------|-----------------------------|
| INPUT     | source | yes      | —       | Input layer                 |
| UNIQUE_ID | field  | yes      | —       | Unique feature identifier   |
| ERRORS    | sink   | yes      | —       | Duplicate geometries errors |
| OUTPUT    | sink   | no       | —       | Duplicate geometries        |

| Parameter         | Type   | Required | Default | Description |
|-------------------|--------|----------|---------|-------------|
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance   |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometryduplicate INPUT=input.gpkg UNIQUE_ID=field1 ERRORS=errors.gpkg
```

This calls **Duplicated geometries**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryduplicatenodes — Duplicated vertices

**group:** Check geometry

This algorithm checks the vertices of line and polygon geometries. Duplicated vertices are errors.

| Parameter         | Type   | Required | Default | Description                  |
|-------------------|--------|----------|---------|------------------------------|
| INPUT             | source | yes      | —       | Input layer                  |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier    |
| ERRORS            | sink   | yes      | —       | Duplicated vertices errors   |
| OUTPUT            | sink   | no       | —       | Duplicated vertices features |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                    |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometryduplicatenodes INPUT=input.gpkg UNIQUE_ID=field1
↳ ERRORS=errors.gpkg
```

This calls **Duplicated vertices**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryfollowboundaries — Polygons exceeding boundaries

**group:** Check geometry

This algorithm checks if the polygons follow the boundaries of the reference layer. Polygons not following reference boundaries are errors.

| Parameter         | Type   | Required | Default | Description                   |
|-------------------|--------|----------|---------|-------------------------------|
| INPUT             | source | yes      | —       | Input layer                   |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier     |
| ERRORS            | sink   | yes      | —       | Errors exceeding boundaries   |
| OUTPUT            | sink   | no       | —       | Features exceeding boundaries |
| REF_LAYER         | source | yes      | —       | Reference layer               |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                     |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometryfollowboundaries INPUT=input.gpkg UNIQUE_ID=field1
↳ REF_LAYER=input.gpkg ERRORS=errors.gpkg
```

This calls **Polygons exceeding boundaries**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; REF\_LAYER=input.gpkg — Reference layer; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrygap — Small gaps

**group:** Check geometry

This algorithm checks the gaps between polygons. Gaps with an area smaller than the gap threshold are errors.

If an allowed gaps layer is given, the gaps contained in polygons from this layer will be ignored. An optional buffer can be applied to the allowed gaps.

The neighbors output layer is needed for the fix geometry (gaps) algorithm. It is a 1-N relational table for correspondence between a gap and the unique id of its neighbor features.

| Parameter           | Type     | Required | Default | Description               |
|---------------------|----------|----------|---------|---------------------------|
| INPUT               | source   | yes      | —       | Input layer               |
| UNIQUE_ID           | field    | yes      | —       | Unique feature identifier |
| GAP_THRESHOLD       | number   | no       | 0       | Gap threshold             |
| ALLOWED_GAPS_LAYER  | vector   | no       | —       | Allowed gaps layer        |
| ALLOWED_GAPS_BUFFER | distance | no       | —       | Allowed gaps buffer       |
| NEIGHBORS           | sink     | yes      | —       | Neighbors layer           |
| ERRORS              | sink     | no       | —       | Gap errors                |
| OUTPUT              | sink     | yes      | —       | Gap features              |
| TOLERANCE *(adv)*   | number   | no       | 8       | Tolerance                 |

**Outputs:** NEIGHBORS (outputVector), ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrygap INPUT=input.gpkg UNIQUE_ID=field1 GAP_THRESHOLD=0
↪ ALLOWED_GAPS_LAYER=input.gpkg ALLOWED_GAPS_BUFFER=10 NEIGHBORS=neighbors.gpkg
```

This calls **Small gaps**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; GAP\_THRESHOLD=0 — Gap threshold; ALLOWED\_GAPS\_LAYER=input.gpkg — Allowed gaps layer; ALLOWED\_GAPS\_BUFFER=10 — Allowed gaps buffer; NEIGHBORS=neighbors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryhole — Holes

**group:** Check geometry

This algorithm checks the holes of polygon geometries. Holes are errors.

| Parameter         | Type   | Required | Default | Description               |
|-------------------|--------|----------|---------|---------------------------|
| INPUT             | source | yes      | —       | Input layer               |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier |
| ERRORS            | sink   | yes      | —       | Holes errors              |
| OUTPUT            | sink   | no       | —       | Polygons with holes       |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                 |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometryhole INPUT=input.gpkg UNIQUE_ID=field1 ERRORS=errors.gpkg
```

This calls **Holes**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrylineintersection — Lines intersecting each other

**group:** Check geometry

This algorithm checks intersections between line geometries. Intersections between two different lines are errors.

| Parameter         | Type   | Required | Default | Description               |
|-------------------|--------|----------|---------|---------------------------|
| INPUT             | source | yes      | —       | Input layer               |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier |
| ERRORS            | sink   | yes      | —       | Intersection errors       |
| OUTPUT            | sink   | no       | —       | Intersecting feature      |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                 |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrylineintersection INPUT=input.gpkg UNIQUE_ID=field1
↳ ERRORS=errors.gpkg
```

This calls **Lines intersecting each other**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrylinelayerintersection — Lines intersecting other layer

**group:** Check geometry

This algorithm checks if the input line layer features intersect with the check layer features. An input feature that intersects with a check layer feature is an error.

| Parameter         | Type   | Required | Default | Description                            |
|-------------------|--------|----------|---------|----------------------------------------|
| INPUT             | source | yes      | —       | Input layer                            |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier              |
| CHECK_LAYER       | source | yes      | —       | Check layer                            |
| ERRORS            | sink   | yes      | —       | Line intersecting other layer errors   |
| OUTPUT            | sink   | no       | —       | Line intersecting other layer features |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                              |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrylinelayerintersection INPUT=input.gpkg UNIQUE_ID=field1
↳ CHECK_LAYER=input.gpkg ERRORS=errors.gpkg
```

This calls **Lines intersecting other layer**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; CHECK\_LAYER=input.gpkg — Check layer; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrymissingvertex — Missing vertices along borders

**group:** Check geometry

This algorithm checks for missing vertices along polygon borders. To be topologically correct, a vertex at the junction of two polygons must be present on both polygons. Missing vertices are errors.

| Parameter         | Type   | Required | Default | Description               |
|-------------------|--------|----------|---------|---------------------------|
| INPUT             | source | yes      | —       | Input layer               |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier |
| ERRORS            | sink   | yes      | —       | Missing vertices errors   |
| OUTPUT            | sink   | no       | —       | Missing vertices features |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                 |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrymissingvertex INPUT=input.gpkg UNIQUE_ID=field1 ERRORS=errors.gpkg
```

This calls **Missing vertices along borders**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrymultipart — Strictly multipart

**group:** Check geometry

This algorithm checks if multipart geometries have more than one part. Multipart geometries with only one part are errors.

| Parameter         | Type   | Required | Default | Description                |
|-------------------|--------|----------|---------|----------------------------|
| INPUT             | source | yes      | —       | Input layer                |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier  |
| ERRORS            | sink   | yes      | —       | One-part geometry errors   |
| OUTPUT            | sink   | no       | —       | One-part geometry features |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                  |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrymultipart INPUT=input.gpkg UNIQUE_ID=field1 ERRORS=errors.gpkg
```

This calls **Strictly multipart**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryoverlap — Overlaps

**group:** Check geometry

This algorithm checks the overlapping areas. Overlapping areas smaller than the minimum overlapping area are errors.

| Parameter | Type   | Required | Default | Description               |
|-----------|--------|----------|---------|---------------------------|
| INPUT     | source | yes      | —       | Input layer               |
| UNIQUE_ID | field  | yes      | —       | Unique feature identifier |

| Parameter         | Type   | Required | Default | Description          |
|-------------------|--------|----------|---------|----------------------|
| ERRORS            | sink   | yes      | —       | Overlap errors       |
| OUTPUT            | sink   | no       | —       | Overlap features     |
| MIN_OVERLAP_AREA  | number | no       | 0       | Minimum overlap area |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance            |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometryoverlap INPUT=input.gpkg UNIQUE_ID=field1 MIN_OVERLAP_AREA=0
↳ ERRORS=errors.gpkg
```

This calls **Overlaps**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; MIN\_OVERLAP\_AREA=0 — Minimum overlap area; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrypointcoveredbyline — Points outside lines

**group:** Check geometry

This algorithm checks if the points in the input layer are covered by a line in the selected line layers. A point not covered by a line is an error.

| Parameter         | Type       | Required | Default | Description                  |
|-------------------|------------|----------|---------|------------------------------|
| INPUT             | source     | yes      | —       | Input layer                  |
| UNIQUE_ID         | field      | yes      | —       | Unique feature identifier    |
| LINES             | multilayer | yes      | —       | Line layers                  |
| ERRORS            | sink       | yes      | —       | Points not covered by a line |
| TOLERANCE *(adv)* | number     | no       | 8       | Tolerance                    |

**Outputs:** ERRORS (outputVector)

### Example usage

```
run native:checkgeometrypointcoveredbyline INPUT=input.gpkg UNIQUE_ID=field1
↳ LINES="a.gpkg;b.gpkg" ERRORS=errors.gpkg
```

This calls **Points outside lines**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; LINES="a.gpkg;b.gpkg" — Line layers; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrypointinpolygon — Points outside polygons

**group:** Check geometry

This algorithm checks if points from the input layer are in polygons from the selected polygon layers. Points that are not fully inside polygons are errors.

| Parameter         | Type       | Required | Default | Description                    |
|-------------------|------------|----------|---------|--------------------------------|
| INPUT             | source     | yes      | —       | Input layer                    |
| UNIQUE_ID         | field      | yes      | —       | Unique feature identifier      |
| POLYGONS          | multilayer | yes      | —       | Polygon layers                 |
| ERRORS            | sink       | yes      | —       | Points outside polygons errors |
| TOLERANCE *(adv)* | number     | no       | 8       | Tolerance                      |



---

**Outputs:** ERRORS (outputVector)

**Example usage**

```
run native:checkgeometrypointinpolygon INPUT=input.gpkg UNIQUE_ID=field1  
↳ POLYGONS="a.gpkg;b.gpkg" ERRORS=errors.gpkg
```

This calls **Points outside polygons**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; POLYGONS="a.gpkg;b.gpkg" — Polygon layers; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrysegmentlength — Small segments

**group:** Check geometry

This algorithm checks the minimum segment length of line or polygon geometries. Segments shorter than the minimum length are errors.

| Parameter          | Type   | Required | Default | Description               |
|--------------------|--------|----------|---------|---------------------------|
| INPUT              | source | yes      | —       | Input layer               |
| UNIQUE_ID          | field  | yes      | —       | Unique feature identifier |
| ERRORS             | sink   | yes      | —       | Short segments errors     |
| OUTPUT             | sink   | no       | —       | Short segments features   |
| MIN_SEGMENT_LENGTH | number | no       | 0       | Minimum segment length    |
| TOLERANCE *(adv)*  | number | no       | 8       | Tolerance                 |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

**Example usage**

```
run native:checkgeometrysegmentlength INPUT=input.gpkg UNIQUE_ID=field1  
↳ MIN_SEGMENT_LENGTH=0 ERRORS=errors.gpkg
```

This calls **Small segments**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; MIN\_SEGMENT\_LENGTH=0 — Minimum segment length; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometryselfcontact — Self-contacts

**group:** Check geometry

This algorithm checks if the geometry has self contact points (in line or polygon). Self contacts are errors.

| Parameter         | Type   | Required | Default | Description               |
|-------------------|--------|----------|---------|---------------------------|
| INPUT             | source | yes      | —       | Input layer               |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier |
| ERRORS            | sink   | yes      | —       | Self contact error points |
| OUTPUT            | sink   | no       | —       | Self contact features     |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                 |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

**Example usage**

```
run native:checkgeometrselfcontact INPUT=input.gpkg UNIQUE_ID=field1 ERRORS=errors.gpkg
```

This calls **Self-contacts**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrselfintersection — Self-intersections

**group:** Check geometry

This algorithm checks self-intersecting geometries. Self-intersecting geometries are errors.

| Parameter         | Type   | Required | Default | Description                |
|-------------------|--------|----------|---------|----------------------------|
| INPUT             | source | yes      | —       | Input layer                |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier  |
| ERRORS            | sink   | yes      | —       | Self-intersecting errors   |
| OUTPUT            | sink   | no       | —       | Self-intersecting features |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                  |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrselfintersection INPUT=input.gpkg UNIQUE_ID=field1
↳ ERRORS=errors.gpkg
```

This calls **Self-intersections**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkgeometrsliverpolygon — Sliver polygons

**group:** Check geometry

This algorithm checks sliver polygons.

The thinness is the ratio between the area of the minimum square containing the polygon and the area of the polygon itself (a square has a thinness value of 1). The thinness value is between 1 and +infinity. If a polygon has an area higher than the maximum area, it is skipped (a maximum area value of 0 means no area check).

Polygons having a thinness higher than the maximum thinness are errors.

To fix sliver polygons, use the “Fix small polygons” algorithm.

| Parameter         | Type   | Required | Default | Description                      |
|-------------------|--------|----------|---------|----------------------------------|
| INPUT             | source | yes      | —       | Input layer                      |
| UNIQUE_ID         | field  | yes      | —       | Unique feature identifier        |
| ERRORS            | sink   | yes      | —       | Sliver polygon errors            |
| OUTPUT            | sink   | no       | —       | Sliver polygon features          |
| MAX_THINNESS      | number | no       | 20      | Maximum thinness                 |
| MAX_AREA          | number | no       | 0       | Maximum area (map units squared) |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                        |

**Outputs:** ERRORS (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:checkgeometrsliverpolygon INPUT=input.gpkg UNIQUE_ID=field1 MAX_THINNESS=20
↳ MAX_AREA=0 ERRORS=errors.gpkg
```

This calls **Sliver polygons**: INPUT=input.gpkg — Input layer; UNIQUE\_ID=field1 — Unique feature identifier; MAX\_THINNESS=20 — Maximum thickness; MAX\_AREA=0 — Maximum area (map units squared); ERRORS=errors.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:checkvalidity — Check validity

**group:** Vector geometry

This algorithm performs a validity check on the geometries of a vector layer.

The geometries are classified in three groups (valid, invalid and error), and a vector layer is generated with the features in each of these categories.

By default the algorithm uses the strict OGC definition of polygon validity, where a polygon is marked as invalid if a self-intersecting ring causes an interior hole. If the 'Ignore ring self intersections' option is checked, then this rule will be ignored and a more lenient validity check will be performed.

The GEOS method is faster and performs better on larger geometries, but is limited to only returning the first error encountered in a geometry. The QGIS method will be slower but reports all errors encountered in the geometry, not just the first.

| Parameter                     | Type    | Required | Default | Description                                                                 |
|-------------------------------|---------|----------|---------|-----------------------------------------------------------------------------|
| INPUT_LAYER                   | source  | yes      | —       | Input layer                                                                 |
| METHOD                        | enum    | no       | 2       | Method — options: 0=The one selected in digitizing settings, 1=QGIS, 2=GEOS |
| IGNORE_RING_SELF_INTERSECTION | boolean | no       | False   | Ignore ring self intersections                                              |
| VALID_OUTPUT                  | sink    | no       | —       | Valid output                                                                |
| INVALID_OUTPUT                | sink    | no       | —       | Invalid output                                                              |
| ERROR_OUTPUT                  | sink    | no       | —       | Error output                                                                |

**Outputs:** VALID\_OUTPUT (outputVector), INVALID\_OUTPUT (outputVector), ERROR\_OUTPUT (outputVector), VALID\_COUNT (outputNumber), INVALID\_COUNT (outputNumber), ERROR\_COUNT (outputNumber)

### Example usage

```
run native:checkvalidity INPUT_LAYER=input.gpkg METHOD=2
↳ IGNORE_RING_SELF_INTERSECTION=false VALID_OUTPUT=valid_output.gpkg
```

This calls **Check validity**: INPUT\_LAYER=input.gpkg — Input layer; METHOD=2 selects *GEOS*; IGNORE\_RING\_SELF\_INTERSECTION=false — Ignore ring self intersections; VALID\_OUTPUT=valid\_output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:climbalongline — Climb along line

**group:** Vector analysis

This algorithm calculates the total climb and descent along line geometries.

Input layer must have Z values present. If Z values are not available, the "Drape" (set Z value from raster) algorithm may be used to add Z values from a DEM layer.

The output layer is a copy of the input layer with additional fields that contain the total climb, total descent, the minimum elevation and the maximum elevation for each line geometry. If the input layer contains fields with the same names as these added fields, they will be renamed (field names will be altered to "name\_2", "name\_3", etc, finding the first non-duplicate name).

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Line layer  |
| OUTPUT    | sink   | yes      | —       | Climb layer |

**Outputs:** OUTPUT (outputVector), TOTALCLIMB (outputNumber), TOTALDESCENT (outputNumber), MINELEVATION (outputNumber), MAXELEVATION (outputNumber)

### Example usage

```
run native:climbalongline INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Climb along line**: INPUT=input.gpkg — Line layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:clip — Clip ★

**group:** Vector overlay · **niva verb:** clip

This algorithm clips a vector layer using the features of an additional polygon layer. Only the parts of the features in the Input layer that fall within the polygons of the Overlay layer will be added to the resulting layer.

The attributes of the features are not modified, although properties such as area or length of the features will be modified by the clipping operation. If such properties are stored as attributes, those attributes will have to be manually updated.

| Parameter | Type   | Required | Default | Description   |
|-----------|--------|----------|---------|---------------|
| INPUT     | source | yes      | —       | Input layer   |
| OVERLAY   | source | yes      | —       | Overlay layer |
| OUTPUT    | sink   | yes      | —       | Clipped       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:clip INPUT=input.gpkg OVERLAY=input.gpkg OUTPUT=output.gpkg
```

This calls **Clip**: INPUT=input.gpkg — Input layer; OVERLAY=input.gpkg — Overlay layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:collect — Collect geometries ★

**group:** Vector geometry · **niva verb:** collect

This algorithm takes a vector layer and collects its geometries into new multipart geometries. One or more attributes can be specified to collect only geometries belonging to the same class (having the same value for the specified attributes), alternatively all geometries can be collected.

All output geometries will be converted to multi geometries, even those with just a single part. This algorithm does not dissolve overlapping geometries - they will be collected together without modifying the shape of each geometry part.

See the 'Promote to multipart' or 'Aggregate' algorithms for alternative options.

| Parameter | Type   | Required | Default | Description      |
|-----------|--------|----------|---------|------------------|
| INPUT     | source | yes      | —       | Input layer      |
| FIELD     | field  | no       | —       | Unique ID fields |
| OUTPUT    | sink   | yes      | —       | Collected        |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:collect INPUT=input.gpkg FIELD=field1 OUTPUT=output.gpkg
```

This calls **Collect geometries**: INPUT=input.gpkg — Input layer; FIELD=field1 — Unique ID fields; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:combinestyles — Combine style databases

**group:** Cartography

This algorithm combines multiple QGIS style databases into a single style database. If any symbols exist with duplicate names between the different source databases these will be renamed to have unique names in the output combined database.

| Parameter | Type            | Required | Default      | Description                                                                              |
|-----------|-----------------|----------|--------------|------------------------------------------------------------------------------------------|
| INPUT     | multilayer      | yes      | —            | Input databases                                                                          |
| OUTPUT    | fileDestination | yes      | —            | Output style database                                                                    |
| OBJECTS   | enum            | no       | [0, 1, 2, 3] | Objects to combine — options: 0=Symbols, 1=Color ramps, 2=Text formats, 3=Label settings |

**Outputs:** OUTPUT (outputFile), SYMBOLS (outputNumber), COLORRAMPS (outputNumber), TEXTFORMATS (outputNumber), LABELSETTINGS (outputNumber)

**Example usage**

```
run native:combinestyles INPUT="a.gpkg;b.gpkg" OBJECTS=0 OUTPUT=output.txt
```

This calls **Combine style databases**: INPUT="a.gpkg;b.gpkg" — Input databases; OBJECTS=0 selects *Symbols*; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:concavehull — Concave hull (by layer)

**group:** Vector geometry

This algorithm computes the concave hull covering all features from an input point layer.

See the 'Concave hull (by feature)' algorithm for a concave hull calculation which covers individual features from a layer.

| Parameter        | Type    | Required | Default | Description                                             |
|------------------|---------|----------|---------|---------------------------------------------------------|
| INPUT            | source  | yes      | —       | Input layer                                             |
| ALPHA            | number  | no       | 0.3     | Threshold (0-1, where 1 is equivalent with Convex Hull) |
| HOLES            | boolean | no       | True    | Allow holes                                             |
| NO_MULTIGEOMETRY | boolean | no       | False   | Split multipart geometry into singleparts               |
| OUTPUT           | sink    | yes      | —       | Concave hull                                            |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:concavehull INPUT=input.gpkg ALPHA=0.3 HOLES=true NO_MULTIGEOMETRY=false  
↪ OUTPUT=output.gpkg
```

This calls **Concave hull (by layer)**: INPUT=input.gpkg — Input layer; ALPHA=0.3 — Threshold (0-1, where 1 is equivalent with Convex Hull); HOLES=true — Allow holes; NO\_MULTIGEOMETRY=false — Split multipart geometry into singleparts; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:concavehullbyfeature — Concave hull (by feature)

**group:** Vector geometry

This algorithm calculates the concave hull for each feature in an input layer.

A concave hull is a polygon which contains all the points of the input geometries, but is a better approximation than the convex hull to the area occupied by the input.

It is frequently used to convert a multi-point into a polygonal area which contains all the points from the input geometry.

See the 'Concave hull (by layer)' algorithm for a concave hull calculation which covers the whole layer or grouped subsets of features.

| Parameter | Type    | Required | Default | Description                                             |
|-----------|---------|----------|---------|---------------------------------------------------------|
| INPUT     | source  | yes      | —       | Input layer                                             |
| ALPHA     | number  | no       | 0.3     | Threshold (0-1, where 1 is equivalent with Convex Hull) |
| HOLES     | boolean | no       | True    | Allow holes                                             |
| OUTPUT    | sink    | yes      | —       | Concave hulls                                           |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:concavehullbyfeature INPUT=input.gpkg ALPHA=0.3 HOLES=true OUTPUT=output.gpkg
```

This calls **Concave hull (by feature)**: INPUT=input.gpkg — Input layer; ALPHA=0.3 — Threshold (0-1, where 1 is equivalent with Convex Hull); HOLES=true — Allow holes; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:condition — Conditional branch

**group:** Modeler tools

This algorithm adds a conditional branch into a model, allowing parts of the model to be executed based on the result of an expression evaluation.

### Example usage

```
run native:condition
```

This calls **Conditional branch** (no required parameters).

## native:convertgeometrytype — Convert geometry type

**group:** Vector geometry

This algorithm generates a new layer based on an existing one, with a different type of geometry.

Not all conversions are possible. For instance, a line layer can be converted to a point layer, but a point layer cannot be converted to a line layer.

See the "Polygonize" or "Lines to polygons" algorithms for alternative options.

| Parameter | Type   | Required | Default | Description                                                                                      |
|-----------|--------|----------|---------|--------------------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                                                                      |
| TYPE      | enum   | yes      | —       | New geometry type — options: 0=Centroids, 1=Nodes, 2=Linestrings, 3=Multilinestrings, 4=Polygons |
| OUTPUT    | sink   | yes      | —       | Converted                                                                                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:convertgeometrytype INPUT=input.gpkg TYPE=0 OUTPUT=output.gpkg
```

This calls **Convert geometry type**: INPUT=input.gpkg — Input layer; TYPE=0 selects *Centroids*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:convertgpsdata — Convert GPS data

**group:** GPS

This algorithm uses the GPSTool tool to convert a GPS data file from a range of formats to the GPX standard format.

| Parameter    | Type            | Required | Default | Description                                             |
|--------------|-----------------|----------|---------|---------------------------------------------------------|
| INPUT        | file            | yes      | —       | Input file                                              |
| FORMAT       | string          | yes      | —       | Format                                                  |
| FEATURE_TYPE | enum            | no       | 0       | Feature type — options: 0=Waypoints, 1=Routes, 2=Tracks |
| OUTPUT       | fileDestination | yes      | —       | Output                                                  |

**Outputs:** OUTPUT (outputFile), OUTPUT\_LAYER (outputVector)

### Example usage

```
run native:convertgpsdata INPUT=input.dat FORMAT=value FEATURE_TYPE=0 OUTPUT=output.txt
```

This calls **Convert GPS data**: INPUT=input.dat — Input file; FORMAT=value — Format; FEATURE\_TYPE=0 selects *Waypoints*; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:convertgpxfeaturetype — Convert GPX feature type

**group:** GPS

This algorithm uses the GPSTool tool to convert GPX features from one type to another (e.g. converting all waypoint features to a route feature).

| Parameter  | Type            | Required | Default | Description                                                                                                              |
|------------|-----------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------|
| INPUT      | file            | yes      | —       | Input file                                                                                                               |
| CONVERSION | enum            | no       | 0       | Conversion — options: 0=Waypoints from a Route, 1=Waypoints from a Track, 2=Route from Waypoints, 3=Track from Waypoints |
| OUTPUT     | fileDestination | yes      | —       | Output                                                                                                                   |

**Outputs:** OUTPUT (outputFile), OUTPUT\_LAYER (outputVector)

### Example usage

```
run native:convertgpxfeaturetype INPUT=input.dat CONVERSION=0 OUTPUT=output.txt
```

This calls **Convert GPX feature type**: INPUT=input.dat — Input file; CONVERSION=0 selects *Waypoints from a Route*; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:converttocurves — Convert to curved geometries

**group:** Vector geometry

This algorithm converts a geometry into its curved geometry equivalent.

Already curved geometries will be retained without change.

| Parameter | Type   | Required | Default | Description                |
|-----------|--------|----------|---------|----------------------------|
| INPUT     | source | yes      | —       | Input layer                |
| DISTANCE  | number | no       | 1e-06   | Maximum distance tolerance |
| ANGLE     | number | no       | 1e-06   | Maximum angle tolerance    |
| OUTPUT    | sink   | yes      | —       | Curves                     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:converttocurves INPUT=input.gpkg DISTANCE=1e-06 ANGLE=1e-06 OUTPUT=output.gpkg
```

This calls **Convert to curved geometries**: INPUT=input.gpkg — Input layer; DISTANCE=1e-06 — Maximum distance tolerance; ANGLE=1e-06 — Maximum angle tolerance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:convexhull — Convex hull ★

**group:** Vector geometry · **niva verb:** convexhull

This algorithm calculates the convex hull for each feature in an input layer.

See the 'Minimum bounding geometry' algorithm for a convex hull calculation which covers the whole layer or grouped subsets of features.

| Parameter | Type   | Required | Default | Description  |
|-----------|--------|----------|---------|--------------|
| INPUT     | source | yes      | —       | Input layer  |
| OUTPUT    | sink   | yes      | —       | Convex hulls |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:convexhull INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Convex hull**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:copylayermetadata — Copy layer metadata

**group:** Metadata tools

This algorithm copies metadata from a source layer to a target layer.

Any existing metadata in the target layer will be replaced.

| Parameter | Type    | Required | Default | Description              |
|-----------|---------|----------|---------|--------------------------|
| SOURCE    | layer   | yes      | —       | Source layer             |
| TARGET    | layer   | yes      | —       | Target layer             |
| DEFAULT   | boolean | no       | False   | Save metadata as default |



---

**Outputs:** OUTPUT (outputLayer)

**Example usage**

```
run native:copylayermetadata SOURCE=input.gpkg TARGET=input.gpkg DEFAULT=false
```

This calls **Copy layer metadata**: SOURCE=input.gpkg — Source layer; TARGET=input.gpkg — Target layer; DEFAULT=false — Save metadata as default. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:countpointsinpolygon — Count points in polygon ★**

**group:** Vector analysis · **niva verb:** countpoints

This algorithm takes a points layer and a polygon layer and counts the number of points from the first one in each polygons of the second one.

A new polygons layer is generated, with the exact same content as the input polygons layer, but containing an additional field with the points count corresponding to each polygon.

An optional weight field can be used to assign weights to each point. If set, the count generated will be the sum of the weight field for each point contained by the polygon.

Alternatively, a unique class field can be specified. If set, points are classified based on the selected attribute, and if several points with the same attribute value are within the polygon, only one of them is counted. The final count of the point in a polygon is, therefore, the count of different classes that are found in it.

Both the weight field and unique class field cannot be specified. If they are, the weight field will take precedence and the unique class field will be ignored.

| Parameter  | Type   | Required | Default   | Description      |
|------------|--------|----------|-----------|------------------|
| POLYGONS   | source | yes      | —         | Polygons         |
| POINTS     | source | yes      | —         | Points           |
| WEIGHT     | field  | no       | —         | Weight field     |
| CLASSFIELD | field  | no       | —         | Class field      |
| FIELD      | string | no       | NUMPOINTS | Count field name |
| OUTPUT     | sink   | yes      | —         | Count            |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:countpointsinpolygon POLYGONS=input.gpkg POINTS=input.gpkg WEIGHT=field1  
↪ CLASSFIELD=field1 FIELD=NUMPOINTS OUTPUT=output.gpkg
```

This calls **Count points in polygon**: POLYGONS=input.gpkg — Polygons; POINTS=input.gpkg — Points; WEIGHT=field1 — Weight field; CLASSFIELD=field1 — Class field; FIELD=NUMPOINTS — Count field name; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:coveragesimplify — Simplify coverage**

**group:** Vector coverage

This algorithm operates on a coverage (represented as a set of polygon features with exactly matching edge geometry) to apply a Visvalingam-Whyatt simplification to the edges, reducing complexity in proportion with the provided tolerance, while retaining a valid coverage (i.e. no edges will cross or touch after the simplification).

Geometries will never be removed, but they may be simplified down to just a triangle. Also, some geometries (such as polygons which have too few non-repeated points) will be returned unchanged. If the input dataset is not a valid coverage due to overlaps, it will still be simplified, but invalid topology such as crossing edges will still be invalid.

| Parameter         | Type     | Required | Default | Description       |
|-------------------|----------|----------|---------|-------------------|
| INPUT             | source   | yes      | —       | Input layer       |
| TOLERANCE         | distance | no       | 1.0     | Tolerance         |
| PRESERVE_BOUNDARY | boolean  | no       | False   | Preserve boundary |
| OUTPUT            | sink     | yes      | —       | Simplified        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:coveragesimplify INPUT=input.gpkg TOLERANCE=1 PRESERVE_BOUNDARY=false
↪ OUTPUT=output.gpkg
```

This calls **Simplify coverage**: INPUT=input.gpkg — Input layer; TOLERANCE=1 — Tolerance; PRESERVE\_BOUNDARY=false — Preserve boundary; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:coverageunion — Dissolve coverage

**group:** Vector coverage

This algorithm operates on a coverage (represented as a set of polygon features with exactly matching edge geometry) to dissolve (union) the geometries.

It provides a heavily optimized approach for unioning these features compared with the standard Dissolve tools.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Dissolved   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:coverageunion INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Dissolve coverage**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:coveragevalidate — Validate coverage

**group:** Vector coverage

This algorithm analyzes a coverage (represented as a set of polygon features with exactly matching edge geometry) to find places where the assumption of exactly matching edges is not met.

Invalidity includes polygons that overlap or that have gaps smaller than the specified gap width.

| Parameter     | Type     | Required | Default | Description   |
|---------------|----------|----------|---------|---------------|
| INPUT         | source   | yes      | —       | Input layer   |
| GAP_WIDTH     | distance | no       | 0.0     | Gap width     |
| INVALID_EDGES | sink     | no       | —       | Invalid edges |

---

**Outputs:** INVALID\_EDGES (outputVector), IS\_VALID (outputBoolean)

**Example usage**

```
run native:coveragevalidate INPUT=input.gpkg GAP_WIDTH=0 INVALID_EDGES=invalid_edges.gpkg
```

This calls **Validate coverage**: INPUT=input.gpkg — Input layer; GAP\_WIDTH=0 — Gap width; INVALID\_EDGES=invalid\_edges.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createattributeindex — Create attribute index

**group:** Vector general

This algorithm creates an index to speed up queries made against a field in a table. Support for index creation is dependent on the layer's data provider and the field type.

| Parameter | Type   | Required | Default | Description        |
|-----------|--------|----------|---------|--------------------|
| INPUT     | vector | yes      | —       | Input layer        |
| FIELD     | field  | yes      | —       | Attribute to index |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:createattributeindex INPUT=input.gpkg FIELD=field1
```

This calls **Create attribute index**: INPUT=input.gpkg — Input layer; FIELD=field1 — Attribute to index. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createconstantrasterlayer — Create constant raster layer

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with a single constant value. Additionally an output data type can be specified. The algorithm will abort if a value has been entered that cannot be represented by the selected output raster data type.

| Parameter                | Type              | Required | Default    | Description                                                                                                                           |
|--------------------------|-------------------|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| EXTENT                   | extent            | yes      | —          | Desired extent                                                                                                                        |
| TARGET_CRS               | crs               | no       | ProjectCrs | Target CRS                                                                                                                            |
| PIXEL_SIZE               | number            | no       | 1          | Pixel size                                                                                                                            |
| NUMBER                   | number            | no       | 1          | Constant value                                                                                                                        |
| OUTPUT_TYPE *(adv)*      | enum              | no       | 5          | Output raster data type — options: 0=Byte, 1=Integer16, 2=Unsigned Integer16, 3=Integer32, 4=Unsigned Integer32, 5=Float32, 6=Float64 |
| CREATE_OPTIONS           | string            | no       | —          | Creation options                                                                                                                      |
| CREATION_OPTIONS *(adv)* | string            | no       | —          | Creation options                                                                                                                      |
| OUTPUT                   | rasterDestination | yes      | —          | Constant                                                                                                                              |

**Outputs:** OUTPUT (outputRaster)

**Example usage**

```
run native:createconstantrasterlayer EXTENT=0,1000,0,1000 TARGET_CRS=EPSG:6346  
↳ PIXEL_SIZE=1 NUMBER=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create constant raster layer**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; NUMBER=1 — Constant value; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createdirectory — Create directory

**group:** Modeler tools

This algorithm creates a new directory on a file system. Directories will be created recursively, creating all required parent directories in order to construct the full specified directory path.

No errors will be raised if the directory already exists.

| Parameter | Type  | Required | Default | Description    |
|-----------|-------|----------|---------|----------------|
| PATH      | layer | yes      | —       | Directory path |

**Outputs:** OUTPUT (outputFolder)

### Example usage

```
run native:createdirectory PATH=input.gpkg
```

This calls **Create directory**: PATH=input.gpkg — Directory path. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:creategrid — Create grid

**group:** Vector creation

This algorithm creates a vector layer with a grid covering a given extent. Elements in the grid can be points, lines or polygons. The size and/or placement of each element in the grid is defined using a horizontal and vertical spacing. The CRS of the output layer must be defined. The grid extent and the spacing values must be expressed in the coordinates and units of this CRS. The top-left point (minX, maxY) is used as the reference point. That means that, at that point, an element is guaranteed to be placed. Unless the width and height of the selected extent is a multiple of the selected spacing, that is not true for the other points that define that extent.

| Parameter | Type     | Required | Default    | Description                                                                                           |
|-----------|----------|----------|------------|-------------------------------------------------------------------------------------------------------|
| TYPE      | enum     | no       | 0          | Grid type — options: 0=Point, 1=Line, 2=Rectangle (Polygon), 3=Diamond (Polygon), 4=Hexagon (Polygon) |
| EXTENT    | extent   | yes      | —          | Grid extent                                                                                           |
| HSPACING  | distance | no       | 1          | Horizontal spacing                                                                                    |
| VSPACING  | distance | no       | 1          | Vertical spacing                                                                                      |
| HOVERLAY  | distance | no       | 0          | Horizontal overlay                                                                                    |
| VOVERLAY  | distance | no       | 0          | Vertical overlay                                                                                      |
| CRS       | crs      | no       | ProjectCrs | Grid CRS                                                                                              |
| OUTPUT    | sink     | yes      | —          | Grid                                                                                                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:creategrid EXTENT=0,1000,0,1000 TYPE=0 HSPACING=1 VSPACING=1 HOVERLAY=0
↳ VOVERLAY=0 CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Create grid**: EXTENT=0,1000,0,1000 — Grid extent; TYPE=0 selects *Point*; HSPACING=1 — Horizontal spacing; VSPACING=1 — Vertical spacing; HOVERLAY=0 — Horizontal overlay; VOVERLAY=0 — Vertical overlay; CRS=EPSG:6346 — Grid CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createpointslayerfromtable — Create points layer from table

**group:** Vector creation

This algorithm generates a point layer based on the coordinates from an input table.

The table must contain a field with the X coordinate of each point and another one with the Y coordinate, as well as optional fields with Z and M values. A CRS for the output layer has to be specified, and the coordinates in the table are assumed to be expressed in the units used by that CRS. The attributes table of the resulting layer will be the input table.

| Parameter  | Type   | Required | Default   | Description       |
|------------|--------|----------|-----------|-------------------|
| INPUT      | source | yes      | —         | Input layer       |
| XFIELD     | field  | yes      | —         | X field           |
| YFIELD     | field  | yes      | —         | Y field           |
| ZFIELD     | field  | no       | —         | Z field           |
| MFIELD     | field  | no       | —         | M field           |
| TARGET_CRS | crs    | no       | EPSG:4326 | Target CRS        |
| OUTPUT     | sink   | yes      | —         | Points from table |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:createpointslayerfromtable INPUT=input.gpkg XFIELD=field1 YFIELD=field1  
↪ ZFIELD=field1 MFIELD=field1 TARGET_CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Create points layer from table**: INPUT=input.gpkg — Input layer; XFIELD=field1 — X field; YFIELD=field1 — Y field; ZFIELD=field1 — Z field; MFIELD=field1 — M field; TARGET\_CRS=EPSG:6346 — Target CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createrandombinomialrasterlayer — Create random raster layer (binomial distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with binomially distributed random values. By default, the values will be chosen given an N of 10 and a probability of 0.5. This can be overridden by using the advanced parameter for N and probability. The raster data type is set to Integer types (Integer16 by default). The binomial distribution random values are defined as positive integer numbers. A floating point raster will represent a cast of integer values to floating point.

| Parameter                | Type              | Required | Default    | Description                                                                                                                   |
|--------------------------|-------------------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| EXTENT                   | extent            | yes      | —          | Desired extent                                                                                                                |
| TARGET_CRS               | crs               | no       | ProjectCrs | Target CRS                                                                                                                    |
| PIXEL_SIZE               | number            | no       | 1          | Pixel size                                                                                                                    |
| OUTPUT_TYPE *(adv)*      | enum              | no       | 0          | Output raster data type — options: 0=Integer16, 1=Unsigned Integer16, 2=Integer32, 3=Unsigned Integer32, 4=Float32, 5=Float64 |
| N *(adv)*                | number            | no       | 10         | N                                                                                                                             |
| PROBABILITY *(adv)*      | number            | no       | 0.5        | Probability                                                                                                                   |
| CREATE_OPTIONS           | string            | no       | —          | Creation options                                                                                                              |
| CREATION_OPTIONS *(adv)* | string            | no       | —          | Creation options                                                                                                              |
| OUTPUT                   | rasterDestination | yes      | —          | Output raster                                                                                                                 |

**Outputs:** OUTPUT (outputRaster)

## Example usage

```
run native:createrandombinomialrasterlayer EXTENT=0,1000,0,1000 TARGET_CRS=EPSG:6346  
↳ PIXEL_SIZE=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create random raster layer (binomial distribution)**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createrandomexponentialrasterlayer — Create random raster layer (exponential distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with exponentially distributed random values. By default, the values will be chosen given a lambda of 1.0. This can be overridden by using the advanced parameter for lambda. The raster data type is set to Float32 by default as the exponential distribution random values are floating point numbers.

| Parameter                | Type              | Required | Default    | Description                                             |
|--------------------------|-------------------|----------|------------|---------------------------------------------------------|
| EXTENT                   | extent            | yes      | —          | Desired extent                                          |
| TARGET_CRS               | crs               | no       | ProjectCrs | Target CRS                                              |
| PIXEL_SIZE               | number            | no       | 1          | Pixel size                                              |
| OUTPUT_TYPE *(adv)*      | enum              | no       | 0          | Output raster data type — options: 0=Float32, 1=Float64 |
| LAMBDA *(adv)*           | number            | no       | 1.0        | Lambda                                                  |
| CREATE_OPTIONS           | string            | no       | —          | Creation options                                        |
| CREATION_OPTIONS *(adv)* | string            | no       | —          | Creation options                                        |
| OUTPUT                   | rasterDestination | yes      | —          | Output raster                                           |

**Outputs:** OUTPUT (outputRaster)

## Example usage

```
run native:createrandomexponentialrasterlayer EXTENT=0,1000,0,1000 TARGET_CRS=EPSG:6346  
↳ PIXEL_SIZE=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create random raster layer (exponential distribution)**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createrandomgammarrasterlayer — Create random raster layer (gamma distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with gamma distributed random values. By default, the values will be chosen given an alpha and beta value of 1.0. This can be overridden by using the advanced parameter for alpha and beta. The raster data type is set to Float32 by default as the gamma distribution random values are floating point numbers.

| Parameter  | Type   | Required | Default    | Description    |
|------------|--------|----------|------------|----------------|
| EXTENT     | extent | yes      | —          | Desired extent |
| TARGET_CRS | crs    | no       | ProjectCrs | Target CRS     |
| PIXEL_SIZE | number | no       | 1          | Pixel size     |

| Parameter                | Type              | Required | Default | Description                                             |
|--------------------------|-------------------|----------|---------|---------------------------------------------------------|
| OUTPUT_TYPE *(adv)*      | enum              | no       | 0       | Output raster data type — options: 0=Float32, 1=Float64 |
| ALPHA *(adv)*            | number            | no       | 1.0     | Alpha                                                   |
| BETA *(adv)*             | number            | no       | 1.0     | Beta                                                    |
| CREATE_OPTIONS           | string            | no       | —       | Creation options                                        |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options                                        |
| OUTPUT                   | rasterDestination | yes      | —       | Output raster                                           |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:createrandomgamm rasterlayer EXTENT=0,1000,0,1000 TARGET_CRS=EPSG:6346
↳ PIXEL_SIZE=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create random raster layer (gamma distribution)**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createrandomgeometricrasterlayer — Create random raster layer (geometric distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with geometrically distributed random values. By default, the values will be chosen given a probability of 0.5. This can be overridden by using the advanced parameter for mean value. The raster data type is set to Integer types (Integer16 by default). The geometric distribution random values are defined as positive integer numbers. A floating point raster will represent a cast of integer values to floating point.

| Parameter                | Type              | Required | Default    | Description                                                                                                                   |
|--------------------------|-------------------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| EXTENT                   | extent            | yes      | —          | Desired extent                                                                                                                |
| TARGET_CRS               | crs               | no       | ProjectCrs | Target CRS                                                                                                                    |
| PIXEL_SIZE               | number            | no       | 1          | Pixel size                                                                                                                    |
| OUTPUT_TYPE *(adv)*      | enum              | no       | 0          | Output raster data type — options: 0=Integer16, 1=Unsigned Integer16, 2=Integer32, 3=Unsigned Integer32, 4=Float32, 5=Float64 |
| PROBABILITY *(adv)*      | number            | no       | 0.5        | Probability                                                                                                                   |
| CREATE_OPTIONS           | string            | no       | —          | Creation options                                                                                                              |
| CREATION_OPTIONS *(adv)* | string            | no       | —          | Creation options                                                                                                              |
| OUTPUT                   | rasterDestination | yes      | —          | Output raster                                                                                                                 |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:createrandomgeometricrasterlayer EXTENT=0,1000,0,1000 TARGET_CRS=EPSG:6346
↳ PIXEL_SIZE=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create random raster layer (geometric distribution)**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createrandomnegativebinomialrasterlayer — Create random raster layer (negative binomial distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with negative binomially distributed random values. By default, the values will be chosen given a distribution parameter  $k$  of 10.0 and a probability of 0.5. This can be overridden by using the advanced parameters for  $k$  and probability. The raster data type is set to Integer types (Integer 16 by default). The negative binomial distribution random values are defined as positive integer numbers. A floating point raster will represent a cast of integer values to floating point.

| Parameter                | Type              | Required | Default    | Description                                                                                                                   |
|--------------------------|-------------------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| EXTENT                   | extent            | yes      | —          | Desired extent                                                                                                                |
| TARGET_CRS               | crs               | no       | ProjectCrs | Target CRS                                                                                                                    |
| PIXEL_SIZE               | number            | no       | 1          | Pixel size                                                                                                                    |
| OUTPUT_TYPE *(adv)*      | enum              | no       | 0          | Output raster data type — options: 0=Integer16, 1=Unsigned Integer16, 2=Integer32, 3=Unsigned Integer32, 4=Float32, 5=Float64 |
| K_PARAMETER *(adv)*      | number            | no       | 10         | Distribution parameter $k$                                                                                                    |
| PROBABILITY *(adv)*      | number            | no       | 0.5        | Probability                                                                                                                   |
| CREATE_OPTIONS           | string            | no       | —          | Creation options                                                                                                              |
| CREATION_OPTIONS *(adv)* | string            | no       | —          | Creation options                                                                                                              |
| OUTPUT                   | rasterDestination | yes      | —          | Output raster                                                                                                                 |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:createrandomnegativebinomialrasterlayer EXTENT=0,1000,0,1000  
↳ TARGET_CRS=EPSG:6346 PIXEL_SIZE=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create random raster layer (negative binomial distribution)**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createrandomnormalrasterlayer — Create random raster layer (normal distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with normally distributed random values. By default, the values will be chosen given a mean of 0.0 and a standard deviation of 1.0. This can be overridden by using the advanced parameters for mean and standard deviation value. The raster data type is set to Float32 by default as the normal distribution random values are floating point numbers.

| Parameter           | Type   | Required | Default    | Description                                             |
|---------------------|--------|----------|------------|---------------------------------------------------------|
| EXTENT              | extent | yes      | —          | Desired extent                                          |
| TARGET_CRS          | crs    | no       | ProjectCrs | Target CRS                                              |
| PIXEL_SIZE          | number | no       | 1          | Pixel size                                              |
| OUTPUT_TYPE *(adv)* | enum   | no       | 0          | Output raster data type — options: 0=Float32, 1=Float64 |
| MEAN *(adv)*        | number | no       | 0          | Mean of normal distribution                             |
| STDDEV *(adv)*      | number | no       | 1          | Standard deviation of normal distribution               |
| CREATE_OPTIONS      | string | no       | —          | Creation options                                        |



| Parameter                   | Type              | Required | Default | Description      |
|-----------------------------|-------------------|----------|---------|------------------|
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Creation options |
| OUTPUT                      | rasterDestination | yes      | —       | Output raster    |

**Outputs:** OUTPUT (outputRaster)

#### Example usage

```
run native:createrandomnormalrasterlayer EXTENT=0,1000,0,1000 TARGET_CRS=EPSG:6346
↳ PIXEL_SIZE=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create random raster layer (normal distribution)**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

### native:createrandompoissonrasterlayer — Create random raster layer (poisson distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with poisson distributed random values. By default, the values will be chosen given a mean of 1.0. This can be overridden by using the advanced parameter for mean value. The raster data type is set to Integer types (Integer16 by default). The poisson distribution random values are positive integer numbers. A floating point raster will represent a cast of integer values to floating point.

| Parameter                   | Type              | Required | Default    | Description                                                                                                                   |
|-----------------------------|-------------------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| EXTENT                      | extent            | yes      | —          | Desired extent                                                                                                                |
| TARGET_CRS                  | crs               | no       | ProjectCrs | Target CRS                                                                                                                    |
| PIXEL_SIZE                  | number            | no       | 1          | Pixel size                                                                                                                    |
| OUTPUT_TYPE *(adv)*         | enum              | no       | 0          | Output raster data type — options: 0=Integer16, 1=Unsigned Integer16, 2=Integer32, 3=Unsigned Integer32, 4=Float32, 5=Float64 |
| MEAN *(adv)*                | number            | no       | 1.0        | Mean                                                                                                                          |
| CREATE_OPTIONS              | string            | no       | —          | Creation options                                                                                                              |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —          | Creation options                                                                                                              |
| OUTPUT                      | rasterDestination | yes      | —          | Output raster                                                                                                                 |

**Outputs:** OUTPUT (outputRaster)

#### Example usage

```
run native:createrandompoissonrasterlayer EXTENT=0,1000,0,1000 TARGET_CRS=EPSG:6346
↳ PIXEL_SIZE=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Create random raster layer (poisson distribution)**: EXTENT=0,1000,0,1000 — Desired extent; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

### native:createrandomuniformrasterlayer — Create random raster layer (uniform distribution)

**group:** Raster creation

This algorithm generates a raster layer for a given extent and cell size filled with random values. By default, the values will range between the minimum and maximum value of the specified output

raster type. This can be overridden by using the advanced parameters for lower and upper bound value. If the bounds have the same value or both are zero (default) the algorithm will create random values in the full value range of the chosen raster data type. Choosing bounds outside the acceptable range of the output raster type will abort the algorithm.

| Parameter                | Type              | Required | Default    | Description                                                                                                                           |
|--------------------------|-------------------|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| EXTENT                   | extent            | yes      | —          | Desired extent                                                                                                                        |
| TARGET_CRS               | crs               | no       | ProjectCrs | Target CRS                                                                                                                            |
| PIXEL_SIZE               | number            | no       | 1          | Pixel size                                                                                                                            |
| OUTPUT_TYPE *(adv)*      | enum              | no       | 5          | Output raster data type — options: 0=Byte, 1=Integer16, 2=Unsigned Integer16, 3=Integer32, 4=Unsigned Integer32, 5=Float32, 6=Float64 |
| LOWER_BOUND *(adv)*      | number            | yes      | —          | Lower bound for random number range                                                                                                   |
| UPPER_BOUND *(adv)*      | number            | yes      | —          | Upper bound for random number range                                                                                                   |
| CREATE_OPTIONS           | string            | no       | —          | Creation options                                                                                                                      |
| CREATION_OPTIONS *(adv)* | string            | no       | —          | Creation options                                                                                                                      |
| OUTPUT                   | rasterDestination | yes      | —          | Output raster                                                                                                                         |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:createrandomuniformrasterlayer EXTENT=0,1000,0,1000 LOWER_BOUND=10
↳ UPPER_BOUND=10 TARGET_CRS=EPSG:6346 PIXEL_SIZE=1 CREATE_OPTIONS=value
↳ OUTPUT=output.tif
```

This calls **Create random raster layer (uniform distribution)**: EXTENT=0,1000,0,1000 — Desired extent; LOWER\_BOUND=10 — Lower bound for random number range; UPPER\_BOUND=10 — Upper bound for random number range; TARGET\_CRS=EPSG:6346 — Target CRS; PIXEL\_SIZE=1 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:createspatialindex — Create spatial index

**group:** Vector general

Creates an index to speed up access to the features in a layer based on their spatial location. Support for spatial index creation is dependent on the layer's data provider.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | vector | yes      | —       | Input layer |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:createspatialindex INPUT=input.gpkg
```

This calls **Create spatial index**: INPUT=input.gpkg — Input layer. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:dbscanclustering — DBSCAN clustering

**group:** Vector analysis

This algorithm clusters point features based on a 2D implementation of Density-based spatial clustering of applications with noise (DBSCAN) algorithm.

The algorithm requires two parameters, a minimum cluster size (“minPts”), and the maximum distance allowed between clustered points (“eps”).

| Parameter               | Type     | Required | Default      | Description                               |
|-------------------------|----------|----------|--------------|-------------------------------------------|
| INPUT                   | source   | yes      | —            | Input layer                               |
| MIN_SIZE                | number   | no       | 5            | Minimum cluster size                      |
| EPS                     | distance | no       | 1            | Maximum distance between clustered points |
| DBSCAN* *(adv)*         | boolean  | no       | False        | Treat border points as noise (DBSCAN*)    |
| FIELD_NAME *(adv)*      | string   | no       | CLUSTER_ID   | Cluster field name                        |
| SIZE_FIELD_NAME *(adv)* | string   | no       | CLUSTER_SIZE | Cluster size field name                   |
| OUTPUT                  | sink     | yes      | —            | Clusters                                  |

**Outputs:** OUTPUT (outputVector), NUM\_CLUSTERS (outputNumber)

### Example usage

```
run native:dbscanclustering INPUT=input.gpkg MIN_SIZE=5 EPS=1 OUTPUT=output.gpkg
```

This calls **DBSCAN clustering**: INPUT=input.gpkg — Input layer; MIN\_SIZE=5 — Minimum cluster size; EPS=1 — Maximum distance between clustered points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:definecurrentprojection — Define projection

**group:** Vector general

This algorithm sets an existing layer's projection to the provided CRS without reprojecting features. Contrary to the “Assign projection” algorithm, it will not output a new layer.

If the input layer is a shapefile, the .prj file will be overwritten — or created if missing — to match the provided CRS.

| Parameter | Type   | Required | Default                                   | Description     |
|-----------|--------|----------|-------------------------------------------|-----------------|
| INPUT     | vector | yes      | —                                         | Input shapefile |
| CRS       | crs    | no       | <QgsCoordinateReferenceSystem: EPSG:4326> | CRS             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:definecurrentprojection INPUT=input.gpkg CRS=EPSG:6346
```

This calls **Define projection**: INPUT=input.gpkg — Input shapefile; CRS=EPSG:6346 — CRS. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:delaunaytriangulation — Delaunay triangulation ★

**group:** Vector geometry · **niva verb:** delaunay

This algorithm creates a polygon layer with the Delaunay triangulation corresponding to a points layer.

| Parameter      | Type    | Required | Default | Description             |
|----------------|---------|----------|---------|-------------------------|
| INPUT          | source  | yes      | —       | Input layer             |
| TOLERANCE      | number  | no       | 0       | Tolerance               |
| ADD_ATTRIBUTES | boolean | no       | True    | Add point IDs to output |
| OUTPUT         | sink    | yes      | —       | Delaunay triangulation  |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:delaunaytriangulation INPUT=input.gpkg TOLERANCE=0 ADD_ATTRIBUTES=true  
↳ OUTPUT=output.gpkg
```

This calls **Delaunay triangulation**: INPUT=input.gpkg — Input layer; TOLERANCE=0 — Tolerance; ADD\_ATTRIBUTES=true — Add point IDs to output; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:deletecolumn — Drop field(s) ★**

**group:** Vector table · **niva verb:** dropfields

This algorithm takes a vector layer and generates a new one that has the exact same content but without the selected columns.

| Parameter | Type   | Required | Default | Description      |
|-----------|--------|----------|---------|------------------|
| INPUT     | source | yes      | —       | Input layer      |
| COLUMN    | field  | yes      | —       | Fields to drop   |
| OUTPUT    | sink   | yes      | —       | Remaining fields |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:deletecolumn INPUT=input.gpkg COLUMN=field1 OUTPUT=output.gpkg
```

This calls **Drop field(s)**: INPUT=input.gpkg — Input layer; COLUMN=field1 — Fields to drop; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:deleteduplicategeometries — Delete duplicate geometries**

**group:** Vector general

This algorithm finds duplicated geometries and removes them.

Attributes are not checked, so in case two features have identical geometries but different attributes, only one of them will be added to the result layer.

Optionally, these duplicate features can be saved to a separate output for analysis.

| Parameter  | Type   | Required | Default | Description |
|------------|--------|----------|---------|-------------|
| INPUT      | source | yes      | —       | Input layer |
| OUTPUT     | sink   | yes      | —       | Cleaned     |
| DUPLICATES | sink   | no       | —       | Duplicates  |

**Outputs:** OUTPUT (outputVector), DUPLICATES (outputVector), RETAINED\_COUNT (outputNumber), DUPLICATE\_COUNT (outputNumber)

**Example usage**

```
run native:deleteduplicategeometries INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Delete duplicate geometries**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:deleteholes — Delete holes

**group:** Vector geometry

This algorithm takes a polygon layer and removes holes in polygons. It creates a new vector layer in which polygons with holes have been replaced by polygons with only their external ring. Attributes are not modified.

An optional minimum area parameter allows removing only holes which are smaller than a specified area threshold. Leaving this parameter as 0.0 results in all holes being removed.

| Parameter | Type   | Required | Default | Description                      |
|-----------|--------|----------|---------|----------------------------------|
| INPUT     | source | yes      | —       | Input layer                      |
| MIN_AREA  | area   | no       | 0.0     | Remove holes with area less than |
| OUTPUT    | sink   | yes      | —       | Cleaned                          |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:deleteholes INPUT=input.gpkg MIN_AREA=0 OUTPUT=output.gpkg
```

This calls **Delete holes**: INPUT=input.gpkg — Input layer; MIN\_AREA=0 — Remove holes with area less than; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:densifygeometries — Densify by count

**group:** Vector geometry

This algorithm takes a polygon or line layer and generates a new one in which the geometries have a larger number of vertices than the original one.

If the geometries have z or m values present then these will be linearly interpolated at the added nodes.

The number of new vertices to add to each feature geometry is specified as an input parameter.

| Parameter | Type   | Required | Default | Description               |
|-----------|--------|----------|---------|---------------------------|
| INPUT     | source | yes      | —       | Input layer               |
| VERTICES  | number | no       | 1       | Number of vertices to add |
| OUTPUT    | sink   | yes      | —       | Densified                 |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:densifygeometries INPUT=input.gpkg VERTICES=1 OUTPUT=output.gpkg
```

This calls **Densify by count**: INPUT=input.gpkg — Input layer; VERTICES=1 — Number of vertices to add; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:densifygeometriesgivenaninterval — Densify by interval ★

**group:** Vector geometry · **niva verb:** densify

This algorithm takes a polygon or line layer and generates a new one in which the geometries have a larger number of vertices than the original one.

Geometries are densified by adding additional vertices on edges that have a maximum distance of the interval parameter in map units.

| Parameter | Type     | Required | Default | Description                      |
|-----------|----------|----------|---------|----------------------------------|
| INPUT     | source   | yes      | —       | Input layer                      |
| INTERVAL  | distance | no       | 1       | Interval between vertices to add |
| OUTPUT    | sink     | yes      | —       | Densified                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:densifygeometriesgivenaninterval INPUT=input.gpkg INTERVAL=1 OUTPUT=output.gpkg
```

This calls **Densify by interval**: INPUT=input.gpkg — Input layer; INTERVAL=1 — Interval between vertices to add; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:detectvectorchanges — Detect dataset changes

**group:** Vector general

This algorithm compares two vector layers, and determines which features are unchanged, added or deleted between the two. It is designed for comparing two different versions of the same dataset.

When comparing features, the original and revised feature geometries will be compared against each other. Depending on the Geometry Comparison Behavior setting, the comparison will either be made using an exact comparison (where geometries must be an exact match for each other, including the order and count of vertices) or a topological comparison only (where geometries are considered equal if all of their component edges overlap. E.g. lines with the same vertex locations but opposite direction will be considered equal by this method). If the topological comparison is selected then any z or m values present in the geometries will not be compared.

By default, the algorithm compares all attributes from the original and revised features. If the Attributes to Consider for Match parameter is changed, then only the selected attributes will be compared (e.g. allowing users to ignore a timestamp or ID field which is expected to change between the revisions).

If any features in the original or revised layers do not have an associated geometry, then care must be taken to ensure that these features have a unique set of attributes selected for comparison. If this condition is not met, warnings will be raised and the resultant outputs may be misleading.

The algorithm outputs three layers, one containing all features which are considered to be unchanged between the revisions, one containing features deleted from the original layer which are not present in the revised layer, and one containing features added to the revised layer which are not present in the original layer.

| Parameter          | Type   | Required | Default | Description                                                                                    |
|--------------------|--------|----------|---------|------------------------------------------------------------------------------------------------|
| ORIGINAL           | source | yes      | —       | Original layer                                                                                 |
| REVISED            | source | yes      | —       | Revised layer                                                                                  |
| COMPARE_ATTRIBUTES | field  | no       | —       | Attributes to consider for match (or none to compare geometry only)                            |
| MATCH_TYPE *(adv)* | enum   | no       | 1       | Geometry comparison behavior — options: 0=Exact Match, 1=Tolerant Match (Topological Equality) |
| UNCHANGED          | sink   | no       | —       | Unchanged features                                                                             |
| ADDED              | sink   | no       | —       | Added features                                                                                 |
| DELETED            | sink   | no       | —       | Deleted features                                                                               |

---

**Outputs:** UNCHANGED (outputVector), ADDED (outputVector), DELETED (outputVector), UNCHANGED\_COUNT (outputNumber), ADDED\_COUNT (outputNumber), DELETED\_COUNT (outputNumber)

### Example usage

```
run native:detectvectorchanges ORIGINAL=input.gpkg REVISED=input.gpkg  
↳ COMPARE_ATTRIBUTES=field1 UNCHANGED=unchanged.gpkg
```

This calls **Detect dataset changes**: ORIGINAL=input.gpkg — Original layer; REVISED=input.gpkg — Revised layer; COMPARE\_ATTRIBUTES=field1 — Attributes to consider for match (or none to compare geometry only); UNCHANGED=unchanged.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:difference — Difference ★

**group:** Vector overlay · **niva verb:** difference

This algorithm extracts features from the Input layer that fall outside, or partially overlap, features in the Overlay layer. Input layer features that partially overlap feature(s) in the Overlay layer are split along those features' boundary and only the portions outside the Overlay layer features are retained.

Attributes are not modified, although properties such as area or length of the features will be modified by the difference operation. If such properties are stored as attributes, those attributes will have to be manually updated.

| Parameter         | Type   | Required | Default | Description   |
|-------------------|--------|----------|---------|---------------|
| INPUT             | source | yes      | —       | Input layer   |
| OVERLAY           | source | yes      | —       | Overlay layer |
| OUTPUT            | sink   | yes      | —       | Difference    |
| GRID_SIZE *(adv)* | number | no       | —       | Grid size     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:difference INPUT=input.gpkg OVERLAY=input.gpkg OUTPUT=output.gpkg
```

This calls **Difference**: INPUT=input.gpkg — Input layer; OVERLAY=input.gpkg — Overlay layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:dissolve — Dissolve ★

**group:** Vector geometry · **niva verb:** dissolve

This algorithm takes a vector layer and combines their features into new features. One or more attributes can be specified to dissolve features belonging to the same class (having the same value for the specified attributes), alternatively all features can be dissolved in a single one.

All output geometries will be converted to multi geometries. In case the input is a polygon layer, common boundaries of adjacent polygons being dissolved will get erased.

If enabled, the optional "Keep disjoint features separate" setting will cause features and parts that do not overlap or touch to be exported as separate features (instead of parts of a single multipart feature).

| Parameter                 | Type    | Required | Default | Description                     |
|---------------------------|---------|----------|---------|---------------------------------|
| INPUT                     | source  | yes      | —       | Input layer                     |
| FIELD                     | field   | no       | —       | Dissolve field(s)               |
| SEPARATE_DISJOINT *(adv)* | boolean | no       | False   | Keep disjoint features separate |
| OUTPUT                    | sink    | yes      | —       | Dissolved                       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:dissolve INPUT=input.gpkg FIELD=field1 OUTPUT=output.gpkg
```

This calls **Dissolve**: INPUT=input.gpkg — Input layer; FIELD=field1 — Dissolve field(s); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:distancetonearesthub — Distance to nearest hub

**group:** Vector analysis

This algorithm computes the distance between features from the source layer to the closest feature from the destination layer.

Distance calculations are based on the feature's bounding box center.

The resulting line layer contains lines linking each origin point with its nearest destination feature.

The resulting point layer contains each origin feature's center point with additional fields indicating the identifier of the nearest destination feature and the distance to it.

| Parameter     | Type   | Required | Default | Description                                                                        |
|---------------|--------|----------|---------|------------------------------------------------------------------------------------|
| INPUT         | source | yes      | —       | Source layer (spokes)                                                              |
| HUBS          | source | yes      | —       | Destination layer (hubs)                                                           |
| FIELD         | field  | yes      | —       | Hub layer name attribute                                                           |
| UNIT          | enum   | no       | 0       | Measurement unit — options: 0=Meters, 1=Feet, 2=Miles, 3=Kilometers, 4=Layer Units |
| OUTPUT_LINES  | sink   | no       | —       | Hub lines                                                                          |
| OUTPUT_POINTS | sink   | no       | —       | Hub points                                                                         |

**Outputs:** OUTPUT\_LINES (outputVector), OUTPUT\_POINTS (outputVector)

### Example usage

```
run native:distancetonearesthub INPUT=input.gpkg HUBS=input.gpkg FIELD=field1 UNIT=0
↳ OUTPUT_LINES=output_lines.gpkg
```

This calls **Distance to nearest hub**: INPUT=input.gpkg — Source layer (spokes); HUBS=input.gpkg — Destination layer (hubs); FIELD=field1 — Hub layer name attribute; UNIT=0 selects *Meters*; OUTPUT\_LINES=output\_lines.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:downloadgpsdata — Download GPS data from device

**group:** GPS

This algorithm uses the GPSTool tool to download data from a GPS device into the GPX standard format.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| DEVICE    | string | yes      | —       | Device      |



| Parameter    | Type            | Required | Default | Description                                             |
|--------------|-----------------|----------|---------|---------------------------------------------------------|
| PORT         | string          | yes      | —       | Port                                                    |
| FEATURE_TYPE | enum            | no       | 0       | Feature type — options: 0=Waypoints, 1=Routes, 2=Tracks |
| OUTPUT       | fileDestination | yes      | —       | Output                                                  |

**Outputs:** OUTPUT (outputFile), OUTPUT\_LAYER (outputVector)

### Example usage

```
run native:downloadgpsdata DEVICE=value PORT=value FEATURE_TYPE=0 OUTPUT=output.txt
```

This calls **Download GPS data from device**: DEVICE=value — Device; PORT=value — Port; FEATURE\_TYPE=0 selects *Waypoints*; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:downloadvectortiles — Download vector tiles

**group:** Vector tiles

This algorithm downloads vector tiles of the input vector tile layer and saves them in the local vector tile file.

| Parameter  | Type                  | Required | Default | Description                    |
|------------|-----------------------|----------|---------|--------------------------------|
| INPUT      | layer                 | yes      | —       | Input layer                    |
| EXTENT     | extent                | yes      | —       | Extent                         |
| MAX_ZOOM   | number                | no       | 10      | Maximum zoom level to download |
| TILE_LIMIT | number                | no       | 100     | Tile limit                     |
| OUTPUT     | vectorTileDestination | yes      | —       | Output                         |

**Outputs:** OUTPUT (outputVectorTile)

### Example usage

```
run native:downloadvectortiles INPUT=input.gpkg EXTENT=0,1000,0,1000 MAX_ZOOM=10
↳ TILE_LIMIT=100 OUTPUT=output.mbtiles
```

This calls **Download vector tiles**: INPUT=input.gpkg — Input layer; EXTENT=0,1000,0,1000 — Extent; MAX\_ZOOM=10 — Maximum zoom level to download; TILE\_LIMIT=100 — Tile limit; OUTPUT=output.mbtiles is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:dropgeometries — Drop geometries

**group:** Vector general

This algorithm removes any geometries from an input layer and returns a layer containing only the feature attributes.

| Parameter | Type   | Required | Default | Description        |
|-----------|--------|----------|---------|--------------------|
| INPUT     | source | yes      | —       | Input layer        |
| OUTPUT    | sink   | yes      | —       | Dropped geometries |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:dropgeometries INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Drop geometries**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:dropmzvalues — Drop M/Z values

**group:** Vector geometry

This algorithm can remove any measure (M) or Z values from input geometries.

| Parameter     | Type    | Required | Default | Description   |
|---------------|---------|----------|---------|---------------|
| INPUT         | source  | yes      | —       | Input layer   |
| DROP_M_VALUES | boolean | no       | False   | Drop M Values |
| DROP_Z_VALUES | boolean | no       | False   | Drop Z Values |
| OUTPUT        | sink    | yes      | —       | Z/M Dropped   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:dropmzvalues INPUT=input.gpkg DROP_M_VALUES=false DROP_Z_VALUES=false  
↳ OUTPUT=output.gpkg
```

This calls **Drop M/Z values**: INPUT=input.gpkg — Input layer; DROP\_M\_VALUES=false — Drop M Values; DROP\_Z\_VALUES=false — Drop Z Values; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:dtmslopebasedfilter — DTM filter (slope-based)

**group:** Raster terrain analysis

This algorithm can be used to filter a Digital Elevation Model in order to classify its cells into ground and object (non-ground) cells.

The tool uses concepts as described by Vosselman (2000) and is based on the assumption that a large height difference between two nearby cells is unlikely to be caused by a steep slope in the terrain. The probability that the higher cell might be non-ground increases when the distance between the two cells decreases. Therefore the filter defines a maximum height difference ( $dz_{max}$ ) between two cells as a function of the distance ( $d$ ) between the cells ( $dz_{max}(d) = d$ ).

A cell is classified as terrain if there is no cell within the kernel radius to which the height difference is larger than the allowed maximum height difference at the distance between these two cells.

The approximate terrain slope ( $s$ ) parameter is used to modify the filter function to match the overall slope in the study area ( $dz_{max}(d) = d * s$ ).

A 5 % confidence interval ( $ci = 1.65 * \sqrt{2 * \text{stddev}}$ ) may be used to modify the filter function even further by either relaxing ( $dz_{max}(d) = d * s + ci$ ) or amplifying ( $dz_{max}(d) = d * s - ci$ ) the filter criterium.

References: Vosselman, G. (2000): Slope based filtering of laser altimetry data. IAPRS, Vol. XXXIII, Part B3, Amsterdam, The Netherlands, 935-942

This algorithm is a port of the SAGA 'DTM Filter (slope-based)' tool.

| Parameter           | Type   | Required | Default | Description                                                      |
|---------------------|--------|----------|---------|------------------------------------------------------------------|
| INPUT               | raster | yes      | —       | Input layer                                                      |
| BAND                | band   | no       | 1       | Band number                                                      |
| RADIUS              | number | no       | 5       | Kernel radius (pixels)                                           |
| TERRAIN_SLOPE       | number | no       | 30      | Terrain slope (% , pixel size/vertical units)                    |
| FILTER_MODIFICATION | enum   | no       | 0       | Filter modification — options: 0=None, 1=Relax filter, 2=Amplify |
| STANDARD_DEVIATION  | number | no       | 0.1     | Standard deviation                                               |
| CREATE_OPTIONS      | string | no       | —       | Creation options                                                 |

| Parameter                   | Type              | Required | Default | Description                       |
|-----------------------------|-------------------|----------|---------|-----------------------------------|
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Creation options                  |
| OUTPUT_GROUND               | rasterDestination | no       | —       | Output layer (ground)             |
| OUTPUT_NONGROUND            | rasterDestination | no       | —       | Output layer (non-ground objects) |

**Outputs:** OUTPUT\_GROUND (outputRaster), OUTPUT\_NONGROUND (outputRaster)

### Example usage

```
run native:dtmslopebasedfilter INPUT=dem.tif BAND=1 RADIUS=5 TERRAIN_SLOPE=30
  ↪ FILTER_MODIFICATION=0 STANDARD_DEVIATION=0.1 CREATE_OPTIONS=value
  ↪ OUTPUT_GROUND=output_ground.tif
```

This calls **DTM filter (slope-based)**: INPUT=dem.tif — Input layer; BAND=1 — Band number; RADIUS=5 — Kernel radius (pixels); TERRAIN\_SLOPE=30 — Terrain slope (%; pixel size/vertical units); FILTER\_MODIFICATION=0 selects *None*; STANDARD\_DEVIATION=0.1 — Standard deviation; CREATE\_OPTIONS=value — Creation options; OUTPUT\_GROUND=output\_ground.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:dxlexport — Export layers to DXF

**group:** Vector general

Exports layers to a DXF file. For each layer, you can choose a field whose values are used to split features in generated destination layers in the DXF output.

If no field is chosen, you can still override the output layer name by directly entering a new output layer name in the Configure Layer panel or by preferring layer title (set in layer properties) to layer name.

| Parameter                    | Type            | Required | Default    | Description                                                                                                                                                                                                                                                                                                                               |
|------------------------------|-----------------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LAYERS                       | dxflayers       | yes      | —          | Input layers                                                                                                                                                                                                                                                                                                                              |
| SYMBOLGY_MODE                | enum            | no       | 0          | Symbology mode — options: 0=No Symbology, 1=Feature Symbology, 2=Symbol Layer Symbology                                                                                                                                                                                                                                                   |
| SYMBOLGY_SCALE               | scale           | no       | 1000000    | Symbology scale                                                                                                                                                                                                                                                                                                                           |
| MAP_THEME                    | maptheme        | no       | —          | Map theme                                                                                                                                                                                                                                                                                                                                 |
| ENCODING                     | enum            | no       | ISO-8859-1 | Encoding — options: 0=ISO-8859-1, 1=ISO-8859-2, 2=ISO-8859-3, 3=ISO-8859-4, 4=ISO-8859-5, 5=ISO-8859-6, 6=ISO-8859-7, 7=ISO-8859-8, 8=ISO-8859-9, 9=Shift_JIS, 10=Big5, 11=CP936, 12=GB2312, 13=ms949, 14=cp850, 15=cp866, 16=cp1250, 17=cp1251, 18=cp1252, 19=cp1253, 20=cp1254, 21=cp1255, 22=cp1256, 23=cp1257, 24=cp1258, 25=macroman |
| CRS                          | crs             | no       | EPSG:4326  | CRS                                                                                                                                                                                                                                                                                                                                       |
| EXTENT                       | extent          | no       | —          | Extent                                                                                                                                                                                                                                                                                                                                    |
| SELECTED_FEATURES_ONLY       | boolean         | no       | False      | Use only selected features                                                                                                                                                                                                                                                                                                                |
| USE_LAYER_TITLE              | boolean         | no       | False      | Use layer title as name                                                                                                                                                                                                                                                                                                                   |
| FORCE_2D                     | boolean         | no       | False      | Force 2D output                                                                                                                                                                                                                                                                                                                           |
| MTEXT                        | boolean         | no       | True       | Export labels as MTEXT elements                                                                                                                                                                                                                                                                                                           |
| EXPORT_LINES_WITH_ZERO_WIDTH | boolean         | yes      | —          | Export lines with zero width                                                                                                                                                                                                                                                                                                              |
| OUTPUT                       | fileDestination | yes      | —          | DXF                                                                                                                                                                                                                                                                                                                                       |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:dxlexport LAYERS=... EXPORT_LINES_WITH_ZERO_WIDTH=false SYMBOLOLOGY_MODE=0
↳ SYMBOLOLOGY_SCALE=1000000 ENCODING=0 CRS=EPSG:6346 OUTPUT=output.txt
```

This calls **Export layers to DXF**: LAYERS=... — Input layers; EXPORT\_LINES\_WITH\_ZERO\_WIDTH=false — Export lines with zero width; SYMBOLOLOGY\_MODE=0 selects *No Symbolology*; SYMBOLOLOGY\_SCALE=1000000 — Symbology scale; ENCODING=0 selects *ISO-8859-1*; CRS=EPSG:6346 — CRS; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:equaltofrequency — Equal to frequency

**group:** Raster analysis

This algorithm evaluates on a cell-by-cell basis the frequency (number of times) the values of an input stack of rasters are equal to the value of a value raster. If multiband rasters are used in the data raster stack, the algorithm will always perform the analysis on the first band of the rasters - use GDAL to use other bands in the analysis. The input value layer serves as reference layer for the sample layers. Any NoData cells in the value raster or the data layer stack will result in a NoData cell in the output raster if the ignore NoData parameter is not checked. The output NoData value can be set manually. The output rasters extent and resolution is defined by the input raster layer and is always of int32 type.

| Parameter                   | Type              | Required | Default | Description          |
|-----------------------------|-------------------|----------|---------|----------------------|
| INPUT_VALUE_RASTER          | raster            | yes      | —       | Input value raster   |
| INPUT_VALUE_RASTER_BAND     | band              | no       | 1       | Value raster band    |
| INPUT_RASTERS               | multilayer        | yes      | —       | Input raster layers  |
| IGNORE_NODATA               | boolean           | no       | False   | Ignore NoData values |
| OUTPUT_NODATA_VALUE *(adv)* | number            | no       | -9999   | Output NoData value  |
| CREATE_OPTIONS              | string            | no       | —       | Creation options     |
| CREATION_OPTIONS *(adv)*    | string            | no       | —       | Creation options     |
| OUTPUT                      | rasterDestination | yes      | —       | Output layer         |

**Outputs:** OUTPUT (outputRaster), OCCURRENCE\_COUNT (outputNumber), FOUND\_LOCATIONS\_COUNT (outputNumber), MEAN\_FREQUENCY\_PER\_LOCATION (outputNumber), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:equaltofrequency INPUT_VALUE_RASTER=dem.tif INPUT_RASTERS="a.gpkg;b.gpkg"
↳ INPUT_VALUE_RASTER_BAND=1 IGNORE_NODATA=false CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Equal to frequency**: INPUT\_VALUE\_RASTER=dem.tif — Input value raster; INPUT\_RASTERS="a.gpkg;b.gpkg" — Input raster layers; INPUT\_VALUE\_RASTER\_BAND=1 — Value raster band; IGNORE\_NODATA=false — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:explodehstorefield — Explode HStore Field

**group:** Vector table

This algorithm creates a copy of the input layer and adds a new field for every unique key in the HStore field. The expected field list is an optional comma separated list. By default, all unique keys are added. If this list is specified, only these fields are added and the HStore field is updated.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |

| Parameter       | Type   | Required | Default | Description                                  |
|-----------------|--------|----------|---------|----------------------------------------------|
| FIELD           | field  | yes      | —       | HStore field                                 |
| EXPECTED_FIELDS | string | no       | —       | Expected list of fields separated by a comma |
| OUTPUT          | sink   | yes      | —       | Exploded                                     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:explodehstorefield INPUT=input.gpkg FIELD=field1 EXPECTED_FIELDS=value
↳ OUTPUT=output.gpkg
```

This calls **Explode HStore Field**: INPUT=input.gpkg — Input layer; FIELD=field1 — HStore field; EXPECTED\_FIELDS=value — Expected list of fields separated by a comma; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:explodelines — Explode lines

**group:** Vector geometry

This algorithm takes a lines layer and creates a new one in which each line is replaced by a set of lines representing the segments in the original line. Each line in the resulting layer contains only a start and an end point, with no intermediate nodes between them.

If the input layer consists of CircularStrings or CompoundCurves, the output layer will be of the same type and contain only single curve segments.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Exploded    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:explodelines INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Explode lines**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:exportaddgeometrycolumns — Add geometry attributes

**group:** Vector geometry

This algorithm computes geometric properties of the features in a vector layer. The algorithm generates a new vector layer with the same content as the input one, but with additional attributes in its attributes table, containing geometric measurements.

Depending on the geometry type of the vector layer, the attributes added to the table will be different.

| Parameter | Type   | Required | Default | Description                                                                                                                               |
|-----------|--------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                                                                                                               |
| METHOD    | enum   | no       | 0       | Calculate using — options: 0=Cartesian Calculations in Layer's CRS, 1=Cartesian Calculations in Project's CRS, 2=Ellipsoidal Calculations |
| OUTPUT    | sink   | yes      | —       | Added geometry info                                                                                                                       |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:exportaddgeometrycolumns INPUT=input.gpkg METHOD=0 OUTPUT=output.gpkg
```

This calls **Add geometry attributes**: INPUT=input.gpkg — Input layer; METHOD=0 selects *Cartesian Calculations in Layer's CRS*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:exportlayermetadata — Export layer metadata

**group:** Metadata tools

This algorithm exports layer's metadata to a QMD file.

| Parameter | Type            | Required | Default | Description |
|-----------|-----------------|----------|---------|-------------|
| INPUT     | layer           | yes      | —       | Layer       |
| OUTPUT    | fileDestination | yes      | —       | Output      |

**Outputs:** OUTPUT (outputFile)

**Example usage**

```
run native:exportlayermetadata INPUT=input.gpkg OUTPUT=output.txt
```

This calls **Export layer metadata**: INPUT=input.gpkg — Layer; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:exportlayersinformation — Export layer(s) information

**group:** Layer tools

This algorithm creates a polygon layer with features corresponding to the extent of selected layer(s).

Additional layer details - CRS, provider name, file path, layer name, subset filter, abstract and attribution - are attached as attributes to each feature.

| Parameter | Type       | Required | Default | Description    |
|-----------|------------|----------|---------|----------------|
| LAYERS    | multilayer | yes      | —       | Input layer(s) |
| OUTPUT    | sink       | yes      | —       | Output         |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:exportlayersinformation LAYERS="a.gpkg;b.gpkg" OUTPUT=output.gpkg
```

This calls **Export layer(s) information**: LAYERS="a.gpkg;b.gpkg" — Input layer(s); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:exportmeshedges — Export mesh edges

**group:** Mesh

This algorithm exports a mesh layer's edges to a line vector layer, with the dataset values on edges as attribute values.

| Parameter      | Type              | Required | Default | Description                                                                                          |
|----------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------|
| INPUT          | mesh              | yes      | —       | Input mesh layer                                                                                     |
| DATASET_GROUPS | meshdatasetgroups | no       | []      | Dataset groups                                                                                       |
| DATASET_TIME   | meshdatasettime   | yes      | —       | Dataset time                                                                                         |
| CRS_OUTPUT     | crs               | no       | —       | Output coordinate system                                                                             |
| VECTOR_OPTION  | enum              | no       | 0       | Export vector option — options: 0=Cartesian (x,y), 1=Polar (magnitude,degree), 2=Cartesian and Polar |
| OUTPUT         | sink              | yes      | —       | Output vector layer                                                                                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:exportmeshedges INPUT=mesh.nc DATASET_TIME=... CRS_OUTPUT=EPSG:6346  
↳ VECTOR_OPTION=0 OUTPUT=output.gpkg
```

This calls **Export mesh edges**: INPUT=mesh.nc — Input mesh layer; DATASET\_TIME=... — Dataset time; CRS\_OUTPUT=EPSG:6346 — Output coordinate system; VECTOR\_OPTION=0 selects *Cartesian* (x,y); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:exportmeshfaces — Export mesh faces

**group:** Mesh

This algorithm exports a mesh layer's faces to a polygon vector layer, with the dataset values on faces as attribute values.

| Parameter      | Type              | Required | Default | Description                                                                                          |
|----------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------|
| INPUT          | mesh              | yes      | —       | Input mesh layer                                                                                     |
| DATASET_GROUPS | meshdatasetgroups | no       | []      | Dataset groups                                                                                       |
| DATASET_TIME   | meshdatasettime   | yes      | —       | Dataset time                                                                                         |
| CRS_OUTPUT     | crs               | no       | —       | Output coordinate system                                                                             |
| VECTOR_OPTION  | enum              | no       | 0       | Export vector option — options: 0=Cartesian (x,y), 1=Polar (magnitude,degree), 2=Cartesian and Polar |
| OUTPUT         | sink              | yes      | —       | Output vector layer                                                                                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:exportmeshfaces INPUT=mesh.nc DATASET_TIME=... CRS_OUTPUT=EPSG:6346  
↳ VECTOR_OPTION=0 OUTPUT=output.gpkg
```

This calls **Export mesh faces**: INPUT=mesh.nc — Input mesh layer; DATASET\_TIME=... — Dataset time; CRS\_OUTPUT=EPSG:6346 — Output coordinate system; VECTOR\_OPTION=0 selects *Cartesian* (x,y); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:exportmeshongrid — Export mesh on grid

**group:** Mesh

This algorithm exports a mesh layer's dataset values to a gridded point vector layer, with the dataset values on each point as attribute values. For data on volume (3D stacked dataset values), the exported dataset values are averaged on faces using the method defined in the mesh layer properties (default is Multi level averaging method). 1D meshes are not supported.

| Parameter      | Type              | Required | Default | Description                                                                                          |
|----------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------|
| INPUT          | mesh              | yes      | —       | Input mesh layer                                                                                     |
| DATASET_GROUPS | meshdatasetgroups | no       | []      | Dataset groups                                                                                       |
| DATASET_TIME   | meshdatasettime   | yes      | —       | Dataset time                                                                                         |
| EXTENT         | extent            | no       | —       | Extent                                                                                               |
| GRID_SPACING   | distance          | no       | 10      | Grid spacing                                                                                         |
| CRS_OUTPUT     | crs               | no       | —       | Output coordinate system                                                                             |
| VECTOR_OPTION  | enum              | no       | 0       | Export vector option — options: 0=Cartesian (x,y), 1=Polar (magnitude,degree), 2=Cartesian and Polar |
| OUTPUT         | sink              | yes      | —       | Output vector layer                                                                                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:exportmeshongrid INPUT=mesh.nc DATASET_TIME=... EXTENT=0,1000,0,1000
↳ GRID_SPACING=10 CRS_OUTPUT=EPSG:6346 VECTOR_OPTION=0 OUTPUT=output.gpkg
```

This calls **Export mesh on grid**: INPUT=mesh.nc — Input mesh layer; DATASET\_TIME=... — Dataset time; EXTENT=0,1000,0,1000 — Extent; GRID\_SPACING=10 — Grid spacing; CRS\_OUTPUT=EPSG:6346 — Output coordinate system; VECTOR\_OPTION=0 selects *Cartesian* (x,y); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:exportmeshvertices — Export mesh vertices

**group:** Mesh

This algorithm exports a mesh layer's vertices to a point vector layer, with the dataset values on vertices as attribute values.

| Parameter      | Type              | Required | Default | Description                                                                                          |
|----------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------|
| INPUT          | mesh              | yes      | —       | Input mesh layer                                                                                     |
| DATASET_GROUPS | meshdatasetgroups | no       | []      | Dataset groups                                                                                       |
| DATASET_TIME   | meshdatasettime   | yes      | —       | Dataset time                                                                                         |
| CRS_OUTPUT     | crs               | no       | —       | Output coordinate system                                                                             |
| VECTOR_OPTION  | enum              | no       | 0       | Export vector option — options: 0=Cartesian (x,y), 1=Polar (magnitude,degree), 2=Cartesian and Polar |
| OUTPUT         | sink              | yes      | —       | Output vector layer                                                                                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:exportmeshvertices INPUT=mesh.nc DATASET_TIME=... CRS_OUTPUT=EPSG:6346
↳ VECTOR_OPTION=0 OUTPUT=output.gpkg
```

This calls **Export mesh vertices**: INPUT=mesh.nc — Input mesh layer; DATASET\_TIME=... — Dataset time; CRS\_OUTPUT=EPSG:6346 — Output coordinate system; VECTOR\_OPTION=0 selects *Cartesian* (x,y); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## native:exporttospreadsheet — Export to spreadsheet

**group:** Layer tools

This algorithm collects a number of existing layers and exports them to a spreadsheet document. Optionally the layers can be appended to an existing spreadsheet as additional sheets.

| Parameter        | Type            | Required | Default | Description                                   |
|------------------|-----------------|----------|---------|-----------------------------------------------|
| LAYERS           | multilayer      | yes      | —       | Input layers                                  |
| USE_ALIAS        | boolean         | no       | False   | Use field aliases as column headings          |
| FORMATTED_VALUES | boolean         | no       | False   | Export formatted values instead of raw values |
| OUTPUT           | fileDestination | yes      | —       | Destination spreadsheet                       |
| OVERWRITE        | boolean         | no       | True    | Overwrite existing spreadsheet                |

**Outputs:** OUTPUT (outputFile), OUTPUT\_LAYERS (outputMultilayer)

### Example usage

```
run native:exporttospreadsheet LAYERS="a.gpkg;b.gpkg" USE_ALIAS=false  
↳ FORMATTED_VALUES=false OVERWRITE=true OUTPUT=output.txt
```

This calls **Export to spreadsheet**: LAYERS="a.gpkg;b.gpkg" — Input layers; USE\_ALIAS=false — Use field aliases as column headings; FORMATTED\_VALUES=false — Export formatted values instead of raw values; OVERWRITE=true — Overwrite existing spreadsheet; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extendlines — Extend lines

**group:** Vector geometry

This algorithm extends line geometries by a specified amount at the start and end of the line. Lines are extended using the bearing of the first and last segment in the line.

| Parameter      | Type     | Required | Default | Description    |
|----------------|----------|----------|---------|----------------|
| INPUT          | source   | yes      | —       | Input layer    |
| START_DISTANCE | distance | no       | 0.0     | Start distance |
| END_DISTANCE   | distance | no       | 0.0     | End distance   |
| OUTPUT         | sink     | yes      | —       | Extended       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extendlines INPUT=input.gpkg START_DISTANCE=0 END_DISTANCE=0 OUTPUT=output.gpkg
```

This calls **Extend lines**: INPUT=input.gpkg — Input layer; START\_DISTANCE=0 — Start distance; END\_DISTANCE=0 — End distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extenttolayer — Create layer from extent

**group:** Vector creation

This algorithm creates a new vector layer that contains a single feature with geometry matching an extent parameter.

It can be used in models to convert an extent into a layer which can be used for other algorithms which require a layer based input.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | extent | yes      | —       | Extent      |
| OUTPUT    | sink   | yes      | —       | Extent      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extenttoayer INPUT=0,1000,0,1000 OUTPUT=output.gpkg
```

This calls **Create layer from extent**: INPUT=0,1000,0,1000 — Extent; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractbinary — Extract binary field

**group:** Vector table

This algorithm extracts contents from a binary field, saving them to individual files.

Filenames can be generated using values taken from an attribute in the source table or based on a more complex expression.

| Parameter | Type              | Required | Default | Description        |
|-----------|-------------------|----------|---------|--------------------|
| INPUT     | source            | yes      | —       | Input layer        |
| FIELD     | field             | yes      | —       | Binary field       |
| FILENAME  | expression        | yes      | —       | File name          |
| FOLDER    | folderDestination | yes      | —       | Destination folder |

**Outputs:** FOLDER (outputFolder)

### Example usage

```
run native:extractbinary INPUT=input.gpkg FIELD=field1 FILENAME="value > 0" FOLDER=folder/
```

This calls **Extract binary field**: INPUT=input.gpkg — Input layer; FIELD=field1 — Binary field; FILENAME="value > 0" — File name; FOLDER=folder/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractbyattribute — Extract by attribute

**group:** Vector selection

This algorithm creates a new vector layer that only contains matching features from an input layer. The criteria for adding features to the resulting layer is defined based on the values of an attribute from the input layer.

| Parameter   | Type   | Required | Default | Description                                                                                                          |
|-------------|--------|----------|---------|----------------------------------------------------------------------------------------------------------------------|
| INPUT       | source | yes      | —       | Input layer                                                                                                          |
| FIELD       | field  | yes      | —       | Selection attribute                                                                                                  |
| OPERATOR    | enum   | no       | 0       | Operator — options: 0==, 1≠, 2>, 3≥, 4<, 5≤, 6=begin with, 7=contains, 8=is null, 9=is not null, 10=does not contain |
| VALUE       | string | no       | —       | Value                                                                                                                |
| OUTPUT      | sink   | yes      | —       | Extracted (attribute)                                                                                                |
| FAIL_OUTPUT | sink   | no       | —       | Extracted (non-matching)                                                                                             |

**Outputs:** OUTPUT (outputVector), FAIL\_OUTPUT (outputVector)

### Example usage

```
run native:extractbyattribute INPUT=input.gpkg FIELD=field1 OPERATOR=0 VALUE=value
↳ OUTPUT=output.gpkg
```

This calls **Extract by attribute**: INPUT=input.gpkg — Input layer; FIELD=field1 — Selection attribute; OPERATOR=0 selects =; VALUE=value — Value; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractbyexpression — Extract by expression ★

**group:** Vector selection · **niva verb:** filter

This algorithm creates a new vector layer that only contains matching features from an input layer. The criteria for adding features to the resulting layer is based on a QGIS expression.

For help with QGIS expression functions, see the inbuilt help for specific functions which is available in the expression builder.

| Parameter   | Type       | Required | Default | Description       |
|-------------|------------|----------|---------|-------------------|
| INPUT       | source     | yes      | —       | Input layer       |
| EXPRESSION  | expression | yes      | —       | Expression        |
| OUTPUT      | sink       | yes      | —       | Matching features |
| FAIL_OUTPUT | sink       | no       | —       | Non-matching      |

**Outputs:** OUTPUT (outputVector), FAIL\_OUTPUT (outputVector)

### Example usage

```
run native:extractbyexpression INPUT=input.gpkg EXPRESSION="value > 0" OUTPUT=output.gpkg
```

This calls **Extract by expression**: INPUT=input.gpkg — Input layer; EXPRESSION="value > 0" — Expression; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractbyextent — Extract/clip by extent

**group:** Vector overlay

This algorithm creates a new vector layer that only contains features which fall within a specified extent. Any features which intersect the extent will be included.

Optionally, feature geometries can also be clipped to the extent. If this option is selected, then the output geometries will automatically be converted to multi geometries to ensure uniform output geometry types.

| Parameter | Type    | Required | Default | Description             |
|-----------|---------|----------|---------|-------------------------|
| INPUT     | source  | yes      | —       | Input layer             |
| EXTENT    | extent  | yes      | —       | Extent                  |
| CLIP      | boolean | no       | False   | Clip features to extent |
| OUTPUT    | sink    | yes      | —       | Extracted               |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractbyextent INPUT=input.gpkg EXTENT=0,1000,0,1000 CLIP=false
↳ OUTPUT=output.gpkg
```

This calls **Extract/clip by extent**: INPUT=input.gpkg — Input layer; EXTENT=0,1000,0,1000 — Extent; CLIP=false — Clip features to extent; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractbylocation — Extract by location ★

**group:** Vector selection · **niva verb:** selectloc

This algorithm creates a new vector layer that only contains matching features from an input layer. The criteria for adding features to the resulting layer is defined based on the spatial relationship between each feature and the features in an additional layer.

| Parameter | Type   | Required | Default | Description                                                                                                                                |
|-----------|--------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Extract features from                                                                                                                      |
| PREDICATE | enum   | no       | [0]     | Where the features (geometric predicate) — options: 0=intersect, 1=contain, 2=disjoint, 3=equal, 4=touch, 5=overlap, 6=are within, 7=cross |
| INTERSECT | source | yes      | —       | By comparing to the features from                                                                                                          |
| OUTPUT    | sink   | yes      | —       | Extracted (location)                                                                                                                       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractbylocation INPUT=input.gpkg INTERSECT=input.gpkg PREDICATE=0
↳ OUTPUT=output.gpkg
```

This calls **Extract by location**: INPUT=input.gpkg — Extract features from; INTERSECT=input.gpkg — By comparing to the features from; PREDICATE=0 selects *intersect*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractlabels — Extract labels

**group:** Cartography

This algorithm extracts label information from a rendered map at a given extent and scale.

If a map theme is provided, the rendered map will match the visibility and symbology of that theme. If left blank, all visible layers from the project will be used.

Extracted label information include: position (served as point geometries), the associated layer name and feature ID, label text, rotation (in degree, clockwise), multiline alignment, and font details.

| Parameter        | Type     | Required | Default | Description             |
|------------------|----------|----------|---------|-------------------------|
| EXTENT           | extent   | yes      | —       | Map extent              |
| SCALE            | scale    | yes      | —       | Map scale               |
| MAP_THEME        | maptheme | no       | —       | Map theme               |
| INCLUDE_UNPLACED | boolean  | no       | True    | Include unplaced labels |
| DPI *(adv)*      | number   | no       | 96.0    | Map resolution (in DPI) |
| OUTPUT           | sink     | yes      | —       | Extracted labels        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractlabels EXTENT=0,1000,0,1000 SCALE=10 INCLUDE_UNPLACED=true
↳ OUTPUT=output.gpkg
```

This calls **Extract labels**: EXTENT=0,1000,0,1000 — Map extent; SCALE=10 — Map scale; INCLUDE\_UNPLACED=true — Include unplaced labels; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractmvalues — Extract M values

**group:** Vector geometry

Extracts m values from geometries into feature attributes.

By default only the m value from the first vertex of each feature is extracted, however the algorithm can optionally calculate statistics on all of the geometry's m values, including sums, means, and minimums and maximums

| Parameter     | Type   | Required | Default | Description                                                                                                                                                                                    |
|---------------|--------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT         | source | yes      | —       | Input layer                                                                                                                                                                                    |
| SUMMARIES     | enum   | no       | [0]     | Summaries to calculate — options: 0=First, 1=Last, 2=Count, 3=Sum, 4=Mean, 5=Median, 6=St dev (pop), 7=Minimum, 8=Maximum, 9=Range, 10=Minority, 11=Majority, 12=Variety, 13=Q1, 14=Q3, 15=IQR |
| COLUMN_PREFIX | string | no       | m_      | Output column prefix                                                                                                                                                                           |
| OUTPUT        | sink   | yes      | —       | Extracted                                                                                                                                                                                      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractmvalues INPUT=input.gpkg SUMMARIES=0 COLUMN_PREFIX=m_ OUTPUT=output.gpkg
```

This calls **Extract M values**: INPUT=input.gpkg — Input layer; SUMMARIES=0 selects *First*; COLUMN\_PREFIX=m\_ — Output column prefix; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractnetworkendpoints — Extract network end points

**group:** Network analysis

This algorithm extracts the end points (nodes) from a network line layer.

Two definitions are available for identifying end points:

1. Nodes with only all incoming or all outgoing edges: Identifies 'Source' or 'Sink' nodes based on the direction of flow. These are nodes where flow can start (only outgoing) or stop (only incoming).
2. Nodes connected to a single edge: Identifies topological 'dead-ends' or 'dangles', regardless of directionality. This checks if the node is connected to only one other distinct node.

| Parameter                 | Type     | Required | Default | Description                                                                                                                              |
|---------------------------|----------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                     | source   | yes      | —       | Vector layer representing network                                                                                                        |
| DIRECTION_FIELD *(adv)*   | field    | no       | —       | Direction field                                                                                                                          |
| VALUE_FORWARD *(adv)*     | string   | no       | —       | Value for forward direction                                                                                                              |
| VALUE_BACKWARD *(adv)*    | string   | no       | —       | Value for backward direction                                                                                                             |
| VALUE_BOTH *(adv)*        | string   | no       | —       | Value for both directions                                                                                                                |
| DEFAULT_DIRECTION *(adv)* | enum     | no       | 2       | Default direction — options: 0=Forward direction, 1=Backward direction, 2=Both directions                                                |
| TOLERANCE *(adv)*         | distance | no       | 0       | Topology tolerance                                                                                                                       |
| ENDPOINT_DEFINITION       | enum     | no       | 1       | End point definition — options: 0=Extract Nodes with only All Incoming or All Outgoing Edges, 1=Extract Nodes Connected to a Single Edge |
| OUTPUT                    | sink     | yes      | —       | Network endpoints                                                                                                                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractnetworkendpoints INPUT=input.gpkg ENDPPOINT_DEFINITION=1
↳ OUTPUT=output.gpkg
```

This calls **Extract network end points**: INPUT=input.gpkg — Vector layer representing network; ENDPPOINT\_DEFINITION=1 selects *Extract Nodes Connected to a Single Edge*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractspecificvertices — Extract specific vertices

**group:** Vector geometry

This algorithm takes a vector layer and generates a point layer with points representing specific vertices in the input geometries. For instance, this algorithm can be used to extract the first or last vertices in the geometry. The attributes associated to each point are the same ones associated to the feature that the point belongs to.

The vertex indices parameter accepts a comma separated string specifying the indices of the vertices to extract. The first vertex corresponds to an index of 0, the second vertex has an index of 1, etc. Negative indices can be used to find vertices at the end of the geometry, e.g., an index of -1 corresponds to the last vertex, -2 corresponds to the second last vertex, etc.

Additional fields are added to the points indicating the specific vertex position (e.g., 0, -1, etc), the original vertex index, the vertex's part and its index within the part (as well as its ring for polygons), distance along the original geometry and bisector angle of vertex for the original geometry.

| Parameter | Type   | Required | Default | Description    |
|-----------|--------|----------|---------|----------------|
| INPUT     | source | yes      | —       | Input layer    |
| VERTICES  | string | no       | 0       | Vertex indices |
| OUTPUT    | sink   | yes      | —       | Vertices       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractspecificvertices INPUT=input.gpkg VERTICES=0 OUTPUT=output.gpkg
```

This calls **Extract specific vertices**: INPUT=input.gpkg — Input layer; VERTICES=0 — Vertex indices; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractvertices — Extract vertices ★

**group:** Vector geometry · **niva verb:** vertices

This algorithm takes a vector layer and generates a point layer with points representing the vertices of the input geometries. The attributes associated to each point are the same ones associated to the feature that the point belongs to.

Additional fields are added to the point indicating the vertex index (beginning at 0), the vertex's part and its index within the part (as well as its ring for polygons), distance along original geometry and bisector angle of vertex for original geometry.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Vertices    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractvertices INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Extract vertices**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractwithindistance — Extract within distance

**group:** Vector selection

This algorithm creates a new vector layer that only contains matching features from an input layer. Features are copied wherever they are within the specified maximum distance from the features in an additional reference layer.

| Parameter | Type     | Required | Default | Description                       |
|-----------|----------|----------|---------|-----------------------------------|
| INPUT     | source   | yes      | —       | Extract features from             |
| REFERENCE | source   | yes      | —       | By comparing to the features from |
| DISTANCE  | distance | no       | 100     | Where the features are within     |
| OUTPUT    | sink     | yes      | —       | Extracted (location)              |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractwithindistance INPUT=input.gpkg REFERENCE=input.gpkg DISTANCE=100
↳ OUTPUT=output.gpkg
```

This calls **Extract within distance**: INPUT=input.gpkg — Extract features from; REFERENCE=input.gpkg — By comparing to the features from; DISTANCE=100 — Where the features are within; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:extractzvalues — Extract Z values

**group:** Vector geometry

Extracts z values from geometries into feature attributes.

By default only the z value from the first vertex of each feature is extracted, however the algorithm can optionally calculate statistics on all of the geometry's z values, including sums, means, and minimums and maximums

| Parameter     | Type   | Required | Default | Description                                                                                                                                                                                    |
|---------------|--------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT         | source | yes      | —       | Input layer                                                                                                                                                                                    |
| SUMMARIES     | enum   | no       | [0]     | Summaries to calculate — options: 0=First, 1=Last, 2=Count, 3=Sum, 4=Mean, 5=Median, 6=St dev (pop), 7=Minimum, 8=Maximum, 9=Range, 10=Minority, 11=Majority, 12=Variety, 13=Q1, 14=Q3, 15=IQR |
| COLUMN_PREFIX | string | no       | z_      | Output column prefix                                                                                                                                                                           |
| OUTPUT        | sink   | yes      | —       | Extracted                                                                                                                                                                                      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:extractzvalues INPUT=input.gpkg SUMMARIES=0 COLUMN_PREFIX=z_ OUTPUT=output.gpkg
```

This calls **Extract Z values**: INPUT=input.gpkg — Input layer; SUMMARIES=0 selects *First*; COLUMN\_PREFIX=z\_ — Output column prefix; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fieldcalculator — Field calculator

**group:** Vector table

This algorithm computes a new vector layer with the same features of the input layer, but either overwriting an existing attribute or adding an additional attribute. The values of this field are computed from each feature using an expression, based on the properties and attributes of the feature. Note that if "Field name" is an existing field in the layer then all the rest of the field settings are ignored.

| Parameter       | Type       | Required | Default | Description                                                                                                                                                                                                     |
|-----------------|------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT           | source     | yes      | —       | Input layer                                                                                                                                                                                                     |
| FIELD_NAME      | string     | yes      | —       | Field name                                                                                                                                                                                                      |
| FIELD_TYPE      | enum       | no       | 0       | Result field type — options: 0=Decimal (double), 1=Integer (32 bit), 2=Text (string), 3=Date, 4=Time, 5=Date & Time, 6=Boolean, 7=Binary Object (BLOB), 8=String List, 9=Integer List, 10=Decimal (double) List |
| FIELD_LENGTH    | number     | no       | 0       | Result field length                                                                                                                                                                                             |
| FIELD_PRECISION | number     | no       | 0       | Result field precision                                                                                                                                                                                          |
| FORMULA         | expression | yes      | —       | Formula                                                                                                                                                                                                         |
| OUTPUT          | sink       | yes      | —       | Calculated                                                                                                                                                                                                      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:fieldcalculator INPUT=input.gpkg FIELD_NAME=value FORMULA="value > 0"  
↪ FIELD_TYPE=0 FIELD_LENGTH=0 FIELD_PRECISION=0 OUTPUT=output.gpkg
```

This calls **Field calculator**: INPUT=input.gpkg — Input layer; FIELD\_NAME=value — Field name; FORMULA="value > 0" — Formula; FIELD\_TYPE=0 selects *Decimal (double)*; FIELD\_LENGTH=0 — Result field length; FIELD\_PRECISION=0 — Result field precision; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:filedownloader — Download file via HTTP(S)

**group:** File tools

This algorithm downloads a URL to the file system with an HTTP(S) GET or POST request

| Parameter      | Type            | Required | Default | Description                     |
|----------------|-----------------|----------|---------|---------------------------------|
| URL            | string          | yes      | —       | URL                             |
| METHOD *(adv)* | enum            | no       | 0       | Method — options: 0=GET, 1=POST |
| DATA *(adv)*   | string          | no       | —       | Data                            |
| OUTPUT         | fileDestination | yes      | —       | File destination                |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:filedownloader URL=value OUTPUT=output.txt
```

This calls **Download file via HTTP(S)**: URL=value — URL; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## native:fillnodata — Fill NoData cells

**group:** Raster tools

This algorithm resets the NoData values in the input raster to a chosen value, resulting in a raster dataset with no NoData pixels. This value can be set by the user using the Fill value parameter. The algorithm respects the input raster data type (eg. a floating point fill value will be truncated when applied to an integer raster).

| Parameter                | Type              | Required | Default | Description      |
|--------------------------|-------------------|----------|---------|------------------|
| INPUT                    | raster            | yes      | —       | Raster input     |
| BAND                     | band              | no       | 1       | Band Number      |
| FILL_VALUE               | number            | no       | 1       | Fill value       |
| CREATE_OPTIONS           | string            | no       | —       | Creation options |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options |
| OUTPUT                   | rasterDestination | yes      | —       | Output raster    |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:fillnodata INPUT=dem.tif BAND=1 FILL_VALUE=1 CREATE_OPTIONS=value  
↪ OUTPUT=output.tif
```

This calls **Fill NoData cells**: INPUT=dem.tif — Raster input; BAND=1 — Band Number; FILL\_VALUE=1 — Fill value; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fillsinkswangliu — Fill sinks (Wang & Liu)

**group:** Raster terrain analysis

This algorithm uses a method proposed by Wang & Liu to identify and fill surface depressions in digital elevation models.

The method was enhanced to allow the creation of hydrologically sound elevation models, i.e. not only to fill the depression(s) but also to preserve a downward slope along the flow path. If desired, this is accomplished by preserving a minimum slope gradient (and thus elevation difference) between cells.

References: Wang, L. & H. Liu (2006): An efficient method for identifying and filling surface depressions in digital elevation models for hydrologic analysis and modelling. International Journal of Geographical Information Science, Vol. 20, No. 2: 193-213.

This algorithm is a port of the SAGA 'Fill Sinks (Wang & Liu)' tool.

| Parameter                | Type              | Required | Default | Description                     |
|--------------------------|-------------------|----------|---------|---------------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                     |
| BAND                     | band              | no       | 1       | Band number                     |
| MIN_SLOPE                | number            | no       | 0.1     | Minimum slope (degrees)         |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options                |
| OUTPUT_FILLED_DEM        | rasterDestination | no       | —       | Output layer (filled DEM)       |
| OUTPUT_FLOW_DIRECTIONS   | rasterDestination | no       | —       | Output layer (flow directions)  |
| OUTPUT_WATERSHED_BASINS  | rasterDestination | no       | —       | Output layer (watershed basins) |

**Outputs:** OUTPUT\_FILLED\_DEM (outputRaster), OUTPUT\_FLOW\_DIRECTIONS (outputRaster), OUTPUT\_WATERSHED\_BASINS (outputRaster)

### Example usage

```
run native:fillsinkswangliu INPUT=dem.tif BAND=1 MIN_SLOPE=0.1
↳ OUTPUT_FILLED_DEM=output_filled_dem.tif
```

This calls **Fill sinks (Wang & Liu)**: INPUT=dem.tif — Input layer; BAND=1 — Band number; MIN\_SLOPE=0.1 — Minimum slope (degrees); OUTPUT\_FILLED\_DEM=output\_filled\_dem.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:filter — Feature filter

**group:** Modeler tools

This algorithm filters features from the input layer and redirects them to one or more outputs.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |

### Example usage

```
run native:filter INPUT=input.gpkg
```

This calls **Feature filter**: INPUT=input.gpkg — Input layer. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:filterbygeometry — Filter by geometry type

**group:** Vector selection

This algorithm filters features by their geometry type. Incoming features will be directed to different outputs based on whether they have a point, line or polygon geometry.

| Parameter   | Type   | Required | Default | Description               |
|-------------|--------|----------|---------|---------------------------|
| INPUT       | source | yes      | —       | Input layer               |
| POINTS      | sink   | no       | —       | Point features            |
| LINES       | sink   | no       | —       | Line features             |
| POLYGONS    | sink   | no       | —       | Polygon features          |
| NO_GEOMETRY | sink   | no       | —       | Features with no geometry |

**Outputs:** POINTS (outputVector), LINES (outputVector), POLYGONS (outputVector), NO\_GEOMETRY (outputVector), POINT\_COUNT (outputNumber), LINE\_COUNT (outputNumber), POLYGON\_COUNT (outputNumber), NO\_GEOMETRY\_COUNT (outputNumber)

### Example usage

```
run native:filterbygeometry INPUT=input.gpkg POINTS=points.gpkg
```

This calls **Filter by geometry type**: INPUT=input.gpkg — Input layer; POINTS=points.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:filterlayersbytype — Filter layers by type

**group:** Modeler tools

This algorithm filters layer by their type. Incoming layers will be directed to different outputs based on whether they are a vector or raster layer.

| Parameter | Type              | Required | Default | Description     |
|-----------|-------------------|----------|---------|-----------------|
| INPUT     | layer             | yes      | —       | Input layer     |
| VECTOR    | vectorDestination | no       | —       | Vector features |
| RASTER    | rasterDestination | no       | —       | Raster layer    |

**Outputs:** VECTOR (outputVector), RASTER (outputRaster)

### Example usage

```
run native:filterlayersbytype INPUT=input.gpkg VECTOR=vector.gpkg
```

This calls **Filter layers by type**: INPUT=input.gpkg — Input layer; VECTOR=vector.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:filterverticesbym — Filter vertices by M value

**group:** Vector geometry

Filters away vertices based on their m-value, returning geometries with only vertex points that have a m-value  $\geq$  the specified minimum value and  $\leq$  the maximum value.

If the minimum value is not specified then only the maximum value is tested, and similarly if the maximum value is not specified then only the minimum value is tested.

Depending on the input geometry attributes and the filters used, the resultant geometries created by this algorithm may no longer be valid.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| MIN       | number | no       | —       | Minimum     |
| MAX       | number | no       | —       | Maximum     |
| OUTPUT    | sink   | yes      | —       | Filtered    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:filterverticesbym INPUT=input.gpkg MIN=10 MAX=10 OUTPUT=output.gpkg
```

This calls **Filter vertices by M value**: INPUT=input.gpkg — Input layer; MIN=10 — Minimum; MAX=10 — Maximum; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:filterverticesbyz — Filter vertices by Z value

**group:** Vector geometry

Filters away vertices based on their z-value, returning geometries with only vertex points that have a z-value  $\geq$  the specified minimum value and  $\leq$  the maximum value.

If the minimum value is not specified then only the maximum value is tested, and similarly if the maximum value is not specified then only the minimum value is tested.

Depending on the input geometry attributes and the filters used, the resultant geometries created by this algorithm may no longer be valid.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| MIN       | number | no       | —       | Minimum     |

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| MAX       | number | no       | —       | Maximum     |
| OUTPUT    | sink   | yes      | —       | Filtered    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:filterverticesbyz INPUT=input.gpkg MIN=10 MAX=10 OUTPUT=output.gpkg
```

This calls **Filter vertices by Z value**: INPUT=input.gpkg — Input layer; MIN=10 — Minimum; MAX=10 — Maximum; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:findprojection — Find projection

**group:** Vector general

Creates a list of possible candidate coordinate reference systems for a layer with an unknown projection.

The expected area which the layer should reside in must be specified via the target area parameter.

The algorithm operates by testing the layer's extent in every known reference system and listing any in which the bounds would fall within the target area if the layer was in this projection.

| Parameter       | Type   | Required | Default | Description           |
|-----------------|--------|----------|---------|-----------------------|
| INPUT           | source | yes      | —       | Input layer           |
| TARGET_AREA     | extent | yes      | —       | Target area for layer |
| TARGET_AREA_CRS | crs    | no       | —       | Target area CRS       |
| OUTPUT          | sink   | yes      | —       | CRS candidates        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:findprojection INPUT=input.gpkg TARGET_AREA=0,1000,0,1000
↳ TARGET_AREA_CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Find projection**: INPUT=input.gpkg — Input layer; TARGET\_AREA=0,1000,0,1000 — Target area for layer; TARGET\_AREA\_CRS=EPSG:6346 — Target area CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometries — Fix geometries ★

**group:** Vector geometry · **niva verb:** fixgeom

This algorithm attempts to create a valid representation of a given invalid geometry without losing any of the input vertices. Already-valid geometries are returned without further intervention. Always outputs multi-geometry layer.

NOTE: M values will be dropped from the output.

| Parameter | Type   | Required | Default | Description                                      |
|-----------|--------|----------|---------|--------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                      |
| METHOD    | enum   | no       | 1       | Repair method — options: 0=Linework, 1=Structure |
| OUTPUT    | sink   | yes      | —       | Fixed geometries                                 |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:fixgeometries INPUT=input.gpkg METHOD=1 OUTPUT=output.gpkg
```

This calls **Fix geometries**: INPUT=input.gpkg — Input layer; METHOD=1 selects *Structure*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometryangle — Delete small angles

**group:** Fix geometry

This algorithm deletes vertices based on an error layer from the "Small angles" algorithm in the "Check geometry" section. When deletion of a vertex results in a duplicate vertex (when a spike vertex is deleted), the duplicate vertex is deleted to keep a single vertex and preserve topology.

| Parameter         | Type   | Required | Default    | Description                                 |
|-------------------|--------|----------|------------|---------------------------------------------|
| INPUT             | source | yes      | —          | Input layer                                 |
| ERRORS            | source | yes      | —          | Error layer                                 |
| UNIQUE_ID         | field  | no       | id         | Field of original feature unique identifier |
| PART_IDX          | field  | no       | gc_partidx | Field of part index                         |
| RING_IDX          | field  | no       | gc_ringidx | Field of ring index                         |
| VERTEX_IDX        | field  | no       | gc_vertidx | Field of vertex index                       |
| OUTPUT            | sink   | yes      | —          | Small angle fixed layer                     |
| REPORT            | sink   | yes      | —          | Report layer from fixing small angles       |
| TOLERANCE *(adv)* | number | no       | 8          | Tolerance                                   |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

**Example usage**

```
run native:fixgeometryangle INPUT=input.gpkg ERRORS=input.gpkg UNIQUE_ID=id  
↪ PART_IDX=gc_partidx RING_IDX=gc_ringidx VERTEX_IDX=gc_vertidx OUTPUT=output.gpkg
```

This calls **Delete small angles**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; UNIQUE\_ID=id — Field of original feature unique identifier; PART\_IDX=gc\_partidx — Field of part index; RING\_IDX=gc\_ringidx — Field of ring index; VERTEX\_IDX=gc\_vertidx — Field of vertex index; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometryarea — Fix small polygons

**group:** Fix geometry

This algorithm merges neighboring polygons according to the chosen method, based on an error layer from the "Small polygons" algorithm in the "Check geometry" section.

| Parameter | Type   | Required | Default | Description                                                                                                                                                                                                              |
|-----------|--------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                                                                                                                                                                                              |
| ERRORS    | source | yes      | —       | Error layer                                                                                                                                                                                                              |
| METHOD    | enum   | yes      | —       | Method — options: 0=Merge with neighboring polygon with longest shared edge, 1=Merge with neighboring polygon with largest area, 2=Merge with neighboring polygon with identical attribute value, if any, or leave as is |

| Parameter         | Type   | Required | Default    | Description                                                                 |
|-------------------|--------|----------|------------|-----------------------------------------------------------------------------|
| MERGE_ATTRIBUTE   | field  | no       | —          | Field to consider when merging polygons with the identical attribute method |
| UNIQUE_ID         | field  | no       | id         | Field of original feature unique identifier                                 |
| PART_IDX          | field  | no       | gc_partidx | Field of part index                                                         |
| RING_IDX          | field  | no       | gc_ringidx | Field of ring index                                                         |
| VERTEX_IDX        | field  | no       | gc_vertidx | Field of vertex index                                                       |
| OUTPUT            | sink   | yes      | —          | Small polygons merged layer                                                 |
| REPORT            | sink   | yes      | —          | Report layer from merging small polygons                                    |
| TOLERANCE *(adv)* | number | no       | 8          | Tolerance                                                                   |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometryarea INPUT=input.gpkg ERRORS=input.gpkg METHOD=0
↳ MERGE_ATTRIBUTE=field1 UNIQUE_ID=id PART_IDX=gc_partidx RING_IDX=gc_ringidx
↳ OUTPUT=output.gpkg
```

This calls **Fix small polygons**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; METHOD=0 selects *Merge with neighboring polygon with longest shared edge*; MERGE\_ATTRIBUTE=field1 — Field to consider when merging polygons with the identical attribute method; UNIQUE\_ID=id — Field of original feature unique identifier; PART\_IDX=gc\_partidx — Field of part index; RING\_IDX=gc\_ringidx — Field of ring index; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometrydeletfeatures — Delete features

**group:** Fix geometry

This algorithm deletes error features listed in the errors layer from an algorithm in the “Check geometry” section. The required inputs are the original layer used in the check algorithm, its unique id field, and its corresponding errors layer.

For instance, it can be used after the following check algorithms to delete error features: Feature inside polygon Degenerate polygons Small segments Duplicated geometries etc.

| Parameter | Type   | Required | Default | Description                                 |
|-----------|--------|----------|---------|---------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                 |
| ERRORS    | source | yes      | —       | Error layer                                 |
| UNIQUE_ID | field  | yes      | —       | Field of original feature unique identifier |
| OUTPUT    | sink   | yes      | —       | Cleaned layer                               |
| REPORT    | sink   | yes      | —       | Report layer from deleting features         |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometrydeletfeatures INPUT=input.gpkg ERRORS=input.gpkg UNIQUE_ID=field1
↳ OUTPUT=output.gpkg
```

This calls **Delete features**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; UNIQUE\_ID=field1 — Field of original feature unique identifier; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometryduplicatenodes — Delete duplicated vertices

**group:** Fix geometry

This algorithm delete duplicate vertices based on an error layer from the "Duplicated vertices" algorithm in the "Check geometry" section.

| Parameter         | Type   | Required | Default    | Description                                 |
|-------------------|--------|----------|------------|---------------------------------------------|
| INPUT             | source | yes      | —          | Input layer                                 |
| ERRORS            | source | yes      | —          | Error layer                                 |
| UNIQUE_ID         | field  | no       | id         | Field of original feature unique identifier |
| PART_IDX          | field  | no       | gc_partidx | Field of part index                         |
| RING_IDX          | field  | no       | gc_ringidx | Field of ring index                         |
| VERTEX_IDX        | field  | no       | gc_vertidx | Field of vertex index                       |
| OUTPUT            | sink   | yes      | —          | Fixed duplicate vertices layer              |
| REPORT            | sink   | yes      | —          | Report layer from fixing duplicate vertices |
| TOLERANCE *(adv)* | number | no       | 8          | Tolerance                                   |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometryduplicatenodes INPUT=input.gpkg ERRORS=input.gpkg UNIQUE_ID=id  
↪ PART_IDX=gc_partidx RING_IDX=gc_ringidx VERTEX_IDX=gc_vertidx OUTPUT=output.gpkg
```

This calls **Delete duplicated vertices**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; UNIQUE\_ID=id — Field of original feature unique identifier; PART\_IDX=gc\_partidx — Field of part index; RING\_IDX=gc\_ringidx — Field of ring index; VERTEX\_IDX=gc\_vertidx — Field of vertex index; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometrygap — Fill gaps

**group:** Fix geometry

This algorithm fills the gaps based on a gap and neighbors layer from the "Small gaps" algorithm in the "Check geometry" section.

3 different fixing methods are available, which will give different results.

| Parameter         | Type   | Required | Default    | Description                                                                                              |
|-------------------|--------|----------|------------|----------------------------------------------------------------------------------------------------------|
| INPUT             | source | yes      | —          | Input layer                                                                                              |
| NEIGHBORS         | source | yes      | —          | Neighbors layer                                                                                          |
| GAPS              | source | yes      | —          | Gaps layer                                                                                               |
| METHOD            | enum   | yes      | —          | Method — options: 0=Add to longest shared edge, 1=Create new feature, 2=Add to largest neighbouring area |
| UNIQUE_ID         | field  | yes      | —          | Field of original feature unique identifier                                                              |
| ERROR_ID_IDX      | field  | no       | gc_errorid | Field of error id                                                                                        |
| OUTPUT            | sink   | yes      | —          | Gaps-filled layer                                                                                        |
| REPORT            | sink   | yes      | —          | Report layer from fixing gaps                                                                            |
| TOLERANCE *(adv)* | number | no       | 8          | Tolerance                                                                                                |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometrygap INPUT=input.gpkg NEIGHBORS=input.gpkg GAPS=input.gpkg METHOD=0  
↪ UNIQUE_ID=field1 ERROR_ID_IDX=gc_errorid OUTPUT=output.gpkg
```

This calls **Fill gaps**: INPUT=input.gpkg — Input layer; NEIGHBORS=input.gpkg — Neighbors layer; GAPS=input.gpkg — Gaps layer; METHOD=0 selects *Add to longest shared edge*; UNIQUE\_ID=field1 — Field of original feature unique identifier; ERROR\_ID\_IDX=gc\_errorid — Field of error id; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometryhole — Fill holes

**group:** Fix geometry

This algorithm deletes holes based on an error layer from the "Holes" algorithm in the "Check geometry" section.

| Parameter         | Type   | Required | Default    | Description                                 |
|-------------------|--------|----------|------------|---------------------------------------------|
| INPUT             | source | yes      | —          | Input layer                                 |
| ERRORS            | source | yes      | —          | Error layer                                 |
| UNIQUE_ID         | field  | no       | id         | Field of original feature unique identifier |
| PART_IDX          | field  | no       | gc_partidx | Field of part index                         |
| RING_IDX          | field  | no       | gc_ringidx | Field of ring index                         |
| VERTEX_IDX        | field  | no       | gc_vertidx | Field of vertex index                       |
| OUTPUT            | sink   | yes      | —          | Holes-filled layer                          |
| REPORT            | sink   | yes      | —          | Report layer from fixing holes              |
| TOLERANCE *(adv)* | number | no       | 8          | Tolerance                                   |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometryhole INPUT=input.gpkg ERRORS=input.gpkg UNIQUE_ID=id
↪ PART_IDX=gc_partidx RING_IDX=gc_ringidx VERTEX_IDX=gc_vertidx OUTPUT=output.gpkg
```

This calls **Fill holes**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; UNIQUE\_ID=id — Field of original feature unique identifier; PART\_IDX=gc\_partidx — Field of part index; RING\_IDX=gc\_ringidx — Field of ring index; VERTEX\_IDX=gc\_vertidx — Field of vertex index; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometrymissingvertex — Add missing vertices along borders

**group:** Fix geometry

This algorithm adds the missing vertices along polygons borders, based on an error layer from the "Missing vertices along borders" algorithm in the "Check geometry" section.

| Parameter         | Type   | Required | Default    | Description                                 |
|-------------------|--------|----------|------------|---------------------------------------------|
| INPUT             | source | yes      | —          | Input layer                                 |
| ERRORS            | source | yes      | —          | Error layer                                 |
| UNIQUE_ID         | field  | no       | id         | Field of original feature unique identifier |
| PART_IDX          | field  | no       | gc_partidx | Field of part index                         |
| RING_IDX          | field  | no       | gc_ringidx | Field of ring index                         |
| VERTEX_IDX        | field  | no       | gc_vertidx | Field of vertex index                       |
| OUTPUT            | sink   | yes      | —          | Border vertices fixed layer                 |
| REPORT            | sink   | yes      | —          | Report layer from fixing border vertices    |
| TOLERANCE *(adv)* | number | no       | 8          | Tolerance                                   |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage



```
run native:fixgeometrymissingvertex INPUT=input.gpkg ERRORS=input.gpkg UNIQUE_ID=id
↳ PART_IDX=gc_partidx RING_IDX=gc_ringidx VERTEX_IDX=gc_vertidx OUTPUT=output.gpkg
```

This calls **Add missing vertices along borders**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; UNIQUE\_ID=id — Field of original feature unique identifier; PART\_IDX=gc\_partidx — Field of part index; RING\_IDX=gc\_ringidx — Field of ring index; VERTEX\_IDX=gc\_vertidx — Field of vertex index; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometrymultipart — Convert to strictly multipart

**group:** Fix geometry

This algorithm converts multipart geometries that consists of only one geometry into singlepart geometries, based on an error layer from the "Strict multipart" algorithm in the "Check geometry" section.

This algorithm does not change the layer geometry type, which will remain multipart.

| Parameter         | Type   | Required | Default | Description                                 |
|-------------------|--------|----------|---------|---------------------------------------------|
| INPUT             | source | yes      | —       | Input layer                                 |
| ERRORS            | source | yes      | —       | Error layer                                 |
| UNIQUE_ID         | field  | no       | id      | Field of original feature unique identifier |
| OUTPUT            | sink   | yes      | —       | Strictly-multipart layer                    |
| REPORT            | sink   | yes      | —       | Report layer from fixing multipart          |
| TOLERANCE *(adv)* | number | no       | 8       | Tolerance                                   |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometrymultipart INPUT=input.gpkg ERRORS=input.gpkg UNIQUE_ID=id
↳ OUTPUT=output.gpkg
```

This calls **Convert to strictly multipart**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; UNIQUE\_ID=id — Field of original feature unique identifier; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometryoverlap — Delete overlaps

**group:** Fix geometry

This algorithm deletes overlap sections based on an error layer from the "Overlap" algorithm in the "Check geometry" section.

| Parameter                  | Type   | Required | Default  | Description                                 |
|----------------------------|--------|----------|----------|---------------------------------------------|
| INPUT                      | source | yes      | —        | Input layer                                 |
| ERRORS                     | source | yes      | —        | Error layer                                 |
| UNIQUE_ID                  | field  | yes      | —        | Field of original feature unique identifier |
| OVERLAP_FEATURE_UNIQUE_IDX | field  | yes      | —        | Field of overlap feature unique identifier  |
| ERROR_VALUE_ID             | field  | no       | gc_error | Field of error value                        |
| OUTPUT                     | sink   | yes      | —        | No-overlap layer                            |
| REPORT                     | sink   | yes      | —        | Report layer from fixing overlaps           |
| TOLERANCE *(adv)*          | number | no       | 8        | Tolerance                                   |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometryoverlap INPUT=input.gpkg ERRORS=input.gpkg UNIQUE_ID=field1
↳ OVERLAP_FEATURE_UNIQUE_IDX=field1 ERROR_VALUE_ID=gc_error OUTPUT=output.gpkg
```

This calls **Delete overlaps**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; UNIQUE\_ID=field1 — Field of original feature unique identifier; OVERLAP\_FEATURE\_UNIQUE\_IDX=field1 — Field of overlap feature unique identifier; ERROR\_VALUE\_ID=gc\_error — Field of error value; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fixgeometryselfintersection — Split self-intersecting geometries

**group:** Fix geometry

This algorithm splits self intersecting lines or polygons according to the chosen method, based on an error layer from the "Self-intersections" algorithm in the "Check geometry" section.

| Parameter         | Type   | Required | Default      | Description                                                                                                     |
|-------------------|--------|----------|--------------|-----------------------------------------------------------------------------------------------------------------|
| INPUT             | source | yes      | —            | Input layer                                                                                                     |
| ERRORS            | source | yes      | —            | Error layer                                                                                                     |
| METHOD            | enum   | yes      | —            | Method — options: 0=Split feature into a multi-part feature, 1=Split feature into multiple single-part features |
| UNIQUE_ID         | field  | yes      | —            | Field of original feature unique identifier                                                                     |
| PART_IDX          | field  | no       | gc_partidx   | Field of part index                                                                                             |
| RING_IDX          | field  | no       | gc_ringidx   | Field of ring index                                                                                             |
| VERTEX_IDX        | field  | no       | gc_vertidx   | Field of vertex index                                                                                           |
| SEGMENT_1         | field  | no       | gc_segment_1 | Field of segment 1                                                                                              |
| SEGMENT_2         | field  | no       | gc_segment_2 | Field of segment 2                                                                                              |
| OUTPUT            | sink   | yes      | —            | Self-intersections fixed layer                                                                                  |
| REPORT            | sink   | yes      | —            | Report layer from fixing self-intersections                                                                     |
| TOLERANCE *(adv)* | number | no       | 8            | Tolerance                                                                                                       |

**Outputs:** OUTPUT (outputVector), REPORT (outputVector)

### Example usage

```
run native:fixgeometryselfintersection INPUT=input.gpkg ERRORS=input.gpkg METHOD=0
↳ UNIQUE_ID=field1 PART_IDX=gc_partidx RING_IDX=gc_ringidx VERTEX_IDX=gc_vertidx
↳ OUTPUT=output.gpkg
```

This calls **Split self-intersecting geometries**: INPUT=input.gpkg — Input layer; ERRORS=input.gpkg — Error layer; METHOD=0 selects *Split feature into a multi-part feature*; UNIQUE\_ID=field1 — Field of original feature unique identifier; PART\_IDX=gc\_partidx — Field of part index; RING\_IDX=gc\_ringidx — Field of ring index; VERTEX\_IDX=gc\_vertidx — Field of vertex index; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:flattenrelationships — Flatten relationship

**group:** Vector general

This algorithm flattens a relationship for a vector layer, exporting a single layer containing one master feature per related feature. This master feature contains all the attributes for the related features.

| Parameter | Type   | Required | Default | Description     |
|-----------|--------|----------|---------|-----------------|
| INPUT     | vector | yes      | —       | Input layer     |
| OUTPUT    | sink   | yes      | —       | Flattened layer |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:flattenrelationships INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Flatten relationship**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:forceccw — Force polygons counter-clockwise**

**group:** Vector geometry

This algorithm forces polygon geometries to respect the convention where the exterior ring is oriented in a counter-clockwise direction and the interior rings in a clockwise direction.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Reoriented  |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:forceccw INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Force polygons counter-clockwise**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:forcecw — Force polygons clockwise**

**group:** Vector geometry

This algorithm forces polygon geometries to respect the convention where the exterior ring is oriented in a clockwise direction and the interior rings in a counter-clockwise direction.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Reoriented  |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:forcecw INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Force polygons clockwise**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:forcerhr — Force right-hand-rule ★

**group:** Vector geometry · **niva verb:** forcerhr

This algorithm forces polygon geometries to respect the Right-Hand-Rule, in which the area that is bounded by a polygon is to the right of the boundary. In particular, the exterior ring is oriented in a clockwise direction and the interior rings in a counter-clockwise direction.

Due to the inconsistency in the definition of the Right-Hand-Rule in some contexts, it is recommended to use the explicit "Force polygons clockwise" processing algorithm instead.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Reoriented  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:forcerhr INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Force right-hand-rule**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fuzzifyrastergaussianmembership — Fuzzify raster (gaussian membership)

**group:** Raster analysis

This algorithm transforms an input raster to a fuzzified raster and thereby assigns values between 0 and 1 following a gaussian fuzzy membership function. The value of 0 implies no membership with the defined fuzzy set, a value of 1 depicts full membership. In between, the degree of membership of raster values follows a gaussian membership function.

The gaussian function is constructed using two user-defined input values which set the midpoint of the gaussian function (midpoint, results to 1) and a predefined function spread which controls the function spread.

This function is typically used when a certain range of raster values around a predefined function midpoint should become members of the fuzzy set.

| Parameter                | Type              | Required | Default | Description       |
|--------------------------|-------------------|----------|---------|-------------------|
| INPUT                    | raster            | yes      | —       | Input Raster      |
| BAND                     | band              | no       | 1       | Band Number       |
| FUZZYMIDPOINT            | number            | no       | 10      | Function midpoint |
| FUZZYSPREAD              | number            | no       | 0.01    | Function spread   |
| CREATE_OPTIONS           | string            | no       | —       | Creation options  |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options  |
| OUTPUT                   | rasterDestination | yes      | —       | Fuzzified raster  |

**Outputs:** EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), OUTPUT (outputRaster)

### Example usage

```
run native:fuzzifyrastergaussianmembership INPUT=dem.tif BAND=1 FUZZYMIDPOINT=10
↳ FUZZYSPREAD=0.01 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Fuzzify raster (gaussian membership)**: INPUT=dem.tif — Input Raster; BAND=1 — Band Number; FUZZYMIDPOINT=10 — Function midpoint; FUZZYSPREAD=0.01 — Function spread; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fuzzifyrasterlargemembership — Fuzzify raster (large membership)

**group:** Raster analysis

This algorithm transforms an input raster to a fuzzified raster and thereby assigns values between 0 and 1 following the 'large' fuzzy membership function. The value of 0 implies no membership with the defined fuzzy set, a value of 1 depicts full membership. In between, the degree of membership of raster values follows the 'large' membership function.

The 'large' function is constructed using two user-defined input raster values which set the point of half membership (midpoint, results to 0.5) and a predefined function spread which controls the function uptake.

This function is typically used when larger input raster values should become members of the fuzzy set more easily than smaller values.

| Parameter                 | Type              | Required | Default | Description       |
|---------------------------|-------------------|----------|---------|-------------------|
| INPUT                     | raster            | yes      | —       | Input Raster      |
| BAND                      | band              | no       | 1       | Band Number       |
| FUZZYMIDPOINT             | number            | no       | 50      | Function midpoint |
| FUZZYSPREAD               | number            | no       | 5       | Function spread   |
| CREATE_OPTIONS            | string            | no       | —       | Creation options  |
| CREATION_OPTIONS *(adv) * | string            | no       | —       | Creation options  |
| OUTPUT                    | rasterDestination | yes      | —       | Fuzzified raster  |

**Outputs:** EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), OUTPUT (outputRaster)

### Example usage

```
run native:fuzzifyrasterlargemembership INPUT=dem.tif BAND=1 FUZZYMIDPOINT=50
  ↪ FUZZYSPREAD=5 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Fuzzify raster (large membership)**: INPUT=dem.tif — Input Raster; BAND=1 — Band Number; FUZZYMIDPOINT=50 — Function midpoint; FUZZYSPREAD=5 — Function spread; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fuzzifyrasterlinearmembership — Fuzzify raster (linear membership)

**group:** Raster analysis

This algorithm transforms an input raster to a fuzzified raster and thereby assigns values between 0 and 1 following a linear fuzzy membership function. The value of 0 implies no membership with the defined fuzzy set, a value of 1 depicts full membership. In between, the degree of membership of raster values follows a linear membership function.

The linear function is constructed using two user-defined input raster values which set the point of full membership (high bound, results to 1) and no membership (low bound, results to 0) respectively. The fuzzy set in between those values is defined as a linear function.

Both increasing and decreasing fuzzy sets can be modeled by swapping the high and low bound parameters.

| Parameter                | Type              | Required | Default | Description                 |
|--------------------------|-------------------|----------|---------|-----------------------------|
| INPUT                    | raster            | yes      | —       | Input Raster                |
| BAND                     | band              | no       | 1       | Band Number                 |
| FUZZYLOWBOUND            | number            | no       | 0       | Low fuzzy membership bound  |
| FUZZYHIGHBOUND           | number            | no       | 1       | High fuzzy membership bound |
| CREATE_OPTIONS           | string            | no       | —       | Creation options            |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options            |
| OUTPUT                   | rasterDestination | yes      | —       | Fuzzified raster            |

**Outputs:** EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), OUTPUT (outputRaster)

### Example usage

```
run native:fuzzifyrasterlinearmembership INPUT=dem.tif BAND=1 FUZZYLOWBOUND=0
↳ FUZZYHIGHBOUND=1 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Fuzzify raster (linear membership)**: INPUT=dem.tif — Input Raster; BAND=1 — Band Number; FUZZYLOWBOUND=0 — Low fuzzy membership bound; FUZZYHIGHBOUND=1 — High fuzzy membership bound; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:fuzzifyrasternearmembership — Fuzzify raster (near membership)

**group:** Raster analysis

The Fuzzify raster (near membership) algorithm transforms an input raster to a fuzzified raster and thereby assigns values between 0 and 1 following the 'near' fuzzy membership function. The value of 0 implies no membership with the defined fuzzy set, a value of 1 depicts full membership. In between, the degree of membership of raster values follows the 'near' membership function.

The 'near' function is constructed using two user-defined input values which set the midpoint of the 'near' function (midpoint, results to 1) and a predefined function spread which controls the function spread.

This function is typically used when a certain range of raster values near a predefined function midpoint should become members of the fuzzy set. The function generally shows a higher rate of decay than the gaussian membership function.

| Parameter                | Type              | Required | Default | Description       |
|--------------------------|-------------------|----------|---------|-------------------|
| INPUT                    | raster            | yes      | —       | Input Raster      |
| BAND                     | band              | no       | 1       | Band Number       |
| FUZZYMIDPOINT            | number            | no       | 50      | Function midpoint |
| FUZZYSPREAD              | number            | no       | 0.01    | Function spread   |
| CREATE_OPTIONS           | string            | no       | —       | Creation options  |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options  |
| OUTPUT                   | rasterDestination | yes      | —       | Fuzzified raster  |

**Outputs:** EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), OUTPUT (outputRaster)

### Example usage

```
run native:fuzzifyrasternearmembership INPUT=dem.tif BAND=1 FUZZYMIDPOINT=50
↳ FUZZYSPREAD=0.01 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Fuzzify raster (near membership)**: INPUT=dem.tif — Input Raster; BAND=1 — Band Number; FUZZYMIDPOINT=50 — Function midpoint; FUZZYSPREAD=0.01 — Function spread; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:fuzzifyrasterpowermembership — Fuzzify raster (power membership)**

**group:** Raster analysis

This algorithm transforms an input raster to a fuzzified raster and thereby assigns values between 0 and 1 following a power function. The value of 0 implies no membership with the defined fuzzy set, a value of 1 depicts full membership. In between, the degree of membership of raster values follows a power function.

The power function is constructed using three user-defined input raster values which set the point of full membership (high bound, results to 1), no membership (low bound, results to 0) and function exponent (only positive) respectively. The fuzzy set in between those the upper and lower bounds values is then defined as a power function.

Both increasing and decreasing fuzzy sets can be modeled by swapping the high and low bound parameters.

| Parameter                | Type              | Required | Default | Description                  |
|--------------------------|-------------------|----------|---------|------------------------------|
| INPUT                    | raster            | yes      | —       | Input Raster                 |
| BAND                     | band              | no       | 1       | Band Number                  |
| FUZZYLOWBOUND            | number            | no       | 0       | Low fuzzy membership bound   |
| FUZZYHIGHBOUND           | number            | no       | 1       | High fuzzy membership bound  |
| FUZZYEXPONENT            | number            | no       | 2       | Membership function exponent |
| CREATE_OPTIONS           | string            | no       | —       | Creation options             |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options             |
| OUTPUT                   | rasterDestination | yes      | —       | Fuzzified raster             |

**Outputs:** EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), OUTPUT (outputRaster)

### **Example usage**

```
run native:fuzzifyrasterpowermembership INPUT=dem.tif BAND=1 FUZZYLOWBOUND=0
↳ FUZZYHIGHBOUND=1 FUZZYEXPONENT=2 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Fuzzify raster (power membership)**: INPUT=dem.tif — Input Raster; BAND=1 — Band Number; FUZZYLOWBOUND=0 — Low fuzzy membership bound; FUZZYHIGHBOUND=1 — High fuzzy membership bound; FUZZYEXPONENT=2 — Membership function exponent; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:fuzzifyrastersmallmembership — Fuzzify raster (small membership)**

**group:** Raster analysis

The Fuzzify raster (small membership) algorithm transforms an input raster to a fuzzified raster and thereby assigns values between 0 and 1 following the 'small' fuzzy membership function. The value of 0 implies no membership with the defined fuzzy set, a value of 1 depicts full membership. In between, the degree of membership of raster values follows the 'small' membership function.

The 'small' function is constructed using two user-defined input raster values which set the point of half membership (midpoint, results to 0.5) and a predefined function spread which controls the function uptake.

This function is typically used when smaller input raster values should become members of the fuzzy set more easily than higher values.

| Parameter                | Type              | Required | Default | Description       |
|--------------------------|-------------------|----------|---------|-------------------|
| INPUT                    | raster            | yes      | —       | Input Raster      |
| BAND                     | band              | no       | 1       | Band Number       |
| FUZZYMIDPOINT            | number            | no       | 50      | Function midpoint |
| FUZZYSPREAD              | number            | no       | 5       | Function spread   |
| CREATE_OPTIONS           | string            | no       | —       | Creation options  |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options  |
| OUTPUT                   | rasterDestination | yes      | —       | Fuzzified raster  |

**Outputs:** EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), OUTPUT (outputRaster)

### Example usage

```
run native:fuzzifyrastersmallmembership INPUT=dem.tif BAND=1 FUZZYMIDPOINT=50
  ↳ FUZZYSPREAD=5 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Fuzzify raster (small membership)**: INPUT=dem.tif — Input Raster; BAND=1 — Band Number; FUZZYMIDPOINT=50 — Function midpoint; FUZZYSPREAD=5 — Function spread; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:generateelevationprofileimage — Generate elevation profile image

**group:** Plots

This algorithm creates an elevation profile image from a list of map layer and an optional terrain.

| Parameter                 | Type       | Required | Default                           | Description                      |
|---------------------------|------------|----------|-----------------------------------|----------------------------------|
| CURVE                     | geometry   | yes      | —                                 | Profile curve                    |
| MAP_LAYERS                | multilayer | yes      | —                                 | Map layers                       |
| WIDTH                     | number     | no       | 400                               | Chart width (in pixels)          |
| HEIGHT                    | number     | no       | 300                               | Chart height (in pixels)         |
| TERRAIN_LAYER             | layer      | no       | —                                 | Terrain layer                    |
| MINIMUM_DISTANCE *(adv)*  | number     | no       | —                                 | Chart minimum distance (X axis)  |
| MAXIMUM_DISTANCE *(adv)*  | number     | no       | —                                 | Chart maximum distance (X axis)  |
| MINIMUM_ELEVATION *(adv)* | number     | no       | —                                 | Chart minimum elevation (Y axis) |
| MAXIMUM_ELEVATION *(adv)* | number     | no       | —                                 | Chart maximum elevation (Y axis) |
| TEXT_COLOR *(adv)*        | color      | no       | <QColor: RGBA 0, 0, 0, 255>       | Chart text color                 |
| TEXT_FONT_FAMILY          | string     | no       | —                                 | Chart text font family           |
| TEXT_FONT_STYLE           | string     | no       | —                                 | Chart text font style            |
| TEXT_FONT_SIZE            | number     | no       | 0                                 | Chart text font size             |
| BACKGROUND_COLOR *(adv)*  | color      | no       | <QColor: RGBA 255, 255, 255, 255> | Chart background color           |
| BORDER_COLOR *(adv)*      | color      | no       | <QColor: RGBA 99, 99, 99, 255>    | Chart border color               |
| TOLERANCE *(adv)*         | number     | no       | 5.0                               | Profile tolerance                |
| DPI *(adv)*               | number     | no       | 96                                | Chart DPI                        |



| Parameter | Type            | Required | Default | Description  |
|-----------|-----------------|----------|---------|--------------|
| OUTPUT    | fileDestination | yes      | —       | Output image |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:generateelevationprofileimage CURVE=... MAP_LAYERS="a.gpkg;b.gpkg" WIDTH=400
↳ HEIGHT=300 TERRAIN_LAYER=input.gpkg TEXT_FONT_FAMILY=value TEXT_FONT_STYLE=value
↳ OUTPUT=output.txt
```

This calls **Generate elevation profile image**: CURVE=... — Profile curve; MAP\_LAYERS="a.gpkg;b.gpkg" — Map layers; WIDTH=400 — Chart width (in pixels); HEIGHT=300 — Chart height (in pixels); TERRAIN\_LAYER=input.gpkg — Terrain layer; TEXT\_FONT\_FAMILY=value — Chart text font family; TEXT\_FONT\_STYLE=value — Chart text font style; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:generatepointspixelcentroidsinsidepolygons — Generate points (pixel centroids) inside polygons

**group:** Vector creation

This algorithm generates pixel centroids for the raster area falling inside polygons. Used to generate points for further raster sampling.

| Parameter    | Type   | Required | Default | Description     |
|--------------|--------|----------|---------|-----------------|
| INPUT_RASTER | raster | yes      | —       | Raster layer    |
| INPUT_VECTOR | source | yes      | —       | Vector layer    |
| OUTPUT       | sink   | yes      | —       | Pixel centroids |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:generatepointspixelcentroidsinsidepolygons INPUT_RASTER=dem.tif
↳ INPUT_VECTOR=input.gpkg OUTPUT=output.gpkg
```

This calls **Generate points (pixel centroids) inside polygons**: INPUT\_RASTER=dem.tif — Raster layer; INPUT\_VECTOR=input.gpkg — Vector layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:geometrybyexpression — Geometry by expression

**group:** Vector geometry

This algorithm updates existing geometries (or creates new geometries) for input features by use of a QGIS expression. This allows complex geometry modifications which can utilize all the flexibility of the QGIS expression engine to manipulate and create geometries for output features.

For help with QGIS expression functions, see the inbuilt help for specific functions which is available in the expression builder.

| Parameter       | Type    | Required | Default | Description                                                |
|-----------------|---------|----------|---------|------------------------------------------------------------|
| INPUT           | source  | yes      | —       | Input layer                                                |
| OUTPUT_GEOMETRY | enum    | no       | 0       | Output geometry type — options: 0=Polygon, 1=Line, 2=Point |
| WITH_Z          | boolean | no       | False   | Output geometry has z dimension                            |
| WITH_M          | boolean | no       | False   | Output geometry has m values                               |

| Parameter  | Type       | Required | Default   | Description         |
|------------|------------|----------|-----------|---------------------|
| EXPRESSION | expression | no       | @geometry | Geometry expression |
| OUTPUT     | sink       | yes      | —         | Modified geometry   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:geometrybyexpression INPUT=input.gpkg OUTPUT_GEOMETRY=0 WITH_Z=false
↳ WITH_M=false EXPRESSION="value > 0" OUTPUT=output.gpkg
```

This calls **Geometry by expression**: INPUT=input.gpkg — Input layer; OUTPUT\_GEOMETRY=0 selects *Polygon*; WITH\_Z=false — Output geometry has z dimension; WITH\_M=false — Output geometry has m values; EXPRESSION="value > 0" — Geometry expression; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:gltftovector — Convert GLTF to vector features

**group:** 3D Tiles

This algorithm converts GLTF content to standard vector layer formats.

| Parameter       | Type | Required | Default | Description     |
|-----------------|------|----------|---------|-----------------|
| INPUT           | file | yes      | —       | Input GLTF      |
| OUTPUT_POLYGONS | sink | no       | —       | Output polygons |
| OUTPUT_LINES    | sink | no       | —       | Output lines    |

**Outputs:** OUTPUT\_POLYGONS (outputVector), OUTPUT\_LINES (outputVector)

### Example usage

```
run native:gltftovector INPUT=input.dat OUTPUT_POLYGONS=output_polygons.gpkg
```

This calls **Convert GLTF to vector features**: INPUT=input.dat — Input GLTF; OUTPUT\_POLYGONS=output\_polygons.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:greaterthanfrequency — Greater than frequency

**group:** Raster analysis

This algorithm evaluates on a cell-by-cell basis the frequency (number of times) the values of an input stack of rasters are greater than the value of a value raster. If multiband rasters are used in the data raster stack, the algorithm will always perform the analysis on the first band of the rasters - use GDAL to use other bands in the analysis. The input value layer serves as reference layer for the sample layers. Any NoData cells in the value raster or the data layer stack will result in a NoData cell in the output raster if the ignore NoData parameter is not checked. The output NoData value can be set manually. The output rasters extent and resolution is defined by the input raster layer and is always of int32 type.

| Parameter                   | Type       | Required | Default | Description          |
|-----------------------------|------------|----------|---------|----------------------|
| INPUT_VALUE_RASTER          | raster     | yes      | —       | Input value raster   |
| INPUT_VALUE_RASTER_BAND     | band       | no       | 1       | Value raster band    |
| INPUT_RASTERS               | multilayer | yes      | —       | Input raster layers  |
| IGNORE_NODATA               | boolean    | no       | False   | Ignore NoData values |
| OUTPUT_NODATA_VALUE *(adv)* | number     | no       | -9999   | Output NoData value  |
| CREATE_OPTIONS              | string     | no       | —       | Creation options     |

| Parameter                | Type              | Required | Default | Description      |
|--------------------------|-------------------|----------|---------|------------------|
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options |
| OUTPUT                   | rasterDestination | yes      | —       | Output layer     |

**Outputs:** OUTPUT (outputRaster), OCCURRENCE\_COUNT (outputNumber), FOUND\_LOCATIONS\_COUNT (outputNumber), MEAN\_FREQUENCY\_PER\_LOCATION (outputNumber), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:greaterthanfrequency INPUT_VALUE_RASTER=dem.tif INPUT_RASTERS="a.gpkg;b.gpkg"
↳ INPUT_VALUE_RASTER_BAND=1 IGNORE_NODATA=false CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Greater than frequency**: INPUT\_VALUE\_RASTER=dem.tif — Input value raster; INPUT\_RASTERS="a.gpkg;b.gpkg" — Input raster layers; INPUT\_VALUE\_RASTER\_BAND=1 — Value raster band; IGNORE\_NODATA=false — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:highestpositioninrasterstack — Highest position in raster stack

**group:** Raster analysis

This algorithm evaluates on a cell-by-cell basis the position of the raster with the highest value in a stack of rasters. Position counts start with 1 and range to the total number of input rasters. The order of the input rasters is relevant for the algorithm. If multiple rasters feature the highest value, the first raster will be used for the position value. If multiband rasters are used in the data raster stack, the algorithm will always perform the analysis on the first band of the rasters - use GDAL to use other bands in the analysis. Any NoData cells in the raster layer stack will result in a NoData cell in the output raster unless the "ignore NoData" parameter is checked. The output NoData value can be set manually. The output rasters extent and resolution is defined by a reference raster layer and is always of int32 type.

| Parameter                   | Type              | Required | Default | Description          |
|-----------------------------|-------------------|----------|---------|----------------------|
| INPUT_RASTERS               | multilayer        | yes      | —       | Input raster layers  |
| REFERENCE_LAYER             | raster            | yes      | —       | Reference layer      |
| IGNORE_NODATA               | boolean           | no       | False   | Ignore NoData values |
| OUTPUT_NODATA_VALUE *(adv)* | number            | no       | -9999   | Output NoData value  |
| CREATE_OPTIONS              | string            | no       | —       | Creation options     |
| CREATION_OPTIONS *(adv)*    | string            | no       | —       | Creation options     |
| OUTPUT                      | rasterDestination | yes      | —       | Output layer         |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:highestpositioninrasterstack INPUT_RASTERS="a.gpkg;b.gpkg"
↳ REFERENCE_LAYER=dem.tif IGNORE_NODATA=false CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Highest position in raster stack**: INPUT\_RASTERS="a.gpkg;b.gpkg" — Input raster layers; REFERENCE\_LAYER=dem.tif — Reference layer; IGNORE\_NODATA=false — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:hillshade — Hillshade ★

**group:** Raster terrain analysis · **niva verb:** hillshade

This algorithm calculates the hillshade of the Digital Terrain Model in input.

The shading of the layer is calculated according to the sun position (azimuth and elevation).

| Parameter                | Type              | Required | Default | Description                |
|--------------------------|-------------------|----------|---------|----------------------------|
| INPUT                    | raster            | yes      | —       | Elevation layer            |
| Z_FACTOR                 | number            | no       | 1.0     | Z factor                   |
| AZIMUTH                  | number            | no       | 300     | Azimuth (horizontal angle) |
| V_ANGLE                  | number            | no       | 40      | Vertical angle             |
| NODATA *(adv)*           | number            | no       | -9999.0 | Output NoData value        |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options           |
| OUTPUT                   | rasterDestination | yes      | —       | Hillshade                  |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:hillshade INPUT=dem.tif Z_FACTOR=1 AZIMUTH=300 V_ANGLE=40 OUTPUT=output.tif
```

This calls **Hillshade**: INPUT=dem.tif — Elevation layer; Z\_FACTOR=1 — Z factor; AZIMUTH=300 — Azimuth (horizontal angle); V\_ANGLE=40 — Vertical angle; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:httprequest — HTTP(S) POST/GET request

**group:** File tools

This algorithm performs a HTTP(S) POST/GET request and returns the HTTP status code and the reply data. If an error occurs then the error code and the message will be returned.

Optionally, the result can be written to a file on disk.

By default the algorithm will warn on errors. Optionally, the algorithm can be set to treat HTTP errors as failures.

| Parameter     | Type            | Required | Default | Description                      |
|---------------|-----------------|----------|---------|----------------------------------|
| URL           | string          | yes      | —       | URL                              |
| METHOD        | enum            | no       | 0       | Method — options: 0=GET, 1=POST  |
| DATA          | string          | no       | —       | POST data                        |
| OUTPUT        | fileDestination | no       | —       | File destination                 |
| AUTH_CONFIG   | authcfg         | no       | —       | Authentication                   |
| FAIL_ON_ERROR | boolean         | no       | False   | Consider HTTP errors as failures |

**Outputs:** OUTPUT (outputFile), ERROR\_CODE (outputNumber), ERROR\_MESSAGE (outputString), STATUS\_CODE (outputNumber), RESULT\_DATA (outputString)

### Example usage

```
run native:httprequest URL=value METHOD=0 DATA=value FAIL_ON_ERROR=false OUTPUT=output.txt
```

This calls **HTTP(S) POST/GET request**: URL=value — URL; METHOD=0 selects *GET*; DATA=value — POST data; FAIL\_ON\_ERROR=false — Consider HTTP errors as failures; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:hublines — Join by lines (hub lines)

**group:** Vector analysis

This algorithm creates hub and spoke diagrams by connecting lines from points on the Spoke layer to matching points in the Hub layer.

Determination of which hub goes with each point is based on a match between the Hub ID field on the hub points and the Spoke ID field on the spoke points.

If input layers are not point layers, a point on the surface of the geometries will be taken as the connecting location.

Optionally, geodesic lines can be created, which represent the shortest path on the surface of an ellipsoid. When geodesic mode is used, it is possible to split the created lines at the antimeridian ( $\pm 180$  degrees longitude), which can improve rendering of the lines. Additionally, the distance between vertices can be specified. A smaller distance results in a denser, more accurate line.

| Parameter                  | Type     | Required | Default | Description                                                 |
|----------------------------|----------|----------|---------|-------------------------------------------------------------|
| HUBS                       | source   | yes      | —       | Hub layer                                                   |
| HUB_FIELD                  | field    | yes      | —       | Hub ID field                                                |
| HUB_FIELDS                 | field    | no       | —       | Hub layer fields to copy (leave empty to copy all fields)   |
| SPOKES                     | source   | yes      | —       | Spoke layer                                                 |
| SPOKE_FIELD                | field    | yes      | —       | Spoke ID field                                              |
| SPOKE_FIELDS               | field    | no       | —       | Spoke layer fields to copy (leave empty to copy all fields) |
| GEODESIC                   | boolean  | no       | False   | Create geodesic lines                                       |
| GEODESIC_DISTANCE *(adv)*  | distance | no       | 1000    | Distance between vertices (geodesic lines only)             |
| ANTIMERIDIAN_SPLIT *(adv)* | boolean  | no       | False   | Split lines at antimeridian ( $\pm 180$ degrees longitude)  |
| OUTPUT                     | sink     | yes      | —       | Hub lines                                                   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:hublines HUBS=input.gpkg HUB_FIELD=field1 SPOKES=input.gpkg SPOKE_FIELD=field1  
↪ HUB_FIELDS=field1 SPOKE_FIELDS=field1 GEODESIC=false OUTPUT=output.gpkg
```

This calls **Join by lines (hub lines)**: HUBS=input.gpkg — Hub layer; HUB\_FIELD=field1 — Hub ID field; SPOKES=input.gpkg — Spoke layer; SPOKE\_FIELD=field1 — Spoke ID field; HUB\_FIELDS=field1 — Hub layer fields to copy (leave empty to copy all fields); SPOKE\_FIELDS=field1 — Spoke layer fields to copy (leave empty to copy all fields); GEODESIC=false — Create geodesic lines; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:importtopostgis — Export to PostgreSQL

**group:** Database

This algorithm exports a vector layer to a PostgreSQL database, creating a new table.

Prior to this a connection between QGIS and the PostgreSQL database has to be created (for example through the QGIS Browser panel).

| Parameter   | Type               | Required | Default | Description                                        |
|-------------|--------------------|----------|---------|----------------------------------------------------|
| INPUT       | source             | yes      | —       | Layer to export                                    |
| DATABASE    | providerconnection | yes      | —       | Database (connection name)                         |
| SCHEMA      | databaseschema     | no       | public  | Schema (schema name)                               |
| TABLENAME   | databasetable      | no       | —       | Table to export to (leave blank to use layer name) |
| PRIMARY_KEY | field              | no       | —       | Primary key field                                  |

| Parameter          | Type    | Required | Default | Description                                        |
|--------------------|---------|----------|---------|----------------------------------------------------|
| GEOMETRY_COLUMN    | string  | no       | geom    | Geometry column                                    |
| ENCODING           | string  | no       | UTF-8   | Encoding                                           |
| OVERWRITE          | boolean | no       | True    | Overwrite                                          |
| CREATEINDEX        | boolean | no       | True    | Create spatial index                               |
| LOWERCASE_NAMES    | boolean | no       | True    | Convert field names to lowercase                   |
| DROP_STRING_LENGTH | boolean | no       | False   | Drop length constraints on character fields        |
| FORCE_SINGLEPART   | boolean | no       | False   | Create single-part geometries instead of multipart |

### Example usage

```
run native:importintopostgis INPUT=input.gpkg DATABASE=... SCHEMA=public PRIMARY_KEY=field1
↳ GEOMETRY_COLUMN=geom ENCODING=UTF-8
```

This calls **Export to PostgreSQL**: INPUT=input.gpkg — Layer to export; DATABASE=... — Database (connection name); SCHEMA=public — Schema (schema name); PRIMARY\_KEY=field1 — Primary key field; GEOMETRY\_COLUMN=geom — Geometry column; ENCODING=UTF-8 — Encoding. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:importintospatialite — Export to SpatialLite

**group:** Database

This algorithm exports a vector layer to a SpatialLite database, creating a new table.

| Parameter          | Type    | Required | Default | Description                                        |
|--------------------|---------|----------|---------|----------------------------------------------------|
| INPUT              | source  | yes      | —       | Layer to export                                    |
| DATABASE           | vector  | yes      | —       | Database layer (or file)                           |
| TABLERNAME         | string  | no       | —       | Table to export to (leave blank to use layer name) |
| PRIMARY_KEY        | field   | no       | —       | Primary key field                                  |
| GEOMETRY_COLUMN    | string  | no       | geom    | Geometry column                                    |
| ENCODING           | string  | no       | UTF-8   | Encoding                                           |
| OVERWRITE          | boolean | no       | True    | Overwrite                                          |
| CREATEINDEX        | boolean | no       | True    | Create spatial index                               |
| LOWERCASE_NAMES    | boolean | no       | True    | Convert field names to lowercase                   |
| DROP_STRING_LENGTH | boolean | no       | False   | Drop length constraints on character fields        |
| FORCE_SINGLEPART   | boolean | no       | False   | Create single-part geometries instead of multipart |

### Example usage

```
run native:importintospatialite INPUT=input.gpkg DATABASE=input.gpkg TABLERNAME=value
↳ PRIMARY_KEY=field1 GEOMETRY_COLUMN=geom ENCODING=UTF-8 OVERWRITE=true
```

This calls **Export to SpatialLite**: INPUT=input.gpkg — Layer to export; DATABASE=input.gpkg — Database layer (or file); TABLERNAME=value — Table to export to (leave blank to use layer name); PRIMARY\_KEY=field1 — Primary key field; GEOMETRY\_COLUMN=geom — Geometry column; ENCODING=UTF-8 — Encoding; OVERWRITE=true — Overwrite. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:importintospatialiteregistered — Export to SpatialLite (registered)

**group:** Database

Exports a vector layer to a SpatialLite database, creating a new table.

Prior to this a connection between QGIS and the SpatialLite database has to be created (for example through the QGIS Browser panel).

| Parameter          | Type               | Required | Default | Description                                        |
|--------------------|--------------------|----------|---------|----------------------------------------------------|
| INPUT              | source             | yes      | —       | Layer to export                                    |
| DATABASE           | providerconnection | yes      | —       | Database (connection name)                         |
| TABLENAME          | databasetable      | no       | —       | Table to export to (leave blank to use layer name) |
| PRIMARY_KEY        | field              | no       | —       | Primary key field                                  |
| GEOMETRY_COLUMN    | string             | no       | geom    | Geometry column                                    |
| ENCODING           | string             | no       | UTF-8   | Encoding                                           |
| OVERWRITE          | boolean            | no       | True    | Overwrite                                          |
| CREATEINDEX        | boolean            | no       | True    | Create spatial index                               |
| LOWERCASE_NAMES    | boolean            | no       | True    | Convert field names to lowercase                   |
| DROP_STRING_LENGTH | boolean            | no       | False   | Drop length constraints on character fields        |
| FORCE_SINGLEPART   | boolean            | no       | False   | Create single-part geometries instead of multipart |

## Example usage

```
run native:importintospatialiterregistered INPUT=input.gpkg DATABASE=... PRIMARY_KEY=field1
↳ GEOMETRY_COLUMN=geom ENCODING=UTF-8 OVERWRITE=true
```

This calls **Export to Spatialite (registered)**: INPUT=input.gpkg — Layer to export; DATABASE=... — Database (connection name); PRIMARY\_KEY=field1 — Primary key field; GEOMETRY\_COLUMN=geom — Geometry column; ENCODING=UTF-8 — Encoding; OVERWRITE=true — Overwrite. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:importphotos — Import geotagged photos

**group:** Vector creation

This algorithm creates a point layer corresponding to the geotagged locations from JPEG or HEIF/HEIC images from a source folder. Optionally the folder can be recursively scanned.

The point layer will contain a single PointZ feature per input file from which the geotags could be read. Any altitude information from the geotags will be used to set the point's Z value.

Optionally, a table of unreadable or non-geotagged photos can also be created.

| Parameter | Type    | Required | Default | Description          |
|-----------|---------|----------|---------|----------------------|
| FOLDER    | file    | yes      | —       | Input folder         |
| RECURSIVE | boolean | no       | False   | Scan recursively     |
| OUTPUT    | sink    | no       | —       | Photos               |
| INVALID   | sink    | no       | —       | Invalid photos table |

**Outputs:** OUTPUT (outputVector), INVALID (outputVector)

## Example usage

```
run native:importphotos FOLDER=input.dat RECURSIVE=false OUTPUT=output.gpkg
```

This calls **Import geotagged photos**: FOLDER=input.dat — Input folder; RECURSIVE=false — Scan recursively; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:interpolatepoint — Interpolate point on line

**group:** Vector geometry

This algorithm creates a point geometry interpolated at a set distance along line or curve geometries.

Z and M values are linearly interpolated from existing values.

If a multipart geometry is encountered, only the first part is considered when interpolating the point.

If the specified distance is greater than the curve's length, the resultant feature will have a null geometry.

| Parameter | Type     | Required | Default | Description         |
|-----------|----------|----------|---------|---------------------|
| INPUT     | source   | yes      | —       | Input layer         |
| DISTANCE  | distance | no       | 0.0     | Distance            |
| OUTPUT    | sink     | yes      | —       | Interpolated points |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:interpolatepoint INPUT=input.gpkg DISTANCE=0 OUTPUT=output.gpkg
```

This calls **Interpolate point on line**: INPUT=input.gpkg — Input layer; DISTANCE=0 — Distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:intersection — Intersection ★

**group:** Vector overlay · **niva verb:** intersect

This algorithm extracts the overlapping portions of features in the Input and Overlay layers. Features in the output Intersection layer are assigned the attributes of the overlapping features from both the Input and Overlay layers.

| Parameter                     | Type   | Required | Default | Description                                             |
|-------------------------------|--------|----------|---------|---------------------------------------------------------|
| INPUT                         | source | yes      | —       | Input layer                                             |
| OVERLAY                       | source | yes      | —       | Overlay layer                                           |
| INPUT_FIELDS                  | field  | no       | —       | Input fields to keep (leave empty to keep all fields)   |
| OVERLAY_FIELDS                | field  | no       | —       | Overlay fields to keep (leave empty to keep all fields) |
| OVERLAY_FIELDS_PREFIX *(adv)* | string | no       | —       | Overlay fields prefix                                   |
| OUTPUT                        | sink   | yes      | —       | Intersection                                            |
| GRID_SIZE *(adv)*             | number | no       | —       | Grid size                                               |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:intersection INPUT=input.gpkg OVERLAY=input.gpkg INPUT_FIELDS=field1  
↪ OVERLAY_FIELDS=field1 OUTPUT=output.gpkg
```

This calls **Intersection**: INPUT=input.gpkg — Input layer; OVERLAY=input.gpkg — Overlay layer; INPUT\_FIELDS=field1 — Input fields to keep (leave empty to keep all fields); OVERLAY\_FIELDS=field1 — Overlay fields to keep (leave empty to keep all fields); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:joinattributesbylocation — Join attributes by location ★

**group:** Vector general · **niva verb:** spatialjoin

This algorithm takes an input vector layer and creates a new vector layer that is an extended version of the input one, with additional attributes in its attribute table.



The additional attributes and their values are taken from a second vector layer. A spatial criteria is applied to select the values from the second layer that are added to each feature from the first layer in the resulting one.

| Parameter           | Type    | Required | Default | Description                                                                                                                                                                                                                        |
|---------------------|---------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT               | source  | yes      | —       | Join to features in                                                                                                                                                                                                                |
| PREDICATE           | enum    | no       | 0       | Features they (geometric predicate) — options: 0=intersect, 1=contain, 2=equal, 3=touch, 4=overlap, 5=are within, 6=cross                                                                                                          |
| JOIN                | source  | yes      | —       | By comparing to                                                                                                                                                                                                                    |
| JOIN_FIELDS         | field   | no       | —       | Fields to add (leave empty to use all fields)                                                                                                                                                                                      |
| METHOD              | enum    | no       | 0       | Join type — options: 0=Create separate feature for each matching feature (one-to-many), 1=Take attributes of the first matching feature only (one-to-one), 2=Take attributes of the feature with largest overlap only (one-to-one) |
| DISCARD_NONMATCHING | boolean | no       | False   | Discard records which could not be joined                                                                                                                                                                                          |
| PREFIX              | string  | no       | —       | Joined field prefix                                                                                                                                                                                                                |
| OUTPUT              | sink    | no       | —       | Joined layer                                                                                                                                                                                                                       |
| NON_MATCHING        | sink    | no       | —       | Unjoinable features from first layer                                                                                                                                                                                               |

**Outputs:** OUTPUT (outputVector), NON\_MATCHING (outputVector), JOINED\_COUNT (outputNumber)

### Example usage

```
run native:joinattributesbylocation INPUT=input.gpkg JOIN=input.gpkg PREDICATE=0
↪ JOIN_FIELDS=field1 METHOD=0 DISCARD_NONMATCHING=false PREFIX=value OUTPUT=output.gpkg
```

This calls **Join attributes by location**: INPUT=input.gpkg — Join to features in; JOIN=input.gpkg — By comparing to; PREDICATE=0 selects *intersect*; JOIN\_FIELDS=field1 — Fields to add (leave empty to use all fields); METHOD=0 selects *Create separate feature for each matching feature (one-to-many)*; DISCARD\_NONMATCHING=false — Discard records which could not be joined; PREFIX=value — joined field prefix; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:joinattributestable — Join attributes by field value ★

**group:** Vector general · **niva verb:** join

This algorithm takes an input vector layer and creates a new vector layer that is an extended version of the input one, with additional attributes in its attribute table.

The additional attributes and their values are taken from a second vector layer. An attribute is selected in each of them to define the join criteria.

| Parameter           | Type    | Required | Default | Description                                                                                                                                               |
|---------------------|---------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT               | source  | yes      | —       | Input layer                                                                                                                                               |
| FIELD               | field   | yes      | —       | Table field                                                                                                                                               |
| INPUT_2             | source  | yes      | —       | Input layer 2                                                                                                                                             |
| FIELD_2             | field   | yes      | —       | Table field 2                                                                                                                                             |
| FIELDS_TO_COPY      | field   | no       | —       | Layer 2 fields to copy (leave empty to copy all fields)                                                                                                   |
| METHOD              | enum    | no       | 1       | Join type — options: 0=Create separate feature for each matching feature (one-to-many), 1=Take attributes of the first matching feature only (one-to-one) |
| DISCARD_NONMATCHING | boolean | no       | False   | Discard records which could not be joined                                                                                                                 |
| PREFIX              | string  | no       | —       | Joined field prefix                                                                                                                                       |
| OUTPUT              | sink    | no       | —       | Joined layer                                                                                                                                              |
| NON_MATCHING        | sink    | no       | —       | Unjoinable features from first layer                                                                                                                      |

**Outputs:** OUTPUT (outputVector), NON\_MATCHING (outputVector), JOINED\_COUNT (outputNumber), UNJOINABLE\_COUNT (outputNumber)

## Example usage

```
run native:joinattributetable INPUT=input.gpkg FIELD=field1 INPUT_2=input.gpkg
↳ FIELD_2=field1 FIELDS_TO_COPY=field1 METHOD=1 DISCARD_NONMATCHING=false
↳ OUTPUT=output.gpkg
```

This calls **Join attributes by field value**: INPUT=input.gpkg — Input layer; FIELD=field1 — Table field; INPUT\_2=input.gpkg — Input layer 2; FIELD\_2=field1 — Table field 2; FIELDS\_TO\_COPY=field1 — Layer 2 fields to copy (leave empty to copy all fields); METHOD=1 selects *Take attributes of the first matching feature only (one-to-one)*; DISCARD\_NONMATCHING=false — Discard records which could not be joined; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:joinbylocationsummary — Join attributes by location (summary)

**group:** Vector general

This algorithm takes an input vector layer and creates a new vector layer that is an extended version of the input one, with additional attributes in its attribute table.

The additional attributes and their values are taken from a second vector layer. A spatial criteria is applied to select the values from the second layer that are added to each feature from the first layer in the resulting one.

The algorithm calculates a statistical summary for the values from matching features in the second layer( e.g. maximum value, mean value, etc ).

| Parameter           | Type    | Required | Default | Description                                                                                                                                                                                                                                                        |
|---------------------|---------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT               | source  | yes      | —       | Join to features in                                                                                                                                                                                                                                                |
| PREDICATE           | enum    | no       | 0       | Where the features — options: 0=intersect, 1=contain, 2=equal, 3=touch, 4=overlap, 5=are within, 6=cross                                                                                                                                                           |
| JOIN                | source  | yes      | —       | By comparing to                                                                                                                                                                                                                                                    |
| JOIN_FIELDS         | field   | no       | —       | Fields to summarise (leave empty to use all fields)                                                                                                                                                                                                                |
| SUMMARIES           | enum    | no       | —       | Summaries to calculate (leave empty to use all available) — options: 0=count, 1=unique, 2=min, 3=max, 4=range, 5=sum, 6=mean, 7=median, 8=stddev, 9=minority, 10=majority, 11=q1, 12=q3, 13=iqr, 14=empty, 15=filled, 16=min_length, 17=max_length, 18=mean_length |
| DISCARD_NONMATCHING | boolean | no       | False   | Discard records which could not be joined                                                                                                                                                                                                                          |
| OUTPUT              | sink    | yes      | —       | Joined layer                                                                                                                                                                                                                                                       |

**Outputs:** OUTPUT (outputVector)

## Example usage

```
run native:joinbylocationsummary INPUT=input.gpkg JOIN=input.gpkg PREDICATE=0
↳ JOIN_FIELDS=field1 SUMMARIES=0 DISCARD_NONMATCHING=false OUTPUT=output.gpkg
```

This calls **Join attributes by location (summary)**: INPUT=input.gpkg — Join to features in; JOIN=input.gpkg — By comparing to; PREDICATE=0 selects *intersect*; JOIN\_FIELDS=field1 — Fields to summarise (leave empty to use all fields); SUMMARIES=0 selects *count*; DISCARD\_NONMATCHING=false — Discard records which could not be joined; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:joinbynearest — Join attributes by nearest

**group:** Vector general

This algorithm takes an input vector layer and creates a new vector layer that is an extended version of the input one, with additional attributes in its attribute table.

The additional attributes and their values are taken from a second vector layer, where features are joined by finding the closest features from each layer. By default only the single nearest feature is joined, but optionally the join can use the n-nearest neighboring features instead. If multiple features are found with identical distances these will all be returned (even if the total number of features exceeds the specified maximum feature count).

If a maximum distance is specified, then only features which are closer than this distance will be matched.

The output features will contain the selected attributes from the nearest feature, along with new attributes for the distance to the near feature, the index of the feature, and the coordinates of the closest point on the input feature (feature\_x, feature\_y) to the matched nearest feature, and the coordinates of the closest point on the matched feature (nearest\_x, nearest\_y).

This algorithm uses purely Cartesian calculations for distance, and does not consider geodetic or ellipsoid properties when determining feature proximity.

| Parameter           | Type     | Required | Default | Description                                             |
|---------------------|----------|----------|---------|---------------------------------------------------------|
| INPUT               | source   | yes      | —       | Input layer                                             |
| INPUT_2             | source   | yes      | —       | Input layer 2                                           |
| FIELDS_TO_COPY      | field    | no       | —       | Layer 2 fields to copy (leave empty to copy all fields) |
| DISCARD_NONMATCHING | boolean  | no       | False   | Discard records which could not be joined               |
| PREFIX              | string   | no       | —       | Joined field prefix                                     |
| NEIGHBORS           | number   | no       | 1       | Maximum nearest neighbors                               |
| MAX_DISTANCE        | distance | no       | —       | Maximum distance                                        |
| OUTPUT              | sink     | no       | —       | Joined layer                                            |
| NON_MATCHING        | sink     | no       | —       | Unjoinable features from first layer                    |

**Outputs:** OUTPUT (outputVector), NON\_MATCHING (outputVector), JOINED\_COUNT (outputNumber), UNJOINABLE\_COUNT (outputNumber)

### Example usage

```
run native:joinbynearest INPUT=input.gpkg INPUT_2=input.gpkg FIELDS_TO_COPY=field1
↳ DISCARD_NONMATCHING=false PREFIX=value NEIGHBORS=1 MAX_DISTANCE=10 OUTPUT=output.gpkg
```

This calls **Join attributes by nearest**: INPUT=input.gpkg — Input layer; INPUT\_2=input.gpkg — Input layer 2; FIELDS\_TO\_COPY=field1 — Layer 2 fields to copy (leave empty to copy all fields); DISCARD\_NONMATCHING=false — Discard records which could not be joined; PREFIX=value — Joined field prefix; NEIGHBORS=1 — Maximum nearest neighbors; MAX\_DISTANCE=10 — Maximum distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:keepnbiggestparts — Keep N biggest parts

**group:** Vector geometry

This algorithm takes a polygon layer and creates a new polygon layer in which multipart geometries have been removed, leaving only the n largest (in terms of area) parts.

| Parameter | Type   | Required | Default | Description   |
|-----------|--------|----------|---------|---------------|
| POLYGONS  | source | yes      | —       | Input layer   |
| PARTS     | number | no       | 1.0     | Parts to keep |
| OUTPUT    | sink   | yes      | —       | Parts         |

**Outputs:** OUTPUT (outputVector)

## Example usage

```
run native:keepnbiggestparts POLYGONS=input.gpkg PARTS=1 OUTPUT=output.gpkg
```

This calls **Keep N biggest parts**: POLYGONS=input.gpkg — Input layer; PARTS=1 — Parts to keep; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:kmeansclustering — K-means clustering

**group:** Vector analysis

This algorithm calculates the 2D distance based k-means cluster number for each input feature.

If input geometries are lines or polygons, the clustering is based on the centroid of the feature.

References: Arthur, David & Vassilvitskii, Sergei. (2007). K-Means++: The Advantages of Careful Seeding. Proc. of the Annu. ACM-SIAM Symp. on Discrete Algorithms. 8.

Bhattacharya, Anup & Eube, Jan & Röglin, Heiko & Schmidt, Melanie. (2019). Noisy, Greedy and Not So Greedy k-means++

| Parameter               | Type   | Required | Default      | Description                                      |
|-------------------------|--------|----------|--------------|--------------------------------------------------|
| INPUT                   | source | yes      | —            | Input layer                                      |
| CLUSTERS                | number | no       | 5            | Number of clusters                               |
| METHOD                  | enum   | no       | 0            | Method — options: 0=Farthest points, 1=K-means++ |
| FIELD_NAME *(adv)*      | string | no       | CLUSTER_ID   | Cluster field name                               |
| SIZE_FIELD_NAME *(adv)* | string | no       | CLUSTER_SIZE | Cluster size field name                          |
| OUTPUT                  | sink   | yes      | —            | Clusters                                         |

**Outputs:** OUTPUT (outputVector)

## Example usage

```
run native:kmeansclustering INPUT=input.gpkg CLUSTERS=5 METHOD=0 OUTPUT=output.gpkg
```

This calls **K-means clustering**: INPUT=input.gpkg — Input layer; CLUSTERS=5 — Number of clusters; METHOD=0 selects *Farthest points*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:layertobookmarks — Convert layer to spatial bookmarks

**group:** Vector general

This algorithm creates spatial bookmarks corresponding to the extent of features contained in a layer.

| Parameter        | Type       | Required | Default | Description                                                           |
|------------------|------------|----------|---------|-----------------------------------------------------------------------|
| INPUT            | source     | yes      | —       | Input layer                                                           |
| DESTINATION      | enum       | no       | 0       | Bookmark destination — options: 0=Project bookmarks, 1=User bookmarks |
| NAME_EXPRESSION  | expression | yes      | —       | Name field                                                            |
| GROUP_EXPRESSION | expression | no       | —       | Group field                                                           |

**Outputs:** COUNT (outputNumber)

## Example usage

```
run native:layertobookmarks INPUT=input.gpkg NAME_EXPRESSION="value > 0" DESTINATION=0  
↳ GROUP_EXPRESSION="value > 0"
```

This calls **Convert layer to spatial bookmarks**: INPUT=input.gpkg — Input layer; NAME\_EXPRESSION="value > 0" — Name field; DESTINATION=0 selects *Project bookmarks*; GROUP\_EXPRESSION="value > 0" — Group field. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:lessthanfrequency — Less than frequency

**group:** Raster analysis

This algorithm evaluates on a cell-by-cell basis the frequency (number of times) the values of an input stack of rasters are less than the value of a value raster. If multiband rasters are used in the data raster stack, the algorithm will always perform the analysis on the first band of the rasters - use GDAL to use other bands in the analysis. The input value layer serves as reference layer for the sample layers. Any NoData cells in the value raster or the data layer stack will result in a NoData cell in the output raster if the ignore NoData parameter is not checked. The output NoData value can be set manually. The output rasters extent and resolution is defined by the input raster layer and is always of int32 type.

| Parameter                   | Type              | Required | Default | Description          |
|-----------------------------|-------------------|----------|---------|----------------------|
| INPUT_VALUE_RASTER          | raster            | yes      | —       | Input value raster   |
| INPUT_VALUE_RASTER_BAND     | band              | no       | 1       | Value raster band    |
| INPUT_RASTERS               | multilayer        | yes      | —       | Input raster layers  |
| IGNORE_NODATA               | boolean           | no       | False   | Ignore NoData values |
| OUTPUT_NODATA_VALUE *(adv)* | number            | no       | -9999   | Output NoData value  |
| CREATE_OPTIONS              | string            | no       | —       | Creation options     |
| CREATION_OPTIONS *(adv)*    | string            | no       | —       | Creation options     |
| OUTPUT                      | rasterDestination | yes      | —       | Output layer         |

**Outputs:** OUTPUT (outputRaster), OCCURRENCE\_COUNT (outputNumber), FOUND\_LOCATIONS\_COUNT (outputNumber), MEAN\_FREQUENCY\_PER\_LOCATION (outputNumber), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:lessthanfrequency INPUT_VALUE_RASTER=dem.tif INPUT_RASTERS="a.gpkg;b.gpkg"
↪ INPUT_VALUE_RASTER_BAND=1 IGNORE_NODATA=false CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Less than frequency**: INPUT\_VALUE\_RASTER=dem.tif — Input value raster; INPUT\_RASTERS="a.gpkg;b.gpkg" — Input raster layers; INPUT\_VALUE\_RASTER\_BAND=1 — Value raster band; IGNORE\_NODATA=false — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:linedensity — Line density

**group:** Interpolation

This algorithm calculates a density measure of linear features which is obtained in a circular neighborhood within each raster cell. First, the length of the segment of each line that is intersected by the circular neighborhood is multiplied with the lines weight factor. In a second step, all length values are summed and divided by the area of the circular neighborhood. This process is repeated for all raster cells.

| Parameter | Type   | Required | Default | Description      |
|-----------|--------|----------|---------|------------------|
| INPUT     | source | yes      | —       | Input line layer |
| WEIGHT    | field  | no       | —       | Weight field     |

| Parameter                | Type              | Required | Default | Description         |
|--------------------------|-------------------|----------|---------|---------------------|
| RADIUS                   | distance          | no       | 10      | Search radius       |
| PIXEL_SIZE               | distance          | no       | 10      | Pixel size          |
| CREATE_OPTIONS           | string            | no       | —       | Creation options    |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options    |
| OUTPUT                   | rasterDestination | yes      | —       | Line density raster |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:linedensity INPUT=input.gpkg WEIGHT=field1 RADIUS=10 PIXEL_SIZE=10
↪ CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Line density**: INPUT=input.gpkg — Input line layer; WEIGHT=field1 — Weight field; RADIUS=10 — Search radius; PIXEL\_SIZE=10 — Pixel size; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:lineintersections — Line intersections

**group:** Vector overlay

This algorithm creates point features where the lines in the Intersect layer intersect the lines in the Input layer.

| Parameter                       | Type   | Required | Default | Description                                               |
|---------------------------------|--------|----------|---------|-----------------------------------------------------------|
| INPUT                           | source | yes      | —       | Input layer                                               |
| INTERSECT                       | source | yes      | —       | Intersect layer                                           |
| INPUT_FIELDS                    | field  | no       | —       | Input fields to keep (leave empty to keep all fields)     |
| INTERSECT_FIELDS                | field  | no       | —       | Intersect fields to keep (leave empty to keep all fields) |
| INTERSECT_FIELDS_PREFIX *(adv)* | string | no       | —       | Intersect fields prefix                                   |
| OUTPUT                          | sink   | yes      | —       | Intersections                                             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:lineintersections INPUT=input.gpkg INTERSECT=input.gpkg INPUT_FIELDS=field1
↪ INTERSECT_FIELDS=field1 OUTPUT=output.gpkg
```

This calls **Line intersections**: INPUT=input.gpkg — Input layer; INTERSECT=input.gpkg — Intersect layer; INPUT\_FIELDS=field1 — Input fields to keep (leave empty to keep all fields); INTERSECT\_FIELDS=field1 — Intersect fields to keep (leave empty to keep all fields); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:linesubstring — Line substring

**group:** Vector geometry

This algorithm returns the portion of a line (or curve) which falls between the specified start and end distances (measured from the beginning of the line).

Z and M values are linearly interpolated from existing values.

If a multipart geometry is encountered, only the first part is considered when calculating the substring.

| Parameter      | Type     | Required | Default | Description    |
|----------------|----------|----------|---------|----------------|
| INPUT          | source   | yes      | —       | Input layer    |
| START_DISTANCE | distance | no       | 0.0     | Start distance |
| END_DISTANCE   | distance | no       | 1.0     | End distance   |
| OUTPUT         | sink     | yes      | —       | Substring      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:linesubstring INPUT=input.gpkg START_DISTANCE=0 END_DISTANCE=1
↪ OUTPUT=output.gpkg
```

This calls **Line substring**: INPUT=input.gpkg — Input layer; START\_DISTANCE=0 — Start distance; END\_DISTANCE=1 — End distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:listuniquevalues — List unique values

**group:** Vector analysis

This algorithm generates a report with information about the unique values found in a given attribute (or attributes) of a vector layer.

| Parameter        | Type            | Required | Default | Description     |
|------------------|-----------------|----------|---------|-----------------|
| INPUT            | source          | yes      | —       | Input layer     |
| FIELDS           | field           | yes      | —       | Target field(s) |
| OUTPUT           | sink            | no       | —       | Unique values   |
| OUTPUT_HTML_FILE | fileDestination | no       | —       | HTML report     |

**Outputs:** OUTPUT (outputVector), OUTPUT\_HTML\_FILE (outputHtml), TOTAL\_VALUES (outputNumber), UNIQUE\_VALUES (outputString)

### Example usage

```
run native:listuniquevalues INPUT=input.gpkg FIELDS=field1 OUTPUT=output.gpkg
```

This calls **List unique values**: INPUT=input.gpkg — Input layer; FIELDS=field1 — Target field(s); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:loadlayer — Load layer into project

**group:** Modeler tools

This algorithm loads a layer to the current project.

| Parameter | Type   | Required | Default | Description       |
|-----------|--------|----------|---------|-------------------|
| INPUT     | layer  | yes      | —       | Layer             |
| NAME      | string | yes      | —       | Loaded layer name |

**Outputs:** OUTPUT (outputLayer)

### Example usage

```
run native:loadlayer INPUT=input.gpkg NAME=value
```

This calls **Load layer into project**: INPUT=input.gpkg — Layer; NAME=value — Loaded layer name. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:lowestpositioninrasterstack — Lowest position in raster stack

**group:** Raster analysis

This algorithm evaluates on a cell-by-cell basis the position of the raster with the lowest value in a stack of rasters. Position counts start with 1 and range to the total number of input rasters. The order of the input rasters is relevant for the algorithm. If multiple rasters feature the lowest value, the first raster will be used for the position value. If multiband rasters are used in the data raster stack, the algorithm will always perform the analysis on the first band of the rasters - use GDAL to use other bands in the analysis. Any NoData cells in the raster layer stack will result in a NoData cell in the output raster unless the "ignore NoData" parameter is checked. The output NoData value can be set manually. The output rasters extent and resolution is defined by a reference raster layer and is always of int32 type.

| Parameter                   | Type              | Required | Default | Description          |
|-----------------------------|-------------------|----------|---------|----------------------|
| INPUT_RASTERS               | multilayer        | yes      | —       | Input raster layers  |
| REFERENCE_LAYER             | raster            | yes      | —       | Reference layer      |
| IGNORE_NODATA               | boolean           | no       | False   | Ignore NoData values |
| OUTPUT_NODATA_VALUE *(adv)* | number            | no       | -9999   | Output NoData value  |
| CREATE_OPTIONS              | string            | no       | —       | Creation options     |
| CREATION_OPTIONS *(adv)*    | string            | no       | —       | Creation options     |
| OUTPUT                      | rasterDestination | yes      | —       | Output layer         |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:lowestpositioninrasterstack INPUT_RASTERS="a.gpkg;b.gpkg"  
↪ REFERENCE_LAYER=dem.tif IGNORE_NODATA=false CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Lowest position in raster stack**: INPUT\_RASTERS="a.gpkg;b.gpkg" — Input raster layers; REFERENCE\_LAYER=dem.tif — Reference layer; IGNORE\_NODATA=false — Ignore NoData values; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:meancoordinates — Mean coordinate(s)

**group:** Vector analysis

This algorithm computes a point layer with the center of mass of geometries in an input layer.

An attribute can be specified as containing weights to be applied to each feature when computing the center of mass.

If an attribute is selected in the parameter, features will be grouped according to values in this field. Instead of a single point with the center of mass of the whole layer, the output layer will contain a center of mass for the features in each category.

| Parameter | Type   | Required | Default | Description      |
|-----------|--------|----------|---------|------------------|
| INPUT     | source | yes      | —       | Input layer      |
| WEIGHT    | field  | no       | —       | Weight field     |
| UID       | field  | no       | —       | Unique ID field  |
| OUTPUT    | sink   | yes      | —       | Mean coordinates |

**Outputs:** OUTPUT (outputVector)

### Example usage



```
run native:meancoordinates INPUT=input.gpkg WEIGHT=field1 UID=field1 OUTPUT=output.gpkg
```

This calls **Mean coordinate(s)**: INPUT=input.gpkg — Input layer; WEIGHT=field1 — Weight field; UID=field1 — Unique ID field; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:mergelines — Merge lines

**group:** Vector geometry

This algorithm joins all connected parts of MultiLineString geometries into single LineString geometries.

If any parts of the input MultiLineString geometries are not connected, the resultant geometry will be a MultiLineString containing any lines which could be merged and any non-connected line parts.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Merged      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:mergelines INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Merge lines**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:mergevectorlayers — Merge vector layers

**group:** Vector general

This algorithm combines multiple vector layers of the same geometry type into a single one.

The attribute table of the resulting layer will contain the fields from all input layers. If fields with the same name but different types are found then the exported field will be automatically converted into a string type field. Optionally, new fields storing the original layer name and source can be added.

If any input layers contain Z or M values, then the output layer will also contain these values. Similarly, if any of the input layers are multi-part, the output layer will also be a multi-part layer.

Optionally, the destination coordinate reference system (CRS) for the merged layer can be set. If it is not set, the CRS will be taken from the first input layer. All layers will all be reprojected to match this CRS.

| Parameter         | Type       | Required | Default | Description                                        |
|-------------------|------------|----------|---------|----------------------------------------------------|
| LAYERS            | multilayer | yes      | —       | Input layers                                       |
| CRS               | crs        | no       | —       | Destination CRS                                    |
| OUTPUT            | sink       | yes      | —       | Merged                                             |
| ADD_SOURCE_FIELDS | boolean    | no       | True    | Add source layer information (layer name and path) |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:mergevectorlayers LAYERS="a.gpkg;b.gpkg" CRS=EPSG:6346 ADD_SOURCE_FIELDS=true  
↳ OUTPUT=output.gpkg
```

This calls **Merge vector layers**: LAYERS="a.gpkg;b.gpkg" — Input layers; CRS=EPSG:6346 — Destination CRS; ADD\_SOURCE\_FIELDS=true — Add source layer information (layer name and path); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:meshcontours — Export contours

**group:** Mesh

This algorithm creates contours as a vector layer from a mesh scalar dataset.

| Parameter          | Type              | Required | Default | Description                      |
|--------------------|-------------------|----------|---------|----------------------------------|
| INPUT              | mesh              | yes      | —       | Input mesh layer                 |
| DATASET_GROUPS     | meshdatasetgroups | no       | []      | Dataset groups                   |
| DATASET_TIME       | meshdatasettime   | yes      | —       | Dataset time                     |
| INCREMENT          | number            | no       | —       | Increment between contour levels |
| MINIMUM            | number            | no       | —       | Minimum contour level            |
| MAXIMUM            | number            | no       | —       | Maximum contour level            |
| CONTOUR_LEVEL_LIST | string            | no       | —       | List of contours level           |
| CRS_OUTPUT         | crs               | no       | —       | Output coordinate system         |
| OUTPUT_LINES       | sink              | yes      | —       | Exported contour lines           |
| OUTPUT_POLYGONS    | sink              | yes      | —       | Exported contour polygons        |

**Outputs:** OUTPUT\_LINES (outputVector), OUTPUT\_POLYGONS (outputVector)

### Example usage

```
run native:meshcontours INPUT=mesh.nc DATASET_TIME=... INCREMENT=10 MINIMUM=10 MAXIMUM=10
↳ CONTOUR_LEVEL_LIST=value OUTPUT_LINES=output_lines.gpkg
```

This calls **Export contours**: INPUT=mesh.nc — Input mesh layer; DATASET\_TIME=... — Dataset time; INCREMENT=10 — Increment between contour levels; MINIMUM=10 — Minimum contour level; MAXIMUM=10 — Maximum contour level; CONTOUR\_LEVEL\_LIST=value — List of contours level; OUTPUT\_LINES=output\_lines.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:meshexportcrosssection — Export cross section dataset values on lines from mesh

**group:** Mesh

This algorithm extracts mesh's dataset values from line contained in a vector layer. Each line is discretized with a resolution distance parameter for extraction of values on its vertices.

| Parameter          | Type              | Required | Default | Description                    |
|--------------------|-------------------|----------|---------|--------------------------------|
| INPUT              | mesh              | yes      | —       | Input mesh layer               |
| DATASET_GROUPS     | meshdatasetgroups | no       | []      | Dataset groups                 |
| DATASET_TIME       | meshdatasettime   | yes      | —       | Dataset time                   |
| INPUT_LINES        | source            | yes      | —       | Lines for data export          |
| RESOLUTION         | distance          | no       | 10.0    | Line segmentation resolution   |
| COORDINATES_DIGITS | number            | no       | 2       | Digits count for coordinates   |
| DATASET_DIGITS     | number            | no       | 2       | Digits count for dataset value |
| OUTPUT             | fileDestination   | yes      | —       | Exported data CSV file         |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:meshexportcrosssection INPUT=mesh.nc DATASET_TIME=... INPUT_LINES=input.gpkg
↳ RESOLUTION=10 COORDINATES_DIGITS=2 DATASET_DIGITS=2 OUTPUT=output.txt
```

This calls **Export cross section dataset values on lines from mesh**: INPUT=mesh.nc — Input mesh layer; DATASET\_TIME=... — Dataset time; INPUT\_LINES=input.gpkg — Lines for data export; RESOLUTION=10 — Line segmentation resolution; COORDINATES\_DIGITS=2 — Digits count for coordinates; DATASET\_DIGITS=2 — Digits count for dataset value; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:meshexporttimeseries — Export time series values from points of a mesh dataset

**group:** Mesh

This algorithm extracts mesh's dataset time series values from points contained in a vector layer. If the time step is kept to its default value (0 hours), the time step used is the one of the two first datasets of the first selected dataset group.

| Parameter          | Type              | Required | Default | Description                    |
|--------------------|-------------------|----------|---------|--------------------------------|
| INPUT              | mesh              | yes      | —       | Input mesh layer               |
| DATASET_GROUPS     | meshdatasetgroups | no       | []      | Dataset groups                 |
| STARTING_TIME      | meshdatasettime   | yes      | —       | Starting time                  |
| FINISHING_TIME     | meshdatasettime   | yes      | —       | Finishing time                 |
| TIME_STEP          | number            | no       | 0       | Time step (hours)              |
| INPUT_POINTS       | source            | yes      | —       | Points for data export         |
| COORDINATES_DIGITS | number            | no       | 2       | Digits count for coordinates   |
| DATASET_DIGITS     | number            | no       | 2       | Digits count for dataset value |
| OUTPUT             | fileDestination   | yes      | —       | Exported data CSV file         |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:meshexporttimeseries INPUT=mesh.nc STARTING_TIME=... FINISHING_TIME=...
↳ INPUT_POINTS=input.gpkg TIME_STEP=0 COORDINATES_DIGITS=2 OUTPUT=output.txt
```

This calls **Export time series values from points of a mesh dataset**: INPUT=mesh.nc — Input mesh layer; STARTING\_TIME=... — Starting time; FINISHING\_TIME=... — Finishing time; INPUT\_POINTS=input.gpkg — Points for data export; TIME\_STEP=0 — Time step (hours); COORDINATES\_DIGITS=2 — Digits count for coordinates; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:meshrasterize — Rasterize mesh dataset

**group:** Mesh

This algorithm creates a raster layer from a mesh dataset. For data on volume (3D stacked dataset values), the exported dataset values are averaged on faces using the method defined in the mesh layer properties (default is Multi level averaging method). 1D meshes are not supported.

| Parameter      | Type              | Required | Default | Description              |
|----------------|-------------------|----------|---------|--------------------------|
| INPUT          | mesh              | yes      | —       | Input mesh layer         |
| DATASET_GROUPS | meshdatasetgroups | no       | []      | Dataset groups           |
| DATASET_TIME   | meshdatasettime   | yes      | —       | Dataset time             |
| EXTENT         | extent            | no       | —       | Extent                   |
| PIXEL_SIZE     | distance          | no       | 1       | Pixel size               |
| CRS_OUTPUT     | crs               | no       | —       | Output coordinate system |

| Parameter                | Type              | Required | Default | Description         |
|--------------------------|-------------------|----------|---------|---------------------|
| CREATE_OPTIONS           | string            | no       | —       | Creation options    |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options    |
| OUTPUT                   | rasterDestination | yes      | —       | Output raster layer |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:meshrasterize INPUT=mesh.nc DATASET_TIME=... EXTENT=0,1000,0,1000 PIXEL_SIZE=1
↳ CRS_OUTPUT=EPSG:6346 CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Rasterize mesh dataset**: INPUT=mesh.nc — Input mesh layer; DATASET\_TIME=... — Dataset time; EXTENT=0,1000,0,1000 — Extent; PIXEL\_SIZE=1 — Pixel size; CRS\_OUTPUT=EPSG:6346 — Output coordinate system; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:minimumboundinggeometry — Minimum bounding geometry

**group:** Vector geometry

This algorithm creates geometries which enclose the features from an input layer.

Numerous enclosing geometry types are supported, including bounding boxes (envelopes), oriented rectangles, circles and convex hulls.

Optionally, the features can be grouped by a field. If set, this causes the output layer to contain one feature per grouped value with a minimal geometry covering just the features with matching values.

| Parameter | Type   | Required | Default | Description                                                                                                                 |
|-----------|--------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                                                                                                 |
| FIELD     | field  | no       | —       | Field (optional, set if features should be grouped by class)                                                                |
| TYPE      | enum   | yes      | —       | Geometry type — options: 0=Envelope (Bounding Box), 1=Minimum Oriented Rectangle, 2=Minimum Enclosing Circle, 3=Convex Hull |
| OUTPUT    | sink   | yes      | —       | Bounding geometry                                                                                                           |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:minimumboundinggeometry INPUT=input.gpkg TYPE=0 FIELD=field1 OUTPUT=output.gpkg
```

This calls **Minimum bounding geometry**: INPUT=input.gpkg — Input layer; TYPE=0 selects *Envelope (Bounding Box)*; FIELD=field1 — Field (optional, set if features should be grouped by class); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:minimumenclosingcircle — Minimum enclosing circles

**group:** Vector geometry

This algorithm calculates the minimum enclosing circle which covers each feature in an input layer.

See the 'Minimum bounding geometry' algorithm for a minimal enclosing circle calculation which covers the whole layer or grouped subsets of features.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |

| Parameter | Type   | Required | Default | Description                   |
|-----------|--------|----------|---------|-------------------------------|
| SEGMENTS  | number | no       | 72      | Number of segments in circles |
| OUTPUT    | sink   | yes      | —       | Minimum enclosing circles     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:minimumenclosingcircle INPUT=input.gpkg SEGMENTS=72 OUTPUT=output.gpkg
```

This calls **Minimum enclosing circles**: INPUT=input.gpkg — Input layer; SEGMENTS=72 — Number of segments in circles; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:modellerrastercalc — Raster calculator

**group:** Raster analysis

This algorithm performs algebraic operations using raster layers.

| Parameter                   | Type              | Required | Default | Description                                         |
|-----------------------------|-------------------|----------|---------|-----------------------------------------------------|
| LAYERS                      | multilayer        | yes      | —       | Input layers                                        |
| EXPRESSION                  | expression        | yes      | —       | Expression                                          |
| EXTENT                      | extent            | no       | —       | Output extent                                       |
| CELL_SIZE                   | number            | no       | —       | Output cell size (leave empty to set automatically) |
| CRS                         | crs               | no       | —       | Output CRS                                          |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Creation options                                    |
| OUTPUT                      | rasterDestination | yes      | —       | Calculated                                          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:modellerrastercalc LAYERS="a.gpkg;b.gpkg" EXPRESSION="value > 0"
↳ EXTENT=0,1000,0,1000 CELL_SIZE=10 CRS=EPSG:6346 OUTPUT=output.tif
```

This calls **Raster calculator**: LAYERS="a.gpkg;b.gpkg" — Input layers; EXPRESSION="value > 0" — Expression; EXTENT=0,1000,0,1000 — Output extent; CELL\_SIZE=10 — Output cell size (leave empty to set automatically); CRS=EPSG:6346 — Output CRS; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:modellervirtualrastercalc — Raster calculator (virtual)

**group:** Raster analysis

This algorithm performs algebraic operations using raster layers and generates in-memory result.

| Parameter  | Type       | Required | Default | Description                                         |
|------------|------------|----------|---------|-----------------------------------------------------|
| LAYERS     | multilayer | yes      | —       | Input layers                                        |
| EXPRESSION | expression | yes      | —       | Expression                                          |
| EXTENT     | extent     | no       | —       | Output extent                                       |
| CELL_SIZE  | number     | no       | —       | Output cell size (leave empty to set automatically) |
| CRS        | crs        | no       | —       | Output CRS                                          |
| LAYER_NAME | string     | no       | —       | Output layer name                                   |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:modellervirtualrastercalc LAYERS="a.gpkg;b.gpkg" EXPRESSION="value > 0"
↳ EXTENT=0,1000,0,1000 CELL_SIZE=10 CRS=EPSG:6346 LAYER_NAME=value
```

This calls **Raster calculator (virtual)**: LAYERS="a.gpkg;b.gpkg" — Input layers; EXPRESSION="value > 0" — Expression; EXTENT=0,1000,0,1000 — Output extent; CELL\_SIZE=10 — Output cell size (leave empty to set automatically); CRS=EPSG:6346 — Output CRS; LAYER\_NAME=value — Output layer name. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:multidifference — Difference (multiple)

**group:** Vector overlay

This algorithm extracts features from the Input layer that fall completely outside or only partially overlap the features from any of the Overlay layer(s). For each overlay layer the difference is calculated between the result of all previous difference operations and this overlay layer. Input layer features that partially overlap feature(s) in the Overlay layers are split along those features' boundary and only the portions outside the Overlay layer features are retained.

Attributes are not modified, although properties such as area or length of the features will be modified by the difference operation. If such properties are stored as attributes, those attributes will have to be manually updated.

| Parameter | Type       | Required | Default | Description    |
|-----------|------------|----------|---------|----------------|
| INPUT     | source     | yes      | —       | Input layer    |
| OVERLAYS  | multilayer | yes      | —       | Overlay layers |
| OUTPUT    | sink       | yes      | —       | Difference     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:multidifference INPUT=input.gpkg OVERLAYS="a.gpkg;b.gpkg" OUTPUT=output.gpkg
```

This calls **Difference (multiple)**: INPUT=input.gpkg — Input layer; OVERLAYS="a.gpkg;b.gpkg" — Overlay layers; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:multiintersection — Intersection (multiple)

**group:** Vector overlay

This algorithm extracts the overlapping portions of features in the Input and all Overlay layers. Features in the output layer are assigned the attributes of the overlapping features from both the Input and Overlay layers.

| Parameter                      | Type       | Required | Default | Description           |
|--------------------------------|------------|----------|---------|-----------------------|
| INPUT                          | source     | yes      | —       | Input layer           |
| OVERLAYS                       | multilayer | yes      | —       | Overlay layers        |
| OVERLAY_FIELDS_PREFIX *(adv) * | string     | no       | —       | Overlay fields prefix |
| OUTPUT                         | sink       | yes      | —       | Intersection          |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:multiintersection INPUT=input.gpkg OVERLAYS="a.gpkg;b.gpkg" OUTPUT=output.gpkg
```

This calls **Intersection (multiple)**: INPUT=input.gpkg — Input layer; OVERLAYS="a.gpkg;b.gpkg" — Overlay layers; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:multiparttosingleparts — Multipart to singleparts ★

**group:** Vector geometry · **niva verb:** explode

This algorithm takes a vector layer with multipart geometries and generates a new one in which all geometries contain a single part. Features with multipart geometries are divided in as many different features as parts the geometry contain, and the same attributes are used for each of them.

| Parameter | Type   | Required | Default | Description  |
|-----------|--------|----------|---------|--------------|
| INPUT     | source | yes      | —       | Input layer  |
| OUTPUT    | sink   | yes      | —       | Single parts |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:multiparttosingleparts INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Multipart to singleparts**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:multiringconstantbuffer — Multi-ring buffer (constant distance)

**group:** Vector geometry

This algorithm computes multi-ring ('donuts') buffer for the features in an input layer, using a fixed or dynamic distance and number of rings.

| Parameter | Type     | Required | Default | Description                           |
|-----------|----------|----------|---------|---------------------------------------|
| INPUT     | source   | yes      | —       | Input layer                           |
| RINGS     | number   | no       | 1       | Number of rings                       |
| DISTANCE  | distance | no       | 1       | Distance between rings                |
| OUTPUT    | sink     | yes      | —       | Multi-ring buffer (constant distance) |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:multiringconstantbuffer INPUT=input.gpkg RINGS=1 DISTANCE=1 OUTPUT=output.gpkg
```

This calls **Multi-ring buffer (constant distance)**: INPUT=input.gpkg — Input layer; RINGS=1 — Number of rings; DISTANCE=1 — Distance between rings; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:multiunion — Union (multiple)

**group:** Vector overlay

This algorithm checks overlaps between features within the Input layer and creates separate features for overlapping and non-overlapping parts. The area of overlap will create as many identical overlapping features as there are features that participate in that overlap.

Multiple Overlay layers can also be used, in which case features from each layer are split at their overlap with features from all other layers, creating a layer containing all the portions from both Input and Overlay layers. The attribute table of the Union layer is filled with attribute values from the respective original layer for non-overlapping features, and attribute values from both layers for overlapping features.

| Parameter                     | Type       | Required | Default | Description           |
|-------------------------------|------------|----------|---------|-----------------------|
| INPUT                         | source     | yes      | —       | Input layer           |
| OVERLAYS                      | multilayer | no       | —       | Overlay layers        |
| OVERLAY_FIELDS_PREFIX *(adv)* | string     | no       | —       | Overlay fields prefix |
| OUTPUT                        | sink       | yes      | —       | Union                 |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:multiunion INPUT=input.gpkg OVERLAYS="a.gpkg;b.gpkg" OUTPUT=output.gpkg
```

This calls **Union (multiple)**: INPUT=input.gpkg — Input layer; OVERLAYS="a.gpkg;b.gpkg" — Overlay layers; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:nearestneighbouranalysis — Nearest neighbour analysis

**group:** Vector analysis

This algorithm performs nearest neighbor analysis for a point layer.

The output describes how the data are distributed (clustered, randomly or distributed).

Output is generated as an HTML file with the computed statistical values.

| Parameter        | Type            | Required | Default | Description       |
|------------------|-----------------|----------|---------|-------------------|
| INPUT            | source          | yes      | —       | Input layer       |
| OUTPUT_HTML_FILE | fileDestination | no       | —       | Nearest neighbour |

**Outputs:** OUTPUT\_HTML\_FILE (outputHtml), OBSERVED\_MD (outputNumber), EXPECTED\_MD (outputNumber), NN\_INDEX (outputNumber), POINT\_COUNT (outputNumber), Z\_SCORE (outputNumber)

#### Example usage

```
run native:nearestneighbouranalysis INPUT=input.gpkg OUTPUT_HTML_FILE=output_html_file.txt
```

This calls **Nearest neighbour analysis**: INPUT=input.gpkg — Input layer; OUTPUT\_HTML\_FILE=output\_html\_file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:offsetline — Offset lines ★

**group:** Vector geometry · **niva verb:** offset

This algorithm offsets lines by a specified distance. Positive distances will offset lines to the left, and negative distances will offset to the right of lines.

The segments parameter controls the number of line segments to use to approximate a quarter circle when creating rounded offsets.

The join style parameter specifies whether round, miter or beveled joins should be used when offsetting corners in a line.

The miter limit parameter is only applicable for miter join styles, and controls the maximum distance from the offset curve to use when creating a mitered join.

| Parameter | Type     | Required | Default | Description |
|-----------|----------|----------|---------|-------------|
| INPUT     | source   | yes      | —       | Input layer |
| DISTANCE  | distance | no       | 10.0    | Distance    |
| SEGMENTS  | number   | no       | 8       | Segments    |



| Parameter   | Type   | Required | Default | Description                                     |
|-------------|--------|----------|---------|-------------------------------------------------|
| JOIN_STYLE  | enum   | no       | 0       | Join style — options: 0=Round, 1=Miter, 2=Bevel |
| MITER_LIMIT | number | no       | 2       | Miter limit                                     |
| OUTPUT      | sink   | yes      | —       | Offset                                          |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:offsetline INPUT=input.gpkg DISTANCE=10 SEGMENTS=8 JOIN_STYLE=0 MITER_LIMIT=2
↳ OUTPUT=output.gpkg
```

This calls **Offset lines**: INPUT=input.gpkg — Input layer; DISTANCE=10 — Distance; SEGMENTS=8 — Segments; JOIN\_STYLE=0 selects *Round*; MITER\_LIMIT=2 — Miter limit; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:openurl — Open file or URL

**group:** File tools

This algorithm opens files in their default associated application, or URLs in the user's default web browser.

| Parameter | Type   | Required | Default | Description      |
|-----------|--------|----------|---------|------------------|
| URL       | string | yes      | —       | URL or file path |

**Outputs:** SUCCESS (outputBoolean)

#### Example usage

```
run native:openurl URL=value
```

This calls **Open file or URL**: URL=value — URL or file path. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:orderbyexpression — Order by expression

**group:** Vector general

This algorithm sorts a vector layer according to an expression. Be careful, it might not work as expected with some providers, the order might not be kept every time.

For help with QGIS expression functions, see the inbuilt help for specific functions which is available in the expression builder.

| Parameter   | Type       | Required | Default | Description      |
|-------------|------------|----------|---------|------------------|
| INPUT       | source     | yes      | —       | Input layer      |
| EXPRESSION  | expression | yes      | —       | Expression       |
| ASCENDING   | boolean    | no       | True    | Sort ascending   |
| NULLS_FIRST | boolean    | no       | False   | Sort nulls first |
| OUTPUT      | sink       | yes      | —       | Ordered          |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:orderbyexpression INPUT=input.gpkg EXPRESSION="value > 0" ASCENDING=true
↳ NULLS_FIRST=false OUTPUT=output.gpkg
```

This calls **Order by expression**: INPUT=input.gpkg — Input layer; EXPRESSION="value > 0" — Expression; ASCENDING=true — Sort ascending; NULLS\_FIRST=false — Sort nulls first; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:orientedminimumboundingbox — Oriented minimum bounding box ★

**group:** Vector geometry · **niva verb:** minrect

This algorithm calculates the minimum area rotated rectangle which covers each feature in an input layer.

For singlepoint point features, the output corresponds to the bounding box of the geometry.

See the 'Minimum bounding geometry' algorithm for an oriented bounding box calculation which covers the whole layer or grouped subsets of features.

| Parameter | Type   | Required | Default | Description    |
|-----------|--------|----------|---------|----------------|
| INPUT     | source | yes      | —       | Input layer    |
| OUTPUT    | sink   | yes      | —       | Bounding boxes |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:orientedminimumboundingbox INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Oriented minimum bounding box**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:orthogonalize — Orthogonalize

**group:** Vector geometry

Takes a line or polygon layer and attempts to orthogonalize all the geometries in the layer. This process shifts the nodes in the geometries to try to make every angle in the geometry either a right angle or a straight line.

The angle tolerance parameter is used to specify the maximum deviation from a right angle or straight line a node can have for it to be adjusted. Smaller tolerances mean that only nodes which are already closer to right angles will be adjusted, and larger tolerances mean that nodes which deviate further from right angles will also be adjusted.

The algorithm is iterative. Setting a larger number for the maximum iterations will result in a more orthogonal geometry at the cost of extra processing time.

| Parameter              | Type   | Required | Default | Description                       |
|------------------------|--------|----------|---------|-----------------------------------|
| INPUT                  | source | yes      | —       | Input layer                       |
| ANGLE_TOLERANCE        | number | no       | 15.0    | Maximum angle tolerance (degrees) |
| MAX_ITERATIONS *(adv)* | number | no       | 1000    | Maximum algorithm iterations      |
| OUTPUT                 | sink   | yes      | —       | Orthogonalized                    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:orthogonalize INPUT=input.gpkg ANGLE_TOLERANCE=15 OUTPUT=output.gpkg
```

This calls **Orthogonalize**: INPUT=input.gpkg — Input layer; ANGLE\_TOLERANCE=15 — Maximum angle tolerance (degrees); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:package — Package layers

**group:** Database

This algorithm collects a number of existing layers and packages them together into a single GeoPackage database.

| Parameter              | Type            | Required | Default | Description                                                      |
|------------------------|-----------------|----------|---------|------------------------------------------------------------------|
| LAYERS                 | multilayer      | yes      | —       | Input layers                                                     |
| OUTPUT                 | fileDestination | yes      | —       | Destination GeoPackage                                           |
| OVERWRITE              | boolean         | no       | False   | Overwrite existing GeoPackage                                    |
| SAVE_STYLES            | boolean         | no       | True    | Save layer styles into GeoPackage                                |
| SAVE_METADATA          | boolean         | no       | True    | Save layer metadata into GeoPackage                              |
| SELECTED_FEATURES_ONLY | boolean         | no       | False   | Save only selected features                                      |
| EXPORT_RELATED_LAYERS  | boolean         | no       | False   | Export related layers following relations defined in the project |
| EXTENT                 | extent          | no       | —       | Extent                                                           |
| CRS *(adv)*            | crs             | no       | —       | Destination CRS                                                  |

**Outputs:** OUTPUT (outputFile), OUTPUT\_LAYERS (outputMultilayer)

### Example usage

```
run native:package LAYERS="a.gpkg;b.gpkg" OVERWRITE=false SAVE_STYLES=true
↳ SAVE_METADATA=true SELECTED_FEATURES_ONLY=false EXPORT_RELATED_LAYERS=false
↳ EXTENT=0,1000,0,1000 OUTPUT=output.txt
```

This calls **Package layers**: LAYERS="a.gpkg;b.gpkg" — Input layers; OVERWRITE=false — Overwrite existing GeoPackage; SAVE\_STYLES=true — Save layer styles into GeoPackage; SAVE\_METADATA=true — Save layer metadata into GeoPackage; SELECTED\_FEATURES\_ONLY=false — Save only selected features; EXPORT\_RELATED\_LAYERS=false — Export related layers following relations defined in the project; EXTENT=0,1000,0,1000 — Extent; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:pixelstopoints — Raster pixels to points

**group:** Vector creation

This algorithm converts a raster layer to a vector layer, by creating point features for each individual pixel's center in the raster layer.

Any NoData pixels are skipped in the output.

| Parameter    | Type   | Required | Default | Description   |
|--------------|--------|----------|---------|---------------|
| INPUT_RASTER | raster | yes      | —       | Raster layer  |
| RASTER_BAND  | band   | no       | 1       | Band number   |
| FIELD_NAME   | string | no       | VALUE   | Field name    |
| OUTPUT       | sink   | yes      | —       | Vector points |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:pixelstopoints INPUT_RASTER=dem.tif RASTER_BAND=1 FIELD_NAME=VALUE
↳ OUTPUT=output.gpkg
```

This calls **Raster pixels to points**: INPUT\_RASTER=dem.tif — Raster layer; RASTER\_BAND=1 — Band number; FIELD\_NAME=VALUE — Field name; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:pixelstopolygons — Raster pixels to polygons

**group:** Vector creation

This algorithm converts a raster layer to a vector layer, by creating polygon features for each individual pixel's extent in the raster layer.

Any NoData pixels are skipped in the output.

| Parameter    | Type   | Required | Default | Description     |
|--------------|--------|----------|---------|-----------------|
| INPUT_RASTER | raster | yes      | —       | Raster layer    |
| RASTER_BAND  | band   | no       | 1       | Band number     |
| FIELD_NAME   | string | no       | VALUE   | Field name      |
| OUTPUT       | sink   | yes      | —       | Vector polygons |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:pixelstopolygons INPUT_RASTER=dem.tif RASTER_BAND=1 FIELD_NAME=VALUE
↳ OUTPUT=output.gpkg
```

This calls **Raster pixels to polygons**: INPUT\_RASTER=dem.tif — Raster layer; RASTER\_BAND=1 — Band number; FIELD\_NAME=VALUE — Field name; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:pointonsurface — Point on surface ★

**group:** Vector geometry · **niva verb:** pointonsurface

This algorithm returns a point guaranteed to lie on the surface of a geometry.

| Parameter | Type    | Required | Default | Description                           |
|-----------|---------|----------|---------|---------------------------------------|
| INPUT     | source  | yes      | —       | Input layer                           |
| ALL_PARTS | boolean | no       | False   | Create point on surface for each part |
| OUTPUT    | sink    | yes      | —       | Point                                 |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:pointonsurface INPUT=input.gpkg ALL_PARTS=false OUTPUT=output.gpkg
```

This calls **Point on surface**: INPUT=input.gpkg — Input layer; ALL\_PARTS=false — Create point on surface for each part; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:pointsalonglines — Points along geometry ★

**group:** Vector geometry · **niva verb:** pointsalong

This algorithm creates a points layer, with points distributed along the lines of an input vector layer. The distance between points (measured along the line) is defined as a parameter.

Start and end offset distances can be defined, so the first and last point will not fall exactly on the line's first and last nodes. These start and end offsets are defined as distances, measured along the line from the first and last nodes of the lines.

| Parameter    | Type     | Required | Default | Description         |
|--------------|----------|----------|---------|---------------------|
| INPUT        | source   | yes      | —       | Input layer         |
| DISTANCE     | distance | no       | 1.0     | Distance            |
| START_OFFSET | distance | no       | 0.0     | Start offset        |
| END_OFFSET   | distance | no       | 0.0     | End offset          |
| OUTPUT       | sink     | yes      | —       | Interpolated points |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:pointsalonglines INPUT=input.gpkg DISTANCE=1 START_OFFSET=0 END_OFFSET=0  
↪ OUTPUT=output.gpkg
```

This calls **Points along geometry**: INPUT=input.gpkg — Input layer; DISTANCE=1 — Distance; START\_OFFSET=0 — Start offset; END\_OFFSET=0 — End offset; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:pointstopath — Points to path

**group:** Vector creation

This algorithm takes a point layer and connects its features to create a new line layer.

An attribute or expression may be specified to define the order the points should be connected. If no order expression is specified, the feature ID is used.

A natural sort can be used when sorting by a string attribute or expression (ie. place 'a9' before 'a10').

An attribute or expression can be selected to group points having the same value into the same resulting line.

| Parameter        | Type              | Required | Default | Description                              |
|------------------|-------------------|----------|---------|------------------------------------------|
| INPUT            | source            | yes      | —       | Input layer                              |
| CLOSE_PATH       | boolean           | no       | False   | Create closed paths                      |
| ORDER_EXPRESSION | expression        | no       | —       | Order expression                         |
| NATURAL_SORT     | boolean           | no       | False   | Sort text containing numbers naturally   |
| GROUP_EXPRESSION | expression        | no       | —       | Path group expression                    |
| OUTPUT           | sink              | yes      | —       | Paths                                    |
| OUTPUT_TEXT_DIR  | folderDestination | no       | —       | Directory for text output                |
| ORDER_FIELD      | field             | no       | —       | Order field                              |
| GROUP_FIELD      | field             | no       | —       | Group field                              |
| DATE_FORMAT      | string            | no       | —       | Date format (if order field is DateTime) |

**Outputs:** OUTPUT (outputVector), OUTPUT\_TEXT\_DIR (outputFolder), NUM\_PATHS (outputNumber)

### Example usage

```
run native:pointstopath INPUT=input.gpkg CLOSE_PATH=false ORDER_EXPRESSION="value > 0"
↳ NATURAL_SORT=false GROUP_EXPRESSION="value > 0" ORDER_FIELD=field1 GROUP_FIELD=field1
↳ OUTPUT=output.gpkg
```

This calls **Points to path**: INPUT=input.gpkg — Input layer; CLOSE\_PATH=false — Create closed paths; ORDER\_EXPRESSION="value > 0" — Order expression; NATURAL\_SORT=false — Sort text containing numbers naturally; GROUP\_EXPRESSION="value > 0" — Path group expression; ORDER\_FIELD=field1 — Order field; GROUP\_FIELD=field1 — Group field; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:pointtolayer — Create layer from point

**group:** Vector creation

This algorithm creates a new vector layer that contains a single feature with geometry matching a point parameter.

It can be used in models to convert a point into a layer which can be used for other algorithms which require a layer based input.

| Parameter | Type  | Required | Default | Description |
|-----------|-------|----------|---------|-------------|
| INPUT     | point | yes      | —       | Point       |
| OUTPUT    | sink  | yes      | —       | Point       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:pointtolayer INPUT=100,100 OUTPUT=output.gpkg
```

This calls **Create layer from point**: INPUT=100,100 — Point; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:poleofinaccessibility — Pole of inaccessibility

**group:** Vector geometry

This algorithm calculates the pole of inaccessibility for a polygon layer, which is the most distant internal point from the boundary of the surface. This algorithm uses the 'polylabel' algorithm (Vladimir Agafonkin, 2016), which is an iterative approach guaranteed to find the true pole of inaccessibility within a specified tolerance (in layer units). More precise tolerances require more iterations and will take longer to calculate.

The distance from the calculated pole to the polygon boundary will be stored as a new attribute in the output layer.

| Parameter | Type     | Required | Default | Description |
|-----------|----------|----------|---------|-------------|
| INPUT     | source   | yes      | —       | Input layer |
| TOLERANCE | distance | no       | 1.0     | Tolerance   |
| OUTPUT    | sink     | yes      | —       | Point       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:poleofinaccessibility INPUT=input.gpkg TOLERANCE=1 OUTPUT=output.gpkg
```

This calls **Pole of inaccessibility**: INPUT=input.gpkg — Input layer; TOLERANCE=1 — Tolerance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:polygonfromlayerextent — Extract layer extent

**group:** Layer tools

This algorithm takes a map layer and generates a new vector layer with the minimum bounding box (rectangle polygon with N-S orientation) that covers the input layer. Optionally, the extent can be enlarged to a rounded value.

| Parameter        | Type     | Required | Default | Description     |
|------------------|----------|----------|---------|-----------------|
| INPUT            | layer    | yes      | —       | Input layer     |
| ROUND_TO *(adv)* | distance | no       | 0       | Round values to |
| OUTPUT           | sink     | yes      | —       | Extent          |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:polygonfromlayerextent INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Extract layer extent**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:polygonize — Polygonize

**group:** Vector geometry

This algorithm creates a polygon layer from the input lines layer.

| Parameter   | Type    | Required | Default | Description                        |
|-------------|---------|----------|---------|------------------------------------|
| INPUT       | source  | yes      | —       | Input layer                        |
| KEEP_FIELDS | boolean | no       | False   | Keep table structure of line layer |
| OUTPUT      | sink    | yes      | —       | Polygons                           |

**Outputs:** OUTPUT (outputVector), NUM\_POLYGONS (outputNumber)

### Example usage

```
run native:polygonize INPUT=input.gpkg KEEP_FIELDS=false OUTPUT=output.gpkg
```

This calls **Polygonize**: INPUT=input.gpkg — Input layer; KEEP\_FIELDS=false — Keep table structure of line layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:polygonstolines — Polygons to lines

**group:** Vector geometry

This algorithm converts polygons to lines.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Lines       |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:polygonstolines INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Polygons to lines**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:postgisexecuteandloadsql — PostgreSQL execute and load SQL**

**group:** Database

This algorithm performs a SQL database query on a PostgreSQL database connected to QGIS and loads the query results as a new layer.

| Parameter      | Type               | Required | Default | Description                |
|----------------|--------------------|----------|---------|----------------------------|
| DATABASE       | providerconnection | yes      | —       | Database (connection name) |
| SQL            | string             | yes      | —       | SQL query                  |
| ID_FIELD       | string             | no       | id      | Unique ID field name       |
| GEOMETRY_FIELD | string             | no       | geom    | Geometry field name        |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:postgisexecuteandloadsql DATABASE=... SQL=value ID_FIELD=id GEOMETRY_FIELD=geom
```

This calls **PostgreSQL execute and load SQL**: DATABASE=... — Database (connection name); SQL=value — SQL query; ID\_FIELD=id — Unique ID field name; GEOMETRY\_FIELD=geom — Geometry field name. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:postgisexecutesql — PostgreSQL execute SQL**

**group:** Database

This algorithm executes a SQL command on a PostgreSQL database.

| Parameter | Type               | Required | Default | Description                |
|-----------|--------------------|----------|---------|----------------------------|
| DATABASE  | providerconnection | yes      | —       | Database (connection name) |
| SQL       | string             | yes      | —       | SQL query                  |

**Example usage**

```
run native:postgisexecutesql DATABASE=... SQL=value
```

This calls **PostgreSQL execute SQL**: DATABASE=... — Database (connection name); SQL=value — SQL query. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## **native:printlayoutmapextenttolayer — Print layout map extent to layer**

**group:** Cartography

This algorithm creates a polygon layer containing the extent of a print layout map item (or items), with attributes specifying the map size (in layout units), scale and rotation.

If the map item parameter is specified, then only the matching map extent will be exported. If it is not specified, all map extents from the layout will be exported.

Optionally, a specific output CRS can be specified. If it is not specified, the original map item CRS will be used.

| Parameter   | Type       | Required | Default | Description  |
|-------------|------------|----------|---------|--------------|
| LAYOUT      | layout     | yes      | —       | Print layout |
| MAP         | layoutitem | no       | —       | Map item     |
| CRS *(adv)* | crs        | no       | —       | Override CRS |
| OUTPUT      | sink       | yes      | —       | Extent       |

**Outputs:** OUTPUT (outputVector), WIDTH (outputNumber), HEIGHT (outputNumber), SCALE (outputNumber), ROTATION (outputNumber)

### **Example usage**

```
run native:printlayoutmapextenttolayer LAYOUT=... OUTPUT=output.gpkg
```

This calls **Print layout map extent to layer**: LAYOUT=... — Print layout; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:printlayouttoimage — Export print layout as image**

**group:** Cartography

This algorithm outputs a print layout as an image file (e.g. PNG or JPEG images).

| Parameter                | Type            | Required | Default | Description                                  |
|--------------------------|-----------------|----------|---------|----------------------------------------------|
| LAYOUT                   | layout          | yes      | —       | Print layout                                 |
| LAYERS *(adv)*           | multilayer      | no       | —       | Map layers to assign to unlocked map item(s) |
| DPI *(adv)*              | number          | no       | —       | DPI (leave blank for default layout DPI)     |
| GEOREFERENCE *(adv)*     | boolean         | no       | True    | Generate world file                          |
| INCLUDE_METADATA *(adv)* | boolean         | no       | True    | Export RDF metadata (title, author, etc.)    |
| ANTIALIAS *(adv)*        | boolean         | no       | True    | Enable antialiasing                          |
| OUTPUT                   | fileDestination | yes      | —       | Image file                                   |

**Outputs:** OUTPUT (outputFile)

### **Example usage**

```
run native:printlayouttoimage LAYOUT=... OUTPUT=output.txt
```

This calls **Export print layout as image**: LAYOUT=... — Print layout; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:printlayouttopdf — Export print layout as PDF**

**group:** Cartography

This algorithm outputs a print layout as a PDF file.

| Parameter                 | Type            | Required | Default | Description                                                                                                                                       |
|---------------------------|-----------------|----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| LAYOUT                    | layout          | yes      | —       | Print layout                                                                                                                                      |
| LAYERS *(adv)*            | multilayer      | no       | —       | Map layers to assign to unlocked map item(s)                                                                                                      |
| DPI *(adv)*               | number          | no       | —       | DPI (leave blank for default layout DPI)                                                                                                          |
| FORCE_VECTOR *(adv)*      | boolean         | no       | False   | Always export as vectors                                                                                                                          |
| FORCE_RASTER *(adv)*      | boolean         | no       | False   | Always export as raster                                                                                                                           |
| GEOREFERENCE *(adv)*      | boolean         | no       | True    | Append georeference information                                                                                                                   |
| INCLUDE_METADATA *(adv)*  | boolean         | no       | True    | Export RDF metadata (title, author, etc.)                                                                                                         |
| DISABLE_TILED *(adv)*     | boolean         | no       | False   | Disable tiled raster layer exports                                                                                                                |
| SIMPLIFY *(adv)*          | boolean         | no       | True    | Simplify geometries to reduce output file size                                                                                                    |
| TEXT_FORMAT *(adv)*       | enum            | no       | 0       | Text export — options: 0=Always Export Text as Paths (Recommended), 1=Always Export Text as Text Objects, 2=Prefer Exporting Text as Text Objects |
| IMAGE_COMPRESSION *(adv)* | enum            | no       | 0       | Image compression — options: 0=Lossy (JPEG), 1=Lossless                                                                                           |
| SEPARATE_LAYERS *(adv)*   | boolean         | no       | False   | Export layers as separate PDF files                                                                                                               |
| OUTPUT                    | fileDestination | yes      | —       | PDF file                                                                                                                                          |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run native:printlayouttopdf LAYOUT=... OUTPUT=output.txt
```

This calls **Export print layout as PDF**: LAYOUT=... — Print layout; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:projectpointcartesian — Project points (Cartesian)

**group:** Vector geometry

This algorithm projects point geometries by a specified distance and bearing (azimuth), creating a new point layer with the projected points.

The distance is specified in layer units, and the bearing in degrees clockwise from North.

| Parameter | Type     | Required | Default | Description                  |
|-----------|----------|----------|---------|------------------------------|
| INPUT     | source   | yes      | —       | Input layer                  |
| BEARING   | number   | no       | 0       | Bearing (degrees from North) |
| DISTANCE  | distance | no       | 1       | Distance                     |
| OUTPUT    | sink     | yes      | —       | Projected                    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:projectpointcartesian INPUT=input.gpkg BEARING=0 DISTANCE=1 OUTPUT=output.gpkg
```

This calls **Project points (Cartesian)**: INPUT=input.gpkg — Input layer; BEARING=0 — Bearing (degrees from North); DISTANCE=1 — Distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:promotetomulti — Promote to multipart ★

**group:** Vector geometry · **niva verb:** promote

This algorithm takes a vector layer with singlepart geometries and generates a new one in which all geometries are multipart. Input features which are already multipart features will remain unchanged.

This algorithm can be used to force geometries to multipart types in order to be compatible with data providers with strict singlepart/multipart compatibility checks.

See the 'Collect geometries' or 'Aggregate' algorithms for alternative options.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Multiparts  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:promotetomulti INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Promote to multipart**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:raiseexception — Raise exception

**group:** Modeler tools

This algorithm raises an exception and cancels a model's execution.

The exception message can be customized, and optionally an expression based condition can be specified. If an expression condition is used, then the exception will only be raised if the expression result is true. A false result indicates that no exception will be raised, and the model execution can continue uninterrupted.

| Parameter | Type       | Required | Default | Description   |
|-----------|------------|----------|---------|---------------|
| MESSAGE   | string     | yes      | —       | Error message |
| CONDITION | expression | no       | —       | Condition     |

### Example usage

```
run native:raiseexception MESSAGE=value CONDITION="value > 0"
```

This calls **Raise exception**: MESSAGE=value — Error message; CONDITION="value > 0" — Condition. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:raisemessage — Raise message

**group:** Modeler tools

This algorithm raises an information message in the log.

The message can be customized, and optionally an expression based condition can be specified. If an expression condition is used, then the message will only be logged if the expression result is true. A false result indicates that no message will be logged.

| Parameter | Type       | Required | Default | Description         |
|-----------|------------|----------|---------|---------------------|
| MESSAGE   | string     | yes      | —       | Information message |
| CONDITION | expression | no       | —       | Condition           |

---

## Example usage

```
run native:raisemessage MESSAGE=value CONDITION="value > 0"
```

This calls **Raise message**: MESSAGE=value — Information message; CONDITION="value > 0" — Condition. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:raisewarning — Raise warning

**group:** Modeler tools

This algorithm raises a warning message in the log.

The warning message can be customized, and optionally an expression based condition can be specified. If an expression condition is used, then the warning will only be logged if the expression result is true. A false result indicates that no warning will be logged.

| Parameter | Type       | Required | Default | Description     |
|-----------|------------|----------|---------|-----------------|
| MESSAGE   | string     | yes      | —       | Warning message |
| CONDITION | expression | no       | —       | Condition       |

## Example usage

```
run native:raisewarning MESSAGE=value CONDITION="value > 0"
```

This calls **Raise warning**: MESSAGE=value — Warning message; CONDITION="value > 0" — Condition. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:randomextract — Random extract ★

**group:** Vector selection · **niva verb:** sample

This algorithm takes a vector layer and generates a new one that contains only a subset of the features in the input layer.

The subset is defined randomly, using a percentage or count value to define the total number of features in the subset.

| Parameter | Type   | Required | Default | Description                                                      |
|-----------|--------|----------|---------|------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                                      |
| METHOD    | enum   | no       | 0       | Method — options: 0=Number of features, 1=Percentage of features |
| NUMBER    | number | no       | 10      | Number/percentage of features                                    |
| OUTPUT    | sink   | yes      | —       | Extracted (random)                                               |

**Outputs:** OUTPUT (outputVector)

## Example usage

```
run native:randomextract INPUT=input.gpkg METHOD=0 NUMBER=10 OUTPUT=output.gpkg
```

This calls **Random extract**: INPUT=input.gpkg — Input layer; METHOD=0 selects *Number of features*; NUMBER=10 — Number/percentage of features; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:randompointsinextent — Random points in extent

**group:** Vector creation

This algorithm creates a new point layer with a given number of random points, all of them within a given extent. A distance factor can be specified, to avoid points being too close to each other. If the minimum distance between points makes it impossible to create new points, either distance can be decreased or the maximum number of attempts may be increased.

| Parameter            | Type     | Required | Default    | Description                                                  |
|----------------------|----------|----------|------------|--------------------------------------------------------------|
| EXTENT               | extent   | yes      | —          | Input extent                                                 |
| POINTS_NUMBER        | number   | no       | 1          | Number of points                                             |
| MIN_DISTANCE         | distance | no       | 0          | Minimum distance between points                              |
| TARGET_CRS           | crs      | no       | ProjectCrs | Target CRS                                                   |
| MAX_ATTEMPTS *(adv)* | number   | no       | 200        | Maximum number of search attempts given the minimum distance |
| OUTPUT               | sink     | yes      | —          | Random points                                                |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:randompointsinextent EXTENT=0,1000,0,1000 POINTS_NUMBER=1 MIN_DISTANCE=0  
↪ TARGET_CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Random points in extent**: EXTENT=0,1000,0,1000 — Input extent; POINTS\_NUMBER=1 — Number of points; MIN\_DISTANCE=0 — Minimum distance between points; TARGET\_CRS=EPSG:6346 — Target CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:randompointsinpolygons — Random points in polygons

**group:** Vector creation

This algorithm creates a point layer, with points placed randomly in the polygons of the Input polygon layer. For each feature in the Input polygon layer, the algorithm attempts to add the specified Number of points for each feature to the output layer. A Minimum distance between points and a Global minimum distance between points can be specified. A point will not be added if there is an already generated point within this (Euclidean) distance from the generated location. With Minimum distance between points, only points in the same polygon feature are considered, while for Global minimum distance between points all previously generated points are considered. If the Global minimum distance between points is set equal to or larger than the (local) Minimum distance between points, the latter has no effect. If the Minimum distance between points is too large, it may not be possible to generate the specified Number of points for each feature, but all the generated points are returned. The Maximum number of attempts per point can be specified. The seed for the random generator can be provided (Random seed - integer, greater than 0). The user can choose not to Include polygon feature attributes in the attributes of the generated point features. The total number of points will be 'number of input features' \* Number of points for each feature if there are no misses. The Number of points for each feature, Minimum distance between points and Maximum number of attempts per point can be data defined. Output from the algorithm: The number of features with an empty or no geometry (FEATURES\_WITH\_EMPTY\_OR\_NO\_GEOMETRY). A point layer containing the random points (OUTPUT). The number of generated features (OUTPUT\_POINTS). The number of missed points (POINTS\_MISSED). The number of features with non-empty geometry and missing points (POLYGONS\_WITH\_MISSED\_POINTS).

| Parameter     | Type   | Required | Default | Description                       |
|---------------|--------|----------|---------|-----------------------------------|
| INPUT         | source | yes      | —       | Input polygon layer               |
| POINTS_NUMBER | number | no       | 1       | Number of points for each feature |

| Parameter                          | Type     | Required | Default | Description                                            |
|------------------------------------|----------|----------|---------|--------------------------------------------------------|
| MIN_DISTANCE                       | distance | no       | 0       | Minimum distance between points                        |
| MIN_DISTANCE_GLOBAL *(adv)*        | distance | no       | 0       | Global minimum distance between points                 |
| MAX_TRIES_PER_POINT *(adv)*        | number   | no       | 10      | Maximum number of search attempts (for Min. dist. > 0) |
| SEED *(adv)*                       | number   | no       | —       | Random seed                                            |
| INCLUDE_POLYGON_ATTRIBUTES *(adv)* | boolean  | no       | True    | Include polygon attributes                             |
| OUTPUT                             | sink     | yes      | —       | Random points in polygons                              |

**Outputs:** OUTPUT (outputVector), OUTPUT\_POINTS (outputNumber), POINTS\_MISSED (outputNumber), POLYGONS\_WITH\_MISSED\_POINTS (outputNumber), FEATURES\_WITH\_EMPTY\_OR\_NO\_GEOMETRY (outputNumber)

### Example usage

```
run native:randompointsinpolygons INPUT=input.gpkg POINTS_NUMBER=1 MIN_DISTANCE=0
↳ OUTPUT=output.gpkg
```

This calls **Random points in polygons**: INPUT=input.gpkg — Input polygon layer; POINTS\_NUMBER=1 — Number of points for each feature; MIN\_DISTANCE=0 — Minimum distance between points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:randompointsonlines — Random points on lines

**group:** Vector creation

This algorithm creates a point layer with points placed randomly on the lines of the Input line layer. The default behavior is that the generated point features inherit the attributes of the line feature on which they were generated. Parameters / options: For each feature in the Input line layer, the algorithm attempts to add the specified Number of points for each feature to the output layer. A Minimum distance between points and a Global minimum distance between points can be specified. A point will not be added if there is an already generated point within this (Euclidean) distance from the generated location. With Minimum distance between points, only points on the same line feature are considered, while for Global minimum distance between points all previously generated points are considered. If the Global minimum distance between points is set larger than the (local) Minimum distance between points, the latter has no effect. If the Minimum distance between points is too large, it may not be possible to generate the specified Number of points for each feature. The Maximum number of attempts per point is only relevant if Minimum distance between points or Global minimum distance between points is greater than 0. The total number of points will be number of input features \* Number of points for each feature if there are no misses and all features have proper geometries. The seed for the random generator can be provided (Random seed - integer, greater than 0). The user can choose not to Include line feature attributes in the generated point features. Output from the algorithm: A point layer containing the random points (OUTPUT). The number of generated features (POINTS\_GENERATED). The number of missed points (POINTS\_MISSED). The number of features with non-empty geometry and missing points (LINES\_WITH\_MISSED\_POINTS). The number of features with an empty or no geometry (LINES\_WITH\_EMPTY\_OR\_NO\_GEOMETRY).

| Parameter                   | Type     | Required | Default | Description                                            |
|-----------------------------|----------|----------|---------|--------------------------------------------------------|
| INPUT                       | source   | yes      | —       | Input line layer                                       |
| POINTS_NUMBER               | number   | no       | 1       | Number of points for each feature                      |
| MIN_DISTANCE                | distance | no       | 0       | Minimum distance between points                        |
| MIN_DISTANCE_GLOBAL *(adv)* | distance | no       | 0       | Global minimum distance between points                 |
| MAX_TRIES_PER_POINT *(adv)* | number   | no       | 10      | Maximum number of search attempts (for Min. dist. > 0) |
| SEED *(adv)*                | number   | no       | —       | Random seed                                            |

| Parameter                          | Type    | Required | Default | Description             |
|------------------------------------|---------|----------|---------|-------------------------|
| INCLUDE_LINE_ATTRIBUTES<br>*(adv)* | boolean | no       | True    | Include line attributes |
| OUTPUT                             | sink    | yes      | —       | Random points on lines  |

**Outputs:** OUTPUT (outputVector), OUTPUT\_POINTS (outputNumber), POINTS\_MISSED (outputNumber), LINES\_WITH\_MISSED\_POINTS (outputNumber), FEATURES\_WITH\_EMPTY\_OR\_NO\_GEOMETRY (outputNumber)

### Example usage

```
run native:randompointsonlines INPUT=input.gpkg POINTS_NUMBER=1 MIN_DISTANCE=0
↳ OUTPUT=output.gpkg
```

This calls **Random points on lines**: INPUT=input.gpkg — Input line layer; POINTS\_NUMBER=1 — Number of points for each feature; MIN\_DISTANCE=0 — Minimum distance between points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:randomselection — Random selection

**group:** Vector selection

This algorithm takes a vector layer and selects a subset of its features. No new layer is generated by this algorithm.

The subset is defined randomly, using a percentage or count value to define the total number of features in the subset.

| Parameter | Type   | Required | Default | Description                                                      |
|-----------|--------|----------|---------|------------------------------------------------------------------|
| INPUT     | vector | yes      | —       | Input layer                                                      |
| METHOD    | enum   | no       | 0       | Method — options: 0=Number of features, 1=Percentage of features |
| NUMBER    | number | no       | 10      | Number/percentage of features                                    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:randomselection INPUT=input.gpkg METHOD=0 NUMBER=10
```

This calls **Random selection**: INPUT=input.gpkg — Input layer; METHOD=0 selects *Number of features*; NUMBER=10 — Number/percentage of features. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterbooleanand — Raster boolean AND

**group:** Raster analysis

This algorithm calculates the boolean AND for a set of input rasters. If all of the input rasters have a non-zero value for a pixel, that pixel will be set to 1 in the output raster. If any of the input rasters have 0 values for the pixel it will be set to 0 in the output raster.

The reference layer parameter specifies an existing raster layer to use as a reference when creating the output raster. The output raster will have the same extent, CRS, and pixel dimensions as this layer.

By default, a NoData pixel in ANY of the input layers will result in a NoData pixel in the output raster. If the 'Treat NoData values as false' option is checked, then NoData inputs will be treated the same as a 0 input value.

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | multilayer        | yes      | —       | Input layers                                                                                                                                         |
| REF_LAYER                | raster            | yes      | —       | Reference layer                                                                                                                                      |
| NODATA_AS_FALSE          | boolean           | no       | False   | Treat NoData values as false                                                                                                                         |
| NO_DATA *(adv)*          | number            | no       | -9999   | Output NoData value                                                                                                                                  |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=Int32, 4=UInt32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| CREATE_OPTIONS           | string            | no       | —       | Creation options                                                                                                                                     |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options                                                                                                                                     |
| OUTPUT                   | rasterDestination | yes      | —       | Output layer                                                                                                                                         |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), NODATA\_PIXEL\_COUNT (outputNumber), TRUE\_PIXEL\_COUNT (outputNumber), FALSE\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:rasterbooleanand INPUT="a.gpkg;b.gpkg" REF_LAYER=dem.tif NODATA_AS_FALSE=false
↳ CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Raster boolean AND**: INPUT="a.gpkg;b.gpkg" — Input layers; REF\_LAYER=dem.tif — Reference layer; NODATA\_AS\_FALSE=false — Treat NoData values as false; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rastercalc — Raster calculator

**group:** Raster analysis

This algorithm performs algebraic operations using raster layers.

| Parameter                | Type              | Required | Default | Description                                         |
|--------------------------|-------------------|----------|---------|-----------------------------------------------------|
| LAYERS                   | multilayer        | yes      | —       | Input layers                                        |
| EXPRESSION               | expression        | yes      | —       | Expression                                          |
| EXTENT                   | extent            | no       | —       | Output extent                                       |
| CELL_SIZE                | number            | no       | —       | Output cell size (leave empty to set automatically) |
| CRS                      | crs               | no       | —       | Output CRS                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options                                    |
| OUTPUT                   | rasterDestination | yes      | —       | Calculated                                          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:rastercalc LAYERS="a.gpkg;b.gpkg" EXPRESSION="value > 0" EXTENT=0,1000,0,1000
↳ CELL_SIZE=10 CRS=EPSG:6346 OUTPUT=output.tif
```

This calls **Raster calculator**: LAYERS="a.gpkg;b.gpkg" — Input layers; EXPRESSION="value > 0" — Expression; EXTENT=0,1000,0,1000 — Output extent; CELL\_SIZE=10 — Output cell size (leave empty to set automatically); CRS=EPSG:6346 — Output CRS; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## native:rasterfeaturepreservingsmoothing — Feature preserving DEM smoothing

**group:** Raster analysis

This algorithm applies the Feature-Preserving DEM Smoothing (FPDEMS) method, as described by Lindsay et al. (2019).

It is effective at removing surface roughness from Digital Elevation Models (DEMs) without significantly altering sharp features such as breaks-in-slope, stream banks, or terrace scarps. This makes it superior to standard low-pass filters (e.g., mean, median, Gaussian) or resampling, which often blur distinct topographic features.

The algorithm works in three steps:

1. Calculating surface normal 3D vectors for each grid cell.
2. Smoothing the normal vector field using a filter that applies more weight to neighbors with similar surface normals (preserving edges).
3. Iteratively updating the elevations in the DEM to match the smoothed normal field.

References: Lindsay, John et al (2019): LiDAR DEM Smoothing and the Preservation of Drainage Features. Remote Sensing, Vol. 11, Issue 16, <https://doi.org/10.3390/rs11161926>

| Parameter                | Type              | Required | Default | Description                           |
|--------------------------|-------------------|----------|---------|---------------------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                           |
| BAND                     | band              | no       | 1       | Band number                           |
| RADIUS                   | number            | no       | 5       | Filter radius (pixels)                |
| THRESHOLD                | number            | no       | 15.0    | Normal difference threshold (degrees) |
| ITERATIONS               | number            | no       | 3       | Elevation update iterations           |
| MAX_ELEVATION_CHANGE     | number            | no       | —       | Maximum elevation change              |
| Z_FACTOR *(adv)*         | number            | no       | 1.0     | Z factor                              |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options                      |
| OUTPUT                   | rasterDestination | yes      | —       | Output layer                          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:rasterfeaturepreservingsmoothing INPUT=dem.tif BAND=1 RADIUS=5 THRESHOLD=15  
↪ ITERATIONS=3 MAX_ELEVATION_CHANGE=10 OUTPUT=output.tif
```

This calls **Feature preserving DEM smoothing**: INPUT=dem.tif — Input layer; BAND=1 — Band number; RADIUS=5 — Filter radius (pixels); THRESHOLD=15 — Normal difference threshold (degrees); ITERATIONS=3 — Elevation update iterations; MAX\_ELEVATION\_CHANGE=10 — Maximum elevation change; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rastergaussianblur — Gaussian blur

**group:** Raster analysis

This algorithm applies a Gaussian blur filter to an input raster layer.

The radius parameter controls the strength of the blur. A larger radius results in a smoother output, at the cost of execution time.

| Parameter                | Type   | Required | Default | Description          |
|--------------------------|--------|----------|---------|----------------------|
| INPUT                    | raster | yes      | —       | Input layer          |
| BAND                     | band   | no       | 1       | Band number          |
| RADIUS                   | number | no       | 2.0     | Blur radius (pixels) |
| CREATION_OPTIONS *(adv)* | string | no       | —       | Creation options     |

| Parameter | Type              | Required | Default | Description  |
|-----------|-------------------|----------|---------|--------------|
| OUTPUT    | rasterDestination | yes      | —       | Output layer |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:rastergaussianblur INPUT=dem.tif BAND=1 RADIUS=2 OUTPUT=output.tif
```

This calls **Gaussian blur**: INPUT=dem.tif — Input layer; BAND=1 — Band number; RADIUS=2 — Blur radius (pixels); OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterize — Convert map to raster

**group:** Raster tools

This algorithm rasterizes map canvas content.

A map theme can be selected to render a predetermined set of layers with a defined style for each layer. Alternatively, a set of layers can be selected if no map theme is set. If neither map theme nor layer is set, all the visible layers in the set extent will be rendered.

The minimum extent entered will internally be extended to a multiple of the tile size.

| Parameter                   | Type              | Required | Default | Description                      |
|-----------------------------|-------------------|----------|---------|----------------------------------|
| EXTENT                      | extent            | yes      | —       | Minimum extent to render         |
| EXTENT_BUFFER               | number            | no       | 0       | Buffer around tiles in map units |
| TILE_SIZE                   | number            | no       | 1024    | Tile size                        |
| MAP_UNITS_PER_PIXEL         | number            | no       | 100     | Map units per pixel              |
| MAKE_BACKGROUND_TRANSPARENT | boolean           | no       | False   | Make background transparent      |
| MAP_THEME                   | maptheme          | no       | —       | Map theme to render              |
| LAYERS                      | multilayer        | no       | —       | Layers to render                 |
| OUTPUT                      | rasterDestination | yes      | —       | Output layer                     |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:rasterize EXTENT=0,1000,0,1000 EXTENT_BUFFER=0 TILE_SIZE=1024
↳ MAP_UNITS_PER_PIXEL=100 MAKE_BACKGROUND_TRANSPARENT=false LAYERS="a.gpkg;b.gpkg"
↳ OUTPUT=output.tif
```

This calls **Convert map to raster**: EXTENT=0,1000,0,1000 — Minimum extent to render; EXTENT\_BUFFER=0 — Buffer around tiles in map units; TILE\_SIZE=1024 — Tile size; MAP\_UNITS\_PER\_PIXEL=100 — Map units per pixel; MAKE\_BACKGROUND\_TRANSPARENT=false — Make background transparent; LAYERS="a.gpkg;b.gpkg" — Layers to render; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterlayerproperties — Raster layer properties

**group:** Raster analysis

This algorithm returns basic properties of the given raster layer, including the extent, size in pixels and dimensions of pixels (in map units).

If an optional band number is specified then the NoData value for the selected band will also be returned.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | raster | yes      | —       | Input layer |
| BAND      | band   | no       | —       | Band number |

**Outputs:** X\_MIN (outputNumber), X\_MAX (outputNumber), Y\_MIN (outputNumber), Y\_MAX (outputNumber), EXTENT (outputString), PIXEL\_WIDTH (outputNumber), PIXEL\_HEIGHT (outputNumber), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), HAS\_NODATA\_VALUE (outputBoolean), NODATA\_VALUE (outputNumber), BAND\_COUNT (outputNumber)

### Example usage

```
run native:rasterlayerproperties INPUT=dem.tif BAND=1
```

This calls **Raster layer properties**: INPUT=dem.tif — Input layer; BAND=1 — Band number. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterlayerstatistics — Raster layer statistics

**group:** Raster analysis

This algorithm computes basic statistics from the values in a given band of the raster layer.

| Parameter        | Type            | Required | Default | Description |
|------------------|-----------------|----------|---------|-------------|
| INPUT            | raster          | yes      | —       | Input layer |
| BAND             | band            | no       | 1       | Band number |
| OUTPUT_HTML_FILE | fileDestination | no       | —       | Statistics  |

**Outputs:** OUTPUT\_HTML\_FILE (outputHtml), COUNT (outputNumber), MIN (outputNumber), MAX (outputNumber), RANGE (outputNumber), SUM (outputNumber), MEAN (outputNumber), STD\_DEV (outputNumber), SUM\_OF\_SQUARES (outputNumber)

### Example usage

```
run native:rasterlayerstatistics INPUT=dem.tif BAND=1
↪ OUTPUT_HTML_FILE=output_html_file.txt
```

This calls **Raster layer statistics**: INPUT=dem.tif — Input layer; BAND=1 — Band number; OUTPUT\_HTML\_FILE=output\_html\_file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterlayeruniquevaluesreport — Raster layer unique values report

**group:** Raster analysis

This algorithm returns the count and area of each unique value in a given raster layer. The area calculation is done in the area unit of the layer's CRS.

| Parameter        | Type            | Required | Default | Description          |
|------------------|-----------------|----------|---------|----------------------|
| INPUT            | raster          | yes      | —       | Input layer          |
| BAND             | band            | no       | 1       | Band number          |
| OUTPUT_HTML_FILE | fileDestination | no       | —       | Unique values report |
| OUTPUT_TABLE     | sink            | no       | —       | Unique values table  |

**Outputs:** OUTPUT\_HTML\_FILE (outputHtml), OUTPUT\_TABLE (outputVector), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), NODATA\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:rasterlayeruniquevaluesreport INPUT=dem.tif BAND=1
↳ OUTPUT_HTML_FILE=output_html_file.txt
```

This calls **Raster layer unique values report**: INPUT=dem.tif — Input layer; BAND=1 — Band number; OUTPUT\_HTML\_FILE=output\_html\_file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterlayerzonalstats — Raster layer zonal statistics

**group:** Raster analysis

This algorithm calculates statistics for a raster layer's values, categorized by zones defined in another raster layer.

If the reference layer parameter is set to "Input layer", then zones are determined by sampling the zone raster layer value at the centroid of each pixel from the source raster layer.

If the reference layer parameter is set to "Zones layer", then the input raster layer will be sampled at the centroid of each pixel from the zones raster layer.

If either the source raster layer or the zone raster layer value is NoData for a pixel, that pixel's value will be skipped and not included in the calculated statistics.

| Parameter         | Type   | Required | Default | Description                                             |
|-------------------|--------|----------|---------|---------------------------------------------------------|
| INPUT             | raster | yes      | —       | Input layer                                             |
| BAND              | band   | no       | 1       | Band number                                             |
| ZONES             | raster | yes      | —       | Zones layer                                             |
| ZONES_BAND        | band   | no       | 1       | Zones band number                                       |
| REF_LAYER *(adv)* | enum   | no       | 0       | Reference layer — options: 0=Input layer, 1=Zones layer |
| OUTPUT_TABLE      | sink   | yes      | —       | Statistics                                              |

**Outputs:** OUTPUT\_TABLE (outputVector), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), NODATA\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:rasterlayerzonalstats INPUT=dem.tif ZONES=dem.tif BAND=1 ZONES_BAND=1
↳ OUTPUT_TABLE=output_table.gpkg
```

This calls **Raster layer zonal statistics**: INPUT=dem.tif — Input layer; ZONES=dem.tif — Zones layer; BAND=1 — Band number; ZONES\_BAND=1 — Zones band number; OUTPUT\_TABLE=output\_table.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterlogicalor — Raster boolean OR

**group:** Raster analysis

This algorithm calculates the boolean OR for a set of input rasters. If any of the input rasters have a non-zero value for a pixel, that pixel will be set to 1 in the output raster. If all the input rasters have 0 values for the pixel it will be set to 0 in the output raster.

The reference layer parameter specifies an existing raster layer to use as a reference when creating the output raster. The output raster will have the same extent, CRS, and pixel dimensions as this layer.

By default, a NoData pixel in ANY of the input layers will result in a NoData pixel in the output raster. If the 'Treat NoData values as false' option is checked, then NoData inputs will be treated the same as a 0 input value.

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | multilayer        | yes      | —       | Input layers                                                                                                                                         |
| REF_LAYER                | raster            | yes      | —       | Reference layer                                                                                                                                      |
| NODATA_AS_FALSE          | boolean           | no       | False   | Treat NoData values as false                                                                                                                         |
| NO_DATA *(adv)*          | number            | no       | -9999   | Output NoData value                                                                                                                                  |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=Int32, 4=UInt32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| CREATE_OPTIONS           | string            | no       | —       | Creation options                                                                                                                                     |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options                                                                                                                                     |
| OUTPUT                   | rasterDestination | yes      | —       | Output layer                                                                                                                                         |

**Outputs:** OUTPUT (outputRaster), EXTENT (outputString), CRS\_AUTHID (outputString), WIDTH\_IN\_PIXELS (outputNumber), HEIGHT\_IN\_PIXELS (outputNumber), TOTAL\_PIXEL\_COUNT (outputNumber), NODATA\_PIXEL\_COUNT (outputNumber), TRUE\_PIXEL\_COUNT (outputNumber), FALSE\_PIXEL\_COUNT (outputNumber)

### Example usage

```
run native:rasterlogicalor INPUT="a.gpkg;b.gpkg" REF_LAYER=dem.tif NODATA_AS_FALSE=false
↳ CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Raster boolean OR**: INPUT="a.gpkg;b.gpkg" — Input layers; REF\_LAYER=dem.tif — Reference layer; NODATA\_AS\_FALSE=false — Treat NoData values as false; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterminmax — Raster minimum/maximum

**group:** Raster analysis

This algorithm extracts extrema (minimum and maximum) values from a given band of the raster layer.

The output is a vector layer containing point features for the selected extrema, at the center of the associated pixel.

If multiple pixels in the raster share the minimum or maximum value, then only one of these pixels will be included in the output.

| Parameter | Type   | Required | Default | Description                                                            |
|-----------|--------|----------|---------|------------------------------------------------------------------------|
| INPUT     | raster | yes      | —       | Input layer                                                            |
| BAND      | band   | no       | 1       | Band number                                                            |
| EXTRACT   | enum   | no       | 0       | Extract extrema — options: 0=Minimum and Maximum, 1=Minimum, 2=Maximum |
| OUTPUT    | sink   | no       | —       | Output                                                                 |

**Outputs:** OUTPUT (outputVector), MINIMUM (outputNumber), MAXIMUM (outputNumber)

### Example usage

```
run native:rasterminmax INPUT=dem.tif BAND=1 EXTRACT=0 OUTPUT=output.gpkg
```

This calls **Raster minimum/maximum**: INPUT=dem.tif — Input layer; BAND=1 — Band number; EXTRACT=0 selects *Minimum and Maximum*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rasterrank — Raster rank

**group:** Raster analysis

This algorithm performs a cell-by-cell analysis in which output values match the rank of a sorted list of overlapping cell values from input layers. The output raster will be multi-band if multiple ranks are provided. If multiband rasters are used in the data raster stack, the algorithm will always perform the analysis on the first band of the rasters.

| Parameter         | Type              | Required | Default | Description                                                                                                                              |
|-------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT_RASTERS     | multilayer        | yes      | —       | Input raster layers                                                                                                                      |
| RANKS             | string            | no       | 1       | Rank(s) (separate multiple ranks using commas)                                                                                           |
| NODATA_HANDLING   | enum              | no       | 0       | NoData value handling — options: 0=Exclude NoData from values lists, 1=Presence of NoData in a values list results in NoData output cell |
| EXTENT *(adv)*    | extent            | no       | —       | Output extent                                                                                                                            |
| CELL_SIZE *(adv)* | number            | no       | —       | Output cell size                                                                                                                         |
| CRS *(adv)*       | crs               | no       | —       | Output CRS                                                                                                                               |
| OUTPUT            | rasterDestination | yes      | —       | Ranked                                                                                                                                   |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:rasterrank INPUT_RASTERS="a.gpkg;b.gpkg" RANKS=value NODATA_HANDLING=0
↳ OUTPUT=output.tif
```

This calls **Raster rank**: INPUT\_RASTERS="a.gpkg;b.gpkg" — Input raster layers; RANKS=value — Rank(s) (separate multiple ranks using commas); NODATA\_HANDLING=0 selects *Exclude NoData from values lists*; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rastersampling — Sample raster values

**group:** Raster analysis

This algorithm creates a new vector layer with the same attributes of the input layer and the raster values corresponding on the point location.

If the raster layer has more than one band, all the band values are sampled.

| Parameter     | Type   | Required | Default | Description          |
|---------------|--------|----------|---------|----------------------|
| INPUT         | source | yes      | —       | Input layer          |
| RASTERCOPY    | raster | yes      | —       | Raster layer         |
| COLUMN_PREFIX | string | no       | SAMPLE_ | Output column prefix |
| OUTPUT        | sink   | yes      | —       | Sampled              |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:rastersampling INPUT=input.gpkg RASTERCOPY=dem.tif COLUMN_PREFIX=SAMPLE_
↳ OUTPUT=output.gpkg
```

This calls **Sample raster values**: INPUT=input.gpkg — Input layer; RASTERCOPY=dem.tif — Raster layer; COLUMN\_PREFIX=SAMPLE\_ — Output column prefix; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rastersurfacevolume — Raster surface volume

**group:** Raster analysis

This algorithm calculates the volume under a raster grid's surface.

Several methods of volume calculation are available, which control whether only values above or below the specified base level are considered, or whether volumes below the base level should be added or subtracted from the total volume.

The algorithm outputs the calculated volume, the total area, and the total number of pixels analysed. If the 'Count Only Above Base Level' or 'Count Only Below Base Level' methods are used, then the calculated area and pixel count only includes pixels which are above or below the specified base level respectively.

Units of the calculated volume are dependent on the coordinate reference system of the input raster file. For a CRS in meters, with a DEM height in meters, the calculated value will be in meters<sup>3</sup>. If instead the input raster is in a geographic coordinate system (e.g. latitude/longitude values), then the result will be in degrees<sup>2</sup> × meters, and an appropriate scaling factor will need to be applied in order to convert to meters<sup>3</sup>.

| Parameter        | Type            | Required | Default | Description                                                                                                                                         |
|------------------|-----------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT            | raster          | yes      | —       | Input layer                                                                                                                                         |
| BAND             | band            | no       | 1       | Band number                                                                                                                                         |
| LEVEL            | number          | no       | 0       | Base level                                                                                                                                          |
| METHOD           | enum            | yes      | —       | Method — options: 0=Count Only Above Base Level, 1=Count Only Below Base Level, 2=Subtract Volumes Below Base Level, 3=Add Volumes Below Base Level |
| OUTPUT_HTML_FILE | fileDestination | no       | —       | Surface volume report                                                                                                                               |
| OUTPUT_TABLE     | sink            | no       | —       | Surface volume table                                                                                                                                |

**Outputs:** OUTPUT\_HTML\_FILE (outputHtml), OUTPUT\_TABLE (outputVector), VOLUME (outputNumber), PIXEL\_COUNT (outputNumber), AREA (outputNumber)

### Example usage

```
run native:rastersurfacevolume INPUT=dem.tif METHOD=0 BAND=1 LEVEL=0  
↳ OUTPUT_HTML_FILE=output_html_file.txt
```

This calls **Raster surface volume**: INPUT=dem.tif — Input layer; METHOD=0 selects *Count Only Above Base Level*; BAND=1 — Band number; LEVEL=0 — Base level; OUTPUT\_HTML\_FILE=output\_html\_file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:reclassifybylayer — Reclassify by layer

**group:** Raster analysis

This algorithm reclassifies a raster band by assigning new class values based on the ranges specified in a vector table.

| Parameter       | Type   | Required | Default | Description                   |
|-----------------|--------|----------|---------|-------------------------------|
| INPUT_RASTER    | raster | yes      | —       | Raster layer                  |
| RASTER_BAND     | band   | no       | 1       | Band number                   |
| INPUT_TABLE     | source | yes      | —       | Layer containing class breaks |
| MIN_FIELD       | field  | yes      | —       | Minimum class value field     |
| MAX_FIELD       | field  | yes      | —       | Maximum class value field     |
| VALUE_FIELD     | field  | yes      | —       | Output value field            |
| NO_DATA *(adv)* | number | no       | -9999   | Output NoData value           |

| Parameter                  | Type              | Required | Default | Description                                                                                                                                          |
|----------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| RANGE_BOUNDARIES *(adv)*   | enum              | no       | 0       | Range boundaries — options: 0=min < value <= max, 1=min <= value < max, 2=min <= value <= max, 3=min < value < max                                   |
| NODATA_FOR_MISSING *(adv)* | boolean           | no       | False   | Use NoData when no range matches value                                                                                                               |
| DATA_TYPE *(adv)*          | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=Int32, 4=UInt32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| CREATE_OPTIONS             | string            | no       | —       | Creation options                                                                                                                                     |
| CREATION_OPTIONS *(adv)*   | string            | no       | —       | Creation options                                                                                                                                     |
| OUTPUT                     | rasterDestination | yes      | —       | Reclassified raster                                                                                                                                  |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:reclassifybylayer INPUT_RASTER=dem.tif INPUT_TABLE=input.gpkg MIN_FIELD=field1
↳ MAX_FIELD=field1 VALUE_FIELD=field1 RASTER_BAND=1 CREATE_OPTIONS=value
↳ OUTPUT=output.tif
```

This calls **Reclassify by layer**: INPUT\_RASTER=dem.tif — Raster layer; INPUT\_TABLE=input.gpkg — Layer containing class breaks; MIN\_FIELD=field1 — Minimum class value field; MAX\_FIELD=field1 — Maximum class value field; VALUE\_FIELD=field1 — Output value field; RASTER\_BAND=1 — Band number; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:reclassifybytable — Reclassify by table

**group:** Raster analysis

This algorithm reclassifies a raster band by assigning new class values based on the ranges specified in a fixed table.

| Parameter                  | Type              | Required | Default | Description                                                                                                                                          |
|----------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT_RASTER               | raster            | yes      | —       | Raster layer                                                                                                                                         |
| RASTER_BAND                | band              | no       | 1       | Band number                                                                                                                                          |
| TABLE                      | matrix            | yes      | —       | Reclassification table                                                                                                                               |
| NO_DATA *(adv)*            | number            | no       | -9999   | Output NoData value                                                                                                                                  |
| RANGE_BOUNDARIES *(adv)*   | enum              | no       | 0       | Range boundaries — options: 0=min < value <= max, 1=min <= value < max, 2=min <= value <= max, 3=min < value < max                                   |
| NODATA_FOR_MISSING *(adv)* | boolean           | no       | False   | Use NoData when no range matches value                                                                                                               |
| DATA_TYPE *(adv)*          | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=Int32, 4=UInt32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| CREATE_OPTIONS             | string            | no       | —       | Creation options                                                                                                                                     |
| CREATION_OPTIONS *(adv)*   | string            | no       | —       | Creation options                                                                                                                                     |
| OUTPUT                     | rasterDestination | yes      | —       | Reclassified raster                                                                                                                                  |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:reclassifybytable INPUT_RASTER=dem.tif TABLE=... RASTER_BAND=1
↳ CREATE_OPTIONS=value OUTPUT=output.tif
```



This calls **Reclassify by table**: INPUT\_RASTER=dem.tif — Raster layer; TABLE=... — Reclassification table; RASTER\_BAND=1 — Band number; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rectanglesovalsdiamonds — Rectangles, ovals, diamonds

**group:** Vector geometry

This algorithm creates rectangle, oval or diamond-shaped polygons from the input point layer using specified width, height and (optional) rotation values. Multipart inputs should be promoted to singleparts first.

| Parameter | Type     | Required | Default | Description                                     |
|-----------|----------|----------|---------|-------------------------------------------------|
| INPUT     | source   | yes      | —       | Input layer                                     |
| SHAPE     | enum     | no       | 0       | Shape — options: 0=Rectangle, 1=Diamond, 2=Oval |
| WIDTH     | distance | no       | 1.0     | Width                                           |
| HEIGHT    | distance | no       | 1.0     | Height                                          |
| ROTATION  | number   | no       | 0.0     | Rotation                                        |
| SEGMENTS  | number   | no       | 36      | Segments                                        |
| OUTPUT    | sink     | yes      | —       | Polygon                                         |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:rectanglesovalsdiamonds INPUT=input.gpkg SHAPE=0 WIDTH=1 HEIGHT=1 ROTATION=0
↪ SEGMENTS=36 OUTPUT=output.gpkg
```

This calls **Rectangles, ovals, diamonds**: INPUT=input.gpkg — Input layer; SHAPE=0 selects *Rectangle*; WIDTH=1 — Width; HEIGHT=1 — Height; ROTATION=0 — Rotation; SEGMENTS=36 — Segments; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:refactorfields — Refactor fields

**group:** Vector table

This algorithm allows editing the structure of the attributes table of a vector layer. Fields can be modified in their type and name, using a fields mapping.

The original layer is not modified. A new layer is generated, which contains a modified attribute table, according to the provided fields mapping.

Rows in orange have constraints in the template layer from which these fields were loaded. Treat this information as a hint during configuration. No constraints will be added on an output layer nor will they be checked or enforced by the algorithm.

| Parameter      | Type           | Required | Default | Description    |
|----------------|----------------|----------|---------|----------------|
| INPUT          | source         | yes      | —       | Input layer    |
| FIELDS_MAPPING | fields_mapping | yes      | —       | Fields mapping |
| OUTPUT         | sink           | yes      | —       | Refactored     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:refactorfields INPUT=input.gpkg FIELDS_MAPPING=... OUTPUT=output.gpkg
```

This calls **Refactor fields**: INPUT=input.gpkg — Input layer; FIELDS\_MAPPING=... — Fields mapping; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:removeduplicatesbyattribute — Delete duplicates by attribute**

**group:** Vector general

This algorithm removes duplicate rows by a field value (or multiple field values). The first matching row will be retained, and duplicates will be discarded.

Optionally, these duplicate records can be saved to a separate output for analysis.

| Parameter  | Type   | Required | Default | Description                  |
|------------|--------|----------|---------|------------------------------|
| INPUT      | source | yes      | —       | Input layer                  |
| FIELDS     | field  | yes      | —       | Field to match duplicates by |
| OUTPUT     | sink   | yes      | —       | Filtered (no duplicates)     |
| DUPLICATES | sink   | no       | —       | Filtered (duplicates)        |

**Outputs:** OUTPUT (outputVector), DUPLICATES (outputVector), RETAINED\_COUNT (outputNumber), DUPLICATE\_COUNT (outputNumber)

### **Example usage**

```
run native:removeduplicatesbyattribute INPUT=input.gpkg FIELDS=field1 OUTPUT=output.gpkg
```

This calls **Delete duplicates by attribute**: INPUT=input.gpkg — Input layer; FIELDS=field1 — Field to match duplicates by; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:removeduplicatevertices — Remove duplicate vertices**

**group:** Vector geometry

This algorithm removes duplicate vertices from features, wherever removing the vertices does not result in a degenerate geometry.

The tolerance parameter specifies the tolerance for coordinates when determining whether vertices are identical.

By default, z values are not considered when detecting duplicate vertices. E.g. two vertices with the same x and y coordinate but different z values will still be considered duplicate and one will be removed. If the Use Z Value parameter is true, then the z values are also tested and vertices with the same x and y but different z will be maintained.

Note that duplicate vertices are not tested between different parts of a multipart geometry. E.g. a multipoint geometry with overlapping points will not be changed by this method.

| Parameter   | Type     | Required | Default | Description |
|-------------|----------|----------|---------|-------------|
| INPUT       | source   | yes      | —       | Input layer |
| TOLERANCE   | distance | no       | 1e-06   | Tolerance   |
| USE_Z_VALUE | boolean  | no       | False   | Use Z Value |
| OUTPUT      | sink     | yes      | —       | Cleaned     |

**Outputs:** OUTPUT (outputVector)

### **Example usage**

```
run native:removeduplicatevertices INPUT=input.gpkg TOLERANCE=1e-06 USE_Z_VALUE=false  
↪ OUTPUT=output.gpkg
```

This calls **Remove duplicate vertices**: INPUT=input.gpkg — Input layer; TOLERANCE=1e-06 — Tolerance; USE\_Z\_VALUE=false — Use Z Value; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:removenullgeometries — Remove null geometries

**group:** Vector geometry

This algorithm removes any features which do not have a geometry from a vector layer. All other features will be copied unchanged.

Optionally, the features with null geometries can be saved to a separate output.

If 'Also remove empty geometries' is checked, the algorithm removes features whose geometries have no coordinates, i.e., geometries that are empty. In that case, also the null output will reflect this option, containing both null and empty geometries.

| Parameter    | Type    | Required | Default | Description                  |
|--------------|---------|----------|---------|------------------------------|
| INPUT        | source  | yes      | —       | Input layer                  |
| REMOVE_EMPTY | boolean | no       | False   | Also remove empty geometries |
| OUTPUT       | sink    | no       | —       | Non null geometries          |
| NULL_OUTPUT  | sink    | no       | —       | Null geometries              |

**Outputs:** OUTPUT (outputVector), NULL\_OUTPUT (outputVector)

### Example usage

```
run native:removenullgeometries INPUT=input.gpkg REMOVE_EMPTY=false OUTPUT=output.gpkg
```

This calls **Remove null geometries**: INPUT=input.gpkg — Input layer; REMOVE\_EMPTY=false — Also remove empty geometries; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:removepartsbyarea — Remove parts by area

**group:** Vector geometry

This algorithm takes a polygon layer and removes polygons which are smaller than a specified area.

If the input geometry is a multipart geometry, then the parts will be filtered by their individual areas. If no parts match the required minimum area, then the feature will be skipped and omitted from the output layer.

If the input geometry is a singlepart geometry, then the feature will be skipped if the geometry's area is below the required size and omitted from the output layer.

The area will be calculated using Cartesian calculations in the source layer's coordinate reference system.

Attributes are not modified.

| Parameter | Type   | Required | Default | Description                      |
|-----------|--------|----------|---------|----------------------------------|
| INPUT     | source | yes      | —       | Input layer                      |
| MIN_AREA  | area   | no       | 0.0     | Remove parts with area less than |
| OUTPUT    | sink   | yes      | —       | Cleaned                          |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:removepartsbyarea INPUT=input.gpkg MIN_AREA=0 OUTPUT=output.gpkg
```

This calls **Remove parts by area**: INPUT=input.gpkg — Input layer; MIN\_AREA=0 — Remove parts with area less than; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:removepartsbylength — Remove parts by length

**group:** Vector geometry

This algorithm takes a line layer and removes lines which are shorter than a specified length.

If the input geometry is a multipart geometry, then the parts will be filtered by their individual lengths. If no parts match the required minimum length, then the feature will be skipped and omitted from the output layer.

If the input geometry is a singlepart geometry, then the feature will be skipped if the geometry's length is below the required size and omitted from the output layer.

The length will be calculated using Cartesian calculations in the source layer's coordinate reference system.

Attributes are not modified.

| Parameter  | Type     | Required | Default | Description                         |
|------------|----------|----------|---------|-------------------------------------|
| INPUT      | source   | yes      | —       | Input layer                         |
| MIN_LENGTH | distance | no       | 0.0     | Remove parts with lengths less than |
| OUTPUT     | sink     | yes      | —       | Cleaned                             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:removepartsbylength INPUT=input.gpkg MIN_LENGTH=0 OUTPUT=output.gpkg
```

This calls **Remove parts by length**: INPUT=input.gpkg — Input layer; MIN\_LENGTH=0 — Remove parts with lengths less than; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:renamelayer — Rename layer

**group:** Modeler tools

This algorithm renames a layer.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | layer  | yes      | —       | Layer       |
| NAME      | string | yes      | —       | New name    |

**Outputs:** OUTPUT (outputLayer)

### Example usage

```
run native:renamelayer INPUT=input.gpkg NAME=value
```

This calls **Rename layer**: INPUT=input.gpkg — Layer; NAME=value — New name. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:renametablefield — Rename field ★

**group:** Vector table · **niva verb:** renamefield

This algorithm renames an existing field from a vector layer.

| Parameter | Type   | Required | Default | Description     |
|-----------|--------|----------|---------|-----------------|
| INPUT     | source | yes      | —       | Input layer     |
| FIELD     | field  | yes      | —       | Field to rename |
| NEW_NAME  | string | yes      | —       | New field name  |
| OUTPUT    | sink   | yes      | —       | Renamed         |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:renametablefield INPUT=input.gpkg FIELD=field1 NEW_NAME=value  
↳ OUTPUT=output.gpkg
```

This calls **Rename field**: INPUT=input.gpkg — Input layer; FIELD=field1 — Field to rename; NEW\_NAME=value — New field name; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:repairshapefile — Repair Shapefile

**group:** Vector general

Repairs a broken Shapefile by recreating missing or broken SHX files.

| Parameter | Type | Required | Default | Description     |
|-----------|------|----------|---------|-----------------|
| INPUT     | file | yes      | —       | Input Shapefile |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:repairshapefile INPUT=input.dat
```

This calls **Repair Shapefile**: INPUT=input.dat — Input Shapefile. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:reprojectlayer — Reproject layer ★

**group:** Vector general · **niva verb:** reproject

This algorithm reprojects a vector layer. It creates a new layer with the same features as the input one, but with geometries reprojected to a new CRS.

Attributes are not modified by this algorithm.

| Parameter                 | Type                | Required | Default   | Description                                    |
|---------------------------|---------------------|----------|-----------|------------------------------------------------|
| INPUT                     | source              | yes      | —         | Input layer                                    |
| TARGET_CRS                | crs                 | no       | EPSG:4326 | Target CRS                                     |
| CONVERT_CURVED_GEOMETRIES | boolean             | no       | False     | Convert curved geometries to straight segments |
| TRANSFORM_Z               | boolean             | no       | False     | Also transform Z coordinates                   |
| OPERATION *(adv)*         | coordinateoperation | no       | —         | Coordinate operation                           |
| OUTPUT                    | sink                | yes      | —         | Reprojected                                    |

---

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:reprojectlayer INPUT=input.gpkg TARGET_CRS=EPSG:6346
↳ CONVERT_CURVED_GEOMETRIES=false TRANSFORM_Z=false OUTPUT=output.gpkg
```

This calls **Reproject layer**: INPUT=input.gpkg — Input layer; TARGET\_CRS=EPSG:6346 — Target CRS; CONVERT\_CURVED\_GEOMETRIES=false — Convert curved geometries to straight segments; TRANSFORM\_Z=false — Also transform Z coordinates; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rescaleraster — Rescale raster

**group:** Raster analysis

This algorithm rescales a raster layer to a new value range, while preserving the shape (distribution) of the raster's histogram (pixel values). Input values are mapped using a linear interpolation from the source raster's minimum and maximum pixel values to the destination minimum and maximum pixel range.

By default the algorithm preserves the original NoData value, but there is an option to override it.

| Parameter                | Type              | Required | Default | Description       |
|--------------------------|-------------------|----------|---------|-------------------|
| INPUT                    | raster            | yes      | —       | Input raster      |
| BAND                     | band              | no       | 1       | Band number       |
| MINIMUM                  | number            | no       | 0       | New minimum value |
| MAXIMUM                  | number            | no       | 255     | New maximum value |
| NODATA                   | number            | no       | —       | New NoData value  |
| CREATE_OPTIONS           | string            | no       | —       | Creation options  |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options  |
| OUTPUT                   | rasterDestination | yes      | —       | Rescaled          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:rescaleraster INPUT=dem.tif BAND=1 MINIMUM=0 MAXIMUM=255 NODATA=10
↳ CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Rescale raster**: INPUT=dem.tif — Input raster; BAND=1 — Band number; MINIMUM=0 — New minimum value; MAXIMUM=255 — New maximum value; NODATA=10 — New NoData value; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:retainfields — Retain fields ★

**group:** Vector table · **niva verb:** keepfields

This algorithm takes a vector layer and generates a new one that retains only the selected fields. All other fields will be dropped.

| Parameter | Type   | Required | Default | Description      |
|-----------|--------|----------|---------|------------------|
| INPUT     | source | yes      | —       | Input layer      |
| FIELDS    | field  | yes      | —       | Fields to retain |
| OUTPUT    | sink   | yes      | —       | Retained fields  |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:retainfields INPUT=input.gpkg FIELDS=field1 OUTPUT=output.gpkg
```

This calls **Retain fields**: INPUT=input.gpkg — Input layer; FIELDS=field1 — Fields to retain; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:reverselinedirection — Reverse line direction

**group:** Vector geometry

This algorithm reverses the direction of curve or LineString geometries.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Reversed    |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:reverselinedirection INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Reverse line direction**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:rotatefeatures — Rotate

**group:** Vector geometry

This algorithm rotates feature geometries by the specified angle clockwise.

Optionally, the rotation can occur around a preset point. If not set the rotation occurs around each feature's centroid.

| Parameter | Type   | Required | Default | Description                  |
|-----------|--------|----------|---------|------------------------------|
| INPUT     | source | yes      | —       | Input layer                  |
| ANGLE     | number | no       | 0.0     | Rotation (degrees clockwise) |
| ANCHOR    | point  | no       | —       | Rotation anchor point        |
| OUTPUT    | sink   | yes      | —       | Rotated                      |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:rotatefeatures INPUT=input.gpkg ANGLE=0 ANCHOR=100,100 OUTPUT=output.gpkg
```

This calls **Rotate**: INPUT=input.gpkg — Input layer; ANGLE=0 — Rotation (degrees clockwise); ANCHOR=100,100 — Rotation anchor point; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:roundness — Roundness

**group:** Vector geometry

Calculates the roundness of each feature and stores it as a new field. The input vector layer must contain polygons.

The roundness of a polygon is defined as  $4\pi \times \text{polygon area} / \text{perimeter}^2$ . The roundness value varies between 0 and 1. A perfect circle has a roundness of 1, while a completely flat polygon has a roundness of 0.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Roundness   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:roundness INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Roundness**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:roundrastervalues — Round raster

**group:** Raster analysis

This algorithm rounds the cell values of a raster dataset to the specified number of decimals. Alternatively, a negative number of decimal places may be used to round values to powers of a base n (specified in the advanced parameter Base n). For example, with a Base value n of 10 and Decimal places of -1 the algorithm rounds cell values to multiples of 10, -2 rounds to multiples of 100, and so on. Arbitrary base values may be chosen, the algorithm applies the same multiplicative principle. Rounding cell values to multiples of a base n may be used to generalize raster layers. The algorithm preserves the data type of the input raster. Therefore byte/integer rasters can only be rounded to multiples of a base n, otherwise a warning is raised and the raster gets copied as byte/integer raster.

| Parameter                | Type              | Required | Default | Description                                                                |
|--------------------------|-------------------|----------|---------|----------------------------------------------------------------------------|
| INPUT                    | raster            | yes      | —       | Input raster                                                               |
| BAND                     | band              | no       | 1       | Band number                                                                |
| ROUNDING_DIRECTION       | enum              | no       | 1       | Rounding direction — options: 0=Round up, 1=Round to nearest, 2=Round down |
| DECIMAL_PLACES           | number            | no       | 2       | Number of decimals places                                                  |
| BASE_N *(adv)*           | number            | no       | 10      | Base n for rounding to multiples of n                                      |
| CREATE_OPTIONS           | string            | no       | —       | Creation options                                                           |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options                                                           |
| OUTPUT                   | rasterDestination | yes      | —       | Output raster                                                              |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:roundrastervalues INPUT=dem.tif BAND=1 ROUNDING_DIRECTION=1 DECIMAL_PLACES=2  
↳ CREATE_OPTIONS=value OUTPUT=output.tif
```

This calls **Round raster**: INPUT=dem.tif — Input raster; BAND=1 — Band number; ROUNDING\_DIRECTION=1 selects *Round to nearest*; DECIMAL\_PLACES=2 — Number of decimals places; CREATE\_OPTIONS=value — Creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## native:ruggednessindex — Ruggedness index

**group:** Raster terrain analysis

This algorithm calculates the quantitative measurement of terrain heterogeneity described by Riley et al. (1999).

It is calculated for every location, by summarizing the change in elevation within the 3x3 pixel grid. Each pixel contains the difference in elevation from a center cell and the 8 cells surrounding it.

| Parameter                | Type              | Required | Default | Description         |
|--------------------------|-------------------|----------|---------|---------------------|
| INPUT                    | raster            | yes      | —       | Elevation layer     |
| Z_FACTOR                 | number            | no       | 1.0     | Z factor            |
| NODATA *(adv)*           | number            | no       | -9999.0 | Output NoData value |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options    |
| OUTPUT                   | rasterDestination | yes      | —       | Ruggedness          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:ruggednessindex INPUT=dem.tif Z_FACTOR=1 OUTPUT=output.tif
```

This calls **Ruggedness index**: INPUT=dem.tif — Elevation layer; Z\_FACTOR=1 — Z factor; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:savefeatures — Save vector features to file

**group:** Vector general

This algorithm saves vector features to a specified file dataset.

For dataset formats supporting layers, an optional layer name parameter can be used to specify a custom string.

Optional GDAL-defined dataset and layer options can be specified. For more information on this, read the online GDAL documentation.

| Parameter                       | Type            | Required | Default | Description                                                                                                                                                                                                                                     |
|---------------------------------|-----------------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                           | source          | yes      | —       | Vector features                                                                                                                                                                                                                                 |
| OUTPUT                          | fileDestination | yes      | —       | Saved features                                                                                                                                                                                                                                  |
| LAYER_NAME *(adv)*              | string          | no       | —       | Layer name                                                                                                                                                                                                                                      |
| DATASOURCE_OPTIONS *(adv)*      | string          | no       | —       | GDAL dataset options (separate individual options with semicolons)                                                                                                                                                                              |
| LAYER_OPTIONS *(adv)*           | string          | no       | —       | GDAL layer options (separate individual options with semicolons)                                                                                                                                                                                |
| ACTION_ON_EXISTING_FILE *(adv)* | enum            | no       | 0       | Action to take on pre-existing file — options: 0=Create or overwrite file, 1=Create or overwrite layer, 2=Append features to existing layer, but do not create new fields, 3=Append features to existing layer, and create new fields if needed |

**Outputs:** OUTPUT (outputFile), FILE\_PATH (outputString), LAYER\_NAME (outputString)

### Example usage

```
run native:savefeatures INPUT=input.gpkg OUTPUT=output.txt
```

This calls **Save vector features to file**: INPUT=input.gpkg — Vector features; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:savelog — Save log to file

**group:** Modeler tools

This algorithm saves the model's execution log to a file. Optionally, the log can be saved in a HTML formatted version.

| Parameter | Type            | Required | Default | Description         |
|-----------|-----------------|----------|---------|---------------------|
| OUTPUT    | fileDestination | yes      | —       | Log file            |
| USE_HTML  | boolean         | no       | False   | Use HTML formatting |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run native:savelog USE_HTML=false OUTPUT=output.txt
```

This calls **Save log to file**: USE\_HTML=false — Use HTML formatting; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:savesselectedfeatures — Extract selected features

**group:** Vector general

This algorithm creates a new layer with all the selected features in a given vector layer.

If the selected layer has no selected features, the newly created layer will be empty.

| Parameter | Type   | Required | Default | Description       |
|-----------|--------|----------|---------|-------------------|
| INPUT     | vector | yes      | —       | Input layer       |
| OUTPUT    | sink   | yes      | —       | Selected features |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:savesselectedfeatures INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Extract selected features**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:segmentizebymaxangle — Segmentize by maximum angle

**group:** Vector geometry

This algorithm segmentizes a geometry by converting curved sections to linear sections.

The segmentization is performed by specifying the maximum allowed radius angle between vertices on the straightened geometry (e.g the angle of the arc created from the original arc center to consecutive output vertices on the linearized geometry).

Non-curved geometries will be retained without change.

| Parameter | Type   | Required | Default | Description                              |
|-----------|--------|----------|---------|------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                              |
| ANGLE     | number | no       | 5.0     | Maximum angle between vertices (degrees) |
| OUTPUT    | sink   | yes      | —       | Segmentized                              |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:segmentizebymaxangle INPUT=input.gpkg ANGLE=5 OUTPUT=output.gpkg
```

This calls **Segmentize by maximum angle**: INPUT=input.gpkg — Input layer; ANGLE=5 — Maximum angle between vertices (degrees); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:segmentizebymaxdistance — Segmentize by maximum distance**

**group:** Vector geometry

This algorithm segmentizes a geometry by converting curved sections to linear sections.

The segmentization is performed by specifying the maximum allowed offset distance between the original curve and the segmentized representation.

Non-curved geometries will be retained without change.

| Parameter | Type     | Required | Default | Description             |
|-----------|----------|----------|---------|-------------------------|
| INPUT     | source   | yes      | —       | Input layer             |
| DISTANCE  | distance | no       | 1.0     | Maximum offset distance |
| OUTPUT    | sink     | yes      | —       | Segmentized             |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run native:segmentizebymaxdistance INPUT=input.gpkg DISTANCE=1 OUTPUT=output.gpkg
```

This calls **Segmentize by maximum distance**: INPUT=input.gpkg — Input layer; DISTANCE=1 — Maximum offset distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## **native:selectbylocation — Select by location**

**group:** Vector selection

This algorithm creates a selection in a vector layer. The criteria for selecting features is based on the spatial relationship between each feature and the features in an additional layer.

| Parameter | Type   | Required | Default | Description                                                                                                                                                             |
|-----------|--------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT     | vector | yes      | —       | Select features from                                                                                                                                                    |
| PREDICATE | enum   | no       | [0]     | Where the features (geometric predicate) — options: 0=intersect, 1=contain, 2=disjoint, 3=equal, 4=touch, 5=overlap, 6=are within, 7=cross                              |
| INTERSECT | source | yes      | —       | By comparing to the features from                                                                                                                                       |
| METHOD    | enum   | no       | 0       | Modify current selection by — options: 0=creating new selection, 1=adding to current selection, 2=selecting within current selection, 3=removing from current selection |

**Example usage**

```
run native:selectbylocation INPUT=input.gpkg INTERSECT=input.gpkg PREDICATE=0 METHOD=0
```

This calls **Select by location**: INPUT=input.gpkg — Select features from; INTERSECT=input.gpkg — By comparing to the features from; PREDICATE=0 selects *intersect*; METHOD=0 selects *creating new selection*. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:selectwithindistance — Select within distance

**group:** Vector selection

This algorithm creates a selection in a vector layer. Features are selected wherever they are within the specified maximum distance from the features in an additional reference layer.

| Parameter | Type     | Required | Default | Description                                                                                                                                                             |
|-----------|----------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT     | vector   | yes      | —       | Select features from                                                                                                                                                    |
| REFERENCE | source   | yes      | —       | By comparing to the features from                                                                                                                                       |
| DISTANCE  | distance | no       | 100     | Where the features are within                                                                                                                                           |
| METHOD    | enum     | no       | 0       | Modify current selection by — options: 0=creating new selection, 1=adding to current selection, 2=selecting within current selection, 3=removing from current selection |

### Example usage

```
run native:selectwithindistance INPUT=input.gpkg REFERENCE=input.gpkg DISTANCE=100
↳ METHOD=0
```

This calls **Select within distance**: INPUT=input.gpkg — Select features from; REFERENCE=input.gpkg — By comparing to the features from; DISTANCE=100 — Where the features are within; METHOD=0 selects *creating new selection*. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:serviceareafromlayer — Service area (from layer)

**group:** Network analysis

This algorithm creates a new vector layer with all the edges or parts of edges of a network line layer that can be reached within a distance or a time, starting from features of a point layer. The distance and the time (both referred to as "travel cost") must be specified respectively in the network layer units or in hours.

| Parameter                 | Type     | Required | Default | Description                                                                               |
|---------------------------|----------|----------|---------|-------------------------------------------------------------------------------------------|
| INPUT                     | source   | yes      | —       | Vector layer representing network                                                         |
| STRATEGY                  | enum     | no       | 0       | Path type to calculate — options: 0=Shortest, 1=Fastest                                   |
| DIRECTION_FIELD *(adv)*   | field    | no       | —       | Direction field                                                                           |
| VALUE_FORWARD *(adv)*     | string   | no       | —       | Value for forward direction                                                               |
| VALUE_BACKWARD *(adv)*    | string   | no       | —       | Value for backward direction                                                              |
| VALUE_BOTH *(adv)*        | string   | no       | —       | Value for both directions                                                                 |
| DEFAULT_DIRECTION *(adv)* | enum     | no       | 2       | Default direction — options: 0=Forward direction, 1=Backward direction, 2=Both directions |
| SPEED_FIELD *(adv)*       | field    | no       | —       | Speed field                                                                               |
| DEFAULT_SPEED *(adv)*     | number   | no       | 50      | Default speed (km/h)                                                                      |
| TOLERANCE *(adv)*         | distance | no       | 0       | Topology tolerance                                                                        |
| START_POINTS              | source   | yes      | —       | Vector layer with start points                                                            |
| TRAVEL_COST               | number   | no       | 0       | Travel cost (distance for 'Shortest', time for 'Fastest')                                 |
| TRAVEL_COST2              | number   | no       | 0       | Travel cost (distance for 'Shortest', time for 'Fastest')                                 |
| INCLUDE_BOUNDS *(adv)*    | boolean  | no       | False   | Include upper/lower bound points                                                          |
| POINT_TOLERANCE *(adv)*   | distance | no       | —       | Maximum point distance from network                                                       |
| OUTPUT_LINES              | sink     | no       | —       | Service area (lines)                                                                      |

| Parameter           | Type | Required | Default | Description                   |
|---------------------|------|----------|---------|-------------------------------|
| OUTPUT              | sink | no       | —       | Service area (boundary nodes) |
| OUTPUT_NON_ROUTABLE | sink | no       | —       | Non-routable features         |

**Outputs:** OUTPUT\_LINES (outputVector), OUTPUT (outputVector), OUTPUT\_NON\_ROUTABLE (outputVector)

### Example usage

```
run native:serviceareafromlayer INPUT=input.gpkg START_POINTS=input.gpkg STRATEGY=0
↳ TRAVEL_COST=0 TRAVEL_COST2=0 OUTPUT_LINES=output_lines.gpkg
```

This calls **Service area (from layer)**: INPUT=input.gpkg — Vector layer representing network; START\_POINTS=input.gpkg — Vector layer with start points; STRATEGY=0 selects *Shortest*; TRAVEL\_COST=0 — Travel cost (distance for 'Shortest', time for 'Fastest'); TRAVEL\_COST2=0 — Travel cost (distance for 'Shortest', time for 'Fastest'); OUTPUT\_LINES=output\_lines.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:serviceareafrompoint — Service area (from point)

**group:** Network analysis

This algorithm creates a new vector layer with all the edges or parts of edges of a network line layer that can be reached within a distance or a time, starting from a point feature. The distance and the time (both referred to as "travel cost") must be specified respectively in the network layer units or in hours.

| Parameter                 | Type     | Required | Default | Description                                                                               |
|---------------------------|----------|----------|---------|-------------------------------------------------------------------------------------------|
| INPUT                     | source   | yes      | —       | Vector layer representing network                                                         |
| STRATEGY                  | enum     | no       | 0       | Path type to calculate — options: 0=Shortest, 1=Fastest                                   |
| DIRECTION_FIELD *(adv)*   | field    | no       | —       | Direction field                                                                           |
| VALUE_FORWARD *(adv)*     | string   | no       | —       | Value for forward direction                                                               |
| VALUE_BACKWARD *(adv)*    | string   | no       | —       | Value for backward direction                                                              |
| VALUE_BOTH *(adv)*        | string   | no       | —       | Value for both directions                                                                 |
| DEFAULT_DIRECTION *(adv)* | enum     | no       | 2       | Default direction — options: 0=Forward direction, 1=Backward direction, 2=Both directions |
| SPEED_FIELD *(adv)*       | field    | no       | —       | Speed field                                                                               |
| DEFAULT_SPEED *(adv)*     | number   | no       | 50      | Default speed (km/h)                                                                      |
| TOLERANCE *(adv)*         | distance | no       | 0       | Topology tolerance                                                                        |
| START_POINT               | point    | yes      | —       | Start point                                                                               |
| TRAVEL_COST               | number   | no       | 0       | Travel cost (distance for 'Shortest', time for 'Fastest')                                 |
| TRAVEL_COST2              | number   | no       | 0       | Travel cost (distance for 'Shortest', time for 'Fastest')                                 |
| POINT_TOLERANCE *(adv)*   | distance | no       | —       | Maximum point distance from network                                                       |
| INCLUDE_BOUNDS *(adv)*    | boolean  | no       | False   | Include upper/lower bound points                                                          |
| OUTPUT_LINES              | sink     | no       | —       | Service area (lines)                                                                      |
| OUTPUT                    | sink     | no       | —       | Service area (boundary nodes)                                                             |

**Outputs:** OUTPUT\_LINES (outputVector), OUTPUT (outputVector)

### Example usage

```
run native:serviceareafrompoint INPUT=input.gpkg START_POINT=100,100 STRATEGY=0
↳ TRAVEL_COST=0 TRAVEL_COST2=0 OUTPUT_LINES=output_lines.gpkg
```

This calls **Service area (from point)**: INPUT=input.gpkg — Vector layer representing network; START\_POINT=100,100 — Start point; STRATEGY=0 selects *Shortest*; TRAVEL\_COST=0 — Travel cost (distance for 'Shortest', time for 'Fastest'); TRAVEL\_COST2=0 — Travel cost (distance for 'Shortest',

time for 'Fastest'); OUTPUT\_LINES=output\_lines.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setlayerencoding — Set layer encoding

**group:** Vector general

This algorithm sets the encoding used for reading a layer's attributes. No permanent changes are made to the layer, rather it affects only how the layer is read during the current session.

Changing the encoding is only supported for some vector layer data sources.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | vector | yes      | —       | Input layer |
| ENCODING  | string | yes      | —       | Encoding    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:setlayerencoding INPUT=input.gpkg ENCODING=value
```

This calls **Set layer encoding**: INPUT=input.gpkg — Input layer; ENCODING=value — Encoding. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setlayermetadata — Set layer metadata

**group:** Metadata tools

This algorithm applies the metadata to a layer. The metadata must be defined as QMD file.

Any existing metadata in the layer will be replaced.

| Parameter | Type    | Required | Default | Description              |
|-----------|---------|----------|---------|--------------------------|
| INPUT     | layer   | yes      | —       | Layer                    |
| METADATA  | file    | yes      | —       | Metadata file            |
| DEFAULT   | boolean | no       | False   | Save metadata as default |

**Outputs:** OUTPUT (outputLayer)

### Example usage

```
run native:setlayermetadata INPUT=input.gpkg METADATA=input.dat DEFAULT=false
```

This calls **Set layer metadata**: INPUT=input.gpkg — Layer; METADATA=input.dat — Metadata file; DEFAULT=false — Save metadata as default. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setlayerstyle — Set layer style

**group:** Cartography

This algorithm applies the style to a layer. The style must be defined as QML file.

| Parameter | Type  | Required | Default | Description |
|-----------|-------|----------|---------|-------------|
| INPUT     | layer | yes      | —       | Layer       |
| STYLE     | file  | yes      | —       | Style file  |

---

**Outputs:** OUTPUT (outputLayer)

**Example usage**

```
run native:setlayerstyle INPUT=input.gpkg STYLE=input.dat
```

This calls **Set layer style**: INPUT=input.gpkg — Layer; STYLE=input.dat — Style file. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setmetadatafields — Set metadata fields

**group:** Metadata tools

This algorithm sets various metadata fields for a layer.

| Parameter         | Type    | Required | Default | Description                 |
|-------------------|---------|----------|---------|-----------------------------|
| INPUT             | layer   | yes      | —       | Layer                       |
| IDENTIFIER        | string  | no       | —       | Identifier                  |
| PARENT_IDENTIFIER | string  | no       | —       | Parent identifier           |
| TITLE             | string  | no       | —       | Title                       |
| TYPE              | string  | no       | —       | Type                        |
| LANGUAGE          | string  | no       | —       | Language                    |
| ENCODING          | string  | no       | —       | Encoding                    |
| ABSTRACT          | string  | no       | —       | Abstract                    |
| CRS               | crs     | no       | —       | Coordinate reference system |
| FEES              | string  | no       | —       | Fees                        |
| IGNORE_EMPTY      | boolean | no       | False   | Ignore empty fields         |

**Outputs:** OUTPUT (outputLayer)

**Example usage**

```
run native:setmetadatafields INPUT=input.gpkg IDENTIFIER=value PARENT_IDENTIFIER=value  
↪ TITLE=value TYPE=value LANGUAGE=value ENCODING=value
```

This calls **Set metadata fields**: INPUT=input.gpkg — Layer; IDENTIFIER=value — Identifier; PARENT\_IDENTIFIER=value — Parent identifier; TITLE=value — Title; TYPE=value — Type; LANGUAGE=value — Language; ENCODING=value — Encoding. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setmfromraster — Set M value from raster

**group:** Vector geometry

This algorithm sets the M value for every vertex in the feature geometry to a value sampled from a band within a raster layer.

The raster values can optionally be scaled by a preset amount and an offset can be algebraically added.

| Parameter | Type   | Required | Default | Description                                   |
|-----------|--------|----------|---------|-----------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                   |
| RASTER    | raster | yes      | —       | Raster layer                                  |
| BAND      | band   | no       | 1       | Band number                                   |
| NODATA    | number | no       | 0.0     | Value for NoData or non-intersecting vertices |
| SCALE     | number | no       | 1.0     | Scale factor                                  |
| OFFSET    | number | no       | 0.0     | Offset                                        |

| Parameter | Type | Required | Default | Description |
|-----------|------|----------|---------|-------------|
| OUTPUT    | sink | yes      | —       | Draped      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:setmfromraster INPUT=input.gpkg RASTER=dem.tif BAND=1 NODATA=0 SCALE=1 OFFSET=0
↳ OUTPUT=output.gpkg
```

This calls **Set M value from raster**: INPUT=input.gpkg — Input layer; RASTER=dem.tif — Raster layer; BAND=1 — Band number; NODATA=0 — Value for NoData or non-intersecting vertices; SCALE=1 — Scale factor; OFFSET=0 — Offset; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setmvalue — Set M value

**group:** Vector geometry

This algorithm sets the M value for geometries in a layer.

If M values already exist in the layer, they will be overwritten with the new value. If no M values exist, the geometry will be upgraded to include M values and the specified value used as the initial M value for all geometries.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| M_VALUE   | number | no       | 0.0     | M Value     |
| OUTPUT    | sink   | yes      | —       | M Added     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:setmvalue INPUT=input.gpkg M_VALUE=0 OUTPUT=output.gpkg
```

This calls **Set M value**: INPUT=input.gpkg — Input layer; M\_VALUE=0 — M Value; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setprojectvariable — Set project variable

**group:** Modeler tools

This algorithm sets an expression variable for the current project.

| Parameter | Type   | Required | Default | Description    |
|-----------|--------|----------|---------|----------------|
| NAME      | string | yes      | —       | Variable name  |
| VALUE     | string | yes      | —       | Variable value |

### Example usage

```
run native:setprojectvariable NAME=value VALUE=value
```

This calls **Set project variable**: NAME=value — Variable name; VALUE=value — Variable value. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## native:setzfromraster — Drape (set Z value from raster)

**group:** Vector geometry

This algorithm sets the z value of every vertex in the feature geometry to a value sampled from a band within a raster layer.

The raster values can optionally be scaled by a preset amount and an offset can be algebraically added.

| Parameter | Type   | Required | Default | Description                                   |
|-----------|--------|----------|---------|-----------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                   |
| RASTER    | raster | yes      | —       | Raster layer                                  |
| BAND      | band   | no       | 1       | Band number                                   |
| NODATA    | number | no       | 0.0     | Value for NoData or non-intersecting vertices |
| SCALE     | number | no       | 1.0     | Scale factor                                  |
| OFFSET    | number | no       | 0.0     | Offset                                        |
| OUTPUT    | sink   | yes      | —       | Draped                                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:setzfromraster INPUT=input.gpkg RASTER=dem.tif BAND=1 NODATA=0 SCALE=1 OFFSET=0  
↪ OUTPUT=output.gpkg
```

This calls **Drape (set Z value from raster)**: INPUT=input.gpkg — Input layer; RASTER=dem.tif — Raster layer; BAND=1 — Band number; NODATA=0 — Value for NoData or non-intersecting vertices; SCALE=1 — Scale factor; OFFSET=0 — Offset; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:setzvalue — Set Z value

**group:** Vector geometry

This algorithm sets the Z value for geometries in a layer.

If Z values already exist in the layer, they will be overwritten with the new value. If no Z values exist, the geometry will be upgraded to include Z values and the specified value used as the initial Z value for all geometries.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| Z_VALUE   | number | no       | 0.0     | Z Value     |
| OUTPUT    | sink   | yes      | —       | Z Added     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:setzvalue INPUT=input.gpkg Z_VALUE=0 OUTPUT=output.gpkg
```

This calls **Set Z value**: INPUT=input.gpkg — Input layer; Z\_VALUE=0 — Z Value; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:shortestline — Shortest line between features

**group:** Vector analysis

This algorithm creates a line layer as the shortest line between the source and the destination layer. By default only the first nearest feature of the destination layer is taken into account. The n-nearest neighboring features number can be specified.

If a maximum distance is specified, then only features which are closer than this distance will be considered.

The output features will contain all the source layer attributes, all the attributes from the n-nearest feature and the additional field of the distance.

This algorithm uses purely Cartesian calculations for distance, and does not consider geodetic or ellipsoid properties when determining feature proximity. The measurement and output coordinate system is based on the coordinate system of the source layer.

| Parameter   | Type     | Required | Default | Description                                                                              |
|-------------|----------|----------|---------|------------------------------------------------------------------------------------------|
| SOURCE      | source   | yes      | —       | Source layer                                                                             |
| DESTINATION | source   | yes      | —       | Destination layer                                                                        |
| METHOD      | enum     | no       | 0       | Method — options: 0=Distance to Nearest Point on feature, 1=Distance to Feature Centroid |
| NEIGHBORS   | number   | no       | 1       | Maximum number of neighbors                                                              |
| DISTANCE    | distance | no       | —       | Maximum distance                                                                         |
| OUTPUT      | sink     | yes      | —       | Shortest lines                                                                           |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:shortestline SOURCE=input.gpkg DESTINATION=input.gpkg METHOD=0 NEIGHBORS=1  
↳ DISTANCE=10 OUTPUT=output.gpkg
```

This calls **Shortest line between features**: SOURCE=input.gpkg — Source layer; DESTINATION=input.gpkg — Destination layer; METHOD=0 selects *Distance to Nearest Point on feature*; NEIGHBORS=1 — Maximum number of neighbors; DISTANCE=10 — Maximum distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:shortestpathlayertopoint — Shortest path (layer to point)

**group:** Network analysis

This algorithm computes optimal (shortest or fastest) routes from multiple start points defined by a vector layer and a given end point.

| Parameter                 | Type     | Required | Default | Description                                                                               |
|---------------------------|----------|----------|---------|-------------------------------------------------------------------------------------------|
| INPUT                     | source   | yes      | —       | Vector layer representing network                                                         |
| STRATEGY                  | enum     | no       | 0       | Path type to calculate — options: 0=Shortest, 1=Fastest                                   |
| DIRECTION_FIELD *(adv)*   | field    | no       | —       | Direction field                                                                           |
| VALUE_FORWARD *(adv)*     | string   | no       | —       | Value for forward direction                                                               |
| VALUE_BACKWARD *(adv)*    | string   | no       | —       | Value for backward direction                                                              |
| VALUE_BOTH *(adv)*        | string   | no       | —       | Value for both directions                                                                 |
| DEFAULT_DIRECTION *(adv)* | enum     | no       | 2       | Default direction — options: 0=Forward direction, 1=Backward direction, 2=Both directions |
| SPEED_FIELD *(adv)*       | field    | no       | —       | Speed field                                                                               |
| DEFAULT_SPEED *(adv)*     | number   | no       | 50      | Default speed (km/h)                                                                      |
| TOLERANCE *(adv)*         | distance | no       | 0       | Topology tolerance                                                                        |

| Parameter               | Type     | Required | Default | Description                         |
|-------------------------|----------|----------|---------|-------------------------------------|
| START_POINTS            | source   | yes      | —       | Vector layer with start points      |
| END_POINT               | point    | yes      | —       | End point                           |
| POINT_TOLERANCE *(adv)* | distance | no       | —       | Maximum point distance from network |
| OUTPUT                  | sink     | yes      | —       | Shortest path                       |
| OUTPUT_NON_ROUTABLE     | sink     | no       | —       | Non-routable features               |

**Outputs:** OUTPUT (outputVector), OUTPUT\_NON\_ROUTABLE (outputVector)

### Example usage

```
run native:shortestpathlayertopoint INPUT=input.gpkg START_POINTS=input.gpkg
↳ END_POINT=100,100 STRATEGY=0 OUTPUT=output.gpkg
```

This calls **Shortest path (layer to point)**: INPUT=input.gpkg — Vector layer representing network; START\_POINTS=input.gpkg — Vector layer with start points; END\_POINT=100,100 — End point; STRATEGY=0 selects *Shortest*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:shortestpathpointtolayer — Shortest path (point to layer)

**group:** Network analysis

This algorithm computes optimal (shortest or fastest) route between a given start point and multiple end points defined by a point vector layer.

| Parameter                 | Type     | Required | Default | Description                                                                               |
|---------------------------|----------|----------|---------|-------------------------------------------------------------------------------------------|
| INPUT                     | source   | yes      | —       | Vector layer representing network                                                         |
| STRATEGY                  | enum     | no       | 0       | Path type to calculate — options: 0=Shortest, 1=Fastest                                   |
| DIRECTION_FIELD *(adv)*   | field    | no       | —       | Direction field                                                                           |
| VALUE_FORWARD *(adv)*     | string   | no       | —       | Value for forward direction                                                               |
| VALUE_BACKWARD *(adv)*    | string   | no       | —       | Value for backward direction                                                              |
| VALUE_BOTH *(adv)*        | string   | no       | —       | Value for both directions                                                                 |
| DEFAULT_DIRECTION *(adv)* | enum     | no       | 2       | Default direction — options: 0=Forward direction, 1=Backward direction, 2=Both directions |
| SPEED_FIELD *(adv)*       | field    | no       | —       | Speed field                                                                               |
| DEFAULT_SPEED *(adv)*     | number   | no       | 50      | Default speed (km/h)                                                                      |
| TOLERANCE *(adv)*         | distance | no       | 0       | Topology tolerance                                                                        |
| START_POINT               | point    | yes      | —       | Start point                                                                               |
| END_POINTS                | source   | yes      | —       | Vector layer with end points                                                              |
| POINT_TOLERANCE *(adv)*   | distance | no       | —       | Maximum point distance from network                                                       |
| OUTPUT                    | sink     | yes      | —       | Shortest path                                                                             |
| OUTPUT_NON_ROUTABLE       | sink     | no       | —       | Non-routable features                                                                     |

**Outputs:** OUTPUT (outputVector), OUTPUT\_NON\_ROUTABLE (outputVector)

### Example usage

```
run native:shortestpathpointtolayer INPUT=input.gpkg START_POINT=100,100
↳ END_POINTS=input.gpkg STRATEGY=0 OUTPUT=output.gpkg
```

This calls **Shortest path (point to layer)**: INPUT=input.gpkg — Vector layer representing network; START\_POINT=100,100 — Start point; END\_POINTS=input.gpkg — Vector layer with end points; STRATEGY=0 selects *Shortest*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:shortestpathpointtopoint — Shortest path (point to point)

**group:** Network analysis

This algorithm computes optimal (shortest or fastest) route between given start and end points.

| Parameter                 | Type     | Required | Default | Description                                                                               |
|---------------------------|----------|----------|---------|-------------------------------------------------------------------------------------------|
| INPUT                     | source   | yes      | —       | Vector layer representing network                                                         |
| STRATEGY                  | enum     | no       | 0       | Path type to calculate — options: 0=Shortest, 1=Fastest                                   |
| DIRECTION_FIELD *(adv)*   | field    | no       | —       | Direction field                                                                           |
| VALUE_FORWARD *(adv)*     | string   | no       | —       | Value for forward direction                                                               |
| VALUE_BACKWARD *(adv)*    | string   | no       | —       | Value for backward direction                                                              |
| VALUE_BOTH *(adv)*        | string   | no       | —       | Value for both directions                                                                 |
| DEFAULT_DIRECTION *(adv)* | enum     | no       | 2       | Default direction — options: 0=Forward direction, 1=Backward direction, 2=Both directions |
| SPEED_FIELD *(adv)*       | field    | no       | —       | Speed field                                                                               |
| DEFAULT_SPEED *(adv)*     | number   | no       | 50      | Default speed (km/h)                                                                      |
| TOLERANCE *(adv)*         | distance | no       | 0       | Topology tolerance                                                                        |
| START_POINT               | point    | yes      | —       | Start point                                                                               |
| END_POINT                 | point    | yes      | —       | End point                                                                                 |
| OUTPUT                    | sink     | yes      | —       | Shortest path                                                                             |
| POINT_TOLERANCE *(adv)*   | distance | no       | —       | Maximum point distance from network                                                       |

**Outputs:** OUTPUT (outputVector), TRAVEL\_COST (outputNumber)

### Example usage

```
run native:shortestpathpointtopoint INPUT=input.gpkg START_POINT=100,100 END_POINT=100,100  
↳ STRATEGY=0 OUTPUT=output.gpkg
```

This calls **Shortest path (point to point)**: INPUT=input.gpkg — Vector layer representing network; START\_POINT=100,100 — Start point; END\_POINT=100,100 — End point; STRATEGY=0 selects *Shortest*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:shpencodinginfo — Extract Shapefile encoding

**group:** Vector general

This algorithm extracts the attribute encoding information embedded in a Shapefile.

Both the encoding specified by an optional .cpg file and any encoding details present in the .dbf LDID header block are considered.

| Parameter | Type | Required | Default | Description |
|-----------|------|----------|---------|-------------|
| INPUT     | file | yes      | —       | Input layer |

**Outputs:** ENCODING (outputString), CPG\_ENCODING (outputString), LDID\_ENCODING (outputString)

### Example usage

```
run native:shpencodinginfo INPUT=input.dat
```

This calls **Extract Shapefile encoding**: INPUT=input.dat — Input layer. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:simplifygeometries — Simplify ★

**group:** Vector geometry · **niva verb:** simplify

This algorithm simplifies the geometries in a line or polygon layer. It creates a new layer with the same features as the ones in the input layer, but with geometries containing a lower number of vertices.

The algorithm gives a choice of simplification methods, including distance based (the "Douglas-Peucker" algorithm), area based ("Visvalingam" algorithm) and snapping geometries to a grid.

| Parameter | Type     | Required | Default | Description                                                                                         |
|-----------|----------|----------|---------|-----------------------------------------------------------------------------------------------------|
| INPUT     | source   | yes      | —       | Input layer                                                                                         |
| METHOD    | enum     | no       | 0       | Simplification method — options: 0=Distance (Douglas-Peucker), 1=Snap to grid, 2=Area (Visvalingam) |
| TOLERANCE | distance | no       | 1.0     | Tolerance                                                                                           |
| OUTPUT    | sink     | yes      | —       | Simplified                                                                                          |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:simplifygeometries INPUT=input.gpkg METHOD=0 TOLERANCE=1 OUTPUT=output.gpkg
```

This calls **Simplify**: INPUT=input.gpkg — Input layer; METHOD=0 selects *Distance (Douglas-Peucker)*; TOLERANCE=1 — Tolerance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:singlesidedbuffer — Single sided buffer

**group:** Vector geometry

This algorithm buffers lines by a specified distance on one side of the line only.

The segments parameter controls the number of line segments to use to approximate a quarter circle when creating rounded buffers. The join style parameter specifies whether round, miter or beveled joins should be used when buffering corners in a line. The miter limit parameter is only applicable for miter join styles, and controls the maximum distance from the buffer to use when creating a mitered join.

| Parameter   | Type     | Required | Default | Description                                     |
|-------------|----------|----------|---------|-------------------------------------------------|
| INPUT       | source   | yes      | —       | Input layer                                     |
| DISTANCE    | distance | no       | 10      | Distance                                        |
| SIDE        | enum     | no       | 0       | Side — options: 0=Left, 1=Right                 |
| SEGMENTS    | number   | no       | 8       | Segments                                        |
| JOIN_STYLE  | enum     | no       | 0       | Join style — options: 0=Round, 1=Miter, 2=Bevel |
| MITER_LIMIT | number   | no       | 2       | Miter limit                                     |
| OUTPUT      | sink     | yes      | —       | Buffered                                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:singlesidedbuffer INPUT=input.gpkg DISTANCE=10 SIDE=0 SEGMENTS=8 JOIN_STYLE=0  
↪ MITER_LIMIT=2 OUTPUT=output.gpkg
```

This calls **Single sided buffer**: INPUT=input.gpkg — Input layer; DISTANCE=10 — Distance; SIDE=0 selects *Left*; SEGMENTS=8 — Segments; JOIN\_STYLE=0 selects *Round*; MITER\_LIMIT=2 — Miter limit; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:slope — Slope

**group:** Raster terrain analysis

This algorithm calculates the angle of inclination of the terrain from an input raster layer. The slope is expressed in degrees.

| Parameter                | Type              | Required | Default | Description         |
|--------------------------|-------------------|----------|---------|---------------------|
| INPUT                    | raster            | yes      | —       | Elevation layer     |
| Z_FACTOR                 | number            | no       | 1.0     | Z factor            |
| NODATA *(adv)*           | number            | no       | -9999.0 | Output NoData value |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options    |
| OUTPUT                   | rasterDestination | yes      | —       | Slope               |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:slope INPUT=dem.tif Z_FACTOR=1 OUTPUT=output.tif
```

This calls **Slope**: INPUT=dem.tif — Elevation layer; Z\_FACTOR=1 — Z factor; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:smoothgeometry — Smooth ★

**group:** Vector geometry · **niva verb:** smooth

This algorithm smooths the geometries in a line or polygon layer. It creates a new layer with the same features as the ones in the input layer, but with geometries containing a higher number of vertices and corners in the geometries smoothed out.

The iterations parameter dictates how many smoothing iterations will be applied to each geometry. A higher number of iterations results in smoother geometries with the cost of greater number of nodes in the geometries.

The offset parameter controls how “tightly” the smoothed geometries follow the original geometries. Smaller values results in a tighter fit, and larger values will create a looser fit.

The maximum angle parameter can be used to prevent smoothing of nodes with large angles. Any node where the angle of the segments to either side is larger than this will not be smoothed. For example, setting the maximum angle to 90 degrees or lower would preserve right angles in the geometry.

If input geometries contain Z or M values, these will also be smoothed and the output geometry will retain the same dimensionality as the input geometry.

| Parameter  | Type   | Required | Default | Description                  |
|------------|--------|----------|---------|------------------------------|
| INPUT      | source | yes      | —       | Input layer                  |
| ITERATIONS | number | no       | 1       | Iterations                   |
| OFFSET     | number | no       | 0.25    | Offset                       |
| MAX_ANGLE  | number | no       | 180.0   | Maximum node angle to smooth |
| OUTPUT     | sink   | yes      | —       | Smoothed                     |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:smoothgeometry INPUT=input.gpkg ITERATIONS=1 OFFSET=0.25 MAX_ANGLE=180
↳ OUTPUT=output.gpkg
```

This calls **Smooth**: INPUT=input.gpkg — Input layer; ITERATIONS=1 — Iterations; OFFSET=0.25 — Offset; MAX\_ANGLE=180 — Maximum node angle to smooth; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:snapgeometries — Snap geometries to layer ★

**group:** Vector geometry · **niva verb:** snap

This algorithm snaps the geometries in a layer. Snapping can be done either to the geometries from another layer, or to geometries within the same layer.

A tolerance is specified in layer units to control how close vertices need to be to the reference layer geometries before they are snapped.

Snapping occurs to both nodes and edges. Depending on the snapping behavior, either nodes or edges will be preferred.

Vertices will be inserted or removed as required to make the geometries match the reference geometries.

| Parameter       | Type     | Required | Default | Description                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|----------|----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT           | source   | yes      | —       | Input layer                                                                                                                                                                                                                                                                                                                                                                                                                           |
| REFERENCE_LAYER | source   | yes      | —       | Reference layer                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TOLERANCE       | distance | no       | 10.0    | Tolerance                                                                                                                                                                                                                                                                                                                                                                                                                             |
| BEHAVIOR        | enum     | no       | [0]     | Behavior — options: 0=Prefer aligning nodes, insert extra vertices where required, 1=Prefer closest point, insert extra vertices where required, 2=Prefer aligning nodes, don't insert new vertices, 3=Prefer closest point, don't insert new vertices, 4=Move end points only, prefer aligning nodes, 5=Move end points only, prefer closest point, 6=Snap end points to end points only, 7=Snap to anchor nodes (single layer only) |
| OUTPUT          | sink     | yes      | —       | Snapped geometry                                                                                                                                                                                                                                                                                                                                                                                                                      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:snapgeometries INPUT=input.gpkg REFERENCE_LAYER=input.gpkg TOLERANCE=10  
↪ BEHAVIOR=0 OUTPUT=output.gpkg
```

This calls **Snap geometries to layer**: INPUT=input.gpkg — Input layer; REFERENCE\_LAYER=input.gpkg — Reference layer; TOLERANCE=10 — Tolerance; BEHAVIOR=0 selects *Prefer aligning nodes, insert extra vertices where required*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:snappointstogrid — Snap points to grid

**group:** Vector geometry

This algorithm modifies the coordinates of geometries in a vector layer, so that all points or vertices are snapped to the closest point of the grid.

If the snapped geometry cannot be calculated (or is totally collapsed) the feature's geometry will be cleared.

Note that snapping to grid may generate an invalid geometry in some corner cases.

Snapping can be performed on the X, Y, Z or M axis. A grid spacing of 0 for any axis will disable snapping for that axis.

| Parameter | Type     | Required | Default | Description    |
|-----------|----------|----------|---------|----------------|
| INPUT     | source   | yes      | —       | Input layer    |
| HSPACING  | distance | no       | 1       | X Grid Spacing |
| VSPACING  | distance | no       | 1       | Y Grid Spacing |
| ZSPACING  | number   | no       | 0       | Z Grid Spacing |
| MSPACING  | number   | no       | 0       | M Grid Spacing |
| OUTPUT    | sink     | yes      | —       | Snapped        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:snappointstogrid INPUT=input.gpkg HSPACING=1 VSPACING=1 ZSPACING=0 MSPACING=0
↳ OUTPUT=output.gpkg
```

This calls **Snap points to grid**: INPUT=input.gpkg — Input layer; HSPACING=1 — X Grid Spacing; VSPACING=1 — Y Grid Spacing; ZSPACING=0 — Z Grid Spacing; MSPACING=0 — M Grid Spacing; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:spatialiteexecutesql — Spatialite execute SQL

**group:** Database

This algorithm executes a SQL command on a Spatialite database. The database is determined by an input layer or file.

| Parameter | Type   | Required | Default | Description              |
|-----------|--------|----------|---------|--------------------------|
| DATABASE  | vector | yes      | —       | Database layer (or file) |
| SQL       | string | yes      | —       | SQL query                |

### Example usage

```
run native:spatialiteexecutesql DATABASE=input.gpkg SQL=value
```

This calls **Spatialite execute SQL**: DATABASE=input.gpkg — Database layer (or file); SQL=value — SQL query. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:spatialiteexecutesqlregistered — Spatialite execute SQL (registered DB)

**group:** Database

This algorithm executes a SQL command on a Spatialite database.

| Parameter | Type               | Required | Default | Description                |
|-----------|--------------------|----------|---------|----------------------------|
| DATABASE  | providerconnection | yes      | —       | Database (connection name) |
| SQL       | string             | yes      | —       | SQL query                  |

### Example usage

```
run native:spatialiteexecutesqlregistered DATABASE=... SQL=value
```

This calls **Spatialite execute SQL (registered DB)**: DATABASE=... — Database (connection name); SQL=value — SQL query. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## native:splitfeaturesbycharacter — Split features by character

**group:** Vector general

This algorithm splits features into multiple output features by splitting a field's value with a specified character.

For instance, if a layer contains features with multiple comma separated values contained in a single field, this algorithm can be used to split these values up across multiple output features.

Geometries and other attributes remain unchanged in the output.

Optionally, the separator string can be a regular expression for added flexibility.

| Parameter     | Type    | Required | Default | Description                      |
|---------------|---------|----------|---------|----------------------------------|
| INPUT         | source  | yes      | —       | Input layer                      |
| FIELD         | field   | yes      | —       | Split using values in field      |
| CHAR          | string  | yes      | —       | Split values using character     |
| REGEX *(adv)* | boolean | yes      | —       | Use regular expression separator |
| OUTPUT        | sink    | yes      | —       | Split                            |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:splitfeaturesbycharacter INPUT=input.gpkg FIELD=field1 CHAR=value REGEX=false  
↪ OUTPUT=output.gpkg
```

This calls **Split features by character**: INPUT=input.gpkg — Input layer; FIELD=field1 — Split using values in field; CHAR=value — Split values using character; REGEX=false — Use regular expression separator; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:splitlinesbylength — Split lines by maximum length

**group:** Vector geometry

This algorithm takes a line (or curve) layer and splits each feature into multiple parts, where each part is of a specified maximum length.

Z and M values at the start and end of the new line substrings are linearly interpolated from existing values.

| Parameter | Type     | Required | Default | Description         |
|-----------|----------|----------|---------|---------------------|
| INPUT     | source   | yes      | —       | Input layer         |
| LENGTH    | distance | no       | 10      | Maximum line length |
| OUTPUT    | sink     | yes      | —       | Split               |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:splitlinesbylength INPUT=input.gpkg LENGTH=10 OUTPUT=output.gpkg
```

This calls **Split lines by maximum length**: INPUT=input.gpkg — Input layer; LENGTH=10 — Maximum line length; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:splitvectorlayer — Split vector layer

**group:** Vector general

This algorithm splits input vector layer into multiple layers by specified unique ID field.

Each of the layers created in the output folder contains all features from the input layer with the same value for the specified attribute. The number of files generated is equal to the number of different values found for the specified attribute.

| Parameter         | Type              | Required | Default | Description                                                                                                                                                                                                               |
|-------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT             | source            | yes      | —       | Input layer                                                                                                                                                                                                               |
| FIELD             | field             | yes      | —       | Unique ID field                                                                                                                                                                                                           |
| PREFIX_FIELD      | boolean           | no       | True    | Add field prefix to file names                                                                                                                                                                                            |
| FILE_TYPE *(adv)* | enum              | no       | 0       | Output file type — options: 0=gpkg, 1=shp, 2=000, 3=csv, 4=dgn, 5=dxg, 6=fgb, 7=gdb, 8=geojson, 9=geojsonl, 10=geojsons, 11=gml, 12=gpkg.zip, 13=gpx, 14=json, 15=kml, 16=ods, 17=sql, 18=sqlite, 19=tab, 20=xlsx, 21=xml |
| OUTPUT            | folderDestination | yes      | —       | Output directory                                                                                                                                                                                                          |

**Outputs:** OUTPUT (outputFolder), OUTPUT\_LAYERS (outputMultilayer)

### Example usage

```
run native:splitvectorlayer INPUT=input.gpkg FIELD=field1 PREFIX_FIELD=true OUTPUT=output/
```

This calls **Split vector layer**: INPUT=input.gpkg — Input layer; FIELD=field1 — Unique ID field; PREFIX\_FIELD=true — Add field prefix to file names; OUTPUT=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:splitwithlines — Split with lines

**group:** Vector overlay

This algorithm splits the lines or polygons in one layer using the lines or polygon rings in another layer to define the breaking points. Intersection between geometries in both layers are considered as split points.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| LINES     | source | yes      | —       | Split layer |
| OUTPUT    | sink   | yes      | —       | Split       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:splitwithlines INPUT=input.gpkg LINES=input.gpkg OUTPUT=output.gpkg
```

This calls **Split with lines**: INPUT=input.gpkg — Input layer; LINES=input.gpkg — Split layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:stdbscanclustering — ST-DBSCAN clustering

**group:** Vector analysis

Clusters point features based on a 2D implementation of spatiotemporal density-based clustering of applications with noise (ST-DBSCAN) algorithm.

For more details, please see the following papers:

- Ester, M., H. P. Kriegel, J. Sander, and X. Xu, "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise". In: Proceedings of the 2nd International Conference on Knowledge Discovery and Data Mining, Portland, OR, AAAI Press, pp. 226-231. 1996
- Birant, Derya, and Alp Kut. "ST-DBSCAN: An algorithm for clustering spatial-temporal data." Data & Knowledge Engineering 60.1 (2007): 208-221.
- Peca, I., Fuchs, G., Vrotsou, K., Andrienko, N. V., & Andrienko, G. L. (2012). Scalable Cluster Analysis of Spatial Events. In EuroVA@ EuroVis.

| Parameter               | Type     | Required | Default      | Description                                    |
|-------------------------|----------|----------|--------------|------------------------------------------------|
| INPUT                   | source   | yes      | —            | Input layer                                    |
| DATETIME_FIELD          | field    | yes      | —            | Date/time field                                |
| MIN_SIZE                | number   | no       | 5            | Minimum cluster size                           |
| EPS                     | distance | no       | 1            | Maximum distance between clustered points      |
| EPS2                    | duration | no       | 0            | Maximum time duration between clustered points |
| DBSCAN* *(adv)*         | boolean  | no       | False        | Treat border points as noise (DBSCAN*)         |
| FIELD_NAME *(adv)*      | string   | no       | CLUSTER_ID   | Cluster field name                             |
| SIZE_FIELD_NAME *(adv)* | string   | no       | CLUSTER_SIZE | Cluster size field name                        |
| OUTPUT                  | sink     | yes      | —            | Clusters                                       |

**Outputs:** OUTPUT (outputVector), NUM\_CLUSTERS (outputNumber)

### Example usage

```
run native:stdbscanclustering INPUT=input.gpkg DATETIME_FIELD=field1 MIN_SIZE=5 EPS=1
↳ EPS2=0 OUTPUT=output.gpkg
```

This calls **ST-DBSCAN clustering**: INPUT=input.gpkg — Input layer; DATETIME\_FIELD=field1 — Date/time field; MIN\_SIZE=5 — Minimum cluster size; EPS=1 — Maximum distance between clustered points; EPS2=0 — Maximum time duration between clustered points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:stringconcatenation — String concatenation

**group:** Modeler tools

This algorithm concatenates two strings together.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT_1   | string | yes      | —       | Input 1     |
| INPUT_2   | string | yes      | —       | Input 2     |

**Outputs:** CONCATENATION (outputString)

### Example usage

```
run native:stringconcatenation INPUT_1=value INPUT_2=value
```

This calls **String concatenation**: INPUT\_1=value — Input 1; INPUT\_2=value — Input 2. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:stylefromproject — Create style database from project

**group:** Cartography

This algorithm extracts all style objects (including symbols, color ramps, text formats and label settings) from a QGIS project.

The extracted symbols are saved to a QGIS style database (XML format), which can be managed and imported via the Style Manager dialog.

| Parameter | Type            | Required | Default      | Description                                                                              |
|-----------|-----------------|----------|--------------|------------------------------------------------------------------------------------------|
| INPUT     | file            | no       | —            | Input project (leave blank to use current)                                               |
| OUTPUT    | fileDestination | yes      | —            | Output style database                                                                    |
| OBJECTS   | enum            | no       | [0, 1, 2, 3] | Objects to extract — options: 0=Symbols, 1=Color ramps, 2=Text formats, 3=Label settings |

**Outputs:** OUTPUT (outputFile), SYMBOLS (outputNumber), COLORRAMPS (outputNumber), TEXTFORMATS (outputNumber), LABELSETTINGS (outputNumber)

### Example usage

```
run native:stylefromproject INPUT=input.dat OBJECTS=0 OUTPUT=output.txt
```

This calls **Create style database from project**: INPUT=input.dat — Input project (leave blank to use current); OBJECTS=0 selects *Symbols*; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:subdivide — Subdivide ★

**group:** Vector geometry · **niva verb:** subdivide

This algorithm subdivides the geometry. The returned geometry will be a collection containing subdivided parts from the original geometry, where no part has more than the specified maximum number of nodes.

This is useful for dividing a complex geometry into less complex parts, which are better able to be spatially indexed and faster to perform further operations such as intersects on. The returned geometry parts may not be valid and may contain self-intersections.

Curved geometries will be segmentized before subdivision.

| Parameter | Type   | Required | Default | Description            |
|-----------|--------|----------|---------|------------------------|
| INPUT     | source | yes      | —       | Input layer            |
| MAX_NODES | number | no       | 256     | Maximum nodes in parts |
| OUTPUT    | sink   | yes      | —       | Subdivided             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:subdivide INPUT=input.gpkg MAX_NODES=256 OUTPUT=output.gpkg
```

This calls **Subdivide**: INPUT=input.gpkg — Input layer; MAX\_NODES=256 — Maximum nodes in parts; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:sumlinelengths — Sum line lengths

**group:** Vector analysis

This algorithm takes a polygon layer and a line layer and measures the total length of lines and the total number of them that cross each polygon.

The resulting layer has the same features as the input polygon layer, but with two additional attributes containing the length and count of the lines across each polygon. The names of these two fields can be configured in the algorithm parameters.

| Parameter   | Type   | Required | Default | Description             |
|-------------|--------|----------|---------|-------------------------|
| POLYGONS    | source | yes      | —       | Polygons                |
| LINES       | source | yes      | —       | Lines                   |
| LEN_FIELD   | string | no       | LENGTH  | Lines length field name |
| COUNT_FIELD | string | no       | COUNT   | Lines count field name  |
| OUTPUT      | sink   | yes      | —       | Line length             |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:sumlinelengths POLYGONS=input.gpkg LINES=input.gpkg LEN_FIELD=LENGTH
↪ COUNT_FIELD=COUNT OUTPUT=output.gpkg
```

This calls **Sum line lengths**: POLYGONS=input.gpkg — Polygons; LINES=input.gpkg — Lines; LEN\_FIELD=LENGTH — Lines length field name; COUNT\_FIELD=COUNT — Lines count field name; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

### native:surfacetopolygon — Surface to polygon

**group:** Mesh

This algorithm exports a polygon layer containing a mesh layer's boundary. It may contain holes and it may be a multi-part polygon.

| Parameter  | Type | Required | Default | Description              |
|------------|------|----------|---------|--------------------------|
| INPUT      | mesh | yes      | —       | Input mesh layer         |
| CRS_OUTPUT | crs  | no       | —       | Output coordinate system |
| OUTPUT     | sink | yes      | —       | Output vector layer      |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:surfacetopolygon INPUT=mesh.nc CRS_OUTPUT=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Surface to polygon**: INPUT=mesh.nc — Input mesh layer; CRS\_OUTPUT=EPSG:6346 — Output coordinate system; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

### native:swapxy — Swap X and Y coordinates ★

**group:** Vector geometry · **niva verb:** swapxy

This algorithm swaps the X and Y coordinate values in input geometries. It can be used to repair geometries which have accidentally had their latitude and longitude values reversed.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Swapped     |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run native:swapxy INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Swap X and Y coordinates**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:symmetricaldifference — Symmetrical difference ★

**group:** Vector overlay · **niva verb:** symdifference

This algorithm extracts the portions of features from both the Input and Overlay layers that do not overlap. Overlapping areas between the two layers are removed. The attribute table of the Symmetrical Difference layer contains original attributes from both the Input and Overlay layers.

| Parameter                     | Type   | Required | Default | Description            |
|-------------------------------|--------|----------|---------|------------------------|
| INPUT                         | source | yes      | —       | Input layer            |
| OVERLAY                       | source | yes      | —       | Overlay layer          |
| OVERLAY_FIELDS_PREFIX *(adv)* | string | no       | —       | Overlay fields prefix  |
| OUTPUT                        | sink   | yes      | —       | Symmetrical difference |
| GRID_SIZE *(adv)*             | number | no       | —       | Grid size              |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:symmetricaldifference INPUT=input.gpkg OVERLAY=input.gpkg OUTPUT=output.gpkg
```

This calls **Symmetrical difference**: INPUT=input.gpkg — Input layer; OVERLAY=input.gpkg — Overlay layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:taperedbuffer — Tapered buffers

**group:** Vector geometry

This algorithm creates tapered buffers along line geometries, using a specified start and end buffer diameter corresponding to the buffer diameter at the start and end of the linestrings.

| Parameter   | Type   | Required | Default | Description |
|-------------|--------|----------|---------|-------------|
| INPUT       | source | yes      | —       | Input layer |
| START_WIDTH | number | no       | 0.0     | Start width |
| END_WIDTH   | number | no       | 1       | End width   |
| SEGMENTS    | number | no       | 16      | Segments    |
| OUTPUT      | sink   | yes      | —       | Buffered    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:taperedbuffer INPUT=input.gpkg START_WIDTH=0 END_WIDTH=1 SEGMENTS=16  
↪ OUTPUT=output.gpkg
```

This calls **Tapered buffers**: INPUT=input.gpkg — Input layer; START\_WIDTH=0 — Start width; END\_WIDTH=1 — End width; SEGMENTS=16 — Segments; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:tilexyzdirectory — Generate XYZ tiles (Directory)

**group:** Raster tools

Generates XYZ tiles of map canvas content and saves them as individual images in a directory.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| EXTENT    | extent | yes      | —       | Extent      |

| Parameter                | Type              | Required | Default                   | Description                                          |
|--------------------------|-------------------|----------|---------------------------|------------------------------------------------------|
| ZOOM_MIN                 | number            | no       | 12                        | Minimum zoom                                         |
| ZOOM_MAX                 | number            | no       | 12                        | Maximum zoom                                         |
| DPI                      | number            | no       | 96                        | DPI                                                  |
| BACKGROUND_COLOR         | color             | no       | <QColor: RGBA 0, 0, 0, 0> | Background color                                     |
| ANTIALIAS                | boolean           | no       | True                      | Enable antialiasing                                  |
| TILE_FORMAT              | enum              | no       | 0                         | Tile format — options: 0=PNG, 1=JPG                  |
| QUALITY                  | number            | no       | 75                        | Quality (JPG only)                                   |
| METATILESIZE             | number            | no       | 4                         | Metatile size                                        |
| TILE_WIDTH               | number            | no       | 256                       | Tile width                                           |
| TILE_HEIGHT              | number            | no       | 256                       | Tile height                                          |
| TMS_CONVENTION           | boolean           | no       | False                     | Use inverted tile Y axis (TMS convention)            |
| HTML_TITLE *(adv)*       | string            | no       | —                         | Leaflet HTML output title                            |
| HTML_ATTRIBUTION *(adv)* | string            | no       | —                         | Leaflet HTML output attribution                      |
| HTML_OSM *(adv)*         | boolean           | no       | False                     | Include OpenStreetMap basemap in Leaflet HTML output |
| OUTPUT_DIRECTORY         | folderDestination | yes      | —                         | Output directory                                     |
| OUTPUT_HTML              | fileDestination   | no       | —                         | Output html (Leaflet)                                |

**Outputs:** OUTPUT\_DIRECTORY (outputFolder), OUTPUT\_HTML (outputHtml)

### Example usage

```
run native:tilexyzdirectory EXTENT=0,1000,0,1000 ZOOM_MIN=12 ZOOM_MAX=12 DPI=96
↳ BACKGROUND_COLOR="#1f78b4" ANTIALIAS=true TILE_FORMAT=0
↳ OUTPUT_DIRECTORY=output_directory/
```

This calls **Generate XYZ tiles (Directory)**: EXTENT=0,1000,0,1000 — Extent; ZOOM\_MIN=12 — Minimum zoom; ZOOM\_MAX=12 — Maximum zoom; DPI=96 — DPI; BACKGROUND\_COLOR="#1f78b4" — Background color; ANTIALIAS=true — Enable antialiasing; TILE\_FORMAT=0 selects PNG; OUTPUT\_DIRECTORY=output\_directory/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:tilexyzmbtiles — Generate XYZ tiles (MBTiles)

**group:** Raster tools

Generates XYZ tiles of map canvas content and saves them as an MBTiles file.

| Parameter        | Type            | Required | Default                   | Description                         |
|------------------|-----------------|----------|---------------------------|-------------------------------------|
| EXTENT           | extent          | yes      | —                         | Extent                              |
| ZOOM_MIN         | number          | no       | 12                        | Minimum zoom                        |
| ZOOM_MAX         | number          | no       | 12                        | Maximum zoom                        |
| DPI              | number          | no       | 96                        | DPI                                 |
| BACKGROUND_COLOR | color           | no       | <QColor: RGBA 0, 0, 0, 0> | Background color                    |
| ANTIALIAS        | boolean         | no       | True                      | Enable antialiasing                 |
| TILE_FORMAT      | enum            | no       | 0                         | Tile format — options: 0=PNG, 1=JPG |
| QUALITY          | number          | no       | 75                        | Quality (JPG only)                  |
| METATILESIZE     | number          | no       | 4                         | Metatile size                       |
| OUTPUT_FILE      | fileDestination | yes      | —                         | Output                              |

**Outputs:** OUTPUT\_FILE (outputFile)

### Example usage

```
run native:tilexyzmbtiles EXTENT=0,1000,0,1000 ZOOM_MIN=12 ZOOM_MAX=12 DPI=96
↳ BACKGROUND_COLOR="#1f78b4" ANTIALIAS=true TILE_FORMAT=0 OUTPUT_FILE=output_file.txt
```

This calls **Generate XYZ tiles (MBTiles)**: EXTENT=0,1000,0,1000 — Extent; ZOOM\_MIN=12 — Minimum zoom; ZOOM\_MAX=12 — Maximum zoom; DPI=96 — DPI; BACKGROUND\_COLOR="#1f78b4" — Background color; ANTIALIAS=true — Enable antialiasing; TILE\_FORMAT=0 selects PNG; OUTPUT\_FILE=output\_file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:tinmeshcreation — TIN Mesh Creation

**group:** Mesh

This algorithm creates a TIN mesh layer from vector layers.

| Parameter   | Type            | Required | Default | Description                                                         |
|-------------|-----------------|----------|---------|---------------------------------------------------------------------|
| SOURCE_DATA | tininputlayers  | yes      | —       | Input layers                                                        |
| MESH_FORMAT | enum            | no       | 0       | Output format — options: 0=2DM, 1=SELAFIN, 2=PLY, 3=Ugrid, 4=Mike21 |
| CRS_OUTPUT  | crs             | no       | —       | Output coordinate system                                            |
| OUTPUT_MESH | fileDestination | yes      | —       | Output file                                                         |

**Outputs:** OUTPUT\_MESH (outputFile)

### Example usage

```
run native:tinmeshcreation SOURCE_DATA=... MESH_FORMAT=0 CRS_OUTPUT=EPSG:6346
↳ OUTPUT_MESH=output_mesh.txt
```

This calls **TIN Mesh Creation**: SOURCE\_DATA=... — Input layers; MESH\_FORMAT=0 selects 2DM; CRS\_OUTPUT=EPSG:6346 — Output coordinate system; OUTPUT\_MESH=output\_mesh.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:totalcurvature — Total curvature

**group:** Raster terrain analysis

This algorithm calculates total curvature of the input raster as described by Wilson, Gallant (2000): terrain analysis.

| Parameter                | Type              | Required | Default | Description         |
|--------------------------|-------------------|----------|---------|---------------------|
| INPUT                    | raster            | yes      | —       | Elevation layer     |
| Z_FACTOR                 | number            | no       | 1.0     | Z factor            |
| NODATA *(adv)*           | number            | no       | -9999.0 | Output NoData value |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Creation options    |
| OUTPUT                   | rasterDestination | yes      | —       | Total curvature     |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:totalcurvature INPUT=dem.tif Z_FACTOR=1 OUTPUT=output.tif
```

This calls **Total curvature**: INPUT=dem.tif — Elevation layer; Z\_FACTOR=1 — Z factor; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## native:transect — Transect

**group:** Vector geometry

This algorithm creates transects on vertices for (multi)linestrings. A transect is a line oriented from an angle (by default perpendicular) to the input polylines (at vertices).

Field(s) from feature(s) are returned in the transect with these new fields:

- TR\_FID: ID of the original feature
- TR\_ID: ID of the transect. Each transect have an unique ID
- TR\_SEGMENT: ID of the segment of the linestring
- TR\_ANGLE: Angle in degrees from the original line at the vertex
- TR\_LENGTH: Total length of the transect returned
- TR\_ORIENT: Side of the transect (only on the left or right of the line, or both side)

| Parameter | Type     | Required | Default | Description                                                     |
|-----------|----------|----------|---------|-----------------------------------------------------------------|
| INPUT     | source   | yes      | —       | Input layer                                                     |
| LENGTH    | distance | no       | 5.0     | Length of the transect                                          |
| ANGLE     | number   | no       | 90.0    | Angle in degrees from the original line at the vertices         |
| SIDE      | enum     | no       | 2       | Side to create the transects — options: 0=Left, 1=Right, 2=Both |
| DIRECTION | enum     | no       | 0       | Direction — options: 0=Right to Left, 1=Left to Right           |
| OUTPUT    | sink     | yes      | —       | Transect                                                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:transect INPUT=input.gpkg LENGTH=5 ANGLE=90 SIDE=2 DIRECTION=0  
↪ OUTPUT=output.gpkg
```

This calls **Transect**: INPUT=input.gpkg — Input layer; LENGTH=5 — Length of the transect; ANGLE=90 — Angle in degrees from the original line at the vertices; SIDE=2 selects *Both*; DIRECTION=0 selects *Right to Left*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:transectfixeddistance — Transect (fixed distance)

**group:** Vector geometry

This algorithm creates transects at fixed distance intervals along (multi)linestrings. A transect is a line oriented from an angle (by default perpendicular) to the input polylines at regular intervals.

Field(s) from feature(s) are returned in the transect with these new fields:

- TR\_FID: ID of the original feature
- TR\_ID: ID of the transect. Each transect have an unique ID
- TR\_SEGMENT: ID of the segment of the linestring
- TR\_ANGLE: Angle in degrees from the original line at the vertex
- TR\_LENGTH: Total length of the transect returned
- TR\_ORIENT: Side of the transect (only on the left or right of the line, or both side)

| Parameter     | Type     | Required | Default | Description                                                     |
|---------------|----------|----------|---------|-----------------------------------------------------------------|
| INPUT         | source   | yes      | —       | Input layer                                                     |
| INTERVAL      | distance | no       | 10.0    | Fixed sampling interval                                         |
| INCLUDE_START | boolean  | no       | True    | Include transect at start of line                               |
| LENGTH        | distance | no       | 5.0     | Length of the transect                                          |
| ANGLE         | number   | no       | 90.0    | Angle in degrees from the original line at the vertices         |
| SIDE          | enum     | no       | 2       | Side to create the transects — options: 0=Left, 1=Right, 2=Both |

| Parameter | Type | Required | Default | Description                                           |
|-----------|------|----------|---------|-------------------------------------------------------|
| DIRECTION | enum | no       | 0       | Direction — options: 0=Right to Left, 1=Left to Right |
| OUTPUT    | sink | yes      | —       | Transect                                              |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:transectfixeddistance INPUT=input.gpkg INTERVAL=10 INCLUDE_START=true LENGTH=5
↳ ANGLE=90 SIDE=2 DIRECTION=0 OUTPUT=output.gpkg
```

This calls **Transect (fixed distance)**: INPUT=input.gpkg — Input layer; INTERVAL=10 — Fixed sampling interval; INCLUDE\_START=true — Include transect at start of line; LENGTH=5 — Length of the transect; ANGLE=90 — Angle in degrees from the original line at the vertices; SIDE=2 selects *Both*; DIRECTION=0 selects *Right to Left*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:transferannotationsfrommain — Transfer annotations from main layer

**group:** Cartography

This algorithm transfers all annotations from the main annotation layer in a project to a new annotation layer.

| Parameter  | Type   | Required | Default     | Description    |
|------------|--------|----------|-------------|----------------|
| LAYER_NAME | string | no       | Annotations | New layer name |

**Outputs:** OUTPUT (outputLayer)

### Example usage

```
run native:transferannotationsfrommain LAYER_NAME=Annotations
```

This calls **Transfer annotations from main layer**: LAYER\_NAME=Annotations — New layer name. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:translategeometry — Translate

**group:** Vector geometry

This algorithm moves the geometries within a layer, by offsetting them with a specified x and y displacement.

Z and M values present in the geometry can also be translated.

| Parameter | Type     | Required | Default | Description                |
|-----------|----------|----------|---------|----------------------------|
| INPUT     | source   | yes      | —       | Input layer                |
| DELTA_X   | distance | no       | 0.0     | Offset distance (x-axis)   |
| DELTA_Y   | distance | no       | 0.0     | Offset distance (y-axis)   |
| DELTA_Z   | number   | no       | 0.0     | Offset distance (z-axis)   |
| DELTA_M   | number   | no       | 0.0     | Offset distance (m values) |
| OUTPUT    | sink     | yes      | —       | Translated                 |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:translategeometry INPUT=input.gpkg DELTA_X=0 DELTA_Y=0 DELTA_Z=0 DELTA_M=0
↳ OUTPUT=output.gpkg
```

This calls **Translate**: INPUT=input.gpkg — Input layer; DELTA\_X=0 — Offset distance (x-axis); DELTA\_Y=0 — Offset distance (y-axis); DELTA\_Z=0 — Offset distance (z-axis); DELTA\_M=0 — Offset distance (m values); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:truncatetable — Truncate table

**group:** Vector general

This algorithm truncates a layer, by deleting all features from within the layer.

Warning — this algorithm modifies the layer in place, and deleted features cannot be restored!

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | vector | yes      | —       | Input layer |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:truncatetable INPUT=input.gpkg
```

This calls **Truncate table**: INPUT=input.gpkg — Input layer. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:union — Union ★

**group:** Vector overlay · **niva verb:** union

This algorithm checks overlaps between features within the Input layer and creates separate features for overlapping and non-overlapping parts. The area of overlap will create as many identical overlapping features as there are features that participate in that overlap.

An Overlay layer can also be used, in which case features from each layer are split at their overlap with features from the other one, creating a layer containing all the portions from both Input and Overlay layers. The attribute table of the Union layer is filled with attribute values from the respective original layer for non-overlapping features, and attribute values from both layers for overlapping features.

| Parameter                     | Type   | Required | Default | Description           |
|-------------------------------|--------|----------|---------|-----------------------|
| INPUT                         | source | yes      | —       | Input layer           |
| OVERLAY                       | source | no       | —       | Overlay layer         |
| OVERLAY_FIELDS_PREFIX *(adv)* | string | no       | —       | Overlay fields prefix |
| OUTPUT                        | sink   | yes      | —       | Union                 |
| GRID_SIZE *(adv)*             | number | no       | —       | Grid size             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:union INPUT=input.gpkg OVERLAY=input.gpkg OUTPUT=output.gpkg
```

This calls **Union**: INPUT=input.gpkg — Input layer; OVERLAY=input.gpkg — Overlay layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:updatelayermetadata — Update layer metadata

**group:** Metadata tools

This algorithm copies all non-empty metadata fields from a source layer to a target layer.

Leaves empty input fields unchanged in the target.

| Parameter | Type  | Required | Default | Description  |
|-----------|-------|----------|---------|--------------|
| SOURCE    | layer | yes      | —       | Source layer |
| TARGET    | layer | yes      | —       | Target layer |

**Outputs:** OUTPUT (outputLayer)

### Example usage

```
run native:updatelayermetadata SOURCE=input.gpkg TARGET=input.gpkg
```

This calls **Update layer metadata**: SOURCE=input.gpkg — Source layer; TARGET=input.gpkg — Target layer. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:uploadgpsdata — Upload GPS data to device

**group:** GPS

This algorithm uses the GPSTabel tool to upload data to a GPS device from the GPX standard format.

| Parameter    | Type   | Required | Default | Description                                             |
|--------------|--------|----------|---------|---------------------------------------------------------|
| INPUT        | file   | yes      | —       | Input file                                              |
| DEVICE       | string | yes      | —       | Device                                                  |
| PORT         | string | yes      | —       | Port                                                    |
| FEATURE_TYPE | enum   | no       | 0       | Feature type — options: 0=Waypoints, 1=Routes, 2=Tracks |

### Example usage

```
run native:uploadgpsdata INPUT=input.dat DEVICE=value PORT=value FEATURE_TYPE=0
```

This calls **Upload GPS data to device**: INPUT=input.dat — Input file; DEVICE=value — Device; PORT=value — Port; FEATURE\_TYPE=0 selects *Waypoints*. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:validatenetwork — Validate network

**group:** Network analysis

This algorithm analyzes a network vector layer to identify data and topology errors that may affect network analysis tools (like shortest path).

Optional checks include:

1. Validating the 'Direction' field to ensure all direction field values in the input layer match the configured forward/backward/both values. Errors will be reported if the direction field value is non-null and does not match one of the configured values.
2. Checking node-to-node separation. This check identifies nodes from the network graph that are closer to other nodes than the specified tolerance distance. This often indicates missed snaps or short segments in the input layer. In the case that a node violates this condition with multiple other nodes, only the closest violation will be reported.

3. Checking node-to-segment separation: This check identifies nodes that are closer to a line segment (e.g. a graph edge) than the specified tolerance distance, without being connected to it. In the case that a node violates this condition with multiple other edges, only the closest violation will be reported.

Topology checks (node-to-node and node-to-segment) can optionally be restricted to only evaluate nodes that are topological dead-ends (connected to only one other distinct node). This is useful for specifically targeting dangles or undershoots.

Two layers are output by this algorithm:

1. An output containing features from the original network layer which failed the direction validation checks.
2. An output representing the problematic node locations with a 'error' field explaining the error. This is a line layer, where the output features join the problematic node to the node or segment which failed the tolerance checks.

| Parameter              | Type     | Required | Default | Description                                             |
|------------------------|----------|----------|---------|---------------------------------------------------------|
| INPUT                  | source   | yes      | —       | Vector layer representing network                       |
| TOLERANCE_NODE_NODE    | distance | no       | —       | Minimum separation between nodes                        |
| TOLERANCE_NODE_SEGMENT | distance | no       | —       | Minimum separation between nodes and non-noded segments |
| ENDPOINTS_ONLY         | boolean  | no       | False   | Only check for errors at end points                     |
| DIRECTION_FIELD        | field    | no       | —       | Direction field                                         |
| VALUE_FORWARD          | string   | no       | —       | Value for forward direction                             |
| VALUE_BACKWARD         | string   | no       | —       | Value for backward direction                            |
| VALUE_BOTH             | string   | no       | —       | Value for both directions                               |
| TOLERANCE *(adv)*      | distance | no       | 0       | Topology tolerance                                      |
| OUTPUT_INVALID_NETWORK | sink     | no       | —       | Invalid network features                                |
| OUTPUT_INVALID_NODES   | sink     | no       | —       | Invalid network nodes                                   |

**Outputs:** OUTPUT\_INVALID\_NETWORK (outputVector), COUNT\_INVALID\_NETWORK\_FEATURES (outputNumber), OUTPUT\_INVALID\_NODES (outputVector), COUNT\_INVALID\_NODES (outputNumber)

### Example usage

```
run native:validatenetwork INPUT=input.gpkg TOLERANCE_NODE_NODE=10
↳ TOLERANCE_NODE_SEGMENT=10 ENDPOINTS_ONLY=false DIRECTION_FIELD=field1
↳ VALUE_FORWARD=value VALUE_BACKWARD=value
↳ OUTPUT_INVALID_NETWORK=output_invalid_network.gpkg
```

This calls **Validate network**: INPUT=input.gpkg — Vector layer representing network; TOLERANCE\_NODE\_NODE=10 — Minimum separation between nodes; TOLERANCE\_NODE\_SEGMENT=10 — Minimum separation between nodes and non-noded segments; ENDPOINTS\_ONLY=false — Only check for errors at end points; DIRECTION\_FIELD=field1 — Direction field; VALUE\_FORWARD=value — Value for forward direction; VALUE\_BACKWARD=value — Value for backward direction; OUTPUT\_INVALID\_NETWORK=output\_invalid\_network.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:virtualrastercalc — Raster calculator (virtual)

**group:** Raster analysis

This algorithm performs algebraic operations using raster layers and generates in-memory result.

| Parameter  | Type       | Required | Default | Description   |
|------------|------------|----------|---------|---------------|
| LAYERS     | multilayer | yes      | —       | Input layers  |
| EXPRESSION | expression | yes      | —       | Expression    |
| EXTENT     | extent     | no       | —       | Output extent |

| Parameter  | Type   | Required | Default | Description                                         |
|------------|--------|----------|---------|-----------------------------------------------------|
| CELL_SIZE  | number | no       | —       | Output cell size (leave empty to set automatically) |
| CRS        | crs    | no       | —       | Output CRS                                          |
| LAYER_NAME | string | no       | —       | Output layer name                                   |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run native:virtualrastercalc LAYERS="a.gpkg;b.gpkg" EXPRESSION="value > 0"
↳ EXTENT=0,1000,0,1000 CELL_SIZE=10 CRS=EPSG:6346 LAYER_NAME=value
```

This calls **Raster calculator (virtual)**: LAYERS="a.gpkg;b.gpkg" — Input layers; EXPRESSION="value > 0" — Expression; EXTENT=0,1000,0,1000 — Output extent; CELL\_SIZE=10 — Output cell size (leave empty to set automatically); CRS=EPSG:6346 — Output CRS; LAYER\_NAME=value — Output layer name. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:voronoipolygons — Voronoi polygons ★

**group:** Vector geometry · **niva verb:** voronoi

This algorithm generates a polygon layer containing the Voronoi diagram corresponding to input points.

| Parameter       | Type    | Required | Default | Description                         |
|-----------------|---------|----------|---------|-------------------------------------|
| INPUT           | source  | yes      | —       | Input layer                         |
| BUFFER          | number  | no       | 0       | Buffer region (% of extent)         |
| TOLERANCE       | number  | no       | 0       | Tolerance                           |
| COPY_ATTRIBUTES | boolean | no       | True    | Copy attributes from input features |
| OUTPUT          | sink    | yes      | —       | Voronoi polygons                    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:voronoipolygons INPUT=input.gpkg BUFFER=0 TOLERANCE=0 COPY_ATTRIBUTES=true
↳ OUTPUT=output.gpkg
```

This calls **Voronoi polygons**: INPUT=input.gpkg — Input layer; BUFFER=0 — Buffer region (% of extent); TOLERANCE=0 — Tolerance; COPY\_ATTRIBUTES=true — Copy attributes from input features; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:wedgebuffers — Create wedge buffers

**group:** Vector geometry

This algorithm creates wedge shaped buffers from input points.

The azimuth parameter gives the angle (in degrees) for the middle of the wedge to point. The buffer width (in degrees) is specified by the width parameter. Note that the wedge will extend to half of the angular width either side of the azimuth direction.

The outer radius of the buffer is specified via outer radius, and optionally an inner radius can also be specified.

The native output from this algorithm are CurvePolygon geometries, but these may be automatically segmented to Polygons depending on the output format.

| Parameter    | Type   | Required | Default | Description                  |
|--------------|--------|----------|---------|------------------------------|
| INPUT        | source | yes      | —       | Input layer                  |
| AZIMUTH      | number | no       | 0       | Azimuth (degrees from North) |
| WIDTH        | number | no       | 45      | Wedge width (in degrees)     |
| OUTER_RADIUS | number | no       | 1       | Outer radius                 |
| INNER_RADIUS | number | no       | 0       | Inner radius                 |
| OUTPUT       | sink   | yes      | —       | Buffers                      |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:wedgebuffers INPUT=input.gpkg AZIMUTH=0 WIDTH=45 OUTER_RADIUS=1 INNER_RADIUS=0
↳ OUTPUT=output.gpkg
```

This calls **Create wedge buffers**: INPUT=input.gpkg — Input layer; AZIMUTH=0 — Azimuth (degrees from North); WIDTH=45 — Wedge width (in degrees); OUTER\_RADIUS=1 — Outer radius; INNER\_RADIUS=0 — Inner radius; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:writevectortiles\_mbtiles — Write Vector Tiles (MBTiles)

**group:** Vector tiles

This algorithm exports one or more vector layers to vector tiles - a data format optimized for fast map rendering and small data size.

| Parameter        | Type                   | Required | Default | Description           |
|------------------|------------------------|----------|---------|-----------------------|
| OUTPUT           | vectorTileDestination  | yes      | —       | Destination MBTiles   |
| LAYERS           | vectortilewriterlayers | yes      | —       | Input layers          |
| MIN_ZOOM         | number                 | no       | 0       | Minimum zoom level    |
| MAX_ZOOM         | number                 | no       | 3       | Maximum zoom level    |
| EXTENT           | extent                 | no       | —       | Extent                |
| META_NAME        | string                 | no       | —       | Metadata: Name        |
| META_DESCRIPTION | string                 | no       | —       | Metadata: Description |
| META_ATTRIBUTION | string                 | no       | —       | Metadata: Attribution |
| META_VERSION     | string                 | no       | —       | Metadata: Version     |
| META_TYPE        | string                 | no       | —       | Metadata: Type        |
| META_CENTER      | string                 | no       | —       | Metadata: Center      |

**Outputs:** OUTPUT (outputVectorTile)

### Example usage

```
run native:writevectortiles_mbtiles LAYERS=... MIN_ZOOM=0 MAX_ZOOM=3 EXTENT=0,1000,0,1000
↳ META_NAME=value META_DESCRIPTION=value META_ATTRIBUTION=value OUTPUT=output.mbtiles
```

This calls **Write Vector Tiles (MBTiles)**: LAYERS=... — Input layers; MIN\_ZOOM=0 — Minimum zoom level; MAX\_ZOOM=3 — Maximum zoom level; EXTENT=0,1000,0,1000 — Extent; META\_NAME=value — Metadata: Name; META\_DESCRIPTION=value — Metadata: Description; META\_ATTRIBUTION=value — Metadata: Attribution; OUTPUT=output.mbtiles is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:writevectortiles\_xyz — Write Vector Tiles (XYZ)

**group:** Vector tiles

This algorithm exports one or more vector layers to vector tiles - a data format optimized for fast map rendering and small data size.

| Parameter        | Type                   | Required | Default         | Description        |
|------------------|------------------------|----------|-----------------|--------------------|
| OUTPUT_DIRECTORY | folderDestination      | yes      | —               | Output directory   |
| XYZ_TEMPLATE     | string                 | no       | {z}/{x}/{y}.pbf | File template      |
| LAYERS           | vectortilewriterlayers | yes      | —               | Input layers       |
| MIN_ZOOM         | number                 | no       | 0               | Minimum zoom level |
| MAX_ZOOM         | number                 | no       | 3               | Maximum zoom level |
| EXTENT           | extent                 | no       | —               | Extent             |

**Outputs:** OUTPUT\_DIRECTORY (outputFolder)

### Example usage

```
run native:writevectortiles_xyz LAYERS=... XYZ_TEMPLATE={z}/{x}/{y}.pbf MIN_ZOOM=0
↳ MAX_ZOOM=3 EXTENT=0,1000,0,1000 OUTPUT_DIRECTORY=output_directory/
```

This calls **Write Vector Tiles (XYZ)**: LAYERS=... — Input layers; XYZ\_TEMPLATE={z}/{x}/{y}.pbf — File template; MIN\_ZOOM=0 — Minimum zoom level; MAX\_ZOOM=3 — Maximum zoom level; EXTENT=0,1000,0,1000 — Extent; OUTPUT\_DIRECTORY=output\_directory/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:zonalhistogram — Zonal histogram

**group:** Raster analysis

This algorithm appends fields representing counts of each unique value from a raster layer contained within zones defined as polygons.

| Parameter     | Type   | Required | Default | Description                   |
|---------------|--------|----------|---------|-------------------------------|
| INPUT_RASTER  | raster | yes      | —       | Raster layer                  |
| RASTER_BAND   | band   | no       | 1       | Band number                   |
| INPUT_VECTOR  | source | yes      | —       | Vector layer containing zones |
| COLUMN_PREFIX | string | no       | HISTO_  | Output column prefix          |
| OUTPUT        | sink   | yes      | —       | Output zones                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:zonalhistogram INPUT_RASTER=dem.tif INPUT_VECTOR=input.gpkg RASTER_BAND=1
↳ COLUMN_PREFIX=HISTO_ OUTPUT=output.gpkg
```

This calls **Zonal histogram**: INPUT\_RASTER=dem.tif — Raster layer; INPUT\_VECTOR=input.gpkg — Vector layer containing zones; RASTER\_BAND=1 — Band number; COLUMN\_PREFIX=HISTO\_ — Output column prefix; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:zonalminmaxpoint — Zonal minimum/maximum point

**group:** Raster analysis

This algorithm extracts point features corresponding to the minimum and maximum pixel values contained within polygon zones.

The output will contain one point feature for the minimum and one for the maximum raster value for every individual zonal feature from a polygon layer.

The created point layer will be in the same spatial reference system as the selected raster layer.



| Parameter    | Type   | Required | Default | Description   |
|--------------|--------|----------|---------|---------------|
| INPUT        | source | yes      | —       | Input layer   |
| INPUT_RASTER | raster | yes      | —       | Raster layer  |
| RASTER_BAND  | band   | no       | 1       | Raster band   |
| OUTPUT       | sink   | yes      | —       | Zonal Extrema |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:zonalminmaxpoint INPUT=input.gpkg INPUT_RASTER=dem.tif RASTER_BAND=1
↳ OUTPUT=output.gpkg
```

This calls **Zonal minimum/maximum point**: INPUT=input.gpkg — Input layer; INPUT\_RASTER=dem.tif — Raster layer; RASTER\_BAND=1 — Raster band; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:zonalstatistics — Zonal statistics (in place)

**group:** Raster analysis

This algorithm calculates statistics of a raster layer for each feature of an overlapping polygon vector layer. The results will be written in place.

| Parameter     | Type   | Required | Default   | Description                                                                                                                                                   |
|---------------|--------|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT_RASTER  | raster | yes      | —         | Raster layer                                                                                                                                                  |
| RASTER_BAND   | band   | no       | 1         | Raster band                                                                                                                                                   |
| INPUT_VECTOR  | vector | yes      | —         | Vector layer containing zones                                                                                                                                 |
| COLUMN_PREFIX | string | no       | _         | Output column prefix                                                                                                                                          |
| STATISTICS    | enum   | no       | [0, 1, 2] | Statistics to calculate — options: 0=Count, 1=Sum, 2=Mean, 3=Median, 4=St dev, 5=Minimum, 6=Maximum, 7=Range, 8=Minority, 9=Majority, 10=Variety, 11=Variance |

**Outputs:** INPUT\_VECTOR (outputVector)

### Example usage

```
run native:zonalstatistics INPUT_RASTER=dem.tif INPUT_VECTOR=input.gpkg RASTER_BAND=1
↳ COLUMN_PREFIX=_ STATISTICS=0
```

This calls **Zonal statistics (in place)**: INPUT\_RASTER=dem.tif — Raster layer; INPUT\_VECTOR=input.gpkg — Vector layer containing zones; RASTER\_BAND=1 — Raster band; COLUMN\_PREFIX=\_ — Output column prefix; STATISTICS=0 selects *Count*. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## native:zonalstatisticsfb — Zonal statistics ★

**group:** Raster analysis · **niva verb:** zonalstats

This algorithm calculates statistics of a raster layer for each feature of an overlapping polygon vector layer.

| Parameter     | Type   | Required | Default | Description          |
|---------------|--------|----------|---------|----------------------|
| INPUT         | source | yes      | —       | Input layer          |
| INPUT_RASTER  | raster | yes      | —       | Raster layer         |
| RASTER_BAND   | band   | no       | 1       | Raster band          |
| COLUMN_PREFIX | string | no       | _       | Output column prefix |

| Parameter  | Type | Required | Default   | Description                                                                                                                                                   |
|------------|------|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STATISTICS | enum | no       | [0, 1, 2] | Statistics to calculate — options: 0=Count, 1=Sum, 2=Mean, 3=Median, 4=St dev, 5=Minimum, 6=Maximum, 7=Range, 8=Minority, 9=Majority, 10=Variety, 11=Variance |
| OUTPUT     | sink | yes      | —         | Zonal Statistics                                                                                                                                              |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run native:zonalstatisticsfb INPUT=input.gpkg INPUT_RASTER=dem.tif RASTER_BAND=1
↪ COLUMN_PREFIX=_ STATISTICS=0 OUTPUT=output.gpkg
```

This calls **Zonal statistics**: INPUT=input.gpkg — Input layer; INPUT\_RASTER=dem.tif — Raster layer; RASTER\_BAND=1 — Raster band; COLUMN\_PREFIX=\_ — Output column prefix; STATISTICS=0 selects *Count*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

# gdal: algorithms

gdal — GDAL/OGR algorithms — 59 algorithms (QGIS 4.0.3). Auto-generated by scripts/gen\_algorithms.py; see [the index](#).

Call any of these with `run <id> KEY=value ...` (the **Name** column is the KEY); each entry includes a worked **Example usage**. A ★ marks an algorithm with a friendly niva alias verb.

## gdal:aspect — Aspect ★

**group:** Raster analysis · **niva verb:** aspect

| Parameter                   | Type              | Required | Default | Description                                              |
|-----------------------------|-------------------|----------|---------|----------------------------------------------------------|
| INPUT                       | raster            | yes      | —       | Input layer                                              |
| BAND                        | band              | no       | 1       | Band number                                              |
| TRIG_ANGLE                  | boolean           | no       | False   | Return trigonometric angle instead of azimuth            |
| ZERO_FLAT                   | boolean           | no       | False   | Return 0 for flat instead of -9999                       |
| COMPUTE_EDGES               | boolean           | no       | False   | Compute edges                                            |
| ZEVENBERGEN                 | boolean           | no       | False   | Use Zevenbergen&Thorne formula instead of the Horn's one |
| OPTIONS                     | string            | no       | —       | Additional creation options                              |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Additional creation options                              |
| EXTRA *(adv)*               | string            | no       | —       | Additional command-line parameters                       |
| OUTPUT                      | rasterDestination | yes      | —       | Aspect                                                   |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:aspect INPUT=dem.tif BAND=1 TRIG_ANGLE=false ZERO_FLAT=false COMPUTE_EDGES=false  
↳ ZEVENBERGEN=false OPTIONS=value OUTPUT=output.tif
```

This calls **Aspect**: INPUT=dem.tif — Input layer; BAND=1 — Band number; TRIG\_ANGLE=false — Return trigonometric angle instead of azimuth; ZERO\_FLAT=false — Return 0 for flat instead of -9999; COMPUTE\_EDGES=false — Compute edges; ZEVENBERGEN=false — Use Zevenbergen&Thorne formula instead of the Horn's one; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:assignprojection — Assign projection

**group:** Raster projections

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | raster | yes      | —       | Input layer |
| CRS       | crs    | yes      | —       | Desired CRS |

---

**Outputs:** OUTPUT (outputRaster)

**Example usage**

```
run gdal:assignprojection INPUT=dem.tif CRS=EPSG:6346
```

This calls **Assign projection**: INPUT=dem.tif — Input layer; CRS=EPSG:6346 — Desired CRS. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:bufferectors — Buffer vectors

**group:** Vector geoprocessing

| Parameter           | Type              | Required | Default  | Description                                                                                 |
|---------------------|-------------------|----------|----------|---------------------------------------------------------------------------------------------|
| INPUT               | source            | yes      | —        | Input layer                                                                                 |
| GEOMETRY            | string            | no       | geometry | Geometry column name                                                                        |
| DISTANCE            | distance          | no       | 10.0     | Buffer distance                                                                             |
| FIELD               | field             | no       | —        | Dissolve by attribute                                                                       |
| DISSOLVE            | boolean           | no       | False    | Dissolve all results                                                                        |
| EXPLODE_COLLECTIONS | boolean           | no       | False    | Produce one feature for each geometry in any kind of geometry collection in the source file |
| OPTIONS *(adv)*     | string            | no       | —        | Additional creation options                                                                 |
| OUTPUT              | vectorDestination | yes      | —        | Buffer                                                                                      |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run gdal:bufferectors INPUT=input.gpkg GEOMETRY=geometry DISTANCE=10 FIELD=field1
↳ DISSOLVE=false EXPLODE_COLLECTIONS=false OUTPUT=output.gpkg
```

This calls **Buffer vectors**: INPUT=input.gpkg — Input layer; GEOMETRY=geometry — Geometry column name; DISTANCE=10 — Buffer distance; FIELD=field1 — Dissolve by attribute; DISSOLVE=false — Dissolve all results; EXPLODE\_COLLECTIONS=false — Produce one feature for each geometry in any kind of geometry collection in the...; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:buildvirtualraster — Build virtual raster

**group:** Raster miscellaneous

| Parameter          | Type       | Required | Default | Description                                                                                                                                                                  |
|--------------------|------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT              | multilayer | yes      | —       | Input layers                                                                                                                                                                 |
| RESOLUTION         | enum       | no       | 0       | Resolution — options: 0=Average, 1=Highest, 2=Lowest                                                                                                                         |
| SEPARATE           | boolean    | no       | True    | Place each input file into a separate band                                                                                                                                   |
| PROJ_DIFFERENCE    | boolean    | no       | False   | Allow projection difference                                                                                                                                                  |
| ADD_ALPHA *(adv)*  | boolean    | no       | False   | Add alpha mask band to VRT when source raster has none                                                                                                                       |
| ASSIGN_CRS *(adv)* | crs        | no       | —       | Override projection for the output file                                                                                                                                      |
| RESAMPLING *(adv)* | enum       | no       | 0       | Resampling algorithm — options: 0=Nearest Neighbour, 1=Bilinear (2x2 Kernel), 2=Cubic (4x4 Kernel), 3=Cubic B-Spline (4x4 Kernel), 4=Lanczos (6x6 Kernel), 5=Average, 6=Mode |
| SRC_NODATA *(adv)* | string     | no       | —       | Nodata value(s) for input bands (space separated)                                                                                                                            |
| EXTRA *(adv)*      | string     | no       | —       | Additional command-line parameters                                                                                                                                           |

| Parameter | Type              | Required | Default | Description |
|-----------|-------------------|----------|---------|-------------|
| OUTPUT    | rasterDestination | yes      | —       | Virtual     |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:buildvirtualraster INPUT="a.gpkg;b.gpkg" RESOLUTION=0 SEPARATE=true
↳ PROJ_DIFFERENCE=false OUTPUT=output.tif
```

This calls **Build virtual raster**: INPUT="a.gpkg;b.gpkg" — Input layers; RESOLUTION=0 selects *Average*; SEPARATE=true — Place each input file into a separate band; PROJ\_DIFFERENCE=false — Allow projection difference; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:buildvirtualvector — Build virtual vector

**group:** Vector miscellaneous

This algorithm creates a virtual layer that contains a set of vector layers.

| Parameter | Type              | Required | Default | Description          |
|-----------|-------------------|----------|---------|----------------------|
| INPUT     | multilayer        | yes      | —       | Input datasources    |
| UNIONED   | boolean           | no       | False   | Create "unioned" VRT |
| OUTPUT    | vectorDestination | yes      | —       | Virtual vector       |

**Outputs:** OUTPUT (outputVector), VRT\_STRING (outputString)

### Example usage

```
run gdal:buildvirtualvector INPUT="a.gpkg;b.gpkg" UNIONED=false OUTPUT=output.gpkg
```

This calls **Build virtual vector**: INPUT="a.gpkg;b.gpkg" — Input datasources; UNIONED=false — Create "unioned" VRT; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:cliprasterbyextent — Clip raster by extent

**group:** Raster extraction

| Parameter                   | Type              | Required | Default | Description                                                                                                                                                                        |
|-----------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                       | raster            | yes      | —       | Input layer                                                                                                                                                                        |
| PROJWIN                     | extent            | yes      | —       | Clipping extent                                                                                                                                                                    |
| OVERCRS                     | boolean           | no       | False   | Override the projection for the output file                                                                                                                                        |
| NODATA                      | number            | no       | —       | Assign a specified NoData value to output bands                                                                                                                                    |
| OPTIONS                     | string            | no       | —       | Additional creation options                                                                                                                                                        |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Additional creation options                                                                                                                                                        |
| DATA_TYPE *(adv)*           | enum              | no       | 0       | Output data type — options: 0=Use Input Layer Data Type, 1=Byte, 2=Int16, 3=UInt16, 4=UInt32, 5=Int32, 6=Float32, 7=Float64, 8=CInt16, 9=CInt32, 10=CFloat32, 11=CFloat64, 12=Int8 |
| EXTRA *(adv)*               | string            | no       | —       | Additional command-line parameters                                                                                                                                                 |
| OUTPUT                      | rasterDestination | yes      | —       | Clipped (extent)                                                                                                                                                                   |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:cliprasterbyextent INPUT=dem.tif PROJWIN=0,1000,0,1000 OVERCRS=false NODATA=10
↳ OPTIONS=value OUTPUT=output.tif
```

This calls **Clip raster by extent**: INPUT=dem.tif — Input layer; PROJWIN=0,1000,0,1000 — Clipping extent; OVERCRS=false — Override the projection for the output file; NODATA=10 — Assign a specified NoData value to output bands; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:cliprasterbymasklayer — Clip raster by mask layer ★

**group:** Raster extraction · **niva verb:** clipraster

| Parameter                | Type              | Required | Default | Description                                                                                                                                                                        |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                                                                                                                                                                        |
| MASK                     | source            | yes      | —       | Mask layer                                                                                                                                                                         |
| SOURCE_CRS               | crs               | no       | —       | Source CRS                                                                                                                                                                         |
| TARGET_CRS               | crs               | no       | —       | Target CRS                                                                                                                                                                         |
| TARGET_EXTENT            | extent            | no       | —       | Target extent                                                                                                                                                                      |
| NODATA                   | number            | no       | —       | Assign a specified NoData value to output bands                                                                                                                                    |
| ALPHA_BAND               | boolean           | no       | False   | Create an output alpha band                                                                                                                                                        |
| CROP_TO_CUTLINE          | boolean           | no       | True    | Match the extent of the clipped raster to the extent of the mask layer                                                                                                             |
| KEEP_RESOLUTION          | boolean           | no       | False   | Keep resolution of input raster                                                                                                                                                    |
| SET_RESOLUTION           | boolean           | no       | False   | Set output file resolution                                                                                                                                                         |
| X_RESOLUTION             | number            | no       | —       | X Resolution to output bands                                                                                                                                                       |
| Y_RESOLUTION             | number            | no       | —       | Y Resolution to output bands                                                                                                                                                       |
| MULTITHREADING *(adv)*   | boolean           | no       | False   | Use multithreaded warping implementation                                                                                                                                           |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                                                        |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                                                        |
| DATA_TYPE *(adv)*        | enum              | no       | 0       | Output data type — options: 0=Use Input Layer Data Type, 1=Byte, 2=Int16, 3=UInt16, 4=UInt32, 5=Int32, 6=Float32, 7=Float64, 8=CInt16, 9=CInt32, 10=CFloat32, 11=CFloat64, 12=Int8 |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                                                 |
| OUTPUT                   | rasterDestination | yes      | —       | Clipped (mask)                                                                                                                                                                     |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:cliprasterbymasklayer INPUT=dem.tif MASK=input.gpkg SOURCE_CRS=EPSG:6346
↳ TARGET_CRS=EPSG:6346 TARGET_EXTENT=0,1000,0,1000 NODATA=10 ALPHA_BAND=false
↳ OUTPUT=output.tif
```

This calls **Clip raster by mask layer**: INPUT=dem.tif — Input layer; MASK=input.gpkg — Mask layer; SOURCE\_CRS=EPSG:6346 — Source CRS; TARGET\_CRS=EPSG:6346 — Target CRS; TARGET\_EXTENT=0,1000,0,1000 — Target extent; NODATA=10 — Assign a specified NoData value to output bands; ALPHA\_BAND=false — Create an output alpha band; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:clipvectorbyextent — Clip vector by extent

**group:** Vector geoprocessing

| Parameter       | Type              | Required | Default | Description                 |
|-----------------|-------------------|----------|---------|-----------------------------|
| INPUT           | source            | yes      | —       | Input layer                 |
| EXTENT          | extent            | yes      | —       | Clipping extent             |
| OPTIONS *(adv)* | string            | no       | —       | Additional creation options |
| OUTPUT          | vectorDestination | yes      | —       | Clipped (extent)            |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run gdal:clipvectorbyextent INPUT=input.gpkg EXTENT=0,1000,0,1000 OUTPUT=output.gpkg
```

This calls **Clip vector by extent**: INPUT=input.gpkg — Input layer; EXTENT=0,1000,0,1000 — Clipping extent; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

### gdal:clipvectorbypolygon — Clip vector by mask layer

**group:** Vector geoprocessing

| Parameter       | Type              | Required | Default | Description                 |
|-----------------|-------------------|----------|---------|-----------------------------|
| INPUT           | source            | yes      | —       | Input layer                 |
| MASK            | source            | yes      | —       | Mask layer                  |
| OPTIONS *(adv)* | string            | no       | —       | Additional creation options |
| OUTPUT          | vectorDestination | yes      | —       | Clipped (mask)              |

**Outputs:** OUTPUT (outputVector)

#### Example usage

```
run gdal:clipvectorbypolygon INPUT=input.gpkg MASK=input.gpkg OUTPUT=output.gpkg
```

This calls **Clip vector by mask layer**: INPUT=input.gpkg — Input layer; MASK=input.gpkg — Mask layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

### gdal:colorrelief — Color relief

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                                                                                                        |
|--------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                                                                                                        |
| BAND                     | band              | no       | 1       | Band number                                                                                                        |
| COMPUTE_EDGES            | boolean           | no       | False   | Compute edges                                                                                                      |
| COLOR_TABLE              | file              | yes      | —       | Color configuration file                                                                                           |
| MATCH_MODE               | enum              | no       | 2       | Matching mode — options: 0=Use strict color matching, 1=Use closest RGBA quadruplet, 2=Use smoothly blended colors |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                        |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                        |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                 |
| OUTPUT                   | rasterDestination | yes      | —       | Color relief                                                                                                       |

**Outputs:** OUTPUT (outputRaster)

#### Example usage

```
run gdal:colorrelief INPUT=dem.tif COLOR_TABLE=input.dat BAND=1 COMPUTE_EDGES=false
↳ MATCH_MODE=2 OPTIONS=value OUTPUT=output.tif
```

This calls **Color relief**: INPUT=dem.tif — Input layer; COLOR\_TABLE=input.dat — Color configuration file; BAND=1 — Band number; COMPUTE\_EDGES=false — Compute edges; MATCH\_MODE=2 selects *Use smoothly blended colors*; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:contour — Contour

**group:** Raster extraction

| Parameter             | Type              | Required | Default | Description                                                     |
|-----------------------|-------------------|----------|---------|-----------------------------------------------------------------|
| INPUT                 | raster            | yes      | —       | Input layer                                                     |
| BAND                  | band              | no       | 1       | Band number                                                     |
| INTERVAL              | number            | no       | 10.0    | Interval between contour lines                                  |
| FIELD_NAME            | string            | no       | ELEV    | Attribute name (if not set, no elevation attribute is attached) |
| CREATE_3D *(adv)*     | boolean           | no       | False   | Produce 3D vector                                               |
| IGNORE_NODATA *(adv)* | boolean           | no       | False   | Treat all raster values as valid                                |
| NODATA *(adv)*        | number            | no       | —       | Input pixel value to treat as NoData                            |
| OFFSET                | number            | no       | 0.0     | Offset from zero relative to which to interpret intervals       |
| EXTRA *(adv)*         | string            | no       | —       | Additional command-line parameters                              |
| OPTIONS               | string            | no       | —       | Additional creation options                                     |
| OUTPUT                | vectorDestination | yes      | —       | Contours                                                        |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:contour INPUT=dem.tif BAND=1 INTERVAL=10 FIELD_NAME=ELEV OFFSET=0 OPTIONS=value
↳ OUTPUT=output.gpkg
```

This calls **Contour**: INPUT=dem.tif — Input layer; BAND=1 — Band number; INTERVAL=10 — Interval between contour lines; FIELD\_NAME=ELEV — Attribute name (if not set, no elevation attribute is attached); OFFSET=0 — Offset from zero relative to which to interpret intervals; OPTIONS=value — Additional creation options; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:contour\_polygon — Contour Polygons

**group:** Raster extraction

| Parameter             | Type    | Required | Default | Description                                               |
|-----------------------|---------|----------|---------|-----------------------------------------------------------|
| INPUT                 | raster  | yes      | —       | Input layer                                               |
| BAND                  | band    | no       | 1       | Band number                                               |
| INTERVAL              | number  | no       | 10.0    | Interval between contour lines                            |
| CREATE_3D *(adv)*     | boolean | no       | False   | Produce 3D vector                                         |
| IGNORE_NODATA *(adv)* | boolean | no       | False   | Treat all raster values as valid                          |
| NODATA *(adv)*        | number  | no       | —       | Input pixel value to treat as NoData                      |
| OFFSET                | number  | no       | 0.0     | Offset from zero relative to which to interpret intervals |
| EXTRA *(adv)*         | string  | no       | —       | Additional command-line parameters                        |
| OPTIONS               | string  | no       | —       | Additional creation options                               |



| Parameter      | Type              | Required | Default  | Description                                             |
|----------------|-------------------|----------|----------|---------------------------------------------------------|
| FIELD_NAME_MIN | string            | no       | ELEV_MIN | Attribute name for minimum elevation of contour polygon |
| FIELD_NAME_MAX | string            | no       | ELEV_MAX | Attribute name for maximum elevation of contour polygon |
| OUTPUT         | vectorDestination | yes      | —        | Contours                                                |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:contour_polygon INPUT=dem.tif BAND=1 INTERVAL=10 OFFSET=0 OPTIONS=value
↳ FIELD_NAME_MIN=ELEV_MIN FIELD_NAME_MAX=ELEV_MAX OUTPUT=output.gpkg
```

This calls **Contour Polygons**: INPUT=dem.tif — Input layer; BAND=1 — Band number; INTERVAL=10 — Interval between contour lines; OFFSET=0 — Offset from zero relative to which to interpret intervals; OPTIONS=value — Additional creation options; FIELD\_NAME\_MIN=ELEV\_MIN — Attribute name for minimum elevation of contour polygon; FIELD\_NAME\_MAX=ELEV\_MAX — Attribute name for maximum elevation of contour polygon; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:convertformat — Convert format

**group:** Vector conversion

The algorithm converts simple features data between file formats.

Use convert all layers to convert a whole dataset. Supported output formats for this option are:

- GPKG
- GML

| Parameter          | Type              | Required | Default | Description                     |
|--------------------|-------------------|----------|---------|---------------------------------|
| INPUT              | source            | yes      | —       | Input layer                     |
| CONVERT_ALL_LAYERS | boolean           | no       | False   | Convert all layers from dataset |
| OPTIONS *(adv)*    | string            | no       | —       | Additional creation options     |
| OUTPUT             | vectorDestination | yes      | —       | Converted                       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:convertformat INPUT=input.gpkg CONVERT_ALL_LAYERS=false OUTPUT=output.gpkg
```

This calls **Convert format**: INPUT=input.gpkg — Input layer; CONVERT\_ALL\_LAYERS=false — Convert all layers from dataset; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:createcog — Create Cloud Optimized GeoTIFF

**group:** Raster conversion

This algorithm creates a Cloud Optimized GeoTIFF (COG) from the input raster layers.

| Parameter                | Type              | Required | Default | Description                 |
|--------------------------|-------------------|----------|---------|-----------------------------|
| LAYERS                   | multilayer        | yes      | —       | Input layers                |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options |
| OUTPUT                   | folderDestination | yes      | —       | Output directory            |

---

**Outputs:** OUTPUT (outputFolder), OUTPUT\_LAYERS (outputMultilayer)

**Example usage**

```
run gdal:createcog LAYERS="a.gpkg;b.gpkg" OUTPUT=output/
```

This calls **Create Cloud Optimized GeoTIFF**: LAYERS="a.gpkg;b.gpkg" — Input layers; OUTPUT=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:dataset\_identify — Dataset identification

**group:** Miscellaneous

This algorithm reports the name of GDAL drivers that can open files contained in a folder, with optional additional details, and write the result into an output vector layer.

| Parameter | Type              | Required | Default | Description                                         |
|-----------|-------------------|----------|---------|-----------------------------------------------------|
| INPUT     | file              | yes      | —       | Input folder                                        |
| RECURSIVE | boolean           | no       | True    | Perform recursive exploration of the input folder   |
| DETAILS   | boolean           | no       | True    | Add details about identified datasets in the output |
| OUTPUT    | vectorDestination | yes      | —       | Output file                                         |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run gdal:dataset_identify INPUT=input.dat RECURSIVE=true DETAILS=true OUTPUT=output.gpkg
```

This calls **Dataset identification**: INPUT=input.dat — Input folder; RECURSIVE=true — Perform recursive exploration of the input folder; DETAILS=true — Add details about identified datasets in the output; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:dissolve — Dissolve

**group:** Vector geoprocessing

| Parameter                    | Type              | Required | Default  | Description                                                                                 |
|------------------------------|-------------------|----------|----------|---------------------------------------------------------------------------------------------|
| INPUT                        | source            | yes      | —        | Input layer                                                                                 |
| FIELD                        | field             | no       | —        | Dissolve field                                                                              |
| GEOMETRY                     | string            | no       | geometry | Geometry column name                                                                        |
| EXPLODE_COLLECTIONS *(adv)*  | boolean           | no       | False    | Produce one feature for each geometry in any kind of geometry collection in the source file |
| KEEP_ATTRIBUTES *(adv)*      | boolean           | no       | False    | Keep input attributes                                                                       |
| COUNT_FEATURES *(adv)*       | boolean           | no       | False    | Count dissolved features                                                                    |
| COMPUTE_AREA *(adv)*         | boolean           | no       | False    | Compute area and perimeter of dissolved features                                            |
| COMPUTE_STATISTICS *(adv)*   | boolean           | no       | False    | Compute min/max/sum/mean for attribute                                                      |
| STATISTICS_ATTRIBUTE *(adv)* | field             | no       | —        | Numeric attribute to calculate statistics on                                                |
| OPTIONS *(adv)*              | string            | no       | —        | Additional creation options                                                                 |
| OUTPUT                       | vectorDestination | yes      | —        | Dissolved                                                                                   |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run gdal:dissolve INPUT=input.gpkg FIELD=field1 GEOMETRY=geometry OUTPUT=output.gpkg
```

This calls **Dissolve**: INPUT=input.gpkg — Input layer; FIELD=field1 — Dissolve field; GEOMETRY=geometry — Geometry column name; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:executesql — Execute SQL

**group:** Vector miscellaneous

| Parameter       | Type              | Required | Default | Description                                        |
|-----------------|-------------------|----------|---------|----------------------------------------------------|
| INPUT           | source            | yes      | —       | Input layer                                        |
| SQL             | string            | yes      | —       | SQL expression                                     |
| DIALECT         | enum              | no       | 0       | SQL dialect — options: 0=None, 1=OGR SQL, 2=SQLite |
| OPTIONS *(adv)* | string            | no       | —       | Additional creation options                        |
| OUTPUT          | vectorDestination | yes      | —       | SQL result                                         |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:executesql INPUT=input.gpkg SQL=value DIALECT=0 OUTPUT=output.gpkg
```

This calls **Execute SQL**: INPUT=input.gpkg — Input layer; SQL=value — SQL expression; DIALECT=0 selects *None*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:extractprojection — Extract projection

**group:** Raster projections

| Parameter       | Type    | Required | Default | Description           |
|-----------------|---------|----------|---------|-----------------------|
| INPUT           | raster  | yes      | —       | Input file            |
| PRJ_FILE_CREATE | boolean | no       | False   | Create also .prj file |

**Outputs:** WORLD\_FILE (outputFile), PRJ\_FILE (outputFile)

### Example usage

```
run gdal:extractprojection INPUT=dem.tif PRJ_FILE_CREATE=false
```

This calls **Extract projection**: INPUT=dem.tif — Input file; PRJ\_FILE\_CREATE=false — Create also .prj file. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:fillnodata — Fill NoData

**group:** Raster analysis

| Parameter  | Type    | Required | Default | Description                                                          |
|------------|---------|----------|---------|----------------------------------------------------------------------|
| INPUT      | raster  | yes      | —       | Input layer                                                          |
| BAND       | band    | no       | 1       | Band number                                                          |
| DISTANCE   | number  | no       | 10      | Maximum distance (in pixels) to search out for values to interpolate |
| ITERATIONS | number  | no       | 0       | Number of smoothing iterations to run after the interpolation        |
| NO_MASK    | boolean | no       | False   | Do not use the default validity mask for the input band              |

| Parameter                | Type              | Required | Default | Description                        |
|--------------------------|-------------------|----------|---------|------------------------------------|
| MASK_LAYER               | raster            | no       | —       | Validity mask                      |
| OPTIONS                  | string            | no       | —       | Additional creation options        |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options        |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters |
| OUTPUT                   | rasterDestination | yes      | —       | Filled                             |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:fillnodata INPUT=dem.tif BAND=1 DISTANCE=10 ITERATIONS=0 NO_MASK=false
↳ MASK_LAYER=dem.tif OPTIONS=value OUTPUT=output.tif
```

This calls **Fill NoData**: INPUT=dem.tif — Input layer; BAND=1 — Band number; DISTANCE=10 — Maximum distance (in pixels) to search out for values to interpolate; ITERATIONS=0 — Number of smoothing iterations to run after the interpolation; NO\_MASK=false — Do not use the default validity mask for the input band; MASK\_LAYER=dem.tif — Validity mask; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gdal2tiles — gdal2tiles

**group:** Raster miscellaneous

| Parameter          | Type              | Required | Default | Description                                                                                                                                                                    |
|--------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT              | raster            | yes      | —       | Input layer                                                                                                                                                                    |
| PROFILE            | enum              | no       | 0       | Tile cutting profile — options: 0=Mercator, 1=Geode-tic, 2=Raster                                                                                                              |
| ZOOM               | string            | no       | —       | Zoom levels to render                                                                                                                                                          |
| VIEWER             | enum              | no       | 0       | Web viewer to generate — options: 0=All, 1=GoogleMaps, 2=OpenLayers, 3=Leaflet, 4=None                                                                                         |
| TITLE              | string            | no       | —       | Title of the map                                                                                                                                                               |
| COPYRIGHT          | string            | no       | —       | Copyright of the map                                                                                                                                                           |
| RESAMPLING *(adv)* | enum              | no       | 0       | Resampling method — options: 0=Average, 1=Nearest Neighbour, 2=Bilinear (2x2 Kernel), 3=Cubic (4x4 Kernel), 4=Cubic B-Spline (4x4 Kernel), 5=Lanczos (6x6 Kernel), 6=Antialias |
| SOURCE_CRS *(adv)* | crs               | no       | —       | The spatial reference system used for the source input data                                                                                                                    |
| NODATA *(adv)*     | number            | no       | 0       | Transparency value to assign to the input data                                                                                                                                 |
| URL *(adv)*        | string            | no       | —       | URL address where the generated tiles are going to be published                                                                                                                |
| GOOGLE_KEY *(adv)* | string            | no       | —       | Google Maps API key ( <a href="http://code.google.com/apis/maps/signup.html">http://code.google.com/apis/maps/signup.html</a> )                                                |
| BING_KEY *(adv)*   | string            | no       | —       | Bing Maps API key ( <a href="https://www.bingmapsportal.com/">https://www.bingmapsportal.com/</a> )                                                                            |
| RESUME *(adv)*     | boolean           | no       | False   | Generate only missing files                                                                                                                                                    |
| KML *(adv)*        | boolean           | no       | False   | Generate KML for Google Earth                                                                                                                                                  |
| NO_KML *(adv)*     | boolean           | no       | False   | Avoid automatic generation of KML files for EPSG:4326                                                                                                                          |
| OUTPUT             | folderDestination | yes      | —       | Output directory                                                                                                                                                               |

**Outputs:** OUTPUT (outputFolder)

### Example usage

```
run gdal:gdal2tiles INPUT=dem.tif PROFILE=0 ZOOM=value VIEWER=0 TITLE=value
↳ COPYRIGHT=value OUTPUT=output/
```

This calls **gdal2tiles**: INPUT=dem.tif — Input layer; PROFILE=0 selects *Mercator*; ZOOM=value — Zoom levels to render; VIEWER=0 selects *All*; TITLE=value — Title of the map; COPYRIGHT=value — Copyright of the map; OUTPUT=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gdal2xyz — gdal2xyz

**group:** Raster conversion

| Parameter     | Type            | Required | Default | Description                             |
|---------------|-----------------|----------|---------|-----------------------------------------|
| INPUT         | raster          | yes      | —       | Input layer                             |
| BAND          | band            | no       | 1       | Band number                             |
| NODATA_INPUT  | number          | no       | —       | Input pixel value to treat as NoData    |
| NODATA_OUTPUT | number          | no       | —       | Assign specified NoData value to output |
| SKIP_NODATA   | boolean         | no       | False   | Do not output NoData values             |
| CSV           | boolean         | no       | False   | Output comma-separated values           |
| OUTPUT        | fileDestination | yes      | —       | XYZ ASCII file                          |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run gdal:gdal2xyz INPUT=dem.tif BAND=1 NODATA_INPUT=10 NODATA_OUTPUT=10 SKIP_NODATA=false
↳ CSV=false OUTPUT=output.txt
```

This calls **gdal2xyz**: INPUT=dem.tif — Input layer; BAND=1 — Band number; NODATA\_INPUT=10 — Input pixel value to treat as NoData; NODATA\_OUTPUT=10 — Assign specified NoData value to output; SKIP\_NODATA=false — Do not output NoData values; CSV=false — Output comma-separated values; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gdalinfo — Raster information

**group:** Raster miscellaneous

| Parameter     | Type            | Required | Default | Description                                                        |
|---------------|-----------------|----------|---------|--------------------------------------------------------------------|
| INPUT         | raster          | yes      | —       | Input layer                                                        |
| MIN_MAX       | boolean         | no       | False   | Force computation of the actual min/max values for each band       |
| STATS         | boolean         | no       | False   | Read and display image statistics (force computation if necessary) |
| NOGCP         | boolean         | no       | False   | Suppress GCP info                                                  |
| NO_METADATA   | boolean         | no       | False   | Suppress metadata info                                             |
| EXTRA *(adv)* | string          | no       | —       | Additional command-line parameters                                 |
| OUTPUT        | fileDestination | yes      | —       | Layer information                                                  |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run gdal:gdalinfo INPUT=dem.tif MIN_MAX=false STATS=false NOGCP=false NO_METADATA=false
↳ OUTPUT=output.txt
```

This calls **Raster information**: INPUT=dem.tif — Input layer; MIN\_MAX=false — Force computation of the actual min/max values for each band; STATS=false — Read and display image statistics (force computation if necessary); NOGCP=false — Suppress GCP info; NO\_METADATA=false — Suppress metadata info; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gridaverage — Grid (Moving average)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | source            | yes      | —       | Point layer                                                                                                                                          |
| Z_FIELD *(adv)*          | field             | no       | —       | Z value from field                                                                                                                                   |
| RADIUS_1                 | number            | no       | 0.0     | The first radius of search ellipse                                                                                                                   |
| RADIUS_2                 | number            | no       | 0.0     | The second radius of search ellipse                                                                                                                  |
| ANGLE                    | number            | no       | 0.0     | Angle of search ellipse rotation in degrees (counter clockwise)                                                                                      |
| MIN_POINTS               | number            | no       | 0       | Minimum number of data points to use                                                                                                                 |
| NODATA                   | number            | no       | 0.0     | NODATA marker to fill empty points                                                                                                                   |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                   |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Interpolated (moving average)                                                                                                                        |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:gridaverage INPUT=input.gpkg RADIUS_1=0 RADIUS_2=0 ANGLE=0 MIN_POINTS=0 NODATA=0  
↳ OPTIONS=value OUTPUT=output.tif
```

This calls **Grid (Moving average)**: INPUT=input.gpkg — Point layer; RADIUS\_1=0 — The first radius of search ellipse; RADIUS\_2=0 — The second radius of search ellipse; ANGLE=0 — Angle of search ellipse rotation in degrees (counter clockwise); MIN\_POINTS=0 — Minimum number of data points to use; NODATA=0 — NODATA marker to fill empty points; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:griddatametrics — Grid (Data metrics)

**group:** Raster analysis

| Parameter                | Type   | Required | Default | Description                                                                                                                 |
|--------------------------|--------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | source | yes      | —       | Point layer                                                                                                                 |
| Z_FIELD *(adv)*          | field  | no       | —       | Z value from field                                                                                                          |
| METRIC                   | enum   | no       | 0       | Data metric to use — options: 0=Minimum, 1=Maximum, 2=Range, 3=Count, 4=Average distance, 5=Average distance between points |
| RADIUS_1                 | number | no       | 0.0     | The first radius of search ellipse                                                                                          |
| RADIUS_2                 | number | no       | 0.0     | The second radius of search ellipse                                                                                         |
| ANGLE                    | number | no       | 0.0     | Angle of search ellipse rotation in degrees (counter clockwise)                                                             |
| MIN_POINTS               | number | no       | 0       | Minimum number of data points to use                                                                                        |
| NODATA                   | number | no       | 0.0     | NODATA marker to fill empty points                                                                                          |
| OPTIONS                  | string | no       | —       | Additional creation options                                                                                                 |
| CREATION_OPTIONS *(adv)* | string | no       | —       | Additional creation options                                                                                                 |
| EXTRA *(adv)*            | string | no       | —       | Additional command-line parameters                                                                                          |

| Parameter         | Type              | Required | Default | Description                                                                                                                                          |
|-------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATA_TYPE *(adv)* | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT            | rasterDestination | yes      | —       | Interpolated (data metrics)                                                                                                                          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:griddatametrics INPUT=input.gpkg METRIC=0 RADIUS_1=0 RADIUS_2=0 ANGLE=0
↳ MIN_POINTS=0 NODATA=0 OUTPUT=output.tif
```

This calls **Grid (Data metrics)**: INPUT=input.gpkg — Point layer; METRIC=0 selects *Minimum*; RADIUS\_1=0 — The first radius of search ellipse; RADIUS\_2=0 — The second radius of search ellipse; ANGLE=0 — Angle of search ellipse rotation in degrees (counter clockwise); MIN\_POINTS=0 — Minimum number of data points to use; NODATA=0 — NODATA marker to fill empty points; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gridinversedistance — Grid (Inverse distance to a power)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | source            | yes      | —       | Point layer                                                                                                                                          |
| Z_FIELD *(adv)*          | field             | no       | —       | Z value from field                                                                                                                                   |
| POWER                    | number            | no       | 2.0     | Weighting power                                                                                                                                      |
| SMOOTHING                | number            | no       | 0.0     | Smoothing                                                                                                                                            |
| RADIUS_1                 | number            | no       | 0.0     | The first radius of search ellipse                                                                                                                   |
| RADIUS_2                 | number            | no       | 0.0     | The second radius of search ellipse                                                                                                                  |
| ANGLE                    | number            | no       | 0.0     | Angle of search ellipse rotation in degrees (counter clockwise)                                                                                      |
| MAX_POINTS               | number            | no       | 0       | Maximum number of data points to use                                                                                                                 |
| MIN_POINTS               | number            | no       | 0       | Minimum number of data points to use                                                                                                                 |
| NODATA                   | number            | no       | 0.0     | NODATA marker to fill empty points                                                                                                                   |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                   |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Interpolated (IDW)                                                                                                                                   |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:gridinversedistance INPUT=input.gpkg POWER=2 SMOOTHING=0 RADIUS_1=0 RADIUS_2=0
↳ ANGLE=0 MAX_POINTS=0 OUTPUT=output.tif
```

This calls **Grid (Inverse distance to a power)**: INPUT=input.gpkg — Point layer; POWER=2 — Weighting power; SMOOTHING=0 — Smoothing; RADIUS\_1=0 — The first radius of search ellipse; RADIUS\_2=0 — The second radius of search ellipse; ANGLE=0 — Angle of search ellipse rotation in degrees (counter clockwise); MAX\_POINTS=0 — Maximum number of data points to use; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gridinversedistancenearestneighbor — Grid (IDW with nearest neighbor searching)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | source            | yes      | —       | Point layer                                                                                                                                          |
| Z_FIELD *(adv)*          | field             | no       | —       | Z value from field                                                                                                                                   |
| POWER                    | number            | no       | 2.0     | Weighting power                                                                                                                                      |
| SMOOTHING                | number            | no       | 0.0     | Smoothing                                                                                                                                            |
| RADIUS                   | number            | no       | 1.0     | The radius of the search circle                                                                                                                      |
| MAX_POINTS               | number            | no       | 12      | Maximum number of data points to use                                                                                                                 |
| MIN_POINTS               | number            | no       | 0       | Minimum number of data points to use                                                                                                                 |
| NODATA                   | number            | no       | 0.0     | NODATA marker to fill empty points                                                                                                                   |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                   |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Interpolated (IDW with NN search)                                                                                                                    |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:gridinversedistancenearestneighbor INPUT=input.gpkg POWER=2 SMOOTHING=0 RADIUS=1  
↳ MAX_POINTS=12 MIN_POINTS=0 NODATA=0 OUTPUT=output.tif
```

This calls **Grid (IDW with nearest neighbor searching)**: INPUT=input.gpkg — Point layer; POWER=2 — Weighting power; SMOOTHING=0 — Smoothing; RADIUS=1 — The radius of the search circle; MAX\_POINTS=12 — Maximum number of data points to use; MIN\_POINTS=0 — Minimum number of data points to use; NODATA=0 — NODATA marker to fill empty points; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gridlinear — Grid (Linear)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | source            | yes      | —       | Point layer                                                                                                                                          |
| Z_FIELD *(adv)*          | field             | no       | —       | Z value from field                                                                                                                                   |
| RADIUS                   | number            | no       | -1.0    | Search distance                                                                                                                                      |
| NODATA                   | number            | no       | 0.0     | NODATA marker to fill empty points                                                                                                                   |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                   |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Interpolated (Linear)                                                                                                                                |

**Outputs:** OUTPUT (outputRaster)

### Example usage



```
run gdal:gridlinear INPUT=input.gpkg RADIUS=-1 NODATA=0 OPTIONS=value OUTPUT=output.tif
```

This calls **Grid (Linear)**: INPUT=input.gpkg — Point layer; RADIUS=-1 — Search distance; NODATA=0 — NODATA marker to fill empty points; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:gridnearestneighbor — Grid (Nearest neighbor)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | source            | yes      | —       | Point layer                                                                                                                                          |
| Z_FIELD *(adv)*          | field             | no       | —       | Z value from field                                                                                                                                   |
| RADIUS_1                 | number            | no       | 0.0     | The first radius of search ellipse                                                                                                                   |
| RADIUS_2                 | number            | no       | 0.0     | The second radius of search ellipse                                                                                                                  |
| ANGLE                    | number            | no       | 0.0     | Angle of search ellipse rotation in degrees (counter clockwise)                                                                                      |
| NODATA                   | number            | no       | 0.0     | NODATA marker to fill empty points                                                                                                                   |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                   |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Interpolated (Nearest neighbor)                                                                                                                      |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:gridnearestneighbor INPUT=input.gpkg RADIUS_1=0 RADIUS_2=0 ANGLE=0 NODATA=0
↳ OPTIONS=value OUTPUT=output.tif
```

This calls **Grid (Nearest neighbor)**: INPUT=input.gpkg — Point layer; RADIUS\_1=0 — The first radius of search ellipse; RADIUS\_2=0 — The second radius of search ellipse; ANGLE=0 — Angle of search ellipse rotation in degrees (counter clockwise); NODATA=0 — NODATA marker to fill empty points; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:hillshade — Hillshade

**group:** Raster analysis

| Parameter        | Type    | Required | Default | Description                                              |
|------------------|---------|----------|---------|----------------------------------------------------------|
| INPUT            | raster  | yes      | —       | Input layer                                              |
| BAND             | band    | no       | 1       | Band number                                              |
| Z_FACTOR         | number  | no       | 1.0     | Z factor (vertical exaggeration)                         |
| SCALE            | number  | no       | 1.0     | Scale (ratio of vertical units to horizontal)            |
| AZIMUTH          | number  | no       | 315.0   | Azimuth of the light                                     |
| ALTITUDE         | number  | no       | 45.0    | Altitude of the light                                    |
| COMPUTE_EDGES    | boolean | no       | False   | Compute edges                                            |
| ZEVENBERGEN      | boolean | no       | False   | Use Zevenbergen&Thorne formula instead of the Horn's one |
| COMBINED         | boolean | no       | False   | Combined shading                                         |
| MULTIDIRECTIONAL | boolean | no       | False   | Multidirectional shading                                 |

| Parameter                   | Type              | Required | Default | Description                        |
|-----------------------------|-------------------|----------|---------|------------------------------------|
| OPTIONS                     | string            | no       | —       | Additional creation options        |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Additional creation options        |
| EXTRA *(adv)*               | string            | no       | —       | Additional command-line parameters |
| OUTPUT                      | rasterDestination | yes      | —       | Hillshade                          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:hillshade INPUT=dem.tif BAND=1 Z_FACTOR=1 SCALE=1 AZIMUTH=315 ALTITUDE=45
↪ COMPUTE_EDGES=false OUTPUT=output.tif
```

This calls **Hillshade**: INPUT=dem.tif — Input layer; BAND=1 — Band number; Z\_FACTOR=1 — Z factor (vertical exaggeration); SCALE=1 — Scale (ratio of vertical units to horizontal); AZIMUTH=315 — Azimuth of the light; ALTITUDE=45 — Altitude of the light; COMPUTE\_EDGES=false — Compute edges; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:importvectorintopostgisdatabaseavailableconnections — Export to PostgreSQL (available connections)

**group:** Vector miscellaneous

Exports a vector layer to an existing PostgreSQL database connection

| Parameter      | Type               | Required | Default | Description                                                                                                                                                                                                                                                                                      |
|----------------|--------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATABASE       | providerconnection | yes      | —       | Database (connection name)                                                                                                                                                                                                                                                                       |
| INPUT          | source             | yes      | —       | Input layer                                                                                                                                                                                                                                                                                      |
| SHAPE_ENCODING | string             | no       | —       | Shape encoding                                                                                                                                                                                                                                                                                   |
| GTYPE          | enum               | no       | 0       | Output geometry type — options: 0=, 1=NONE, 2=GEOMETRY, 3=POINT, 4=LINestring, 5=POLYGON, 6=GEOMETRYCOLLECTION, 7=MULTIPOINT, 8=MULTIPOLYGON, 9=MULTILINestring, 10=CIRCULARSTRING, 11=COMPOUNDCURVE, 12=CURVEPOLYGON, 13=MULTICURVE, 14=MULTISURFACE, 15=CONVERT_TO_LINEAR, 16=CONVERT_TO_CURVE |
| A_SRS          | crs                | no       | —       | Assign an output CRS                                                                                                                                                                                                                                                                             |
| T_SRS          | crs                | no       | —       | Reproject to this CRS on output                                                                                                                                                                                                                                                                  |
| S_SRS          | crs                | no       | —       | Override source CRS                                                                                                                                                                                                                                                                              |
| SCHEMA         | databaseschema     | no       | public  | Schema (schema name)                                                                                                                                                                                                                                                                             |
| TABLE          | databasetable      | no       | —       | Table to import to (leave blank to use layer name)                                                                                                                                                                                                                                               |
| PK             | string             | no       | id      | Primary key (new field)                                                                                                                                                                                                                                                                          |
| PRIMARY_KEY    | field              | no       | —       | Primary key (existing field, used if the above option is left empty)                                                                                                                                                                                                                             |
| GEOCOLUMN      | string             | no       | geom    | Geometry column name                                                                                                                                                                                                                                                                             |
| DIM            | enum               | no       | 0       | Vector dimensions — options: 0=2, 1=3, 2=4                                                                                                                                                                                                                                                       |
| SIMPLIFY       | string             | no       | —       | Distance tolerance for simplification                                                                                                                                                                                                                                                            |
| SEGMENTIZE     | string             | no       | —       | Maximum distance between 2 nodes (densification)                                                                                                                                                                                                                                                 |
| SPAT           | extent             | no       | —       | Select features by extent (defined in input layer CRS)                                                                                                                                                                                                                                           |
| CLIP           | boolean            | no       | False   | Clip the input layer using the above (rectangle) extent                                                                                                                                                                                                                                          |
| WHERE          | string             | no       | —       | Select features using a SQL "WHERE" statement (Ex: column='value')                                                                                                                                                                                                                               |
| GT             | string             | no       | —       | Group N features per transaction (Default: 20000)                                                                                                                                                                                                                                                |
| OVERWRITE      | boolean            | no       | True    | Overwrite existing table                                                                                                                                                                                                                                                                         |
| APPEND         | boolean            | no       | False   | Append to existing table                                                                                                                                                                                                                                                                         |
| ADDFIELDS      | boolean            | no       | False   | Append and add new fields to existing table                                                                                                                                                                                                                                                      |
| LAUNDER        | boolean            | no       | False   | Do not launder columns/table names                                                                                                                                                                                                                                                               |
| INDEX          | boolean            | no       | False   | Do not create spatial index                                                                                                                                                                                                                                                                      |

| Parameter      | Type    | Required | Default | Description                                                |
|----------------|---------|----------|---------|------------------------------------------------------------|
| SKIPFAILURES   | boolean | no       | False   | Continue after a failure, skipping the failed feature      |
| MAKEVALID      | boolean | no       | False   | Validate geometries based on Simple Features specification |
| PROMOTETOMULTI | boolean | no       | True    | Promote to Multipart                                       |
| PRECISION      | boolean | no       | True    | Keep width and precision of input attributes               |
| OPTIONS        | string  | no       | —       | Additional creation options                                |

## Example usage

```
run gdal:importvectorintopostgisdatabaseavailableconnections DATABASE=... INPUT=input.gpkg
↳ SHAPE_ENCODING=value GTYPE=0 A_SRS=EPSG:6346 T_SRS=EPSG:6346 S_SRS=EPSG:6346
```

This calls **Export to PostgreSQL (available connections)**: DATABASE=... — Database (connection name); INPUT=input.gpkg — Input layer; SHAPE\_ENCODING=value — Shape encoding; GTYPE=0 selects \*\*; A\_SRS=EPSG:6346 — Assign an output CRS; T\_SRS=EPSG:6346 — Reproject to this CRS on output; S\_SRS=EPSG:6346 — Override source CRS. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:importvectorintopostgisdatabasenewconnection — Export to PostgreSQL (new connection)

**group:** Vector miscellaneous

Exports a vector layer to a new PostgreSQL database connection

| Parameter      | Type    | Required | Default   | Description                                                                                                                                                                                                                                                                                      |
|----------------|---------|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT          | source  | yes      | —         | Input layer                                                                                                                                                                                                                                                                                      |
| SHAPE_ENCODING | string  | no       | —         | Shape encoding                                                                                                                                                                                                                                                                                   |
| GTYPE          | enum    | no       | 0         | Output geometry type — options: 0=, 1=NONE, 2=GEOMETRY, 3=POINT, 4=LINestring, 5=POLYGON, 6=GEOMETRYCOLLECTION, 7=MULTIPOINT, 8=MULTIPOLYGON, 9=MULTILINestring, 10=CIRCULARSTRING, 11=COMPOUNDCURVE, 12=CURVEPOLYGON, 13=MULTICURVE, 14=MULTISURFACE, 15=CONVERT_TO_LINEAR, 16=CONVERT_TO_CURVE |
| A_SRS          | crs     | no       | —         | Assign an output CRS                                                                                                                                                                                                                                                                             |
| T_SRS          | crs     | no       | —         | Reproject to this CRS on output                                                                                                                                                                                                                                                                  |
| S_SRS          | crs     | no       | —         | Override source CRS                                                                                                                                                                                                                                                                              |
| HOST           | string  | no       | localhost | Host                                                                                                                                                                                                                                                                                             |
| PORT           | string  | no       | 5432      | Port                                                                                                                                                                                                                                                                                             |
| USER           | string  | no       | —         | Username                                                                                                                                                                                                                                                                                         |
| DBNAME         | string  | no       | —         | Database name                                                                                                                                                                                                                                                                                    |
| PASSWORD       | string  | no       | —         | Password                                                                                                                                                                                                                                                                                         |
| SCHEMA         | string  | no       | public    | Schema name                                                                                                                                                                                                                                                                                      |
| TABLE          | string  | no       | —         | Table name, leave blank to use input name                                                                                                                                                                                                                                                        |
| PK             | string  | no       | id        | Primary key (new field)                                                                                                                                                                                                                                                                          |
| PRIMARY_KEY    | field   | no       | —         | Primary key (existing field, used if the above option is left empty)                                                                                                                                                                                                                             |
| GEOCOLUMN      | string  | no       | geom      | Geometry column name                                                                                                                                                                                                                                                                             |
| DIM            | enum    | no       | 0         | Vector dimensions — options: 0=2, 1=3, 2=4                                                                                                                                                                                                                                                       |
| SIMPLIFY       | string  | no       | —         | Distance tolerance for simplification                                                                                                                                                                                                                                                            |
| SEGMENTIZE     | string  | no       | —         | Maximum distance between 2 nodes (densification)                                                                                                                                                                                                                                                 |
| SPAT           | extent  | no       | —         | Select features by extent (defined in input layer CRS)                                                                                                                                                                                                                                           |
| CLIP           | boolean | no       | False     | Clip the input layer using the above (rectangle) extent                                                                                                                                                                                                                                          |
| FIELDS         | field   | no       | —         | Fields to include (leave empty to use all fields)                                                                                                                                                                                                                                                |
| WHERE          | string  | no       | —         | Select features using a SQL "WHERE" statement (Ex: column='value')                                                                                                                                                                                                                               |
| GT             | string  | no       | —         | Group N features per transaction (Default: 20000)                                                                                                                                                                                                                                                |
| OVERWRITE      | boolean | no       | True      | Overwrite existing table                                                                                                                                                                                                                                                                         |

| Parameter      | Type    | Required | Default | Description                                                |
|----------------|---------|----------|---------|------------------------------------------------------------|
| APPEND         | boolean | no       | False   | Append to existing table                                   |
| ADDFIELDS      | boolean | no       | False   | Append and add new fields to existing table                |
| LAUNDER        | boolean | no       | False   | Do not launder columns/table names                         |
| INDEX          | boolean | no       | False   | Do not create spatial index                                |
| SKIPFAILURES   | boolean | no       | False   | Continue after a failure, skipping the failed feature      |
| MAKEVALID      | boolean | no       | False   | Validate geometries based on Simple Features specification |
| PROMOTETOMULTI | boolean | no       | True    | Promote to Multipart                                       |
| PRECISION      | boolean | no       | True    | Keep width and precision of input attributes               |
| OPTIONS        | string  | no       | —       | Additional creation options                                |

## Example usage

```
run gdal:importvectorintopostgisdatabasenewconnection INPUT=input.gpkg
↳ SHAPE_ENCODING=value GTYPE=0 A_SRS=EPSG:6346 T_SRS=EPSG:6346 S_SRS=EPSG:6346
↳ HOST=localhost
```

This calls **Export to PostgreSQL (new connection)**: INPUT=input.gpkg — Input layer; SHAPE\_ENCODING=value — Shape encoding; GTYPE=0 selects **\*\***; A\_SRS=EPSG:6346 — Assign an output CRS; T\_SRS=EPSG:6346 — Reproject to this CRS on output; S\_SRS=EPSG:6346 — Override source CRS; HOST=localhost — Host. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:merge — Merge

**group:** Raster miscellaneous

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | multilayer        | yes      | —       | Input layers                                                                                                                                         |
| PCT                      | boolean           | no       | False   | Grab pseudocolor table from first layer                                                                                                              |
| SEPARATE                 | boolean           | no       | False   | Place each input file into a separate band                                                                                                           |
| NODATA_INPUT *(adv)*     | number            | no       | —       | Input pixel value to treat as NoData                                                                                                                 |
| NODATA_OUTPUT *(adv)*    | number            | no       | —       | Assign specified NoData value to output                                                                                                              |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                   |
| DATA_TYPE                | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Merged                                                                                                                                               |

**Outputs:** OUTPUT (outputRaster)

## Example usage

```
run gdal:merge INPUT="a.gpkg;b.gpkg" PCT=false SEPARATE=false OPTIONS=value DATA_TYPE=5
↳ OUTPUT=output.tif
```

This calls **Merge**: INPUT="a.gpkg;b.gpkg" — Input layers; PCT=false — Grab pseudocolor table from first layer; SEPARATE=false — Place each input file into a separate band; OPTIONS=value — Additional creation options; DATA\_TYPE=5 selects *Float32*; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:nearblack — Near black

**group:** Raster analysis

| Parameter                   | Type              | Required | Default | Description                                            |
|-----------------------------|-------------------|----------|---------|--------------------------------------------------------|
| INPUT                       | raster            | yes      | —       | Input layer                                            |
| NEAR                        | number            | no       | 15      | How far from black (white)                             |
| WHITE                       | boolean           | no       | False   | Search for nearly white pixels instead of nearly black |
| OPTIONS                     | string            | no       | —       | Additional creation options                            |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Additional creation options                            |
| EXTRA *(adv)*               | string            | no       | —       | Additional command-line parameters                     |
| OUTPUT                      | rasterDestination | yes      | —       | Nearblack                                              |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:nearblack INPUT=dem.tif NEAR=15 WHITE=false OPTIONS=value OUTPUT=output.tif
```

This calls **Near black**: INPUT=dem.tif — Input layer; NEAR=15 — How far from black (white); WHITE=false — Search for nearly white pixels instead of nearly black; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:offsetcurve — Offset curve

**group:** Vector geoprocessing

| Parameter       | Type              | Required | Default  | Description                                                   |
|-----------------|-------------------|----------|----------|---------------------------------------------------------------|
| INPUT           | source            | yes      | —        | Input layer                                                   |
| GEOMETRY        | string            | no       | geometry | Geometry column name                                          |
| DISTANCE        | distance          | no       | 10       | Offset distance (left-sided: positive, right-sided: negative) |
| OPTIONS *(adv)* | string            | no       | —        | Additional creation options                                   |
| OUTPUT          | vectorDestination | yes      | —        | Offset curve                                                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:offsetcurve INPUT=input.gpkg GEOMETRY=geometry DISTANCE=10 OUTPUT=output.gpkg
```

This calls **Offset curve**: INPUT=input.gpkg — Input layer; GEOMETRY=geometry — Geometry column name; DISTANCE=10 — Offset distance (left-sided: positive, right-sided: negative); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:ogrinfo — Vector information

**group:** Vector miscellaneous

| Parameter     | Type            | Required | Default | Description                                 |
|---------------|-----------------|----------|---------|---------------------------------------------|
| INPUT         | vector          | yes      | —       | Input layer                                 |
| ALL_LAYERS    | boolean         | no       | False   | Enable listing of all layers in the dataset |
| SUMMARY_ONLY  | boolean         | no       | True    | Summary output only                         |
| NO_METADATA   | boolean         | no       | False   | Suppress metadata info                      |
| EXTRA *(adv)* | string          | no       | —       | Additional command-line parameters          |
| OUTPUT        | fileDestination | yes      | —       | Layer information                           |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run gdal:ogrinfo INPUT=input.gpkg ALL_LAYERS=false SUMMARY_ONLY=true NO_METADATA=false
↳ OUTPUT=output.txt
```

This calls **Vector information**: INPUT=input.gpkg — Input layer; ALL\_LAYERS=false — Enable listing of all layers in the dataset; SUMMARY\_ONLY=true — Summary output only; NO\_METADATA=false — Suppress metadata info; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:ogrinfojson — Vector information (JSON)

**group:** Vector miscellaneous

| Parameter     | Type            | Required | Default | Description                                 |
|---------------|-----------------|----------|---------|---------------------------------------------|
| INPUT         | vector          | yes      | —       | Input layer                                 |
| ALL_LAYERS    | boolean         | no       | False   | Enable listing of all layers in the dataset |
| FEATURES      | boolean         | no       | False   | Enable listing of features                  |
| NO_METADATA   | boolean         | no       | False   | Suppress metadata info                      |
| EXTRA *(adv)* | string          | no       | —       | Additional command-line parameters          |
| OUTPUT        | fileDestination | yes      | —       | Layer information                           |

**Outputs:** OUTPUT (outputFile)

### Example usage

```
run gdal:ogrinfojson INPUT=input.gpkg ALL_LAYERS=false FEATURES=false NO_METADATA=false
↳ OUTPUT=output.txt
```

This calls **Vector information (JSON)**: INPUT=input.gpkg — Input layer; ALL\_LAYERS=false — Enable listing of all layers in the dataset; FEATURES=false — Enable listing of features; NO\_METADATA=false — Suppress metadata info; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:onesidebuffer — One side buffer

**group:** Vector geoprocessing

| Parameter           | Type              | Required | Default  | Description                                                                                 |
|---------------------|-------------------|----------|----------|---------------------------------------------------------------------------------------------|
| INPUT               | source            | yes      | —        | Input layer                                                                                 |
| GEOMETRY            | string            | no       | geometry | Geometry column name                                                                        |
| DISTANCE            | distance          | no       | 10       | Buffer distance                                                                             |
| BUFFER_SIDE         | enum              | no       | 0        | Buffer side — options: 0=Right, 1=Left                                                      |
| FIELD               | field             | no       | —        | Dissolve by attribute                                                                       |
| DISSOLVE            | boolean           | no       | False    | Dissolve all results                                                                        |
| EXPLODE_COLLECTIONS | boolean           | no       | False    | Produce one feature for each geometry in any kind of geometry collection in the source file |
| OPTIONS *(adv)*     | string            | no       | —        | Additional creation options                                                                 |
| OUTPUT              | vectorDestination | yes      | —        | One-sided buffer                                                                            |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:onesidebuffer INPUT=input.gpkg GEOMETRY=geometry DISTANCE=10 BUFFER_SIDE=0
↳ FIELD=field1 DISSOLVE=false EXPLODE_COLLECTIONS=false OUTPUT=output.gpkg
```

This calls **One side buffer**: INPUT=input.gpkg — Input layer; GEOMETRY=geometry — Geometry column name; DISTANCE=10 — Buffer distance; BUFFER\_SIDE=0 selects *Right*; FIELD=field1 — Dissolve by attribute; DISSOLVE=false — Dissolve all results; EXPLODE\_COLLECTIONS=false —

Produce one feature for each geometry in any kind of geometry collection in the...; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:overviews — Build overviews (pyramids)

**group:** Raster miscellaneous

| Parameter          | Type    | Required | Default | Description                                                                                                                                                                                                        |
|--------------------|---------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT              | raster  | yes      | —       | Input layer                                                                                                                                                                                                        |
| CLEAN              | boolean | no       | False   | Remove all existing overviews                                                                                                                                                                                      |
| LEVELS *(adv)*     | string  | no       | —       | Overview levels (e.g. 2 4 8 16)                                                                                                                                                                                    |
| RESAMPLING *(adv)* | enum    | no       | —       | Resampling method — options: 0=Nearest Neighbour (default), 1=Average, 2=Gaussian, 3=Cubic (4x4 Kernel), 4=Cubic B-Spline (4x4 Kernel), 5=Lanczos (6x6 Kernel), 6=Average MP, 7=Average in Mag/Phase Space, 8=Mode |
| FORMAT *(adv)*     | enum    | no       | 0       | Overviews format — options: 0=Internal (if possible), 1=External (GTiff .ovr), 2=External (ERDAS Imagine .aux)                                                                                                     |
| EXTRA *(adv)*      | string  | no       | —       | Additional command-line parameters                                                                                                                                                                                 |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:overviews INPUT=dem.tif CLEAN=false
```

This calls **Build overviews (pyramids)**: INPUT=dem.tif — Input layer; CLEAN=false — Remove all existing overviews. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:pansharp — Pansharpening

**group:** Raster miscellaneous

| Parameter                | Type              | Required | Default | Description                                                                                                                                                          |
|--------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPECTRAL                 | raster            | yes      | —       | Spectral dataset                                                                                                                                                     |
| PANCHROMATIC             | raster            | yes      | —       | Panchromatic dataset                                                                                                                                                 |
| RESAMPLING *(adv)*       | enum              | no       | 2       | Resampling algorithm — options: 0=Nearest Neighbour, 1=Bilinear (2x2 Kernel), 2=Cubic (4x4 Kernel), 3=Cubic B-Spline (4x4 Kernel), 4=Lanczos (6x6 Kernel), 5=Average |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                                   |
| OUTPUT                   | rasterDestination | yes      | —       | Output                                                                                                                                                               |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:pansharp SPECTRAL=dem.tif PANCHROMATIC=dem.tif OPTIONS=value OUTPUT=output.tif
```

This calls **Pansharpening**: SPECTRAL=dem.tif — Spectral dataset; PANCHROMATIC=dem.tif — Panchromatic dataset; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:pcttorgb — PCT to RGB

**group:** Raster conversion

| Parameter | Type              | Required | Default | Description          |
|-----------|-------------------|----------|---------|----------------------|
| INPUT     | raster            | yes      | —       | Input layer          |
| BAND      | band              | no       | 1       | Band number          |
| RGBA      | boolean           | no       | False   | Generate a RGBA file |
| OUTPUT    | rasterDestination | yes      | —       | PCT to RGB           |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:pcttorgb INPUT=dem.tif BAND=1 RGBA=false OUTPUT=output.tif
```

This calls **PCT to RGB**: INPUT=dem.tif — Input layer; BAND=1 — Band number; RGBA=false — Generate a RGBA file; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:pointsalonglines — Points along lines

**group:** Vector geoprocessing

| Parameter       | Type              | Required | Default  | Description                                                     |
|-----------------|-------------------|----------|----------|-----------------------------------------------------------------|
| INPUT           | source            | yes      | —        | Input layer                                                     |
| GEOMETRY        | string            | no       | geometry | Geometry column name                                            |
| DISTANCE        | number            | no       | 0.5      | Distance from line start represented as fraction of line length |
| OPTIONS *(adv)* | string            | no       | —        | Additional creation options                                     |
| OUTPUT          | vectorDestination | yes      | —        | Points along lines                                              |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:pointsalonglines INPUT=input.gpkg GEOMETRY=geometry DISTANCE=0.5
↳ OUTPUT=output.gpkg
```

This calls **Points along lines**: INPUT=input.gpkg — Input layer; GEOMETRY=geometry — Geometry column name; DISTANCE=0.5 — Distance from line start represented as fraction of line length; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:polygonize — Polygonize (raster to vector) ★

**group:** Raster conversion · **niva verb:** polygonize

| Parameter           | Type              | Required | Default | Description                        |
|---------------------|-------------------|----------|---------|------------------------------------|
| INPUT               | raster            | yes      | —       | Input layer                        |
| BAND                | band              | no       | 1       | Band number                        |
| FIELD               | string            | no       | DN      | Name of the field to create        |
| EIGHT_CONNECTEDNESS | boolean           | no       | False   | Use 8-connectedness                |
| EXTRA *(adv)*       | string            | no       | —       | Additional command-line parameters |
| OUTPUT              | vectorDestination | yes      | —       | Vectorized                         |

**Outputs:** OUTPUT (outputVector)

### Example usage



```
run gdal:polygonize INPUT=dem.tif BAND=1 FIELD=DN EIGHT_CONNECTEDNESS=false
↳ OUTPUT=output.gpkg
```

This calls **Polygonize (raster to vector)**: INPUT=dem.tif — Input layer; BAND=1 — Band number; FIELD=DN — Name of the field to create; EIGHT\_CONNECTEDNESS=false — Use 8-connectedness; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:proximity — Proximity (raster distance)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                                                                                                                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                                                                                                                                          |
| BAND                     | band              | no       | 1       | Band number                                                                                                                                          |
| VALUES                   | string            | no       | —       | List of target pixels                                                                                                                                |
| UNITS                    | enum              | no       | 1       | Distance units — options: 0=Georeferenced coordinates, 1=Pixel coordinates                                                                           |
| MAX_DISTANCE             | number            | no       | 0.0     | The maximum distance to generate                                                                                                                     |
| REPLACE                  | number            | no       | 0.0     | Value to apply to pixels within the maximum distance of target pixels                                                                                |
| NODATA                   | number            | no       | 0.0     | Nodata value to use for the destination proximity raster                                                                                             |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                   |
| DATA_TYPE *(adv)*        | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Proximity map                                                                                                                                        |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:proximity INPUT=dem.tif BAND=1 VALUES=value UNITS=1 MAX_DISTANCE=0 REPLACE=0
↳ NODATA=0 OUTPUT=output.tif
```

This calls **Proximity (raster distance)**: INPUT=dem.tif — Input layer; BAND=1 — Band number; VALUES=value — List of target pixels; UNITS=1 selects *Pixel coordinates*; MAX\_DISTANCE=0 — The maximum distance to generate; REPLACE=0 — Value to apply to pixels within the maximum distance of target pixels; NODATA=0 — Nodata value to use for the destination proximity raster; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:rastercalculator — Raster calculator

**group:** Raster miscellaneous

| Parameter | Type   | Required | Default | Description                 |
|-----------|--------|----------|---------|-----------------------------|
| INPUT_A   | raster | yes      | —       | Input layer A               |
| BAND_A    | band   | yes      | —       | Number of raster band for A |
| INPUT_B   | raster | no       | —       | Input layer B               |
| BAND_B    | band   | no       | —       | Number of raster band for B |
| INPUT_C   | raster | no       | —       | Input layer C               |
| BAND_C    | band   | no       | —       | Number of raster band for C |
| INPUT_D   | raster | no       | —       | Input layer D               |

| Parameter                | Type              | Required | Default | Description                                                                                                                                            |
|--------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| BAND_D                   | band              | no       | —       | Number of raster band for D                                                                                                                            |
| INPUT_E                  | raster            | no       | —       | Input layer E                                                                                                                                          |
| BAND_E                   | band              | no       | —       | Number of raster band for E                                                                                                                            |
| INPUT_F                  | raster            | no       | —       | Input layer F                                                                                                                                          |
| BAND_F                   | band              | no       | —       | Number of raster band for F                                                                                                                            |
| FORMULA                  | string            | no       | A*2     | Calculation in gdalnumeric syntax using +/.* or any numpy array functions (i.e. logical_and())                                                         |
| NO_DATA                  | number            | no       | —       | Set output NoData value                                                                                                                                |
| EXTENT_OPT               | enum              | no       | 0       | Handling of extent differences — options: 0=Ignore, 1=Fail, 2=Union, 3=Intersect                                                                       |
| PROJWIN                  | extent            | no       | —       | Output extent                                                                                                                                          |
| RTYPE                    | enum              | no       | 5       | Output raster type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                            |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                            |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                     |
| OUTPUT                   | rasterDestination | yes      | —       | Calculated                                                                                                                                             |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:rastercalculator INPUT_A=dem.tif BAND_A=1 INPUT_B=dem.tif BAND_B=1
↳ INPUT_C=dem.tif BAND_C=1 INPUT_D=dem.tif OUTPUT=output.tif
```

This calls **Raster calculator**: INPUT\_A=dem.tif — Input layer A; BAND\_A=1 — Number of raster band for A; INPUT\_B=dem.tif — Input layer B; BAND\_B=1 — Number of raster band for B; INPUT\_C=dem.tif — Input layer C; BAND\_C=1 — Number of raster band for C; INPUT\_D=dem.tif — Input layer D; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:rasterize — Rasterize (vector to raster)

**group:** Vector conversion

| Parameter                | Type    | Required | Default | Description                                                                                                                                          |
|--------------------------|---------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                    | source  | yes      | —       | Input layer                                                                                                                                          |
| FIELD                    | field   | no       | —       | Field to use for a burn-in value                                                                                                                     |
| BURN                     | number  | no       | 0.0     | A fixed value to burn                                                                                                                                |
| USE_Z                    | boolean | no       | False   | Burn value extracted from the "Z" values of the feature                                                                                              |
| UNITS                    | enum    | yes      | —       | Output raster size units — options: 0=Pixels, 1=Georeferenced units                                                                                  |
| WIDTH                    | number  | no       | 0.0     | Width/Horizontal resolution                                                                                                                          |
| HEIGHT                   | number  | no       | 0.0     | Height/Vertical resolution                                                                                                                           |
| EXTENT                   | extent  | no       | —       | Output extent                                                                                                                                        |
| NODATA                   | number  | no       | —       | Assign a specified NoData value to output bands                                                                                                      |
| OPTIONS                  | string  | no       | —       | Additional creation options                                                                                                                          |
| CREATION_OPTIONS *(adv)* | string  | no       | —       | Additional creation options                                                                                                                          |
| DATA_TYPE *(adv)*        | enum    | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8 |
| INIT *(adv)*             | number  | no       | —       | Pre-initialize the output image with value                                                                                                           |
| INVERT *(adv)*           | boolean | no       | False   | Invert rasterization                                                                                                                                 |

| Parameter     | Type              | Required | Default | Description                        |
|---------------|-------------------|----------|---------|------------------------------------|
| EXTRA *(adv)* | string            | no       | —       | Additional command-line parameters |
| OUTPUT        | rasterDestination | yes      | —       | Rasterized                         |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:rasterize INPUT=input.gpkg UNITS=0 FIELD=field1 BURN=0 USE_Z=false WIDTH=0
↳ HEIGHT=0 OUTPUT=output.tif
```

This calls **Rasterize (vector to raster)**: INPUT=input.gpkg — Input layer; UNITS=0 selects *Pixels*; FIELD=field1 — Field to use for a burn-in value; BURN=0 — A fixed value to burn; USE\_Z=false — Burn value extracted from the "Z" values of the feature; WIDTH=0 — Width/Horizontal resolution; HEIGHT=0 — Height/Vertical resolution; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:rasterize\_over — Rasterize (overwrite with attribute)

**group:** Vector conversion

| Parameter     | Type    | Required | Default | Description                                  |
|---------------|---------|----------|---------|----------------------------------------------|
| INPUT         | source  | yes      | —       | Input vector layer                           |
| INPUT_RASTER  | raster  | yes      | —       | Input raster layer                           |
| FIELD         | field   | yes      | —       | Field to use for burn in value               |
| ADD *(adv)*   | boolean | no       | False   | Add burn in values to existing raster values |
| EXTRA *(adv)* | string  | no       | —       | Additional command-line parameters           |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:rasterize_over INPUT=input.gpkg INPUT_RASTER=dem.tif FIELD=field1
```

This calls **Rasterize (overwrite with attribute)**: INPUT=input.gpkg — Input vector layer; INPUT\_RASTER=dem.tif — Input raster layer; FIELD=field1 — Field to use for burn in value. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:rasterize\_over\_fixed\_value — Rasterize (overwrite with fixed value)

**group:** Vector conversion

| Parameter     | Type    | Required | Default | Description                                  |
|---------------|---------|----------|---------|----------------------------------------------|
| INPUT         | source  | yes      | —       | Input vector layer                           |
| INPUT_RASTER  | raster  | yes      | —       | Input raster layer                           |
| BURN          | number  | no       | 0.0     | A fixed value to burn                        |
| ADD *(adv)*   | boolean | no       | False   | Add burn in values to existing raster values |
| EXTRA *(adv)* | string  | no       | —       | Additional command-line parameters           |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:rasterize_over_fixed_value INPUT=input.gpkg INPUT_RASTER=dem.tif BURN=0
```

This calls **Rasterize (overwrite with fixed value)**: INPUT=input.gpkg — Input vector layer; INPUT\_RASTER=dem.tif — Input raster layer; BURN=0 — A fixed value to burn. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:rearrange\_bands — Rearrange bands

**group:** Raster conversion

This algorithm creates a new raster using selected band(s) from a given raster layer.

The algorithm also makes it possible to reorder the bands for the newly-created raster.

| Parameter                   | Type              | Required | Default | Description                                                                                                                                                                        |
|-----------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                       | raster            | yes      | —       | Input layer                                                                                                                                                                        |
| BANDS                       | band              | yes      | —       | Selected band(s)                                                                                                                                                                   |
| OPTIONS                     | string            | no       | —       | Additional creation options                                                                                                                                                        |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Additional creation options                                                                                                                                                        |
| DATA_TYPE *(adv)*           | enum              | no       | 0       | Output data type — options: 0=Use Input Layer Data Type, 1=Byte, 2=Int16, 3=UInt16, 4=UInt32, 5=Int32, 6=Float32, 7=Float64, 8=CInt16, 9=CInt32, 10=CFloat32, 11=CFloat64, 12=Int8 |
| OUTPUT                      | rasterDestination | yes      | —       | Converted                                                                                                                                                                          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:rearrange_bands INPUT=dem.tif BANDS=1 OPTIONS=value OUTPUT=output.tif
```

This calls **Rearrange bands**: INPUT=dem.tif — Input layer; BANDS=1 — Selected band(s); OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:retile — Retile

**group:** Raster miscellaneous

| Parameter                   | Type              | Required | Default | Description                                                                                                                                            |
|-----------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                       | multilayer        | yes      | —       | Input files                                                                                                                                            |
| TILE_SIZE_X                 | number            | no       | 256     | Tile width                                                                                                                                             |
| TILE_SIZE_Y                 | number            | no       | 256     | Tile height                                                                                                                                            |
| OVERLAP                     | number            | no       | 0       | Overlap in pixels between consecutive tiles                                                                                                            |
| LEVELS                      | number            | no       | 1       | Number of pyramids levels to build                                                                                                                     |
| OPTIONS                     | string            | no       | —       | Additional creation options                                                                                                                            |
| CREATION_OPTIONS<br>*(adv)* | string            | no       | —       | Additional creation options                                                                                                                            |
| SOURCE_CRS *(adv)*          | crs               | no       | —       | Source coordinate reference system                                                                                                                     |
| RESAMPLING *(adv)*          | enum              | no       | 0       | Resampling method — options: 0=Nearest Neighbour, 1=Bilinear (2x2 Kernel), 2=Cubic (4x4 Kernel), 3=Cubic B-Spline (4x4 Kernel), 4=Lanczos (6x6 Kernel) |
| DELIMITER *(adv)*           | string            | no       | ;       | Column delimiter used in the CSV file                                                                                                                  |
| EXTRA *(adv)*               | string            | no       | —       | Additional command-line parameters                                                                                                                     |
| DATA_TYPE *(adv)*           | enum              | no       | 5       | Output data type — options: 0=Byte, 1=Int16, 2=UInt16, 3=UInt32, 4=Int32, 5=Float32, 6=Float64, 7=CInt16, 8=CInt32, 9=CFloat32, 10=CFloat64, 11=Int8   |
| ONLY_PYRAMIDS *(adv)*       | boolean           | no       | False   | Build only the pyramids                                                                                                                                |
| DIR_FOR_ROW *(adv)*         | boolean           | no       | False   | Use separate directory for each tiles row                                                                                                              |
| OUTPUT                      | folderDestination | yes      | —       | Output directory                                                                                                                                       |
| OUTPUT_CSV                  | fileDestination   | no       | —       | CSV file containing the tile(s) georeferencing information                                                                                             |

**Outputs:** OUTPUT (outputFolder), OUTPUT\_CSV (outputFile)

## Example usage

```
run gdal:retiler INPUT="a.gpkg;b.gpkg" TILE_SIZE_X=256 TILE_SIZE_Y=256 OVERLAP=0 LEVELS=1  
↳ OPTIONS=value OUTPUT=output/
```

This calls **Retiler**: INPUT="a.gpkg;b.gpkg" — Input files; TILE\_SIZE\_X=256 — Tile width; TILE\_SIZE\_Y=256 — Tile height; OVERLAP=0 — Overlap in pixels between consecutive tiles; LEVELS=1 — Number of pyramids levels to build; OPTIONS=value — Additional creation options; OUTPUT=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:rgbtopct — RGB to PCT

**group:** Raster conversion

| Parameter | Type              | Required | Default | Description      |
|-----------|-------------------|----------|---------|------------------|
| INPUT     | raster            | yes      | —       | Input layer      |
| NCOLORS   | number            | no       | 2       | Number of colors |
| OUTPUT    | rasterDestination | yes      | —       | RGB to PCT       |

**Outputs:** OUTPUT (outputRaster)

## Example usage

```
run gdal:rgbtopct INPUT=dem.tif NCOLORS=2 OUTPUT=output.tif
```

This calls **RGB to PCT**: INPUT=dem.tif — Input layer; NCOLORS=2 — Number of colors; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:roughness — Roughness

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                 |
|--------------------------|-------------------|----------|---------|-----------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                 |
| BAND                     | band              | no       | 1       | Band number                 |
| COMPUTE_EDGES            | boolean           | no       | False   | Compute edges               |
| OPTIONS                  | string            | no       | —       | Additional creation options |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options |
| OUTPUT                   | rasterDestination | yes      | —       | Roughness                   |

**Outputs:** OUTPUT (outputRaster)

## Example usage

```
run gdal:roughness INPUT=dem.tif BAND=1 COMPUTE_EDGES=false OPTIONS=value  
↳ OUTPUT=output.tif
```

This calls **Roughness**: INPUT=dem.tif — Input layer; BAND=1 — Band number; COMPUTE\_EDGES=false — Compute edges; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:sieve — Sieve

**group:** Raster analysis

| Parameter           | Type              | Required | Default | Description                                             |
|---------------------|-------------------|----------|---------|---------------------------------------------------------|
| INPUT               | raster            | yes      | —       | Input layer                                             |
| THRESHOLD           | number            | no       | 10      | Threshold                                               |
| EIGHT_CONNECTEDNESS | boolean           | no       | False   | Use 8-connectedness                                     |
| NO_MASK             | boolean           | no       | False   | Do not use the default validity mask for the input band |
| MASK_LAYER          | raster            | no       | —       | Validity mask                                           |
| EXTRA *(adv)*       | string            | no       | —       | Additional command-line parameters                      |
| OUTPUT              | rasterDestination | yes      | —       | Sieved                                                  |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:sieve INPUT=dem.tif THRESHOLD=10 EIGHT_CONNECTEDNESS=false NO_MASK=false
↳ MASK_LAYER=dem.tif OUTPUT=output.tif
```

This calls **Sieve**: INPUT=dem.tif — Input layer; THRESHOLD=10 — Threshold; EIGHT\_CONNECTEDNESS=false — Use 8-connectedness; NO\_MASK=false — Do not use the default validity mask for the input band; MASK\_LAYER=dem.tif — Validity mask; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:slope — Slope ★

**group:** Raster analysis · **niva verb:** slope

| Parameter                | Type              | Required | Default | Description                                              |
|--------------------------|-------------------|----------|---------|----------------------------------------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                                              |
| BAND                     | band              | no       | 1       | Band number                                              |
| SCALE                    | number            | no       | 1.0     | Ratio of vertical units to horizontal                    |
| AS_PERCENT               | boolean           | no       | False   | Express slope as percent instead of degrees              |
| COMPUTE_EDGES            | boolean           | no       | False   | Compute edges                                            |
| ZEVENBERGEN              | boolean           | no       | False   | Use Zevenbergen&Thorne formula instead of the Horn's one |
| OPTIONS                  | string            | no       | —       | Additional creation options                              |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                              |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                       |
| OUTPUT                   | rasterDestination | yes      | —       | Slope                                                    |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:slope INPUT=dem.tif BAND=1 SCALE=1 AS_PERCENT=false COMPUTE_EDGES=false
↳ ZEVENBERGEN=false OPTIONS=value OUTPUT=output.tif
```

This calls **Slope**: INPUT=dem.tif — Input layer; BAND=1 — Band number; SCALE=1 — Ratio of vertical units to horizontal; AS\_PERCENT=false — Express slope as percent instead of degrees; COMPUTE\_EDGES=false — Compute edges; ZEVENBERGEN=false — Use Zevenbergen&Thorne formula instead of the Horn's one; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:tileindex — Tile index

**group:** Raster miscellaneous

| Parameter              | Type              | Required | Default  | Description                                                                                                           |
|------------------------|-------------------|----------|----------|-----------------------------------------------------------------------------------------------------------------------|
| LAYERS                 | multilayer        | yes      | —        | Input files                                                                                                           |
| PATH_FIELD_NAME        | string            | no       | location | Field name to hold the file path to the indexed rasters                                                               |
| ABSOLUTE_PATH          | boolean           | no       | False    | Store absolute path to the indexed rasters                                                                            |
| PROJ_DIFFERENCE        | boolean           | no       | False    | Skip files with different projection reference                                                                        |
| TARGET_CRS *(adv)*     | crs               | no       | —        | Transform geometries to the given CRS                                                                                 |
| CRS_FIELD_NAME *(adv)* | string            | no       | —        | The name of the field to store the SRS of each tile                                                                   |
| CRS_FORMAT *(adv)*     | enum              | no       | 0        | The format in which the CRS of each tile must be written — options: 0=Auto, 1=Well-known text (WKT), 2=EPSG, 3=Proj.4 |
| OUTPUT                 | vectorDestination | yes      | —        | Tile index                                                                                                            |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run gdal:tileindex LAYERS="a.gpkg;b.gpkg" PATH_FIELD_NAME=location ABSOLUTE_PATH=false
↳ PROJ_DIFFERENCE=false OUTPUT=output.gpkg
```

This calls **Tile index**: LAYERS="a.gpkg;b.gpkg" — Input files; PATH\_FIELD\_NAME=location — Field name to hold the file path to the indexed rasters; ABSOLUTE\_PATH=false — Store absolute path to the indexed rasters; PROJ\_DIFFERENCE=false — Skip files with different projection reference; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:tpitopographicpositionindex — Topographic Position Index (TPI)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                 |
|--------------------------|-------------------|----------|---------|-----------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                 |
| BAND                     | band              | no       | 1       | Band number                 |
| COMPUTE_EDGES            | boolean           | no       | False   | Compute edges               |
| OPTIONS                  | string            | no       | —       | Additional creation options |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options |
| OUTPUT                   | rasterDestination | yes      | —       | Topographic Position Index  |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:tpitopographicpositionindex INPUT=dem.tif BAND=1 COMPUTE_EDGES=false
↳ OPTIONS=value OUTPUT=output.tif
```

This calls **Topographic Position Index (TPI)**: INPUT=dem.tif — Input layer; BAND=1 — Band number; COMPUTE\_EDGES=false — Compute edges; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:translate — Translate (convert format)

**group:** Raster conversion

| Parameter  | Type   | Required | Default | Description                                     |
|------------|--------|----------|---------|-------------------------------------------------|
| INPUT      | raster | yes      | —       | Input layer                                     |
| TARGET_CRS | crs    | no       | —       | Override the projection for the output file     |
| NODATA     | number | no       | —       | Assign a specified NoData value to output bands |

| Parameter                | Type              | Required | Default | Description                                                                                                                                                                        |
|--------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COPY_SUBDATASETS         | boolean           | no       | False   | Copy all subdatasets of this file to individual output files                                                                                                                       |
| OPTIONS                  | string            | no       | —       | Additional creation options                                                                                                                                                        |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                                                                                                                                                        |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                                                                                                                                                 |
| DATA_TYPE *(adv)*        | enum              | no       | 0       | Output data type — options: 0=Use Input Layer Data Type, 1=Byte, 2=Int16, 3=UInt16, 4=UInt32, 5=Int32, 6=Float32, 7=Float64, 8=CInt16, 9=CInt32, 10=CFloat32, 11=CFloat64, 12=Int8 |
| OUTPUT                   | rasterDestination | yes      | —       | Converted                                                                                                                                                                          |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:translate INPUT=dem.tif TARGET_CRS=EPSG:6346 NODATA=10 COPY_SUBDATASETS=false
↳ OPTIONS=value OUTPUT=output.tif
```

This calls **Translate (convert format)**: INPUT=dem.tif — Input layer; TARGET\_CRS=EPSG:6346 — Override the projection for the output file; NODATA=10 — Assign a specified NoData value to output bands; COPY\_SUBDATASETS=false — Copy all subdatasets of this file to individual output files; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:triterrainruggednessindex — Terrain Ruggedness Index (TRI)

**group:** Raster analysis

| Parameter                | Type              | Required | Default | Description                 |
|--------------------------|-------------------|----------|---------|-----------------------------|
| INPUT                    | raster            | yes      | —       | Input layer                 |
| BAND                     | band              | no       | 1       | Band number                 |
| COMPUTE_EDGES            | boolean           | no       | False   | Compute edges               |
| OPTIONS                  | string            | no       | —       | Additional creation options |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options |
| OUTPUT                   | rasterDestination | yes      | —       | Terrain Ruggedness Index    |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:triterrainruggednessindex INPUT=dem.tif BAND=1 COMPUTE_EDGES=false OPTIONS=value
↳ OUTPUT=output.tif
```

This calls **Terrain Ruggedness Index (TRI)**: INPUT=dem.tif — Input layer; BAND=1 — Band number; COMPUTE\_EDGES=false — Compute edges; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:viewshed — Viewshed

**group:** Raster miscellaneous

| Parameter | Type   | Required | Default | Description       |
|-----------|--------|----------|---------|-------------------|
| INPUT     | raster | yes      | —       | Input layer       |
| BAND      | band   | no       | 1       | Band number       |
| OBSERVER  | point  | yes      | —       | Observer location |



| Parameter                | Type              | Required | Default | Description                                          |
|--------------------------|-------------------|----------|---------|------------------------------------------------------|
| OBSERVER_HEIGHT          | number            | no       | 1.0     | Observer height, DEM units                           |
| TARGET_HEIGHT            | number            | no       | 1.0     | Target height, DEM units                             |
| MAX_DISTANCE             | distance          | no       | 100.0   | Maximum distance from observer to compute visibility |
| OPTIONS                  | string            | no       | —       | Additional creation options                          |
| CREATION_OPTIONS *(adv)* | string            | no       | —       | Additional creation options                          |
| EXTRA *(adv)*            | string            | no       | —       | Additional command-line parameters                   |
| OUTPUT                   | rasterDestination | yes      | —       | Output                                               |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:viewshed INPUT=dem.tif OBSERVER=100,100 BAND=1 OBSERVER_HEIGHT=1 TARGET_HEIGHT=1
↳ MAX_DISTANCE=100 OPTIONS=value OUTPUT=output.tif
```

This calls **Viewshed**: INPUT=dem.tif — Input layer; OBSERVER=100,100 — Observer location; BAND=1 — Band number; OBSERVER\_HEIGHT=1 — Observer height, DEM units; TARGET\_HEIGHT=1 — Target height, DEM units; MAX\_DISTANCE=100 — Maximum distance from observer to compute visibility; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## gdal:warpproject — Warp (reproject) ★

**group:** Raster projections · **niva verb:** warp

| Parameter                 | Type              | Required | Default | Description                                                                                                                                                                                                                                                      |
|---------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                     | raster            | yes      | —       | Input layer                                                                                                                                                                                                                                                      |
| SOURCE_CRS                | crs               | no       | —       | Source CRS                                                                                                                                                                                                                                                       |
| TARGET_CRS                | crs               | no       | —       | Target CRS                                                                                                                                                                                                                                                       |
| RESAMPLING                | enum              | no       | 0       | Resampling method to use — options: 0=Nearest Neighbour, 1=Bilinear (2x2 Kernel), 2=Cubic (4x4 Kernel), 3=Cubic B-Spline (4x4 Kernel), 4=Lanczos (6x6 Kernel), 5=Average, 6=Mode, 7=Maximum, 8=Minimum, 9=Median, 10=First Quartile (Q1), 11=Third Quartile (Q3) |
| NODATA                    | number            | no       | —       | Nodata value for output bands                                                                                                                                                                                                                                    |
| TARGET_RESOLUTION         | number            | no       | —       | Output file resolution in target georeferenced units                                                                                                                                                                                                             |
| OPTIONS                   | string            | no       | —       | Additional creation options                                                                                                                                                                                                                                      |
| CREATION_OPTIONS *(adv)*  | string            | no       | —       | Additional creation options                                                                                                                                                                                                                                      |
| DATA_TYPE *(adv)*         | enum              | no       | 0       | Output data type — options: 0=Use Input Layer Data Type, 1=Byte, 2=Int16, 3=UInt16, 4=UInt32, 5=Int32, 6=Float32, 7=Float64, 8=CInt16, 9=CInt32, 10=CFloat32, 11=CFloat64, 12=Int8                                                                               |
| TARGET_EXTENT *(adv)*     | extent            | no       | —       | Georeferenced extents of output file to be created                                                                                                                                                                                                               |
| TARGET_EXTENT_CRS *(adv)* | crs               | no       | —       | CRS of the target raster extent                                                                                                                                                                                                                                  |
| MULTITHREADING *(adv)*    | boolean           | no       | False   | Use multithreaded warping implementation                                                                                                                                                                                                                         |
| EXTRA *(adv)*             | string            | no       | —       | Additional command-line parameters                                                                                                                                                                                                                               |
| OUTPUT                    | rasterDestination | yes      | —       | Reprojected                                                                                                                                                                                                                                                      |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run gdal:warpproject INPUT=dem.tif SOURCE_CRS=EPSG:6346 TARGET_CRS=EPSG:6346
↳ RESAMPLING=0 NODATA=10 TARGET_RESOLUTION=10 OPTIONS=value OUTPUT=output.tif
```

This calls **Warp (reproject)**: INPUT=dem.tif — Input layer; SOURCE\_CRS=EPSG:6346 — Source CRS; TARGET\_CRS=EPSG:6346 — Target CRS; RESAMPLING=0 selects *Nearest Neighbour*; NODATA=10 — Nodata value for output bands; TARGET\_RESOLUTION=10 — Output file resolution in target geo-referenced units; OPTIONS=value — Additional creation options; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

# qgis: algorithms

qgis — QGIS (Python) algorithms — 39 algorithms (QGIS 4.0.3). Auto-generated by scripts/gen\_algorithms.py; see [the index](#).

Call any of these with run <id> KEY=value ... (the **Name** column is the KEY); each entry includes a worked **Example usage**. A ★ marks an algorithm with a friendly niva alias verb.

## qgis:advancedpythonfieldcalculator — Advanced Python field calculator

**group:** Vector table

This algorithm adds a new attribute to a vector layer, with values calculated by applying an expression to each feature. The expression is defined as a Python function.

| Parameter       | Type   | Required | Default  | Description                                                                                                                                                                                              |
|-----------------|--------|----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT           | source | yes      | —        | Input layer                                                                                                                                                                                              |
| FIELD_NAME      | string | no       | NewField | Result field name                                                                                                                                                                                        |
| FIELD_TYPE      | enum   | yes      | —        | Field type — options: 0=Integer (32 bit), 1=Decimal (double), 2=Text (string), 3=Boolean, 4=Date, 5=Time, 6=Date & Time, 7=Binary Object (BLOB), 8=String List, 9=Integer List, 10=Decimal (double) List |
| FIELD_LENGTH    | number | no       | 10       | Field length                                                                                                                                                                                             |
| FIELD_PRECISION | number | no       | 3        | Field precision                                                                                                                                                                                          |
| GLOBAL          | string | no       | —        | Global expression                                                                                                                                                                                        |
| FORMULA         | string | no       | value =  | Formula                                                                                                                                                                                                  |
| OUTPUT          | sink   | yes      | —        | Calculated                                                                                                                                                                                               |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:advancedpythonfieldcalculator INPUT=input.gpkg FIELD_TYPE=0 FIELD_NAME=NewField  
↪ FIELD_LENGTH=10 FIELD_PRECISION=3 GLOBAL=value FORMULA="value =" OUTPUT=output.gpkg
```

This calls **Advanced Python field calculator**: INPUT=input.gpkg — Input layer; FIELD\_TYPE=0 selects *Integer (32 bit)*; FIELD\_NAME=NewField — Result field name; FIELD\_LENGTH=10 — Field length; FIELD\_PRECISION=3 — Field precision; GLOBAL=value — Global expression; FORMULA="value =" — Formula; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:barplot — Bar plot

**group:** Plots

This algorithm creates a bar plot from a category and a layer field.

| Parameter   | Type            | Required | Default | Description         |
|-------------|-----------------|----------|---------|---------------------|
| INPUT       | source          | yes      | —       | Input layer         |
| NAME_FIELD  | field           | yes      | —       | Category name field |
| VALUE_FIELD | field           | yes      | —       | Value field         |
| OUTPUT      | fileDestination | yes      | —       | Bar plot            |
| TITLE       | string          | no       | —       | Title               |
| XAXIS_TITLE | string          | no       | —       | X-axis title        |
| YAXIS_TITLE | string          | no       | —       | Y-axis title        |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run qgis:barplot INPUT=input.gpkg NAME_FIELD=field1 VALUE_FIELD=field1 TITLE=value
↳ XAXIS_TITLE=value YAXIS_TITLE=value OUTPUT=output.txt
```

This calls **Bar plot**: INPUT=input.gpkg — Input layer; NAME\_FIELD=field1 — Category name field; VALUE\_FIELD=field1 — Value field; TITLE=value — Title; XAXIS\_TITLE=value — X-axis title; YAXIS\_TITLE=value — Y-axis title; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:boxplot — Box plot

**group:** Plots

This algorithm creates a box plot from a category and a layer field.

| Parameter   | Type            | Required | Default | Description                                                                                                            |
|-------------|-----------------|----------|---------|------------------------------------------------------------------------------------------------------------------------|
| INPUT       | source          | yes      | —       | Input layer                                                                                                            |
| NAME_FIELD  | field           | yes      | —       | Category name field                                                                                                    |
| VALUE_FIELD | field           | yes      | —       | Value field                                                                                                            |
| MSD         | enum            | no       | 0       | Additional Statistic Lines — options: 0=Show Mean, 1=Show Standard Deviation, 2=Don't show Mean and Standard Deviation |
| TITLE       | string          | no       | —       | Title                                                                                                                  |
| XAXIS_TITLE | string          | no       | —       | X-axis title                                                                                                           |
| YAXIS_TITLE | string          | no       | —       | Y-axis title                                                                                                           |
| OUTPUT      | fileDestination | yes      | —       | Box plot                                                                                                               |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run qgis:boxplot INPUT=input.gpkg NAME_FIELD=field1 VALUE_FIELD=field1 MSD=0 TITLE=value
↳ XAXIS_TITLE=value YAXIS_TITLE=value OUTPUT=output.txt
```

This calls **Box plot**: INPUT=input.gpkg — Input layer; NAME\_FIELD=field1 — Category name field; VALUE\_FIELD=field1 — Value field; MSD=0 selects *Show Mean*; TITLE=value — Title; XAXIS\_TITLE=value — X-axis title; YAXIS\_TITLE=value — Y-axis title; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:distancematrix — Distance matrix

**group:** Vector analysis

This algorithm creates a table containing a distance matrix, with distances between all the points in a points layer.

| Parameter      | Type   | Required | Default | Description                                                                                                                                                 |
|----------------|--------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT          | source | yes      | —       | Input point layer                                                                                                                                           |
| INPUT_FIELD    | field  | yes      | —       | Input unique ID field                                                                                                                                       |
| TARGET         | source | yes      | —       | Target point layer                                                                                                                                          |
| TARGET_FIELD   | field  | yes      | —       | Target unique ID field                                                                                                                                      |
| MATRIX_TYPE    | enum   | no       | 0       | Output matrix type — options: 0=Linear (N*k x 3) distance matrix, 1=Standard (N x T) distance matrix, 2=Summary distance matrix (mean, std. dev., min, max) |
| NEAREST_POINTS | number | no       | 0       | Use only the nearest (k) target points                                                                                                                      |
| OUTPUT         | sink   | yes      | —       | Distance matrix                                                                                                                                             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:distancematrix INPUT=input.gpkg INPUT_FIELD=field1 TARGET=input.gpkg
↳ TARGET_FIELD=field1 MATRIX_TYPE=0 NEAREST_POINTS=0 OUTPUT=output.gpkg
```

This calls **Distance matrix**: INPUT=input.gpkg — Input point layer; INPUT\_FIELD=field1 — Input unique ID field; TARGET=input.gpkg — Target point layer; TARGET\_FIELD=field1 — Target unique ID field; MATRIX\_TYPE=0 selects *Linear* (Nk x 3) distance matrix\*; NEAREST\_POINTS=0 — Use only the nearest (k) target points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:distancetonearesthublinetohub — Distance to nearest hub (line to hub)

**group:** Vector analysis

Given an origin and a destination layers, this algorithm computes the distance between origin features and their closest destination one. Distance calculations are based on the feature's center. The resulting layer contains lines linking each origin point with its nearest destination feature.

| Parameter | Type   | Required | Default | Description                                                                        |
|-----------|--------|----------|---------|------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Source points layer                                                                |
| HUBS      | source | yes      | —       | Destination hubs layer                                                             |
| FIELD     | field  | yes      | —       | Hub layer name attribute                                                           |
| UNIT      | enum   | yes      | —       | Measurement unit — options: 0=Meters, 1=Feet, 2=Miles, 3=Kilometers, 4=Layer units |
| OUTPUT    | sink   | yes      | —       | Hub distance                                                                       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:distancetonearesthublinetohub INPUT=input.gpkg HUBS=input.gpkg FIELD=field1
↳ UNIT=0 OUTPUT=output.gpkg
```

This calls **Distance to nearest hub (line to hub)**: INPUT=input.gpkg — Source points layer; HUBS=input.gpkg — Destination hubs layer; FIELD=field1 — Hub layer name attribute; UNIT=0 selects *Meters*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:distancetonearesthubpoints — Distance to nearest hub (points)

**group:** Vector analysis

Given an origin and a destination layers, this algorithm computes the distance between origin features and their closest destination one. Distance calculations are based on the feature's center.

The resulting layer contains origin features center point with an additional field indicating the identifier of the nearest destination feature and the distance to it.

| Parameter | Type   | Required | Default | Description                                                                        |
|-----------|--------|----------|---------|------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Source points layer                                                                |
| HUBS      | source | yes      | —       | Destination hubs layer                                                             |
| FIELD     | field  | yes      | —       | Hub layer name attribute                                                           |
| UNIT      | enum   | yes      | —       | Measurement unit — options: 0=Meters, 1=Feet, 2=Miles, 3=Kilometers, 4=Layer units |
| OUTPUT    | sink   | yes      | —       | Hub distance                                                                       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:distancetonearesthubpoints INPUT=input.gpkg HUBS=input.gpkg FIELD=field1 UNIT=0
↳ OUTPUT=output.gpkg
```

This calls **Distance to nearest hub (points)**: INPUT=input.gpkg — Source points layer; HUBS=input.gpkg — Destination hubs layer; FIELD=field1 — Hub layer name attribute; UNIT=0 selects *Meters*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:eliminateselectedpolygons — Eliminate selected polygons

**group:** Vector geometry

This algorithm combines selected polygons from the input layer with certain adjacent polygons by erasing their common boundary. The adjacent polygon can be either the one with the largest or smallest area or the one sharing the largest common boundary with the polygon to be eliminated. Eliminate is normally used to get rid of sliver polygons, i.e. tiny polygons that are a result of polygon intersection processes where boundaries of the inputs are similar but not identical.

| Parameter | Type   | Required | Default | Description                                                                                                                  |
|-----------|--------|----------|---------|------------------------------------------------------------------------------------------------------------------------------|
| INPUT     | vector | yes      | —       | Input layer                                                                                                                  |
| MODE      | enum   | yes      | —       | Merge selection with the neighbouring polygon with the — options: 0=Largest Area, 1=Smallest Area, 2=Largest Common Boundary |
| OUTPUT    | sink   | yes      | —       | Eliminated                                                                                                                   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:eliminateselectedpolygons INPUT=input.gpkg MODE=0 OUTPUT=output.gpkg
```

This calls **Eliminate selected polygons**: INPUT=input.gpkg — Input layer; MODE=0 selects *Largest Area*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:executesql — Execute SQL

**group:** Vector general

This algorithm runs a query with SQL syntax. Input data sources are identified with input1, input2, ..., inputN and a simple query will look like: 'SELECT \* FROM input1'. The result of the query will be added as a new layer.

| Parameter            | Type        | Required | Default | Description                                                                                                                             |
|----------------------|-------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
| INPUT_DATASOURCES    | multilayer  | no       | —       | Input data sources (called input1, .., inputN in the query)                                                                             |
| INPUT_QUERY          | execute_sql | yes      | —       | SQL query                                                                                                                               |
| INPUT_UID_FIELD      | string      | no       | —       | Unique identifier field                                                                                                                 |
| INPUT_GEOMETRY_FIELD | string      | no       | —       | Geometry field                                                                                                                          |
| INPUT_GEOMETRY_TYPE  | enum        | no       | 0       | Geometry type — options: 0=Autodetect, 1=No geometry, 2=Point, 3=LineString, 4=Polygon, 5=MultiPoint, 6=MultiLineString, 7=MultiPolygon |
| INPUT_GEOMETRY_CRS   | crs         | no       | —       | CRS                                                                                                                                     |
| OUTPUT               | sink        | yes      | —       | SQL Output                                                                                                                              |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:executesql INPUT_QUERY=value INPUT_DATASOURCES="a.gpkg;b.gpkg"
↪ INPUT_UID_FIELD=value INPUT_GEOMETRY_FIELD=value INPUT_GEOMETRY_TYPE=0
↪ INPUT_GEOMETRY_CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Execute SQL**: INPUT\_QUERY=value — SQL query; INPUT\_DATASOURCES="a.gpkg;b.gpkg" — Input data sources (called input1, .., inputN in the query); INPUT\_UID\_FIELD=value — Unique identifier field; INPUT\_GEOMETRY\_FIELD=value — Geometry field; INPUT\_GEOMETRY\_TYPE=0 selects *Autodetect*; INPUT\_GEOMETRY\_CRS=EPSG:6346 — CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:generatepointspixelcentroidsalongline — Generate points (pixel centroids) along line

**group:** Vector creation

This algorithm generates a point vector layer from an input raster and line layer. The points correspond to the pixel centroids that intersect the line layer.

| Parameter    | Type   | Required | Default | Description        |
|--------------|--------|----------|---------|--------------------|
| INPUT_RASTER | raster | yes      | —       | Raster layer       |
| INPUT_VECTOR | source | yes      | —       | Vector layer       |
| OUTPUT       | sink   | yes      | —       | Points along lines |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:generatepointspixelcentroidsalongline INPUT_RASTER=dem.tif
↪ INPUT_VECTOR=input.gpkg OUTPUT=output.gpkg
```

This calls **Generate points (pixel centroids) along line**: INPUT\_RASTER=dem.tif — Raster layer; INPUT\_VECTOR=input.gpkg — Vector layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:heatmapkerneldensityestimation — Heatmap (Kernel Density Estimation)

**group:** Interpolation

This algorithm creates a density (heatmap) raster of an input point vector layer using kernel density estimation. Heatmaps allow easy identification of hotspots and clustering of points. The density

is calculated based on the number of points in a location, with larger numbers of clustered points resulting in larger values.

| Parameter               | Type              | Required | Default | Description                                                                             |
|-------------------------|-------------------|----------|---------|-----------------------------------------------------------------------------------------|
| INPUT                   | source            | yes      | —       | Point layer                                                                             |
| RADIUS                  | distance          | no       | 100.0   | Radius                                                                                  |
| RADIUS_FIELD<br>*(adv)* | field             | no       | —       | Radius from field                                                                       |
| PIXEL_SIZE              | number            | no       | 0.1     | Output raster size                                                                      |
| WEIGHT_FIELD<br>*(adv)* | field             | no       | —       | Weight from field                                                                       |
| KERNEL *(adv)*          | enum              | no       | 0       | Kernel shape — options: 0=Quartic, 1=Triangular, 2=Uniform, 3=Triweight, 4=Epanechnikov |
| DECAY *(adv)*           | number            | no       | 0.0     | Decay ratio (Triangular kernels only)                                                   |
| OUTPUT_VALUE<br>*(adv)* | enum              | no       | 0       | Output value scaling — options: 0=Raw, 1=Scaled                                         |
| OUTPUT                  | rasterDestination | yes      | —       | Heatmap                                                                                 |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run qgis:heatmapkerneldensityestimation INPUT=input.gpkg RADIUS=100 PIXEL_SIZE=0.1
↳ OUTPUT=output.tif
```

This calls **Heatmap (Kernel Density Estimation)**: INPUT=input.gpkg — Point layer; RADIUS=100 — Radius; PIXEL\_SIZE=0.1 — Output raster size; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:hypsometriccurves — Hypsometric curves

**group:** Raster terrain analysis

This algorithm computes hypsometric curves for an input Digital Elevation Model (DEM). Curves are produced as table files in an output folder specified by the user.

| Parameter        | Type              | Required | Default | Description                             |
|------------------|-------------------|----------|---------|-----------------------------------------|
| INPUT_DEM        | raster            | yes      | —       | DEM to analyze                          |
| BOUNDARY_LAYER   | source            | yes      | —       | Boundary layer                          |
| STEP             | number            | no       | 100.0   | Step                                    |
| USE_PERCENTAGE   | boolean           | no       | False   | Use % of area instead of absolute value |
| OUTPUT_DIRECTORY | folderDestination | yes      | —       | Hypsometric curves                      |

**Outputs:** OUTPUT\_DIRECTORY (outputFolder)

### Example usage

```
run qgis:hypsometriccurves INPUT_DEM=dem.tif BOUNDARY_LAYER=input.gpkg STEP=100
↳ USE_PERCENTAGE=false OUTPUT_DIRECTORY=output_directory/
```

This calls **Hypsometric curves**: INPUT\_DEM=dem.tif — DEM to analyze; BOUNDARY\_LAYER=input.gpkg — Boundary layer; STEP=100 — Step; USE\_PERCENTAGE=false — Use % of area instead of absolute value; OUTPUT\_DIRECTORY=output\_directory/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## qgis:idwinterpolation — IDW interpolation

**group:** Interpolation

This algorithm generates an Inverse Distance Weighted (IDW) interpolation of a point vector layer. Sample points are weighted during interpolation such that the influence of one point relative to another declines with distance from the unknown point you want to create.

| Parameter            | Type                   | Required | Default | Description            |
|----------------------|------------------------|----------|---------|------------------------|
| INTERPOLATION_DATA   | idw_interpolation_data | yes      | —       | Input layer(s)         |
| DISTANCE_COEFFICIENT | number                 | no       | 2.0     | Distance coefficient P |
| EXTENT               | extent                 | yes      | —       | Extent                 |
| PIXEL_SIZE           | number                 | no       | 0.1     | Output raster size     |
| COLUMNS              | number                 | no       | —       | Number of columns      |
| ROWS                 | number                 | no       | —       | Number of rows         |
| OUTPUT               | rasterDestination      | yes      | —       | Interpolated           |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run qgis:idwinterpolation INTERPOLATION_DATA=... EXTENT=0,1000,0,1000 DISTANCE_COEFFICIENT=2  
↪ PIXEL_SIZE=0.1 COLUMNS=10 ROWS=10 OUTPUT=output.tif
```

This calls **IDW interpolation**: INTERPOLATION\_DATA=... — Input layer(s); EXTENT=0,1000,0,1000 — Extent; DISTANCE\_COEFFICIENT=2 — Distance coefficient P; PIXEL\_SIZE=0.1 — Output raster size; COLUMNS=10 — Number of columns; ROWS=10 — Number of rows; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:knearestconcavehull — Concave hull (k-nearest neighbor)

**group:** Vector geometry

This algorithm generates a concave hull polygon from a set of points. If the input layer is a line or polygon layer, it will use the nodes. The number of neighbours to consider determines the concaveness of the output polygon. A lower number will result in a concave hull that follows the points very closely, while a higher number will have a smoother shape. The minimum number of neighbour points to consider is 3. A value equal to or greater than the number of points will result in a convex hull. If a field is selected, the algorithm will group the features in the input layer using unique values in that field and generate individual polygons in the output layer for each group.

| Parameter  | Type   | Required | Default | Description                                                                                            |
|------------|--------|----------|---------|--------------------------------------------------------------------------------------------------------|
| INPUT      | source | yes      | —       | Input layer                                                                                            |
| KNEIGHBORS | number | no       | 3       | Number of neighboring points to consider (a lower number is more concave, a higher number is smoother) |
| FIELD      | field  | no       | —       | Field (set if creating concave hulls by class)                                                         |
| OUTPUT     | sink   | yes      | —       | Concave hull                                                                                           |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:knearestconcavehull INPUT=input.gpkg KNEIGHBORS=3 FIELD=field1 OUTPUT=output.gpkg
```

This calls **Concave hull (k-nearest neighbor)**: INPUT=input.gpkg — Input layer; KNEIGHBORS=3 — Number of neighboring points to consider (a lower number is more concave, a...; FIELD=field1 — Field (set if creating concave hulls by class); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:linestopolygons — Lines to polygons

**group:** Vector geometry

This algorithm generates a polygon layer using as polygon rings the lines from an input line layer. All attributes from the input line layer will be copied to the output layer.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Polygons    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:linestopolygons INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Lines to polygons**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:meanandstandarddeviationplot — Mean and standard deviation plot

**group:** Plots

This algorithm creates a box plot with mean and standard deviation values.

| Parameter   | Type            | Required | Default | Description         |
|-------------|-----------------|----------|---------|---------------------|
| INPUT       | source          | yes      | —       | Input table         |
| NAME_FIELD  | field           | yes      | —       | Category name field |
| VALUE_FIELD | field           | yes      | —       | Value field         |
| OUTPUT      | fileDestination | yes      | —       | Plot                |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run qgis:meanandstandarddeviationplot INPUT=input.gpkg NAME_FIELD=field1
↪ VALUE_FIELD=field1 OUTPUT=output.txt
```

This calls **Mean and standard deviation plot**: INPUT=input.gpkg — Input table; NAME\_FIELD=field1 — Category name field; VALUE\_FIELD=field1 — Value field; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:pointsdisplacement — Points displacement

**group:** Vector geometry

This algorithm offsets point features by moving nearby points by a preset amount to minimize overlapping features.

| Parameter  | Type     | Required | Default | Description                                |
|------------|----------|----------|---------|--------------------------------------------|
| INPUT      | source   | yes      | —       | Input layer                                |
| PROXIMITY  | distance | no       | 1.0     | Minimum distance to other points           |
| DISTANCE   | distance | no       | 1.0     | Displacement distance                      |
| HORIZONTAL | boolean  | yes      | —       | Horizontal distribution for two point case |
| OUTPUT     | sink     | yes      | —       | Displaced                                  |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run qgis:pointsdisplacement INPUT=input.gpkg HORIZONTAL=false PROXIMITY=1 DISTANCE=1
↳ OUTPUT=output.gpkg
```

This calls **Points displacement**: INPUT=input.gpkg — Input layer; HORIZONTAL=false — Horizontal distribution for two point case; PROXIMITY=1 — Minimum distance to other points; DISTANCE=1 — Displacement distance; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:polarplot — Polar plot

**group:** Plots

This algorithm generates a polar plot based on the value of an input vector layer.

| Parameter   | Type            | Required | Default | Description |
|-------------|-----------------|----------|---------|-------------|
| INPUT       | source          | yes      | —       | Input layer |
| VALUE_FIELD | field           | yes      | —       | Value field |
| OUTPUT      | fileDestination | yes      | —       | Polar plot  |

**Outputs:** OUTPUT (outputHtml)

**Example usage**

```
run qgis:polarplot INPUT=input.gpkg VALUE_FIELD=field1 OUTPUT=output.txt
```

This calls **Polar plot**: INPUT=input.gpkg — Input layer; VALUE\_FIELD=field1 — Value field; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:randomextractwithinsubsets — Random extract within subsets

**group:** Vector selection

This algorithm takes a vector layer and generates a new one that contains only a subset of the features in the input layer. The subset is defined randomly, using a percentage or count value to define the total number of features in the subset. The percentage/count value is not applied to the whole layer, but instead to each category. Categories are defined according to a given attribute, which is also specified as an input parameter for the algorithm.

| Parameter | Type   | Required | Default | Description                                                                        |
|-----------|--------|----------|---------|------------------------------------------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                                                        |
| FIELD     | field  | yes      | —       | ID field                                                                           |
| METHOD    | enum   | no       | 0       | Method — options: 0=Number of selected features, 1=Percentage of selected features |
| NUMBER    | number | no       | 10      | Number/percentage of selected features                                             |
| OUTPUT    | sink   | yes      | —       | Extracted (random stratified)                                                      |

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run qgis:randomextractwithinsubsets INPUT=input.gpkg FIELD=field1 METHOD=0 NUMBER=10
↳ OUTPUT=output.gpkg
```

This calls **Random extract within subsets**: INPUT=input.gpkg — Input layer; FIELD=field1 — ID field; METHOD=0 selects *Number of selected features*; NUMBER=10 — Number/percentage of selected features; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:randompointsalongline — Random points along line

**group:** Vector creation

This algorithm creates a point layer with a given number of points placed on the lines of the input layer. The location of each point is determined by randomly selecting a feature, then a segment of the line geometry of that feature, and finally a random position on that segment. A minimum distance between the points can be specified (Euclidean distance).

| Parameter     | Type     | Required | Default | Description                     |
|---------------|----------|----------|---------|---------------------------------|
| INPUT         | source   | yes      | —       | Input layer                     |
| POINTS_NUMBER | number   | no       | 1       | Number of points                |
| MIN_DISTANCE  | distance | no       | 0       | Minimum distance between points |
| OUTPUT        | sink     | yes      | —       | Random points                   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:randompointsalongline INPUT=input.gpkg POINTS_NUMBER=1 MIN_DISTANCE=0
↳ OUTPUT=output.gpkg
```

This calls **Random points along line**: INPUT=input.gpkg — Input layer; POINTS\_NUMBER=1 — Number of points; MIN\_DISTANCE=0 — Minimum distance between points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:randompointsinlayerbounds — Random points in layer bounds

**group:** Vector creation

This algorithm creates a new point layer with a given number of random points, all of them within the extent of a given layer. A distance factor can be specified, to avoid points being too close to each other.

| Parameter     | Type     | Required | Default | Description                     |
|---------------|----------|----------|---------|---------------------------------|
| INPUT         | source   | yes      | —       | Input layer                     |
| POINTS_NUMBER | number   | no       | 1       | Number of points                |
| MIN_DISTANCE  | distance | no       | 0       | Minimum distance between points |
| OUTPUT        | sink     | yes      | —       | Random points                   |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:randompointsinlayerbounds INPUT=input.gpkg POINTS_NUMBER=1 MIN_DISTANCE=0
↳ OUTPUT=output.gpkg
```

This calls **Random points in layer bounds**: INPUT=input.gpkg — Input layer; POINTS\_NUMBER=1 — Number of points; MIN\_DISTANCE=0 — Minimum distance between points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:randompointsinsidepolygons — Random points inside polygons

**group:** Vector creation

This algorithm creates a new point layer with random points inside the polygons of a given layer. The number of points in each polygon can be defined as a fixed count or as a density value. The count/density value could also be taken from an attribute or an expression specified using the 'Data defined override' functionality, so it can be different for each polygon in the input layer. A minimum distance can be specified, to avoid points being too close to each other.

| Parameter    | Type       | Required | Default | Description                                                   |
|--------------|------------|----------|---------|---------------------------------------------------------------|
| INPUT        | source     | yes      | —       | Input layer                                                   |
| STRATEGY     | enum       | no       | 0       | Sampling strategy — options: 0=Points count, 1=Points density |
| VALUE        | number     | no       | 1       | Point count or density                                        |
| EXPRESSION   | expression | no       | —       | Expression                                                    |
| MIN_DISTANCE | distance   | no       | —       | Minimum distance between points                               |
| OUTPUT       | sink       | yes      | —       | Random points                                                 |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:randompointsinsidepolygons INPUT=input.gpkg STRATEGY=0 VALUE=1 EXPRESSION="value > 0" MIN_DISTANCE=10 OUTPUT=output.gpkg
```

This calls **Random points inside polygons**: INPUT=input.gpkg — Input layer; STRATEGY=0 selects *Points count*; VALUE=1 — Point count or density; EXPRESSION="value > 0" — Expression; MIN\_DISTANCE=10 — Minimum distance between points; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:randomselectionwithinsubsets — Random selection within subsets

**group:** Vector selection

This algorithm takes a vector layer and selects a subset of its features. No new layer is generated by this algorithm. The subset is defined randomly, using a percentage or count value to define the total number of features in the subset. The percentage/count value is not applied to the whole layer, but instead to each category. Categories are defined according to a given attribute, which is also specified as an input parameter for the algorithm.

| Parameter | Type   | Required | Default | Description                                                                        |
|-----------|--------|----------|---------|------------------------------------------------------------------------------------|
| INPUT     | vector | yes      | —       | Input layer                                                                        |
| FIELD     | field  | yes      | —       | ID field                                                                           |
| METHOD    | enum   | no       | 0       | Method — options: 0=Number of selected features, 1=Percentage of selected features |
| NUMBER    | number | no       | 10      | Number/percentage of selected features                                             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:randomselectionwithinsubsets INPUT=input.gpkg FIELD=field1 METHOD=0 NUMBER=10
```

This calls **Random selection within subsets**: INPUT=input.gpkg — Input layer; FIELD=field1 — ID field; METHOD=0 selects *Number of selected features*; NUMBER=10 — Number/percentage of selected features. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:rastercalculator — Raster calculator

**group:** Raster analysis

This algorithm allows performing algebraic operations using raster layers. The resulting layer will have its values computed according to an expression. The expression can contain numerical values, operators and references to any of the layers in the current project. The following functions are also supported:

- `sin()`, `cos()`, `tan()`, `atan2()`, `ln()`, `log10()` The extent, cell size, and output CRS can be defined by the user. If the extent is not specified, the minimum extent that covers selected reference layer(s) will be used. If the cell size is not specified, the minimum cell size of selected reference layer(s) will be used. If the output CRS is not specified, the CRS of the first reference layer will be used. The cell size is assumed to be the same in both X and Y axes. Layers are referred by their name as displayed in the layer list and the number of the band to use (based on 1), using the pattern 'layer\_name@band number'. For instance, the first band from a layer named DEM will be referred as DEM@1. When using the calculator in the batch interface or from the console, the files to use have to be specified. The corresponding layers are referred using the base name of the file (without the full path). For instance, if using a layer at path/to/my/rasterfile.tif, the first band of that layer will be referred as rasterfile.tif@1.

| Parameter  | Type                   | Required | Default | Description                                                       |
|------------|------------------------|----------|---------|-------------------------------------------------------------------|
| EXPRESSION | raster_calc_expression | yes      | —       | Expression                                                        |
| LAYERS     | multilayer             | no       | —       | Reference layer(s) (used for automated extent, cellsize, and CRS) |
| CELLSIZE   | number                 | no       | 0.0     | Cell size (use 0 or empty to set it automatically)                |
| EXTENT     | extent                 | no       | —       | Output extent                                                     |
| CRS        | crs                    | no       | —       | Output CRS                                                        |
| OUTPUT     | rasterDestination      | yes      | —       | Output                                                            |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run qgis:rastercalculator EXPRESSION=... LAYERS="a.gpkg;b.gpkg" CELLSIZE=0  
↳ EXTENT=0,1000,0,1000 CRS=EPSG:6346 OUTPUT=output.tif
```

This calls **Raster calculator**: EXPRESSION=... — Expression; LAYERS="a.gpkg;b.gpkg" — Reference layer(s) (used for automated extent, cellsize, and CRS); CELLSIZE=0 — Cell size (use 0 or empty to set it automatically); EXTENT=0,1000,0,1000 — Output extent; CRS=EPSG:6346 — Output CRS; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:rasterlayerhistogram — Raster layer histogram

**group:** Plots

This algorithm generates a histogram with the values of a raster layer. The raster layer must have a single band.

| Parameter | Type            | Required | Default | Description    |
|-----------|-----------------|----------|---------|----------------|
| INPUT     | raster          | yes      | —       | Input layer    |
| BAND      | band            | no       | 1       | Band number    |
| BINS      | number          | no       | 10      | number of bins |
| OUTPUT    | fileDestination | yes      | —       | Histogram      |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run qgis:rasterlayerhistogram INPUT=dem.tif BAND=1 BINS=10 OUTPUT=output.txt
```

This calls **Raster layer histogram**: INPUT=dem.tif — Input layer; BAND=1 — Band number; BINS=10 — number of bins; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:rectanglesovalsdiamondsvariable — Rectangles, ovals, diamonds (variable)

**group:** Vector geometry

This algorithm creates rectangle, oval or diamond-shaped polygons from the input point layer using specified width, height and (optional) rotation values. Multipart inputs should be promoted to singleparts first.

| Parameter | Type   | Required | Default | Description                                               |
|-----------|--------|----------|---------|-----------------------------------------------------------|
| INPUT     | source | yes      | —       | Input layer                                               |
| SHAPE     | enum   | yes      | —       | Buffer shape — options: 0=Rectangles, 1=Diamonds, 2=Ovals |
| WIDTH     | field  | yes      | —       | Width field                                               |
| HEIGHT    | field  | yes      | —       | Height field                                              |
| ROTATION  | field  | no       | —       | Rotation field                                            |
| SEGMENTS  | number | no       | 36      | Number of segments                                        |
| OUTPUT    | sink   | yes      | —       | Output                                                    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:rectanglesovalsdiamondsvariable INPUT=input.gpkg SHAPE=0 WIDTH=field1  
↪ HEIGHT=field1 ROTATION=field1 SEGMENTS=36 OUTPUT=output.gpkg
```

This calls **Rectangles, ovals, diamonds (variable)**: INPUT=input.gpkg — Input layer; SHAPE=0 selects *Rectangles*; WIDTH=field1 — Width field; HEIGHT=field1 — Height field; ROTATION=field1 — Rotation field; SEGMENTS=36 — Number of segments; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:regularpoints — Regular points

**group:** Vector creation

This algorithm creates a point layer with a given number of regular points, all of them within a given extent.

| Parameter  | Type     | Required | Default    | Description                          |
|------------|----------|----------|------------|--------------------------------------|
| EXTENT     | extent   | yes      | —          | Input extent                         |
| SPACING    | distance | no       | 100        | Point spacing/count                  |
| INSET      | distance | no       | 0.0        | Initial inset from corner (LH side)  |
| RANDOMIZE  | boolean  | no       | False      | Apply random offset to point spacing |
| IS_SPACING | boolean  | no       | True       | Use point spacing                    |
| CRS        | crs      | no       | ProjectCrs | Output layer CRS                     |
| OUTPUT     | sink     | yes      | —          | Regular points                       |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:regularpoints EXTENT=0,1000,0,1000 SPACING=100 INSET=0 RANDOMIZE=false  
↪ IS_SPACING=true CRS=EPSG:6346 OUTPUT=output.gpkg
```

This calls **Regular points**: EXTENT=0,1000,0,1000 — Input extent; SPACING=100 — Point spacing/count; INSET=0 — Initial inset from corner (LH side); RANDOMIZE=false — Apply random offset to point spacing; IS\_SPACING=true — Use point spacing; CRS=EPSG:6346 — Output layer CRS; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:relief — Relief

**group:** Raster terrain analysis

This algorithm creates a shaded relief layer from digital elevation data.

| Parameter              | Type              | Required | Default | Description                           |
|------------------------|-------------------|----------|---------|---------------------------------------|
| INPUT                  | raster            | yes      | —       | Elevation layer                       |
| Z_FACTOR               | number            | no       | 1.0     | Z factor                              |
| AUTO_COLORS            | boolean           | no       | False   | Generate relief classes automatically |
| COLORS                 | relief_colors     | no       | —       | Relief colors                         |
| OUTPUT                 | rasterDestination | yes      | —       | Relief                                |
| FREQUENCY_DISTRIBUTION | fileDestination   | no       | —       | Frequency distribution                |

**Outputs:** OUTPUT (outputRaster), FREQUENCY\_DISTRIBUTION (outputFile)

### Example usage

```
run qgis:relief INPUT=dem.tif Z_FACTOR=1 AUTO_COLORS=false OUTPUT=output.tif
```

This calls **Relief**: INPUT=dem.tif — Elevation layer; Z\_FACTOR=1 — Z factor; AUTO\_COLORS=false — Generate relief classes automatically; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:scatter3dplot — Vector layer scatterplot 3D

**group:** Plots

This algorithm creates a 3D scatter plot for a vector layer.

| Parameter   | Type            | Required | Default | Description    |
|-------------|-----------------|----------|---------|----------------|
| INPUT       | source          | yes      | —       | Input layer    |
| XFIELD      | field           | yes      | —       | X attribute    |
| YFIELD      | field           | yes      | —       | Y attribute    |
| ZFIELD      | field           | yes      | —       | Z attribute    |
| TITLE       | string          | no       | —       | Title          |
| XAXIS_TITLE | string          | no       | —       | X-axis title   |
| YAXIS_TITLE | string          | no       | —       | Y-axis title   |
| ZAXIS_TITLE | string          | no       | —       | Z-axis title   |
| OUTPUT      | fileDestination | yes      | —       | Scatterplot 3D |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run qgis:scatter3dplot INPUT=input.gpkg XFIELD=field1 YFIELD=field1 ZFIELD=field1
  TITLE=value XAXIS_TITLE=value YAXIS_TITLE=value OUTPUT=output.txt
```

This calls **Vector layer scatterplot 3D**: INPUT=input.gpkg — Input layer; XFIELD=field1 — X attribute; YFIELD=field1 — Y attribute; ZFIELD=field1 — Z attribute; TITLE=value — Title; XAXIS\_TITLE=value — X-axis title; YAXIS\_TITLE=value — Y-axis title; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## qgis:selectbyattribute — Select by attribute

**group:** Vector selection

This algorithm creates a selection in a vector layer. The criteria for selected features is defined based on the values of an attribute from the input layer.

| Parameter | Type   | Required | Default | Description                                                                                                                                                             |
|-----------|--------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT     | vector | yes      | —       | Input layer                                                                                                                                                             |
| FIELD     | field  | yes      | —       | Selection attribute                                                                                                                                                     |
| OPERATOR  | enum   | no       | 0       | Operator — options: 0=, 1=≠, 2=>, 3=≥, 4=<, 5=≤, 6=begins with, 7=contains, 8=is null, 9=is not null, 10=does not contain                                               |
| VALUE     | string | no       | —       | Value                                                                                                                                                                   |
| METHOD    | enum   | no       | 0       | Modify current selection by — options: 0=creating new selection, 1=adding to current selection, 2=removing from current selection, 3=selecting within current selection |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:selectbyattribute INPUT=input.gpkg FIELD=field1 OPERATOR=0 VALUE=value METHOD=0
```

This calls **Select by attribute**: INPUT=input.gpkg — Input layer; FIELD=field1 — Selection attribute; OPERATOR=0 selects =; VALUE=value — Value; METHOD=0 selects *creating new selection*. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:selectbyexpression — Select by expression

**group:** Vector selection

This algorithm creates a selection in a vector layer. The criteria for selecting features is based on a QGIS expression. For help with QGIS expression functions, see the inbuilt help for specific functions which is available in the expression builder.

| Parameter  | Type       | Required | Default | Description                                                                                                                                                             |
|------------|------------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT      | vector     | yes      | —       | Input layer                                                                                                                                                             |
| EXPRESSION | expression | yes      | —       | Expression                                                                                                                                                              |
| METHOD     | enum       | no       | 0       | Modify current selection by — options: 0=creating new selection, 1=adding to current selection, 2=removing from current selection, 3=selecting within current selection |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:selectbyexpression INPUT=input.gpkg EXPRESSION="value > 0" METHOD=0
```

This calls **Select by expression**: INPUT=input.gpkg — Input layer; EXPRESSION="value > 0" — Expression; METHOD=0 selects *creating new selection*. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:setstyleforrasterlayer — Set style for raster layer

**group:** Raster tools

This algorithm sets the style of a raster layer. The style must be defined in a QML file.

| Parameter | Type   | Required | Default | Description  |
|-----------|--------|----------|---------|--------------|
| INPUT     | raster | yes      | —       | Raster layer |

| Parameter | Type | Required | Default | Description |
|-----------|------|----------|---------|-------------|
| STYLE     | file | yes      | —       | Style file  |

**Outputs:** INPUT (outputRaster)

### Example usage

```
run qgis:setstyleforrasterlayer INPUT=dem.tif STYLE=input.dat
```

This calls **Set style for raster layer**: INPUT=dem.tif — Raster layer; STYLE=input.dat — Style file. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:setstyleforvectorlayer — Set style for vector layer

**group:** Vector general

This algorithm sets the style of a vector layer. The style must be defined in a QML file.

| Parameter | Type   | Required | Default | Description  |
|-----------|--------|----------|---------|--------------|
| INPUT     | vector | yes      | —       | Vector layer |
| STYLE     | file   | yes      | —       | Style file   |

**Outputs:** INPUT (outputVector)

### Example usage

```
run qgis:setstyleforvectorlayer INPUT=input.gpkg STYLE=input.dat
```

This calls **Set style for vector layer**: INPUT=input.gpkg — Vector layer; STYLE=input.dat — Style file. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:statisticsbycategories — Statistics by categories

**group:** Vector analysis

This algorithm calculates statistics of fields depending on a parent class.

| Parameter             | Type   | Required | Default | Description                                                           |
|-----------------------|--------|----------|---------|-----------------------------------------------------------------------|
| INPUT                 | source | yes      | —       | Input vector layer                                                    |
| VALUES_FIELD_NAME     | field  | no       | —       | Field to calculate statistics on (if empty, only count is calculated) |
| CATEGORIES_FIELD_NAME | field  | yes      | —       | Field(s) with categories                                              |
| OUTPUT                | sink   | yes      | —       | Statistics by category                                                |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:statisticsbycategories INPUT=input.gpkg CATEGORIES_FIELD_NAME=field1
VALUES_FIELD_NAME=field1 OUTPUT=output.gpkg
```

This calls **Statistics by categories**: INPUT=input.gpkg — Input vector layer; CATEGORIES\_FIELD\_NAME=field1 — Field(s) with categories; VALUES\_FIELD\_NAME=field1 — Field to calculate statistics on (if empty, only count is calculated); OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:texttfloat — Text to float

**group:** Vector table

This algorithm modifies the type of a given attribute in a vector layer, converting a text attribute containing numeric strings into a numeric attribute.

| Parameter | Type   | Required | Default | Description                        |
|-----------|--------|----------|---------|------------------------------------|
| INPUT     | source | yes      | —       | Input layer                        |
| FIELD     | field  | yes      | —       | Text attribute to convert to float |
| OUTPUT    | sink   | yes      | —       | Float from text                    |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:texttfloat INPUT=input.gpkg FIELD=field1 OUTPUT=output.gpkg
```

This calls **Text to float**: INPUT=input.gpkg — Input layer; FIELD=field1 — Text attribute to convert to float; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:tininterpolation — TIN interpolation

**group:** Interpolation

This algorithm generates a Triangulated Irregular Network (TIN) interpolation of a point vector layer. With the TIN method you can create a surface formed by triangles of nearest neighbor points. To do this, circumcircles around selected sample points are created and their intersections are connected to a network of non overlapping and as compact as possible triangles. The resulting surfaces are not smooth. The algorithm creates both the raster layer of the interpolated values and the vector line layer with the triangulation boundaries.

| Parameter             | Type                   | Required | Default | Description                                                        |
|-----------------------|------------------------|----------|---------|--------------------------------------------------------------------|
| INTERPOLATION_ - DATA | idw_interpolation_data | yes      | —       | Input layer(s)                                                     |
| METHOD                | enum                   | no       | 0       | Interpolation method — options: 0=Linear, 1=Clough-Toucher (cubic) |
| EXTENT                | extent                 | yes      | —       | Extent                                                             |
| PIXEL_SIZE            | number                 | no       | 0.1     | Output raster size                                                 |
| COLUMNS               | number                 | no       | —       | Number of columns                                                  |
| ROWS                  | number                 | no       | —       | Number of rows                                                     |
| OUTPUT                | rasterDestination      | yes      | —       | Interpolated                                                       |
| TRIANGULATION         | sink                   | no       | —       | Triangulation                                                      |

**Outputs:** OUTPUT (outputRaster), TRIANGULATION (outputVector)

### Example usage

```
run qgis:tininterpolation INTERPOLATION_DATA=... EXTENT=0,1000,0,1000 METHOD=0  
↳ PIXEL_SIZE=0.1 COLUMNS=10 ROWS=10 OUTPUT=output.tif
```

This calls **TIN interpolation**: INTERPOLATION\_DATA=... — Input layer(s); EXTENT=0,1000,0,1000 — Extent; METHOD=0 selects *Linear*; PIXEL\_SIZE=0.1 — Output raster size; COLUMNS=10 — Number of columns; ROWS=10 — Number of rows; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:topologicalcoloring — Topological coloring

**group:** Cartography

This algorithm assigns a color index to polygon features in such a way that no adjacent polygons share the same color index, whilst minimizing the number of colors required. An optional minimum distance between features assigned the same color can be set to prevent nearby (but non-touching) features from being assigned equal colors. The algorithm allows choice of method to use when assigning colors. The default method attempts to assign colors so that the count of features assigned to each individual color index is balanced. The 'by assigned area' mode instead assigns colors so that the total area of features assigned to each color is balanced. This mode can be useful to help avoid large features resulting in one of the colors appearing more dominant on a colored map. The 'by distance between colors' mode will assign colors in order to maximize the distance between features of the same color. This mode helps to create a more uniform distribution of colors across a map. A minimum number of colors can be specified if desired. The color index is saved to a new attribute named `color_id`.

| Parameter    | Type     | Required | Default | Description                                                                                              |
|--------------|----------|----------|---------|----------------------------------------------------------------------------------------------------------|
| INPUT        | source   | yes      | —       | Input layer                                                                                              |
| MIN_COLORS   | number   | no       | 4       | Minimum number of colors                                                                                 |
| MIN_DISTANCE | distance | no       | 0.0     | Minimum distance between features                                                                        |
| BALANCE      | enum     | no       | 0       | Balance color assignment — options: 0=By feature count, 1=By assigned area, 2=By distance between colors |
| OUTPUT       | sink     | yes      | —       | Colored                                                                                                  |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:topologicalcoloring INPUT=input.gpkg MIN_COLORS=4 MIN_DISTANCE=0 BALANCE=0  
↳ OUTPUT=output.gpkg
```

This calls **Topological coloring**: INPUT=input.gpkg — Input layer; MIN\_COLORS=4 — Minimum number of colors; MIN\_DISTANCE=0 — Minimum distance between features; BALANCE=0 selects *By feature count*; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:variabledistancebuffer — Variable distance buffer

**group:** Vector geometry

This algorithm computes a buffer area for all the features in an input layer. The size of the buffer for a given feature is defined by an attribute, so it allows different features to have different buffer sizes.

| Parameter     | Type    | Required | Default | Description                                        |
|---------------|---------|----------|---------|----------------------------------------------------|
| INPUT         | source  | yes      | —       | Input layer                                        |
| FIELD         | field   | yes      | —       | Distance field                                     |
| SEGMENTS      | number  | no       | 5       | Segments                                           |
| DISSOLVE      | boolean | no       | False   | Dissolve result                                    |
| END_CAP_STYLE | enum    | no       | 0       | End cap style — options: 0=Round, 1=Flat, 2=Square |
| JOIN_STYLE    | enum    | no       | 0       | Join style — options: 0=Round, 1=Miter, 2=Bevel    |
| MITER_LIMIT   | number  | no       | 2       | Miter limit                                        |
| OUTPUT        | sink    | yes      | —       | Buffer                                             |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run qgis:variabledistancebuffer INPUT=input.gpkg FIELD=field1 SEGMENTS=5 DISSOLVE=false
↳ END_CAP_STYLE=0 JOIN_STYLE=0 MITER_LIMIT=2 OUTPUT=output.gpkg
```

This calls **Variable distance buffer**: INPUT=input.gpkg — Input layer; FIELD=field1 — Distance field; SEGMENTS=5 — Segments; DISSOLVE=false — Dissolve result; END\_CAP\_STYLE=0 selects *Round*; JOIN\_STYLE=0 selects *Round*; MITER\_LIMIT=2 — Miter limit; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:vectorlayerhistogram — Vector layer histogram

**group:** Plots

This algorithm generates a histogram with the values of the attribute of a vector layer. The attribute to use for computing the histogram must be a numeric attribute.

| Parameter | Type            | Required | Default | Description    |
|-----------|-----------------|----------|---------|----------------|
| INPUT     | source          | yes      | —       | Input layer    |
| FIELD     | field           | yes      | —       | Attribute      |
| BINS      | number          | no       | 10      | number of bins |
| OUTPUT    | fileDestination | yes      | —       | Histogram      |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run qgis:vectorlayerhistogram INPUT=input.gpkg FIELD=field1 BINS=10 OUTPUT=output.txt
```

This calls **Vector layer histogram**: INPUT=input.gpkg — Input layer; FIELD=field1 — Attribute; BINS=10 — number of bins; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## qgis:vectorlayerscatterplot — Vector layer scatterplot

**group:** Plots

This algorithm creates a simple X - Y scatter plot for a vector layer.

| Parameter   | Type            | Required | Default | Description                      |
|-------------|-----------------|----------|---------|----------------------------------|
| INPUT       | source          | yes      | —       | Input layer                      |
| XFIELD      | field           | yes      | —       | X attribute                      |
| YFIELD      | field           | yes      | —       | Y attribute                      |
| HOVERTEXT   | expression      | no       | —       | Hover text                       |
| TITLE       | string          | no       | —       | Title                            |
| XAXIS_TITLE | string          | no       | —       | X-axis title                     |
| YAXIS_TITLE | string          | no       | —       | Y-axis title                     |
| XAXIS_LOG   | boolean         | no       | False   | Use logarithmic scale for x-axis |
| YAXIS_LOG   | boolean         | no       | False   | Use logarithmic scale for y-axis |
| OUTPUT      | fileDestination | yes      | —       | Scatterplot                      |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run qgis:vectorlayerscatterplot INPUT=input.gpkg XFIELD=field1 YFIELD=field1
↳ HOVERTEXT="value > 0" TITLE=value XAXIS_TITLE=value YAXIS_TITLE=value
↳ OUTPUT=output.txt
```

This calls **Vector layer scatterplot**: INPUT=input.gpkg — Input layer; XFIELD=field1 — X attribute; YFIELD=field1 — Y attribute; HOVERTEXT="value > 0" — Hover text; TITLE=value — Title; XAXIS\_TITLE=value — X-axis title; YAXIS\_TITLE=value — Y-axis title; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

# pdal: algorithms

pdal — PDAL point-cloud algorithms — 24 algorithms (QGIS 4.0.3). Auto-generated by scripts/gen\_algorithms.py; see [the index](#).

Call any of these with `run <id> KEY=value ...` (the **Name** column is the KEY); each entry includes a worked **Example usage**. A ★ marks an algorithm with a friendly niva alias verb.

## pdal:assignprojection — Assign projection

**group:** Point cloud data management

This algorithm assigns point cloud CRS if it is not present or wrong.

| Parameter                    | Type                  | Required | Default   | Description                                       |
|------------------------------|-----------------------|----------|-----------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —         | Input layer                                       |
| CRS                          | crs                   | no       | EPSG:4326 | Desired CRS                                       |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC      | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —         | Output layer                                      |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:assignprojection INPUT=cloud.copc.laz CRS=EPSG:6346 OUTPUT=output.laz
```

This calls **Assign projection**: INPUT=cloud.copc.laz — Input layer; CRS=EPSG:6346 — Desired CRS; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:boundary — Boundary

**group:** Point cloud extraction

This algorithm exports a polygon file containing point cloud layer boundary. It may contain holes and it may be a multi-part polygon.

| Parameter                    | Type              | Required | Default | Description                                                  |
|------------------------------|-------------------|----------|---------|--------------------------------------------------------------|
| INPUT                        | pointcloud        | yes      | —       | Input layer                                                  |
| RESOLUTION                   | number            | no       | —       | Resolution of cells used to calculate boundary               |
| THRESHOLD                    | number            | no       | —       | Minimal number of points in a cell to consider cell occupied |
| FILTER_EXPRESSION<br>*(adv)* | expression        | no       | —       | Filter expression                                            |
| FILTER_EXTENT *(adv)*        | extent            | no       | —       | Cropping extent                                              |
| OUTPUT                       | vectorDestination | yes      | —       | Boundary                                                     |

---

**Outputs:** OUTPUT (outputVector)

**Example usage**

```
run pdal:boundary INPUT=cloud.copc.laz RESOLUTION=10 THRESHOLD=10 OUTPUT=output.gpkg
```

This calls **Boundary**: INPUT=cloud.copc.laz — Input layer; RESOLUTION=10 — Resolution of cells used to calculate boundary; THRESHOLD=10 — Minimal number of points in a cell to consider cell occupied; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:classifyground — Classify ground points

**group:** Point cloud data management

This algorithm classifies ground points using the Simple Morphological Filter (SMRF) algorithm.

Cell Size is the cell size for processing grid in map units, where smaller values give finer detail but may increase noise. Scalar is the threshold for steeper slopes; a higher value is needed if the terrain is rough, otherwise real ground might be misclassified. Slope is the slope threshold measured as rise over run, indicating how much slope is tolerated as ground and should be higher for steep terrain. Threshold is the elevation threshold for separating ground from objects; higher values allow larger deviations from the ground. Window Size is the maximum filter window size, where higher values better identify large buildings or objects while smaller values protect smaller features.

| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| CELL_SIZE                    | number                | no       | 1.0     | Grid Cell Size                                    |
| SCALAR                       | number                | no       | 1.25    | Scalar                                            |
| SLOPE                        | number                | no       | 0.15    | Slope                                             |
| THRESHOLD                    | number                | no       | 0.5     | Threshold                                         |
| WINDOW_SIZE                  | number                | no       | 18.0    | Window Size                                       |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Classified Ground                                 |

**Outputs:** OUTPUT (outputPointCloud)

**Example usage**

```
run pdal:classifyground INPUT=cloud.copc.laz CELL_SIZE=1 SCALAR=1.25 SLOPE=0.15  
↪ THRESHOLD=0.5 WINDOW_SIZE=18 OUTPUT=output.laz
```

This calls **Classify ground points**: INPUT=cloud.copc.laz — Input layer; CELL\_SIZE=1 — Grid Cell Size; SCALAR=1.25 — Scalar; SLOPE=0.15 — Slope; THRESHOLD=0.5 — Threshold; WINDOW\_SIZE=18 — Window Size; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:clip — Clip point cloud

**group:** Point cloud data management

This algorithm clips point cloud with clipping polygons, the resulting point cloud contains points that are inside these polygons.



| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| OVERLAY                      | vector                | yes      | —       | Clipping polygons                                 |
| FILTER_EXPRESSION<br>*(adv)* | expression            | no       | —       | Filter expression                                 |
| FILTER_EXTENT *(adv)*        | extent                | no       | —       | Cropping extent                                   |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Clipped                                           |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:clip INPUT=cloud.copc.laz OVERLAY=input.gpkg OUTPUT=output.laz
```

This calls **Clip point cloud**: INPUT=cloud.copc.laz — Input layer; OVERLAY=input.gpkg — Clipping polygons; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:compare — Compare point clouds

**group:** Point cloud data management

This algorithm compares two point clouds using an M3C2 (Multiscale Model-to-Model Cloud Comparison) algorithm and outputs a subset (filtered using Poisson sampling, based on Subsampling cell size parameter ) of the original point cloud.

The output point cloud will have several new dimensions: m3c2\_distance, m3c2\_uncertainty, m3c2\_significant, m3c2\_std\_dev1, m3c2\_std\_dev2, m3c2\_count1 and m3c2\_count2.

The M3C2 algorithm calculates the distance between two point clouds by considering the local orientation of the surface. It estimates surface normals at a user-defined scale and measures the average distance between clouds within a cylindrical projection along those normals.

The approach is highly robust against sensor noise and surface roughness, making it the standard for detecting change in complex natural environments. It also provides a sign (indicating whether a surface has moved in or out) and a statistically significant level of detection to distinguish real change from measurement error.

References: Lague, Dimitri, Nicolas Brodu, and Jérôme Leroux. Accurate 3D Comparison of Complex Topography with Terrestrial Laser Scanner: Application to the Rangitikei Canyon (N-Z). arXiv, 2013, <https://arxiv.org/abs/1302.1183>.

| Parameter                    | Type                  | Required | Default | Description                                            |
|------------------------------|-----------------------|----------|---------|--------------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                            |
| INPUT_COMPARE                | pointcloud            | yes      | —       | Compare layer                                          |
| SUBSAMPLING_CELL_SIZE        | distance              | no       | 0.0     | Subsampling cell size                                  |
| NORMAL_RADIUS                | distance              | no       | 2.0     | Normal Radius                                          |
| CYLINDER_RADIUS              | distance              | no       | 2.0     | Cylinder Radius                                        |
| CYLINDER_HALF_LENGTH         | distance              | no       | 5.0     | Cylinder Half-Length                                   |
| REGISTRATION_ERROR           | number                | no       | 0.0     | Registration Error                                     |
| CYLINDER_ORIENTATION         | enum                  | no       | 0       | Cylinder Orientation — options: 0=Up, 1=Origin, 2=None |
| FILTER_EXPRESSION<br>*(adv)* | expression            | no       | —       | Filter expression                                      |
| FILTER_EXTENT *(adv)*        | extent                | no       | —       | Cropping extent                                        |
| OUTPUT                       | pointCloudDestination | yes      | —       | Comparison Point Cloud                                 |

**Outputs:** OUTPUT (outputPointCloud)

## Example usage

```
run pdal:compare INPUT=cloud.copc.laz INPUT_COMPARE=cloud.copc.laz SUBSAMPLING_CELL_SIZE=0
↪ NORMAL_RADIUS=2 CYLINDER_RADIUS=2 CYLINDER_HALF_LENGTH=5 REGISTRATION_ERROR=0
↪ OUTPUT=output.laz
```

This calls **Compare point clouds**: INPUT=cloud.copc.laz — Input layer; INPUT\_COMPARE=cloud.copc.laz — Compare layer; SUBSAMPLING\_CELL\_SIZE=0 — Subsampling cell size; NORMAL\_RADIUS=2 — Normal Radius; CYLINDER\_RADIUS=2 — Cylinder Radius; CYLINDER\_HALF\_LENGTH=5 — Cylinder Half-Length; REGISTRATION\_ERROR=0 — Registration Error; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:convertformat — Convert point cloud format

**group:** Point cloud conversion

This algorithm converts point cloud to a different file format, e. g. creates compressed LAZ.

| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Converted                                         |

**Outputs:** OUTPUT (outputPointCloud)

## Example usage

```
run pdal:convertformat INPUT=cloud.copc.laz OUTPUT=output.laz
```

This calls **Convert point cloud format**: INPUT=cloud.copc.laz — Input layer; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:createcopc — Create COPC

**group:** Point cloud data management

This algorithm creates a COPC file for each input point cloud file.

| Parameter | Type              | Required | Default | Description      |
|-----------|-------------------|----------|---------|------------------|
| LAYERS    | multilayer        | yes      | —       | Input layers     |
| OUTPUT    | folderDestination | no       | —       | Output directory |

**Outputs:** OUTPUT (outputFolder), OUTPUT\_LAYERS (outputMultilayer)

## Example usage

```
run pdal:createcopc LAYERS="a.gpkg;b.gpkg" OUTPUT=output/
```

This calls **Create COPC**: LAYERS="a.gpkg;b.gpkg" — Input layers; OUTPUT=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:density — Point cloud density

**group:** Point cloud extraction

This algorithm exports a raster file where each cell contains number of points that are in that cell's area.

| Parameter                 | Type              | Required | Default | Description                          |
|---------------------------|-------------------|----------|---------|--------------------------------------|
| INPUT                     | pointcloud        | yes      | —       | Input layer                          |
| RESOLUTION                | number            | no       | 1       | Resolution of the density raster     |
| TILE_SIZE                 | number            | no       | 1000    | Tile size for parallel runs          |
| FILTER_EXPRESSION *(adv)* | expression        | no       | —       | Filter expression                    |
| FILTER_EXTENT *(adv)*     | extent            | no       | —       | Cropping extent                      |
| ORIGIN_X *(adv)*          | number            | no       | —       | X origin of a tile for parallel runs |
| ORIGIN_Y *(adv)*          | number            | no       | —       | Y origin of a tile for parallel runs |
| OUTPUT                    | rasterDestination | yes      | —       | Density                              |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run pdal:density INPUT=cloud.copc.laz RESOLUTION=1 TILE_SIZE=1000 OUTPUT=output.tif
```

This calls **Point cloud density**: INPUT=cloud.copc.laz — Input layer; RESOLUTION=1 — Resolution of the density raster; TILE\_SIZE=1000 — Tile size for parallel runs; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:exportraster — Export point cloud to raster

**group:** Point cloud conversion

This algorithm exports point cloud data to a 2D raster grid having cell size of given resolution, writing values from the specified attribute.

| Parameter                 | Type              | Required | Default | Description                          |
|---------------------------|-------------------|----------|---------|--------------------------------------|
| INPUT                     | pointcloud        | yes      | —       | Input layer                          |
| ATTRIBUTE                 | attribute         | no       | Z       | Attribute                            |
| RESOLUTION                | number            | no       | 1       | Resolution of the density raster     |
| TILE_SIZE                 | number            | no       | 1000    | Tile size for parallel runs          |
| FILTER_EXPRESSION *(adv)* | expression        | no       | —       | Filter expression                    |
| FILTER_EXTENT *(adv)*     | extent            | no       | —       | Cropping extent                      |
| ORIGIN_X *(adv)*          | number            | no       | —       | X origin of a tile for parallel runs |
| ORIGIN_Y *(adv)*          | number            | no       | —       | Y origin of a tile for parallel runs |
| OUTPUT                    | rasterDestination | yes      | —       | Exported                             |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run pdal:exportraster INPUT=cloud.copc.laz ATTRIBUTE=Z RESOLUTION=1 TILE_SIZE=1000  
↪ OUTPUT=output.tif
```

This calls **Export point cloud to raster**: INPUT=cloud.copc.laz — Input layer; ATTRIBUTE=Z — Attribute; RESOLUTION=1 — Resolution of the density raster; TILE\_SIZE=1000 — Tile size for parallel runs; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:exportrastertin — Export point cloud to raster (using triangulation)

**group:** Point cloud conversion

This algorithm exports point cloud data to a 2D raster grid using a triangulation of points and then interpolating cell values from triangles.

| Parameter                 | Type              | Required | Default | Description                          |
|---------------------------|-------------------|----------|---------|--------------------------------------|
| INPUT                     | pointcloud        | yes      | —       | Input layer                          |
| RESOLUTION                | number            | no       | 1       | Resolution of the density raster     |
| TILE_SIZE                 | number            | no       | 1000    | Tile size for parallel runs          |
| FILTER_EXPRESSION *(adv)* | expression        | no       | —       | Filter expression                    |
| FILTER_EXTENT *(adv)*     | extent            | no       | —       | Cropping extent                      |
| ORIGIN_X *(adv)*          | number            | no       | —       | X origin of a tile for parallel runs |
| ORIGIN_Y *(adv)*          | number            | no       | —       | Y origin of a tile for parallel runs |
| OUTPUT                    | rasterDestination | yes      | —       | Exported (using triangulation)       |

**Outputs:** OUTPUT (outputRaster)

### Example usage

```
run pdal:exportrastertin INPUT=cloud.copc.laz RESOLUTION=1 TILE_SIZE=1000  
↪ OUTPUT=output.tif
```

This calls **Export point cloud to raster (using triangulation)**: INPUT=cloud.copc.laz — Input layer; RESOLUTION=1 — Resolution of the density raster; TILE\_SIZE=1000 — Tile size for parallel runs; OUTPUT=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:exportvector — Export point cloud to vector

**group:** Point cloud conversion

This algorithm exports point cloud data to a vector layer with 3D points (a GeoPackage), optionally with extra attributes.

| Parameter                 | Type              | Required | Default | Description       |
|---------------------------|-------------------|----------|---------|-------------------|
| INPUT                     | pointcloud        | yes      | —       | Input layer       |
| ATTRIBUTE                 | attribute         | no       | —       | Attribute         |
| FILTER_EXPRESSION *(adv)* | expression        | no       | —       | Filter expression |
| FILTER_EXTENT *(adv)*     | extent            | no       | —       | Cropping extent   |
| OUTPUT                    | vectorDestination | yes      | —       | Exported          |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run pdal:exportvector INPUT=cloud.copc.laz ATTRIBUTE=field1 OUTPUT=output.gpkg
```

This calls **Export point cloud to vector**: INPUT=cloud.copc.laz — Input layer; ATTRIBUTE=field1 — Attribute; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:filter — Filter point cloud

**group:** Point cloud extraction

This algorithm extracts point from the input point cloud which match PDAL expression and/or are inside of a cropping rectangle.

| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| FILTER_EXPRESSION            | expression            | yes      | —       | Filter expression                                 |
| FILTER_EXTENT                | extent                | no       | —       | Cropping extent                                   |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Filtered                                          |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:filter INPUT=cloud.copc.laz FILTER_EXPRESSION="value > 0"
↪ FILTER_EXTENT=0,1000,0,1000 OUTPUT=output.laz
```

This calls **Filter point cloud**: INPUT=cloud.copc.laz — Input layer; FILTER\_EXPRESSION="value > 0" — Filter expression; FILTER\_EXTENT=0,1000,0,1000 — Cropping extent; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:filternoiseradius — Filter noise (using radius)

**group:** Point cloud data management

This algorithm filters noise in a point cloud using radius algorithm.

Points are marked as noise if they have fewer than the minimum number of neighbors within the specified radius.

| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| REMOVE_NOISE_POINTS          | boolean               | no       | False   | Remove noise points                               |
| MIN_K                        | number                | no       | 2       | Minimum number of neighbors in radius             |
| RADIUS                       | number                | no       | 1.0     | Radius                                            |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Filtered noise                                    |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:filternoiseradius INPUT=cloud.copc.laz REMOVE_NOISE_POINTS=false MIN_K=2 RADIUS=1
↪ OUTPUT=output.laz
```

This calls **Filter noise (using radius)**: INPUT=cloud.copc.laz — Input layer; REMOVE\_NOISE\_POINTS=false — Remove noise points; MIN\_K=2 — Minimum number of neighbors in radius; RADIUS=1 — Radius; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:filternoisestatistical — Filter noise

**group:** Point cloud data management

This algorithm filters noise in a point cloud using a statistical outlier removal algorithm.

For each point, the algorithm computes the mean distance to its K nearest neighbors. Points whose mean distance exceeds a threshold (mean distance + multiplier × standard deviation) are classified as noise.

| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| REMOVE_NOISE_POINTS          | boolean               | no       | False   | Remove noise points                               |
| MEAN_K                       | number                | no       | 8       | Mean number of neighbors                          |
| MULTIPLIER                   | number                | no       | 2.0     | Standard deviation multiplier                     |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Filtered noise                                    |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:filternoisestatistical INPUT=cloud.copc.laz REMOVE_NOISE_POINTS=false MEAN_K=8
↳ MULTIPLIER=2 OUTPUT=output.laz
```

This calls **Filter noise**: INPUT=cloud.copc.laz — Input layer; REMOVE\_NOISE\_POINTS=false — Remove noise points; MEAN\_K=8 — Mean number of neighbors; MULTIPLIER=2 — Standard deviation multiplier; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:heightabovegroundbynearestneighbor — Height above ground

**group:** Point cloud data management

This algorithm calculates the height of points above the ground surface in a point cloud using a nearest neighbor algorithm.

For each point, the algorithm finds the specified number of nearest ground-classified points (classification value 2) and interpolates the ground elevation from them using inverse distance weighting.

The output adds a HeightAboveGround dimension to the point cloud. If 'Replace Z values' is enabled, the Z coordinate will be replaced with the height above ground value.

Maximum Distance parameter can limit the search radius (0 = no limit).

| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| REPLACE_Z                    | boolean               | no       | True    | Replace Z values with height above ground         |
| COUNT                        | number                | no       | 1       | Number of neighbors for terrain interpolation     |
| MAX_DISTANCE                 | number                | no       | 0.0     | Maximum search distance                           |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Height above ground                               |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:heightabovegroundbynearestneighbor INPUT=cloud.copc.laz REPLACE_Z=true COUNT=1
↳ MAX_DISTANCE=0 OUTPUT=output.laz
```

This calls **Height above ground**: INPUT=cloud.copc.laz — Input layer; REPLACE\_Z=true — Replace Z values with height above ground; COUNT=1 — Number of neighbors for terrain interpolation; MAX\_DISTANCE=0 — Maximum search distance; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:heightabovegroundtriangulation — Height above ground (using triangulation)

**group:** Point cloud data management

This algorithm calculates the height of points above the ground surface in a point cloud using a Delaunay triangulation algorithm.

The algorithm uses ground-classified points (classification value 2) to create a triangulated irregular network (TIN) from specified number of neighbors, then computes the height above this surface for all points.

The output adds a HeightAboveGround dimension to the point cloud. If 'Replace Z values' is enabled, the Z coordinate will be replaced with the height above ground value.

| Parameter                    | Type                  | Required | Default | Description                                         |
|------------------------------|-----------------------|----------|---------|-----------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                         |
| REPLACE_Z                    | boolean               | no       | True    | Replace Z values with height above ground           |
| COUNT                        | number                | no       | 10      | Number of ground neighbors for terrain construction |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS   |
| OUTPUT                       | pointCloudDestination | yes      | —       | Height above ground                                 |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:heightabovegroundtriangulation INPUT=cloud.copc.laz REPLACE_Z=true COUNT=10  
↪ OUTPUT=output.laz
```

This calls **Height above ground (using triangulation)**: INPUT=cloud.copc.laz — Input layer; REPLACE\_Z=true — Replace Z values with height above ground; COUNT=10 — Number of ground neighbors for terrain construction; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:info — Point cloud information

**group:** Point cloud data management

This algorithm outputs basic metadata from the point cloud file.

| Parameter | Type            | Required | Default | Description       |
|-----------|-----------------|----------|---------|-------------------|
| INPUT     | pointcloud      | yes      | —       | Input layer       |
| OUTPUT    | fileDestination | yes      | —       | Layer information |

**Outputs:** OUTPUT (outputHtml)

### Example usage

```
run pdal:info INPUT=cloud.copc.laz OUTPUT=output.txt
```

This calls **Point cloud information**: INPUT=cloud.copc.laz — Input layer; OUTPUT=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:merge — Merge point cloud

**group:** Point cloud data management

This algorithm merges multiple point cloud files to a single one.

| Parameter                 | Type                  | Required | Default | Description       |
|---------------------------|-----------------------|----------|---------|-------------------|
| LAYERS                    | multilayer            | yes      | —       | Input layers      |
| FILTER_EXPRESSION *(adv)* | expression            | no       | —       | Filter expression |
| FILTER_EXTENT *(adv)*     | extent                | no       | —       | Cropping extent   |
| OUTPUT                    | pointCloudDestination | yes      | —       | Merged            |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:merge LAYERS="a.gpkg;b.gpkg" OUTPUT=output.laz
```

This calls **Merge point cloud**: LAYERS="a.gpkg;b.gpkg" — Input layers; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:reproject — Reproject point cloud

**group:** Point cloud data management

This algorithm reprojects point cloud to a different coordinate reference system.

| Parameter                 | Type                  | Required | Default   | Description                                       |
|---------------------------|-----------------------|----------|-----------|---------------------------------------------------|
| INPUT                     | pointcloud            | yes      | —         | Input layer                                       |
| CRS                       | crs                   | no       | EPSG:4326 | Target CRS                                        |
| OPERATION *(adv)*         | coordinateoperation   | no       | —         | Coordinate operation                              |
| VPC_OUTPUT_FORMAT *(adv)* | enum                  | no       | COPC      | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                    | pointCloudDestination | yes      | —         | Reprojected                                       |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:reproject INPUT=cloud.copc.laz CRS=EPSG:6346 OUTPUT=output.laz
```

This calls **Reproject point cloud**: INPUT=cloud.copc.laz — Input layer; CRS=EPSG:6346 — Target CRS; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:thinbydecimate — Thin (by skipping points)

**group:** Point cloud data management

This algorithm creates a thinned version of the point cloud by keeping only every N-th point.

| Parameter                 | Type       | Required | Default | Description                                       |
|---------------------------|------------|----------|---------|---------------------------------------------------|
| INPUT                     | pointcloud | yes      | —       | Input layer                                       |
| POINTS_NUMBER             | number     | no       | 1       | Number of points to skip                          |
| FILTER_EXPRESSION *(adv)* | expression | no       | —       | Filter expression                                 |
| FILTER_EXTENT *(adv)*     | extent     | no       | —       | Cropping extent                                   |
| VPC_OUTPUT_FORMAT *(adv)* | enum       | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |



| Parameter | Type                  | Required | Default | Description             |
|-----------|-----------------------|----------|---------|-------------------------|
| OUTPUT    | pointCloudDestination | yes      | —       | Thinned (by decimation) |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:thinbydecimate INPUT=cloud.copc.laz POINTS_NUMBER=1 OUTPUT=output.laz
```

This calls **Thin (by skipping points)**: INPUT=cloud.copc.laz — Input layer; POINTS\_NUMBER=1 — Number of points to skip; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:thinbyradius — Thin (by sampling radius)

**group:** Point cloud data management

This algorithm creates a thinned version of the point cloud by performing sampling by distance point.

| Parameter                 | Type                  | Required | Default | Description                                       |
|---------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                     | pointcloud            | yes      | —       | Input layer                                       |
| SAMPLING_RADIUS           | number                | no       | 1.0     | Sampling radius (in map units)                    |
| FILTER_EXPRESSION *(adv)* | expression            | no       | —       | Filter expression                                 |
| FILTER_EXTENT *(adv)*     | extent                | no       | —       | Cropping extent                                   |
| VPC_OUTPUT_FORMAT *(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                    | pointCloudDestination | yes      | —       | Thinned (by radius)                               |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:thinbyradius INPUT=cloud.copc.laz SAMPLING_RADIUS=1 OUTPUT=output.laz
```

This calls **Thin (by sampling radius)**: INPUT=cloud.copc.laz — Input layer; SAMPLING\_RADIUS=1 — Sampling radius (in map units); OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:tile — Create tiles from point cloud

**group:** Point cloud data management

This algorithm creates tiles from an input point cloud.

| Parameter                 | Type              | Required | Default | Description                                       |
|---------------------------|-------------------|----------|---------|---------------------------------------------------|
| LAYERS                    | multilayer        | yes      | —       | Input layers                                      |
| LENGTH                    | number            | no       | 1000.0  | Tile length                                       |
| CRS *(adv)*               | crs               | no       | —       | Assign CRS                                        |
| VPC_OUTPUT_FORMAT *(adv)* | enum              | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                    | folderDestination | yes      | —       | Output directory                                  |

**Outputs:** OUTPUT (outputFolder)

### Example usage

```
run pdal:tile LAYERS="a.gpkg;b.gpkg" LENGTH=1000 OUTPUT=output/
```

This calls **Create tiles from point cloud**: LAYERS="a.gpkg;b.gpkg" — Input layers; LENGTH=1000 — Tile length; OUTPUT=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:transformpointcloud — Transform point cloud

**group:** Point cloud data management

This algorithm transforms point cloud coordinates using translation, rotation, and scaling operations using a 4x4 transformation matrix.

The algorithm applies transformations in the following order: scaling, rotation (using Euler angles), then translation.

Rotation angles are specified in degrees around the X, Y, and Z axes.

All parameters are combined into a 4x4 transformation matrix that is passed to PDAL Wrench.

| Parameter                    | Type                  | Required | Default | Description                                       |
|------------------------------|-----------------------|----------|---------|---------------------------------------------------|
| INPUT                        | pointcloud            | yes      | —       | Input layer                                       |
| TRANSLATE_X                  | number                | no       | 0.0     | X Translation                                     |
| TRANSLATE_Y                  | number                | no       | 0.0     | Y Translation                                     |
| TRANSLATE_Z                  | number                | no       | 0.0     | Z Translation                                     |
| SCALE_X                      | number                | no       | 1.0     | X Scale                                           |
| SCALE_Y                      | number                | no       | 1.0     | Y Scale                                           |
| SCALE_Z                      | number                | no       | 1.0     | Z Scale                                           |
| ROTATE_X                     | number                | no       | 0.0     | X Rotation                                        |
| ROTATE_Y                     | number                | no       | 0.0     | Y Rotation                                        |
| ROTATE_Z                     | number                | no       | 0.0     | Z Rotation                                        |
| VPC_OUTPUT_FORMAT<br>*(adv)* | enum                  | no       | COPC    | VPC Output Format — options: 0=COPC, 1=LAZ, 2=LAS |
| OUTPUT                       | pointCloudDestination | yes      | —       | Transformed                                       |

**Outputs:** OUTPUT (outputPointCloud)

### Example usage

```
run pdal:transformpointcloud INPUT=cloud.copc.laz TRANSLATE_X=0 TRANSLATE_Y=0
↪ TRANSLATE_Z=0 SCALE_X=1 SCALE_Y=1 SCALE_Z=1 OUTPUT=output.laz
```

This calls **Transform point cloud**: INPUT=cloud.copc.laz — Input layer; TRANSLATE\_X=0 — X Translation; TRANSLATE\_Y=0 — Y Translation; TRANSLATE\_Z=0 — Z Translation; SCALE\_X=1 — X Scale; SCALE\_Y=1 — Y Scale; SCALE\_Z=1 — Z Scale; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## pdal:virtualpointcloud — Build virtual point cloud (VPC)

**group:** Point cloud data management

This algorithm creates a virtual point cloud from input data.

| Parameter    | Type                  | Required | Default | Description                             |
|--------------|-----------------------|----------|---------|-----------------------------------------|
| LAYERS       | multilayer            | yes      | —       | Input layers                            |
| BOUNDARY     | boolean               | no       | False   | Calculate boundary polygons             |
| STATISTICS   | boolean               | no       | False   | Calculate statistics                    |
| OVERVIEW     | boolean               | no       | False   | Build overview point cloud              |
| CONVERT_COPC | boolean               | no       | False   | Convert individual files to COPC format |
| OUTPUT       | pointCloudDestination | yes      | —       | Virtual point cloud                     |

---

**Outputs:** OUTPUT (outputPointCloud)

**Example usage**

```
run pdal:virtualpointcloud LAYERS="a.gpkg;b.gpkg" BOUNDARY=false STATISTICS=false  
↳ OVERVIEW=false CONVERT_COPC=false OUTPUT=output.laz
```

This calls **Build virtual point cloud (VPC)**: LAYERS="a.gpkg;b.gpkg" — Input layers; BOUNDARY=false — Calculate boundary polygons; STATISTICS=false — Calculate statistics; OVERVIEW=false — Build overview point cloud; CONVERT\_COPC=false — Convert individual files to COPC format; OUTPUT=output.laz is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

# 3d: algorithms

3d — 3D algorithms — 1 algorithms (QGIS 4.0.3). Auto-generated by scripts/gen\_algorithms.py; see [the index](#).

Call any of these with `run <id> KEY=value ...` (the **Name** column is the KEY); each entry includes a worked **Example usage**. A ★ marks an algorithm with a friendly niva alias verb.

## 3d:tessellate — Tessellate

**group:** Vector geometry

This algorithm tessellates a polygon geometry layer, dividing the geometries into triangular components.

The output layer consists of multipolygon geometries for each input feature, with each multipolygon consisting of multiple triangle component polygons.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| INPUT     | source | yes      | —       | Input layer |
| OUTPUT    | sink   | yes      | —       | Tessellated |

**Outputs:** OUTPUT (outputVector)

### Example usage

```
run 3d:tessellate INPUT=input.gpkg OUTPUT=output.gpkg
```

This calls **Tessellate**: INPUT=input.gpkg — Input layer; OUTPUT=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

# grass: algorithms

grass — GRASS GIS algorithms — 307 algorithms (QGIS 4.0.3). Auto-generated by scripts/gen\_algorithms.py; see [the index](#).

Call any of these with run <id> KEY=value ... (the **Name** column is the KEY); each entry includes a worked **Example usage**. A ★ marks an algorithm with a friendly niva alias verb.

## grass:g.extension.list — g.extension.list

**group:** General (g.\*)

g.extension.list - List GRASS addons.

| Parameter                         | Type            | Required | Default          | Description                                                                                                                                                                                                         |
|-----------------------------------|-----------------|----------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| list                              | enum            | no       | 0                | List — options: 0=Locally installed extensions, 1=Extensions available in the official GRASS GIS Addons repository, 2=Extensions available in the official GRASS GIS Addons repository including module description |
| html                              | fileDestination | no       | addons_list.html | List of addons                                                                                                                                                                                                      |
| GRASS_REGION_PARAMETER<br>*(adv)* | extent          | no       | —                | GRASS GIS region extent                                                                                                                                                                                             |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:g.extension.list list=0 html=html.txt
```

This calls **g.extension.list**: list=0 selects *Locally installed extensions*; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:g.extension.manage — g.extension.manage

**group:** General (g.\*)

g.extension.manage - Install or uninstall GRASS addons.

| Parameter                         | Type    | Required | Default | Description                                                   |
|-----------------------------------|---------|----------|---------|---------------------------------------------------------------|
| extension                         | string  | yes      | —       | Name of Extension                                             |
| operation                         | enum    | yes      | —       | Operation — options: 0=add, 1=remove                          |
| -f                                | boolean | no       | False   | Force (required for removal)                                  |
| -t                                | boolean | no       | False   | Operate on toolboxes instead of single modules (experimental) |
| GRASS_REGION_PARAMETER<br>*(adv)* | extent  | no       | —       | GRASS GIS region extent                                       |

## Example usage

```
run grass:g.extension.manage extension=value operation=0
```

This calls **g.extension.manage**: extension=value — Name of Extension; operation=0 selects *add*. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:g.version — g.version

**group:** General (g.\*)

g.version - Display GRASS GIS version info. Prints only version if run with no options checked.

| Parameter                      | Type            | Required | Default                 | Description                                             |
|--------------------------------|-----------------|----------|-------------------------|---------------------------------------------------------|
| -c                             | boolean         | no       | False                   | Print copyright message                                 |
| -x                             | boolean         | no       | False                   | Print citation options                                  |
| -b                             | boolean         | no       | False                   | Print build information                                 |
| -r                             | boolean         | no       | True                    | Print GIS library revision number and date              |
| -e                             | boolean         | no       | True                    | Print info for additional libraries                     |
| -g                             | boolean         | no       | False                   | Print in shell script style (with Git commit reference) |
| --verbose                      | boolean         | no       | False                   | Print verbose output                                    |
| html                           | fileDestination | no       | grass_version_info.html | Output file                                             |
| GRASS_REGION_PARAMETER *(adv)* | extent          | no       | —                       | GRASS GIS region extent                                 |

**Outputs:** html (outputHtml)

## Example usage

```
run grass:g.version html=html.txt
```

This calls **g.version**: html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.albedo — i.albedo

**group:** Imagery (i.\*)

Computes broad band albedo from surface reflectance.

| Parameter                                | Type              | Required | Default | Description                                     |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Name of input raster maps                       |
| -m                                       | boolean           | no       | False   | MODIS (7 input bands:1,2,3,4,5,6,7)             |
| -n                                       | boolean           | no       | False   | NOAA AVHRR (2 input bands:1,2)                  |
| -l                                       | boolean           | no       | False   | Landsat 5+7 (6 input bands:1,2,3,4,5,7)         |
| -8                                       | boolean           | no       | False   | Landsat 8 (7 input bands:1,2,3,4,5,6,7)         |
| -a                                       | boolean           | no       | False   | ASTER (6 input bands:1,3,5,6,8,9)               |
| -c                                       | boolean           | no       | False   | Aggressive mode (Landsat)                       |
| -d                                       | boolean           | no       | False   | Soft mode (MODIS)                               |
| output                                   | rasterDestination | yes      | —       | Albedo                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)       |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.albedo input="a.gpkg;b.gpkg" output=output.tif
```

This calls **i.albedo**: input="a.gpkg;b.gpkg" — Name of input raster maps; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.aster.toar — i.aster.toar

**group:** Imagery (i.\*)

Calculates Top of Atmosphere Radiance/Reflectance/Brightness Temperature from ASTER DN.

| Parameter                                | Type              | Required | Default | Description                                     |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Names of ASTER DN layers (15 layers)            |
| dayofyear                                | number            | no       | 0.0     | Day of Year of satellite overpass [0-366]       |
| sun_elevation                            | number            | yes      | —       | Sun elevation angle (degrees, < 90.0)           |
| -r                                       | boolean           | no       | False   | Output is radiance (W/m2)                       |
| -a                                       | boolean           | no       | False   | VNIR is High Gain                               |
| -b                                       | boolean           | no       | False   | SWIR is High Gain                               |
| -c                                       | boolean           | no       | False   | VNIR is Low Gain 1                              |
| -d                                       | boolean           | no       | False   | SWIR is Low Gain 1                              |
| -e                                       | boolean           | no       | False   | SWIR is Low Gain 2                              |
| output                                   | folderDestination | yes      | —       | Output Directory                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:i.aster.toar input="a.gpkg;b.gpkg" sun_elevation=10 dayofyear=0 output=output/
```

This calls **i.aster.toar**: input="a.gpkg;b.gpkg" — Names of ASTER DN layers (15 layers); sun\_elevation=10 — Sun elevation angle (degrees, < 90.0); dayofyear=0 — Day of Year of satellite overpass [0-366]; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.atcorr — i.atcorr

**group:** Imagery (i.\*)

Performs atmospheric correction using the 6S algorithm.

| Parameter  | Type   | Required | Default | Description                                  |
|------------|--------|----------|---------|----------------------------------------------|
| input      | raster | yes      | —       | Name of input raster map                     |
| range      | range  | no       | 0,255   | Input imagery range [0,255]                  |
| elevation  | raster | no       | —       | Input altitude raster map in m (optional)    |
| visibility | raster | no       | —       | Input visibility raster map in km (optional) |
| parameters | file   | yes      | —       | Name of input text file                      |
| rescale    | range  | no       | 0,255   | Rescale output raster map [0,255]            |

| Parameter                                | Type              | Required | Default | Description                                                     |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------|
| output                                   | rasterDestination | yes      | —       | Atmospheric correction                                          |
| -i *(adv)*                               | boolean           | no       | False   | Output raster map as integer                                    |
| -r *(adv)*                               | boolean           | no       | False   | Input raster map converted to reflectance (default is radiance) |
| -a *(adv)*                               | boolean           | no       | False   | Input from ETM+ image taken after July 1, 2000                  |
| -b *(adv)*                               | boolean           | no       | False   | Input from ETM+ image taken before July 1, 2000                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                 |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                       |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.atcorr input=dem.tif parameters=input.dat range=0,255 elevation=dem.tif
↪ visibility=dem.tif rescale=0,255 output=output.tif
```

This calls **i.atcorr**: input=dem.tif — Name of input raster map; parameters=input.dat — Name of input text file; range=0,255 — Input imagery range [0,255]; elevation=dem.tif — Input altitude raster map in m (optional); visibility=dem.tif — Input visibility raster map in km (optional); rescale=0,255 — Rescale output raster map [0,255]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.biomass — i.biomass

**group:** Imagery (i.\*)

Computes biomass growth, precursor of crop yield calculation.

| Parameter                                | Type              | Required | Default | Description                                              |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------|
| fpar                                     | raster            | yes      | —       | Name of fPAR raster map                                  |
| lightuse_efficiency                      | raster            | yes      | —       | Name of light use efficiency raster map (UZB:cotton=1.9) |
| latitude                                 | raster            | yes      | —       | Name of degree latitude raster map [dd.ddd]              |
| dayofyear                                | raster            | yes      | —       | Name of Day of Year raster map [1-366]                   |
| transmissivity_singleway                 | raster            | yes      | —       | Name of single-way transmissivity raster map [0.0-1.0]   |
| water_availability                       | raster            | yes      | —       | Value of water availability raster map [0.0-1.0]         |
| output                                   | rasterDestination | yes      | —       | Biomass                                                  |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                  |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)          |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)         |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.biomass fpar=dem.tif lightuse_efficiency=dem.tif latitude=dem.tif
↪ dayofyear=dem.tif transmissivity_singleway=dem.tif water_availability=dem.tif
↪ output=output.tif
```



This calls **i.biomass**: fpar=dem.tif — Name of fPAR raster map; lightuse\_efficiency=dem.tif — Name of light use efficiency raster map (UZH:cotton=1.9); latitude=dem.tif — Name of degree latitude raster map [dd.ddd]; dayofyear=dem.tif — Name of Day of Year raster map [1-366]; transmissivity\_singleway=dem.tif — Name of single-way transmissivity raster map [0.0-1.0]; water\_availability=dem.tif — Value of water availability raster map [0.0-1.0]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.cca — i.cca

**group:** Imagery (i.\*)

Canonical components analysis (CCA) program for image processing.

| Parameter                                | Type              | Required | Default | Description                                     |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Input rasters (2 to 8)                          |
| signature                                | file              | yes      | —       | File containing spectral signatures             |
| output                                   | folderDestination | yes      | —       | Output Directory                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:i.cca input="a.gpkg;b.gpkg" signature=input.dat output=output/
```

This calls **i.cca**: input="a.gpkg;b.gpkg" — Input rasters (2 to 8); signature=input.dat — File containing spectral signatures; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.cluster — i.cluster

**group:** Imagery (i.\*)

Generates spectral signatures for land cover types in an image using a clustering algorithm.

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | multilayer      | yes      | —       | Input rasters                                   |
| classes                                  | number          | yes      | —       | Initial number of classes (1-255)               |
| seed                                     | file            | no       | —       | Name of file containing initial signatures      |
| sample                                   | string          | no       | —       | Sampling intervals (by row and col)             |
| iterations                               | number          | no       | 30.0    | Maximum number of iterations                    |
| convergence                              | number          | no       | 98.0    | Percent convergence                             |
| separation                               | number          | no       | 0.0     | Cluster separation                              |
| min_size                                 | number          | no       | 17.0    | Minimum number of pixels in a class             |
| signaturefile                            | fileDestination | yes      | —       | Signature File                                  |
| reportfile                               | fileDestination | no       | —       | Final Report File                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** signaturefile (outputFile), reportfile (outputFile)

### Example usage

```
run grass:i.cluster input="a.gpkg;b.gpkg" classes=10 seed=input.dat sample=value
↳ iterations=30 convergence=98 separation=0 signaturefile=signaturefile.txt
```

This calls **i.cluster**: input="a.gpkg;b.gpkg" — Input rasters; classes=10 — Initial number of classes (1-255); seed=input.dat — Name of file containing initial signatures; sample=value — Sampling intervals (by row and col); iterations=30 — Maximum number of iterations; convergence=98 — Percent convergence; separation=0 — Cluster separation; signaturefile=signaturefile.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.colors.enhance — i.colors.enhance

**group:** Imagery (i.\*)

Performs auto-balancing of colors for RGB images.

| Parameter                                | Type              | Required | Default | Description                                         |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------|
| red                                      | raster            | yes      | —       | Name of red channel                                 |
| green                                    | raster            | yes      | —       | Name of green channel                               |
| blue                                     | raster            | yes      | —       | Name of blue channel                                |
| strength                                 | number            | no       | 98.0    | Cropping intensity (upper brightness level)         |
| -f *(adv)*                               | boolean           | no       | False   | Extend colors to full range of data on each channel |
| -p *(adv)*                               | boolean           | no       | False   | Preserve relative colors, adjust brightness only    |
| -r *(adv)*                               | boolean           | no       | False   | Reset to standard color range                       |
| -s *(adv)*                               | boolean           | no       | False   | Process bands serially (default: run in parallel)   |
| redoutput                                | rasterDestination | yes      | —       | Enhanced Red                                        |
| greenoutput                              | rasterDestination | yes      | —       | Enhanced Green                                      |
| blueoutput                               | rasterDestination | yes      | —       | Enhanced Blue                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                             |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)     |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)           |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)    |

**Outputs:** redoutput (outputRaster), greenoutput (outputRaster), blueoutput (outputRaster)

### Example usage

```
run grass:i.colors.enhance red=dem.tif green=dem.tif blue=dem.tif strength=98
↳ redoutput=redoutput.tif
```

This calls **i.colors.enhance**: red=dem.tif — Name of red channel; green=dem.tif — Name of green channel; blue=dem.tif — Name of blue channel; strength=98 — Cropping intensity (upper brightness level); redoutput=redoutput.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.eb.eta — i.eb.eta

**group:** Imagery (i.\*)

Actual evapotranspiration for diurnal period (Bastiaanssen, 1995).

| Parameter           | Type              | Required | Default | Description                                  |
|---------------------|-------------------|----------|---------|----------------------------------------------|
| netradiationdiurnal | raster            | yes      | —       | Name of the diurnal net radiation map [W/m2] |
| evaporativefraction | raster            | yes      | —       | Name of the evaporative fraction map         |
| temperature         | raster            | yes      | —       | Name of the surface skin temperature [K]     |
| output              | rasterDestination | yes      | —       | Evapotranspiration                           |

| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.eb.eta netradiationdiurnal=dem.tif evaporativefraction=dem.tif
↪ temperature=dem.tif output=output.tif
```

This calls **i.eb.eta**: netradiationdiurnal=dem.tif — Name of the diurnal net radiation map [W/m<sup>2</sup>]; evaporativefraction=dem.tif — Name of the evaporative fraction map; temperature=dem.tif — Name of the surface skin temperature [K]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.eb.evapfr — i.eb.evapfr

**group:** Imagery (i.\*)

Computes evaporative fraction (Bastiaanssen, 1995) and root zone soil moisture (Makin, Molden and Bastiaanssen, 2001).

| Parameter                                | Type              | Required | Default | Description                                               |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------|
| netradiation                             | raster            | yes      | —       | Name of Net Radiation raster map [W/m <sup>2</sup> ]      |
| soilheatflux                             | raster            | yes      | —       | Name of soil heat flux raster map [W/m <sup>2</sup> ]     |
| sensibleheatflux                         | raster            | yes      | —       | Name of sensible heat flux raster map [W/m <sup>2</sup> ] |
| evaporativefraction                      | rasterDestination | yes      | —       | Evaporative Fraction                                      |
| soilmoisture                             | rasterDestination | no       | —       | Root Zone Soil Moisture                                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)           |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                 |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)          |

**Outputs:** evaporativefraction (outputRaster), soilmoisture (outputRaster)

### Example usage

```
run grass:i.eb.evapfr netradiation=dem.tif soilheatflux=dem.tif sensibleheatflux=dem.tif
↪ evaporativefraction=evaporativefraction.tif
```

This calls **i.eb.evapfr**: netradiation=dem.tif — Name of Net Radiation raster map [W/m<sup>2</sup>]; soilheatflux=dem.tif — Name of soil heat flux raster map [W/m<sup>2</sup>]; sensibleheatflux=dem.tif — Name of sensible heat flux raster map [W/m<sup>2</sup>]; evaporativefraction=evaporativefraction.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.eb.hsebal01.coords — i.eb.hsebal01.coords

**group:** Imagery (i.\*)

i.eb.hsebal01.coords - Computes sensible heat flux iteration SEBAL 01. Inline coordinates

| Parameter                                | Type              | Required | Default | Description                                                      |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| netradiation                             | raster            | yes      | —       | Name of instantaneous net radiation raster map [W/m2]            |
| soilheatflux                             | raster            | yes      | —       | Name of instantaneous soil heat flux raster map [W/m2]           |
| aerodynresistance                        | raster            | yes      | —       | Name of aerodynamic resistance to heat momentum raster map [s/m] |
| temperaturemeansealevel                  | raster            | yes      | —       | Name of altitude corrected surface temperature raster map [K]    |
| frictionvelocitystar                     | number            | no       | 0.32407 | Value of the height independent friction velocity (u*) [m/s]     |
| vapourpressureactual                     | raster            | yes      | —       | Name of the actual vapour pressure (e_act) map [KPa]             |
| row_wet_pixel                            | number            | no       | —       | Row value of the wet pixel                                       |
| column_wet_pixel                         | number            | no       | —       | Column value of the wet pixel                                    |
| row_dry_pixel                            | number            | no       | —       | Row value of the dry pixel                                       |
| column_dry_pixel                         | number            | no       | —       | Column value of the dry pixel                                    |
| -a *(adv)*                               | boolean           | no       | False   | Automatic wet/dry pixel (careful!)                               |
| -c *(adv)*                               | boolean           | no       | False   | Dry/Wet pixels coordinates are in image projection, not row/col  |
| output                                   | rasterDestination | yes      | —       | Sensible Heat Flux                                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                 |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.eb.hsebal01.coords netradiation=dem.tif soilheatflux=dem.tif
↪ aerodynresistance=dem.tif temperaturemeansealevel=dem.tif vapourpressureactual=dem.tif
↪ frictionvelocitystar=0.32407 row_wet_pixel=10 output=output.tif
```

This calls **i.eb.hsebal01.coords**: netradiation=dem.tif — Name of instantaneous net radiation raster map [W/m2]; soilheatflux=dem.tif — Name of instantaneous soil heat flux raster map [W/m2]; aerodynresistance=dem.tif — Name of aerodynamic resistance to heat momentum raster map [s/m]; temperaturemeansealevel=dem.tif — Name of altitude corrected surface temperature raster map [K]; vapourpressureactual=dem.tif — Name of the actual vapour pressure (e\_act) map [KPa]; frictionvelocitystar=0.32407 — Value of the height independent friction velocity (u\*) [m/s]; row\_wet\_pixel=10 — Row value of the wet pixel; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.eb.netrad — i.eb.netrad

**group:** Imagery (i.\*)

Net radiation approximation (Bastiaanssen, 1995).

| Parameter               | Type   | Required | Default | Description                                                                         |
|-------------------------|--------|----------|---------|-------------------------------------------------------------------------------------|
| albedo                  | raster | yes      | —       | Name of albedo raster map [0.0;1.0]                                                 |
| ndvi                    | raster | yes      | —       | Name of NDVI raster map [-1.0;+1.0]                                                 |
| temperature             | raster | yes      | —       | Name of surface temperature raster map [K]                                          |
| localutctime            | raster | yes      | —       | Name of time of satellite overpass raster map [local time in UTC]                   |
| temperaturedifference2m | raster | yes      | —       | Name of the difference map of temperature from surface skin to about 2 m height [K] |

| Parameter                               | Type              | Required | Default | Description                                              |
|-----------------------------------------|-------------------|----------|---------|----------------------------------------------------------|
| emissivity                              | raster            | yes      | —       | Name of the emissivity map [-]                           |
| transmissivity_singleway                | raster            | yes      | —       | Name of the single-way atmospheric transmissivitymap [-] |
| dayofyear                               | raster            | yes      | —       | Name of the Day Of Year (DOY) map [-]                    |
| sunzenithangle                          | raster            | yes      | —       | Name of the sun zenith angle map [degrees]               |
| output                                  | rasterDestination | yes      | —       | Net Radiation                                            |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                  |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)          |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)         |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.eb.netrad albedo=dem.tif ndvi=dem.tif temperature=dem.tif localutctime=dem.tif
↳ temperaturedifference2m=dem.tif emissivity=dem.tif transmissivity_singleway=dem.tif
↳ output=output.tif
```

This calls **i.eb.netrad**: albedo=dem.tif — Name of albedo raster map [0.0;1.0]; ndvi=dem.tif — Name of NDVI raster map [-1.0;+1.0]; temperature=dem.tif — Name of surface temperature raster map [K]; localutctime=dem.tif — Name of time of satellite overpass raster map [local time in UTC]; temperaturedifference2m=dem.tif — Name of the difference map of temperature from surface skin to about 2 m height...; emissivity=dem.tif — Name of the emissivity map [-]; transmissivity\_singleway=dem.tif — Name of the single-way atmospheric transmissivitymap [-]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.eb.soilheatflux — i.eb.soilheatflux

**group:** Imagery (i.\*)

Soil heat flux approximation (Bastiaanssen, 1995).

| Parameter                               | Type              | Required | Default | Description                                                       |
|-----------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------|
| albedo                                  | raster            | yes      | —       | Name of albedo raster map [0.0;1.0]                               |
| ndvi                                    | raster            | yes      | —       | Name of NDVI raster map [-1.0;+1.0]                               |
| temperature                             | raster            | yes      | —       | Name of Surface temperature raster map [K]                        |
| netradiation                            | raster            | yes      | —       | Name of Net Radiation raster map [W/m2]                           |
| localutctime                            | raster            | yes      | —       | Name of time of satellite overpass raster map [local time in UTC] |
| -r                                      | boolean           | no       | False   | HAPEX-Sahel empirical correction (Roerink, 1995)                  |
| output                                  | rasterDestination | yes      | —       | Soil Heat Flux                                                    |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                           |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                   |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                         |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                  |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.eb.soilheatflux albedo=dem.tif ndvi=dem.tif temperature=dem.tif
↳ netradiation=dem.tif localutctime=dem.tif output=output.tif
```

This calls **i.eb.soilheatflux**: albedo=dem.tif — Name of albedo raster map [0.0;1.0]; ndvi=dem.tif — Name of NDVI raster map [-1.0;+1.0]; temperature=dem.tif — Name of Surface temperature raster map [K]; netradiation=dem.tif — Name of Net Radiation raster map [W/m2]; localutctime=dem.tif — Name of time of satellite overpass raster map [local time in UTC]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.emissivity — i.emissivity

**group:** Imagery (i.\*)

Computes emissivity from NDVI, generic method for sparse land.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of NDVI raster map [-]                      |
| output                                   | rasterDestination | yes      | —       | Emissivity                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.emissivity input=dem.tif output=output.tif
```

This calls **i.emissivity**: input=dem.tif — Name of NDVI raster map [-]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.evapo.mh — i.evapo.mh

**group:** Imagery (i.\*)

Computes evapotranspiration calculation modified or original Hargreaves formulation, 2001.

| Parameter                                | Type              | Required | Default | Description                                             |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------|
| netradiation_diurnal                     | raster            | yes      | —       | Name of input diurnal net radiation raster map [W/m2/d] |
| average_temperature                      | raster            | yes      | —       | Name of input average air temperature raster map [C]    |
| minimum_temperature                      | raster            | yes      | —       | Name of input minimum air temperature raster map [C]    |
| maximum_temperature                      | raster            | yes      | —       | Name of input maximum air temperature raster map [C]    |
| precipitation                            | raster            | no       | —       | Name of precipitation raster map [mm/month]             |
| -z *(adv)*                               | boolean           | no       | False   | Set negative ETa to zero                                |
| -h *(adv)*                               | boolean           | no       | False   | Use original Hargreaves (1985)                          |
| -s *(adv)*                               | boolean           | no       | False   | Use Hargreaves-Samani (1985)                            |
| output                                   | rasterDestination | yes      | —       | Evapotranspiration                                      |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)         |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)               |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)        |

---

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.evapo.mh netradiation_diurnal=dem.tif average_temperature=dem.tif
↪ minimum_temperature=dem.tif maximum_temperature=dem.tif precipitation=dem.tif
↪ output=output.tif
```

This calls **i.evapo.mh**: netradiation\_diurnal=dem.tif — Name of input diurnal net radiation raster map [W/m2/d]; average\_temperature=dem.tif — Name of input average air temperature raster map [C]; minimum\_temperature=dem.tif — Name of input minimum air temperature raster map [C]; maximum\_temperature=dem.tif — Name of input maximum air temperature raster map [C]; precipitation=dem.tif — Name of precipitation raster map [mm/month]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.evapo.pm — i.evapo.pm

**group:** Imagery (i.\*)

Computes potential evapotranspiration calculation with hourly Penman-Monteith.

| Parameter                               | Type              | Required | Default | Description                                            |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------------|
| elevation                               | raster            | yes      | —       | Name of input elevation raster map [m a.s.l.]          |
| temperature                             | raster            | yes      | —       | Name of input temperature raster map [C]               |
| relativehumidity                        | raster            | yes      | —       | Name of input relative humidity raster map [%]         |
| windspeed                               | raster            | yes      | —       | Name of input wind speed raster map [m/s]              |
| netradiation                            | raster            | yes      | —       | Name of input net solar radiation raster map [MJ/m2/h] |
| cropheight                              | raster            | yes      | —       | Name of input crop height raster map [m]               |
| -z *(adv)*                              | boolean           | no       | False   | Set negative ETa to zero                               |
| -n *(adv)*                              | boolean           | no       | False   | Use Night-time                                         |
| output                                  | rasterDestination | yes      | —       | Evapotranspiration                                     |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)        |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)              |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)       |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.evapo.pm elevation=dem.tif temperature=dem.tif relativehumidity=dem.tif
↪ windspeed=dem.tif netradiation=dem.tif cropheight=dem.tif output=output.tif
```

This calls **i.evapo.pm**: elevation=dem.tif — Name of input elevation raster map [m a.s.l.]; temperature=dem.tif — Name of input temperature raster map [C]; relativehumidity=dem.tif — Name of input relative humidity raster map [%]; windspeed=dem.tif — Name of input wind speed raster map [m/s]; netradiation=dem.tif — Name of input net solar radiation raster map [MJ/m2/h]; cropheight=dem.tif — Name of input crop height raster map [m]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.evapo.pt — i.evapo.pt

**group:** Imagery (i.\*)

Computes evapotranspiration calculation Priestley and Taylor formulation, 1972.

| Parameter                                | Type              | Required | Default | Description                                               |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------|
| net_radiation                            | raster            | yes      | —       | Name of input net radiation raster map [W/m2]             |
| soil_heatflux                            | raster            | yes      | —       | Name of input soil heat flux raster map [W/m2]            |
| air_temperature                          | raster            | yes      | —       | Name of input air temperature raster map [K]              |
| atmospheric_pressure                     | raster            | yes      | —       | Name of input atmospheric pressure raster map [millibars] |
| priestley_taylor_coeff                   | number            | no       | 1.26    | Priestley-Taylor coefficient                              |
| -z *(adv)*                               | boolean           | no       | False   | Set negative ETa to zero                                  |
| output                                   | rasterDestination | yes      | —       | Evapotranspiration                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)           |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                 |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)          |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.evapo.pt net_radiation=dem.tif soil_heatflux=dem.tif air_temperature=dem.tif  
↪ atmospheric_pressure=dem.tif priestley_taylor_coeff=1.26 output=output.tif
```

This calls **i.evapo.pt**: net\_radiation=dem.tif — Name of input net radiation raster map [W/m2]; soil\_heatflux=dem.tif — Name of input soil heat flux raster map [W/m2]; air\_temperature=dem.tif — Name of input air temperature raster map [K]; atmospheric\_pressure=dem.tif — Name of input atmospheric pressure raster map [millibars]; priestley\_taylor\_coeff=1.26 — Priestley-Taylor coefficient; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.evapo.time — i.evapo.time

**group:** Imagery (i.\*)

Computes temporal integration of satellite ET actual (ETa) following the daily ET reference (ETo) from meteorological station(s).

| Parameter                      | Type              | Required | Default | Description                                                            |
|--------------------------------|-------------------|----------|---------|------------------------------------------------------------------------|
| eta                            | multilayer        | yes      | —       | Names of satellite ETa raster maps [mm/d or cm/d]                      |
| eta_doy                        | multilayer        | yes      | —       | Names of satellite ETa Day of Year (DOY) raster maps [0-400] [-]       |
| eto                            | multilayer        | yes      | —       | Names of meteorological station ETo raster maps [0-400] [mm/d or cm/d] |
| eto_doy_min                    | number            | no       | 1.0     | Value of DOY for ETo first day                                         |
| start_period                   | number            | no       | 1.0     | Value of DOY for the first day of the period studied                   |
| end_period                     | number            | no       | 1.0     | Value of DOY for the last day of the period studied                    |
| output                         | rasterDestination | yes      | —       | Temporal integration                                                   |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                                                |



| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.evapo.time eta="a.gpkg;b.gpkg" eta_doy="a.gpkg;b.gpkg" eto="a.gpkg;b.gpkg"
↳ eto_doy_min=1 start_period=1 end_period=1 output=output.tif
```

This calls **i.evapo.time**: eta="a.gpkg;b.gpkg" — Names of satellite ETa raster maps [mm/d or cm/d]; eta\_doy="a.gpkg;b.gpkg" — Names of satellite ETa Day of Year (DOY) raster maps [0-400] [-]; eto="a.gpkg;b.gpkg" — Names of meteorological station ETo raster maps [0-400] [mm/d or cm/d]; eto\_doy\_min=1 — Value of DOY for ETo first day; start\_period=1 — Value of DOY for the first day of the period studied; end\_period=1 — Value of DOY for the last day of the period studied; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.fft — i.fft

**group:** Imagery (i.\*)

Fast Fourier Transform (FFT) for image processing.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                         |
| real                                     | rasterDestination | yes      | —       | Real part arrays                                 |
| imaginary                                | rasterDestination | yes      | —       | Imaginary part arrays                            |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** real (outputRaster), imaginary (outputRaster)

### Example usage

```
run grass:i.fft input=dem.tif real=real.tif
```

This calls **i.fft**: input=dem.tif — Name of input raster map; real=real.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.gensig — i.gensig

**group:** Imagery (i.\*)

Generates statistics for i.maxlik from raster map.

| Parameter     | Type            | Required | Default | Description               |
|---------------|-----------------|----------|---------|---------------------------|
| trainingmap   | raster          | yes      | —       | Ground truth training map |
| input         | multilayer      | yes      | —       | Input rasters             |
| signaturefile | fileDestination | yes      | —       | Signature File            |

| Parameter                                | Type   | Required | Default | Description                                     |
|------------------------------------------|--------|----------|---------|-------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** signaturefile (outputFile)

### Example usage

```
run grass:i.gensig trainingmap=dem.tif input="a.gpkg;b.gpkg"
↪ signaturefile=signaturefile.txt
```

This calls **i.gensig**: trainingmap=dem.tif — Ground truth training map; input="a.gpkg;b.gpkg" — Input rasters; signaturefile=signaturefile.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.gensigset — i.gensigset

**group:** Imagery (i.\*)

Generates statistics for i.smap from raster map.

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| trainingmap                              | raster          | yes      | —       | Ground truth training map                       |
| input                                    | multilayer      | yes      | —       | Input rasters                                   |
| maxsig                                   | number          | no       | 5.0     | Maximum number of sub-signatures in any class   |
| signaturefile                            | fileDestination | yes      | —       | Signature File                                  |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** signaturefile (outputFile)

### Example usage

```
run grass:i.gensigset trainingmap=dem.tif input="a.gpkg;b.gpkg" maxsig=5
↪ signaturefile=signaturefile.txt
```

This calls **i.gensigset**: trainingmap=dem.tif — Ground truth training map; input="a.gpkg;b.gpkg" — Input rasters; maxsig=5 — Maximum number of sub-signatures in any class; signaturefile=signaturefile.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.group — i.group

**group:** Imagery (i.\*)

Regroup multiple mono-band rasters into a single multiband raster.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Input rasters                                    |
| group                                    | rasterDestination | yes      | —       | Multiband raster                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

---

**Outputs:** group (outputRaster)

**Example usage**

```
run grass:i.group input="a.gpkg;b.gpkg" group=group.tif
```

This calls **i.group**: input="a.gpkg;b.gpkg" — Input rasters; group=group.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.his.rgb — i.his.rgb

**group:** Imagery (i.\*)

Transforms raster maps from HIS (Hue-Intensity-Saturation) color space to RGB (Red-Green-Blue) color space.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| hue                                     | raster            | yes      | —       | Name of input raster map (hue)                   |
| intensity                               | raster            | yes      | —       | Name of input raster map (intensity)             |
| saturation                              | raster            | yes      | —       | Name of input raster map (saturation)            |
| red                                     | rasterDestination | yes      | —       | Red                                              |
| green                                   | rasterDestination | yes      | —       | Green                                            |
| blue                                    | rasterDestination | yes      | —       | Blue                                             |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** red (outputRaster), green (outputRaster), blue (outputRaster)

**Example usage**

```
run grass:i.his.rgb hue=dem.tif intensity=dem.tif saturation=dem.tif red=red.tif
```

This calls **i.his.rgb**: hue=dem.tif — Name of input raster map (hue); intensity=dem.tif — Name of input raster map (intensity); saturation=dem.tif — Name of input raster map (saturation); red=red.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.ifft — i.ifft

**group:** Imagery (i.\*)

Inverse Fast Fourier Transform (IFFT) for image processing.

| Parameter                               | Type              | Required | Default | Description                                          |
|-----------------------------------------|-------------------|----------|---------|------------------------------------------------------|
| real                                    | raster            | yes      | —       | Name of input raster map (image fft, real part)      |
| imaginary                               | raster            | yes      | —       | Name of input raster map (image fft, imaginary part) |
| output                                  | rasterDestination | yes      | —       | Inverse Fast Fourier Transform                       |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                              |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)      |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)            |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.ifft real=dem.tif imaginary=dem.tif output=output.tif
```

This calls **i.ifft**: real=dem.tif — Name of input raster map (image fft, real part); imaginary=dem.tif — Name of input raster map (image fft, imaginary part); output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.image.mosaic — i.image.mosaic

**group:** Imagery (i.\*)

Mosaics several images and extends colormap.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Input rasters                                    |
| output                                   | rasterDestination | yes      | —       | Mosaic Raster                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.image.mosaic input="a.gpkg;b.gpkg" output=output.tif
```

This calls **i.image.mosaic**: input="a.gpkg;b.gpkg" — Input rasters; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.in.spotvgt — i.in.spotvgt

**group:** Imagery (i.\*)

Imports SPOT VGT NDVI data into a raster map.

| Parameter                                | Type              | Required | Default | Description                                                       |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------|
| input                                    | file              | yes      | —       | Name of input SPOT VGT NDVI HDF file                              |
| -a *(adv)*                               | boolean           | no       | False   | Also import quality map (SM status map layer) and filter NDVI map |
| output                                   | rasterDestination | yes      | —       | SPOT NDVI Raster                                                  |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                           |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                   |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                         |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                  |

---

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:i.in.spotvgt input=input.dat output=output.tif
```

This calls **i.in.spotvgt**: input=input.dat — Name of input SPOT VGT NDVI HDF file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.landsat.acca — i.landsat.acca

**group:** Imagery (i.\*)

Performs Landsat TM/ETM+ Automatic Cloud Cover Assessment (ACCA).

| Parameter                                | Type              | Required | Default | Description                                                                   |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------------------|
| rasters                                  | multilayer        | yes      | —       | Landsat input rasters                                                         |
| b56composite                             | number            | no       | 225.0   | B56composite (step 6)                                                         |
| b45ratio                                 | number            | no       | 1.0     | B45ratio: Desert detection (step 10)                                          |
| histogram                                | number            | no       | 100.0   | Number of classes in the cloud temperature histogram                          |
| -5 *(adv)*                               | boolean           | no       | False   | Data is Landsat-5 TM                                                          |
| -f *(adv)*                               | boolean           | no       | False   | Apply post-processing filter to remove small holes                            |
| -x *(adv)*                               | boolean           | no       | False   | Always use cloud signature (step 14)                                          |
| -2 *(adv)*                               | boolean           | no       | False   | Bypass second-pass processing, and merge warm (not ambiguous) and cold clouds |
| -s *(adv)*                               | boolean           | no       | False   | Include a category for cloud shadows                                          |
| output                                   | rasterDestination | yes      | —       | ACCA Raster                                                                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                       |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                               |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                     |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                              |

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:i.landsat.acca rasters="a.gpkg;b.gpkg" b56composite=225 b45ratio=1 histogram=100  
↪ output=output.tif
```

This calls **i.landsat.acca**: rasters="a.gpkg;b.gpkg" — Landsat input rasters; b56composite=225 — B56composite (step 6); b45ratio=1 — B45ratio: Desert detection (step 10); histogram=100 — Number of classes in the cloud temperature histogram; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.landsat.toar — i.landsat.toar

**group:** Imagery (i.\*)

Calculates top-of-atmosphere radiance or reflectance and temperature for Landsat MSS/TM/ETM+/OLI

| Parameter                               | Type              | Required | Default | Description                                                                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------|
| rasters                                 | multilayer        | yes      | —       | Landsat input rasters                                                                            |
| metfile                                 | file              | no       | —       | Name of Landsat metadata file (.met or MTL.txt)                                                  |
| sensor                                  | enum              | no       | 7       | Spacecraft sensor — options: 0=mss1, 1=mss2, 2=mss3, 3=mss4, 4=mss5, 5=tm4, 6=tm5, 7=tm7, 8=oli8 |
| method                                  | enum              | no       | 0       | Atmospheric correction method — options: 0=uncorrected, 1=dos1, 2=dos2, 3=dos2b, 4=dos3, 5=dos4  |
| date                                    | string            | no       | —       | Image acquisition date (yyyy-mm-dd)                                                              |
| sun_elevation                           | number            | no       | —       | Sun elevation in degrees                                                                         |
| product_date                            | string            | no       | —       | Image creation date (yyyy-mm-dd)                                                                 |
| gain                                    | string            | no       | —       | Gain (H/L) of all Landsat ETM+ bands (1-5,61,62,7,8)                                             |
| percent                                 | number            | no       | 0.01    | Percent of solar radiance in path radiance                                                       |
| pixel                                   | number            | no       | 1000.0  | Minimum pixels to consider digital number as dark object                                         |
| rayleigh                                | number            | no       | 0.0     | Rayleigh atmosphere (diffuse sky irradiance)                                                     |
| scale                                   | number            | no       | 1.0     | Scale factor for output                                                                          |
| -r *(adv)*                              | boolean           | no       | False   | Output at-sensor radiance instead of reflectance for all bands                                   |
| -n *(adv)*                              | boolean           | no       | False   | Input raster maps use as extension the number of the band instead the code                       |
| output                                  | folderDestination | yes      | —       | Output Directory                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                  |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:i.landsat.toar rasters="a.gpkg;b.gpkg" metfile=input.dat sensor=7 method=0
↳ date=value sun_elevation=10 product_date=value output=output/
```

This calls **i.landsat.toar**: rasters="a.gpkg;b.gpkg" — Landsat input rasters; metfile=input.dat — Name of Landsat metadata file (.met or MTL.txt); sensor=7 selects *tm7*; method=0 selects *uncorrected*; date=value — Image acquisition date (yyyy-mm-dd); sun\_elevation=10 — Sun elevation in degrees; product\_date=value — Image creation date (yyyy-mm-dd); output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.maxlik — i.maxlik

**group:** Imagery (i.\*)

Classifies the cell spectral reflectances in imagery data.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | multilayer        | yes      | —       | Input rasters                                    |
| signaturefile                           | file              | yes      | —       | Name of input file containing signatures         |
| output                                  | rasterDestination | yes      | —       | Classification                                   |
| reject                                  | rasterDestination | no       | —       | Reject Threshold                                 |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster), reject (outputRaster)

### Example usage

```
run grass:i.maxlik input="a.gpkg;b.gpkg" signaturefile=input.dat output=output.tif
```

This calls **i.maxlik**: input="a.gpkg;b.gpkg" — Input rasters; signaturefile=input.dat — Name of input file containing signatures; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.modis.qc — i.modis.qc

**group:** Imagery (i.\*)

Extracts quality control parameters from MODIS QC layers.

| Parameter                                | Type              | Required | Default | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input surface reflectance QC layer [bit array]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| productname                              | enum              | no       | 8       | Name of MODIS product type — options: 0=mod09Q1, 1=mod09A1, 2=mod09A1s, 3=mod09CMG, 4=mod09CMGs, 5=mod09CMGi, 6=mod11A1, 7=mod11A2, 8=mod13A2, 9=mcd43B2, 10=mcd43B2q, 11=mod13Q1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| qcname                                   | enum              | no       | 5       | Name of QC type to extract — options: 0=adjcorr, 1=atcorr, 2=cloud, 3=data_quality, 4=diff_orbit_from_500m, 5=modland_qa, 6=mandatory_qa_11A1, 7=data_quality_flag_11A1, 8=emis_error_11A1, 9=lst_error_11A1, 10=data_quality_flag_11A2, 11=emis_error_11A2, 12=mandatory_qa_11A2, 13=lst_error_11A2, 14=aerosol_quantity, 15=brdf_correction_performed, 16=cirrus_detected, 17=cloud_shadow, 18=cloud_state, 19=internal_cloud_algorithm, 20=internal_fire_algorithm, 21=internal_snow_mask, 22=land_water, 23=mod35_snow_ice, 24=pixel_adjacent_to_cloud, 25=icm_cloudy, 26=icm_clear, 27=icm_high_clouds, 28=icm_low_clouds, 29=icm_snow, 30=icm_fire, 31=icm_sun_glint, 32=icm_dust, 33=icm_cloud_shadow, 34=icm_pixel_is_adjacent_to_cloud, 35=icm_cirrus, 36=icm_pan_flag, 37=icm_criteria_for_aerosol_retrieval, 38=icm_aot_has_clim_val, 39=modland_qa, 40=vi_usefulness, 41=aerosol_quantity, 42=pixel_adjacent_to_cloud, 43=brdf_correction_performed, 44=mixed_clouds, 45=land_water, 46=possible_snow_ice, 47=possible_shadow, 48=platform, 49=land_water, 50=sun_z_angle_at_local_noon, 51=brdf_correction_performed |
| band                                     | enum              | no       | [0, 1]  | Band number of MODIS product (mod09Q1=[1,2], mod09A1=[1-7], m[o/y]d09CMG=[1-7], mcd43B2q=[1-7]) — options: 0=1, 1=2, 2=3, 3=4, 4=5, 5=6, 6=7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| output                                   | rasterDestination | yes      | —       | QC Classification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_OPT *(adv)*  | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.modis.qc input=dem.tif productname=8 qcname=5 band=0 output=output.tif
```

This calls **i.modis.qc**: input=dem.tif — Name of input surface reflectance QC layer [bit array]; productname=8 selects *mod13A2*; qcname=5 selects *modland\_qa*; band=0 selects 1; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.oif — i.oif

**group:** Imagery (i.\*)

Calculates Optimum-Index-Factor table for spectral bands

| Parameter                                | Type            | Required | Default | Description                                       |
|------------------------------------------|-----------------|----------|---------|---------------------------------------------------|
| input                                    | multilayer      | yes      | —       | Name of input raster map(s)                       |
| -g *(adv)*                               | boolean         | no       | False   | Print in shell script style                       |
| -s *(adv)*                               | boolean         | no       | False   | Process bands serially (default: run in parallel) |
| output                                   | fileDestination | yes      | —       | OIF File                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                           |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)   |

**Outputs:** output (outputFile)

### Example usage

```
run grass:i.oif input="a.gpkg;b.gpkg" output=output.txt
```

This calls **i.oif**: input="a.gpkg;b.gpkg" — Name of input raster map(s); output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.pansharpen — i.pansharpen

**group:** Imagery (i.\*)

Image fusion algorithms to sharpen multispectral with high-res panchromatic channels

| Parameter   | Type              | Required | Default | Description                                                            |
|-------------|-------------------|----------|---------|------------------------------------------------------------------------|
| red         | raster            | yes      | —       | Name of red channel                                                    |
| green       | raster            | yes      | —       | Name of green channel                                                  |
| blue        | raster            | yes      | —       | Name of blue channel                                                   |
| pan         | raster            | yes      | —       | Name of raster map to be used for high resolution panchromatic channel |
| method      | enum              | no       | 1       | Method — options: 0=brovey, 1=ihs, 2=pca                               |
| -l *(adv)*  | boolean           | no       | False   | Rebalance blue channel for LANDSAT                                     |
| -s *(adv)*  | boolean           | no       | False   | Process bands serially (default: run in parallel)                      |
| redoutput   | rasterDestination | yes      | —       | Enhanced Red                                                           |
| greenoutput | rasterDestination | yes      | —       | Enhanced Green                                                         |



| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| blueoutput                               | rasterDestination | yes      | —       | Enhanced Blue                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** redoutput (outputRaster), greenoutput (outputRaster), blueoutput (outputRaster)

### Example usage

```
run grass:i.pansharpen red=dem.tif green=dem.tif blue=dem.tif pan=dem.tif method=1
↪ redoutput=redoutput.tif
```

This calls **i.pansharpen**: red=dem.tif — Name of red channel; green=dem.tif — Name of green channel; blue=dem.tif — Name of blue channel; pan=dem.tif — Name of raster map to be used for high resolution panchromatic channel; method=1 selects *ihs*; redoutput=redoutput.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.pca — i.pca

**group:** Imagery (i.\*)

Principal components analysis (PCA) for image processing.

| Parameter                                | Type              | Required | Default | Description                                               |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Name of two or more input raster maps                     |
| rescale                                  | range             | no       | 0,255   | Rescaling range for output maps. For no rescaling use 0,0 |
| percent                                  | number            | no       | 99.0    | Cumulative percent importance for filtering               |
| -n *(adv)*                               | boolean           | no       | False   | Normalize (center and scale) input maps                   |
| -f *(adv)*                               | boolean           | no       | False   | Output will be filtered input bands                       |
| output                                   | folderDestination | yes      | —       | Output Directory                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)           |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:i.pca input="a.gpkg;b.gpkg" rescale=0,255 percent=99 output=output/
```

This calls **i.pca**: input="a.gpkg;b.gpkg" — Name of two or more input raster maps; rescale=0,255 — Rescaling range for output maps. For no rescaling use 0,0; percent=99 — Cumulative percent importance for filtering; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.rgb.his — i.rgb.his

**group:** Imagery (i.\*)

Transforms raster maps from RGB (Red-Green-Blue) color space to HIS (Hue-Intensity-Saturation) color space.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| red                                     | raster            | no       | —       | Name for input raster map (red)                  |
| green                                   | raster            | no       | —       | Name for input raster map (green)                |
| blue                                    | raster            | no       | —       | Name for input raster map (blue)                 |
| hue                                     | rasterDestination | no       | False   | Hue                                              |
| intensity                               | rasterDestination | no       | False   | Intensity                                        |
| saturation                              | rasterDestination | no       | False   | Saturation                                       |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** hue (outputRaster), intensity (outputRaster), saturation (outputRaster)

### Example usage

```
run grass:i.rgb.his red=dem.tif green=dem.tif blue=dem.tif hue=hue.tif
```

This calls **i.rgb.his**: red=dem.tif — Name for input raster map (red); green=dem.tif — Name for input raster map (green); blue=dem.tif — Name for input raster map (blue); hue=hue.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.segment — i.segment

**group:** Imagery (i.\*)

Identifies segments (objects) from imagery data.

| Parameter                               | Type              | Required | Default | Description                                                                          |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------|
| input                                   | multilayer        | yes      | —       | Input rasters                                                                        |
| threshold                               | number            | no       | 0.5     | Difference threshold between 0 and 1                                                 |
| method                                  | enum              | no       | 0       | Segmentation method — options: 0=region_growing                                      |
| similarity                              | enum              | no       | 0       | Similarity calculation method — options: 0=euclidean, 1=manhattan                    |
| minsize                                 | number            | no       | 1.0     | Minimum number of cells in a segment                                                 |
| memory                                  | number            | no       | 300.0   | Amount of memory to use in MB                                                        |
| iterations                              | number            | no       | 20.0    | Maximum number of iterations                                                         |
| seeds                                   | raster            | no       | —       | Name for input raster map with starting seeds                                        |
| bounds                                  | raster            | no       | —       | Name of input bounding/constraining raster map                                       |
| -d *(adv)*                              | boolean           | no       | False   | Use 8 neighbors (3x3 neighborhood) instead of the default 4 neighbors for each pixel |
| -w *(adv)*                              | boolean           | no       | False   | Weighted input, do not perform the default scaling of input raster maps              |
| output                                  | rasterDestination | yes      | —       | Segmented Raster                                                                     |
| goodness                                | rasterDestination | no       | —       | Goodness Raster                                                                      |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                              |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                      |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                            |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                     |

**Outputs:** output (outputRaster), goodness (outputRaster)

### Example usage

```
run grass:i.segment input="a.gpkg;b.gpkg" threshold=0.5 method=0 similarity=0 minsize=1
↳ memory=300 iterations=20 output=output.tif
```

This calls **i.segment**: input="a.gpkg;b.gpkg" — Input rasters; threshold=0.5 — Difference threshold between 0 and 1; method=0 selects *region\_growing*; similarity=0 selects *euclidean*; minsize=1 — Minimum number of cells in a segment; memory=300 — Amount of memory to use in MB; iterations=20 — Maximum number of iterations; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.smap — i.smap

**group:** Imagery (i.\*)

Performs contextual image classification using sequential maximum a posteriori (SMAP) estimation.

| Parameter                               | Type              | Required | Default | Description                                         |
|-----------------------------------------|-------------------|----------|---------|-----------------------------------------------------|
| input                                   | multilayer        | yes      | —       | Input rasters                                       |
| signaturefile                           | file              | yes      | —       | Name of input file containing signatures            |
| blocksize                               | number            | no       | 1024.0  | Size of submatrix to process at one time            |
| -m *(adv)*                              | boolean           | no       | False   | Use maximum likelihood estimation (instead of smap) |
| output                                  | rasterDestination | yes      | —       | Classification                                      |
| goodness                                | rasterDestination | no       | —       | Goodness_of_fit                                     |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                             |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)     |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)           |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)    |

**Outputs:** output (outputRaster), goodness (outputRaster)

### Example usage

```
run grass:i.smap input="a.gpkg;b.gpkg" signaturefile=input.dat blocksize=1024
↳ output=output.tif
```

This calls **i.smap**: input="a.gpkg;b.gpkg" — Input rasters; signaturefile=input.dat — Name of input file containing signatures; blocksize=1024 — Size of submatrix to process at one time; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.tasscap — i.tasscap

**group:** Imagery (i.\*)

Performs Tasseled Cap (Kauth Thomas) transformation.

| Parameter                      | Type              | Required | Default | Description                                                                                           |
|--------------------------------|-------------------|----------|---------|-------------------------------------------------------------------------------------------------------|
| input                          | multilayer        | yes      | —       | Input rasters. Landsat4-7: bands 1,2,3,4,5,7; Landsat8: bands 2,3,4,5,6,7; MODIS: bands 1,2,3,4,5,6,7 |
| sensor                         | enum              | no       | 0       | Satellite sensor — options: 0=landsat4_tm, 1=landsat5_tm, 2=landsat7_etm, 3=landsat8_oli, 4=modis     |
| output                         | folderDestination | yes      | —       | Output Directory                                                                                      |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                                                                               |

| Parameter                                | Type   | Required | Default | Description                                     |
|------------------------------------------|--------|----------|---------|-------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:i.tasscap input="a.gpkg;b.gpkg" sensor=0 output=output/
```

This calls **i.tasscap**: input="a.gpkg;b.gpkg" — Input rasters. Landsat4-7: bands 1,2,3,4,5,7; Landsat8: bands 2,3,4,5,6,7,...; sensor=0 selects *landsat4\_tm*; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.topo.coor.ill — i.topo.coor.ill

**group:** Imagery (i.\*)

i.topo.coor.ill - Creates illumination model for topographic correction of reflectance.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| basemap                                  | raster            | yes      | —       | Name of elevation raster map                     |
| zenith                                   | number            | no       | 0.0     | Solar zenith in degrees                          |
| azimuth                                  | number            | no       | 0.0     | Solar azimuth in degrees                         |
| output                                   | rasterDestination | yes      | —       | Illumination Model                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.topo.coor.ill basemap=dem.tif zenith=0 azimuth=0 output=output.tif
```

This calls **i.topo.coor.ill**: basemap=dem.tif — Name of elevation raster map; zenith=0 — Solar zenith in degrees; azimuth=0 — Solar azimuth in degrees; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.topo.corr — i.topo.corr

**group:** Imagery (i.\*)

Computes topographic correction of reflectance.

| Parameter  | Type              | Required | Default | Description                                                                          |
|------------|-------------------|----------|---------|--------------------------------------------------------------------------------------|
| input      | multilayer        | yes      | —       | Name of reflectance raster maps to be corrected topographically                      |
| basemap    | raster            | yes      | —       | Name of illumination input base raster map                                           |
| zenith     | number            | no       | 0.0     | Solar zenith in degrees                                                              |
| method     | enum              | no       | 0       | Topographic correction method — options: 0=cosine, 1=minnaert, 2=c-factor, 3=percent |
| -s *(adv)* | boolean           | no       | False   | Scale output to input and copy color rules                                           |
| output     | folderDestination | yes      | —       | Output Directory                                                                     |

| Parameter                                | Type   | Required | Default | Description                                     |
|------------------------------------------|--------|----------|---------|-------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:i.topo.corr input="a.gpkg;b.gpkg" basemap=dem.tif zenith=0 method=0
↪ output=output/
```

This calls **i.topo.corr**: input="a.gpkg;b.gpkg" — Name of reflectance raster maps to be corrected topographically; basemap=dem.tif — Name of illumination input base raster map; zenith=0 — Solar zenith in degrees; method=0 selects *cosine*; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.vi — i.vi

**group:** Imagery (i.\*)

Calculates different types of vegetation indices.

| Parameter                                | Type              | Required | Default | Description                                                                                                                                                           |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| red                                      | raster            | no       | —       | Name of input red channel surface reflectance map [0.0-1.0]                                                                                                           |
| viname                                   | enum              | no       | 10      | Type of vegetation index — options: 0=arvi, 1=dvi, 2=evi, 3=evi2, 4=gvi, 5=gari, 6=gemi, 7=ipvi, 8=msavi, 9=msavi2, 10=ndvi, 11=pvi, 12=savi, 13=sr, 14=vari, 15=wdvi |
| nir                                      | raster            | no       | —       | Name of input nir channel surface reflectance map [0.0-1.0]                                                                                                           |
| green                                    | raster            | no       | —       | Name of input green channel surface reflectance map [0.0-1.0]                                                                                                         |
| blue                                     | raster            | no       | —       | Name of input blue channel surface reflectance map [0.0-1.0]                                                                                                          |
| band5                                    | raster            | no       | —       | Name of input 5th channel surface reflectance map [0.0-1.0]                                                                                                           |
| band7                                    | raster            | no       | —       | Name of input 7th channel surface reflectance map [0.0-1.0]                                                                                                           |
| soil_line_slope                          | number            | no       | —       | Value of the slope of the soil line (MSAVI2 only)                                                                                                                     |
| soil_line_intercept                      | number            | no       | —       | Value of the factor of reduction of soil noise (MSAVI2 only)                                                                                                          |
| soil_noise_reduction                     | number            | no       | —       | Value of the slope of the soil line (MSAVI2 only)                                                                                                                     |
| storage_bit                              | enum              | no       | 1       | Maximum bits for digital numbers — options: 0=7, 1=8, 2=9, 3=10, 4=16                                                                                                 |
| output                                   | rasterDestination | yes      | —       | Vegetation Index                                                                                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                                               |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                                       |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                                                                             |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                                                      |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.vi red=dem.tif vname=10 nir=dem.tif green=dem.tif blue=dem.tif band5=dem.tif
↳ band7=dem.tif output=output.tif
```

This calls **i.vi**: red=dem.tif — Name of input red channel surface reflectance map [0.0-1.0]; vname=10 selects *ndvi*; nir=dem.tif — Name of input nir channel surface reflectance map [0.0-1.0]; green=dem.tif — Name of input green channel surface reflectance map [0.0-1.0]; blue=dem.tif — Name of input blue channel surface reflectance map [0.0-1.0]; band5=dem.tif — Name of input 5th channel surface reflectance map [0.0-1.0]; band7=dem.tif — Name of input 7th channel surface reflectance map [0.0-1.0]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:i.zc — i.zc

**group:** Imagery (i.\*)

Zero-crossing “edge detection” raster function for image processing.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                         |
| width                                    | number            | no       | 9.0     | x-y extent of the Gaussian filter                |
| threshold                                | number            | no       | 10.0    | Sensitivity of Gaussian filter                   |
| orientations                             | number            | no       | 1.0     | Number of azimuth directions categorized         |
| output                                   | rasterDestination | yes      | —       | Zero crossing                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:i.zc input=dem.tif width=9 threshold=10 orientations=1 output=output.tif
```

This calls **i.zc**: input=dem.tif — Name of input raster map; width=9 — x-y extent of the Gaussian filter; threshold=10 — Sensitivity of Gaussian filter; orientations=1 — Number of azimuth directions categorized; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:m.cogo — m.cogo

**group:** Miscellaneous (m.\*)

A simple utility for converting bearing and distance measurements to coordinates and vice versa. It assumes a Cartesian coordinate system

| Parameter                      | Type            | Required | Default | Description                                               |
|--------------------------------|-----------------|----------|---------|-----------------------------------------------------------|
| input                          | file            | yes      | —       | Name of input file                                        |
| output                         | fileDestination | yes      | —       | Output text file                                          |
| coordinates                    | point           | no       | 0.0,0.0 | Starting coordinate pair                                  |
| -l *(adv)*                     | boolean         | no       | False   | Lines are labelled                                        |
| -q *(adv)*                     | boolean         | no       | False   | Suppress warnings                                         |
| -r *(adv)*                     | boolean         | no       | False   | Convert from coordinates to bearing and distance          |
| -c *(adv)*                     | boolean         | no       | False   | Repeat the starting coordinate at the end to close a loop |
| GRASS_REGION_PARAMETER *(adv)* | extent          | no       | —       | GRASS GIS region extent                                   |

**Outputs:** output (outputFile)

### Example usage

```
run grass:m.cogo input=input.dat coordinates=100,100 output=output.txt
```

This calls **m.cogo**: input=input.dat — Name of input file; coordinates=100,100 — Starting coordinate pair; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:nviz — nviz

**group:** Visualization(NVIZ)

Visualization and animation tool for GRASS data.

| Parameter                               | Type       | Required | Default | Description                                     |
|-----------------------------------------|------------|----------|---------|-------------------------------------------------|
| elevation                               | multilayer | yes      | —       | Name of elevation raster map                    |
| color                                   | multilayer | yes      | —       | Name of raster map(s) for Color                 |
| vector                                  | multilayer | yes      | —       | Name of vector lines/areas overlay map(s)       |
| point                                   | multilayer | no       | —       | Name of vector points overlay file(s)           |
| volume                                  | multilayer | no       | —       | Name of existing 3d raster map                  |
| GRASS_REGION_PARAMETER *(adv)*          | extent     | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number     | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*  | number     | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)          |
| GRASS_MIN_AREA_PARAMETER *(adv)*        | number     | no       | 0.0001  | v.in.ogr min area                               |

### Example usage

```
run grass:nviz elevation="a.gpkg;b.gpkg" color="a.gpkg;b.gpkg" vector="a.gpkg;b.gpkg"  
↪ point="a.gpkg;b.gpkg" volume="a.gpkg;b.gpkg"
```

This calls **nviz**: elevation="a.gpkg;b.gpkg" — Name of elevation raster map; color="a.gpkg;b.gpkg" — Name of raster map(s) for Color; vector="a.gpkg;b.gpkg" — Name of vector lines/areas overlay map(s); point="a.gpkg;b.gpkg" — Name of vector points overlay file(s); volume="a.gpkg;b.gpkg" — Name of existing 3d raster map. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.basins.fill — r.basins.fill

**group:** Raster (r.\*)

Generates watershed subbasins raster map.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| cnetwork                                | raster            | yes      | —       | Input coded stream network raster layer          |
| tnetwork                                | raster            | yes      | —       | Input thinned ridge network raster layer         |
| number                                  | number            | no       | 1.0     | Number of passes through the dataset             |
| output                                  | rasterDestination | yes      | —       | Watersheds                                       |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

---

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.basins.fill cnetwork=dem.tif tnetwork=dem.tif number=1 output=output.tif
```

This calls **r.basins.fill**: cnetwork=dem.tif — Input coded stream network raster layer; tnetwork=dem.tif — Input thinned ridge network raster layer; number=1 — Number of passes through the dataset; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.blend.combine — r.blend.combine

**group:** Raster (r.\*)

r.blend.combine - Blends color components of two raster maps by a given ratio and export into a unique raster.

| Parameter                               | Type              | Required | Default | Description                                       |
|-----------------------------------------|-------------------|----------|---------|---------------------------------------------------|
| first                                   | raster            | yes      | —       | Name of first raster map for blending             |
| second                                  | raster            | yes      | —       | Name of second raster map for blending            |
| percent                                 | number            | no       | 50.0    | Percentage weight of first map for color blending |
| output                                  | rasterDestination | yes      | —       | Blended                                           |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                           |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)   |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)         |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)  |

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.blend.combine first=dem.tif second=dem.tif percent=50 output=output.tif
```

This calls **r.blend.combine**: first=dem.tif — Name of first raster map for blending; second=dem.tif — Name of second raster map for blending; percent=50 — Percentage weight of first map for color blending; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.blend.rgb — r.blend.rgb

**group:** Raster (r.\*)

r.blend.rgb - Blends color components of two raster maps by a given ratio and exports into three rasters.

| Parameter                      | Type              | Required | Default | Description                                       |
|--------------------------------|-------------------|----------|---------|---------------------------------------------------|
| first                          | raster            | yes      | —       | Name of first raster map for blending             |
| second                         | raster            | yes      | —       | Name of second raster map for blending            |
| percent                        | number            | no       | 50.0    | Percentage weight of first map for color blending |
| output_red                     | rasterDestination | yes      | —       | Blended Red                                       |
| output_green                   | rasterDestination | yes      | —       | Blended Green                                     |
| output_blue                    | rasterDestination | yes      | —       | Blended Blue                                      |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                           |



| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output\_red (outputRaster), output\_green (outputRaster), output\_blue (outputRaster)

### Example usage

```
run grass:r.blend.rgb first=dem.tif second=dem.tif percent=50 output_red=output_red.tif
```

This calls **r.blend.rgb**: first=dem.tif — Name of first raster map for blending; second=dem.tif — Name of second raster map for blending; percent=50 — Percentage weight of first map for color blending; output\_red=output\_red.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.buffer — r.buffer

**group:** Raster (r.\*)

Creates a raster map layer showing buffer zones surrounding cells that contain non-NULL category values.

| Parameter                                | Type              | Required | Default | Description                                                                        |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input raster layer                                                                 |
| distances                                | string            | yes      | —       | Distance zone(s) (e.g. 100,200,300)                                                |
| units                                    | enum              | no       | 0       | Units of distance — options: 0=meters, 1=kilometers, 2=feet, 3=miles, 4=naut-miles |
| -z                                       | boolean           | no       | False   | Ignore zero (0) data cells instead of NULL cells                                   |
| output                                   | rasterDestination | yes      | —       | Buffer                                                                             |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                            |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                    |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                          |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                   |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.buffer input=dem.tif distances=value units=0 output=output.tif
```

This calls **r.buffer**: input=dem.tif — Input raster layer; distances=value — Distance zone(s) (e.g. 100,200,300); units=0 selects *meters*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.buffer.lowmem — r.buffer.lowmem

**group:** Raster (r.\*)

Creates a raster map layer showing buffer zones surrounding cells that contain non-NULL category values (low-memory alternative).

| Parameter | Type   | Required | Default | Description        |
|-----------|--------|----------|---------|--------------------|
| input     | raster | yes      | —       | Input raster layer |

| Parameter                                | Type              | Required | Default | Description                                                                        |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------|
| distances                                | string            | yes      | —       | Distance zone(s) (e.g. 100,200,300)                                                |
| units                                    | enum              | no       | 0       | Units of distance — options: 0=meters, 1=kilometers, 2=feet, 3=miles, 4=naut-miles |
| -z                                       | boolean           | no       | False   | Ignore zero (0) data cells instead of NULL cells                                   |
| output                                   | rasterDestination | yes      | —       | Buffer                                                                             |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                            |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                    |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                          |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                   |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.buffer.lowmem input=dem.tif distances=value units=0 output=output.tif
```

This calls **r.buffer.lowmem**: input=dem.tif — Input raster layer; distances=value — Distance zone(s) (e.g. 100,200,300); units=0 selects *meters*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.carve — r.carve

**group:** Raster (r.\*)

Takes vector stream data, transforms it to raster and subtracts depth from the output DEM.

| Parameter                                | Type              | Required | Default | Description                                                                                             |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| raster                                   | raster            | yes      | —       | Elevation                                                                                               |
| vector                                   | source            | yes      | —       | Vector layer containing stream(s)                                                                       |
| width                                    | number            | no       | —       | Stream width (in meters). Default is raster cell width                                                  |
| depth                                    | number            | no       | —       | Additional stream depth (in meters)                                                                     |
| -n                                       | boolean           | no       | False   | No flat areas allowed in flow direction                                                                 |
| output                                   | rasterDestination | yes      | —       | Modified elevation                                                                                      |
| points                                   | vectorDestination | yes      | —       | Adjusted stream points                                                                                  |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DS_CO *(adv)*               | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputRaster), points (outputVector)

### Example usage

```
run grass:r.carve raster=dem.tif vector=input.gpkg width=10 depth=10 output=output.tif
```

This calls **r.carve**: raster=dem.tif — Elevation; vector=input.gpkg — Vector layer containing stream(s); width=10 — Stream width (in meters). Default is raster cell width; depth=10 — Additional stream depth (in meters); output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.category — r.category

**group:** Raster (r.\*)

Manages category values and labels associated with user-specified raster map layers.

| Parameter                                | Type              | Required | Default | Description                                                                                              |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------|
| map                                      | raster            | yes      | —       | Name of raster map                                                                                       |
| separator                                | string            | no       | tab     | Field separator (Special characters: pipe, comma, space, tab, newline)                                   |
| rules                                    | file              | no       | —       | File containing category label rules                                                                     |
| txtrules                                 | string            | no       | —       | Inline category label rules                                                                              |
| raster                                   | raster            | no       | —       | Raster map from which to copy category table                                                             |
| format *(adv)*                           | string            | no       | —       | Default label or format string for dynamic labeling. Used when no explicit label exists for the category |
| coefficients *(adv)*                     | string            | no       | —       | Dynamic label coefficients. Two pairs of category multiplier and offsets, for \$1 and \$2                |
| output                                   | rasterDestination | yes      | —       | Category                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                  |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                          |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                         |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.category map=dem.tif separator=tab rules=input.dat txtrules=value
↪ raster=dem.tif output=output.tif
```

This calls **r.category**: map=dem.tif — Name of raster map; separator=tab — Field separator (Special characters: pipe, comma, space, tab, newline); rules=input.dat — File containing category label rules; txtrules=value — Inline category label rules; raster=dem.tif — Raster map from which to copy category table; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.category.out — r.category.out

**group:** Raster (r.\*)

r.category.out - Exports category values and labels associated with user-specified raster map layers.

| Parameter | Type   | Required | Default | Description                                                         |
|-----------|--------|----------|---------|---------------------------------------------------------------------|
| map       | raster | yes      | —       | Name of raster map                                                  |
| cats      | string | no       | —       | Category values (for Integer rasters). Example: 1,3,7-9,13          |
| values    | string | no       | —       | Comma separated value list (for float rasters). Example: 1.4,3.8,13 |

| Parameter                               | Type            | Required | Default | Description                                                            |
|-----------------------------------------|-----------------|----------|---------|------------------------------------------------------------------------|
| separator                               | string          | no       | tab     | Field separator (Special characters: pipe, comma, space, tab, newline) |
| html                                    | fileDestination | yes      | —       | Category                                                               |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                                                |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                        |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.category.out map=dem.tif cats=value values=value separator=tab html=html.txt
```

This calls **r.category.out**: map=dem.tif — Name of raster map; cats=value — Category values (for Integer rasters). Example: 1,3,7-9,13; values=value — Comma separated value list (for float rasters). Example: 1.4,3.8,13; separator=tab — Field separator (Special characters: pipe, comma, space, tab, newline); html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.circle — r.circle

**group:** Raster (r.\*)

Creates a raster map containing concentric rings around a given point.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| coordinates                             | point             | no       | 0,0     | The coordinate of the center (east,north)        |
| min                                     | number            | no       | —       | Minimum radius for ring/circle map (in meters)   |
| max                                     | number            | no       | —       | Maximum radius for ring/circle map (in meters)   |
| multiplier                              | number            | no       | —       | Data value multiplier                            |
| -b                                      | boolean           | no       | False   | Generate binary raster map                       |
| output                                  | rasterDestination | yes      | —       | Circles                                          |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.circle coordinates=100,100 min=10 max=10 multiplier=10 output=output.tif
```

This calls **r.circle**: coordinates=100,100 — The coordinate of the center (east,north); min=10 — Minimum radius for ring/circle map (in meters); max=10 — Maximum radius for ring/circle map (in meters); multiplier=10 — Data value multiplier; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.clump — r.clump

**group:** Raster (r.\*)

Recategorizes data in a raster map by grouping cells that form physically discrete areas into unique categories.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Input layer                                      |
| title                                   | string            | yes      | —       | Title for output raster map                      |
| -d *(adv)*                              | boolean           | no       | False   | Clump also diagonal cells                        |
| output                                  | rasterDestination | yes      | —       | Clumps                                           |
| threshold                               | number            | no       | 0.0     | Threshold to identify similar cells              |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.clump input=dem.tif title=value threshold=0 output=output.tif
```

This calls **r.clump**: input=dem.tif — Input layer; title=value — Title for output raster map; threshold=0 — Threshold to identify similar cells; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.coin — r.coin

**group:** Raster (r.\*)

Tabulates the mutual occurrence (coincidence) of categories for two raster map layers.

| Parameter                               | Type            | Required | Default     | Description                                                       |
|-----------------------------------------|-----------------|----------|-------------|-------------------------------------------------------------------|
| first                                   | raster          | yes      | —           | Name of first raster map                                          |
| second                                  | raster          | yes      | —           | Name of second raster map                                         |
| units                                   | enum            | yes      | —           | Unit of measure — options: 0=c, 1=p, 2=x, 3=y, 4=a, 5=h, 6=k, 7=m |
| -w                                      | boolean         | no       | False       | Wide report, 132 columns (default: 80)                            |
| html                                    | fileDestination | no       | report.html | Coincidence report                                                |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —           | GRASS GIS region extent                                           |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default)                   |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.coin first=dem.tif second=dem.tif units=0 html=html.txt
```

This calls **r.coin**: first=dem.tif — Name of first raster map; second=dem.tif — Name of second raster map; units=0 selects c; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.colors — r.colors

**group:** Raster (r.\*)

Creates/modifies the color table associated with a raster map.

| Parameter | Type       | Required | Default | Description            |
|-----------|------------|----------|---------|------------------------|
| map       | multilayer | yes      | —       | Name of raster maps(s) |

| Parameter                                | Type              | Required | Default | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| color                                    | enum              | no       | 0       | Name of color table — options: 0=not selected, 1=aspect, 2=aspectcolr, 3=bcyr, 4=bgyr, 5=blues, 6=byg, 7=byr, 8=celsius, 9=corine, 10=curvature, 11=differences, 12=elevation, 13=etopo2, 14=evi, 15=fahrenheit, 16=gdd, 17=greens, 18=grey, 19=grey.eq, 20=grey.log, 21=grey1.0, 22=grey255, 23=gyr, 24=haxby, 25=kelvin, 26=ndvi, 27=ndwi, 28=oranges, 29=population, 30=population_dens, 31=precipitation, 32=precipitation_daily, 33=precipitation_monthly, 34=rainbow, 35=ramp, 36=random, 37=reds, 38=rstcurv, 39=ryb, 40=ryg, 41=sepia, 42=slope, 43=srtm, 44=srtm_plus, 45=terrain, 46=wave |
| rules_txt                                | string            | no       | —       | Color rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| rules                                    | file              | no       | —       | Color rules file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| raster                                   | raster            | no       | —       | Raster map from which to copy color table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| -r                                       | boolean           | no       | False   | Remove existing color table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| -w                                       | boolean           | no       | False   | Only write new color table if it does not already exist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -n                                       | boolean           | no       | False   | Invert colors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -g                                       | boolean           | no       | False   | Logarithmic scaling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| -a                                       | boolean           | no       | False   | Logarithmic-absolute scaling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -e                                       | boolean           | no       | False   | Histogram equalization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| output_dir                               | folderDestination | yes      | —       | Output Directory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**Outputs:** output\_dir (outputFolder)

### Example usage

```
run grass:r.colors map="a.gpkg;b.gpkg" color=0 rules_txt=value rules=input.dat
↳ raster=dem.tif output_dir=output_dir/
```

This calls **r.colors**: map="a.gpkg;b.gpkg" — Name of raster maps(s); color=0 selects *not selected*; rules\_txt=value — Color rules; rules=input.dat — Color rules file; raster=dem.tif — Raster map from which to copy color table; output\_dir=output\_dir/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.colors.out — r.colors.out

**group:** Raster (r.\*)

Exports the color table associated with a raster map.

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| map                                      | raster          | yes      | —       | Name of raster map                              |
| -p *(adv)*                               | boolean         | no       | False   | Output values as percentages                    |
| rules                                    | fileDestination | yes      | —       | Color Table                                     |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** rules (outputFile)

### Example usage

```
run grass:r.colors.out map=dem.tif rules=rules.txt
```

This calls **r.colors.out**: map=dem.tif — Name of raster map; rules=rules.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.colors.stddev — r.colors.stddev

**group:** Raster (r.\*)

Sets color rules based on stddev from a raster map's mean value.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| map                                      | raster            | yes      | —       | Name of raster map                               |
| -b *(adv)*                               | boolean           | no       | False   | Color using standard deviation bands             |
| -z *(adv)*                               | boolean           | no       | False   | Force center at zero                             |
| output                                   | rasterDestination | yes      | —       | Stddev Colors                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.colors.stddev map=dem.tif output=output.tif
```

This calls **r.colors.stddev**: map=dem.tif — Name of raster map; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.composite — r.composite

**group:** Raster (r.\*)

Combines red, green and blue raster maps into a single composite raster map.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| red                                      | raster            | yes      | —       | Red                                              |
| green                                    | raster            | yes      | —       | Green                                            |
| blue                                     | raster            | yes      | —       | Blue                                             |
| levels                                   | number            | no       | 32.0    | Number of levels to be used for each component   |
| level_red                                | number            | no       | —       | Number of levels to be used for                  |
| level_green                              | number            | no       | —       | Number of levels to be used for                  |
| level_blue                               | number            | no       | —       | Number of levels to be used for                  |
| -d                                       | boolean           | no       | False   | Dither                                           |
| -c                                       | boolean           | no       | False   | Use closest color                                |
| output                                   | rasterDestination | yes      | —       | Composite                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

---

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.composite red=dem.tif green=dem.tif blue=dem.tif levels=32 level_red=10  
↪ level_green=10 level_blue=10 output=output.tif
```

This calls **r.composite**: red=dem.tif — Red; green=dem.tif — Green; blue=dem.tif — Blue; levels=32 — Number of levels to be used for each component; level\_red=10 — Number of levels to be used for; level\_green=10 — Number of levels to be used for; level\_blue=10 — Number of levels to be used for; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:r.contour — r.contour**

**group:** Raster (r.\*)

Produces a vector map of specified contours from a raster map.

| Parameter                               | Type              | Required | Default | Description                                                                                             |
|-----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster                                                                                            |
| step                                    | number            | no       | —       | Increment between contour levels                                                                        |
| levels                                  | string            | no       | —       | List of contour levels                                                                                  |
| minlevel                                | number            | no       | —       | Minimum contour level                                                                                   |
| maxlevel                                | number            | no       | —       | Maximum contour level                                                                                   |
| cut                                     | number            | no       | 0.0     | Minimum number of points for a contour line (0 -> no limit)                                             |
| output                                  | vectorDestination | yes      | —       | Contours                                                                                                |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*     | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*               | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*       | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

**Example usage**

```
run grass:r.contour input=dem.tif step=10 levels=value minlevel=10 maxlevel=10 cut=0  
↪ output=output.gpkg
```

This calls **r.contour**: input=dem.tif — Input raster; step=10 — Increment between contour levels; levels=value — List of contour levels; minlevel=10 — Minimum contour level; maxlevel=10 — Maximum contour level; cut=0 — Minimum number of points for a contour line (0 -> no limit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:r.cost — r.cost**

**group:** Raster (r.\*)

Creates a raster layer of cumulative cost of moving across a raster layer whose cell values represent cost.



| Parameter                                | Type              | Required | Default | Description                                                      |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Unit cost layer                                                  |
| start_coordinates                        | point             | no       | —       | Coordinates of starting point(s) (E,N)                           |
| stop_coordinates                         | point             | no       | —       | Coordinates of stopping point(s) (E,N)                           |
| -k                                       | boolean           | no       | False   | Use the 'Knight's move'; slower, but more accurate               |
| -n                                       | boolean           | no       | True    | Keep null values in output raster layer                          |
| start_points                             | source            | no       | —       | Start points                                                     |
| stop_points                              | source            | no       | —       | Stop points                                                      |
| start_raster                             | raster            | no       | —       | Name of starting raster points map                               |
| max_cost                                 | number            | no       | 0.0     | Maximum cumulative cost                                          |
| null_cost                                | number            | no       | —       | Cost assigned to null cells. By default, null cells are excluded |
| memory                                   | number            | no       | 300.0   | Maximum memory to be used in MB                                  |
| output                                   | rasterDestination | yes      | —       | Cumulative cost                                                  |
| nearest                                  | rasterDestination | no       | —       | Cost allocation map                                              |
| outdir                                   | rasterDestination | no       | —       | Movement directions                                              |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                           |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                                |

**Outputs:** output (outputRaster), nearest (outputRaster), outdir (outputRaster)

### Example usage

```
run grass:r.cost input=dem.tif start_coordinates=100,100 stop_coordinates=100,100
↪ start_points=input.gpkg stop_points=input.gpkg start_raster=dem.tif max_cost=0
↪ output=output.tif
```

This calls **r.cost**: input=dem.tif — Unit cost layer; start\_coordinates=100,100 — Coordinates of starting point(s) (E,N); stop\_coordinates=100,100 — Coordinates of stopping point(s) (E,N); start\_points=input.gpkg — Start points; stop\_points=input.gpkg — Stop points; start\_raster=dem.tif — Name of starting raster points map; max\_cost=0 — Maximum cumulative cost; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.covar — r.covar

**group:** Raster (r.\*)

Outputs a covariance/correlation matrix for user-specified raster layer(s).

| Parameter                                | Type            | Required | Default     | Description                                     |
|------------------------------------------|-----------------|----------|-------------|-------------------------------------------------|
| map                                      | multilayer      | yes      | —           | Input layers                                    |
| -r                                       | boolean         | no       | True        | Print correlation matrix                        |
| html                                     | fileDestination | no       | report.html | Covariance report                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.covar map="a.gpkg;b.gpkg" html=html.txt
```

This calls **r.covar**: map="a.gpkg;b.gpkg" — Input layers; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.cross — r.cross

**group:** Raster (r.\*)

Creates a cross product of the category values from multiple raster map layers.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Input raster layers                              |
| -z                                       | boolean           | no       | False   | Non-zero data only                               |
| output                                   | rasterDestination | yes      | —       | Cross product                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.cross input="a.gpkg;b.gpkg" output=output.tif
```

This calls **r.cross**: input="a.gpkg;b.gpkg" — Input raster layers; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.describe — r.describe

**group:** Raster (r.\*)

Prints terse list of category values found in a raster layer.

| Parameter                                | Type            | Required | Default     | Description                                     |
|------------------------------------------|-----------------|----------|-------------|-------------------------------------------------|
| map                                      | raster          | yes      | —           | input raster layer                              |
| null_value                               | string          | no       | *           | String representing NULL value                  |
| nsteps                                   | number          | no       | 255.0       | Number of quantization steps                    |
| -r                                       | boolean         | no       | False       | Only print the range of the data                |
| -n                                       | boolean         | no       | False       | Suppress reporting of any NULLs                 |
| -d                                       | boolean         | no       | False       | Use the current region                          |
| -i                                       | boolean         | no       | False       | Read floating-point map as integer              |
| html                                     | fileDestination | no       | report.html | Categories                                      |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.describe map=dem.tif null_value=* nsteps=255 html=html.txt
```

This calls **r.distance**: map=dem.tif — input raster layer; null\_value=\* — String representing NULL value; nsteps=255 — Number of quantization steps; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.distance — r.distance

**group:** Raster (r.\*)

Locates the closest points between objects in two raster maps.

| Parameter                                | Type            | Required | Default | Description                                                            |
|------------------------------------------|-----------------|----------|---------|------------------------------------------------------------------------|
| map                                      | multilayer      | yes      | —       | Name of two input raster for computing inter-class distances           |
| separator                                | string          | no       | :       | Field separator (Special characters: pipe, comma, space, tab, newline) |
| sort                                     | enum            | no       | 0       | Sort output by distance — options: 0=asc, 1=desc                       |
| -l *(adv)*                               | boolean         | no       | False   | Include category labels in the output                                  |
| -o *(adv)*                               | boolean         | no       | False   | Report zero distance if rasters are overlapping                        |
| -n *(adv)*                               | boolean         | no       | False   | Report null objects as *                                               |
| html                                     | fileDestination | yes      | —       | Distance                                                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                                                |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                        |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.distance map="a.gpkg;b.gpkg" separator=: sort=0 html=html.txt
```

This calls **r.distance**: map="a.gpkg;b.gpkg" — Name of two input raster for computing inter-class distances; separator=: — Field separator (Special characters: pipe, comma, space, tab, newline); sort=0 selects asc; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.drain — r.drain

**group:** Raster (r.\*)

Traces a flow through an elevation model on a raster map.

| Parameter                      | Type              | Required | Default | Description                                                                       |
|--------------------------------|-------------------|----------|---------|-----------------------------------------------------------------------------------|
| input                          | raster            | yes      | —       | Elevation                                                                         |
| direction                      | raster            | no       | —       | Name of input movement direction map associated with the cost surface             |
| start_coordinates              | point             | no       | —       | Map coordinates of starting point(s) (E,N)                                        |
| start_points                   | source            | no       | —       | Vector layer containing starting point(s)                                         |
| -c                             | boolean           | no       | False   | Copy input cell values on output                                                  |
| -a                             | boolean           | no       | False   | Accumulate input values along the path                                            |
| -n                             | boolean           | no       | False   | Count cell numbers along the path                                                 |
| -d                             | boolean           | no       | False   | The input raster map is a cost surface (direction surface must also be specified) |
| output                         | rasterDestination | yes      | —       | Least cost path                                                                   |
| drain                          | vectorDestination | yes      | —       | Drain                                                                             |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                                                           |

| Parameter                                | Type    | Required | Default | Description                                                                                             |
|------------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number  | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string  | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*         | string  | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number  | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DS_CO *(adv)*               | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputRaster), drain (outputVector)

### Example usage

```
run grass:r.drain input=dem.tif direction=dem.tif start_coordinates=100,100
↪ start_points=input.gpkg output=output.tif
```

This calls **r.drain**: input=dem.tif — Elevation; direction=dem.tif — Name of input movement direction map associated with the cost surface; start\_coordinates=100,100 — Map coordinates of starting point(s) (E,N); start\_points=input.gpkg — Vector layer containing starting point(s); output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.fill.dir — r.fill.dir

**group:** Raster (r.\*)

Filters and generates a depressionless elevation layer and a flow direction layer from a given elevation raster layer.

| Parameter                                | Type              | Required | Default | Description                                                           |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Elevation                                                             |
| format                                   | enum              | no       | 0       | Output aspect direction format — options: 0=grass, 1=agnps, 2=answers |
| -f *(adv)*                               | boolean           | no       | False   | Find unresolved areas only                                            |
| output                                   | rasterDestination | yes      | —       | Depressionless DEM                                                    |
| direction                                | rasterDestination | yes      | —       | Flow direction                                                        |
| areas                                    | rasterDestination | yes      | —       | Problem areas                                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                               |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                       |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                             |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                      |

**Outputs:** output (outputRaster), direction (outputRaster), areas (outputRaster)

### Example usage

```
run grass:r.fill.dir input=dem.tif format=0 output=output.tif
```

This calls **r.fill.dir**: input=dem.tif — Elevation; format=0 selects *grass*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.fill.stats — r.fill.stats

**group:** Raster (r.\*)

Rapidly fills 'no data' cells (NULLs) of a raster map with interpolated values (IDW).

| Parameter                               | Type              | Required | Default | Description                                                                                    |
|-----------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer with data gaps to fill                                                      |
| -k                                      | boolean           | no       | False   | Preserve original cell values (By default original values are smoothed)                        |
| mode                                    | enum              | no       | 0       | Statistic for interpolated cell values — options: 0=wmean, 1=mean, 2=median, 3=mode            |
| -m                                      | boolean           | no       | False   | Interpret distance as map units, not number of cells (Do not select with geodetic coordinates) |
| distance                                | number            | no       | 3.0     | Distance threshold (default: in cells) for interpolation                                       |
| minimum                                 | number            | no       | —       | Minimum input data value to include in interpolation                                           |
| maximum                                 | number            | no       | —       | Maximum input data value to include in interpolation                                           |
| power                                   | number            | no       | 2.0     | Power coefficient for IDW interpolation                                                        |
| cells                                   | number            | no       | 8.0     | Minimum number of data cells within search radius                                              |
| output                                  | rasterDestination | yes      | —       | Output Map                                                                                     |
| uncertainty                             | rasterDestination | no       | —       | Uncertainty Map                                                                                |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                        |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                      |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                               |

**Outputs:** output (outputRaster), uncertainty (outputRaster)

### Example usage

```
run grass:r.fill.stats input=dem.tif mode=0 distance=3 minimum=10 maximum=10 power=2  
↪ cells=8 output=output.tif
```

This calls **r.fill.stats**: input=dem.tif — Input raster layer with data gaps to fill; mode=0 selects *wmean*; distance=3 — Distance threshold (default: in cells) for interpolation; minimum=10 — Minimum input data value to include in interpolation; maximum=10 — Maximum input data value to include in interpolation; power=2 — Power coefficient for IDW interpolation; cells=8 — Minimum number of data cells within search radius; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.fillnulls — r.fillnulls

**group:** Raster (r.\*)

Fills no-data areas in raster maps using spline interpolation.

| Parameter | Type   | Required | Default | Description                                                         |
|-----------|--------|----------|---------|---------------------------------------------------------------------|
| input     | raster | yes      | —       | Input raster layer to fill                                          |
| method    | enum   | no       | 2       | Interpolation method to use — options: 0=bilinear, 1=bicubic, 2=rst |
| tension   | number | no       | 40.0    | Spline tension parameter                                            |
| smooth    | number | no       | 0.1     | Spline smoothing parameter                                          |

| Parameter                                | Type              | Required | Default | Description                                                       |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------|
| edge                                     | number            | no       | 3.0     | Width of hole edge used for interpolation (in cells)              |
| npmin                                    | number            | no       | 600.0   | Minimum number of points for approximation in a segment (>segmax) |
| segmax                                   | number            | no       | 300.0   | Maximum number of points in a segment                             |
| lambda                                   | number            | no       | 0.01    | Tykhonov regularization parameter (affects smoothing)             |
| output                                   | rasterDestination | yes      | —       | Filled                                                            |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                           |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                   |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                         |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                  |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.fillnulls input=dem.tif method=2 tension=40 smooth=0.1 edge=3 npmin=600
↳ segmax=300 output=output.tif
```

This calls **r.fillnulls**: input=dem.tif — Input raster layer to fill; method=2 selects *rst*; tension=40 — Spline tension parameter; smooth=0.1 — Spline smoothing parameter; edge=3 — Width of hole edge used for interpolation (in cells); npmin=600 — Minimum number of points for approximation in a segment (>segmax); segmax=300 — Maximum number of points in a segment; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.flow — r.flow

**group:** Raster (r.\*)

Construction of flowlines, flowpath lengths, and flowaccumulation (contributing areas) from a raster digital elevation model (DEM).

| Parameter                                | Type              | Required | Default | Description                                                      |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| elevation                                | raster            | yes      | —       | Elevation                                                        |
| aspect                                   | raster            | no       | —       | Aspect                                                           |
| barrier                                  | raster            | no       | —       | Barrier                                                          |
| skip                                     | number            | no       | —       | Number of cells between flowlines                                |
| bound                                    | number            | no       | —       | Maximum number of segments per flowline                          |
| -u                                       | boolean           | no       | False   | Compute upslope flowlines instead of default downhill flowlines  |
| -3                                       | boolean           | no       | False   | 3-D lengths instead of 2-D                                       |
| -m *(adv)*                               | boolean           | no       | False   | Use less memory, at a performance penalty                        |
| flowline                                 | vectorDestination | no       | —       | Flow line                                                        |
| flowlength                               | rasterDestination | yes      | —       | Flow path length                                                 |
| flowaccumulation                         | rasterDestination | yes      | —       | Flow accumulation                                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                 |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —       | v.out.ogr output data source options (dsco)                      |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —       | v.out.ogr output layer options (lco)                             |

| Parameter                         | Type    | Required | Default | Description                                                                                             |
|-----------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_VECTOR_EXPORT_NOCAT *(adv)* | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** flowline (outputVector), flowlength (outputRaster), flowaccumulation (outputRaster)

### Example usage

```
run grass:r.flow elevation=dem.tif aspect=dem.tif barrier=dem.tif skip=10 bound=10
↳ flowline=flowline.gpkg
```

This calls **r.flow**: elevation=dem.tif — Elevation; aspect=dem.tif — Aspect; barrier=dem.tif — Barrier; skip=10 — Number of cells between flowlines; bound=10 — Maximum number of segments per flowline; flowline=flowline.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.geomorphon — r.geomorphon

**group:** Raster (r.\*)

Calculates geomorphons (terrain forms) and associated geometry using machine vision approach.

| Parameter                                | Type              | Required | Default | Description                                          |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------|
| elevation                                | raster            | yes      | —       | Name of input elevation raster map                   |
| search                                   | number            | no       | 3.0     | Outer search radius                                  |
| skip                                     | number            | no       | 0.0     | Inner search radius                                  |
| flat                                     | number            | no       | 1.0     | Flatness threshold (degrees)                         |
| dist                                     | number            | no       | 0.0     | Flatness distance, zero for none                     |
| forms                                    | rasterDestination | yes      | —       | Most common geomorphic forms                         |
| -m *(adv)*                               | boolean           | no       | False   | Use meters to define search units (default is cells) |
| -e *(adv)*                               | boolean           | no       | False   | Use extended form correction                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                              |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)      |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)            |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)     |

**Outputs:** forms (outputRaster)

### Example usage

```
run grass:r.geomorphon elevation=dem.tif search=3 skip=0 flat=1 dist=0 forms=forms.tif
```

This calls **r.geomorphon**: elevation=dem.tif — Name of input elevation raster map; search=3 — Outer search radius; skip=0 — Inner search radius; flat=1 — Flatness threshold (degrees); dist=0 — Flatness distance, zero for none; forms=forms.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.grow — r.grow

**group:** Raster (r.\*)

Generates a raster layer with contiguous areas grown by one cell.

| Parameter                                | Type              | Required | Default | Description                                                    |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------|
| input                                    | raster            | yes      | —       | input raster layer                                             |
| radius                                   | number            | no       | 1.01    | Radius of buffer in raster cells                               |
| metric                                   | enum              | no       | 0       | Metric — options: 0=euclidean, 1=maximum, 2=manhattan          |
| old                                      | number            | no       | —       | Value to write for input cells which are non-NULL (-1 => NULL) |
| new                                      | number            | no       | —       | Value to write for "grown" cells                               |
| -m *(adv)*                               | boolean           | no       | False   | Radius is in map units rather than cells                       |
| output                                   | rasterDestination | yes      | —       | Expanded                                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                        |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                      |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)               |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.grow input=dem.tif radius=1.01 metric=0 old=10 new=10 output=output.tif
```

This calls **r.grow**: input=dem.tif — input raster layer; radius=1.01 — Radius of buffer in raster cells; metric=0 selects *euclidean*; old=10 — Value to write for input cells which are non-NULL (-1 => NULL); new=10 — Value to write for "grown" cells; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.grow.distance — r.grow.distance

**group:** Raster (r.\*)

Generates a raster layer of distance to features in input layer.

| Parameter                                | Type              | Required | Default | Description                                                                  |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input input raster layer                                                     |
| metric                                   | enum              | no       | 0       | Metric — options: 0=euclidean, 1=squared, 2=maximum, 3=manhattan, 4=geodesic |
| -m *(adv)*                               | boolean           | no       | False   | Output distances in meters instead of map units                              |
| -n *(adv)*                               | boolean           | no       | False   | Calculate distance to nearest NULL cell                                      |
| distance                                 | rasterDestination | yes      | —       | Distance                                                                     |
| value                                    | rasterDestination | yes      | —       | Value of nearest cell                                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                      |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                              |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                    |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                             |

**Outputs:** distance (outputRaster), value (outputRaster)

### Example usage

```
run grass:r.grow.distance input=dem.tif metric=0 distance=distance.tif
```

This calls **r.grow.distance**: input=dem.tif — Input input raster layer; metric=0 selects *euclidean*; distance=distance.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## grass:r.gwflow — r.gwflow

**group:** Raster (r.\*)

Numerical calculation program for transient, confined and unconfined groundwater flow in two dimensions.

| Parameter                                | Type              | Required | Default  | Description                                                                                                   |
|------------------------------------------|-------------------|----------|----------|---------------------------------------------------------------------------------------------------------------|
| phead                                    | raster            | yes      | —        | The initial piezometric head in [m]                                                                           |
| status                                   | raster            | yes      | —        | Boundary condition status, 0-inactive, 1-active, 2-dirichlet                                                  |
| hc_x                                     | raster            | yes      | —        | X-part of the hydraulic conductivity tensor in [m/s]                                                          |
| hc_y                                     | raster            | yes      | —        | Y-part of the hydraulic conductivity tensor in [m/s]                                                          |
| q                                        | raster            | no       | —        | Water sources and sinks in [m <sup>3</sup> /s]                                                                |
| s                                        | raster            | yes      | —        | Specific yield in [1/m]                                                                                       |
| recharge                                 | raster            | no       | —        | Recharge map e.g: 610 <sup>-9</sup> per cell in [m <sup>3</sup> /sm <sup>2</sup> ]                            |
| top                                      | raster            | yes      | —        | Top surface of the aquifer in [m]                                                                             |
| bottom                                   | raster            | yes      | —        | Bottom surface of the aquifer in [m]                                                                          |
| type                                     | enum              | no       | 0        | The type of groundwater flow — options: 0=confined, 1=unconfined                                              |
| river_bed                                | raster            | no       | —        | The height of the river bed in [m]                                                                            |
| river_head                               | raster            | no       | —        | Water level (head) of the river with leakage connection in [m]                                                |
| river_leak                               | raster            | no       | —        | The leakage coefficient of the river bed in [1/s]                                                             |
| drain_bed                                | raster            | no       | —        | The height of the drainage bed in [m]                                                                         |
| drain_leak                               | raster            | no       | —        | The leakage coefficient of the drainage bed in [1/s]                                                          |
| dttime                                   | number            | no       | 86400.0  | The calculation time in seconds                                                                               |
| maxit                                    | number            | no       | 100000.0 | Maximum number of iteration used to solve the linear equation system                                          |
| error                                    | number            | no       | 1e-06    | Error break criteria for iterative solvers (jacobi, sor, cg or bicgstab)                                      |
| solver                                   | enum              | no       | 0        | The type of solver which should solve the symmetric linear equation system — options: 0=cg, 1=pcg, 2=cholesky |
| relax                                    | string            | no       | 1        | The relaxation parameter used by the jacobi and sor solver for speedup or stabilizing                         |
| -f                                       | boolean           | no       | False    | Allocate a full quadratic linear equation system, default is a sparse linear equation system                  |
| output                                   | rasterDestination | yes      | —        | Groundwater flow                                                                                              |
| vx                                       | rasterDestination | yes      | —        | Groundwater filter velocity vector part in x direction [m/s]                                                  |
| vy                                       | rasterDestination | yes      | —        | Groundwater filter velocity vector part in y direction [m/s]                                                  |
| budget                                   | rasterDestination | yes      | —        | Groundwater budget for each cell [m <sup>3</sup> /s]                                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —        | GRASS GIS region extent                                                                                       |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0      | GRASS GIS region cellsize (leave 0 for default)                                                               |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —        | Output Rasters format options (createopt)                                                                     |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —        | Output Rasters format metadata options (metaopt)                                                              |

**Outputs:** output (outputRaster), vx (outputRaster), vy (outputRaster), budget (outputRaster)

### Example usage

```
run grass:r.gwflow phead=dem.tif status=dem.tif hc_x=dem.tif hc_y=dem.tif s=dem.tif  
↪ top=dem.tif bottom=dem.tif output=output.tif
```

This calls **r.gwflow**: phead=dem.tif — The initial piezometric head in [m]; status=dem.tif — Boundary condition status, 0-inactive, 1-active, 2-dirichlet; hc\_x=dem.tif — X-part of the hydraulic conductivity tensor in [m/s]; hc\_y=dem.tif — Y-part of the hydraulic conductivity tensor in [m/s]; s=dem.tif — Specific yield in [1/m]; top=dem.tif — Top surface of the aquifer in [m]; bottom=dem.tif — Bottom surface of the aquifer in [m]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.his — r.his

**group:** Raster (r.\*)

Generates red, green and blue raster layers combining hue, intensity and saturation (HIS) values from user-specified input raster layers.

| Parameter                                | Type              | Required | Default | Description                                                                                 |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------|
| hue                                      | raster            | yes      | —       | Hue                                                                                         |
| intensity                                | raster            | yes      | —       | Intensity                                                                                   |
| saturation                               | raster            | yes      | —       | Saturation                                                                                  |
| bgcolor                                  | string            | no       | —       | Color to use instead of NULL values. Either a standard color name, R:G:B triplet, or "none" |
| -c                                       | boolean           | no       | False   | Use colors from color tables for NULL values                                                |
| red                                      | rasterDestination | yes      | —       | Red                                                                                         |
| green                                    | rasterDestination | yes      | —       | Green                                                                                       |
| blue                                     | rasterDestination | yes      | —       | Blue                                                                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                     |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                             |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                   |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                            |

**Outputs:** red (outputRaster), green (outputRaster), blue (outputRaster)

### Example usage

```
run grass:r.his hue=dem.tif intensity=dem.tif saturation=dem.tif bgcolor=value red=red.tif
```

This calls **r.his**: hue=dem.tif — Hue; intensity=dem.tif — Intensity; saturation=dem.tif — Saturation; bgcolor=value — Color to use instead of NULL values. Either a standard color name, R:G:B...; red=red.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.horizon — r.horizon

**group:** Raster (r.\*)

Horizon angle computation from a digital elevation model.

| Parameter  | Type   | Required | Default | Description                                                                      |
|------------|--------|----------|---------|----------------------------------------------------------------------------------|
| elevation  | raster | yes      | —       | Name of input elevation raster map                                               |
| direction  | number | no       | —       | Direction in which you want to know the horizon height                           |
| step       | number | no       | —       | Angle step size for multidirectional horizon                                     |
| start      | number | no       | 0.0     | Start angle for multidirectional horizon                                         |
| end        | number | no       | 360.0   | End angle for multidirectional horizon                                           |
| bufferzone | number | no       | —       | For horizon rasters, read from the DEM an extra buffer around the present region |

| Parameter                                | Type              | Required | Default | Description                                                                         |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------------------------|
| e_buff                                   | number            | no       | —       | For horizon rasters, read from the DEM an extra buffer eastward the present region  |
| w_buff                                   | number            | no       | —       | For horizon rasters, read from the DEM an extra buffer westward the present region  |
| n_buff                                   | number            | no       | —       | For horizon rasters, read from the DEM an extra buffer northward the present region |
| s_buff                                   | number            | no       | —       | For horizon rasters, read from the DEM an extra buffer southward the present region |
| maxdistance                              | number            | no       | —       | The maximum distance to consider when finding the horizon height                    |
| distance                                 | number            | no       | 1.0     | Sampling distance step coefficient                                                  |
| -d                                       | boolean           | no       | False   | Write output in degrees (default is radians)                                        |
| -c                                       | boolean           | no       | False   | Write output in compass orientation (default is CCW, East=0)                        |
| output                                   | folderDestination | yes      | —       | Folder to get horizon rasters                                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                             |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                     |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:r.horizon elevation=dem.tif direction=10 step=10 start=0 end=360 bufferzone=10
↳ e_buff=10 output=output/
```

This calls **r.horizon**: elevation=dem.tif — Name of input elevation raster map; direction=10 — Direction in which you want to know the horizon height; step=10 — Angle step size for multidirectional horizon; start=0 — Start angle for multidirectional horizon; end=360 — End angle for multidirectional horizon; bufferzone=10 — For horizon rasters, read from the DEM an extra buffer around the present region; e\_buff=10 — For horizon rasters, read from the DEM an extra buffer eastward the present...; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.horizon.height — r.horizon.height

**group:** Raster (r.\*)

r.horizon.height - Horizon angle computation from a digital elevation model.

| Parameter   | Type   | Required | Default | Description                                                                         |
|-------------|--------|----------|---------|-------------------------------------------------------------------------------------|
| elevation   | raster | yes      | —       | Name of input elevation raster map                                                  |
| coordinates | point  | no       | 0,0     | Coordinate for which you want to calculate the horizon                              |
| direction   | number | no       | —       | Direction in which you want to know the horizon height                              |
| step        | number | no       | —       | Angle step size for multidirectional horizon                                        |
| start       | number | no       | 0.0     | Start angle for multidirectional horizon                                            |
| end         | number | no       | 360.0   | End angle for multidirectional horizon                                              |
| bufferzone  | number | no       | —       | For horizon rasters, read from the DEM an extra buffer around the present region    |
| e_buff      | number | no       | —       | For horizon rasters, read from the DEM an extra buffer eastward the present region  |
| w_buff      | number | no       | —       | For horizon rasters, read from the DEM an extra buffer westward the present region  |
| n_buff      | number | no       | —       | For horizon rasters, read from the DEM an extra buffer northward the present region |
| s_buff      | number | no       | —       | For horizon rasters, read from the DEM an extra buffer southward the present region |

| Parameter                                | Type            | Required | Default     | Description                                                      |
|------------------------------------------|-----------------|----------|-------------|------------------------------------------------------------------|
| maxdistance                              | number          | no       | —           | The maximum distance to consider when finding the horizon height |
| distance                                 | number          | no       | 1.0         | Sampling distance step coefficient                               |
| -d                                       | boolean         | no       | False       | Write output in degrees (default is radians)                     |
| -c                                       | boolean         | no       | False       | Write output in compass orientation (default is CCW, East=0)     |
| html                                     | fileDestination | no       | report.html | Horizon                                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default)                  |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.horizon.height elevation=dem.tif coordinates=100,100 direction=10 step=10
↳ start=0 end=360 bufferzone=10 html=html.txt
```

This calls **r.horizon.height**: elevation=dem.tif — Name of input elevation raster map; coordinates=100,100 — Coordinate for which you want to calculate the horizon; direction=10 — Direction in which you want to know the horizon height; step=10 — Angle step size for multidirectional horizon; start=0 — Start angle for multidirectional horizon; end=360 — End angle for multidirectional horizon; bufferzone=10 — For horizon rasters, read from the DEM an extra buffer around the present region; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.in.lidar — r.in.lidar

**group:** Raster (r.\*)

Creates a raster map from LAS LiDAR points using univariate statistics.

| Parameter       | Type    | Required | Default | Description                                                                                                                                                                   |
|-----------------|---------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input           | file    | yes      | —       | LAS input file                                                                                                                                                                |
| method          | enum    | no       | 5       | Statistic to use for raster values — options: 0=n, 1=min, 2=max, 3=range, 4=sum, 5=mean, 6=stddev, 7=variance, 8=coeff_var, 9=median, 10=percentile, 11=skewness, 12=trimmean |
| type            | enum    | no       | 1       | Storage type for resultant raster map — options: 0=CELL, 1=FCELL, 2=DCELL                                                                                                     |
| base_raster     | raster  | no       | —       | Subtract raster values from the Z coordinates                                                                                                                                 |
| zrange          | range   | no       | —       | Filter range for z data (min, max)                                                                                                                                            |
| zscale          | number  | no       | 1.0     | Scale to apply to z data                                                                                                                                                      |
| intensity_range | range   | no       | —       | Filter range for intensity values (min, max)                                                                                                                                  |
| intensity_scale | number  | no       | 1.0     | Scale to apply to intensity values                                                                                                                                            |
| percent         | number  | no       | 100.0   | Percent of map to keep in memory                                                                                                                                              |
| pth             | number  | no       | —       | pth percentile of the values (between 1 and 100)                                                                                                                              |
| trim            | number  | no       | —       | Discard percent of the smallest and percent of the largest observations (0-50)                                                                                                |
| resolution      | number  | no       | —       | Output raster resolution                                                                                                                                                      |
| return_filter   | string  | no       | —       | Only import points of selected return type<br>Options: first, last, mid                                                                                                       |
| class_filter    | string  | no       | —       | Only import points of selected class(es)<br>(comma separated integers)                                                                                                        |
| -e *(adv)*      | boolean | no       | False   | Use the extent of the input for the raster extent                                                                                                                             |

| Parameter                                | Type              | Required | Default | Description                                                     |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------|
| -n *(adv)*                               | boolean           | no       | False   | Set computation region to match the new raster map              |
| -o *(adv)*                               | boolean           | no       | False   | Override projection check (use current location's projection)   |
| -i *(adv)*                               | boolean           | no       | False   | Use intensity values rather than Z values                       |
| -j *(adv)*                               | boolean           | no       | False   | Use Z values for filtering, but intensity values for statistics |
| -d *(adv)*                               | boolean           | no       | False   | Use base raster resolution instead of computational region      |
| -v *(adv)*                               | boolean           | no       | False   | Use only valid points                                           |
| output                                   | rasterDestination | yes      | —       | Lidar Raster                                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                 |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                       |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.in.lidar input=input.dat method=5 type=1 base_raster=dem.tif zscale=1
↳ output=output.tif
```

This calls **r.in.lidar**: input=input.dat — LAS input file; method=5 selects *mean*; type=1 selects *FCELL*; base\_raster=dem.tif — Subtract raster values from the Z coordinates; zscale=1 — Scale to apply to z data; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.in.lidar.info — r.in.lidar.info

**group:** Raster (r.\*)

r.in.lidar.info - Extract information from LAS file

| Parameter                      | Type            | Required | Default     | Description             |
|--------------------------------|-----------------|----------|-------------|-------------------------|
| input                          | file            | yes      | —           | LAS input file          |
| html                           | fileDestination | no       | report.html | LAS information         |
| GRASS_REGION_PARAMETER *(adv)* | extent          | no       | —           | GRASS GIS region extent |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.in.lidar.info input=input.dat html=html.txt
```

This calls **r.in.lidar.info**: input=input.dat — LAS input file; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.info — r.info

**group:** Raster (r.\*)

Output basic information about a raster layer.

| Parameter | Type   | Required | Default | Description  |
|-----------|--------|----------|---------|--------------|
| map       | raster | yes      | —       | Raster layer |

| Parameter                                | Type            | Required | Default     | Description                                               |
|------------------------------------------|-----------------|----------|-------------|-----------------------------------------------------------|
| -r                                       | boolean         | no       | False       | Print range only                                          |
| -g                                       | boolean         | no       | False       | Print raster array information in shell script style      |
| -h                                       | boolean         | no       | False       | Print raster history instead of info                      |
| -e                                       | boolean         | no       | False       | Print extended metadata information in shell script style |
| html                                     | fileDestination | no       | report.html | Basic information                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default)           |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.info map=dem.tif html=html.txt
```

This calls **r.info**: map=dem.tif — Raster layer; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.kappa — r.kappa

**group:** Raster (r.\*)

Calculate error matrix and kappa parameter for accuracy assessment of classification result.

| Parameter                                | Type            | Required | Default             | Description                                     |
|------------------------------------------|-----------------|----------|---------------------|-------------------------------------------------|
| classification                           | raster          | yes      | —                   | Raster layer containing classification result   |
| reference                                | raster          | yes      | —                   | Raster layer containing reference classes       |
| title                                    | string          | no       | ACCURACY ASSESSMENT | Title for error matrix and kappa                |
| -h                                       | boolean         | no       | False               | No header in the report                         |
| -w                                       | boolean         | no       | False               | Wide report (132 columns)                       |
| output                                   | fileDestination | yes      | —                   | Error matrix and kappa                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —                   | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0                 | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.kappa classification=dem.tif reference=dem.tif title="ACCURACY ASSESSMENT"
↪ output=output.txt
```

This calls **r.kappa**: classification=dem.tif — Raster layer containing classification result; reference=dem.tif — Raster layer containing reference classes; title="ACCURACY ASSESSMENT" — Title for error matrix and kappa; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.lake — r.lake

**group:** Raster (r.\*)

Fills lake at given point to given level.

| Parameter                                | Type              | Required | Default | Description                                               |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------|
| elevation                                | raster            | yes      | —       | Elevation                                                 |
| water_level                              | number            | yes      | —       | Water level                                               |
| coordinates                              | point             | no       | —       | Seed point coordinates                                    |
| seed                                     | raster            | no       | —       | Raster layer with starting point(s) (at least 1 cell > 0) |
| -n                                       | boolean           | no       | False   | Use negative depth values for lake raster layer           |
| lake                                     | rasterDestination | yes      | —       | Lake                                                      |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)           |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                 |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)          |

**Outputs:** lake (outputRaster)

### Example usage

```
run grass:r.lake elevation=dem.tif water_level=10 coordinates=100,100 seed=dem.tif
↳ lake=lake.tif
```

This calls **r.lake**: elevation=dem.tif — Elevation; water\_level=10 — Water level; coordinates=100,100 — Seed point coordinates; seed=dem.tif — Raster layer with starting point(s) (at least 1 cell > 0); lake=lake.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.latlong — r.latlong

**group:** Raster (r.\*)

Creates a latitude/longitude raster map.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                         |
| -l                                       | boolean           | no       | False   | Longitude output                                 |
| output                                   | rasterDestination | yes      | —       | LatLong                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.latlong input=dem.tif output=output.tif
```

This calls **r.latlong**: input=dem.tif — Name of input raster map; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.cwed — r.li.cwed

**group:** Raster (r.\*)

Calculates contrast weighted edge density index on a raster map

| Parameter                                | Type              | Required | Default | Description                                                  |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                                     |
| config_txt                               | string            | no       | —       | Landscape structure configuration                            |
| config                                   | file              | no       | —       | Landscape structure configuration file                       |
| path                                     | file              | yes      | —       | Name of file that contains the weight to calculate the index |
| output                                   | rasterDestination | yes      | —       | CWED                                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                      |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)              |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                    |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)             |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.cwed input=dem.tif path=input.dat config_txt=value config=input.dat
↳ output=output.tif
```

This calls **r.li.cwed**: input=dem.tif — Name of input raster map; path=input.dat — Name of file that contains the weight to calculate the index; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.cwed.ascii — r.li.cwed.ascii

**group:** Raster (r.\*)

r.li.cwed.ascii - Calculates contrast weighted edge density index on a raster map

| Parameter                                | Type            | Required | Default | Description                                                  |
|------------------------------------------|-----------------|----------|---------|--------------------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                                     |
| config_txt                               | string          | no       | —       | Landscape structure configuration                            |
| config                                   | file            | no       | —       | Landscape structure configuration file                       |
| path                                     | file            | yes      | —       | Name of file that contains the weight to calculate the index |
| output_txt                               | fileDestination | yes      | —       | CWED                                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                                      |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)              |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.cwed.ascii input=dem.tif path=input.dat config_txt=value config=input.dat
↳ output_txt=output_txt.txt
```

This calls **r.li.cwed.ascii**: input=dem.tif — Name of input raster map; path=input.dat — Name of file that contains the weight to calculate the index; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## grass:r.li.dominance — r.li.dominance

**group:** Raster (r.\*)

Calculates dominance's diversity index on a raster map

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                         |
| config_txt                               | string            | no       | —       | Landscape structure configuration                |
| config                                   | file              | no       | —       | Landscape structure configuration file           |
| output                                   | rasterDestination | yes      | —       | Dominance                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.dominance input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.dominance**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.dominance.ascii — r.li.dominance.ascii

**group:** Raster (r.\*)

r.li.dominance.ascii - Calculates dominance's diversity index on a raster map

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                        |
| config_txt                               | string          | no       | —       | Landscape structure configuration               |
| config                                   | file            | no       | —       | Landscape structure configuration file          |
| output_txt                               | fileDestination | yes      | —       | Dominance                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.dominance.ascii input=dem.tif config_txt=value config=input.dat  
↪ output_txt=output_txt.txt
```

This calls **r.li.dominance.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.edgedensity — r.li.edgedensity

**group:** Raster (r.\*)

Calculates edge density index on a raster map, using a 4 neighbour algorithm

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input raster map                         |
| config_txt                              | string            | no       | —       | Landscape structure configuration                |
| config                                  | file              | no       | —       | Landscape structure configuration file           |
| patch_type                              | string            | no       | —       | The value of the patch type                      |
| -b                                      | boolean           | no       | False   | Exclude border edges                             |
| output                                  | rasterDestination | yes      | —       | Edge Density                                     |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.edgedensity input=dem.tif config_txt=value config=input.dat
↳ patch_type=value output=output.tif
```

This calls **r.li.edgedensity**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; patch\_type=value — The value of the patch type; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.edgedensity.ascii — r.li.edgedensity.ascii

**group:** Raster (r.\*)

r.li.edgedensity.ascii - Calculates edge density index on a raster map, using a 4 neighbour algorithm

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| config_txt                              | string          | no       | —       | Landscape structure configuration               |
| config                                  | file            | no       | —       | Landscape structure configuration file          |
| patch_type                              | string          | no       | —       | The value of the patch type                     |
| -b                                      | boolean         | no       | False   | Exclude border edges                            |
| output_txt                              | fileDestination | yes      | —       | Edge Density                                    |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.edgedensity.ascii input=dem.tif config_txt=value config=input.dat
↳ patch_type=value output_txt=output_txt.txt
```

This calls **r.li.edgedensity.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; patch\_type=value — The value of the patch type; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.mpa — r.li.mpa

**group:** Raster (r.\*)

Calculates mean pixel attribute index on a raster map

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input raster map                         |
| config_txt                              | string            | no       | —       | Landscape structure configuration                |
| config                                  | file              | no       | —       | Landscape structure configuration file           |
| output                                  | rasterDestination | yes      | —       | Mean Pixel Attribute                             |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.mpa input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.mpa**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.mpa.ascii — r.li.mpa.ascii

**group:** Raster (r.\*)

r.li.mpa.ascii - Calculates mean pixel attribute index on a raster map

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| config_txt                              | string          | no       | —       | Landscape structure configuration               |
| config                                  | file            | no       | —       | Landscape structure configuration file          |
| output_txt                              | fileDestination | yes      | —       | Mean Pixel Attribute                            |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.mpa.ascii input=dem.tif config_txt=value config=input.dat  
↪ output_txt=output_txt.txt
```

This calls **r.li.mpa.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.mps — r.li.mps

**group:** Raster (r.\*)

Calculates mean patch size index on a raster map, using a 4 neighbour algorithm

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input raster map                         |
| config_txt                              | string            | no       | —       | Landscape structure configuration                |
| config                                  | file              | no       | —       | Landscape structure configuration file           |
| output                                  | rasterDestination | yes      | —       | Mean Patch Size                                  |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.mps input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.mps**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.mps.ascii — r.li.mps.ascii

**group:** Raster (r.\*)

r.li.mps.ascii - Calculates mean patch size index on a raster map, using a 4 neighbour algorithm

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| config_txt                              | string          | no       | —       | Landscape structure configuration               |
| config                                  | file            | yes      | —       | Landscape structure configuration file          |
| output_txt                              | fileDestination | yes      | —       | Mean Patch Size                                 |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.mps.ascii input=dem.tif config=input.dat config_txt=value
↪ output_txt=output_txt.txt
```

This calls **r.li.mps.ascii**: input=dem.tif — Name of input raster map; config=input.dat — Landscape structure configuration file; config\_txt=value — Landscape structure configuration; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.padc — r.li.padc

**group:** Raster (r.\*)

Calculates coefficient of variation of patch area on a raster map

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                         |
| config_txt                               | string            | no       | —       | Landscape structure configuration                |
| config                                   | file              | no       | —       | Landscape structure configuration file           |
| output                                   | rasterDestination | yes      | —       | PADCV                                            |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.padcw input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.padcw**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.padcw.ascii — r.li.padcw.ascii

**group:** Raster (r.\*)

r.li.padcw.ascii - Calculates coefficient of variation of patch area on a raster map

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                        |
| config_txt                               | string          | no       | —       | Landscape structure configuration               |
| config                                   | file            | no       | —       | Landscape structure configuration file          |
| output_txt                               | fileDestination | yes      | —       | PADCV                                           |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.padcw.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.padcw.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.padrng — r.li.padrng

**group:** Raster (r.\*)

Calculates range of patch area size on a raster map

| Parameter  | Type   | Required | Default | Description                       |
|------------|--------|----------|---------|-----------------------------------|
| input      | raster | yes      | —       | Name of input raster map          |
| config_txt | string | no       | —       | Landscape structure configuration |

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| config                                   | file              | no       | —       | Landscape structure configuration file           |
| output                                   | rasterDestination | yes      | —       | Pad Range                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.padrage input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.padrage**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.padrage.ascii — r.li.padrage.ascii

**group:** Raster (r.\*)

r.li.padrage.ascii - Calculates range of patch area size on a raster map

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                        |
| config_txt                               | string          | no       | —       | Landscape structure configuration               |
| config                                   | file            | no       | —       | Landscape structure configuration file          |
| output_txt                               | fileDestination | yes      | —       | Pad Range                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.padrage.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.padrage.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.padsd — r.li.padsd

**group:** Raster (r.\*)

Calculates standard deviation of patch area a raster map

| Parameter  | Type              | Required | Default | Description                            |
|------------|-------------------|----------|---------|----------------------------------------|
| input      | raster            | yes      | —       | Name of input raster map               |
| config_txt | string            | no       | —       | Landscape structure configuration      |
| config     | file              | no       | —       | Landscape structure configuration file |
| output     | rasterDestination | yes      | —       | Patch Area SD                          |

| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.padsd input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.padsd**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.padsd.ascii — r.li.padsd.ascii

**group:** Raster (r.\*)

r.li.padsd.ascii - Calculates standard deviation of patch area a raster map

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                        |
| config_txt                               | string          | no       | —       | Landscape structure configuration               |
| config                                   | file            | no       | —       | Landscape structure configuration file          |
| output_txt                               | fileDestination | yes      | —       | Patch Area SD                                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.padsd.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.padsd.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.patchdensity — r.li.patchdensity

**group:** Raster (r.\*)

Calculates patch density index on a raster map, using a 4 neighbour algorithm

| Parameter                      | Type              | Required | Default | Description                            |
|--------------------------------|-------------------|----------|---------|----------------------------------------|
| input                          | raster            | yes      | —       | Name of input raster map               |
| config_txt                     | string            | no       | —       | Landscape structure configuration      |
| config                         | file              | no       | —       | Landscape structure configuration file |
| output                         | rasterDestination | yes      | —       | Patch Density                          |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                |

| Parameter                               | Type   | Required | Default | Description                                      |
|-----------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.patchdensity input=dem.tif config_txt=value config=input.dat
↪ output=output.tif
```

This calls **r.li.patchdensity**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.patchdensity.ascii — r.li.patchdensity.ascii

**group:** Raster (r.\*)

r.li.patchdensity.ascii - Calculates patch density index on a raster map, using a 4 neighbour algorithm

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| config_txt                              | string          | no       | —       | Landscape structure configuration               |
| config                                  | file            | no       | —       | Landscape structure configuration file          |
| output_txt                              | fileDestination | yes      | —       | Patch Density                                   |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.patchdensity.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.patchdensity.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.patchnum — r.li.patchnum

**group:** Raster (r.\*)

Calculates patch number index on a raster map, using a 4 neighbour algorithm.

| Parameter                      | Type              | Required | Default | Description                            |
|--------------------------------|-------------------|----------|---------|----------------------------------------|
| input                          | raster            | yes      | —       | Name of input raster map               |
| config_txt                     | string            | no       | —       | Landscape structure configuration      |
| config                         | file              | no       | —       | Landscape structure configuration file |
| output                         | rasterDestination | yes      | —       | Patch Number                           |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                |



| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.patchnum input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.patchnum**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.patchnum.ascii — r.li.patchnum.ascii

**group:** Raster (r.\*)

r.li.patchnum.ascii - Calculates patch number index on a raster map, using a 4 neighbour algorithm.

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                        |
| config_txt                               | string          | no       | —       | Landscape structure configuration               |
| config                                   | file            | no       | —       | Landscape structure configuration file          |
| output_txt                               | fileDestination | yes      | —       | Patch Number                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.patchnum.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.patchnum.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.pielou — r.li.pielou

**group:** Raster (r.\*)

Calculates Pielou's diversity index on a raster map

| Parameter                                | Type              | Required | Default | Description                                     |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                        |
| config_txt                               | string            | no       | —       | Landscape structure configuration               |
| config                                   | file              | no       | —       | Landscape structure configuration file          |
| output                                   | rasterDestination | yes      | —       | Pielou                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_OPT *(adv)*  | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.pielou input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.pielou**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.pielou.ascii — r.li.pielou.ascii

**group:** Raster (r.\*)

r.li.pielou.ascii - Calculates Pielou's diversity index on a raster map

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| config_txt                              | string          | no       | —       | Landscape structure configuration               |
| config                                  | file            | no       | —       | Landscape structure configuration file          |
| output_txt                              | fileDestination | yes      | —       | Pielou                                          |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.pielou.ascii input=dem.tif config_txt=value config=input.dat
```

```
↪ output_txt=output_txt.txt
```

This calls **r.li.pielou.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.renyi — r.li.renyi

**group:** Raster (r.\*)

Calculates Renyi's diversity index on a raster map

| Parameter                               | Type              | Required | Default | Description                                         |
|-----------------------------------------|-------------------|----------|---------|-----------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input raster map                            |
| config_txt                              | string            | no       | —       | Landscape structure configuration                   |
| config                                  | file              | no       | —       | Landscape structure configuration file              |
| alpha                                   | string            | yes      | —       | Alpha value is the order of the generalized entropy |
| output                                  | rasterDestination | yes      | —       | Renyi                                               |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                             |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)     |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_OPT *(adv)*  | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.renyi input=dem.tif alpha=value config_txt=value config=input.dat
↪ output=output.tif
```

This calls **r.li.renyi**: input=dem.tif — Name of input raster map; alpha=value — Alpha value is the order of the generalized entropy; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.renyi.ascii — r.li.renyi.ascii

**group:** Raster (r.\*)

r.li.renyi.ascii - Calculates Renyi's diversity index on a raster map

| Parameter                                | Type            | Required | Default | Description                                         |
|------------------------------------------|-----------------|----------|---------|-----------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                            |
| config_txt                               | string          | no       | —       | Landscape structure configuration                   |
| config                                   | file            | no       | —       | Landscape structure configuration file              |
| alpha                                    | string          | yes      | —       | Alpha value is the order of the generalized entropy |
| output_txt                               | fileDestination | yes      | —       | Renyi                                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                             |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)     |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.renyi.ascii input=dem.tif alpha=value config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.renyi.ascii**: input=dem.tif — Name of input raster map; alpha=value — Alpha value is the order of the generalized entropy; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.richness — r.li.richness

**group:** Raster (r.\*)

Calculates richness index on a raster map

| Parameter  | Type              | Required | Default | Description                            |
|------------|-------------------|----------|---------|----------------------------------------|
| input      | raster            | yes      | —       | Name of input raster map               |
| config_txt | string            | no       | —       | Landscape structure configuration      |
| config     | file              | no       | —       | Landscape structure configuration file |
| output     | rasterDestination | yes      | —       | Richness                               |

| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.richness input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.richness**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.richness.ascii — r.li.richness.ascii

**group:** Raster (r.\*)

r.li.richness.ascii - Calculates richness index on a raster map

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                        |
| config_txt                               | string          | no       | —       | Landscape structure configuration               |
| config                                   | file            | no       | —       | Landscape structure configuration file          |
| output_txt                               | fileDestination | yes      | —       | Richness                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.richness.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.richness.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.shannon — r.li.shannon

**group:** Raster (r.\*)

Calculates Shannon's diversity index on a raster map

| Parameter                      | Type              | Required | Default | Description                            |
|--------------------------------|-------------------|----------|---------|----------------------------------------|
| input                          | raster            | yes      | —       | Name of input raster map               |
| config_txt                     | string            | no       | —       | Landscape structure configuration      |
| config                         | file              | no       | —       | Landscape structure configuration file |
| output                         | rasterDestination | yes      | —       | Shannon                                |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                |

| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.shannon input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.shannon**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.shannon.ascii — r.li.shannon.ascii

**group:** Raster (r.\*)

r.li.shannon.ascii - Calculates Shannon's diversity index on a raster map

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Name of input raster map                        |
| config_txt                               | string          | no       | —       | Landscape structure configuration               |
| config                                   | file            | no       | —       | Landscape structure configuration file          |
| output_txt                               | fileDestination | yes      | —       | Shannon                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.shannon.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.shannon.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.shape — r.li.shape

**group:** Raster (r.\*)

Calculates shape index on a raster map

| Parameter                                | Type              | Required | Default | Description                                     |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                        |
| config_txt                               | string            | no       | —       | Landscape structure configuration               |
| config                                   | file              | no       | —       | Landscape structure configuration file          |
| output                                   | rasterDestination | yes      | —       | Shape                                           |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_OPT *(adv)*  | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.shape input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.shape**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.shape.ascii — r.li.shape.ascii

**group:** Raster (r.\*)

r.li.shape.ascii - Calculates shape index on a raster map

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| config_txt                              | string          | no       | —       | Landscape structure configuration               |
| config                                  | file            | no       | —       | Landscape structure configuration file          |
| output_txt                              | fileDestination | yes      | —       | Shape                                           |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.shape.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.shape.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.simpson — r.li.simpson

**group:** Raster (r.\*)

Calculates Simpson's diversity index on a raster map

| Parameter                               | Type              | Required | Default | Description                                     |
|-----------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input raster map                        |
| config_txt                              | string            | no       | —       | Landscape structure configuration               |
| config                                  | file              | no       | —       | Landscape structure configuration file          |
| output                                  | rasterDestination | yes      | —       | Simpson                                         |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)       |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.li.simpson input=dem.tif config_txt=value config=input.dat output=output.tif
```

This calls **r.li.simpson**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.li.simpson.ascii — r.li.simpson.ascii

**group:** Raster (r.\*)

r.li.simpson.ascii - Calculates Simpson's diversity index on a raster map

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| config_txt                              | string          | no       | —       | Landscape structure configuration               |
| config                                  | file            | no       | —       | Landscape structure configuration file          |
| output_txt                              | fileDestination | yes      | —       | Simpson                                         |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output\_txt (outputFile)

### Example usage

```
run grass:r.li.simpson.ascii input=dem.tif config_txt=value config=input.dat
↪ output_txt=output_txt.txt
```

This calls **r.li.simpson.ascii**: input=dem.tif — Name of input raster map; config\_txt=value — Landscape structure configuration; config=input.dat — Landscape structure configuration file; output\_txt=output\_txt.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.mapcalc.simple — r.mapcalc.simple

**group:** Raster (r.\*)

Calculate new raster map from a r.mapcalc expression.

| Parameter                      | Type              | Required | Default | Description             |
|--------------------------------|-------------------|----------|---------|-------------------------|
| a                              | raster            | yes      | —       | Raster layer A          |
| b                              | raster            | no       | —       | Raster layer B          |
| c                              | raster            | no       | —       | Raster layer C          |
| d                              | raster            | no       | —       | Raster layer D          |
| e                              | raster            | no       | —       | Raster layer E          |
| f                              | raster            | no       | —       | Raster layer F          |
| expression                     | string            | no       | A*2     | Formula                 |
| output                         | rasterDestination | yes      | —       | Calculated              |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent |

| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.mapcalc.simple a=dem.tif b=dem.tif c=dem.tif d=dem.tif e=dem.tif f=dem.tif
↪ expression=A*2 output=output.tif
```

This calls **r.mapcalc.simple**: a=dem.tif — Raster layer A; b=dem.tif — Raster layer B; c=dem.tif — Raster layer C; d=dem.tif — Raster layer D; e=dem.tif — Raster layer E; f=dem.tif — Raster layer F; expression=A\*2 — Formula; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.mask.rast — r.mask.rast

**group:** Raster (r.\*)

r.mask.rast - Creates a MASK for limiting raster operation.

| Parameter                                | Type              | Required | Default | Description                                           |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------|
| raster                                   | raster            | yes      | —       | Name of raster map to use as mask                     |
| input                                    | raster            | yes      | —       | Name of raster map to which apply the mask            |
| maskcats                                 | string            | no       | *       | Raster values to use for mask. Format: 1 2 3 thru 7 * |
| -i *(adv)*                               | boolean           | no       | False   | Create inverse mask                                   |
| output                                   | rasterDestination | yes      | —       | Masked                                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                               |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)       |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)             |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)      |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.mask.rast raster=dem.tif input=dem.tif maskcats=* output=output.tif
```

This calls **r.mask.rast**: raster=dem.tif — Name of raster map to use as mask; input=dem.tif — Name of raster map to which apply the mask; maskcats=\* — Raster values to use for mask. Format: 1 2 3 thru 7\*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.mask.vect — r.mask.vect

**group:** Raster (r.\*)

r.mask.vect - Creates a MASK for limiting raster operation with a vector layer.

| Parameter    | Type   | Required | Default | Description                                |
|--------------|--------|----------|---------|--------------------------------------------|
| vector       | source | yes      | —       | Name of vector map to use as mask          |
| input        | raster | yes      | —       | Name of raster map to which apply the mask |
| cats *(adv)* | string | no       | —       | Category values. Example: 1,3,7-9,13       |



| Parameter                                | Type              | Required | Default | Description                                               |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------|
| where *(adv)*                            | string            | no       | —       | WHERE conditions of SQL statement without 'where' keyword |
| -i *(adv)*                               | boolean           | no       | False   | Create inverse mask                                       |
| output                                   | rasterDestination | yes      | —       | Masked                                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)           |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                 |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)          |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                    |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                         |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.mask.vect vector=input.gpkg input=dem.tif output=output.tif
```

This calls **r.mask.vect**: vector=input.gpkg — Name of vector map to use as mask; input=dem.tif — Name of raster map to which apply the mask; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.mfilter — r.mfilter

**group:** Raster (r.\*)

Performs raster map matrix filter.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Input layer                                      |
| filter                                   | file              | yes      | —       | Filter file                                      |
| repeat                                   | number            | no       | 1.0     | Number of times to repeat the filter             |
| -z                                       | boolean           | no       | False   | Apply filter only to zero data values            |
| output                                   | rasterDestination | yes      | —       | Filtered                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.mfilter input=dem.tif filter=input.dat repeat=1 output=output.tif
```

This calls **r.mfilter**: input=dem.tif — Input layer; filter=input.dat — Filter file; repeat=1 — Number of times to repeat the filter; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.mode — r.mode

**group:** Raster (r.\*)

Finds the mode of values in a cover layer within areas assigned the same category value in a user-specified base layer.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| base                                     | raster            | yes      | —       | Base layer to be reclassified                    |
| cover                                    | raster            | yes      | —       | Categories layer                                 |
| output                                   | rasterDestination | yes      | —       | Mode                                             |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.mode base=dem.tif cover=dem.tif output=output.tif
```

This calls **r.mode**: base=dem.tif — Base layer to be reclassified; cover=dem.tif — Categories layer; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.neighbors — r.neighbors

**group:** Raster (r.\*)

Makes each cell category value a function of the category values assigned to the cells around it

| Parameter                                | Type              | Required | Default | Description                                                                                                                                                                                                       |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input raster layer                                                                                                                                                                                                |
| selection                                | raster            | no       | —       | Raster layer to select the cells which should be processed                                                                                                                                                        |
| method                                   | enum              | no       | 0       | Neighborhood operation — options: 0=average, 1=median, 2=mode, 3=minimum, 4=maximum, 5=range, 6=stddev, 7=sum, 8=count, 9=variance, 10=diversity, 11=interspersions, 12=quart1, 13=quart3, 14=perc90, 15=quantile |
| size                                     | number            | no       | 3.0     | Neighborhood size (must be odd)                                                                                                                                                                                   |
| gauss                                    | number            | no       | —       | Sigma (in cells) for Gaussian filter                                                                                                                                                                              |
| quantile                                 | string            | no       | —       | Quantile to calculate for method=quantile                                                                                                                                                                         |
| -c                                       | boolean           | no       | False   | Use circular neighborhood                                                                                                                                                                                         |
| -a *(adv)*                               | boolean           | no       | False   | Do not align output with the input                                                                                                                                                                                |
| weight *(adv)*                           | file              | no       | —       | File containing weights                                                                                                                                                                                           |
| output                                   | rasterDestination | yes      | —       | Neighbors                                                                                                                                                                                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                                                                                           |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                                                                                   |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                                                                                                                         |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                                                                                                  |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.neighbors input=dem.tif selection=dem.tif method=0 size=3 gauss=10
↳ quantile=value output=output.tif
```

This calls **r.neighbors**: input=dem.tif — Input raster layer; selection=dem.tif — Raster layer to select the cells which should be processed; method=0 selects *average*; size=3 — Neighborhood size (must be odd); gauss=10 — Sigma (in cells) for Gaussian filter; quantile=value — Quantile

to calculate for method=quantile; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.null — r.null

**group:** Raster (r.\*)

Manages NULL-values of given raster map.

| Parameter                                | Type              | Required | Default | Description                                                       |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------|
| map                                      | raster            | yes      | —       | Name of raster map for which to edit null values                  |
| setnull                                  | string            | no       | —       | List of cell values to be set to NULL                             |
| null                                     | number            | no       | —       | The value to replace the null value by                            |
| -f *(adv)*                               | boolean           | no       | False   | Only do the work if the map is floating-point                     |
| -i *(adv)*                               | boolean           | no       | False   | Only do the work if the map is integer                            |
| -n *(adv)*                               | boolean           | no       | False   | Only do the work if the map doesn't have a NULL-value bitmap file |
| -c *(adv)*                               | boolean           | no       | False   | Create NULL-value bitmap file validating all data cells           |
| -r *(adv)*                               | boolean           | no       | False   | Remove NULL-value bitmap file                                     |
| output                                   | rasterDestination | yes      | —       | NullRaster                                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                           |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                   |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                         |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                  |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.null map=dem.tif setnull=value null=10 output=output.tif
```

This calls **r.null**: map=dem.tif — Name of raster map for which to edit null values; setnull=value — List of cell values to be set to NULL; null=10 — The value to replace the null value by; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.ascii — r.out.ascii

**group:** Raster (r.\*)

Export a raster layer into a GRASS ASCII text file

| Parameter                                | Type            | Required | Default | Description                                                                     |
|------------------------------------------|-----------------|----------|---------|---------------------------------------------------------------------------------|
| input                                    | raster          | yes      | —       | Input raster                                                                    |
| precision                                | number          | no       | —       | Number of significant digits                                                    |
| width                                    | number          | no       | —       | Number of values printed before wrapping a line (only SURFER or MODFLOW format) |
| null_value                               | string          | no       | *       | String to represent null cell (GRASS grid only)                                 |
| -s *(adv)*                               | boolean         | no       | False   | Write SURFER (Golden Software) ASCII grid                                       |
| -m *(adv)*                               | boolean         | no       | False   | Write MODFLOW (USGS) ASCII array                                                |
| -i *(adv)*                               | boolean         | no       | False   | Force output of integer values                                                  |
| output                                   | fileDestination | yes      | —       | GRASS Ascii                                                                     |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                                                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                 |

---

**Outputs:** output (outputFile)

**Example usage**

```
run grass:r.out.ascii input=dem.tif precision=10 width=10 null_value=* output=output.txt
```

This calls **r.out.ascii**: input=dem.tif — Input raster; precision=10 — Number of significant digits; width=10 — Number of values printed before wrapping a line (only SURFER or MODFLOW format); null\_value=\* — String to represent null cell (GRASS grid only); output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:r.out.gridatb — r.out.gridatb**

**group:** Raster (r.\*)

Exports GRASS raster map to GRIDATB.FOR map file (TOPMODEL)

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Name of input raster map                        |
| output                                  | fileDestination | yes      | —       | GRIDATB                                         |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

**Example usage**

```
run grass:r.out.gridatb input=dem.tif output=output.txt
```

This calls **r.out.gridatb**: input=dem.tif — Name of input raster map; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:r.out.mat — r.out.mat**

**group:** Raster (r.\*)

Exports a GRASS raster to a binary MAT-File

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                   | raster          | yes      | —       | Input raster                                    |
| output                                  | fileDestination | yes      | —       | MAT File                                        |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

**Example usage**

```
run grass:r.out.mat input=dem.tif output=output.txt
```

This calls **r.out.mat**: input=dem.tif — Input raster; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.mpeg — r.out.mpeg

**group:** Raster (r.\*)

Converts raster map series to MPEG movie

| Parameter                               | Type            | Required | Default | Description                                              |
|-----------------------------------------|-----------------|----------|---------|----------------------------------------------------------|
| view1                                   | multilayer      | yes      | —       | Name of input raster map(s) for view no.1                |
| view2                                   | multilayer      | no       | —       | Name of input raster map(s) for view no.2                |
| view3                                   | multilayer      | no       | —       | Name of input raster map(s) for view no.3                |
| view4                                   | multilayer      | no       | —       | Name of input raster map(s) for view no.4                |
| quality                                 | number          | no       | 3.0     | Quality factor (1 = highest quality, lowest compression) |
| output                                  | fileDestination | yes      | —       | MPEG file                                                |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                                  |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)          |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.mpeg view1="a.gpkg;b.gpkg" view2="a.gpkg;b.gpkg" view3="a.gpkg;b.gpkg"  
↪ view4="a.gpkg;b.gpkg" quality=3 output=output.txt
```

This calls **r.out.mpeg**: view1="a.gpkg;b.gpkg" — Name of input raster map(s) for view no.1; view2="a.gpkg;b.gpkg" — Name of input raster map(s) for view no.2; view3="a.gpkg;b.gpkg" — Name of input raster map(s) for view no.3; view4="a.gpkg;b.gpkg" — Name of input raster map(s) for view no.4; quality=3 — Quality factor (1 = highest quality, lowest compression); out=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.png — r.out.png

**group:** Raster (r.\*)

Export a GRASS raster map as a non-georeferenced PNG image

| Parameter                               | Type            | Required | Default | Description                                                     |
|-----------------------------------------|-----------------|----------|---------|-----------------------------------------------------------------|
| input                                   | raster          | yes      | —       | Input raster                                                    |
| compression                             | number          | no       | 6.0     | Compression level of PNG file (0 = none, 1 = fastest, 9 = best) |
| -t *(adv)*                              | boolean         | no       | False   | Make NULL cells transparent                                     |
| -w *(adv)*                              | boolean         | no       | False   | Output world file                                               |
| output                                  | fileDestination | yes      | —       | PNG File                                                        |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                 |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.png input=dem.tif compression=6 output=output.txt
```

This calls **r.out.png**: input=dem.tif — Input raster; compression=6 — Compression level of PNG file (0 = none, 1 = fastest, 9 = best); output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.pov — r.out.pov

**group:** Raster (r.\*)

Converts a raster map layer into a height-field file for POV-Ray

| Parameter                                | Type            | Required | Default | Description                                        |
|------------------------------------------|-----------------|----------|---------|----------------------------------------------------|
| input                                    | raster          | yes      | —       | Input raster                                       |
| hftype                                   | number          | no       | 0.0     | Height-field type (0=actual heights 1=normalized)  |
| bias                                     | number          | no       | —       | Elevation bias                                     |
| scale                                    | number          | no       | —       | Vertical scaling factor                            |
| output                                   | fileDestination | yes      | —       | Name of output povray file (TGA height field file) |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                            |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)    |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.pov input=dem.tif hftype=0 bias=10 scale=10 output=output.txt
```

This calls **r.out.pov**: input=dem.tif — Input raster; hftype=0 — Height-field type (0=actual heights 1=normalized); bias=10 — Elevation bias; scale=10 — Vertical scaling factor; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.ppm — r.out.ppm

**group:** Raster (r.\*)

Converts a raster layer to a PPM image file at the pixel resolution of the currently defined region.

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | raster          | yes      | —       | Input raster layer                              |
| -g                                       | boolean         | no       | True    | Output greyscale instead of color               |
| output                                   | fileDestination | yes      | —       | PPM                                             |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.ppm input=dem.tif output=output.txt
```

This calls **r.out.ppm**: input=dem.tif — Input raster layer; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.ppm3 — r.out.ppm3

**group:** Raster (r.\*)

Converts 3 GRASS raster layers (R,G,B) to a PPM image file

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| red                                     | raster          | yes      | —       | Name of raster map to be used for               |
| green                                   | raster          | yes      | —       | Name of raster map to be used for               |
| blue                                    | raster          | yes      | —       | Name of raster map to be used for               |
| -c                                      | boolean         | no       | False   | Add comments to describe the region             |
| output                                  | fileDestination | yes      | —       | Name for new PPM file                           |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.ppm3 red=dem.tif green=dem.tif blue=dem.tif output=output.txt
```

This calls **r.out.ppm3**: red=dem.tif — Name of raster map to be used for; green=dem.tif — Name of raster map to be used for; blue=dem.tif — Name of raster map to be used for; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.vrml — r.out.vrml

**group:** Raster (r.\*)

Export a raster layer to the Virtual Reality Modeling Language (VRML)

| Parameter                               | Type            | Required | Default | Description                                     |
|-----------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| elevation                               | raster          | yes      | —       | Elevation layer                                 |
| color                                   | raster          | yes      | —       | Color layer                                     |
| exaggeration                            | number          | no       | 1.0     | Vertical exaggeration                           |
| output                                  | fileDestination | yes      | —       | VRML                                            |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.vrml elevation=dem.tif color=dem.tif exaggeration=1 output=output.txt
```

This calls **r.out.vrml**: elevation=dem.tif — Elevation layer; color=dem.tif — Color layer; exaggeration=1 — Vertical exaggeration; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.vtk — r.out.vtk

**group:** Raster (r.\*)

Converts raster maps into the VTK-ASCII format

| Parameter | Type       | Required | Default   | Description                                           |
|-----------|------------|----------|-----------|-------------------------------------------------------|
| input     | multilayer | yes      | —         | Input raster                                          |
| elevation | raster     | no       | —         | Input elevation raster map                            |
| null      | number     | no       | -99999.99 | Value to represent no data cell                       |
| z         | number     | no       | 0.0       | Constant elevation (if no elevation map is specified) |

| Parameter                                | Type            | Required | Default | Description                                                    |
|------------------------------------------|-----------------|----------|---------|----------------------------------------------------------------|
| rgbmaps                                  | multilayer      | no       | —       | Three (r,g,b) raster maps to create RGB values                 |
| vectormaps                               | multilayer      | no       | —       | Three (x,y,z) raster maps to create vector values              |
| zscale                                   | number          | no       | 1.0     | Scale factor for elevation                                     |
| precision                                | number          | no       | 12.0    | Number of significant digits                                   |
| -p *(adv)*                               | boolean         | no       | False   | Create VTK point data instead of VTK cell data                 |
| -s *(adv)*                               | boolean         | no       | False   | Use structured grid for elevation (not recommended)            |
| -t *(adv)*                               | boolean         | no       | False   | Use polydata-trianglestrips for elevation grid creation        |
| -v *(adv)*                               | boolean         | no       | False   | Use polydata-vertices for elevation grid creation              |
| -o *(adv)*                               | boolean         | no       | False   | Scale factor affects the origin (if no elevation map is given) |
| -c *(adv)*                               | boolean         | no       | False   | Correct the coordinates to match the VTK-OpenGL precision      |
| output                                   | fileDestination | yes      | —       | VTK File                                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                                        |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.vtk input="a.gpkg;b.gpkg" elevation=dem.tif null=-100000 z=0
↪ rgbmaps="a.gpkg;b.gpkg" vectormaps="a.gpkg;b.gpkg" zscale=1 output=output.txt
```

This calls **r.out.vtk**: input="a.gpkg;b.gpkg" — Input raster; elevation=dem.tif — Input elevation raster map; null=-100000 — Value to represent no data cell; z=0 — Constant elevation (if no elevation map is specified); rgbmaps="a.gpkg;b.gpkg" — Three (r,g,b) raster maps to create RGB values; vectormaps="a.gpkg;b.gpkg" — Three (x,y,z) raster maps to create vector values; zscale=1 — Scale factor for elevation; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.out.xyz — r.out.xyz

**group:** Raster (r.\*)

Exports a raster map to a text file as x,y,z values based on cell centers

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| input                                    | multilayer      | yes      | —       | Input raster(s)                                 |
| separator                                | string          | no       | pipe    | Field separator                                 |
| -i *(adv)*                               | boolean         | no       | False   | Include no data values                          |
| output                                   | fileDestination | yes      | —       | XYZ File                                        |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.out.xyz input="a.gpkg;b.gpkg" separator=pipe output=output.txt
```

This calls **r.out.xyz**: input="a.gpkg;b.gpkg" — Input raster(s); separator=pipe — Field separator; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.



## grass:r.param.scale — r.param.scale

**group:** Raster (r.\*)

Extracts terrain parameters from a DEM.

| Parameter                                | Type              | Required | Default | Description                                                                                                                                                |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                                                                                                                                   |
| slope_tolerance                          | number            | no       | 1.0     | Slope tolerance that defines a 'flat' surface (degrees)                                                                                                    |
| curvature_tolerance                      | number            | no       | 0.0001  | Curvature tolerance that defines 'planar' surface                                                                                                          |
| size                                     | number            | no       | 3.0     | Size of processing window (odd number only, max: 499)                                                                                                      |
| method                                   | enum              | no       | 0       | Morphometric parameter in 'size' window to calculate — options: 0=elev, 1=slope, 2=aspect, 3=profc, 4=planc, 5=longc, 6=crosc, 7=minic, 8=maxic, 9=feature |
| exponent                                 | number            | no       | 0.0     | Exponent for distance weighting (0.0-4.0)                                                                                                                  |
| zscale                                   | number            | no       | 1.0     | Vertical scaling factor                                                                                                                                    |
| -c                                       | boolean           | no       | False   | Constrain model through central window cell                                                                                                                |
| output                                   | rasterDestination | yes      | —       | Morphometric parameter                                                                                                                                     |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                                    |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                            |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                                                                  |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                                           |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.param.scale input=dem.tif slope_tolerance=1 curvature_tolerance=0.0001 size=3  
↪ method=0 exponent=0 zscale=1 output=output.tif
```

This calls **r.param.scale**: input=dem.tif — Name of input raster map; slope\_tolerance=1 — Slope tolerance that defines a 'flat' surface (degrees); curvature\_tolerance=0.0001 — Curvature tolerance that defines 'planar' surface; size=3 — Size of processing window (odd number only, max: 499); method=0 selects *elev*; exponent=0 — Exponent for distance weighting (0.0-4.0); zscale=1 — Vertical scaling factor; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.patch — r.patch

**group:** Raster (r.\*)

Creates a composite raster layer by using one (or more) layer(s) to fill in areas of "no data" in another map layer.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Raster layers to be patched together             |
| -z                                       | boolean           | no       | False   | Use zero (0) for transparency instead of NULL    |
| output                                   | rasterDestination | yes      | —       | Patched                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

---

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.patch input="a.gpkg;b.gpkg" output=output.tif
```

This calls **r.patch**: input="a.gpkg;b.gpkg" — Raster layers to be patched together; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.path — r.path

**group:** Raster (r.\*)

Traces paths from starting points following input directions.

| Parameter                               | Type              | Required | Default | Description                                                                                             |
|-----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input direction                                                                                 |
| format                                  | enum              | no       | 0       | Format of the input direction map — options: 0=auto, 1=degree, 2=45degree, 3=bitmask                    |
| values                                  | raster            | no       | —       | Name of input raster values to be used for output                                                       |
| raster_path                             | rasterDestination | yes      | —       | Name for output raster path map                                                                         |
| vector_path                             | vectorDestination | yes      | —       | Name for output vector path map                                                                         |
| start_points                            | source            | yes      | —       | Vector layer containing starting point(s)                                                               |
| -c                                      | boolean           | no       | False   | Copy input cell values on output                                                                        |
| -a                                      | boolean           | no       | False   | Accumulate input values along the path                                                                  |
| -n                                      | boolean           | no       | False   | Count cell numbers along the path                                                                       |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*  | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*        | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*     | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*               | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*       | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** raster\_path (outputRaster), vector\_path (outputVector)

**Example usage**

```
run grass:r.path input=dem.tif start_points=input.gpkg format=0 values=dem.tif
↪ raster_path=raster_path.tif
```

This calls **r.path**: input=dem.tif — Name of input direction; start\_points=input.gpkg — Vector layer containing starting point(s); format=0 selects *auto*; values=dem.tif — Name of input raster values to be used for output; raster\_path=raster\_path.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.path.coordinate.txt — r.path.coordinate.txt

**group:** Raster (r.\*)

r.path.coordinate.txt - Traces paths from starting points following input directions.

| Parameter                               | Type              | Required | Default | Description                                                                                             |
|-----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input direction                                                                                 |
| format                                  | enum              | no       | 0       | Format of the input direction map — options: 0=auto, 1=degree, 2=45degree, 3=bitmask                    |
| values                                  | raster            | no       | —       | Name of input raster values to be used for output                                                       |
| raster_path                             | rasterDestination | yes      | —       | Name for output raster path map                                                                         |
| vector_path                             | vectorDestination | yes      | —       | Name for output vector path map                                                                         |
| start_coordinates                       | point             | yes      | —       | Map coordinate of starting point (E,N)                                                                  |
| -c                                      | boolean           | no       | False   | Copy input cell values on output                                                                        |
| -a                                      | boolean           | no       | False   | Accumulate input values along the path                                                                  |
| -n                                      | boolean           | no       | False   | Count cell numbers along the path                                                                       |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*     | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*               | string            | no       | —       | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*                | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*       | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** raster\_path (outputRaster), vector\_path (outputVector)

### Example usage

```
run grass:r.path.coordinate.txt input=dem.tif start_coordinates=100,100 format=0  
↪ values=dem.tif raster_path=raster_path.tif
```

This calls **r.path.coordinate.txt**: input=dem.tif — Name of input direction; start\_coordinates=100,100 — Map coordinate of starting point (E,N); format=0 selects *auto*; values=dem.tif — Name of input raster values to be used for output; raster\_path=raster\_path.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.plane — r.plane

**group:** Raster (r.\*)

Creates raster plane layer given dip (inclination), aspect (azimuth) and one point.

| Parameter | Type              | Required | Default | Description                                                      |
|-----------|-------------------|----------|---------|------------------------------------------------------------------|
| dip       | number            | no       | 0.0     | Dip of plane                                                     |
| azimuth   | number            | no       | 0.0     | Azimuth of the plane                                             |
| easting   | number            | yes      | —       | Easting coordinate of a point on the plane                       |
| northing  | number            | yes      | —       | Northing coordinate of a point on the plane                      |
| elevation | number            | yes      | —       | Elevation coordinate of a point on the plane                     |
| type      | enum              | no       | 1       | Data type of resulting layer — options: 0=CELL, 1=FCELL, 2=DCELL |
| output    | rasterDestination | yes      | —       | Plane                                                            |

| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.plane easting=10 northing=10 elevation=10 dip=0 azimuth=0 type=1
↳ output=output.tif
```

This calls **r.plane**: easting=10 — Easting coordinate of a point on the plane; northing=10 — Northing coordinate of a point on the plane; elevation=10 — Elevation coordinate of a point on the plane; dip=0 — Dip of plane; azimuth=0 — Azimuth of the plane; type=1 selects *FCELL*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.profile — r.profile

**group:** Raster (r.\*)

Outputs the raster layer values lying on user-defined line(s).

| Parameter                                | Type            | Required | Default | Description                                                            |
|------------------------------------------|-----------------|----------|---------|------------------------------------------------------------------------|
| input                                    | raster          | yes      | —       | Input raster layer                                                     |
| coordinates                              | string          | no       | 0,0,1,1 | Profile coordinate pairs                                               |
| resolution                               | number          | no       | —       | Resolution along profile                                               |
| null_value                               | string          | no       | *       | Character to represent no data cell                                    |
| file                                     | file            | no       | —       | Name of input file containing coordinate pairs                         |
| -g                                       | boolean         | no       | False   | Output easting and northing in first two columns of four column output |
| -c                                       | boolean         | no       | False   | Output RRR:GGG:BBB color values for each profile point                 |
| output                                   | fileDestination | yes      | —       | Profile                                                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                                                |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                        |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.profile input=dem.tif coordinates=0,0,1,1 resolution=10 null_value=*
↳ file=input.dat output=output.txt
```

This calls **r.profile**: input=dem.tif — Input raster layer; coordinates=0,0,1,1 — Profile coordinate pairs; resolution=10 — Resolution along profile; null\_value=\* — Character to represent no data cell; file=input.dat — Name of input file containing coordinate pairs; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.proj — r.proj

**group:** Raster (r.\*)

Re-projects a raster layer to another coordinate reference system

| Parameter                                | Type              | Required | Default | Description                                                                                                                |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input raster to reproject                                                                                                  |
| crs                                      | crs               | yes      | —       | New coordinate reference system                                                                                            |
| method                                   | enum              | no       | 0       | Interpolation method to use — options: 0=nearest, 1=bilinear, 2=bicubic, 3=lanczos, 4=bilinear_f, 5=bicubic_f, 6=lanczos_f |
| memory                                   | number            | no       | 300.0   | Maximum memory to be used (in MB)                                                                                          |
| resolution                               | number            | no       | —       | Resolution of output raster map                                                                                            |
| -n *(adv)*                               | boolean           | no       | False   | Do not perform region cropping optimization                                                                                |
| output                                   | rasterDestination | yes      | —       | Reprojected raster                                                                                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                    |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                            |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                                  |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                           |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.proj input=dem.tif crs=EPSG:6346 method=0 memory=300 resolution=10
↳ output=output.tif
```

This calls **r.proj**: input=dem.tif — Input raster to reproject; crs=EPSG:6346 — New coordinate reference system; method=0 selects *nearest*; memory=300 — Maximum memory to be used (in MB); resolution=10 — Resolution of output raster map; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.quant — r.quant

**group:** Raster (r.\*)

Produces the quantization file for a floating-point map.

| Parameter                                | Type              | Required | Default | Description                                     |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Raster layer(s) to be quantized                 |
| basemap                                  | raster            | no       | —       | Base layer to take quant rules from             |
| fprange                                  | range             | no       | —       | Floating point range: dmin,dmax                 |
| range                                    | range             | no       | —       | Integer range: min,max                          |
| -t                                       | boolean           | no       | False   | Truncate floating point data                    |
| -r                                       | boolean           | no       | False   | Round floating point data                       |
| output                                   | folderDestination | yes      | —       | Quantized raster(s)                             |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)          |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                               |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:r.quant input="a.gpkg;b.gpkg" basemap=dem.tif output=output/
```

This calls **r.quant**: input="a.gpkg;b.gpkg" — Raster layer(s) to be quantized; basemap=dem.tif — Base layer to take quant rules from; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.quantile — r.quantile

**group:** Raster (r.\*)

Compute quantiles using two passes.

| Parameter                                | Type            | Required | Default     | Description                                               |
|------------------------------------------|-----------------|----------|-------------|-----------------------------------------------------------|
| input                                    | raster          | yes      | —           | Input raster layer                                        |
| quantiles                                | number          | no       | 4.0         | Number of quantiles                                       |
| percentiles                              | string          | no       | —           | List of percentiles                                       |
| bins                                     | number          | no       | 1000000.0   | Number of bins to use                                     |
| -r *(adv)*                               | boolean         | no       | False       | Generate recode rules based on quantile-defined intervals |
| file                                     | fileDestination | no       | report.html | Quantiles                                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default)           |

**Outputs:** file (outputHtml)

### Example usage

```
run grass:r.quantile input=dem.tif quantiles=4 percentiles=value bins=1000000  
↪ file=file.txt
```

This calls **r.quantile**: input=dem.tif — Input raster layer; quantiles=4 — Number of quantiles; percentiles=value — List of percentiles; bins=1000000 — Number of bins to use; file=file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.quantile.plain — r.quantile.plain

**group:** Raster (r.\*)

r.quantile.plain - Compute quantiles using two passes and save them as plain text.

| Parameter                                | Type            | Required | Default    | Description                                               |
|------------------------------------------|-----------------|----------|------------|-----------------------------------------------------------|
| input                                    | raster          | yes      | —          | Input raster layer                                        |
| quantiles                                | number          | no       | 4.0        | Number of quantiles                                       |
| percentiles                              | string          | no       | —          | List of percentiles                                       |
| bins                                     | number          | no       | 1000000.0  | Number of bins to use                                     |
| -r *(adv)*                               | boolean         | no       | False      | Generate recode rules based on quantile-defined intervals |
| file                                     | fileDestination | no       | report.txt | Quantiles                                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —          | GRASS GIS region extent                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0        | GRASS GIS region cellsize (leave 0 for default)           |

**Outputs:** file (outputFile)

### Example usage

```
run grass:r.quantile.plain input=dem.tif quantiles=4 percentiles=value bins=1000000  
↪ file=file.txt
```

This calls **r.quantile.plain**: input=dem.tif — Input raster layer; quantiles=4 — Number of quantiles; percentiles=value — List of percentiles; bins=1000000 — Number of bins to use; file=file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.random — r.random

**group:** Raster (r.\*)

Creates a raster layer and vector point map containing randomly located points.

| Parameter                                | Type              | Required | Default | Description                                                                                             |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input raster layer                                                                                      |
| cover                                    | raster            | yes      | —       | Input cover raster layer                                                                                |
| npoints                                  | number            | yes      | —       | The number of points to allocate                                                                        |
| -z                                       | boolean           | no       | False   | Generate points also for NULL category                                                                  |
| -d                                       | boolean           | no       | False   | Generate vector points as 3D points                                                                     |
| -b                                       | boolean           | no       | False   | Do not build topology                                                                                   |
| raster                                   | rasterDestination | yes      | —       | Random raster                                                                                           |
| vector                                   | vectorDestination | yes      | —       | Random vector                                                                                           |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** raster (outputRaster), vector (outputVector)

### Example usage

```
run grass:r.random input=dem.tif cover=dem.tif npoints=10 raster=raster.tif
```

This calls **r.random**: input=dem.tif — Input raster layer; cover=dem.tif — Input cover raster layer; npoints=10 — The number of points to allocate; raster=raster.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.random.cells — r.random.cells

**group:** Raster (r.\*)

Generates random cell values with spatial dependence.

| Parameter                                | Type              | Required | Default | Description                                                    |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------|
| distance                                 | number            | no       | 0.0     | Maximum distance of spatial correlation (value(s) >= 0.0)      |
| ncells                                   | number            | no       | —       | Maximum number of cells to be created                          |
| seed *(adv)*                             | number            | no       | —       | Random seed (SEED_MIN >= value >= SEED_MAX) (default [random]) |
| output                                   | rasterDestination | yes      | —       | Random                                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                        |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                      |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)               |

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**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.random.cells distance=0 ncells=10 output=output.tif
```

This calls **r.random.cells**: distance=0 — Maximum distance of spatial correlation (value(s) >= 0.0); ncells=10 — Maximum number of cells to be created; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:r.random.surface — r.random.surface**

**group:** Raster (r.\*)

Generates random surface(s) with spatial dependence.

| Parameter                                | Type              | Required | Default | Description                                            |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------------|
| distance                                 | number            | no       | 0.0     | Maximum distance of spatial correlation                |
| exponent                                 | number            | no       | 1.0     | Distance decay exponent                                |
| flat                                     | number            | no       | 0.0     | Distance filter remains flat before beginning exponent |
| seed                                     | number            | no       | —       | Random seed (SEED_MIN >= value >= SEED_MAX)            |
| high                                     | number            | no       | 255.0   | Maximum cell value of distribution                     |
| -u                                       | boolean           | no       | False   | Uniformly distributed cell values                      |
| output                                   | rasterDestination | yes      | —       | Random_Surface                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)        |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)              |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)       |

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.random.surface distance=0 exponent=1 flat=0 seed=10 high=255 output=output.tif
```

This calls **r.random.surface**: distance=0 — Maximum distance of spatial correlation; exponent=1 — Distance decay exponent; flat=0 — Distance filter remains flat before beginning exponent; seed=10 — Random seed (SEED\_MIN >= value >= SEED\_MAX); high=255 — Maximum cell value of distribution; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:r.reclass — r.reclass**

**group:** Raster (r.\*)

Creates a new map layer whose category values are based upon a reclassification of the categories in an existing raster map layer.

| Parameter | Type   | Required | Default | Description                                |
|-----------|--------|----------|---------|--------------------------------------------|
| input     | raster | yes      | —       | Input raster layer                         |
| rules     | file   | no       | —       | File containing reclass rules              |
| txtrules  | string | no       | —       | Reclass rules text (if rule file not used) |



| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| output                                   | rasterDestination | yes      | —       | Reclassified                                     |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.reclass input=dem.tif rules=input.dat txrules=value output=output.tif
```

This calls **r.reclass**: input=dem.tif — Input raster layer; rules=input.dat — File containing reclass rules; txrules=value — Reclass rules text (if rule file not used); output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.reclass.area — r.reclass.area

**group:** Raster (r.\*)

Reclassifies a raster layer, greater or less than user specified area size (in hectares)

| Parameter                                | Type              | Required | Default | Description                                                           |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input raster layer                                                    |
| value                                    | number            | no       | 1.0     | Value option that sets the area size limit [hectares]                 |
| mode                                     | enum              | no       | 0       | Lesser or greater than specified value — options: 0=lesser, 1=greater |
| method                                   | enum              | no       | 0       | Method used for reclassification — options: 0=reclass, 1=rmare        |
| -c *(adv)*                               | boolean           | no       | False   | Input map is clumped                                                  |
| -d *(adv)*                               | boolean           | no       | False   | Clumps including diagonal neighbors                                   |
| output                                   | rasterDestination | yes      | —       | Reclassified                                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                               |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                       |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                             |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                      |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.reclass.area input=dem.tif value=1 mode=0 method=0 output=output.tif
```

This calls **r.reclass.area**: input=dem.tif — Input raster layer; value=1 — Value option that sets the area size limit [hectares]; mode=0 selects *lesser*; method=0 selects *reclass*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.recode — r.recode

**group:** Raster (r.\*)

Recodes categorical raster maps.

| Parameter                                | Type              | Required | Default   | Description                                      |
|------------------------------------------|-------------------|----------|-----------|--------------------------------------------------|
| input                                    | raster            | yes      | —         | Input layer                                      |
| rules                                    | file              | no       | NoneFalse | File containing recode rules                     |
| -d *(adv)*                               | boolean           | no       | False     | Force output to 'double' raster map type (DCELL) |
| -a *(adv)*                               | boolean           | no       | False     | Align the current region to the input raster map |
| output                                   | rasterDestination | yes      | —         | Recoded                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —         | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0       | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —         | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —         | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.recode input=dem.tif rules=input.dat output=output.tif
```

This calls **r.recode**: input=dem.tif — Input layer; rules=input.dat — File containing recode rules; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.regression.line — r.regression.line

**group:** Raster (r.\*)

Calculates linear regression from two raster layers :  $y = a + b \cdot x$ .

| Parameter                                | Type            | Required | Default     | Description                                     |
|------------------------------------------|-----------------|----------|-------------|-------------------------------------------------|
| mapx                                     | raster          | yes      | —           | Layer for x coefficient                         |
| mapy                                     | raster          | yes      | —           | Layer for y coefficient                         |
| html                                     | fileDestination | no       | report.html | Regression coefficients                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.regression.line mapx=dem.tif mapy=dem.tif html=html.txt
```

This calls **r.regression.line**: mapx=dem.tif — Layer for x coefficient; mapy=dem.tif — Layer for y coefficient; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.regression.multi — r.regression.multi

**group:** Raster (r.\*)

Calculates multiple linear regression from raster maps.

| Parameter | Type              | Required | Default | Description              |
|-----------|-------------------|----------|---------|--------------------------|
| mapx      | multilayer        | yes      | —       | Map(s) for x coefficient |
| mapy      | raster            | yes      | —       | Map for y coefficient    |
| residuals | rasterDestination | yes      | —       | Residual Map             |
| estimates | rasterDestination | yes      | —       | Estimates Map            |

| Parameter                                | Type            | Required | Default     | Description                                      |
|------------------------------------------|-----------------|----------|-------------|--------------------------------------------------|
| html                                     | fileDestination | no       | report.html | Regression coefficients                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string          | no       | —           | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string          | no       | —           | Output Rasters format metadata options (metaopt) |

**Outputs:** residuals (outputRaster), estimates (outputRaster), html (outputHtml)

### Example usage

```
run grass:r.regression.multi mapx="a.gpkg;b.gpkg" mapy=dem.tif residuals=residuals.tif
```

This calls **r.regression.multi**: mapx="a.gpkg;b.gpkg" — Map(s) for x coefficient; mapy=dem.tif — Map for y coefficient; residuals=residuals.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.relief — r.relief

**group:** Raster (r.\*)

Creates shaded relief from an elevation layer (DEM).

| Parameter                                | Type              | Required | Default | Description                                                          |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input elevation layer                                                |
| altitude                                 | number            | no       | 30.0    | Altitude of the sun in degrees above the horizon                     |
| azimuth                                  | number            | no       | 270.0   | Azimuth of the sun in degrees to the east of north                   |
| zscale                                   | number            | no       | 1.0     | Factor for exaggerating relief                                       |
| scale                                    | number            | no       | 1.0     | Scale factor for converting horizontal units to elevation units      |
| units                                    | enum              | no       | 0       | Elevation units (overrides scale factor) — options: 0=intl, 1=survey |
| output                                   | rasterDestination | yes      | —       | Output shaded relief layer                                           |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                              |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                      |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                            |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                     |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.relief input=dem.tif altitude=30 azimuth=270 zscale=1 scale=1 units=0
↪ output=output.tif
```

This calls **r.relief**: input=dem.tif — Input elevation layer; altitude=30 — Altitude of the sun in degrees above the horizon; azimuth=270 — Azimuth of the sun in degrees to the east of north; zscale=1 — Factor for exaggerating relief; scale=1 — Scale factor for converting horizontal units to elevation units; units=0 selects *intl*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.relief.scaling — r.relief.scaling

**group:** Raster (r.\*)

r.relief.scaling - Creates shaded relief from an elevation layer (DEM).

| Parameter                               | Type              | Required | Default | Description                                                          |
|-----------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Input elevation layer                                                |
| altitude                                | number            | no       | 30.0    | Altitude of the sun in degrees above the horizon                     |
| azimuth                                 | number            | no       | 270.0   | Azimuth of the sun in degrees to the east of north                   |
| zscale                                  | number            | no       | 1.0     | Factor for exaggerating relief                                       |
| scale                                   | number            | no       | 1.0     | Scale factor for converting horizontal units to elevation units      |
| units                                   | enum              | yes      | —       | Elevation units (overrides scale factor) — options: 0=intl, 1=survey |
| output                                  | rasterDestination | yes      | —       | Output shaded relief layer                                           |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                              |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                      |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                            |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                     |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.relief.scaling input=dem.tif units=0 altitude=30 azimuth=270 zscale=1 scale=1
↳ output=output.tif
```

This calls **r.relief.scaling**: input=dem.tif — Input elevation layer; units=0 selects *intl*; altitude=30 — Altitude of the sun in degrees above the horizon; azimuth=270 — Azimuth of the sun in degrees to the east of north; zscale=1 — Factor for exaggerating relief; scale=1 — Scale factor for converting horizontal units to elevation units; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.report — r.report

**group:** Raster (r.\*)

Reports statistics for raster layers.

| Parameter   | Type       | Required | Default | Description                                                      |
|-------------|------------|----------|---------|------------------------------------------------------------------|
| map         | multilayer | yes      | —       | Raster layer(s) to report on                                     |
| units       | enum       | no       | 1       | Units — options: 0=mi, 1=me, 2=k, 3=a, 4=h, 5=c, 6=p             |
| null_value  | string     | no       | *       | Character representing no data cell value                        |
| page_length | number     | no       | 0.0     | Page length                                                      |
| page_width  | number     | no       | 79.0    | Page width                                                       |
| nsteps      | number     | no       | 255.0   | Number of fp subranges to collect stats from                     |
| sort        | enum       | no       | 0       | Sort output statistics by cell counts — options: 0=asc, 1=desc   |
| -h          | boolean    | no       | False   | Suppress page headers                                            |
| -f          | boolean    | no       | False   | Use formfeeds between pages                                      |
| -e          | boolean    | no       | False   | Scientific format                                                |
| -n          | boolean    | no       | False   | Do not report no data cells                                      |
| -a          | boolean    | no       | False   | Do not report cells where all maps have no data                  |
| -c          | boolean    | no       | False   | Report for cats floating-point ranges (floating-point maps only) |
| -i          | boolean    | no       | False   | Read floating-point map as integer (use map's quant rules)       |

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| output                                   | fileDestination | no       | —       | Name for output file to hold the report         |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.report map="a.gpkg;b.gpkg" units=1 null_value=* page_length=0 page_width=79
↳ nsteps=255 sort=0 output=output.txt
```

This calls **r.report**: map="a.gpkg;b.gpkg" — Raster layer(s) to report on; units=1 selects *me*; null\_value=\* — Character representing no data cell value; page\_length=0 — Page length; page\_width=79 — Page width; nsteps=255 — Number of fp subranges to collect stats from; sort=0 selects *asc*; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.resamp.bspline — r.resamp.bspline

**group:** Raster (r.\*)

Performs bilinear or bicubic spline interpolation with Tykhonov regularization.

| Parameter                                | Type              | Required | Default | Description                                                                                             |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Input raster layer                                                                                      |
| mask                                     | raster            | no       | —       | Name of raster map to use for masking. Only cells that are not NULL and not zero are interpolated       |
| method                                   | enum              | no       | 1       | Sampling interpolation method — options: 0=bilinear, 1=bicubic                                          |
| ew_step                                  | number            | no       | —       | Length (float) of each spline step in the east-west direction                                           |
| ns_step                                  | number            | no       | —       | Length (float) of each spline step in the north-south direction                                         |
| lambda                                   | number            | no       | 0.01    | Tykhonov regularization parameter (affects smoothing)                                                   |
| memory                                   | number            | no       | 300.0   | Maximum memory to be used (in MB). Cache size for raster rows                                           |
| -n *(adv)*                               | boolean           | no       | False   | Only interpolate null cells in input raster map                                                         |
| -c *(adv)*                               | boolean           | no       | False   | Find the best Tykhonov regularizing parameter using a "leave-one-out" cross validation method           |
| output                                   | rasterDestination | yes      | —       | Resampled BSpline                                                                                       |
| grid                                     | vectorDestination | yes      | —       | Interpolation Grid                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputRaster), grid (outputVector)

## Example usage

```
run grass:r.resamp.bspline input=dem.tif mask=dem.tif method=1 ew_step=10 ns_step=10
↳ lambda=0.01 memory=300 output=output.tif
```

This calls **r.resamp.bspline**: `input=dem.tif` — Input raster layer; `mask=dem.tif` — Name of raster map to use for masking. Only cells that are not NULL and not...; `method=1` selects *bicubic*; `ew_step=10` — Length (float) of each spline step in the east-west direction; `ns_step=10` — Length (float) of each spline step in the north-south direction; `lambda=0.01` — Tykhonov regularization parameter (affects smoothing); `memory=300` — Maximum memory to be used (in MB). Cache size for raster rows; `output=output.tif` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.resamp.filter — r.resamp.filter

**group:** Raster (r.\*)

Resamples raster map layers using an analytic kernel.

| Parameter                               | Type              | Required | Default | Description                                                                                                                                              |
|-----------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer                                                                                                                                       |
| filter                                  | enum              | no       | [0]     | Filter kernel(s) — options: 0=box, 1=bartlett, 2=gauss, 3=normal, 4=hermite, 5=sinc, 6=lanczos1, 7=lanczos2, 8=lanczos3, 9=hann, 10=hamming, 11=blackman |
| radius                                  | string            | no       | —       | Filter radius for each filter (comma separated list of float if multiple)                                                                                |
| x_radius                                | string            | no       | —       | Filter radius (horizontal) for each filter (comma separated list of float if multiple)                                                                   |
| y_radius                                | string            | no       | —       | Filter radius (vertical) for each filter (comma separated list of float if multiple)                                                                     |
| -n *(adv)*                              | boolean           | no       | False   | Propagate NULLs                                                                                                                                          |
| output                                  | rasterDestination | yes      | —       | Resampled Filter                                                                                                                                         |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                                                                  |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                          |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                                                                                |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                                         |

**Outputs:** output (outputRaster)

## Example usage

```
run grass:r.resamp.filter input=dem.tif filter=0 radius=value x_radius=value
↳ y_radius=value output=output.tif
```

This calls **r.resamp.filter**: `input=dem.tif` — Input raster layer; `filter=0` selects *box*; `radius=value` — Filter radius for each filter (comma separated list of float if multiple); `x_radius=value` — Filter radius (horizontal) for each filter (comma separated list of float if...; `y_radius=value` — Filter radius (vertical) for each filter (comma separated list of float if...; `output=output.tif` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.resamp.interp — r.resamp.interp

**group:** Raster (r.\*)

Resamples raster map to a finer grid using interpolation.

| Parameter                               | Type              | Required | Default | Description                                                                          |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer                                                                   |
| method                                  | enum              | no       | 1       | Sampling interpolation method — options: 0=nearest, 1=bilinear, 2=bicubic, 3=lanczos |
| output                                  | rasterDestination | yes      | —       | Resampled interpolated                                                               |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                              |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                      |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                            |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                     |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.resamp.interp input=dem.tif method=1 output=output.tif
```

This calls **r.resamp.interp**: input=dem.tif — Input raster layer; method=1 selects *bilinear*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.resamp.rst — r.resamp.rst

**group:** Raster (r.\*)

Reinterpolates using regularized spline with tension and smoothing.

| Parameter                               | Type              | Required | Default | Description                                                  |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Raster layer                                                 |
| smooth                                  | raster            | no       | —       | Input raster map containing smoothing                        |
| maskmap                                 | raster            | no       | —       | Input raster map to be used as mask                          |
| ew_res                                  | number            | yes      | —       | Desired east-west resolution                                 |
| ns_res                                  | number            | yes      | —       | Desired north-south resolution                               |
| overlap                                 | number            | no       | 3.0     | Rows/columns overlap for segmentation                        |
| zscale                                  | number            | no       | 1.0     | Multiplier for z-values                                      |
| tension                                 | number            | no       | 40.0    | Spline tension value                                         |
| theta                                   | number            | no       | —       | Anisotropy angle (in degrees counterclockwise from East)     |
| scalex                                  | number            | no       | —       | Anisotropy scaling factor                                    |
| -t                                      | boolean           | no       | False   | Use dnorm independent tension                                |
| -d                                      | boolean           | no       | False   | Output partial derivatives instead of topographic parameters |
| elevation                               | rasterDestination | yes      | —       | Resampled RST                                                |
| slope                                   | rasterDestination | yes      | —       | Slope raster                                                 |
| aspect                                  | rasterDestination | yes      | —       | Aspect raster                                                |
| pcurvature                              | rasterDestination | yes      | —       | Profile curvature raster                                     |
| tcurvature                              | rasterDestination | yes      | —       | Tangential curvature raster                                  |
| mcurvature                              | rasterDestination | yes      | —       | Mean curvature raster                                        |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                      |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)              |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                    |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)             |

**Outputs:** elevation (outputRaster), slope (outputRaster), aspect (outputRaster), pcurvature (outputRaster), tcurvature (outputRaster), mcurvature (outputRaster)

### Example usage

```
run grass:r.resamp.rst input=dem.tif ew_res=10 ns_res=10 smooth=dem.tif maskmap=dem.tif
↳ overlap=3 zscale=1 elevation=elevation.tif
```

This calls **r.resamp.rst**: input=dem.tif — Raster layer; ew\_res=10 — Desired east-west resolution; ns\_res=10 — Desired north-south resolution; smooth=dem.tif — Input raster map containing smoothing; maskmap=dem.tif — Input raster map to be used as mask; overlap=3 — Rows/columns overlap for segmentation; zscale=1 — Multiplier for z-values; elevation=elevation.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.resamp.stats — r.resamp.stats

**group:** Raster (r.\*)

Resamples raster layers to a coarser grid using aggregation.

| Parameter                               | Type              | Required | Default | Description                                                                                                                                              |
|-----------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer                                                                                                                                       |
| method                                  | enum              | no       | 0       | Aggregation method — options: 0=average, 1=median, 2=mode, 3=minimum, 4=maximum, 5=quart1, 6=quart3, 7=perc90, 8=sum, 9=variance, 10=stddev, 11=quantile |
| quantile                                | number            | no       | 0.5     | Quantile to calculate for method=quantile                                                                                                                |
| -n                                      | boolean           | no       | False   | Propagate NULLs                                                                                                                                          |
| -w                                      | boolean           | no       | False   | Weight according to area (slower)                                                                                                                        |
| output                                  | rasterDestination | yes      | —       | Resampled aggregated                                                                                                                                     |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                                                                  |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                          |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                                                                                |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                                         |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.resamp.stats input=dem.tif method=0 quantile=0.5 output=output.tif
```

This calls **r.resamp.stats**: input=dem.tif — Input raster layer; method=0 selects *average*; quantile=0.5 — Quantile to calculate for method=quantile; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.resample — r.resample

**group:** Raster (r.\*)

GRASS raster map layer data resampling capability using nearest neighbors.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer                               |
| output                                  | rasterDestination | yes      | —       | Resampled NN                                     |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |



---

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.resample input=dem.tif output=output.tif
```

This calls **r.resample**: input=dem.tif — Input raster layer; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.rescale — r.rescale

**group:** Raster (r.\*)

Rescales the range of category values in a raster layer.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer                               |
| from                                    | range             | no       | —       | The input data range to be rescaled              |
| to                                      | range             | yes      | —       | The output data range                            |
| output                                  | rasterDestination | yes      | —       | Rescaled                                         |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.rescale input=dem.tif to=... output=output.tif
```

This calls **r.rescale**: input=dem.tif — Input raster layer; to=... — The output data range; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.rescale.eq — r.rescale.eq

**group:** Raster (r.\*)

Rescales histogram equalized the range of category values in a raster layer.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer                               |
| from                                    | range             | no       | —       | The input data range to be rescaled              |
| to                                      | range             | yes      | —       | The output data range                            |
| output                                  | rasterDestination | yes      | —       | Rescaled equalized                               |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

**Example usage**

```
run grass:r.rescale.eq input=dem.tif to=... output=output.tif
```

This calls **r.rescale.eq**: input=dem.tif — Input raster layer; to=... — The output data range; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.rgb — r.rgb

**group:** Raster (r.\*)

Splits a raster map into red, green and blue maps.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                         |
| red                                      | rasterDestination | yes      | —       | Red                                              |
| green                                    | rasterDestination | yes      | —       | Green                                            |
| blue                                     | rasterDestination | yes      | —       | Blue                                             |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** red (outputRaster), green (outputRaster), blue (outputRaster)

### Example usage

```
run grass:r.rgb input=dem.tif red=red.tif
```

This calls **r.rgb**: input=dem.tif — Name of input raster map; red=red.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.ros — r.ros

**group:** Raster (r.\*)

Generates rate of spread raster maps.

| Parameter     | Type              | Required | Default | Description                                                |
|---------------|-------------------|----------|---------|------------------------------------------------------------|
| model         | raster            | yes      | —       | Raster map containing fuel models                          |
| moisture_1h   | raster            | no       | —       | Raster map containing the 1-hour fuel moisture (%)         |
| moisture_10h  | raster            | no       | —       | Raster map containing the 10-hour fuel moisture (%)        |
| moisture_100h | raster            | no       | —       | Raster map containing the 100-hour fuel moisture (%)       |
| moisture_live | raster            | yes      | —       | Raster map containing live fuel moisture (%)               |
| velocity      | raster            | no       | —       | Raster map containing midflame wind velocities (ft/min)    |
| direction     | raster            | no       | —       | Name of raster map containing wind directions (degree)     |
| slope         | raster            | no       | —       | Name of raster map containing slope (degree)               |
| aspect        | raster            | no       | —       | Raster map containing aspect (degree, CCW from E)          |
| elevation     | raster            | no       | —       | Raster map containing elevation (m, required for spotting) |
| base_ros      | rasterDestination | yes      | —       | Base ROS                                                   |

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| max_ros                                  | rasterDestination | yes      | —       | Max ROS                                          |
| direction_ros                            | rasterDestination | yes      | —       | Direction ROS                                    |
| spotting_distance                        | rasterDestination | yes      | —       | Spotting Distance                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** base\_ros (outputRaster), max\_ros (outputRaster), direction\_ros (outputRaster), spotting\_distance (outputRaster)

### Example usage

```
run grass:r.ros model=dem.tif moisture_live=dem.tif moisture_1h=dem.tif
↪ moisture_10h=dem.tif moisture_100h=dem.tif velocity=dem.tif direction=dem.tif
↪ base_ros=base_ros.tif
```

This calls **r.ros**: model=dem.tif — Raster map containing fuel models; moisture\_live=dem.tif — Raster map containing live fuel moisture (%); moisture\_1h=dem.tif — Raster map containing the 1-hour fuel moisture (%); moisture\_10h=dem.tif — Raster map containing the 10-hour fuel moisture (%); moisture\_100h=dem.tif — Raster map containing the 100-hour fuel moisture (%); velocity=dem.tif — Raster map containing midflame wind velocities (ft/min); direction=dem.tif — Name of raster map containing wind directions (degree); base\_ros=base\_ros.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.series — r.series

**group:** Raster (r.\*)

Makes each output cell value a function of the values assigned to the corresponding cells in the input raster layers. Input raster layers/bands must be separated in different data sources.

| Parameter                                | Type              | Required | Default | Description                                                                                                                                                                                                                                                                                          |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Input raster layer(s)                                                                                                                                                                                                                                                                                |
| -n                                       | boolean           | no       | False   | Propagate NULLs                                                                                                                                                                                                                                                                                      |
| method                                   | enum              | no       | [0]     | Aggregate operation — options: 0=average, 1=count, 2=median, 3=mode, 4=minimum, 5=min_raster, 6=maximum, 7=max_raster, 8=stddev, 9=range, 10=sum, 11=variance, 12=diversity, 13=slope, 14=offset, 15=detect-coeff, 16=quart1, 17=quart3, 18=perc90, 19=skewness, 20=kurtosis, 21=quantile, 22=tvalue |
| quantile                                 | string            | no       | —       | Quantile to calculate for method=quantile                                                                                                                                                                                                                                                            |
| weights                                  | string            | no       | —       | Weighting factor for each input map, default value is 1.0                                                                                                                                                                                                                                            |
| range *(adv)*                            | range             | no       | —       | Ignore values outside this range (lo,hi)                                                                                                                                                                                                                                                             |
| output                                   | rasterDestination | yes      | —       | Aggregated                                                                                                                                                                                                                                                                                           |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                                                                                                                                                                              |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                                                                                                                                                                      |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                                                                                                                                                                                                            |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                                                                                                                                                                                     |

**Outputs:** output (outputRaster)

## Example usage

```
run grass:r.series input="a.gpkg;b.gpkg" method=0 quantile=value weights=value  
↳ output=output.tif
```

This calls **r.series**: input="a.gpkg;b.gpkg" — Input raster layer(s); method=0 selects *average*; quantile=value — Quantile to calculate for method=quantile; weights=value — Weighting factor for each input map, default value is 1.0; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.series.accumulate — r.series.accumulate

**group:** Raster (r.\*)

Makes each output cell value an accumulation function of the values assigned to the corresponding cells in the input raster map layers.

| Parameter                               | Type              | Required | Default | Description                                                                                                                   |
|-----------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------|
| input                                   | multilayer        | yes      | —       | Input raster layer(s)                                                                                                         |
| lower                                   | raster            | no       | —       | Raster map specifying the lower accumulation limit                                                                            |
| upper                                   | raster            | no       | —       | Raster map specifying the upper accumulation limit                                                                            |
| method                                  | enum              | no       | 0       | This method will be applied to compute the accumulative values from the input maps — options: 0=gdd, 1=bedd, 2=huglin, 3=mean |
| scale                                   | number            | no       | 1.0     | Scale factor for input                                                                                                        |
| shift                                   | number            | no       | 0.0     | Shift factor for input                                                                                                        |
| range                                   | range             | no       | —       | Ignore values outside this range (min,max)                                                                                    |
| limits                                  | range             | no       | 10,30   | Lower and upper accumulation limits (lower,upper)                                                                             |
| -n                                      | boolean           | no       | False   | Propagate NULLs                                                                                                               |
| -f *(adv)*                              | boolean           | no       | False   | Create a FCELL map (floating point single precision) as output                                                                |
| output                                  | rasterDestination | yes      | —       | Accumulated                                                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                                       |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                               |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                                                     |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                              |

**Outputs:** output (outputRaster)

## Example usage

```
run grass:r.series.accumulate input="a.gpkg;b.gpkg" lower=dem.tif upper=dem.tif method=0  
↳ scale=1 shift=0 output=output.tif
```

This calls **r.series.accumulate**: input="a.gpkg;b.gpkg" — Input raster layer(s); lower=dem.tif — Raster map specifying the lower accumulation limit; upper=dem.tif — Raster map specifying the upper accumulation limit; method=0 selects *gdd*; scale=1 — Scale factor for input; shift=0 — Shift factor for input; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.series.interp — r.series.interp

**group:** Raster (r.\*)

Interpolates raster maps located (temporal or spatial) in between input raster maps at specific sampling positions.

| Parameter                                | Type              | Required | Default | Description                                                                                                                            |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| input                                    | multilayer        | yes      | —       | Input raster layer(s)                                                                                                                  |
| datapos                                  | string            | no       | —       | Data point position for each input map                                                                                                 |
| infile                                   | file              | no       | —       | Input file with one input raster map name and data point position per line, field separator between name and sample point is 'pipe'    |
| output                                   | string            | no       | —       | Name for output raster map (comma separated list if multiple)                                                                          |
| samplingpos                              | string            | no       | —       | Sampling point position for each output map (comma separated list)                                                                     |
| outfile                                  | file              | no       | —       | Input file with one output raster map name and sample point position per line, field separator between name and sample point is 'pipe' |
| method                                   | enum              | no       | 0       | Interpolation method, currently only linear interpolation is supported — options: 0=linear                                             |
| output_dir                               | folderDestination | yes      | —       | Interpolated rasters                                                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                        |

**Outputs:** output\_dir (outputFolder)

### Example usage

```
run grass:r.series.interp input="a.gpkg;b.gpkg" datapos=value infile=input.dat
↪ output=value samplingpos=value outfile=input.dat method=0 output_dir=output_dir/
```

This calls **r.series.interp**: input="a.gpkg;b.gpkg" — Input raster layer(s); datapos=value — Data point position for each input map; infile=input.dat — Input file with one input raster map name and data point position per line,...; output=value — Name for output raster map (comma separated list if multiple); samplingpos=value — Sampling point position for each output map (comma separated list); outfile=input.dat — Input file with one output raster map name and sample point position per line,...; method=0 selects *linear*; output\_dir=output\_dir/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.shade — r.shade

**group:** Raster (r.\*)

Drapes a color raster over an shaded relief or aspect map.

| Parameter                                | Type              | Required | Default | Description                                                                                 |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------|
| shade                                    | raster            | yes      | —       | Name of shaded relief or aspect raster map                                                  |
| color                                    | raster            | yes      | —       | Name of raster to drape over relief raster map                                              |
| brighten                                 | number            | no       | 0.0     | Percent to brighten                                                                         |
| bgcolor                                  | string            | no       | —       | Color to use instead of NULL values. Either a standard color name, R:G:B triplet, or "none" |
| -c *(adv)*                               | boolean           | no       | False   | Use colors from color tables for NULL values                                                |
| output                                   | rasterDestination | yes      | —       | Shaded                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                     |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                             |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                   |

| Parameter                        | Type   | Required | Default | Description                                      |
|----------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_META *(adv)* | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.shade shade=dem.tif color=dem.tif brighten=0 bgcolor=value output=output.tif
```

This calls **r.shade**: shade=dem.tif — Name of shaded relief or aspect raster map; color=dem.tif — Name of raster to drape over relief raster map; brighten=0 — Percent to brighten; bgcolor=value — Color to use instead of NULL values. Either a standard color name, R:G:B...; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.sim.sediment — r.sim.sediment

**group:** Raster (r.\*)

Sediment transport and erosion/deposition simulation using path sampling method (SIMWE).

| Parameter                                | Type              | Required | Default | Description                                                  |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------------------|
| elevation                                | raster            | yes      | —       | Name of the elevation raster map [m]                         |
| water_depth                              | raster            | yes      | —       | Name of the water depth raster map [m]                       |
| dx                                       | raster            | yes      | —       | Name of the x-derivatives raster map [m/m]                   |
| dy                                       | raster            | yes      | —       | Name of the y-derivatives raster map [m/m]                   |
| detachment_coeff                         | raster            | yes      | —       | Name of the detachment capacity coefficient raster map [s/m] |
| transport_coeff                          | raster            | yes      | —       | Name of the transport capacity coefficient raster map [s]    |
| shear_stress                             | raster            | yes      | —       | Name of the critical shear stress raster map [Pa]            |
| man                                      | raster            | no       | —       | Name of the Mannings n raster map                            |
| man_value                                | number            | no       | 0.1     | Name of the Mannings n value                                 |
| observation                              | source            | no       | —       | Sampling locations vector points                             |
| nwalkers                                 | number            | no       | —       | Number of walkers                                            |
| niterations                              | number            | no       | 10.0    | Time used for iterations [minutes]                           |
| output_step                              | number            | no       | 2.0     | Time interval for creating output maps [minutes]             |
| diffusion_coeff                          | number            | no       | 0.8     | Water diffusion constant                                     |
| transport_capacity                       | rasterDestination | yes      | —       | Transport capacity [kg/ms]                                   |
| tlimit_erosion_deposition                | rasterDestination | yes      | —       | Transport limited erosion-deposition [kg/m2s]                |
| sediment_concentration                   | rasterDestination | yes      | —       | Sediment concentration [particle/m3]                         |
| sediment_flux                            | rasterDestination | yes      | —       | Sediment flux [kg/ms]                                        |
| erosion_deposition                       | rasterDestination | yes      | —       | Erosion-deposition [kg/m2s]                                  |
| walkers_output                           | vectorDestination | no       | —       | Name of the output walkers vector points layer               |
| logfile                                  | fileDestination   | no       | —       | Name for sampling points output text file.                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                      |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)              |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                    |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)             |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                       |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                            |

| Parameter                              | Type    | Required | Default | Description                                                                                             |
|----------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_OUTPUT_TYPE_PARAMETER<br>*(adv)* | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string  | no       | —       | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT<br>*(adv)*   | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** transport\_capacity (outputRaster), tlimit\_erosion\_deposition (outputRaster), sediment\_concentration (outputRaster), sediment\_flux (outputRaster), erosion\_deposition (outputRaster), walkers\_output (outputVector), logfile (outputFile)

### Example usage

```
run grass:r.sim.sediment elevation=dem.tif water_depth=dem.tif dx=dem.tif dy=dem.tif
↳ detachment_coeff=dem.tif transport_coeff=dem.tif shear_stress=dem.tif
↳ transport_capacity=transport_capacity.tif
```

This calls **r.sim.sediment**: elevation=dem.tif — Name of the elevation raster map [m]; water\_depth=dem.tif — Name of the water depth raster map [m]; dx=dem.tif — Name of the x-derivatives raster map [m/m]; dy=dem.tif — Name of the y-derivatives raster map [m/m]; detachment\_coeff=dem.tif — Name of the detachment capacity coefficient raster map [s/m]; transport\_coeff=dem.tif — Name of the transport capacity coefficient raster map [s]; shear\_stress=dem.tif — Name of the critical shear stress raster map [Pa]; transport\_capacity=transport\_capacity.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.sim.water — r.sim.water

**group:** Raster (r.\*)

Overland flow hydrologic simulation using path sampling method (SIMWE).

| Parameter       | Type    | Required | Default | Description                                                      |
|-----------------|---------|----------|---------|------------------------------------------------------------------|
| elevation       | raster  | yes      | —       | Name of the elevation raster map [m]                             |
| dx              | raster  | yes      | —       | Name of the x-derivatives raster map [m/m]                       |
| dy              | raster  | yes      | —       | Name of the y-derivatives raster map [m/m]                       |
| rain            | raster  | no       | —       | Name of the rainfall excess rate (rain-infil) raster map [mm/hr] |
| rain_value      | number  | no       | 50.0    | Rainfall excess rate unique value [mm/hr]                        |
| infil           | raster  | no       | —       | Name of the runoff infiltration rate raster map [mm/hr]          |
| infil_value     | number  | no       | 0.0     | Runoff infiltration rate unique value [mm/hr]                    |
| man             | raster  | no       | —       | Name of the Mannings n raster map                                |
| man_value       | number  | no       | 0.1     | Manning's n unique value                                         |
| flow_control    | raster  | no       | —       | Name of the flow controls raster map (permeability ratio 0-1)    |
| observation     | source  | no       | —       | Sampling locations vector points                                 |
| nwalkers        | number  | no       | —       | Number of walkers, default is twice the number of cells          |
| niterations     | number  | no       | 10.0    | Time used for iterations [minutes]                               |
| output_step     | number  | no       | 2.0     | Time interval for creating output maps [minutes]                 |
| diffusion_coeff | number  | no       | 0.8     | Water diffusion constant                                         |
| hmax            | number  | no       | 4.0     | Threshold water depth [m]                                        |
| halpha          | number  | no       | 4.0     | Diffusion increase constant                                      |
| hbeta           | number  | no       | 0.5     | Weighting factor for water flow velocity vector                  |
| -t              | boolean | no       | False   | Time-series output                                               |

| Parameter                               | Type              | Required | Default | Description                                                                                             |
|-----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| depth                                   | rasterDestination | yes      | —       | Water depth [m]                                                                                         |
| discharge                               | rasterDestination | yes      | —       | Water discharge [m3/s]                                                                                  |
| error                                   | rasterDestination | yes      | —       | Simulation error [m]                                                                                    |
| walkers_output                          | vectorDestination | no       | —       | Name of the output walkers vector points layer                                                          |
| logfile                                 | fileDestination   | no       | —       | Name for sampling points output text file.                                                              |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*  | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*        | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*     | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*               | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*       | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** depth (outputRaster), discharge (outputRaster), error (outputRaster), walkers\_output (outputVector), logfile (outputFile)

### Example usage

```
run grass:r.sim.water elevation=dem.tif dx=dem.tif dy=dem.tif rain=dem.tif rain_value=50
↳ infil=dem.tif infil_value=0 depth=depth.tif
```

This calls **r.sim.water**: elevation=dem.tif — Name of the elevation raster map [m]; dx=dem.tif — Name of the x-derivatives raster map [m/m]; dy=dem.tif — Name of the y-derivatives raster map [m/m]; rain=dem.tif — Name of the rainfall excess rate (rain-infil) raster map [mm/hr]; rain\_value=50 — Rainfall excess rate unique value [mm/hr]; infil=dem.tif — Name of the runoff infiltration rate raster map [mm/hr]; infil\_value=0 — Runoff infiltration rate unique value [mm/hr]; depth=depth.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.slope.aspect — r.slope.aspect

**group:** Raster (r.\*)

Generates raster layers of slope, aspect, curvatures and partial derivatives from a elevation raster layer.

| Parameter | Type    | Required | Default | Description                                                                |
|-----------|---------|----------|---------|----------------------------------------------------------------------------|
| elevation | raster  | yes      | —       | Elevation                                                                  |
| format    | enum    | no       | 0       | Format for reporting the slope — options: 0=degrees, 1=percent             |
| precision | enum    | no       | 0       | Type of output aspect and slope layer — options: 0=FCELL, 1=CELL, 2=DCELL  |
| -a        | boolean | no       | True    | Do not align the current region to the elevation layer                     |
| -e        | boolean | no       | False   | Compute output at edges and near NULL values                               |
| -n        | boolean | no       | False   | Create aspect as degrees clockwise from North (azimuth), with flat = -9999 |



| Parameter                                | Type              | Required | Default | Description                                                  |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------------------|
| zscale                                   | number            | no       | 1.0     | Multiplicative factor to convert elevation units to meters   |
| min_slope                                | number            | no       | 0.0     | Minimum slope val. (in percent) for which aspect is computed |
| slope                                    | rasterDestination | no       | —       | Slope                                                        |
| aspect                                   | rasterDestination | no       | —       | Aspect                                                       |
| pcurvature                               | rasterDestination | no       | —       | Profile curvature                                            |
| tcurvature                               | rasterDestination | no       | —       | Tangential curvature                                         |
| dx                                       | rasterDestination | no       | —       | First order partial derivative dx (E-W slope)                |
| dy                                       | rasterDestination | no       | —       | First order partial derivative dy (N-S slope)                |
| dxx                                      | rasterDestination | no       | —       | Second order partial derivative dxx                          |
| dyy                                      | rasterDestination | no       | —       | Second order partial derivative dyy                          |
| dxy                                      | rasterDestination | no       | —       | Second order partial derivative dxy                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                      |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)              |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                    |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)             |

**Outputs:** slope (outputRaster), aspect (outputRaster), pcurvature (outputRaster), tcurvature (outputRaster), dx (outputRaster), dy (outputRaster), dxx (outputRaster), dyy (outputRaster), dxy (outputRaster)

### Example usage

```
run grass:r.slope.aspect elevation=dem.tif format=0 precision=0 zscale=1 min_slope=0
↪ slope=slope.tif
```

This calls **r.slope.aspect**: elevation=dem.tif — Elevation; format=0 selects *degrees*; precision=0 selects *FCELL*; zscale=1 — Multiplicative factor to convert elevation units to meters; min\_slope=0 — Minimum slope val. (in percent) for which aspect is computed; slope=slope.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.solute.transport — r.solute.transport

**group:** Raster (r.\*)

Numerical calculation program for transient, confined and unconfined solute transport in two dimensions

| Parameter | Type   | Required | Default | Description                                                                                                        |
|-----------|--------|----------|---------|--------------------------------------------------------------------------------------------------------------------|
| c         | raster | yes      | —       | The initial concentration in [kg/m <sup>3</sup> ]                                                                  |
| phead     | raster | yes      | —       | The piezometric head in [m]                                                                                        |
| hc_x      | raster | yes      | —       | The x-part of the hydraulic conductivity tensor in [m/s]                                                           |
| hc_y      | raster | yes      | —       | The y-part of the hydraulic conductivity tensor in [m/s]                                                           |
| status    | raster | yes      | —       | The status for each cell, = 0 - inactive cell, 1 - active cell, 2 - dirichlet- and 3 - transfer boundary condition |
| diff_x    | raster | yes      | —       | The x-part of the diffusion tensor in [m <sup>2</sup> /s]                                                          |
| diff_y    | raster | yes      | —       | The y-part of the diffusion tensor in [m <sup>2</sup> /s]                                                          |
| q         | raster | no       | —       | Groundwater sources and sinks in [m <sup>3</sup> /s]                                                               |
| cin       | raster | no       | —       | Concentration sources and sinks bounded to a water source or sink in [kg/s]                                        |
| cs        | raster | yes      | —       | Concentration of inner sources and inner sinks in [kg/s] (i.e. a chemical reaction)                                |

| Parameter                                | Type              | Required | Default | Description                                                                                                            |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------|
| rd                                       | raster            | yes      | —       | Retardation factor [-]                                                                                                 |
| nf                                       | raster            | yes      | —       | Effective porosity [-]                                                                                                 |
| top                                      | raster            | yes      | —       | Top surface of the aquifer in [m]                                                                                      |
| bottom                                   | raster            | yes      | —       | Bottom surface of the aquifer in [m]                                                                                   |
| dttime                                   | number            | no       | 86400.0 | Calculation time (in seconds)                                                                                          |
| maxit                                    | number            | no       | 10000.0 | Maximum number of iteration used to solve the linear equation system                                                   |
| error                                    | number            | no       | 1e-06   | Error break criteria for iterative solver                                                                              |
| solver                                   | enum              | no       | 4       | The type of solver which should solve the linear equation system — options: 0=gauss, 1=lu, 2=jacobi, 3=sor, 4=bicgstab |
| relax                                    | number            | no       | 1.0     | The relaxation parameter used by the jacobi and sor solver for speedup or stabilizing                                  |
| al                                       | number            | no       | 0.0     | The longitudinal dispersivity length. [m]                                                                              |
| at                                       | number            | no       | 0.0     | The transversal dispersivity length. [m]                                                                               |
| loops                                    | number            | no       | 1.0     | Use this number of time loops if the CFL flag is off. The timestep will become dt/loops.                               |
| stab                                     | enum              | no       | 0       | Set the flow stabilizing scheme (full or exponential upwinding). — options: 0=full, 1=exp                              |
| -c *(adv)*                               | boolean           | no       | False   | Use the Courant-Friedrichs-Lewy criteria for time step calculation                                                     |
| -f *(adv)*                               | boolean           | no       | False   | Use a full filled quadratic linear equation system, default is a sparse linear equation system.                        |
| output                                   | rasterDestination | yes      | —       | Solute Transport                                                                                                       |
| vx                                       | rasterDestination | no       | —       | Calculate and store the groundwater filter velocity vector part in x direction [m/s]                                   |
| vy                                       | rasterDestination | no       | —       | Calculate and store the groundwater filter velocity vector part in y direction [m/s]                                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                        |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                              |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                       |

**Outputs:** output (outputRaster), vx (outputRaster), vy (outputRaster)

### Example usage

```
run grass:r.solute.transport c=dem.tif phead=dem.tif hc_x=dem.tif hc_y=dem.tif
↪ status=dem.tif diff_x=dem.tif diff_y=dem.tif output=output.tif
```

This calls **r.solute.transport**: c=dem.tif — The initial concentration in [kg/m<sup>3</sup>]; phead=dem.tif — The piezometric head in [m]; hc\_x=dem.tif — The x-part of the hydraulic conductivity tensor in [m/s]; hc\_y=dem.tif — The y-part of the hydraulic conductivity tensor in [m/s]; status=dem.tif — The status for each cell, = 0 - inactive cell, 1 - active cell, 2 - dirichlet-...; diff\_x=dem.tif — The x-part of the diffusion tensor in [m<sup>2</sup>/s]; diff\_y=dem.tif — The y-part of the diffusion tensor in [m<sup>2</sup>/s]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

### grass:r.spread — r.spread

**group:** Raster (r.\*)

Simulates elliptically anisotropic spread.

| Parameter                                | Type              | Required | Default | Description                                                                                                    |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------|
| base_ros                                 | raster            | yes      | —       | Raster map containing base ROS (cm/min)                                                                        |
| max_ros                                  | raster            | yes      | —       | Raster map containing maximal ROS (cm/min)                                                                     |
| direction_ros                            | raster            | yes      | —       | Raster map containing directions of maximal ROS (degree)                                                       |
| start                                    | raster            | yes      | —       | Raster map containing starting sources                                                                         |
| spotting_distance                        | raster            | no       | —       | Raster map containing maximal spotting distance (m, required with -s)                                          |
| wind_speed                               | raster            | no       | —       | Raster map containing midflame wind speed (ft/min, required with -s)                                           |
| fuel_moisture                            | raster            | no       | —       | Raster map containing fine fuel moisture of the cell receiving a spotting firebrand (% , required with -s)     |
| backdrop                                 | raster            | no       | —       | Name of raster map as a display backdrop                                                                       |
| least_size                               | enum              | no       | 0       | Basic sampling window size needed to meet certain accuracy (3) — options: 0=3, 1=5, 2=7, 3=9, 4=11, 5=13, 6=15 |
| comp_dens                                | number            | no       | 0.5     | Sampling density for additional computing (range: 0.0 - 1.0 (0.5))                                             |
| init_time                                | number            | no       | 0.0     | Initial time for current simulation (0) (min)                                                                  |
| lag                                      | number            | no       | —       | Simulating time duration LAG (fill the region) (min)                                                           |
| -s *(adv)*                               | boolean           | no       | False   | Consider spotting effect (for wildfires)                                                                       |
| -i *(adv)*                               | boolean           | no       | False   | Use start raster map values in output spread time raster map                                                   |
| output                                   | rasterDestination | yes      | —       | Spread Time                                                                                                    |
| x_output                                 | rasterDestination | yes      | —       | X Back Coordinates                                                                                             |
| y_output                                 | rasterDestination | yes      | —       | Y Back Coordinates                                                                                             |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                        |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                      |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                               |

**Outputs:** output (outputRaster), x\_output (outputRaster), y\_output (outputRaster)

### Example usage

```
run grass:r.spread base_ros=dem.tif max_ros=dem.tif direction_ros=dem.tif start=dem.tif
↳ spotting_distance=dem.tif wind_speed=dem.tif fuel_moisture=dem.tif output=output.tif
```

This calls **r.spread**: base\_ros=dem.tif — Raster map containing base ROS (cm/min); max\_ros=dem.tif — Raster map containing maximal ROS (cm/min); direction\_ros=dem.tif — Raster map containing directions of maximal ROS (degree); start=dem.tif — Raster map containing starting sources; spotting\_distance=dem.tif — Raster map containing maximal spotting distance (m, required with -s); wind\_speed=dem.tif — Raster map containing midflame wind speed (ft/min, required with -s); fuel\_moisture=dem.tif — Raster map containing fine fuel moisture of the cell receiving a spotting...; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.spreadpath — r.spreadpath

**group:** Raster (r.\*)

Recursively traces the least cost path backwards to cells from which the cumulative cost was determined.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| x_input                                  | raster            | yes      | —       | x_input                                          |
| y_input                                  | raster            | yes      | —       | y_input                                          |
| coordinates                              | point             | no       | 0,0     | coordinate                                       |
| output                                   | rasterDestination | yes      | —       | Backward least cost                              |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.spreadpath x_input=dem.tif y_input=dem.tif coordinates=100,100
↪ output=output.tif
```

This calls **r.spreadpath**: x\_input=dem.tif — x\_input; y\_input=dem.tif — y\_input; coordinates=100,100 — coordinate; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.statistics — r.statistics

**group:** Raster (r.\*)

Calculates category or object oriented statistics.

| Parameter                                | Type              | Required | Default | Description                                                                                                                                |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------|
| base                                     | raster            | yes      | —       | Base raster layer                                                                                                                          |
| cover                                    | raster            | yes      | —       | Cover raster layer                                                                                                                         |
| method                                   | enum              | no       | 0       | method — options: 0=diversity, 1=average, 2=mode, 3=median, 4=avedev, 5=std-dev, 6=variance, 7=skewness, 8=kurtosis, 9=min, 10=max, 11=sum |
| -c                                       | boolean           | no       | False   | Cover values extracted from the category labels of the cover map                                                                           |
| routput                                  | rasterDestination | yes      | —       | Statistics                                                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                    |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                            |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                                                  |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                           |

**Outputs:** routput (outputRaster)

### Example usage

```
run grass:r.statistics base=dem.tif cover=dem.tif method=0 routput=routput.tif
```

This calls **r.statistics**: base=dem.tif — Base raster layer; cover=dem.tif — Cover raster layer; method=0 selects *diversity*; routput=routput.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.stats — r.stats

**group:** Raster (r.\*)

Generates area statistics for raster layers.

| Parameter                                | Type            | Required | Default     | Description                                                    |
|------------------------------------------|-----------------|----------|-------------|----------------------------------------------------------------|
| input                                    | multilayer      | yes      | —           | Name of input raster map                                       |
| separator                                | string          | no       | space       | Output field separator                                         |
| null_value                               | string          | no       | *           | String representing no data cell value                         |
| nsteps                                   | number          | no       | 255.0       | Number of floating-point subranges to collect stats from       |
| sort                                     | enum            | no       | 0           | Sort output statistics by cell counts — options: 0=asc, 1=desc |
| -l                                       | boolean         | no       | True        | One cell (range) per line                                      |
| -A                                       | boolean         | no       | False       | Print averaged values instead of intervals                     |
| -a                                       | boolean         | no       | False       | Print area totals                                              |
| -c                                       | boolean         | no       | False       | Print cell counts                                              |
| -p                                       | boolean         | no       | False       | Print APPROXIMATE percents (total percent may not be 100%)     |
| -l                                       | boolean         | no       | False       | Print category labels                                          |
| -g                                       | boolean         | no       | False       | Print grid coordinates (east and north)                        |
| -x                                       | boolean         | no       | False       | Print x and y (column and row)                                 |
| -r                                       | boolean         | no       | False       | Print raw indexes of fp ranges (fp maps only)                  |
| -n                                       | boolean         | no       | False       | Suppress reporting of any NULLs                                |
| -N                                       | boolean         | no       | False       | Suppress reporting of NULLs when all values are NULL           |
| -C                                       | boolean         | no       | False       | Report for cats fp ranges (fp maps only)                       |
| -i                                       | boolean         | no       | False       | Read fp map as integer (use map's quant rules)                 |
| html                                     | fileDestination | no       | report.html | Statistics                                                     |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                                        |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default)                |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.stats input="a.gpkg;b.gpkg" separator=space null_value=* nsteps=255 sort=0
↳ html=html.txt
```

This calls **r.stats**: input="a.gpkg;b.gpkg" — Name of input raster map; separator=space — Output field separator; null\_value=\* — String representing no data cell value; nsteps=255 — Number of floating-point subranges to collect stats from; sort=0 selects asc; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.stats.quantile.out — r.stats.quantile.out

**group:** Raster (r.\*)

r.stats.quantile.out - Compute category quantiles using two passes and output statistics

| Parameter                                | Type            | Required | Default | Description                                           |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------------|
| base                                     | raster          | yes      | —       | Name of base raster map                               |
| cover                                    | raster          | yes      | —       | Name of cover raster map                              |
| quantiles                                | number          | no       | —       | Number of quantiles                                   |
| percentiles                              | string          | no       | —       | List of percentiles                                   |
| bins                                     | number          | no       | 1000.0  | Number of bins to use                                 |
| -r *(adv)*                               | boolean         | no       | False   | Create reclass map with statistics as category labels |
| file                                     | fileDestination | yes      | —       | Statistics File                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                               |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)       |

---

**Outputs:** file (outputFile)

**Example usage**

```
run grass:r.stats.quantile.out base=dem.tif cover=dem.tif quantiles=10 percentiles=value  
↪ bins=1000 file=file.txt
```

This calls **r.stats.quantile.out**: base=dem.tif — Name of base raster map; cover=dem.tif — Name of cover raster map; quantiles=10 — Number of quantiles; percentiles=value — List of percentiles; bins=1000 — Number of bins to use; file=file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.stats.quantile.rast — r.stats.quantile.rast

**group:** Raster (r.\*)

r.stats.quantile.rast - Compute category quantiles using two passes and output rasters.

| Parameter                                | Type              | Required | Default | Description                                           |
|------------------------------------------|-------------------|----------|---------|-------------------------------------------------------|
| base                                     | raster            | yes      | —       | Name of base raster map                               |
| cover                                    | raster            | yes      | —       | Name of cover raster map                              |
| quantiles                                | number            | no       | —       | Number of quantiles                                   |
| percentiles                              | string            | no       | —       | List of percentiles                                   |
| bins                                     | number            | no       | 1000.0  | Number of bins to use                                 |
| -r *(adv)*                               | boolean           | no       | False   | Create reclass map with statistics as category labels |
| output                                   | folderDestination | yes      | —       | Output Directory                                      |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                               |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)       |

**Outputs:** output (outputFolder)

**Example usage**

```
run grass:r.stats.quantile.rast base=dem.tif cover=dem.tif quantiles=10 percentiles=value  
↪ bins=1000 output=output/
```

This calls **r.stats.quantile.rast**: base=dem.tif — Name of base raster map; cover=dem.tif — Name of cover raster map; quantiles=10 — Number of quantiles; percentiles=value — List of percentiles; bins=1000 — Number of bins to use; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.stats.zonal — r.stats.zonal

**group:** Raster (r.\*)

Calculates category or object oriented statistics (accumulator-based statistics)

| Parameter | Type   | Required | Default | Description                                                                                                                                                                                                 |
|-----------|--------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base      | raster | yes      | —       | Base raster                                                                                                                                                                                                 |
| cover     | raster | yes      | —       | Cover raster                                                                                                                                                                                                |
| method    | enum   | no       | 0       | Method of object-based statistic — options: 0=count, 1=sum, 2=min, 3=max, 4=range, 5=average, 6=avedev, 7=variance, 8=stddev, 9=skewness, 10=kurtosis, 11=variance2, 12=stddev2, 13=skewness2, 14=kurtosis2 |

| Parameter                                | Type              | Required | Default | Description                                                      |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| -c *(adv)*                               | boolean           | no       | False   | Cover values extracted from the category labels of the cover map |
| -r *(adv)*                               | boolean           | no       | False   | Create reclass map with statistics as category labels            |
| output                                   | rasterDestination | yes      | —       | Resultant raster                                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                 |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.stats.zonal base=dem.tif cover=dem.tif method=0 output=output.tif
```

This calls **r.stats.zonal**: base=dem.tif — Base raster; cover=dem.tif — Cover raster; method=0 selects *count*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.stream.extract — r.stream.extract

**group:** Raster (r.\*)

Stream network extraction

| Parameter                                | Type              | Required | Default | Description                                                                                             |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| elevation                                | raster            | yes      | —       | Input map: elevation map                                                                                |
| accumulation                             | raster            | no       | —       | Input map: accumulation map                                                                             |
| depression                               | raster            | no       | —       | Input map: map with real depressions                                                                    |
| threshold                                | number            | no       | 1.0     | Minimum flow accumulation for streams                                                                   |
| mexp                                     | number            | no       | 0.0     | Montgomery exponent for slope                                                                           |
| stream_length                            | number            | no       | 0.0     | Delete stream segments shorter than cells                                                               |
| d8cut                                    | number            | no       | —       | Use SFD above this threshold                                                                            |
| memory *(adv)*                           | number            | no       | 300.0   | Maximum memory to be used (in MB)                                                                       |
| stream_raster                            | rasterDestination | no       | —       | Unique stream ids (rast)                                                                                |
| stream_vector                            | vectorDestination | no       | —       | Unique stream ids (vect)                                                                                |
| direction                                | rasterDestination | no       | —       | Flow direction                                                                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** stream\_raster (outputRaster), stream\_vector (outputVector), direction (outputRaster)

### Example usage

```
run grass:r.stream.extract elevation=dem.tif accumulation=dem.tif depression=dem.tif
↳ threshold=1 mexp=0 stream_length=0 d8cut=10 stream_raster=stream_raster.tif
```

This calls **r.stream.extract**: elevation=dem.tif — Input map: elevation map; accumulation=dem.tif — Input map: accumulation map; depression=dem.tif — Input map: map with real depressions; threshold=1 — Minimum flow accumulation for streams; mexp=0 — Montgomery exponent for slope; stream\_length=0 — Delete stream segments shorter than cells; d8cut=10 — Use SFD above this threshold; stream\_raster=stream\_raster.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.sun.incidentout — r.sun.incidentout

**group:** Raster (r.\*)

r.sun.incidentout - Solar irradiance and irradiation model ( for the set local time).

| Parameter             | Type              | Required | Default | Description                                                                                     |
|-----------------------|-------------------|----------|---------|-------------------------------------------------------------------------------------------------|
| elevation             | raster            | yes      | —       | Elevation layer [meters]                                                                        |
| aspect                | raster            | yes      | —       | Aspect layer [decimal degrees]                                                                  |
| aspect_value          | number            | no       | 270.0   | A single value of the orientation (aspect), 270 is south                                        |
| slope                 | raster            | yes      | —       | Name of the input slope raster map (terrain slope or solar panel inclination) [decimal degrees] |
| slope_value           | number            | no       | 0.0     | A single value of inclination (slope)                                                           |
| linke                 | raster            | no       | —       | Name of the Linke atmospheric turbidity coefficient input raster map                            |
| albedo                | raster            | no       | —       | Name of the ground albedo coefficient input raster map                                          |
| albedo_value          | number            | no       | 0.2     | A single value of the ground albedo coefficient                                                 |
| lat                   | raster            | no       | —       | Name of input raster map containing latitudes [decimal degrees]                                 |
| long                  | raster            | no       | —       | Name of input raster map containing longitudes [decimal degrees]                                |
| coeff_bh              | raster            | no       | —       | Name of real-sky beam radiation coefficient input raster map                                    |
| coeff_dh              | raster            | no       | —       | Name of real-sky diffuse radiation coefficient input raster map                                 |
| horizon_basemap       | raster            | no       | —       | The horizon information input map base-name                                                     |
| horizon_step          | number            | no       | —       | Angle step size for multidirectional horizon [degrees]                                          |
| day                   | number            | no       | 1.0     | No. of day of the year (1-365)                                                                  |
| step *(adv)*          | number            | no       | 0.5     | Time step when computing all-day radiation sums [decimal hours]                                 |
| declination *(adv)*   | number            | no       | —       | Declination value (overriding the internally computed value) [radians]                          |
| distance_step *(adv)* | number            | no       | 1.0     | Sampling distance step coefficient (0.5-1.5)                                                    |
| npartitions *(adv)*   | number            | no       | 1.0     | Read the input files in this number of chunks                                                   |
| civil_time *(adv)*    | number            | no       | —       | Civil time zone value, if none, the time will be local solar time                               |
| time                  | number            | yes      | —       | Local (solar) time (decimal hours)                                                              |
| -p                    | boolean           | no       | False   | Do not incorporate the shadowing effect of terrain                                              |
| -m *(adv)*            | boolean           | no       | False   | Use the low-memory version of the program                                                       |
| incidentout           | rasterDestination | no       | —       | incidence angle raster map                                                                      |
| beam_rad              | rasterDestination | no       | —       | Beam irradiance [W.m-2]                                                                         |
| diff_rad              | rasterDestination | no       | —       | Diffuse irradiance [W.m-2]                                                                      |
| refl_rad              | rasterDestination | no       | —       | Ground reflected irradiance [W.m-2]                                                             |
| glob_rad              | rasterDestination | no       | —       | Global (total) irradiance/irradiation [W.m-2]                                                   |



| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** incidout (outputRaster), beam\_rad (outputRaster), diff\_rad (outputRaster), refl\_rad (outputRaster), glob\_rad (outputRaster)

### Example usage

```
run grass:r.sun.incidout elevation=dem.tif aspect=dem.tif slope=dem.tif time=10
↪ aspect_value=270 slope_value=0 linke=dem.tif incidout=incidout.tif
```

This calls **r.sun.incidout**: elevation=dem.tif — Elevation layer [meters]; aspect=dem.tif — Aspect layer [decimal degrees]; slope=dem.tif — Name of the input slope raster map (terrain slope or solar panel inclination)...; time=10 — Local (solar) time (decimal hours); aspect\_value=270 — A single value of the orientation (aspect), 270 is south; slope\_value=0 — A single value of inclination (slope); linke=dem.tif — Name of the Linke atmospheric turbidity coefficient input raster map; incidout=incidout.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.sun.insoltime — r.sun.insoltime

**group:** Raster (r.\*)

r.sun.insoltime - Solar irradiance and irradiation model (daily sums).

| Parameter       | Type   | Required | Default | Description                                                                                     |
|-----------------|--------|----------|---------|-------------------------------------------------------------------------------------------------|
| elevation       | raster | yes      | —       | Elevation layer [meters]                                                                        |
| aspect          | raster | yes      | —       | Aspect layer [decimal degrees]                                                                  |
| aspect_value    | number | no       | 270.0   | A single value of the orientation (aspect), 270 is south                                        |
| slope           | raster | yes      | —       | Name of the input slope raster map (terrain slope or solar panel inclination) [decimal degrees] |
| slope_value     | number | no       | 0.0     | A single value of inclination (slope)                                                           |
| linke           | raster | no       | —       | Name of the Linke atmospheric turbidity coefficient input raster map                            |
| albedo          | raster | no       | —       | Name of the ground albedo coefficient input raster map                                          |
| albedo_value    | number | no       | 0.2     | A single value of the ground albedo coefficient                                                 |
| lat             | raster | no       | —       | Name of input raster map containing latitudes [decimal degrees]                                 |
| long            | raster | no       | —       | Name of input raster map containing longitudes [decimal degrees]                                |
| coeff_bh        | raster | no       | —       | Name of real-sky beam radiation coefficient input raster map                                    |
| coeff_dh        | raster | no       | —       | Name of real-sky diffuse radiation coefficient input raster map                                 |
| horizon_basemap | raster | no       | —       | The horizon information input map base-name                                                     |
| horizon_step    | number | no       | —       | Angle step size for multidirectional horizon [degrees]                                          |
| day             | number | no       | 1.0     | No. of day of the year (1-365)                                                                  |
| step *(adv)*    | number | no       | 0.5     | Time step when computing all-day radiation sums [decimal hours]                                 |

| Parameter                                | Type              | Required | Default | Description                                                            |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------|
| declination *(adv)*                      | number            | no       | —       | Declination value (overriding the internally computed value) [radians] |
| distance_step *(adv)*                    | number            | no       | 1.0     | Sampling distance step coefficient (0.5-1.5)                           |
| npartitions *(adv)*                      | number            | no       | 1.0     | Read the input files in this number of chunks                          |
| civil_time *(adv)*                       | number            | no       | —       | Civil time zone value, if none, the time will be local solar time      |
| -p                                       | boolean           | no       | False   | Do not incorporate the shadowing effect of terrain                     |
| -m *(adv)*                               | boolean           | no       | False   | Use the low-memory version of the program                              |
| insol_time                               | rasterDestination | no       | —       | Insolation time [h]                                                    |
| beam_rad                                 | rasterDestination | no       | —       | Irradiation raster map [Wh.m-2.day-1]                                  |
| diff_rad                                 | rasterDestination | no       | —       | Irradiation raster map [Wh.m-2.day-1]                                  |
| refl_rad                                 | rasterDestination | no       | —       | Irradiation raster map [Wh.m-2.day-1]                                  |
| glob_rad                                 | rasterDestination | no       | —       | Irradiance/irradiation raster map [Wh.m-2.day-1]                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                        |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                              |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                       |

**Outputs:** insol\_time (outputRaster), beam\_rad (outputRaster), diff\_rad (outputRaster), refl\_rad (outputRaster), glob\_rad (outputRaster)

### Example usage

```
run grass:r.sun.insoltime elevation=dem.tif aspect=dem.tif slope=dem.tif aspect_value=270
↳ slope_value=0 linke=dem.tif albedo=dem.tif insol_time=insol_time.tif
```

This calls **r.sun.insoltime**: elevation=dem.tif — Elevation layer [meters]; aspect=dem.tif — Aspect layer [decimal degrees]; slope=dem.tif — Name of the input slope raster map (terrain slope or solar panel inclination)...; aspect\_value=270 — A single value of the orientation (aspect), 270 is south; slope\_value=0 — A single value of inclination (slope); linke=dem.tif — Name of the Linke atmospheric turbidity coefficient input raster map; albedo=dem.tif — Name of the ground albedo coefficient input raster map; insol\_time=insol\_time.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.sunhours — r.sunhours

**group:** Raster (r.\*)

Calculates solar elevation, solar azimuth, and sun hours.

| Parameter                      | Type              | Required | Default | Description                                              |
|--------------------------------|-------------------|----------|---------|----------------------------------------------------------|
| year                           | number            | no       | 2017.0  | Year                                                     |
| month                          | number            | no       | 1.0     | Month                                                    |
| day                            | number            | no       | 1.0     | Day                                                      |
| hour                           | number            | no       | 12.0    | Hour                                                     |
| minute                         | number            | no       | 0.0     | Minutes                                                  |
| second                         | number            | no       | 0.0     | Seconds                                                  |
| -t *(adv)*                     | boolean           | no       | False   | Time is local sidereal time, not Greenwich standard time |
| -s *(adv)*                     | boolean           | no       | False   | Do not use SOLPOS algorithm of NREL                      |
| elevation                      | rasterDestination | yes      | —       | Solar Elevation Angle                                    |
| azimuth                        | rasterDestination | yes      | —       | Solar Azimuth Angle                                      |
| sunhour                        | rasterDestination | yes      | —       | Sunshine Hours                                           |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                                  |

| Parameter                                | Type   | Required | Default | Description                                      |
|------------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** elevation (outputRaster), azimuth (outputRaster), sunhour (outputRaster)

### Example usage

```
run grass:r.sunhours year=2017 month=1 day=1 hour=12 minute=0 second=0
↪ elevation=elevation.tif
```

This calls **r.sunhours**: year=2017 — Year; month=1 — Month; day=1 — Day; hour=12 — Hour; minute=0 — Minutes; second=0 — Seconds; elevation=elevation.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.sunmask.datetime — r.sunmask.datetime

**group:** Raster (r.\*)

r.sunmask.datetime - Calculates cast shadow areas from sun position and elevation raster map.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| elevation                                | raster            | yes      | —       | Elevation raster layer [meters]                  |
| year                                     | number            | no       | 2000.0  | year                                             |
| month                                    | number            | no       | 1.0     | month                                            |
| day                                      | number            | no       | 1.0     | day                                              |
| hour                                     | number            | no       | 1.0     | hour                                             |
| minute                                   | number            | no       | 0.0     | minute                                           |
| second                                   | number            | no       | 0.0     | second                                           |
| timezone                                 | number            | no       | 0.0     | East positive, offset from GMT                   |
| east                                     | string            | yes      | —       | Easting coordinate (point of interest)           |
| north                                    | string            | yes      | —       | Northing coordinate (point of interest)          |
| -z                                       | boolean           | no       | True    | Do not ignore zero elevation                     |
| -s                                       | boolean           | no       | False   | Calculate sun position only and exit             |
| output                                   | rasterDestination | yes      | —       | Shadows                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.sunmask.datetime elevation=dem.tif east=value north=value year=2000 month=1
↪ day=1 hour=1 output=output.tif
```

This calls **r.sunmask.datetime**: elevation=dem.tif — Elevation raster layer [meters]; east=value — Easting coordinate (point of interest); north=value — Northing coordinate (point of interest); year=2000 — year; month=1 — month; day=1 — day; hour=1 — hour; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.sunmask.position — r.sunmask.position

**group:** Raster (r.\*)

r.sunmask.position - Calculates cast shadow areas from sun position and elevation raster map.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| elevation                                | raster            | yes      | —       | Elevation raster layer [meters]                  |
| altitude                                 | number            | no       | —       | Altitude of the sun in degrees above the horizon |
| azimuth                                  | number            | no       | —       | Azimuth of the sun in degrees from north         |
| east                                     | string            | no       | False   | Easting coordinate (point of interest)           |
| north                                    | string            | no       | False   | Northing coordinate (point of interest)          |
| -z                                       | boolean           | no       | True    | Do not ignore zero elevation                     |
| -s                                       | boolean           | no       | False   | Calculate sun position only and exit             |
| output                                   | rasterDestination | yes      | —       | Shadows                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.sunmask.position elevation=dem.tif altitude=10 azimuth=10 east=False  
↪ north=False output=output.tif
```

This calls **r.sunmask.position**: elevation=dem.tif — Elevation raster layer [meters]; altitude=10 — Altitude of the sun in degrees above the horizon; azimuth=10 — Azimuth of the sun in degrees from north; east=False — Easting coordinate (point of interest); north=False — Northing coordinate (point of interest); output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.surf.area — r.surf.area

**group:** Raster (r.\*)

Surface area estimation for rasters.

| Parameter                                | Type            | Required | Default     | Description                                                                   |
|------------------------------------------|-----------------|----------|-------------|-------------------------------------------------------------------------------|
| map                                      | raster          | yes      | —           | Input layer                                                                   |
| vscale                                   | number          | no       | 1.0         | Vertical scale                                                                |
| units                                    | enum            | no       | 1           | Units — options: 0=miles, 1=feet, 2=meters, 3=kilometers, 4=acres, 5=hectares |
| html                                     | fileDestination | no       | report.html | Area                                                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —           | GRASS GIS region extent                                                       |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0         | GRASS GIS region cellsize (leave 0 for default)                               |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.surf.area map=dem.tif vscale=1 units=1 html=html.txt
```

This calls **r.surf.area**: map=dem.tif — Input layer; vscale=1 — Vertical scale; units=1 selects feet; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.surf.contour — r.surf.contour

**group:** Raster (r.\*)

Surface generation program from rasterized contours.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Raster layer with rasterized contours            |
| output                                  | rasterDestination | yes      | —       | DTM from contours                                |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.surf.contour input=dem.tif output=output.tif
```

This calls **r.surf.contour**: input=dem.tif — Raster layer with rasterized contours; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.surf.fractal — r.surf.fractal

**group:** Raster (r.\*)

Creates a fractal surface of a given fractal dimension.

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| dimension                               | number            | no       | 2.05    | Fractal dimension of surface ( $2 < D < 3$ )     |
| number                                  | number            | no       | 0.0     | Number of intermediate images to produce         |
| output                                  | rasterDestination | yes      | —       | Fractal Surface                                  |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.surf.fractal dimension=2.05 number=0 output=output.tif
```

This calls **r.surf.fractal**: dimension=2.05 — Fractal dimension of surface ( $2 < D < 3$ ); number=0 — Number of intermediate images to produce; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.surf.gauss — r.surf.gauss

**group:** Raster (r.\*)

Creates a raster layer of Gaussian deviates.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| mean                                     | number            | no       | 0.0     | Distribution mean                                |
| sigma                                    | number            | no       | 1.0     | Standard deviation                               |
| output                                   | rasterDestination | yes      | —       | Gaussian deviates                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.surf.gauss mean=0 sigma=1 output=output.tif
```

This calls **r.surf.gauss**: mean=0 — Distribution mean; sigma=1 — Standard deviation; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.surf.idw — r.surf.idw

**group:** Raster (r.\*)

Surface interpolation utility for raster layers.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster layer                       |
| npnts                                    | number            | no       | 12.0    | Number of interpolation points                   |
| -e                                       | boolean           | no       | False   | Output is the interpolation error                |
| output                                   | rasterDestination | yes      | —       | Interpolated IDW                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.surf.idw input=dem.tif npnts=12 output=output.tif
```

This calls **r.surf.idw**: input=dem.tif — Name of input raster layer; npnts=12 — Number of interpolation points; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.surf.random — r.surf.random

**group:** Raster (r.\*)

Produces a raster layer of uniform random deviates whose range can be expressed by the user.

| Parameter | Type    | Required | Default | Description                    |
|-----------|---------|----------|---------|--------------------------------|
| min       | number  | no       | 0.0     | Minimum random value           |
| max       | number  | no       | 100.0   | Maximum random value           |
| -i        | boolean | no       | False   | Create an integer raster layer |

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| output                                   | rasterDestination | yes      | —       | Random                                           |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.surf.random min=0 max=100 output=output.tif
```

This calls **r.surf.random**: min=0 — Minimum random value; max=100 — Maximum random value; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.terraflow — r.terraflow

**group:** Raster (r.\*)

Flow computation for massive grids.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| elevation                                | raster            | yes      | —       | Name of elevation raster map                     |
| -s                                       | boolean           | no       | False   | SFD (D8) flow (default is MFD)                   |
| d8cut                                    | number            | no       | —       | Routing using SFD (D8) direction                 |
| memory                                   | number            | no       | 300.0   | Maximum memory to be used (in MB)                |
| filled                                   | rasterDestination | yes      | —       | Filled (flooded) elevation                       |
| direction                                | rasterDestination | yes      | —       | Flow direction                                   |
| swatershed                               | rasterDestination | yes      | —       | Sink-watershed                                   |
| accumulation                             | rasterDestination | yes      | —       | Flow accumulation                                |
| tci                                      | rasterDestination | yes      | —       | Topographic convergence index (tci)              |
| stats                                    | fileDestination   | yes      | —       | Runtime statistics                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** filled (outputRaster), direction (outputRaster), swatershed (outputRaster), accumulation (outputRaster), tci (outputRaster), stats (outputFile)

### Example usage

```
run grass:r.terraflow elevation=dem.tif d8cut=10 memory=300 filled=filled.tif
```

This calls **r.terraflow**: elevation=dem.tif — Name of elevation raster map; d8cut=10 — Routing using SFD (D8) direction; memory=300 — Maximum memory to be used (in MB); filled=filled.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.texture — r.texture

**group:** Raster (r.\*)

Generate images with textural features from a raster map.

| Parameter                                | Type              | Required | Default | Description                                                                                                                                |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map                                                                                                                   |
| method                                   | enum              | no       | [0]     | Textural measurement method(s) — options: 0=asm, 1=contrast, 2=corr, 3=var, 4=idm, 5=sa, 6=se, 7=sv, 8=entr, 9=dv, 10=de, 11=moc1, 12=moc2 |
| size                                     | number            | no       | 3.0     | The size of moving window (odd and $\geq 3$ )                                                                                              |
| distance                                 | number            | no       | 1.0     | The distance between two samples ( $\geq 1$ )                                                                                              |
| -s *(adv)*                               | boolean           | no       | False   | Separate output for each angle (0, 45, 90, 135)                                                                                            |
| -a *(adv)*                               | boolean           | no       | False   | Calculate all textural measurements                                                                                                        |
| output                                   | folderDestination | yes      | —       | Texture files directory                                                                                                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                    |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                            |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:r.texture input=dem.tif method=0 size=3 distance=1 output=output/
```

This calls **r.texture**: input=dem.tif — Name of input raster map; method=0 selects *asm*; size=3 — The size of moving window (odd and  $\geq 3$ ); distance=1 — The distance between two samples ( $\geq 1$ ); output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.thin — r.thin

**group:** Raster (r.\*)

Thins non-zero cells that denote linear features in a raster layer.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Input raster layer to thin                       |
| iterations                               | number            | no       | 200.0   | Maximum number of iterations                     |
| output                                   | rasterDestination | yes      | —       | Thinned                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.thin input=dem.tif iterations=200 output=output.tif
```

This calls **r.thin**: input=dem.tif — Input raster layer to thin; iterations=200 — Maximum number of iterations; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.tile — r.tile

**group:** Raster (r.\*)

Splits a raster map into tiles



| Parameter                               | Type              | Required | Default | Description                                     |
|-----------------------------------------|-------------------|----------|---------|-------------------------------------------------|
| input                                   | raster            | yes      | —       | Name of input raster map                        |
| width                                   | number            | no       | 1024.0  | Width of tiles (columns)                        |
| height                                  | number            | no       | 1024.0  | Height of tiles (rows)                          |
| overlap                                 | number            | no       | —       | Overlap of tiles                                |
| output                                  | folderDestination | yes      | —       | Tiles Directory                                 |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** output (outputFolder)

### Example usage

```
run grass:r.tile input=dem.tif width=1024 height=1024 overlap=10 output=output/
```

This calls **r.tile**: input=dem.tif — Name of input raster map; width=1024 — Width of tiles (columns); height=1024 — Height of tiles (rows); overlap=10 — Overlap of tiles; output=output/ is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.tileset — r.tileset

**group:** Raster (r.\*)

Produces tilings of the source projection for use in the destination region and projection.

| Parameter                      | Type            | Required | Default | Description                                                      |
|--------------------------------|-----------------|----------|---------|------------------------------------------------------------------|
| sourceproj                     | crs             | yes      | —       | Source projection                                                |
| sourcescale                    | number          | no       | 1.0     | Conversion factor from units to meters in source projection      |
| destproj                       | crs             | no       | —       | Destination projection                                           |
| destscale                      | number          | no       | 1.0     | Conversion factor from units to meters in destination projection |
| maxcols                        | number          | no       | 1024.0  | Maximum number of columns for a tile in the source projection    |
| maxrows                        | number          | no       | 1024.0  | Maximum number of rows for a tile in the source projection       |
| overlap                        | number          | no       | 0.0     | Number of cells tiles should overlap in each direction           |
| separator                      | string          | no       | pipe    | Output field separator                                           |
| -g *(adv)*                     | boolean         | no       | False   | Produces shell script output                                     |
| -w *(adv)*                     | boolean         | no       | False   | Produces web map server query string output                      |
| html                           | fileDestination | yes      | —       | Tileset                                                          |
| GRASS_REGION_PARAMETER *(adv)* | extent          | no       | —       | GRASS GIS region extent                                          |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.tileset sourceproj=EPSG:6346 sourcescale=1 destproj=EPSG:6346 destscale=1
↳ maxcols=1024 maxrows=1024 overlap=0 html=html.txt
```

This calls **r.tileset**: sourceproj=EPSG:6346 — Source projection; sourcescale=1 — Conversion factor from units to meters in source projection; destproj=EPSG:6346 — Destination projection; destscale=1 — Conversion factor from units to meters in destination projection; maxcols=1024 — Maximum number of columns for a tile in the source projection; maxrows=1024 — Maximum number of rows for a tile in the source projection; overlap=0 — Number of cells tiles should overlap in each direction; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.to.vect — r.to.vect

**group:** Raster (r.\*)

Converts a raster into a vector layer.

| Parameter                               | Type              | Required | Default | Description                                                                                             |
|-----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                   | raster            | yes      | —       | Input raster layer                                                                                      |
| type                                    | enum              | no       | 2       | Feature type — options: 0=line, 1=point, 2=area                                                         |
| column                                  | string            | no       | value   | Name of attribute column to store value                                                                 |
| -s                                      | boolean           | no       | False   | Smooth corners of area features                                                                         |
| -v                                      | boolean           | no       | False   | Use raster values as categories instead of unique sequence                                              |
| -z                                      | boolean           | no       | False   | Write raster values as z coordinate                                                                     |
| -b                                      | boolean           | no       | False   | Do not build vector topology                                                                            |
| -t                                      | boolean           | no       | False   | Do not create attribute table                                                                           |
| output                                  | vectorDestination | yes      | —       | Vectorized                                                                                              |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*     | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*               | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*       | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:r.to.vect input=dem.tif type=2 column=value output=output.gpkg
```

This calls **r.to.vect**: input=dem.tif — Input raster layer; type=2 selects *area*; column=value — Name of attribute column to store value; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.topidx — r.topidx

**group:** Raster (r.\*)

Creates topographic index layer from elevation raster layer

| Parameter                               | Type              | Required | Default | Description                                      |
|-----------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                   | raster            | yes      | —       | Input elevation layer                            |
| output                                  | rasterDestination | yes      | —       | Topographic index                                |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.topidx input=dem.tif output=output.tif
```

This calls **r.topidx**: input=dem.tif — Input elevation layer; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.topmodel — r.topmodel

**group:** Raster (r.\*)

Simulates TOPMODEL which is a physically based hydrologic model.

| Parameter                      | Type            | Required | Default | Description                                                               |
|--------------------------------|-----------------|----------|---------|---------------------------------------------------------------------------|
| parameters                     | file            | yes      | —       | Name of TOPMODEL parameters file                                          |
| topidxstats                    | file            | yes      | —       | Name of topographic index statistics file                                 |
| input                          | file            | yes      | —       | Name of rainfall and potential evapotranspiration data file               |
| timestep                       | number          | no       | —       | Time step. Generate output for this time step                             |
| topidxclass                    | number          | no       | —       | Topographic index class. Generate output for this topographic index class |
| output                         | fileDestination | yes      | —       | TOPMODEL output                                                           |
| GRASS_REGION_PARAMETER *(adv)* | extent          | no       | —       | GRASS GIS region extent                                                   |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.topmodel parameters=input.dat topidxstats=input.dat input=input.dat
↳ timestep=10 topidxclass=10 output=output.txt
```

This calls **r.topmodel**: parameters=input.dat — Name of TOPMODEL parameters file; topidxstats=input.dat — Name of topographic index statistics file; input=input.dat — Name of rainfall and potential evapotranspiration data file; timestep=10 — Time step. Generate output for this time step; topidxclass=10 — Topographic index class. Generate output for this topographic index class; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.topmodel.topidxstats — r.topmodel.topidxstats

**group:** Raster (r.\*)

r.topmodel.topidxstats - Builds a TOPMODEL topographic index statistics file.

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| topidx                                   | raster          | yes      | —       | Name of input topographic index raster map      |
| ntopidxclasses                           | number          | no       | 30.0    | Number of topographic index classes             |
| outtopidxstats                           | fileDestination | yes      | —       | TOPMODEL topographic index statistics file      |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |

**Outputs:** outtopidxstats (outputFile)

### Example usage

```
run grass:r.topmodel.topidxstats topidx=dem.tif ntopidxclasses=30
↳ outtopidxstats=outtopidxstats.txt
```

This calls **r.topmodel.topidxstats**: topidx=dem.tif — Name of input topographic index raster map; ntopidxclasses=30 — Number of topographic index classes; outtopidxstats=outtopidxstats.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.transect — r.transect

**group:** Raster (r.\*)

Outputs raster map layer values lying along user defined transect line(s).

| Parameter                               | Type            | Required | Default | Description                                                                        |
|-----------------------------------------|-----------------|----------|---------|------------------------------------------------------------------------------------|
| map                                     | raster          | yes      | —       | Raster map to be queried                                                           |
| line                                    | string          | yes      | —       | Transect definition: east,north,azimuth,distance[,east,north,azimuth,distance,...] |
| null_value                              | string          | no       | *       | String representing NULL value                                                     |
| -g *(adv)*                              | boolean         | no       | False   | Output easting and northing in first two columns of four column output             |
| html                                    | fileDestination | yes      | —       | Transect file                                                                      |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                                                            |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                    |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.transect map=dem.tif line=value null_value=* html=html.txt
```

This calls **r.transect**: map=dem.tif — Raster map to be queried; line=value — Transect definition:...; null\_value=\* — String representing NULL value; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.univar — r.univar

**group:** Raster (r.\*)

Calculates univariate statistics from the non-null cells of a raster map.

| Parameter                               | Type            | Required | Default | Description                                                                                    |
|-----------------------------------------|-----------------|----------|---------|------------------------------------------------------------------------------------------------|
| map                                     | multilayer      | yes      | —       | Name of raster map(s)                                                                          |
| zones                                   | raster          | no       | —       | Raster map used for zoning, must be of type CELL                                               |
| percentile                              | string          | no       | —       | Percentile to calculate (comma separated list if multiple) (requires extended statistics flag) |
| separator                               | string          | no       | pipe    | Field separator. Special characters: pipe, comma, space, tab, newline                          |
| -e *(adv)*                              | boolean         | no       | False   | Calculate extended statistics                                                                  |
| output                                  | fileDestination | yes      | —       | Univariate results                                                                             |
| GRASS_REGION_PARAMETER *(adv)*          | extent          | no       | —       | GRASS GIS region extent                                                                        |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.univar map="a.gpkg;b.gpkg" zones=dem.tif percentile=value separator=pipe  
↪ output=output.txt
```

This calls **r.univar**: map="a.gpkg;b.gpkg" — Name of raster map(s); zones=dem.tif — Raster map used for zoning, must be of type CELL; percentile=value — Percentile to calculate (comma separated list if multiple) (requires extended...; separator=pipe — Field separator. Special characters: pipe, comma, space, tab, newline; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.uslek — r.uslek

**group:** Raster (r.\*)

Computes USLE Soil Erodibility Factor (K).

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| psand                                    | raster            | yes      | —       | Name of soil sand fraction raster map [0.0-1.0]  |
| pclay                                    | raster            | yes      | —       | Name of soil clay fraction raster map [0.0-1.0]  |
| psilt                                    | raster            | yes      | —       | Name of soil silt fraction raster map [0.0-1.0]  |
| pomat                                    | raster            | yes      | —       | Name of soil organic matter raster map [0.0-1.0] |
| output                                   | rasterDestination | yes      | —       | USLE R Raster                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.uslek psand=dem.tif pclay=dem.tif psilt=dem.tif pomat=dem.tif  
↪ output=output.tif
```

This calls **r.uslek**: psand=dem.tif — Name of soil sand fraction raster map [0.0-1.0]; pclay=dem.tif — Name of soil clay fraction raster map [0.0-1.0]; psilt=dem.tif — Name of soil silt fraction raster map [0.0-1.0]; pomat=dem.tif — Name of soil organic matter raster map [0.0-1.0]; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.usler — r.usler

**group:** Raster (r.\*)

Computes USLE R factor, Rainfall erosivity index.

| Parameter                                | Type              | Required | Default | Description                                                                |
|------------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of annual precipitation raster map [mm/year]                          |
| method                                   | enum              | no       | 0       | Name of USLE R equation — options: 0=roose, 1=morgan, 2=foster, 3=elswaify |
| output                                   | rasterDestination | yes      | —       | USLE R Raster                                                              |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                    |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                            |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                  |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                           |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.usler input=dem.tif method=0 output=output.tif
```

This calls **r.usler**: input=dem.tif — Name of annual precipitation raster map [mm/year]; method=0 selects *roose*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.viewshed — r.viewshed

**group:** Raster (r.\*)

Computes the viewshed of a point on an elevation raster map.

| Parameter                                | Type              | Required | Default | Description                                                           |
|------------------------------------------|-------------------|----------|---------|-----------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Elevation                                                             |
| coordinates                              | point             | no       | 0.0,0.0 | Coordinate identifying the viewing position                           |
| observer_elevation                       | number            | no       | 1.75    | Viewing elevation above the ground                                    |
| target_elevation                         | number            | no       | 0.0     | Offset for target elevation above the ground                          |
| max_distance                             | number            | no       | -1.0    | Maximum visibility radius. By default infinity (-1)                   |
| refraction_coeff                         | number            | no       | 0.14286 | Refraction coefficient                                                |
| memory                                   | number            | no       | 500.0   | Amount of memory to use in MB                                         |
| -c *(adv)*                               | boolean           | no       | False   | Consider earth curvature (current ellipsoid)                          |
| -r *(adv)*                               | boolean           | no       | False   | Consider the effect of atmospheric refraction                         |
| -b *(adv)*                               | boolean           | no       | False   | Output format is invisible = 0, visible = 1                           |
| -e *(adv)*                               | boolean           | no       | False   | Output format is invisible = NULL, else current elev - viewpoint_elev |
| output                                   | rasterDestination | yes      | —       | Intervisibility                                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                               |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                       |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                             |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                      |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.viewshed input=dem.tif coordinates=100,100 observer_elevation=1.75
↪ target_elevation=0 max_distance=-1 refraction_coeff=0.14286 memory=500
↪ output=output.tif
```

This calls **r.viewshed**: input=dem.tif — Elevation; coordinates=100,100 — Coordinate identifying the viewing position; observer\_elevation=1.75 — Viewing elevation above the ground; target\_elevation=0 — Offset for target elevation above the ground; max\_distance=-1 — Maximum visibility radius. By default infinity (-1); refraction\_coeff=0.14286 — Refraction coefficient; memory=500 — Amount of memory to use in MB; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.volume — r.volume

**group:** Raster (r.\*)

Calculates the volume of data "clumps".

| Parameter                                | Type              | Required | Default | Description                                                                  |
|------------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------|
| input                                    | raster            | yes      | —       | Name of input raster map representing data that will be summed within clumps |
| clump                                    | raster            | yes      | —       | Clumps layer (preferably the output of r.clump)                              |
| -f *(adv)*                               | boolean           | no       | False   | Generate unformatted report                                                  |
| centroids                                | vectorDestination | yes      | —       | Centroids                                                                    |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                      |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                              |

| Parameter                           | Type    | Required | Default | Description                                                                                             |
|-------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** centroids (outputVector)

### Example usage

```
run grass:r.volume input=dem.tif clump=dem.tif centroids=centroids.gpkg
```

This calls **r.volume**: input=dem.tif — Name of input raster map representing data that will be summed within clumps; clump=dem.tif — Clumps layer (preferably the output of r.clump); centroids=centroids.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.walk.coords — r.walk.coords

**group:** Raster (r.\*)

r.walk.coords - Creates a raster map showing the anisotropic cumulative cost of moving between different geographic locations on an input raster map whose cell category values represent cost from a list of coordinates.

| Parameter                                | Type              | Required | Default                 | Description                                                        |
|------------------------------------------|-------------------|----------|-------------------------|--------------------------------------------------------------------|
| elevation                                | raster            | yes      | —                       | Name of input elevation raster map                                 |
| friction                                 | raster            | yes      | —                       | Name of input raster map containing friction costs                 |
| start_coordinates                        | string            | yes      | —                       | Coordinates of starting point(s) (a list of E,N)                   |
| stop_coordinates                         | string            | no       | —                       | Coordinates of stopping point(s) (a list of E,N)                   |
| walk_coeff                               | string            | no       | 0.72,6.0,1.9998,-1.9998 | Coefficients for walking energy formula parameters a,b,c,d         |
| lambda                                   | number            | no       | 1.0                     | Lambda coefficients for combining walking energy and friction cost |
| slope_factor                             | number            | no       | -0.2125                 | Slope factor determines travel energy cost per height step         |
| max_cost                                 | number            | no       | 0.0                     | Maximum cumulative cost                                            |
| null_cost                                | number            | no       | —                       | Cost assigned to null cells. By default, null cells are excluded   |
| memory *(adv)*                           | number            | no       | 300.0                   | Maximum memory to be used in MB                                    |
| -k *(adv)*                               | boolean           | no       | False                   | Use the 'Knight's move'; slower, but more accurate                 |
| -n *(adv)*                               | boolean           | no       | False                   | Keep null values in output raster layer                            |
| output                                   | rasterDestination | yes      | —                       | Cumulative cost                                                    |
| outdir                                   | rasterDestination | yes      | —                       | Movement Directions                                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —                       | GRASS GIS region extent                                            |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0                     | GRASS GIS region cellsize (leave 0 for default)                    |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —                       | Output Rasters format options (createopt)                          |
| GRASS_RASTER_FORMAT_METADATA *(adv)*     | string            | no       | —                       | Output Rasters format metadata options (metaopt)                   |

**Outputs:** output (outputRaster), outdir (outputRaster)

## Example usage

```
run grass:r.walk.coords elevation=dem.tif friction=dem.tif start_coordinates=value
↳ stop_coordinates=value walk_coeff=0.72,6.0,1.9998,-1.9998 lambda=1
↳ slope_factor=-0.2125 output=output.tif
```

This calls **r.walk.coords**: elevation=dem.tif — Name of input elevation raster map; friction=dem.tif — Name of input raster map containing friction costs; start\_coordinates=value — Coordinates of starting point(s) (a list of E,N); stop\_coordinates=value — Coordinates of stopping point(s) (a list of E,N); walk\_coeff=0.72,6.0,1.9998,-1.9998 — Coefficients for walking energy formula parameters a,b,c,d; lambda=1 — Lambda coefficients for combining walking energy and friction cost; slope\_factor=-0.2125 — Slope factor determines travel energy cost per height step; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.walk.points — r.walk.points

**group:** Raster (r.\*)

r.walk.points - Creates a raster map showing the anisotropic cumulative cost of moving between different geographic locations on an input raster map whose cell category values represent cost from point vector layers.

| Parameter                                | Type              | Required | Default                 | Description                                                        |
|------------------------------------------|-------------------|----------|-------------------------|--------------------------------------------------------------------|
| elevation                                | raster            | yes      | —                       | Name of input elevation raster map                                 |
| friction                                 | raster            | yes      | —                       | Name of input raster map containing friction costs                 |
| start_points                             | source            | yes      | —                       | Start points                                                       |
| stop_points                              | source            | no       | —                       | Stop points                                                        |
| walk_coeff                               | string            | no       | 0.72,6.0,1.9998,-1.9998 | Coefficients for walking energy formula parameters a,b,c,d         |
| lambda                                   | number            | no       | 1.0                     | Lambda coefficients for combining walking energy and friction cost |
| slope_factor                             | number            | no       | -0.2125                 | Slope factor determines travel energy cost per height step         |
| max_cost                                 | number            | no       | 0.0                     | Maximum cumulative cost                                            |
| null_cost                                | number            | no       | —                       | Cost assigned to null cells. By default, null cells are excluded   |
| memory *(adv)*                           | number            | no       | 300.0                   | Maximum memory to be used in MB                                    |
| -k *(adv)*                               | boolean           | no       | False                   | Use the 'Knight's move'; slower, but more accurate                 |
| -n *(adv)*                               | boolean           | no       | False                   | Keep null values in output raster layer                            |
| output                                   | rasterDestination | yes      | —                       | Cumulative cost                                                    |
| outdir                                   | rasterDestination | yes      | —                       | Movement Directions                                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —                       | GRASS GIS region extent                                            |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0                     | GRASS GIS region cellsize (leave 0 for default)                    |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —                       | Output Rasters format options (createopt)                          |
| GRASS_RASTER_FORMAT_METADATA *(adv)*     | string            | no       | —                       | Output Rasters format metadata options (metaopt)                   |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0                    | v.in.ogr snap tolerance (-1 = no snap)                             |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001                  | v.in.ogr min area                                                  |

**Outputs:** output (outputRaster), outdir (outputRaster)

## Example usage



```
run grass:r.walk.points elevation=dem.tif friction=dem.tif start_points=input.gpkg
↳ stop_points=input.gpkg walk_coeff=0.72,6.0,1.9998,-1.9998 lambda=1
↳ slope_factor=-0.2125 output=output.tif
```

This calls **r.walk.points**: elevation=dem.tif — Name of input elevation raster map; friction=dem.tif — Name of input raster map containing friction costs; start\_points=input.gpkg — Start points; stop\_points=input.gpkg — Stop points; walk\_coeff=0.72,6.0,1.9998,-1.9998 — Coefficients for walking energy formula parameters a,b,c,d; lambda=1 — Lambda coefficients for combining walking energy and friction cost; slope\_factor=-0.2125 — Slope factor determines travel energy cost per height step; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.walk.rast — r.walk.rast

**group:** Raster (r.\*)

r.walk.rast - Creates a raster map showing the anisotropic cumulative cost of moving between different geographic locations on an input raster map whose cell category values represent cost from a raster.

| Parameter                                | Type              | Required | Default                 | Description                                                                 |
|------------------------------------------|-------------------|----------|-------------------------|-----------------------------------------------------------------------------|
| elevation                                | raster            | yes      | —                       | Name of input elevation raster map                                          |
| friction                                 | raster            | yes      | —                       | Name of input raster map containing friction costs                          |
| start_raster                             | raster            | yes      | —                       | Name of starting raster points map (all non-NULL cells are starting points) |
| walk_coeff                               | string            | no       | 0.72,6.0,1.9998,-1.9998 | Coefficients for walking energy formula parameters a,b,c,d                  |
| lambda                                   | number            | no       | 1.0                     | Lambda coefficients for combining walking energy and friction cost          |
| slope_factor                             | number            | no       | -0.2125                 | Slope factor determines travel energy cost per height step                  |
| max_cost                                 | number            | no       | 0.0                     | Maximum cumulative cost                                                     |
| null_cost                                | number            | no       | —                       | Cost assigned to null cells. By default, null cells are excluded            |
| memory *(adv)*                           | number            | no       | 300.0                   | Maximum memory to be used in MB                                             |
| -k *(adv)*                               | boolean           | no       | False                   | Use the 'Knight's move'; slower, but more accurate                          |
| -n *(adv)*                               | boolean           | no       | False                   | Keep null values in output raster layer                                     |
| output                                   | rasterDestination | yes      | —                       | Cumulative cost                                                             |
| outdir                                   | rasterDestination | yes      | —                       | Movement Directions                                                         |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —                       | GRASS GIS region extent                                                     |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0                     | GRASS GIS region cellsize (leave 0 for default)                             |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —                       | Output Rasters format options (createopt)                                   |
| GRASS_RASTER_FORMAT_METADATA *(adv)*     | string            | no       | —                       | Output Rasters format metadata options (metaopt)                            |

**Outputs:** output (outputRaster), outdir (outputRaster)

### Example usage

```
run grass:r.walk.rast elevation=dem.tif friction=dem.tif start_raster=dem.tif
↳ walk_coeff=0.72,6.0,1.9998,-1.9998 lambda=1 slope_factor=-0.2125 max_cost=0
↳ output=output.tif
```

This calls **r.walk.rast**: elevation=dem.tif — Name of input elevation raster map; friction=dem.tif — Name of input raster map containing friction costs; start\_raster=dem.tif —

Name of starting raster points map (all non-NULL cells are starting points); `walk_coeff=0.72,6 .0,1.9998,-1.9998` — Coefficients for walking energy formula parameters a,b,c,d; `lambda=1` — Lambda coefficients for combining walking energy and friction cost; `slope_factor=-0.2125` — Slope factor determines travel energy cost per height step; `max_cost=0` — Maximum cumulative cost; `output=output.tif` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.water.outlet — r.water.outlet

**group:** Raster (r.\*)

Watershed basin creation program.

| Parameter                                | Type              | Required | Default | Description                                      |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------|
| input                                    | raster            | yes      | —       | Drainage direction raster                        |
| coordinates                              | point             | no       | 0.0,0.0 | Coordinates of outlet point                      |
| output                                   | rasterDestination | yes      | —       | Basin                                            |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                          |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)  |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)        |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt) |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:r.water.outlet input=dem.tif coordinates=100,100 output=output.tif
```

This calls **r.water.outlet**: `input=dem.tif` — Drainage direction raster; `coordinates=100,100` — Coordinates of outlet point; `output=output.tif` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.watershed — r.watershed

**group:** Raster (r.\*)

Watershed basin analysis program.

| Parameter        | Type    | Required | Default | Description                                                                 |
|------------------|---------|----------|---------|-----------------------------------------------------------------------------|
| elevation        | raster  | yes      | —       | Elevation                                                                   |
| depression       | raster  | no       | —       | Locations of real depressions                                               |
| flow             | raster  | no       | —       | Amount of overland flow per cell                                            |
| disturbed_land   | raster  | no       | —       | Percent of disturbed land, for USLE                                         |
| blocking         | raster  | no       | —       | Terrain blocking overland surface flow, for USLE                            |
| threshold        | number  | no       | —       | Minimum size of exterior watershed basin                                    |
| max_slope_length | number  | no       | —       | Maximum length of surface flow, for USLE                                    |
| convergence      | number  | no       | 5.0     | Convergence factor for MFD (1-10)                                           |
| memory           | number  | no       | 300.0   | Maximum memory to be used with -m flag (in MB)                              |
| -s               | boolean | no       | False   | Enable Single Flow Direction (D8) flow (default is Multiple Flow Direction) |
| -m               | boolean | no       | False   | Enable disk swap memory option (-m): Operation is slow                      |
| -4               | boolean | no       | False   | Allow only horizontal and vertical flow of water                            |

| Parameter                                | Type              | Required | Default | Description                                                   |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------|
| -a                                       | boolean           | no       | False   | Use positive flow accumulation even for likely underestimates |
| -b                                       | boolean           | no       | False   | Beautify flat areas                                           |
| accumulation                             | rasterDestination | no       | —       | Number of cells that drain through each cell                  |
| drainage                                 | rasterDestination | no       | —       | Drainage direction                                            |
| basin                                    | rasterDestination | no       | —       | Unique label for each watershed basin                         |
| stream                                   | rasterDestination | no       | —       | Stream segments                                               |
| half_basin                               | rasterDestination | no       | —       | Half-basins                                                   |
| length_slope                             | rasterDestination | no       | —       | Slope length and steepness (LS) factor for USLE               |
| slope_steepness                          | rasterDestination | no       | —       | Slope steepness (S) factor for USLE                           |
| tci                                      | rasterDestination | no       | —       | Topographic index $\ln(a / \tan(b))$                          |
| spi                                      | rasterDestination | no       | —       | Stream power index $a * \tan(b)$                              |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                       |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)               |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                     |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)              |

**Outputs:** accumulation (outputRaster), drainage (outputRaster), basin (outputRaster), stream (outputRaster), half\_basin (outputRaster), length\_slope (outputRaster), slope\_steepness (outputRaster), tci (outputRaster), spi (outputRaster)

### Example usage

```
run grass:r.watershed elevation=dem.tif depression=dem.tif flow=dem.tif
↳ disturbed_land=dem.tif blocking=dem.tif threshold=10 max_slope_length=10
↳ accumulation=accumulation.tif
```

This calls **r.watershed**: elevation=dem.tif — Elevation; depression=dem.tif — Locations of real depressions; flow=dem.tif — Amount of overland flow per cell; disturbed\_land=dem.tif — Percent of disturbed land, for USLE; blocking=dem.tif — Terrain blocking overland surface flow, for USLE; threshold=10 — Minimum size of exterior watershed basin; max\_slope\_length=10 — Maximum length of surface flow, for USLE; accumulation=accumulation.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.what.color — r.what.color

**group:** Raster (r.\*)

Queries colors for a raster map layer.

| Parameter                                | Type            | Required | Default  | Description                                       |
|------------------------------------------|-----------------|----------|----------|---------------------------------------------------|
| input                                    | raster          | yes      | —        | Raster map to query colors                        |
| value                                    | string          | no       | —        | Values to query colors for (comma separated list) |
| format                                   | string          | no       | %d:%d:%d | Output format (printf-style)                      |
| html                                     | fileDestination | yes      | —        | Colors file                                       |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —        | GRASS GIS region extent                           |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0      | GRASS GIS region cellsize (leave 0 for default)   |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:r.what.color input=dem.tif value=value format=%d:%d:%d html=html.txt
```

This calls **r.what.color**: input=dem.tif — Raster map to query colors; value=value — Values to query colors for (comma separated list); format=%d:%d:%d — Output format (printf-style); html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.what.coords — r.what.coords

**group:** Raster (r.\*)

r.what.coords - Queries raster maps on their category values and category labels on a point.

| Parameter                                | Type            | Required | Default  | Description                                                                                         |
|------------------------------------------|-----------------|----------|----------|-----------------------------------------------------------------------------------------------------|
| map                                      | raster          | yes      | —        | Name of raster map                                                                                  |
| coordinates                              | point           | no       | 0.0, 0.0 | Coordinates for query (east, north)                                                                 |
| null_value                               | string          | no       | *        | String representing NULL value                                                                      |
| separator                                | string          | no       | pipe     | Field separator. Special characters: pipe, comma, space, tab, newlineString representing NULL value |
| cache                                    | number          | no       | 500.0    | Size of point cache                                                                                 |
| -n *(adv)*                               | boolean         | no       | False    | Output header row                                                                                   |
| -f *(adv)*                               | boolean         | no       | False    | Show the category labels of the grid cell(s)                                                        |
| -r *(adv)*                               | boolean         | no       | False    | Output color values as RRR:GGG:BBB                                                                  |
| -i *(adv)*                               | boolean         | no       | False    | Output integer category values, not cell values                                                     |
| -c *(adv)*                               | boolean         | no       | False    | Turn on cache reporting                                                                             |
| output                                   | fileDestination | yes      | —        | Raster Value File                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —        | GRASS GIS region extent                                                                             |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0      | GRASS GIS region cellsize (leave 0 for default)                                                     |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.what.coords map=dem.tif coordinates=100,100 null_value=* separator=pipe
↳ cache=500 output=output.txt
```

This calls **r.what.coords**: map=dem.tif — Name of raster map; coordinates=100,100 — Coordinates for query (east, north); null\_value=\* — String representing NULL value; separator=pipe — Field separator. Special characters: pipe, comma, space, tab, newlineString...; cache=500 — Size of point cache; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:r.what.points — r.what.points

**group:** Raster (r.\*)

r.what.points - Queries raster maps on their category values and category labels on a layer of points.

| Parameter  | Type    | Required | Default | Description                                                           |
|------------|---------|----------|---------|-----------------------------------------------------------------------|
| map        | raster  | yes      | —       | Name of raster map                                                    |
| points     | source  | yes      | —       | Name of vector points layer for query                                 |
| null_value | string  | no       | *       | String representing NULL value                                        |
| separator  | string  | no       | pipe    | Field separator. Special characters: pipe, comma, space, tab, newline |
| cache      | number  | no       | 500.0   | Size of point cache                                                   |
| -n *(adv)* | boolean | no       | False   | Output header row                                                     |
| -f *(adv)* | boolean | no       | False   | Show the category labels of the grid cell(s)                          |

| Parameter                                | Type            | Required | Default | Description                                     |
|------------------------------------------|-----------------|----------|---------|-------------------------------------------------|
| -r *(adv)*                               | boolean         | no       | False   | Output color values as RRR:GGG:BBB              |
| -i *(adv)*                               | boolean         | no       | False   | Output integer category values, not cell values |
| -c *(adv)*                               | boolean         | no       | False   | Turn on cache reporting                         |
| output                                   | fileDestination | yes      | —       | Raster Values File                              |
| GRASS_REGION_PARAMETER *(adv)*           | extent          | no       | —       | GRASS GIS region extent                         |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number          | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default) |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number          | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)          |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number          | no       | 0.0001  | v.in.ogr min area                               |

**Outputs:** output (outputFile)

### Example usage

```
run grass:r.what.points map=dem.tif points=input.gpkg null_value=* separator=pipe
↪ cache=500 output=output.txt
```

This calls **r.what.points**: map=dem.tif — Name of raster map; points=input.gpkg — Name of vector points layer for query; null\_value=\* — String representing NULL value; separator=pipe — Field separator. Special characters: pipe, comma, space, tab, newline; cache=500 — Size of point cache; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.buffer — v.buffer

**group:** Vector (v.\*)

Creates a buffer around vector features of given type.

| Parameter                              | Type              | Required | Default   | Description                                                                         |
|----------------------------------------|-------------------|----------|-----------|-------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —         | Input vector layer                                                                  |
| cats                                   | string            | no       | —         | Category values                                                                     |
| where                                  | string            | no       | —         | WHERE conditions of SQL statement without 'where' keyword                           |
| type                                   | enum              | no       | [0, 1, 4] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area       |
| distance                               | number            | no       | —         | Buffer distance in map units                                                        |
| minordistance                          | number            | no       | —         | Buffer distance along minor axis in map units                                       |
| angle                                  | number            | no       | 0.0       | Angle of major axis in degrees                                                      |
| layer                                  | string            | no       | -1        | Layer number or name ('-1' for all layers)                                          |
| column                                 | field             | no       | —         | Name of column to use for buffer distances                                          |
| scale                                  | number            | no       | 1.0       | Scaling factor for attribute column values                                          |
| tolerance                              | number            | no       | 0.01      | Maximum distance between theoretical arc and polygon segments as multiple of buffer |
| -s *(adv)*                             | boolean           | no       | False     | Make outside corners straight                                                       |
| -c *(adv)*                             | boolean           | no       | False     | Do not make caps at the ends of polylines                                           |
| -t *(adv)*                             | boolean           | no       | False     | Transfer categories and attributes                                                  |
| output                                 | vectorDestination | yes      | —         | Buffer                                                                              |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —         | GRASS GIS region extent                                                             |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0      | v.in.ogr snap tolerance (-1 = no snap)                                              |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001    | v.in.ogr min area                                                                   |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0         | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                    |
| GRASS_VECTOR_DS_CO *(adv)*             | string            | no       | —         | v.out.ogr output data source options (dsco)                                         |

| Parameter                         | Type    | Required | Default | Description                                                                                             |
|-----------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_VECTOR_LCO *(adv)*          | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)* | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

#### Example usage

```
run grass:v.buffer input=input.gpkg cats=value where=value type=0 distance=10
↳ minordistance=10 angle=0 output=output.gpkg
```

This calls **v.buffer**: input=input.gpkg — Input vector layer; cats=value — Category values; where=value — WHERE conditions of SQL statement without 'where' keyword; type=0 selects *point*; distance=10 — Buffer distance in map units; minordistance=10 — Buffer distance along minor axis in map units; angle=0 — Angle of major axis in degrees; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.build.check — v.build.check

**group:** Vector (v.\*)

v.build.check - Checks for topological errors.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| map                                    | source            | yes      | —       | Name of vector map                                                                                      |
| error                                  | vectorDestination | yes      | —       | Topological errors                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** error (outputVector)

#### Example usage

```
run grass:v.build.check map=input.gpkg error=error.gpkg
```

This calls **v.build.check**: map=input.gpkg — Name of vector map; error=error.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.build.polylines — v.build.polylines

**group:** Vector (v.\*)

Builds polylines from lines or boundaries.

| Parameter | Type   | Required | Default | Description              |
|-----------|--------|----------|---------|--------------------------|
| input     | source | yes      | —       | Name of input vector map |

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| cats                                   | enum              | no       | 0       | Category number mode — options: 0=no, 1=first, 2=multi, 3=same                                          |
| type                                   | enum              | no       | [0, 1]  | Input feature type — options: 0=line, 1=boundary                                                        |
| output                                 | vectorDestination | yes      | —       | Polylines                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.build.polylines input=input.gpkg cats=0 type=0 output=output.gpkg
```

This calls **v.build.polylines**: input=input.gpkg — Name of input vector map; cats=0 selects *no*; type=0 selects *line*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.class — v.class

**group:** Vector (v.\*)

Classifies attribute data, e.g. for thematic mapping.

| Parameter                              | Type            | Required | Default     | Description                                                               |
|----------------------------------------|-----------------|----------|-------------|---------------------------------------------------------------------------|
| map                                    | source          | yes      | —           | Input vector layer                                                        |
| column                                 | field           | yes      | —           | Column name or expression                                                 |
| where                                  | string          | no       | —           | WHERE conditions of SQL statement without 'where' keyword                 |
| algorithm                              | enum            | no       | 0           | Algorithm to use for classification — options: 0=int, 1=std, 2=qua, 3=equ |
| nbclasses                              | number          | no       | 3.0         | Number of classes to define                                               |
| -g                                     | boolean         | no       | True        | Print only class breaks (without min and max)                             |
| html                                   | fileDestination | no       | report.html | Classification                                                            |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —           | GRASS GIS region extent                                                   |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0        | v.in.ogr snap tolerance (-1 = no snap)                                    |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001      | v.in.ogr min area                                                         |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:v.class map=input.gpkg column=field1 where=value algorithm=0 nbclasses=3
↳ html=html.txt
```

This calls **v.class**: map=input.gpkg — Input vector layer; column=field1 — Column name or expression; where=value — WHERE conditions of SQL statement without 'where' keyword; algorithm=0 selects *int*; nbclasses=3 — Number of classes to define; html=html.txt is where the

result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.clean — v.clean

**group:** Vector (v.\*)

Toolset for cleaning topology of vector map.

| Parameter                              | Type              | Required | Default               | Description                                                                                                                                                 |
|----------------------------------------|-------------------|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —                     | Layer to clean                                                                                                                                              |
| type                                   | enum              | no       | [0, 1, 2, 3, 4, 5, 6] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area, 5=face, 6=kernel                                                             |
| tool                                   | enum              | no       | [0]                   | Cleaning tool — options: 0=break, 1=snap, 2=rmdangle, 3=chdangle, 4=rmbridge, 5=chbridge, 6=rmdupl, 7=rmdac, 8=bpol, 9=prune, 10=rmarea, 11=rmline, 12=rmsa |
| threshold                              | string            | no       | —                     | Threshold (comma separated for each tool)                                                                                                                   |
| -b *(adv)*                             | boolean           | no       | False                 | Do not build topology for the output vector                                                                                                                 |
| -c *(adv)*                             | boolean           | no       | False                 | Combine tools with recommended follow-up tools                                                                                                              |
| output                                 | vectorDestination | yes      | —                     | Cleaned                                                                                                                                                     |
| error                                  | vectorDestination | yes      | —                     | Errors                                                                                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —                     | GRASS GIS region extent                                                                                                                                     |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0                  | v.in.ogr snap tolerance (-1 = no snap)                                                                                                                      |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001                | v.in.ogr min area                                                                                                                                           |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0                     | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                                                            |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —                     | v.out.ogr output data source options (dsco)                                                                                                                 |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —                     | v.out.ogr output layer options (lco)                                                                                                                        |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False                 | Also export features without category (not labeled). Otherwise only features with category are exported                                                     |

**Outputs:** output (outputVector), error (outputVector)

### Example usage

```
run grass:v.clean input=input.gpkg type=0 tool=0 threshold=value output=output.gpkg
```

This calls **v.clean**: input=input.gpkg — Layer to clean; type=0 selects *point*; tool=0 selects *break*; threshold=value — Threshold (comma separated for each tool); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.cluster — v.cluster

**group:** Vector (v.\*)

Performs cluster identification

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| input     | source | yes      | —       | Input layer |



| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| distance                               | number            | no       | —       | Maximum distance to neighbors                                                                           |
| min                                    | number            | no       | —       | Minimum number of points to create a cluster                                                            |
| method                                 | enum              | no       | [0]     | Clustering method — options: 0=dbscan, 1=dbscan2, 2=density, 3=optics, 4=optics2                        |
| -2 *(adv)*                             | boolean           | no       | False   | Force 2D clustering                                                                                     |
| -b *(adv)*                             | boolean           | no       | False   | Do not build topology                                                                                   |
| -t *(adv)*                             | boolean           | no       | False   | Do not create attribute table                                                                           |
| output                                 | vectorDestination | yes      | —       | Clustered                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.cluster input=input.gpkg distance=10 min=10 method=0 output=output.gpkg
```

This calls **v.cluster**: input=input.gpkg — Input layer; distance=10 — Maximum distance to neighbors; min=10 — Minimum number of points to create a cluster; method=0 selects *dbscan*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.db.select — v.db.select

**group:** Vector (v.\*)

Prints vector map attributes

| Parameter                              | Type            | Required | Default | Description                                                                   |
|----------------------------------------|-----------------|----------|---------|-------------------------------------------------------------------------------|
| map                                    | source          | yes      | —       | Input vector map                                                              |
| layer                                  | number          | no       | 1.0     | Layer Number                                                                  |
| columns                                | string          | no       | —       | Name of attribute column(s), comma separated                                  |
| -c *(adv)*                             | boolean         | no       | False   | Do not include column names in output                                         |
| separator                              | string          | no       | ,       | Output field separator                                                        |
| where *(adv)*                          | string          | no       | —       | WHERE conditions of SQL statement without 'where' keyword                     |
| group *(adv)*                          | string          | no       | —       | GROUP BY conditions of SQL statement without 'group by' keyword               |
| vertical_separator *(adv)*             | string          | no       | —       | Output vertical record separator                                              |
| null_value *(adv)*                     | string          | no       | —       | Null value indicator                                                          |
| -v *(adv)*                             | boolean         | no       | False   | Vertical output (instead of horizontal)                                       |
| -r *(adv)*                             | boolean         | no       | False   | Print minimal region extent of selected vector features instead of attributes |
| file                                   | fileDestination | yes      | —       | Attributes                                                                    |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —       | GRASS GIS region extent                                                       |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                        |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001  | v.in.ogr min area                                                             |

**Outputs:** file (outputFile)

### Example usage

```
run grass:v.db.select map=input.gpkg layer=1 columns=value separator=, file=file.txt
```

This calls **v.db.select**: map=input.gpkg — Input vector map; layer=1 — Layer Number; columns=value — Name of attribute column(s), comma separated; separator=, — Output field separator; file=file.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.decimate — v.decimate

**group:** Vector (v.\*)

Decimates a point cloud

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector                                                                                            |
| zrange                                 | range             | no       | —       | Filter range for z data (min,max)                                                                       |
| cats                                   | string            | no       | —       | Category values                                                                                         |
| skip                                   | number            | no       | —       | Throw away every n-th point                                                                             |
| preserve                               | number            | no       | —       | Preserve only every n-th point                                                                          |
| offset                                 | number            | no       | —       | Skip first n points                                                                                     |
| limit                                  | number            | no       | —       | Copy only n points                                                                                      |
| zdiff                                  | number            | no       | —       | Minimal difference of z values                                                                          |
| cell_limit                             | number            | no       | —       | Preserve only n points per grid cell                                                                    |
| -g *(adv)*                             | boolean           | no       | False   | Apply grid-based decimation                                                                             |
| -f *(adv)*                             | boolean           | no       | False   | Use only first point in grid cell during grid-based decimation                                          |
| -c *(adv)*                             | boolean           | no       | False   | Only one point per cat in grid cell                                                                     |
| -z *(adv)*                             | boolean           | no       | False   | Use z in grid decimation                                                                                |
| -x *(adv)*                             | boolean           | no       | False   | Store only the coordinates, throw away categories                                                       |
| -b *(adv)*                             | boolean           | no       | False   | Do not build topology                                                                                   |
| output                                 | vectorDestination | yes      | —       | Output vector map                                                                                       |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.decimate input=input.gpkg cats=value skip=10 preserve=10 offset=10 limit=10  
↪ output=output.gpkg
```

This calls **v.decimate**: input=input.gpkg — Input vector; cats=value — Category values; skip=10 — Throw away every n-th point; preserve=10 — Preserve only every n-th point; offset=10 — Skip first n points; limit=10 — Copy only n points; output=output.gpkg is where the

result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.delaunay — v.delaunay

**group:** Vector (v.\*)

Creates a Delaunay triangulation from an input vector map containing points or centroids.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector layer                                                                                      |
| -r                                     | boolean           | no       | False   | Use only points in current region                                                                       |
| -l                                     | boolean           | no       | False   | Output triangulation as a graph (lines), not areas                                                      |
| output                                 | vectorDestination | yes      | —       | Delaunay triangulation                                                                                  |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.delaunay input=input.gpkg output=output.gpkg
```

This calls **v.delaunay**: input=input.gpkg — Input vector layer; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.dissolve — v.dissolve

**group:** Vector (v.\*)

Dissolves boundaries between adjacent areas sharing a common category number or attribute.

| Parameter                              | Type              | Required | Default | Description                                                      |
|----------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector layer                                               |
| column                                 | field             | no       | —       | Name of column used to dissolve common boundaries                |
| output                                 | vectorDestination | yes      | —       | Dissolved                                                        |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                           |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                      |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                             |

| Parameter                            | Type    | Required | Default | Description                                                                                             |
|--------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_VECTOR_EXPORT_NOCAT<br>*(adv)* | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.dissolve input=input.gpkg column=field1 output=output.gpkg
```

This calls **v.dissolve**: input=input.gpkg — Input vector layer; column=field1 — Name of column used to dissolve common boundaries; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.distance — v.distance

**group:** Vector (v.\*)

Finds the nearest element in vector map 'to' for elements in vector map 'from'.

| Parameter                              | Type              | Required | Default   | Description                                                                                                                                       |
|----------------------------------------|-------------------|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| from                                   | source            | yes      | —         | 'from' vector map                                                                                                                                 |
| from_type *(adv)*                      | enum              | no       | [0, 1, 3] | 'from' feature type — options: 0=point, 1=line, 2=boundary, 3=area, 4=centroid                                                                    |
| to                                     | source            | yes      | —         | 'to' vector map                                                                                                                                   |
| to_type *(adv)*                        | enum              | no       | [0, 1, 3] | 'to' feature type — options: 0=point, 1=line, 2=boundary, 3=area, 4=centroid                                                                      |
| dmax                                   | number            | no       | -1.0      | Maximum distance or -1.0 for no limit                                                                                                             |
| dmin                                   | number            | no       | -1.0      | Minimum distance or -1.0 for no limit                                                                                                             |
| upload                                 | enum              | no       | [0]       | 'upload': Values describing the relation between two nearest features — options: 0=cat, 1=dist, 2=to_x, 3=to_y, 4=to_along, 5=to_angle, 6=to_attr |
| column                                 | field             | yes      | —         | Column name(s) where values specified by 'upload' option will be uploaded                                                                         |
| to_column                              | field             | no       | —         | Column name of nearest feature (used with upload=to_attr)                                                                                         |
| from_output                            | vectorDestination | yes      | —         | Nearest                                                                                                                                           |
| output                                 | vectorDestination | yes      | —         | Distance                                                                                                                                          |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —         | GRASS GIS region extent                                                                                                                           |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0      | v.in.ogr snap tolerance (-1 = no snap)                                                                                                            |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001    | v.in.ogr min area                                                                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0         | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                                                  |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —         | v.out.ogr output data source options (dsco)                                                                                                       |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —         | v.out.ogr output layer options (lco)                                                                                                              |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False     | Also export features without category (not labeled). Otherwise only features with category are exported                                           |

**Outputs:** from\_output (outputVector), output (outputVector)

### Example usage

```
run grass:v.distance from=input.gpkg to=input.gpkg column=field1 dmax=-1 dmin=-1 upload=0
↳ to_column=field1 from_output=from_output.gpkg
```

This calls **v.distance**: from=input.gpkg — 'from' vector map; to=input.gpkg — 'to' vector map; column=field1 — Column name(s) where values specified by 'upload' option will be uploaded;

dmax=-1 — Maximum distance or -1.0 for no limit; dmin=-1 — Minimum distance or -1.0 for no limit; upload=0 selects *cat*; to\_column=field1 — Column name of nearest feature (used with upload=to\_attr); from\_output=from\_output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.drape — v.drape

**group:** Vector (v.\*)

Converts 2D vector features to 3D by sampling of elevation raster map.

| Parameter                                | Type              | Required | Default      | Description                                                                                             |
|------------------------------------------|-------------------|----------|--------------|---------------------------------------------------------------------------------------------------------|
| input                                    | source            | yes      | —            | Input vector layer                                                                                      |
| where                                    | string            | no       | —            | WHERE conditions of SQL statement without 'where' keyword                                               |
| type                                     | enum              | no       | [0, 1, 3, 4] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid                                   |
| elevation                                | raster            | yes      | —            | Elevation raster map for height extraction                                                              |
| method                                   | enum              | no       | 0            | Sampling method — options: 0=nearest, 1=bilinear, 2=bicubic                                             |
| scale                                    | number            | no       | 1.0          | Scale factor sampled raster values                                                                      |
| null_value                               | number            | no       | —            | Height for sampled raster NULL values                                                                   |
| output                                   | vectorDestination | yes      | —            | 3D vector                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —            | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0          | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0         | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001       | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0            | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DS_CO *(adv)*               | string            | no       | —            | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —            | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False        | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.drape input=input.gpkg elevation=dem.tif where=value type=0 method=0 scale=1
↪ null_value=10 output=output.gpkg
```

This calls **v.drape**: input=input.gpkg — Input vector layer; elevation=dem.tif — Elevation raster map for height extraction; where=value — WHERE conditions of SQL statement without 'where' keyword; type=0 selects *point*; method=0 selects *nearest*; scale=1 — Scale factor sampled raster values; null\_value=10 — Height for sampled raster NULL values; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.edit — v.edit

**group:** Vector (v.\*)

Edits a vector map, allows adding, deleting and modifying selected vector features.

| Parameter                              | Type              | Required | Default      | Description                                                                                                                                                                                                          |
|----------------------------------------|-------------------|----------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| map                                    | source            | yes      | —            | Name of vector layer                                                                                                                                                                                                 |
| type                                   | enum              | no       | [0, 1, 2, 3] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid                                                                                                                                                |
| tool                                   | enum              | no       | 0            | Tool — options: 0=create, 1=add, 2=delete, 3=copy, 4=move, 5=flip, 6=catadd, 7=catdel, 8=merge, 9=break, 10=snap, 11=connect, 12=chtype, 13=vertexadd, 14=vertexdel, 15=vertex-move, 16=areadel, 17=zbulk, 18=select |
| input                                  | file              | no       | —            | ASCII file for add tool                                                                                                                                                                                              |
| move                                   | string            | no       | —            | Difference in x,y,z direction for moving feature or vertex                                                                                                                                                           |
| threshold                              | string            | no       | —            | Threshold distance (coords,snap,query)                                                                                                                                                                               |
| ids                                    | string            | no       | —            | Feature ids                                                                                                                                                                                                          |
| cats                                   | string            | no       | —            | Category values                                                                                                                                                                                                      |
| coords                                 | string            | no       | —            | List of point coordinates                                                                                                                                                                                            |
| bbox                                   | extent            | no       | —            | Bounding box for selecting features                                                                                                                                                                                  |
| polygon                                | string            | no       | —            | Polygon for selecting features                                                                                                                                                                                       |
| where                                  | string            | no       | —            | WHERE conditions of SQL statement without 'where' keyword                                                                                                                                                            |
| query                                  | enum              | no       | —            | Query tool — options: 0=length, 1=dangle                                                                                                                                                                             |
| bgmap                                  | source            | no       | —            | Name of background vector map                                                                                                                                                                                        |
| snap                                   | enum              | no       | 0            | Snap added or modified features in the given threshold to the nearest existing feature — options: 0=no, 1=node, 2=vertex                                                                                             |
| zbulk                                  | string            | no       | —            | Starting value and step for z bulk-labeling. Pair: value,step (e.g. 1100,10)                                                                                                                                         |
| -r                                     | boolean           | no       | False        | Reverse selection                                                                                                                                                                                                    |
| -c                                     | boolean           | no       | False        | Close added boundaries (using threshold distance)                                                                                                                                                                    |
| -n                                     | boolean           | no       | False        | Do not expect header of input data                                                                                                                                                                                   |
| -b                                     | boolean           | no       | False        | Do not build topology                                                                                                                                                                                                |
| -l                                     | boolean           | no       | False        | Modify only first found feature in bounding box                                                                                                                                                                      |
| output                                 | vectorDestination | yes      | —            | Edited                                                                                                                                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —            | GRASS GIS region extent                                                                                                                                                                                              |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0         | v.in.ogr snap tolerance (-1 = no snap)                                                                                                                                                                               |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001       | v.in.ogr min area                                                                                                                                                                                                    |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0            | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                                                                                                                     |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —            | v.out.ogr output data source options (dsko)                                                                                                                                                                          |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —            | v.out.ogr output layer options (lco)                                                                                                                                                                                 |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False        | Also export features without category (not labeled). Otherwise only features with category are exported                                                                                                              |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.edit map=input.gpkg type=0 tool=0 input=input.dat move=value threshold=value
↳ ids=value output=output.gpkg
```

This calls **v.edit**: map=input.gpkg — Name of vector layer; type=0 selects *point*; tool=0 selects *create*; input=input.dat — ASCII file for add tool; move=value — Difference in x,y,z direction for moving feature or vertex; threshold=value — Threshold distance (coords,snap,query); ids=value — Feature ids; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.extract — v.extract

**group:** Vector (v.\*)

Selects vector objects from a vector layer and creates a new layer containing only the selected objects.

| Parameter                              | Type              | Required | Default            | Description                                                                                             |
|----------------------------------------|-------------------|----------|--------------------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —                  | Vector layer                                                                                            |
| where                                  | string            | no       | —                  | WHERE conditions of SQL statement without 'where' keyword                                               |
| type                                   | enum              | no       | [0, 1, 3, 4, 5, 6] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area, 5=face                   |
| file                                   | file              | no       | —                  | Input text file with category numbers/number ranges to be extracted                                     |
| random                                 | number            | no       | —                  | Number of random categories matching vector objects to extract                                          |
| new                                    | number            | no       | -1.0               | Desired new category value (enter -1 to keep original categories)                                       |
| -d *(adv)*                             | boolean           | no       | True               | Dissolve common boundaries                                                                              |
| -t *(adv)*                             | boolean           | no       | False              | Do not copy attributes                                                                                  |
| -r *(adv)*                             | boolean           | no       | False              | Reverse selection                                                                                       |
| output                                 | vectorDestination | yes      | —                  | Selected                                                                                                |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —                  | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0               | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001             | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0                  | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —                  | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —                  | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False              | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.extract input=input.gpkg where=value type=0 file=input.dat random=10 new=-1  
↪ output=output.gpkg
```

This calls **v.extract**: input=input.gpkg — Vector layer; where=value — WHERE conditions of SQL statement without 'where' keyword; type=0 selects *point*; file=input.dat — Input text file with category numbers/number ranges to be extracted; random=10 — Number of random categories matching vector objects to extract; new=-1 — Desired new category value (enter -1 to keep original categories); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.extrude — v.extrude

**group:** Vector (v.\*)

Extrudes flat vector object to 3D with defined height.

| Parameter | Type   | Required | Default | Description                 |
|-----------|--------|----------|---------|-----------------------------|
| input     | source | yes      | —       | Name of input 2D vector map |

| Parameter                                | Type              | Required | Default   | Description                                                                                             |
|------------------------------------------|-------------------|----------|-----------|---------------------------------------------------------------------------------------------------------|
| where                                    | string            | no       | —         | WHERE conditions of SQL statement without 'where' keyword                                               |
| type                                     | enum              | no       | [0, 1, 2] | Input feature type — options: 0=point, 1=line, 2=area                                                   |
| zshift                                   | number            | no       | 0.0       | Shifting value for z coordinates                                                                        |
| height                                   | number            | no       | —         | Fixed height for 3D vector objects                                                                      |
| height_column                            | field             | no       | —         | Name of attribute column with object heights                                                            |
| elevation                                | raster            | no       | —         | Elevation raster for height extraction                                                                  |
| method                                   | enum              | no       | 0         | Sampling interpolation method — options: 0=nearest, 1=bilinear, 2=bicubic                               |
| scale                                    | number            | no       | 1.0       | Scale factor sampled raster values                                                                      |
| null_value                               | number            | no       | —         | Height for sampled raster NULL values                                                                   |
| -t *(adv)*                               | boolean           | no       | False     | Trace elevation                                                                                         |
| output                                   | vectorDestination | yes      | —         | 3D Vector                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —         | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0       | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0      | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001    | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0         | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —         | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —         | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False     | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.extrude input=input.gpkg where=value type=0 zshift=0 height=10
↪ height_column=field1 elevation=dem.tif output=output.gpkg
```

This calls **v.extrude**: input=input.gpkg — Name of input 2D vector map; where=value — WHERE conditions of SQL statement without 'where' keyword; type=0 selects *point*; zshift=0 — Shifting value for z coordinates; height=10 — Fixed height for 3D vector objects; height\_column=field1 — Name of attribute column with object heights; elevation=dem.tif — Elevation raster for height extraction; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.generalize — v.generalize

**group:** Vector (v.\*)

Vector based generalization.

| Parameter | Type   | Required | Default   | Description                                               |
|-----------|--------|----------|-----------|-----------------------------------------------------------|
| input     | source | yes      | —         | Input layer                                               |
| type      | enum   | no       | [0, 1, 2] | Input feature type — options: 0=line, 1=boundary, 2=area  |
| cats      | string | no       | —         | Category values                                           |
| where     | string | no       | —         | WHERE conditions of SQL statement without 'where' keyword |



| Parameter                              | Type              | Required | Default | Description                                                                                                                                                                                                          |
|----------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| method                                 | enum              | no       | 0       | Generalization algorithm — options: 0=douglas, 1=douglas_reduction, 2=lang, 3=reduction, 4=reumann, 5=boyle, 6=sliding_averaging, 7=distance_weighting, 8=chaiken, 9=hermite, 10=snakes, 11=network, 12=displacement |
| threshold                              | number            | no       | 1.0     | Maximal tolerance value                                                                                                                                                                                              |
| look_ahead                             | number            | no       | 7.0     | Look-ahead parameter                                                                                                                                                                                                 |
| reduction                              | number            | no       | 50.0    | Percentage of the points in the output of 'douglas_reduction' algorithm                                                                                                                                              |
| slide                                  | number            | no       | 0.5     | Slide of computed point toward the original point                                                                                                                                                                    |
| angle_thresh                           | number            | no       | 3.0     | Minimum angle between two consecutive segments in Hermite method                                                                                                                                                     |
| degree_thresh                          | number            | no       | 0.0     | Degree threshold in network generalization                                                                                                                                                                           |
| closeness_thresh                       | number            | no       | 0.0     | Closeness threshold in network generalization                                                                                                                                                                        |
| betweeness_thresh                      | number            | no       | 0.0     | Betweenness threshold in network generalization                                                                                                                                                                      |
| alpha                                  | number            | no       | 1.0     | Snakes alpha parameter                                                                                                                                                                                               |
| beta                                   | number            | no       | 1.0     | Snakes beta parameter                                                                                                                                                                                                |
| iterations                             | number            | no       | 1.0     | Number of iterations                                                                                                                                                                                                 |
| -t *(adv)*                             | boolean           | no       | False   | Do not copy attributes                                                                                                                                                                                               |
| -l *(adv)*                             | boolean           | no       | True    | Disable loop support                                                                                                                                                                                                 |
| output                                 | vectorDestination | yes      | —       | Generalized                                                                                                                                                                                                          |
| error                                  | vectorDestination | yes      | —       | Errors                                                                                                                                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                                                                                                                              |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                                                                                                                               |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                                                                                                                                    |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                                                                                                                     |
| GRASS_VECTOR_DS_CO *(adv)*             | string            | no       | —       | v.out.ogr output data source options (dsco)                                                                                                                                                                          |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                                                                                                                                 |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported                                                                                                              |

**Outputs:** output (outputVector), error (outputVector)

### Example usage

```
run grass:v.generalize input=input.gpkg type=0 cats=value where=value method=0 threshold=1
↳ look_ahead=7 output=output.gpkg
```

This calls **v.generalize**: input=input.gpkg — Input layer; type=0 selects *line*; cats=value — Category values; where=value — WHERE conditions of SQL statement without 'where' keyword; method=0 selects *douglas*; threshold=1 — Maximal tolerance value; look\_ahead=7 — Look-ahead parameter; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.hull — v.hull

**group:** Vector (v.\*)

Produces a convex hull for a given vector map.

| Parameter | Type   | Required | Default | Description |
|-----------|--------|----------|---------|-------------|
| input     | source | yes      | —       | Input layer |

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| where                                  | string            | no       | —       | WHERE conditions of SQL statement without 'where' keyword                                               |
| -f                                     | boolean           | no       | False   | Create a 'flat' 2D hull even if the input is 3D points                                                  |
| output                                 | vectorDestination | yes      | —       | Convex hull                                                                                             |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSICO *(adv)*             | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.hull input=input.gpkg where=value output=output.gpkg
```

This calls **v.hull**: input=input.gpkg — Input layer; where=value — WHERE conditions of SQL statement without 'where' keyword; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.ascii — v.in.ascii

**group:** Vector (v.\*)

Creates a vector map from an ASCII points file or ASCII vector file.

| Parameter                      | Type              | Required | Default | Description                                                                                                   |
|--------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------------|
| input                          | file              | yes      | —       | ASCII file to be imported                                                                                     |
| format                         | enum              | no       | 0       | Input file format — options: 0=point, 1=standard                                                              |
| separator                      | string            | no       | pipe    | Field separator                                                                                               |
| text                           | string            | no       | —       | Text delimiter                                                                                                |
| skip                           | number            | no       | 0.0     | Number of header lines to skip at top of input file                                                           |
| columns                        | string            | no       | —       | Column definition in SQL style (example: 'x double precision, y double precision, cat int, name varchar(10)') |
| x                              | number            | no       | 1.0     | Number of column used as x coordinate                                                                         |
| y                              | number            | no       | 2.0     | Number of column used as y coordinate                                                                         |
| z                              | number            | no       | 0.0     | Number of column used as z coordinate                                                                         |
| cat                            | number            | no       | 0.0     | Number of column used as category                                                                             |
| -z *(adv)*                     | boolean           | no       | False   | Create 3D vector map                                                                                          |
| -n *(adv)*                     | boolean           | no       | False   | Do not expect a header when reading in standard format                                                        |
| -t *(adv)*                     | boolean           | no       | False   | Do not create table in points mode                                                                            |
| -b *(adv)*                     | boolean           | no       | False   | Do not build topology in points mode                                                                          |
| -r *(adv)*                     | boolean           | no       | False   | Only import points falling within current region (points mode)                                                |
| -i *(adv)*                     | boolean           | no       | False   | Ignore broken line(s) in points mode                                                                          |
| output                         | vectorDestination | yes      | —       | ASCII                                                                                                         |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent                                                                                       |

| Parameter                           | Type    | Required | Default | Description                                                                                             |
|-------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.ascii input=input.dat format=0 separator=pipe text=value skip=0
↳ columns=value x=1 output=output.gpkg
```

This calls **v.in.ascii**: input=input.dat — ASCII file to be imported; format=0 selects *point*; separator=pipe — Field separator; text=value — Text delimiter; skip=0 — Number of header lines to skip at top of input file; columns=value — Column definition in SQL style (example: 'x double precision, y double...; x=1 — Number of column used as x coordinate; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.dxf — v.in.dxf

**group:** Vector (v.\*)

Converts files in DXF format to GRASS vector map format.

| Parameter                           | Type              | Required | Default | Description                                                                                             |
|-------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                               | file              | yes      | —       | Name of input DXF file                                                                                  |
| layers                              | string            | no       | —       | List of layers to import                                                                                |
| -e                                  | boolean           | no       | True    | Ignore the map extent of DXF file                                                                       |
| -t                                  | boolean           | no       | False   | Do not create attribute tables                                                                          |
| -f                                  | boolean           | no       | True    | Import polyface meshes as 3D wire frame                                                                 |
| -l                                  | boolean           | no       | False   | List available layers and exit                                                                          |
| -i                                  | boolean           | no       | False   | Invert selection by layers (don't import layers in list)                                                |
| -1                                  | boolean           | no       | True    | Import all objects into one layer                                                                       |
| output                              | vectorDestination | yes      | —       | Converted                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*      | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.dxf input=input.dat layers=value output=output.gpkg
```

This calls **v.in.dxf**: input=input.dat — Name of input DXF file; layers=value — List of layers to import; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.e00 — v.in.e00

**group:** Vector (v.\*)

Imports E00 file into a vector map

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | file              | yes      | —       | Name of input E00 file                                                                                  |
| type                                   | enum              | no       | [0]     | Input feature type — options: 0=point, 1=line, 2=area                                                   |
| output                                 | vectorDestination | yes      | —       | Name of output vector                                                                                   |
| GRASS_REGION_PARAMETER<br>*(adv)*      | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER<br>*(adv)* | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT<br>*(adv)*   | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.e00 input=input.dat type=0 output=output.gpkg
```

This calls **v.in.e00**: input=input.dat — Name of input E00 file; type=0 selects *point*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.geonames — v.in.geonames

**group:** Vector (v.\*)

Imports geonames.org country files into a GRASS vector points map.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | file              | yes      | —       | Uncompressed geonames file from (with .txt extension)                                                   |
| output                                 | vectorDestination | yes      | —       | Geonames                                                                                                |
| GRASS_REGION_PARAMETER<br>*(adv)*      | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER<br>*(adv)* | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT<br>*(adv)*   | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.geonames input=input.dat output=output.gpkg
```

This calls **v.in.geonames**: input=input.dat — Uncompressed geonames file from (with .txt extension); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.lidar — v.in.lidar

**group:** Vector (v.\*)

Converts LAS LiDAR point clouds to a GRASS vector map with libLAS.

| Parameter                           | Type              | Required | Default | Description                                                                                             |
|-------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                               | file              | yes      | —       | LiDAR input files in LAS format (*.las or *.laz)                                                        |
| spatial                             | extent            | no       | —       | Import subregion only                                                                                   |
| zrange                              | range             | no       | —       | Filter range for z data                                                                                 |
| return_filter                       | enum              | no       | —       | Only import points of selected return type — options: 0=first, 1=last, 2=mid                            |
| class_filter                        | string            | no       | —       | Only import points of selected class(es) (comma separated integers)                                     |
| skip                                | number            | no       | —       | Do not import every n-th point                                                                          |
| preserve                            | number            | no       | —       | Import only every n-th point                                                                            |
| offset                              | number            | no       | —       | Skip first n points                                                                                     |
| limit                               | number            | no       | —       | Import only n points                                                                                    |
| -t *(adv)*                          | boolean           | no       | False   | Do not create attribute table                                                                           |
| -c *(adv)*                          | boolean           | no       | False   | Do not automatically add unique ID as category to each point                                            |
| -b *(adv)*                          | boolean           | no       | False   | Do not build topology                                                                                   |
| output                              | vectorDestination | yes      | —       | Lidar                                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*      | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.lidar input=input.dat spatial=0,1000,0,1000 return_filter=0  
↪ class_filter=value skip=10 preserve=10 output=output.gpkg
```

This calls **v.in.lidar**: input=input.dat — LiDAR input files in LAS format (\*.las or \*.laz); spatial=0,1000,0,1000 — Import subregion only; return\_filter=0 selects *first*; class\_filter=value — Only import points of selected class(es) (comma separated integers); skip=10 — Do not import every n-th point; preserve=10 — Import only every n-th point; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.lines — v.in.lines

**group:** Vector (v.\*)

Import ASCII x,y[,z] coordinates as a series of lines.

| Parameter                      | Type              | Required | Default | Description               |
|--------------------------------|-------------------|----------|---------|---------------------------|
| input                          | file              | yes      | —       | ASCII file to be imported |
| separator                      | string            | no       | pipe    | Field separator           |
| -z *(adv)*                     | boolean           | no       | False   | Create 3D vector map      |
| output                         | vectorDestination | yes      | —       | Lines                     |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —       | GRASS GIS region extent   |

| Parameter                           | Type    | Required | Default | Description                                                                                             |
|-------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.lines input=input.dat separator=pipe output=output.gpkg
```

This calls **v.in.lines**: input=input.dat — ASCII file to be imported; separator=pipe — Field separator; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.mapgen — v.in.mapgen

**group:** Vector (v.\*)

Imports Mapgen or Matlab-ASCII vector maps into GRASS.

| Parameter                           | Type              | Required | Default | Description                                                                                             |
|-------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                               | file              | yes      | —       | Name of input file in Mapgen/Matlab format                                                              |
| -z *(adv)*                          | boolean           | no       | False   | Create 3D vector map                                                                                    |
| -f *(adv)*                          | boolean           | no       | False   | Input map is in Matlab format                                                                           |
| output                              | vectorDestination | yes      | —       | Mapgen                                                                                                  |
| GRASS_REGION_PARAMETER *(adv)*      | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.mapgen input=input.dat output=output.gpkg
```

This calls **v.in.mapgen**: input=input.dat — Name of input file in Mapgen/Matlab format; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.in.wfs — v.in.wfs

**group:** Vector (v.\*)

Import GetFeature from WFS

| Parameter | Type   | Required | Default | Description                         |
|-----------|--------|----------|---------|-------------------------------------|
| url       | string | no       | http:// | GetFeature URL starting with 'http' |
| srs       | crs    | no       | —       | Alternate spatial reference system  |

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| name                                   | string            | no       | —       | Comma separated names of data layers to download                                                        |
| maximum_features                       | number            | no       | —       | Maximum number of features to download                                                                  |
| start_index                            | number            | no       | —       | Skip earlier feature IDs and start downloading at this one                                              |
| output                                 | vectorDestination | yes      | —       | Converted                                                                                               |
| GRASS_REGION_PARAMETER<br>*(adv)*      | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER<br>*(adv)* | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsc)                                                              |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT<br>*(adv)*   | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.in.wfs url=http:// srs=EPSG:6346 name=value maximum_features=10 start_index=10
↳ output=output.gpkg
```

This calls **v.in.wfs**: url=http:// — GetFeature URL starting with 'http'; srs=EPSG:6346 — Alternate spatial reference system; name=value — Comma separated names of data layers to download; maximum\_features=10 — Maximum number of features to download; start\_index=10 — Skip earlier feature IDs and start downloading at this one; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.info — v.info

**group:** Vector (v.\*)

Outputs basic information about a user-specified vector map.

| Parameter                                 | Type            | Required | Default     | Description                                                            |
|-------------------------------------------|-----------------|----------|-------------|------------------------------------------------------------------------|
| map                                       | source          | yes      | —           | Name of input vector map                                               |
| -c                                        | boolean         | no       | False       | Print types/names of table columns for specified layer instead of info |
| -g                                        | boolean         | no       | False       | Print map region only                                                  |
| -e                                        | boolean         | no       | False       | Print extended metadata info in shell script style                     |
| -t                                        | boolean         | no       | False       | Print topology information only                                        |
| html                                      | fileDestination | no       | report.html | Information report                                                     |
| GRASS_REGION_PARAMETER *(adv)*            | extent          | no       | —           | GRASS GIS region extent                                                |
| GRASS_SNAP_TOLERANCE_PARAMETER<br>*(adv)* | number          | no       | -1.0        | v.in.ogr snap tolerance (-1 = no snap)                                 |
| GRASS_MIN_AREA_PARAMETER<br>*(adv)*       | number          | no       | 0.0001      | v.in.ogr min area                                                      |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:v.info map=input.gpkg html=html.txt
```

This calls **v.info**: map=input.gpkg — Name of input vector map; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.kcv — v.kcv

**group:** Vector (v.\*)

Randomly partition points into test/train sets.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| map                                    | source            | yes      | —       | Input layer                                                                                             |
| npartitions                            | number            | no       | 10.0    | Number of partitions                                                                                    |
| column                                 | string            | no       | part    | Name for new column to which partition number is written                                                |
| output                                 | vectorDestination | yes      | —       | Partition                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.kcv map=input.gpkg npartitions=10 column=part output=output.gpkg
```

This calls **v.kcv**: map=input.gpkg — Input layer; npartitions=10 — Number of partitions; column=part — Name for new column to which partition number is written; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.kernel.rast — v.kernel.rast

**group:** Vector (v.\*)

v.kernel.rast - Generates a raster density map from vector points map.

| Parameter                               | Type              | Required | Default | Description                                                                                                      |
|-----------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------|
| input                                   | source            | yes      | —       | Name of input vector map with training points                                                                    |
| radius                                  | number            | no       | 10.0    | Kernel radius in map units                                                                                       |
| dsize                                   | number            | no       | 0.0     | Discretization error in map units                                                                                |
| segmax                                  | number            | no       | 100.0   | Maximum length of segment on network                                                                             |
| distmax                                 | number            | no       | 100.0   | Maximum distance from point to network                                                                           |
| multiplier                              | number            | no       | 1.0     | Multiply the density result by this number                                                                       |
| node                                    | enum              | no       | 0       | Node method — options: 0=none, 1=split                                                                           |
| kernel                                  | enum              | no       | 5       | Kernel function — options: 0=uniform, 1=triangular, 2=epanechnikov, 3=quartic, 4=triweight, 5=gaussian, 6=cosine |
| -o *(adv)*                              | boolean           | no       | False   | Try to calculate an optimal radius with given 'radius' taken as maximum (experimental)                           |
| output                                  | rasterDestination | yes      | —       | Kernel                                                                                                           |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                                                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                  |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                                                        |



| Parameter                              | Type   | Required | Default | Description                                      |
|----------------------------------------|--------|----------|---------|--------------------------------------------------|
| GRASS_RASTER_FORMAT_META *(adv)*       | string | no       | —       | Output Rasters format metadata options (metaopt) |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)           |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number | no       | 0.0001  | v.in.ogr min area                                |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:v.kernel.rast input=input.gpkg radius=10 dsize=0 segmax=100 distmax=100
↳ multiplier=1 node=0 output=output.tif
```

This calls **v.kernel.rast**: input=input.gpkg — Name of input vector map with training points; radius=10 — Kernel radius in map units; dsize=0 — Discretization error in map units; segmax=100 — Maximum length of segment on network; distmax=100 — Maximum distance from point to network; multiplier=1 — Multiply the density result by this number; node=0 selects *none*; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.kernel.vector — v.kernel.vector

**group:** Vector (v.\*)

v.kernel.vector - Generates a vector density map from vector points on a vector network.

| Parameter                              | Type              | Required | Default | Description                                                                                                      |
|----------------------------------------|-------------------|----------|---------|------------------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Name of input vector map with training points                                                                    |
| net                                    | source            | yes      | —       | Name of input network vector map                                                                                 |
| radius                                 | number            | no       | 10.0    | Kernel radius in map units                                                                                       |
| dsize                                  | number            | no       | 0.0     | Discretization error in map units                                                                                |
| segmax                                 | number            | no       | 100.0   | Maximum length of segment on network                                                                             |
| distmax                                | number            | no       | 100.0   | Maximum distance from point to network                                                                           |
| multiplier                             | number            | no       | 1.0     | Multiply the density result by this number                                                                       |
| node                                   | enum              | no       | 0       | Node method — options: 0=none, 1=split                                                                           |
| kernel                                 | enum              | no       | 5       | Kernel function — options: 0=uniform, 1=triangular, 2=epanechnikov, 3=quartic, 4=triweight, 5=gaussian, 6=cosine |
| -o *(adv)*                             | boolean           | no       | False   | Try to calculate an optimal radius with given 'radius' taken as maximum (experimental)                           |
| -n *(adv)*                             | boolean           | no       | False   | Normalize values by sum of density multiplied by length of each segment.                                         |
| -m *(adv)*                             | boolean           | no       | False   | Multiply the result by number of input points                                                                    |
| output                                 | vectorDestination | yes      | —       | Kernel                                                                                                           |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                          |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                           |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                                |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                 |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                                      |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                             |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported          |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.kernel.vector input=input.gpkg net=input.gpkg radius=10 dsize=0 segmax=100  
↳ distmax=100 multiplier=1 output=output.gpkg
```

This calls **v.kernel.vector**: input=input.gpkg — Name of input vector map with training points; net=input.gpkg — Name of input network vector map; radius=10 — Kernel radius in map units; dsize=0 — Discretization error in map units; segmax=100 — Maximum length of segment on network; distmax=100 — Maximum distance from point to network; multiplier=1 — Multiply the density result by this number; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.lidar.correction — v.lidar.correction

**group:** Vector (v.\*)

Correction of the v.lidar.growing output. It is the last of the three algorithms for LIDAR filtering.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector layer (v.lidar.growing output)                                                             |
| ew_step                                | number            | no       | 25.0    | Length of each spline step in the east-west direction                                                   |
| ns_step                                | number            | no       | 25.0    | Length of each spline step in the north-south direction                                                 |
| lambda_c                               | number            | no       | 1.0     | Regularization weight in reclassification evaluation                                                    |
| tch                                    | number            | no       | 2.0     | High threshold for object to terrain reclassification                                                   |
| tcl                                    | number            | no       | 1.0     | Low threshold for terrain to object reclassification                                                    |
| -e                                     | boolean           | no       | False   | Estimate point density and distance                                                                     |
| output                                 | vectorDestination | yes      | —       | Classified                                                                                              |
| terrain                                | vectorDestination | yes      | —       | Only 'terrain' points                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), terrain (outputVector)

## Example usage

```
run grass:v.lidar.correction input=input.gpkg ew_step=25 ns_step=25 lambda_c=1 tch=2 tcl=1  
↳ output=output.gpkg
```

This calls **v.lidar.correction**: input=input.gpkg — Input vector layer (v.lidar.growing output); ew\_step=25 — Length of each spline step in the east-west direction; ns\_step=25 — Length of each spline step in the north-south direction; lambda\_c=1 — Regularization weight in reclassification evaluation; tch=2 — High threshold for object to terrain reclassification; tcl=1 — Low threshold for terrain to object reclassification; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.lidar.edgedetection — v.lidar.edgedetection

**group:** Vector (v.\*)

Detects the object's edges from a LIDAR data set.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector layer                                                                                      |
| ew_step                                | number            | no       | 4.0     | Length of each spline step in the east-west direction                                                   |
| ns_step                                | number            | no       | 4.0     | Length of each spline step in the north-south direction                                                 |
| lambda_g                               | number            | no       | 0.01    | Regularization weight in gradient evaluation                                                            |
| tgh                                    | number            | no       | 6.0     | High gradient threshold for edge classification                                                         |
| tgl                                    | number            | no       | 3.0     | Low gradient threshold for edge classification                                                          |
| theta_g                                | number            | no       | 0.26    | Angle range for same direction detection                                                                |
| lambda_r                               | number            | no       | 2.0     | Regularization weight in residual evaluation                                                            |
| -e                                     | boolean           | no       | False   | Estimate point density and distance                                                                     |
| output                                 | vectorDestination | yes      | —       | Edges                                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.lidar.edgedetection input=input.gpkg ew_step=4 ns_step=4 lambda_g=0.01 tgh=6  
↪ tgl=3 theta_g=0.26 output=output.gpkg
```

This calls **v.lidar.edgedetection**: input=input.gpkg — Input vector layer; ew\_step=4 — Length of each spline step in the east-west direction; ns\_step=4 — Length of each spline step in the north-south direction; lambda\_g=0.01 — Regularization weight in gradient evaluation; tgh=6 — High gradient threshold for edge classification; tgl=3 — Low gradient threshold for edge classification; theta\_g=0.26 — Angle range for same direction detection; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.lidar.growing — v.lidar.growing

**group:** Vector (v.\*)

Building contour determination and Region Growing algorithm for determining the building inside

| Parameter | Type   | Required | Default | Description                                           |
|-----------|--------|----------|---------|-------------------------------------------------------|
| input     | source | yes      | —       | Input vector (v.lidar.edgedetection output)           |
| first     | source | yes      | —       | First pulse vector layer                              |
| tj        | number | no       | 0.2     | Threshold for cell object frequency in region growing |
| td        | number | no       | 0.6     | Threshold for double pulse in region growing          |

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| output                                 | vectorDestination | yes      | —       | Buildings                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.lidar.growing input=input.gpkg first=input.gpkg tj=0.2 td=0.6
↳ output=output.gpkg
```

This calls **v.lidar.growing**: input=input.gpkg — Input vector (v.lidar.edgedetection output); first=input.gpkg — First pulse vector layer; tj=0.2 — Threshold for cell object frequency in region growing; td=0.6 — Threshold for double pulse in region growing; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.mkgrid — v.mkgrid

**group:** Vector (v.\*)

Creates a GRASS vector layer of a user-defined grid.

| Parameter                           | Type              | Required | Default | Description                                                                                             |
|-------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| grid                                | string            | no       | 10,10   | Number of rows and columns in grid                                                                      |
| position                            | enum              | no       | 0       | Where to place the grid — options: 0=coor                                                               |
| coordinates                         | point             | no       | —       | Lower left easting and northing coordinates of map                                                      |
| box                                 | string            | yes      | —       | Width and height of boxes in grid                                                                       |
| angle                               | number            | no       | 0.0     | Angle of rotation (in degrees counter-clockwise)                                                        |
| breaks                              | number            | no       | 0.0     | Number of vertex points per grid cell                                                                   |
| type                                | enum              | no       | 2       | Output feature type — options: 0=point, 1=line, 2=area                                                  |
| -h                                  | boolean           | no       | False   | Create hexagons (default: rectangles)                                                                   |
| -a                                  | boolean           | no       | False   | Allow asymmetric hexagons                                                                               |
| map                                 | vectorDestination | yes      | —       | Grid                                                                                                    |
| GRASS_REGION_PARAMETER *(adv)*      | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** map (outputVector)

### Example usage

```
run grass:v.mkgrid box=value grid=10,10 position=0 coordinates=100,100 angle=0 breaks=0
↳ type=2 map=map.gpkg
```

This calls **v.mkgrid**: `box=value` — Width and height of boxes in grid; `grid=10,10` — Number of rows and columns in grid; `position=0` selects *coord*; `coordinates=100,100` — Lower left easting and northing coordinates of map; `angle=0` — Angle of rotation (in degrees counter-clockwise); `breaks=0` — Number of vertex points per grid cell; `type=2` selects *area*; `map=map.gpkg` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.neighbors — v.neighbors

**group:** Vector (v.\*)

Makes each cell value a function of attribute values and stores in an output raster map.

| Parameter                                | Type              | Required | Default | Description                                                                                                                                                                    |
|------------------------------------------|-------------------|----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                    | source            | yes      | —       | Input vector layer                                                                                                                                                             |
| method                                   | enum              | no       | 0       | Method for aggregate statistics (count if non given) — options: 0=count, 1=sum, 2=average, 3=median, 4=mode, 5=minimum, 6=maximum, 7=range, 8=stddev, 9=variance, 10=diversity |
| points_column                            | field             | no       | —       | Column name of points map to use for statistics                                                                                                                                |
| size                                     | number            | no       | 0.1     | Neighborhood diameter in map units                                                                                                                                             |
| output                                   | rasterDestination | yes      | —       | Neighborhood                                                                                                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                                                                                        |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                                                                                                |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                                                                                                      |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                                                                                               |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                                                                                         |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                                                                                                                                              |

**Outputs:** output (outputRaster)

### Example usage

```
run grass:v.neighbors input=input.gpkg method=0 points_column=field1 size=0.1
↳ output=output.tif
```

This calls **v.neighbors**: `input=input.gpkg` — Input vector layer; `method=0` selects *count*; `points_column=field1` — Column name of points map to use for statistics; `size=0.1` — Neighborhood diameter in map units; `output=output.tif` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net — v.net

**group:** Vector (v.\*)

Performs network maintenance

| Parameter | Type   | Required | Default | Description                      |
|-----------|--------|----------|---------|----------------------------------|
| input     | source | no       | —       | Input vector line layer (arcs)   |
| points    | source | no       | —       | Input vector point layer (nodes) |
| file      | file   | no       | —       | Name of input arcs file          |

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| operation                              | enum              | no       | 0       | Operation to be performed — options: 0=nodes, 1=connect, 2=arcs                                         |
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)                                           |
| arc_type                               | enum              | no       | [0, 1]  | Arc type — options: 0=line, 1=boundary                                                                  |
| -s *(adv)*                             | boolean           | no       | False   | Snap points to network                                                                                  |
| -c *(adv)*                             | boolean           | no       | False   | Assign unique categories to new points                                                                  |
| output                                 | vectorDestination | yes      | —       | Network                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net input=input.gpkg points=input.gpkg file=input.dat operation=0 threshold=50
↪ arc_type=0 output=output.gpkg
```

This calls **v.net**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Input vector point layer (nodes); file=input.dat — Name of input arcs file; operation=0 selects *nodes*; threshold=50 — Threshold for connecting centers to the network (in map unit); arc\_type=0 selects *line*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.alloc — v.net.alloc

**group:** Vector (v.\*)

Allocates subnets for nearest centers

| Parameter                      | Type              | Required | Default  | Description                                                        |
|--------------------------------|-------------------|----------|----------|--------------------------------------------------------------------|
| input                          | source            | yes      | —        | Input vector line layer (arcs)                                     |
| points                         | source            | yes      | —        | Centers point layer (nodes)                                        |
| threshold                      | number            | no       | 50.0     | Threshold for connecting centers to the network (in map unit)      |
| center_cats *(adv)*            | string            | no       | 1-100000 | Category values                                                    |
| arc_type *(adv)*               | enum              | no       | [0, 1]   | Arc type — options: 0=line, 1=boundary                             |
| method *(adv)*                 | enum              | no       | 0        | Use costs from centers or costs to centers — options: 0=from, 1=to |
| arc_column *(adv)*             | field             | no       | —        | Arc forward/both direction(s) cost column (number)                 |
| arc_backward_column *(adv)*    | field             | no       | —        | Arc backward direction cost column (number)                        |
| node_column *(adv)*            | field             | no       | —        | Node cost column (number)                                          |
| -g *(adv)*                     | boolean           | no       | False    | Use geodesic calculation for longitude-latitude locations          |
| output                         | vectorDestination | yes      | —        | Network Allocation                                                 |
| GRASS_REGION_PARAMETER *(adv)* | extent            | no       | —        | GRASS GIS region extent                                            |

| Parameter                              | Type    | Required | Default | Description                                                                                             |
|----------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number  | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.alloc input=input.gpkg points=input.gpkg threshold=50 output=output.gpkg
```

This calls **v.net.alloc**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Centers point layer (nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.allpairs — v.net.allpairs

**group:** Vector (v.\*)

Computes the shortest path between all pairs of nodes in the network

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector line layer (arcs)                                                                          |
| points                                 | source            | yes      | —       | Centers point layer (nodes)                                                                             |
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)                                           |
| cats *(adv)*                           | string            | no       | 1-10000 | Category values                                                                                         |
| where *(adv)*                          | string            | no       | —       | WHERE condition of SQL statement without 'where' keyword                                                |
| arc_column *(adv)*                     | field             | no       | —       | Arc forward/both direction(s) cost column (number)                                                      |
| arc_backward_column *(adv)*            | field             | no       | —       | Arc backward direction cost column (number)                                                             |
| node_column *(adv)*                    | field             | no       | —       | Node cost column (number)                                                                               |
| -g *(adv)*                             | boolean           | no       | False   | Use geodesic calculation for longitude-latitude locations                                               |
| output                                 | vectorDestination | yes      | —       | Network_Allpairs                                                                                        |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.net.allpairs input=input.gpkg points=input.gpkg threshold=50
↳ output=output.gpkg
```

This calls **v.net.allpairs**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Centers point layer (nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.bridge — v.net.bridge

**group:** Vector (v.\*)

Computes bridges and articulation points in the network.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector line layer (network)                                                                       |
| points                                 | source            | no       | —       | Centers point layer (nodes)                                                                             |
| method                                 | enum              | no       | 0       | Feature type — options: 0=bridge, 1=articulation                                                        |
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)                                           |
| arc_column *(adv)*                     | field             | no       | —       | Arc forward/both direction(s) cost column (name)                                                        |
| arc_backward_column *(adv)*            | field             | no       | —       | Arc backward direction cost column (name)                                                               |
| node_column *(adv)*                    | field             | no       | —       | Node cost column (number)                                                                               |
| output                                 | vectorDestination | yes      | —       | Bridge                                                                                                  |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.net.bridge input=input.gpkg points=input.gpkg method=0 threshold=50
↳ output=output.gpkg
```

This calls **v.net.bridge**: input=input.gpkg — Input vector line layer (network); points=input.gpkg — Centers point layer (nodes); method=0 selects *bridge*; threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.centraliity — v.net.centraliity

**group:** Vector (v.\*)

Computes degree, centrality, betweenness, closeness and eigenvector centrality measures in the network.



| Parameter                              | Type              | Required | Default     | Description                                                                                             |
|----------------------------------------|-------------------|----------|-------------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —           | Input vector line layer (network)                                                                       |
| degree                                 | string            | no       | degree      | Name of output degree centrality column                                                                 |
| closeness                              | string            | no       | closeness   | Name of output closeness centrality column                                                              |
| betweenness                            | string            | no       | betweenness | Name of output betweenness centrality column                                                            |
| eigenvector                            | string            | no       | eigenvector | Name of output eigenvector centrality column                                                            |
| iterations *(adv)*                     | number            | no       | 1000.0      | Maximum number of iterations to compute eigenvector centrality                                          |
| error *(adv)*                          | number            | no       | 0.1         | Cumulative error tolerance for eigenvector centrality                                                   |
| cats *(adv)*                           | string            | no       | —           | Category values                                                                                         |
| where *(adv)*                          | string            | no       | —           | WHERE conditions of SQL statement without 'where' keyword                                               |
| arc_column *(adv)*                     | field             | no       | —           | Arc forward/both direction(s) cost column (number)                                                      |
| arc_backward_column *(adv)*            | field             | no       | —           | Arc backward direction cost column (number)                                                             |
| node_column *(adv)*                    | field             | no       | —           | Node cost column (number)                                                                               |
| -a *(adv)*                             | boolean           | no       | True        | Add points on nodes                                                                                     |
| -g *(adv)*                             | boolean           | no       | False       | Use geodesic calculation for longitude-latitude locations                                               |
| output                                 | vectorDestination | yes      | —           | Network Centrality                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —           | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0        | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001      | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0           | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —           | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —           | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False       | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.centralities input=input.gpkg degree=degree closeness=closeness
↪ betweenness=betweenness eigenvector=eigenvector output=output.gpkg
```

This calls **v.net.centralities**: input=input.gpkg — Input vector line layer (network); degree=degree — Name of output degree centrality column; closeness=closeness — Name of output closeness centrality column; betweenness=betweenness — Name of output betweenness centrality column; eigenvector=eigenvector — Name of output eigenvector centrality column; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.components — v.net.components

**group:** Vector (v.\*)

Computes strongly and weakly connected components in the network.

| Parameter | Type   | Required | Default | Description                       |
|-----------|--------|----------|---------|-----------------------------------|
| input     | source | yes      | —       | Input vector line layer (network) |
| points    | source | no       | —       | Centers point layer (nodes)       |

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)                                           |
| method                                 | enum              | no       | 0       | Type of components — options: 0=weak, 1=strong                                                          |
| arc_column *(adv)*                     | field             | no       | —       | Arc forward/both direction(s) cost column (number)                                                      |
| arc_backward_column *(adv)*            | field             | no       | —       | Arc backward direction cost column (number)                                                             |
| node_column *(adv)*                    | field             | no       | —       | Node cost column (number)                                                                               |
| -a *(adv)*                             | boolean           | no       | True    | Add points on nodes                                                                                     |
| output                                 | vectorDestination | yes      | —       | Network_Components_Line                                                                                 |
| output_point                           | vectorDestination | yes      | —       | Network_Components_Point                                                                                |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), output\_point (outputVector)

### Example usage

```
run grass:v.net.components input=input.gpkg points=input.gpkg threshold=50 method=0
↳ output=output.gpkg
```

This calls **v.net.components**: input=input.gpkg — Input vector line layer (network); points=input.gpkg — Centers point layer (nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); method=0 selects *weak*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.connectivity — v.net.connectivity

**group:** Vector (v.\*)

Computes vertex connectivity between two sets of nodes in the network.

| Parameter                   | Type   | Required | Default | Description                                                    |
|-----------------------------|--------|----------|---------|----------------------------------------------------------------|
| input                       | source | yes      | —       | Input vector line layer (network)                              |
| points                      | source | yes      | —       | Input vector point layer (first set of nodes)                  |
| threshold                   | number | no       | 50.0    | Threshold for connecting centers to the network (in map unit)  |
| set1_cats                   | string | no       | —       | Set1 Category values                                           |
| set1_where                  | string | no       | —       | Set1 WHERE conditions of SQL statement without 'where' keyword |
| set2_cats                   | string | no       | —       | Set2 Category values                                           |
| set2_where                  | string | no       | —       | Set2 WHERE conditions of SQL statement without 'where' keyword |
| arc_column *(adv)*          | field  | no       | —       | Arc forward/both direction(s) cost column (number)             |
| arc_backward_column *(adv)* | field  | no       | —       | Arc backward direction cost column (number)                    |
| node_column *(adv)*         | field  | no       | —       | Node cost column (number)                                      |

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| output                                 | vectorDestination | yes      | —       | Network_Connectivity                                                                                    |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.connectivity input=input.gpkg points=input.gpkg threshold=50
↳ set1_cats=value set1_where=value set2_cats=value set2_where=value output=output.gpkg
```

This calls **v.net.connectivity**: input=input.gpkg — Input vector line layer (network); points=input.gpkg — Input vector point layer (first set of nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); set1\_cats=value — Set1 Category values; set1\_where=value — Set1 WHERE conditions of SQL statement without 'where' keyword; set2\_cats=value — Set2 Category values; set2\_where=value — Set2 WHERE conditions of SQL statement without 'where' keyword; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.distance — v.net.distance

**group:** Vector (v.\*)

Computes shortest distance via the network between the given sets of features.

| Parameter                   | Type              | Required | Default | Description                                                        |
|-----------------------------|-------------------|----------|---------|--------------------------------------------------------------------|
| input                       | source            | yes      | —       | Input vector line layer (network)                                  |
| flayer                      | source            | yes      | —       | Input vector from points layer (from)                              |
| tlayer                      | source            | yes      | —       | Input vector to layer (to)                                         |
| threshold                   | number            | no       | 50.0    | Threshold for connecting nodes to the network (in map unit)        |
| arc_type *(adv)*            | enum              | no       | [0, 1]  | Arc type — options: 0=line, 1=boundary                             |
| from_cats *(adv)*           | string            | no       | —       | From Category values                                               |
| from_where *(adv)*          | string            | no       | —       | From WHERE conditions of SQL statement without 'where' keyword     |
| to_type *(adv)*             | enum              | no       | [0]     | To feature type — options: 0=point, 1=line, 2=boundary             |
| to_cats *(adv)*             | string            | no       | —       | To Category values                                                 |
| to_where *(adv)*            | string            | no       | —       | To WHERE conditions of SQL statement without 'where' keyword       |
| arc_column *(adv)*          | field             | no       | —       | Arc forward/both direction(s) cost column (number)                 |
| arc_backward_column *(adv)* | field             | no       | —       | Arc backward direction cost column (number)                        |
| node_column *(adv)*         | field             | no       | —       | Node cost column (number)                                          |
| -g *(adv)*                  | boolean           | no       | False   | Use geodesic calculation for longitude-latitude locations          |
| -l *(adv)*                  | boolean           | no       | False   | Write each output path as one line, not as original input segments |
| output                      | vectorDestination | yes      | —       | Network_Distance                                                   |

| Parameter                              | Type    | Required | Default | Description                                                                                             |
|----------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*         | extent  | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number  | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.distance input=input.gpkg flayer=input.gpkg tlayer=input.gpkg threshold=50
↳ output=output.gpkg
```

This calls **v.net.distance**: input=input.gpkg — Input vector line layer (network); flayer=input.gpkg — Input vector from points layer (from); tlayer=input.gpkg — Input vector to layer (to); threshold=50 — Threshold for connecting nodes to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.flow — v.net.flow

**group:** Vector (v.\*)

Computes the maximum flow between two sets of nodes in the network.

| Parameter                              | Type              | Required | Default | Description                                                      |
|----------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector line layer (network)                                |
| points                                 | source            | yes      | —       | Input vector point layer (flow nodes)                            |
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)    |
| source_cats                            | string            | no       | —       | Source Category values                                           |
| source_where                           | string            | no       | —       | Source WHERE conditions of SQL statement without 'where' keyword |
| sink_cats                              | string            | no       | —       | Sink Category values                                             |
| sink_where                             | string            | no       | —       | Sink WHERE conditions of SQL statement without 'where' keyword   |
| arc_column *(adv)*                     | field             | no       | —       | Arc forward/both direction(s) cost column (number)               |
| arc_backward_column *(adv)*            | field             | no       | —       | Arc backward direction cost column (number)                      |
| node_column *(adv)*                    | field             | no       | —       | Node cost column (number)                                        |
| output                                 | vectorDestination | yes      | —       | Network_Flow                                                     |
| cut                                    | vectorDestination | yes      | —       | Network_Cut                                                      |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                           |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                      |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                             |

| Parameter                         | Type    | Required | Default | Description                                                                                             |
|-----------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_VECTOR_EXPORT_NOCAT *(adv)* | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), cut (outputVector)

### Example usage

```
run grass:v.net.flow input=input.gpkg points=input.gpkg threshold=50 source_cats=value
↳ source_where=value sink_cats=value sink_where=value output=output.gpkg
```

This calls **v.net.flow**: input=input.gpkg — Input vector line layer (network); points=input.gpkg — Input vector point layer (flow nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); source\_cats=value — Source Category values; source\_where=value — Source WHERE conditions of SQL statement without 'where' keyword; sink\_cats=value — Sink Category values; sink\_where=value — Sink WHERE conditions of SQL statement without 'where' keyword; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.iso — v.net.iso

**group:** Vector (v.\*)

Splits network by cost isolines.

| Parameter                              | Type              | Required | Default        | Description                                                                                             |
|----------------------------------------|-------------------|----------|----------------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —              | Input vector line layer (arcs)                                                                          |
| points                                 | source            | yes      | —              | Centers point layer (nodes)                                                                             |
| threshold                              | number            | no       | 50.0           | Threshold for connecting centers to the network (in map unit)                                           |
| arc_type *(adv)*                       | enum              | no       | [0, 1]         | Arc type — options: 0=line, 1=boundary                                                                  |
| center_cats *(adv)*                    | string            | no       | 1-100000       | Category values                                                                                         |
| costs                                  | string            | no       | 1000,2000,3000 | Costs for isolines                                                                                      |
| arc_column *(adv)*                     | field             | no       | —              | Arc forward/both direction(s) cost column (number)                                                      |
| arc_backward_column *(adv)*            | field             | no       | —              | Arc backward direction cost column (number)                                                             |
| node_column *(adv)*                    | field             | no       | —              | Node cost column (number)                                                                               |
| -g *(adv)*                             | boolean           | no       | False          | Use geodesic calculation for longitude-latitude locations                                               |
| output                                 | vectorDestination | yes      | —              | Network_Iso                                                                                             |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —              | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0           | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001         | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0              | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —              | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —              | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False          | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.iso input=input.gpkg points=input.gpkg threshold=50 costs=1000,2000,3000
↳ output=output.gpkg
```

This calls **v.net.iso**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Centers point layer (nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); costs=1000,2000,3000 — Costs for isolines; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.nreport — v.net.nreport

**group:** Vector (v.\*)

v.net.nreport - Reports nodes information of a network

| Parameter                              | Type            | Required | Default | Description                            |
|----------------------------------------|-----------------|----------|---------|----------------------------------------|
| input                                  | source          | yes      | —       | Input vector line layer (arcs)         |
| output                                 | fileDestination | yes      | —       | NReport                                |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —       | GRASS GIS region extent                |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap) |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001  | v.in.ogr min area                      |

**Outputs:** output (outputHtml)

### Example usage

```
run grass:v.net.nreport input=input.gpkg output=output.txt
```

This calls **v.net.nreport**: input=input.gpkg — Input vector line layer (arcs); output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.path — v.net.path

**group:** Vector (v.\*)

Finds shortest path on vector network

| Parameter                              | Type              | Required | Default | Description                                                        |
|----------------------------------------|-------------------|----------|---------|--------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector line layer (arcs)                                     |
| points                                 | source            | yes      | —       | Centers point layer (nodes)                                        |
| file                                   | file              | yes      | —       | Name of file containing start and end points                       |
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)      |
| arc_type *(adv)*                       | enum              | no       | [0, 1]  | Arc type — options: 0=line, 1=boundary                             |
| arc_column *(adv)*                     | field             | no       | —       | Arc forward/both direction(s) cost column (number)                 |
| arc_backward_column *(adv)*            | field             | no       | —       | Arc backward direction cost column (number)                        |
| node_column *(adv)*                    | field             | no       | —       | Node cost column (number)                                          |
| dmax *(adv)*                           | number            | no       | 1000.0  | Maximum distance to the network                                    |
| -g *(adv)*                             | boolean           | no       | False   | Use geodesic calculation for longitude-latitude locations          |
| -s *(adv)*                             | boolean           | no       | False   | Write output as original input segments, not each path as one line |
| output                                 | vectorDestination | yes      | —       | Network_Path                                                       |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                            |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                             |

| Parameter                           | Type    | Required | Default | Description                                                                                             |
|-------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_MIN_AREA_PARAMETER *(adv)*    | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.path input=input.gpkg points=input.gpkg file=input.dat threshold=50
↳ output=output.gpkg
```

This calls **v.net.path**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Centers point layer (nodes); file=input.dat — Name of file containing start and end points; threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.report — v.net.report

**group:** Vector (v.\*)

v.net.report - Reports lines information of a network

| Parameter                              | Type            | Required | Default | Description                            |
|----------------------------------------|-----------------|----------|---------|----------------------------------------|
| input                                  | source          | yes      | —       | Input vector line layer (arcs)         |
| html                                   | fileDestination | yes      | —       | Report                                 |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —       | GRASS GIS region extent                |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap) |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001  | v.in.ogr min area                      |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:v.net.report input=input.gpkg html=html.txt
```

This calls **v.net.report**: input=input.gpkg — Input vector line layer (arcs); html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.salesman — v.net.salesman

**group:** Vector (v.\*)

Creates a cycle connecting given nodes (Traveling salesman problem)

| Parameter        | Type   | Required | Default | Description                                                   |
|------------------|--------|----------|---------|---------------------------------------------------------------|
| input            | source | yes      | —       | Input vector line layer (arcs)                                |
| points           | source | yes      | —       | Centers point layer (nodes)                                   |
| threshold        | number | no       | 50.0    | Threshold for connecting centers to the network (in map unit) |
| arc_type *(adv)* | enum   | no       | [0, 1]  | Arc type — options: 0=line, 1=boundary                        |

| Parameter                              | Type              | Required | Default  | Description                                                                                             |
|----------------------------------------|-------------------|----------|----------|---------------------------------------------------------------------------------------------------------|
| center_cats *(adv)*                    | string            | no       | 1-100000 | Category values                                                                                         |
| arc_column *(adv)*                     | field             | no       | —        | Arc forward/both direction(s) cost column (number)                                                      |
| arc_backward_column *(adv)*            | field             | no       | —        | Arc backward direction cost column (number)                                                             |
| -g *(adv)*                             | boolean           | no       | False    | Use geodesic calculation for longitude-latitude locations                                               |
| output                                 | vectorDestination | yes      | —        | Network_Salesman                                                                                        |
| sequence                               | fileDestination   | no       | —        | Output file holding node sequence                                                                       |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —        | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0     | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001   | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0        | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —        | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —        | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False    | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), sequence (outputFile)

### Example usage

```
run grass:v.net.salesman input=input.gpkg points=input.gpkg threshold=50
↳ output=output.gpkg
```

This calls **v.net.salesman**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Centers point layer (nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.spanningtree — v.net.spanningtree

**group:** Vector (v.\*)

Computes minimum spanning tree for the network.

| Parameter                              | Type              | Required | Default | Description                                                      |
|----------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector line layer (arcs)                                   |
| points                                 | source            | no       | —       | Input point layer (nodes)                                        |
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)    |
| arc_column *(adv)*                     | field             | no       | —       | Arc forward/both direction(s) cost column (number)               |
| node_column *(adv)*                    | field             | no       | —       | Node cost column (number)                                        |
| -g *(adv)*                             | boolean           | no       | False   | Use geodesic calculation for longitude-latitude locations        |
| output                                 | vectorDestination | yes      | —       | SpanningTree                                                     |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                           |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                      |



| Parameter                         | Type    | Required | Default | Description                                                                                             |
|-----------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_VECTOR_LCO *(adv)*          | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)* | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.spanningtree input=input.gpkg points=input.gpkg threshold=50
↳ output=output.gpkg
```

This calls **v.net.spanningtree**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Input point layer (nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.steiner — v.net.steiner

**group:** Vector (v.\*)

Creates Steiner tree for the network and given terminals

| Parameter                              | Type              | Required | Default  | Description                                                                                             |
|----------------------------------------|-------------------|----------|----------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —        | Input vector line layer (arcs)                                                                          |
| points                                 | source            | yes      | —        | Centers point layer (nodes)                                                                             |
| threshold                              | number            | no       | 50.0     | Threshold for connecting centers to the network (in map unit)                                           |
| arc_type *(adv)*                       | enum              | no       | [0, 1]   | Arc type — options: 0=line, 1=boundary                                                                  |
| terminal_cats *(adv)*                  | string            | no       | 1-100000 | Category values                                                                                         |
| acolumn *(adv)*                        | field             | no       | —        | Arc forward/both direction(s) cost column (number)                                                      |
| npoints *(adv)*                        | number            | no       | -1.0     | Number of Steiner points                                                                                |
| -g *(adv)*                             | boolean           | no       | False    | Use geodesic calculation for longitude-latitude locations                                               |
| output                                 | vectorDestination | yes      | —        | Network Steiner                                                                                         |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —        | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0     | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001   | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0        | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —        | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —        | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False    | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.steiner input=input.gpkg points=input.gpkg threshold=50 output=output.gpkg
```

This calls **v.net.steiner**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Centers point layer (nodes); threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.timetable — v.net.timetable

**group:** Vector (v.\*)

Finds shortest path using timetables.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector line layer (arcs)                                                                          |
| points                                 | source            | yes      | —       | Centers point layer (nodes)                                                                             |
| walk_layer                             | source            | no       | —       | Layer number or name with walking connections                                                           |
| threshold                              | number            | no       | 50.0    | Threshold for connecting centers to the network (in map unit)                                           |
| arc_column *(adv)*                     | field             | no       | —       | Arc forward/both direction(s) cost column (number)                                                      |
| arc_backward_column *(adv)*            | field             | no       | —       | Arc backward direction cost column (number)                                                             |
| node_column *(adv)*                    | field             | no       | —       | Node cost column (number)                                                                               |
| route_id *(adv)*                       | field             | no       | —       | Name of column with route ids                                                                           |
| stop_time *(adv)*                      | field             | no       | —       | Name of column with stop timestamps                                                                     |
| to_stop *(adv)*                        | field             | no       | —       | Name of column with stop ids                                                                            |
| walk_length *(adv)*                    | field             | no       | —       | Name of column with walk lengths                                                                        |
| output                                 | vectorDestination | yes      | —       | Network Timetable                                                                                       |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.timetable input=input.gpkg points=input.gpkg walk_layer=input.gpkg  
↪ threshold=50 output=output.gpkg
```

This calls **v.net.timetable**: input=input.gpkg — Input vector line layer (arcs); points=input.gpkg — Centers point layer (nodes); walk\_layer=input.gpkg — Layer number or name with walking connections; threshold=50 — Threshold for connecting centers to the network (in map unit); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.net.visibility — v.net.visibility

**group:** Vector (v.\*)

Performs visibility graph construction.

| Parameter   | Type              | Required | Default | Description                                       |
|-------------|-------------------|----------|---------|---------------------------------------------------|
| input       | source            | yes      | —       | Input vector line layer (arcs)                    |
| coordinates | string            | no       | —       | Coordinates                                       |
| visibility  | source            | no       | —       | Input vector line layer containing visible points |
| output      | vectorDestination | yes      | —       | Network Visibility                                |

| Parameter                              | Type    | Required | Default | Description                                                                                             |
|----------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*         | extent  | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number  | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.net.visibility input=input.gpkg coordinates=value visibility=input.gpkg
↪ output=output.gpkg
```

This calls **v.net.visibility**: input=input.gpkg — Input vector line layer (arcs); coordinates=value — Coordinates; visibility=input.gpkg — Input vector line layer containing visible points; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.normal — v.normal

**group:** Vector (v.\*)

Tests for normality for points.

| Parameter                              | Type            | Required | Default     | Description                            |
|----------------------------------------|-----------------|----------|-------------|----------------------------------------|
| map                                    | source          | yes      | —           | point vector defining sample points    |
| tests                                  | string          | no       | 1-3         | Lists of tests (1-15): e.g. 1,3-8,13   |
| column                                 | field           | yes      | —           | Attribute column                       |
| -r                                     | boolean         | no       | True        | Use only points in current region      |
| -l                                     | boolean         | no       | False       | lognormal                              |
| html                                   | fileDestination | no       | report.html | Normality                              |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —           | GRASS GIS region extent                |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0        | v.in.ogr snap tolerance (-1 = no snap) |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001      | v.in.ogr min area                      |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:v.normal map=input.gpkg column=field1 tests=1-3 html=html.txt
```

This calls **v.normal**: map=input.gpkg — point vector defining sample points; column=field1 — Attribute column; tests=1-3 — Lists of tests (1-15): e.g. 1,3-8,13; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.out.ascii — v.out.ascii

**group:** Vector (v.\*)

Exports a vector map to a GRASS ASCII vector representation.

| Parameter                              | Type            | Required | Default               | Description                                                                                     |
|----------------------------------------|-----------------|----------|-----------------------|-------------------------------------------------------------------------------------------------|
| input                                  | source          | yes      | —                     | Name of input vector map                                                                        |
| type                                   | enum            | no       | [0, 1, 2, 3, 4, 5, 6] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area, 5=face, 6=kernel |
| columns                                | field           | no       | —                     | Name of attribute column(s) to be exported                                                      |
| where                                  | string          | no       | —                     | WHERE conditions of SQL statement without 'where' keyword                                       |
| format                                 | enum            | no       | 0                     | Output format — options: 0=point, 1=standard, 2=wkt                                             |
| separator                              | enum            | no       | 0                     | Field separator — options: 0=pipe, 1=comma, 2=space, 3=tab, 4=newline                           |
| precision                              | number          | no       | 8.0                   | Number of significant digits (floating point only)                                              |
| -o *(adv)*                             | boolean         | no       | False                 | Create old (version 4) ASCII file                                                               |
| -c *(adv)*                             | boolean         | no       | False                 | Include column names in output (points mode)                                                    |
| output                                 | fileDestination | yes      | —                     | Name for output ASCII file or ASCII vector name if '-o' is defined                              |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —                     | GRASS GIS region extent                                                                         |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0                  | v.in.ogr snap tolerance (-1 = no snap)                                                          |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001                | v.in.ogr min area                                                                               |

**Outputs:** output (outputFile)

### Example usage

```
run grass:v.out.ascii input=input.gpkg type=0 columns=field1 where=value format=0
↪ separator=0 precision=8 output=output.txt
```

This calls **v.out.ascii**: input=input.gpkg — Name of input vector map; type=0 selects *point*; columns=field1 — Name of attribute column(s) to be exported; where=value — WHERE conditions of SQL statement without 'where' keyword; format=0 selects *point*; separator=0 selects *pipe*; precision=8 — Number of significant digits (floating point only); output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.out.dxf — v.out.dxf

**group:** Vector (v.\*)

Exports GRASS vector map layers to DXF file format.

| Parameter                              | Type            | Required | Default | Description                            |
|----------------------------------------|-----------------|----------|---------|----------------------------------------|
| input                                  | source          | yes      | —       | Name of input vector map               |
| output                                 | fileDestination | yes      | —       | DXF vector                             |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —       | GRASS GIS region extent                |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap) |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001  | v.in.ogr min area                      |

**Outputs:** output (outputFile)

### Example usage

```
run grass:v.out.dxf input=input.gpkg output=output.txt
```

This calls **v.out.dxf**: input=input.gpkg — Name of input vector map; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.out.postgis — v.out.postgis

**group:** Vector (v.\*)

Exports a vector map layer to PostGIS feature table.

| Parameter                              | Type    | Required | Default         | Description                                                                                             |
|----------------------------------------|---------|----------|-----------------|---------------------------------------------------------------------------------------------------------|
| input                                  | source  | yes      | —               | Name of input vector map                                                                                |
| type                                   | enum    | no       | [7]             | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area, 5=face, 6=kernel, 7=auto |
| output                                 | string  | no       | PG:dbname=grass | Name for output PostGIS datasource                                                                      |
| output_layer                           | string  | no       | —               | Name for output PostGIS layer                                                                           |
| options                                | string  | no       | —               | Creation options                                                                                        |
| -t *(adv)*                             | boolean | no       | False           | Do not export attribute table                                                                           |
| -l *(adv)*                             | boolean | no       | False           | Export PostGIS topology instead of simple features                                                      |
| -2 *(adv)*                             | boolean | no       | False           | Force 2D output even if input is 3D                                                                     |
| GRASS_REGION_PARAMETER *(adv)*         | extent  | no       | —               | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number  | no       | -1.0            | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number  | no       | 0.0001          | v.in.ogr min area                                                                                       |

### Example usage

```
run grass:v.out.postgis input=input.gpkg type=0 output=PG:dbname=grass output_layer=value
↳ options=value
```

This calls **v.out.postgis**: input=input.gpkg — Name of input vector map; type=0 selects *point*; output=PG:dbname=grass — Name for output PostGIS datasource; output\_layer=value — Name for output PostGIS layer; options=value — Creation options. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.out.pov — v.out.pov

**group:** Vector (v.\*)

Converts to POV-Ray format, GRASS x,y,z -> POV-Ray x,z,y

| Parameter                              | Type            | Required | Default      | Description                                                                                     |
|----------------------------------------|-----------------|----------|--------------|-------------------------------------------------------------------------------------------------|
| input                                  | source          | yes      | —            | Name of input vector map                                                                        |
| type                                   | enum            | no       | [0, 1, 4, 5] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area, 5=face, 6=kernel |
| size                                   | number          | no       | 10.0         | Radius of sphere for points and tube for lines                                                  |
| zmod                                   | string          | no       | —            | Modifier for z coordinates, this string is appended to each z coordinate                        |
| objmod                                 | string          | no       | —            | Object modifier (OBJECT_MODIFIER in POV-Ray documentation)                                      |
| output                                 | fileDestination | yes      | —            | POV vector                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —            | GRASS GIS region extent                                                                         |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0         | v.in.ogr snap tolerance (-1 = no snap)                                                          |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001       | v.in.ogr min area                                                                               |

---

**Outputs:** output (outputFile)

**Example usage**

```
run grass:v.out.pov input=input.gpkg type=0 size=10 zmod=value objmod=value  
↪ output=output.txt
```

This calls **v.out.pov**: input=input.gpkg — Name of input vector map; type=0 selects *point*; size=10 — Radius of sphere for points and tube for lines; zmod=value — Modifier for z coordinates, this string is appended to each z coordinate; objmod=value — Object modifier (OBJECT\_MODIFIER in POV-Ray documentation); output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:v.out.svg — v.out.svg**

**group:** Vector (v.\*)

Exports a vector map to SVG file.

| Parameter                              | Type            | Required | Default | Description                                    |
|----------------------------------------|-----------------|----------|---------|------------------------------------------------|
| input                                  | source          | yes      | —       | Name of input vector map                       |
| type                                   | enum            | no       | 0       | Output type — options: 0=poly, 1=line, 2=point |
| precision                              | number          | no       | 6.0     | Coordinate precision                           |
| attribute                              | field           | no       | —       | Attribute(s) to include in output SVG          |
| output                                 | fileDestination | yes      | —       | SVG File                                       |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —       | GRASS GIS region extent                        |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)         |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001  | v.in.ogr min area                              |

**Outputs:** output (outputFile)

**Example usage**

```
run grass:v.out.svg input=input.gpkg type=0 precision=6 attribute=field1 output=output.txt
```

This calls **v.out.svg**: input=input.gpkg — Name of input vector map; type=0 selects *poly*; precision=6 — Coordinate precision; attribute=field1 — Attribute(s) to include in output SVG; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

**grass:v.out.vtk — v.out.vtk**

**group:** Vector (v.\*)

Converts a vector map to VTK ASCII output.

| Parameter  | Type    | Required | Default               | Description                                                                                     |
|------------|---------|----------|-----------------------|-------------------------------------------------------------------------------------------------|
| input      | source  | yes      | —                     | Name of input vector map                                                                        |
| type       | enum    | no       | [0, 1, 2, 3, 4, 5, 6] | Input feature type — options: 0=point, 1=kernel, 2=centroid, 3=line, 4=boundary, 5=area, 6=face |
| precision  | number  | no       | 2.0                   | Number of significant digits (floating point only)                                              |
| zscale     | number  | no       | 1.0                   | Scale factor for elevation                                                                      |
| -c *(adv)* | boolean | no       | False                 | Correct the coordinates to fit the VTK-OpenGL precision                                         |

| Parameter                              | Type            | Required | Default | Description                                                   |
|----------------------------------------|-----------------|----------|---------|---------------------------------------------------------------|
| -n *(adv)*                             | boolean         | no       | False   | Export numeric attribute table fields as VTK scalar variables |
| output                                 | fileDestination | yes      | —       | VTK File                                                      |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —       | GRASS GIS region extent                                       |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                        |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001  | v.in.ogr min area                                             |

**Outputs:** output (outputFile)

### Example usage

```
run grass:v.out.vtk input=input.gpkg type=0 precision=2 zscale=1 output=output.txt
```

This calls **v.out.vtk**: input=input.gpkg — Name of input vector map; type=0 selects *point*; precision=2 — Number of significant digits (floating point only); zscale=1 — Scale factor for elevation; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.outlier — v.outlier

**group:** Vector (v.\*)

Removes outliers from vector point data.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector layer                                                                                      |
| ew_step                                | number            | no       | 10.0    | Interpolation spline step value in east direction                                                       |
| ns_step                                | number            | no       | 10.0    | Interpolation spline step value in north direction                                                      |
| lambda                                 | number            | no       | 0.1     | Tykhonov regularization weight                                                                          |
| threshold                              | number            | no       | 50.0    | Threshold for the outliers                                                                              |
| filter                                 | enum              | no       | 0       | Filtering option — options: 0=both, 1=positive, 2=negative                                              |
| -e                                     | boolean           | no       | False   | Estimate point density and distance                                                                     |
| output                                 | vectorDestination | yes      | —       | Layer without outliers                                                                                  |
| outlier                                | vectorDestination | yes      | —       | Outliers                                                                                                |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), outlier (outputVector)

### Example usage

```
run grass:v.outlier input=input.gpkg ew_step=10 ns_step=10 lambda=0.1 threshold=50
↳ filter=0 output=output.gpkg
```

This calls **v.outlier**: input=input.gpkg — Input vector layer; ew\_step=10 — Interpolation spline step value in east direction; ns\_step=10 — Interpolation spline step value in north direction; lambda=0.1 — Tykhonov regularization weight; threshold=50 — Threshold for the outliers; filter=0 selects *both*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.overlay — v.overlay

**group:** Vector (v.\*)

Overlays two vector maps.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| ainput                                 | source            | yes      | —       | Input layer (A)                                                                                         |
| atype                                  | enum              | no       | 0       | Input layer (A) Type — options: 0=area, 1=line, 2=auto                                                  |
| binput                                 | source            | yes      | —       | Input layer (B)                                                                                         |
| btype                                  | enum              | no       | 0       | Input layer (B) Type — options: 0=area                                                                  |
| operator                               | enum              | no       | 0       | Operator to use — options: 0=and, 1=or, 2=not, 3=xor                                                    |
| snap                                   | number            | no       | 1e-08   | Snapping threshold for boundaries                                                                       |
| -t                                     | boolean           | no       | False   | Do not create attribute table                                                                           |
| output                                 | vectorDestination | yes      | —       | Overlay                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.overlay ainput=input.gpkg binput=input.gpkg atype=0 btype=0 operator=0
↳ snap=1e-08 output=output.gpkg
```

This calls **v.overlay**: ainput=input.gpkg — Input layer (A); binput=input.gpkg — Input layer (B); atype=0 selects *area*; btype=0 selects *area*; operator=0 selects *and*; snap=1e-08 — Snapping threshold for boundaries; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.pack — v.pack

**group:** Vector (v.\*)

Exports a vector map as GRASS GIS specific archive file.

| Parameter                      | Type            | Required | Default     | Description                      |
|--------------------------------|-----------------|----------|-------------|----------------------------------|
| input                          | source          | yes      | —           | Name of input vector map to pack |
| -c *(adv)*                     | boolean         | no       | False       | Switch the compression off       |
| output                         | fileDestination | no       | output.pack | Packed archive                   |
| GRASS_REGION_PARAMETER *(adv)* | extent          | no       | —           | GRASS GIS region extent          |



| Parameter                              | Type   | Required | Default | Description                            |
|----------------------------------------|--------|----------|---------|----------------------------------------|
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap) |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number | no       | 0.0001  | v.in.ogr min area                      |

**Outputs:** output (outputFile)

### Example usage

```
run grass:v.pack input=input.gpkg output=output.txt
```

This calls **v.pack**: input=input.gpkg — Name of input vector map to pack; output=output.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.parallel — v.parallel

**group:** Vector (v.\*)

Creates parallel line to input vector lines.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input lines                                                                                             |
| distance                               | number            | no       | 1.0     | Offset along major axis in map units                                                                    |
| minordistance                          | number            | no       | —       | Offset along minor axis in map units                                                                    |
| angle                                  | number            | no       | 0.0     | Angle of major axis in degrees                                                                          |
| side                                   | enum              | no       | 0       | Side — options: 0=left, 1=right, 2=both                                                                 |
| tolerance                              | number            | no       | —       | Tolerance of arc polylines in map units                                                                 |
| -r                                     | boolean           | no       | False   | Make outside corners round                                                                              |
| -b                                     | boolean           | no       | False   | Create buffer-like parallel lines                                                                       |
| output                                 | vectorDestination | yes      | —       | Parallel lines                                                                                          |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.parallel input=input.gpkg distance=1 minordistance=10 angle=0 side=0
↳ tolerance=10 output=output.gpkg
```

This calls **v.parallel**: input=input.gpkg — Input lines; distance=1 — Offset along major axis in map units; minordistance=10 — Offset along minor axis in map units; angle=0 — Angle of major axis in degrees; side=0 selects *left*; tolerance=10 — Tolerance of arc polylines in map units; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.patch — v.patch

**group:** Vector (v.\*)

Create a new vector map layer by combining other vector map layers.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | multilayer        | yes      | —       | Input layers                                                                                            |
| -e                                     | boolean           | no       | True    | Copy also attribute table                                                                               |
| output                                 | vectorDestination | yes      | —       | Combined                                                                                                |
| bbox                                   | vectorDestination | yes      | —       | Bounding boxes                                                                                          |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), bbox (outputVector)

### Example usage

```
run grass:v.patch input="a.gpkg;b.gpkg" output=output.gpkg
```

This calls **v.patch**: input="a.gpkg;b.gpkg" — Input layers; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.perturb — v.perturb

**group:** Vector (v.\*)

Random location perturbations of GRASS vector points

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Vector points to be spatially perturbed                                                                 |
| distribution                           | enum              | no       | 0       | Distribution of perturbation — options: 0=uniform, 1=normal                                             |
| parameters                             | string            | no       | —       | Parameter(s) of distribution (uniform: maximum; normal: mean and stddev)                                |
| minimum                                | number            | no       | 0.0     | Minimum deviation in map units                                                                          |
| seed                                   | number            | no       | 0.0     | Seed for random number generation                                                                       |
| output                                 | vectorDestination | yes      | —       | Perturbed                                                                                               |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.perturb input=input.gpkg distribution=0 parameters=value minimum=0 seed=0
↳ output=output.gpkg
```

This calls **v.perturb**: input=input.gpkg — Vector points to be spatially perturbed; distribution=0 selects *uniform*; parameters=value — Parameter(s) of distribution (uniform: maximum; normal: mean and stddev); minimum=0 — Minimum deviation in map units; seed=0 — Seed for random number generation; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.proj — v.proj

**group:** Vector (v.\*)

Re-projects a vector layer to another coordinate reference system

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector to reproject                                                                               |
| crs                                    | crs               | yes      | —       | New coordinate reference system                                                                         |
| smax                                   | number            | no       | 10000.0 | Maximum segment length in meters in output vector map                                                   |
| -z *(adv)*                             | boolean           | no       | False   | Assume z coordinate is ellipsoidal height and transform if possible                                     |
| -w *(adv)*                             | boolean           | no       | False   | Disable wrapping to -180,180 for latlon output                                                          |
| output                                 | vectorDestination | yes      | —       | Output vector map                                                                                       |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.proj input=input.gpkg crs=EPSG:6346 smax=10000 output=output.gpkg
```

This calls **v.proj**: input=input.gpkg — Input vector to reproject; crs=EPSG:6346 — New coordinate reference system; smax=10000 — Maximum segment length in meters in output vector map; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.qcount — v.qcount

**group:** Vector (v.\*)

Indices for quadrat counts of vector point lists.

| Parameter | Type              | Required | Default | Description         |
|-----------|-------------------|----------|---------|---------------------|
| input     | source            | yes      | —       | Vector points layer |
| nquadrats | number            | no       | 4.0     | Number of quadrats  |
| radius    | number            | no       | 10.0    | Quadrat radius      |
| output    | vectorDestination | yes      | —       | Quadrats            |

| Parameter                              | Type    | Required | Default | Description                                                                                             |
|----------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*         | extent  | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number  | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.qcount input=input.gpkg nquadrats=4 radius=10 output=output.gpkg
```

This calls **v.qcount**: input=input.gpkg — Vector points layer; nquadrats=4 — Number of quadrats; radius=10 — Quadrat radius; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.random — v.random

**group:** Vector (v.\*)

Randomly generate a 2D/3D vector points map.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| npoints                                | number            | no       | 100.0   | Number of points to be created                                                                          |
| restrict                               | source            | no       | —       | Restrict points to areas in input vector                                                                |
| where                                  | string            | no       | —       | WHERE conditions of SQL statement without 'where' keyword                                               |
| zmin                                   | number            | no       | 0.0     | Minimum z height for 3D output                                                                          |
| zmax                                   | number            | no       | 0.0     | Maximum z height for 3D output                                                                          |
| seed                                   | number            | no       | —       | Seed for random number generation                                                                       |
| column                                 | string            | no       | z       | Column for Z values                                                                                     |
| column_type                            | enum              | no       | 0       | Type of column for z values — options: 0=integer, 1=double precision                                    |
| -z                                     | boolean           | no       | False   | Create 3D output                                                                                        |
| -a                                     | boolean           | no       | False   | Generate n points for each individual area                                                              |
| output                                 | vectorDestination | yes      | —       | Random                                                                                                  |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.random npoints=100 restrict=input.gpkg where=value zmin=0 zmax=0 seed=10
↳ column=z output=output.gpkg
```

This calls **v.random**: npoints=100 — Number of points to be created; restrict=input.gpkg — Restrict points to areas in input vector; where=value — WHERE conditions of SQL statement without 'where' keyword; zmin=0 — Minimum z height for 3D output; zmax=0 — Maximum z height for 3D output; seed=10 — Seed for random number generation; column=z — Column for Z values; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.rast.stats — v.rast.stats

**group:** Vector (v.\*)

Calculates univariate statistics from a raster map based on vector polygons and uploads statistics to new attribute columns.

| Parameter                                | Type              | Required | Default                                    | Description                                                                                                                                                                               |
|------------------------------------------|-------------------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| map                                      | source            | yes      | —                                          | Name of vector polygon map                                                                                                                                                                |
| raster                                   | raster            | yes      | —                                          | Name of raster map to calculate statistics from                                                                                                                                           |
| column_prefix                            | string            | yes      | —                                          | Column prefix for new attribute columns                                                                                                                                                   |
| method                                   | enum              | no       | [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12] | The methods to use — options: 0=number, 1=minimum, 2=maximum, 3=range, 4=average, 5=stddev, 6=variance, 7=coeff_var, 8=sum, 9=first_quartile, 10=median, 11=third_quartile, 12=percentile |
| percentile                               | number            | no       | 90.0                                       | Percentile to calculate                                                                                                                                                                   |
| output                                   | vectorDestination | yes      | —                                          | Rast stats                                                                                                                                                                                |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —                                          | GRASS GIS region extent                                                                                                                                                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0                                        | GRASS GIS region cellsize (leave 0 for default)                                                                                                                                           |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0                                       | v.in.ogr snap tolerance (-1 = no snap)                                                                                                                                                    |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001                                     | v.in.ogr min area                                                                                                                                                                         |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0                                          | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                                                                                          |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —                                          | v.out.ogr output data source options (dsko)                                                                                                                                               |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —                                          | v.out.ogr output layer options (lco)                                                                                                                                                      |
| GRASS_VECTOR_EXPORT_NO-CAT *(adv)*       | boolean           | no       | False                                      | Also export features without category (not labeled). Otherwise only features with category are exported                                                                                   |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.rast.stats map=input.gpkg raster=dem.tif column_prefix=value method=0
↳ percentile=90 output=output.gpkg
```

This calls **v.rast.stats**: map=input.gpkg — Name of vector polygon map; raster=dem.tif — Name of raster map to calculate statistics from; column\_prefix=value — Column prefix for new attribute columns; method=0 selects *number*; percentile=90 — Percentile to calculate; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.reclass — v.reclass

**group:** Vector (v.\*)

Changes vector category values for an existing vector map according to results of SQL queries or a value in attribute table column.

| Parameter                              | Type              | Required | Default      | Description                                                                                             |
|----------------------------------------|-------------------|----------|--------------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —            | Input layer                                                                                             |
| type                                   | enum              | no       | [0, 1, 2, 3] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid                                   |
| column                                 | field             | no       | —            | The name of the column whose values are to be used as new categories                                    |
| rules                                  | file              | no       | —            | Reclass rule file                                                                                       |
| output                                 | vectorDestination | yes      | —            | Reclassified                                                                                            |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —            | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0         | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001       | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0            | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —            | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —            | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False        | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.reclass input=input.gpkg type=0 column=field1 rules=input.dat  
↪ output=output.gpkg
```

This calls **v.reclass**: input=input.gpkg — Input layer; type=0 selects *point*; column=field1 — The name of the column whose values are to be used as new categories; rules=input.dat — Reclass rule file; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.rectify — v.rectify

**group:** Vector (v.\*)

Rectifies a vector by computing a coordinate transformation for each object in the vector based on the control points.

| Parameter                              | Type              | Required | Default | Description                            |
|----------------------------------------|-------------------|----------|---------|----------------------------------------|
| input                                  | source            | yes      | —       | Name of input vector map               |
| inline_points                          | string            | no       | —       | Inline control points                  |
| points                                 | file              | no       | —       | Name of input file with control points |
| order                                  | number            | no       | 1.0     | Rectification polynomial order         |
| separator                              | string            | no       | pipe    | Field separator for RMS report         |
| -3 *(adv)*                             | boolean           | no       | False   | Perform 3D transformation              |
| -o *(adv)*                             | boolean           | no       | False   | Perform orthogonal 3D transformation   |
| -b *(adv)*                             | boolean           | no       | False   | Do not build topology                  |
| output                                 | vectorDestination | yes      | —       | Rectified                              |
| rmsfile                                | fileDestination   | yes      | —       | Root Mean Square errors file           |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap) |

| Parameter                              | Type    | Required | Default | Description                                                                                             |
|----------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_MIN_AREA_PARAMETER<br>*(adv)*    | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER<br>*(adv)* | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT<br>*(adv)*   | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), rmsfile (outputFile)

### Example usage

```
run grass:v.rectify input=input.gpkg inline_points=value points=input.dat order=1
↪ separator=pipe output=output.gpkg
```

This calls **v.rectify**: input=input.gpkg — Name of input vector map; inline\_points=value — Inline control points; points=input.dat — Name of input file with control points; order=1 — Rectification polynomial order; separator=pipe — Field separator for RMS report; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.report — v.report

**group:** Vector (v.\*)

Reports geometry statistics for vectors.

| Parameter                              | Type            | Required | Default     | Description                                                                              |
|----------------------------------------|-----------------|----------|-------------|------------------------------------------------------------------------------------------|
| map                                    | source          | yes      | —           | Input layer                                                                              |
| option                                 | enum            | no       | 0           | Value to calculate — options: 0=area, 1=length, 2=coord                                  |
| units                                  | enum            | no       | 2           | units — options: 0=miles, 1=feet, 2=meters, 3=kilometers, 4=acres, 5=hectares, 6=percent |
| sort                                   | enum            | no       | 0           | Sort the result (ascending, descending) — options: 0=asc, 1=desc                         |
| html                                   | fileDestination | no       | report.html | Report                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —           | GRASS GIS region extent                                                                  |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0        | v.in.ogr snap tolerance (-1 = no snap)                                                   |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001      | v.in.ogr min area                                                                        |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:v.report map=input.gpkg option=0 units=2 sort=0 html=html.txt
```

This calls **v.report**: map=input.gpkg — Input layer; option=0 selects *area*; units=2 selects *meters*; sort=0 selects *asc*; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.sample — v.sample

**group:** Vector (v.\*)

Samples a raster layer at vector point locations.

| Parameter                                | Type              | Required | Default | Description                                                                                             |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                    | source            | yes      | —       | Vector layer defining sample points                                                                     |
| column                                   | field             | yes      | —       | Vector layer attribute column to use for comparison                                                     |
| raster                                   | raster            | yes      | —       | Raster map to be sampled                                                                                |
| zscale                                   | number            | no       | 1.0     | Sampled raster values will be multiplied by this factor                                                 |
| method                                   | enum              | no       | 0       | Sampling interpolation method — options: 0=nearest, 1=bilinear, 2=bicubic                               |
| output                                   | vectorDestination | no       | —       | Sampled                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —       | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.sample input=input.gpkg column=field1 raster=dem.tif zscale=1 method=0
↪ output=output.gpkg
```

This calls **v.sample**: input=input.gpkg — Vector layer defining sample points; column=field1 — Vector layer attribute column to use for comparison; raster=dem.tif — Raster map to be sampled; zscale=1 — Sampled raster values will be multiplied by this factor; method=0 selects *nearest*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.segment — v.segment

**group:** Vector (v.\*)

Creates points/segments from input vector lines and positions.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input lines layer                                                                                       |
| rules                                  | file              | yes      | —       | File containing segment rules                                                                           |
| output                                 | vectorDestination | yes      | —       | Segments                                                                                                |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)



## Example usage

```
run grass:v.segment input=input.gpkg rules=input.dat output=output.gpkg
```

This calls **v.segment**: input=input.gpkg — Input lines layer; rules=input.dat — File containing segment rules; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.select — v.select

**group:** Vector (v.\*)

Selects features from vector map (A) by features from other vector map (B).

| Parameter                              | Type              | Required | Default   | Description                                                                                                                               |
|----------------------------------------|-------------------|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| ainput                                 | source            | yes      | —         | Input layer (A)                                                                                                                           |
| atype                                  | enum              | no       | [0, 1, 4] | Input layer (A) Type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area                                                           |
| binput                                 | source            | yes      | —         | Input layer (B)                                                                                                                           |
| btype                                  | enum              | no       | [0, 1, 4] | Input layer (B) Type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area                                                           |
| operator                               | enum              | no       | 0         | Operator to use — options: 0=overlap, 1>equals, 2=disjoint, 3=intersect, 4=touches, 5=crosses, 6=within, 7=contains, 8=overlaps, 9=relate |
| relate                                 | string            | no       | —         | Intersection Matrix Pattern used for 'relate' operator                                                                                    |
| -t                                     | boolean           | no       | False     | Do not create attribute table                                                                                                             |
| -c                                     | boolean           | no       | False     | Do not skip features without category                                                                                                     |
| -r                                     | boolean           | no       | False     | Reverse selection                                                                                                                         |
| output                                 | vectorDestination | yes      | —         | Selected                                                                                                                                  |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —         | GRASS GIS region extent                                                                                                                   |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0      | v.in.ogr snap tolerance (-1 = no snap)                                                                                                    |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001    | v.in.ogr min area                                                                                                                         |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0         | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                                          |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —         | v.out.ogr output data source options (dsco)                                                                                               |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —         | v.out.ogr output layer options (lco)                                                                                                      |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False     | Also export features without category (not labeled). Otherwise only features with category are exported                                   |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.select ainput=input.gpkg binput=input.gpkg atype=0 btype=0 operator=0  
↪ relate=value output=output.gpkg
```

This calls **v.select**: ainput=input.gpkg — Input layer (A); binput=input.gpkg — Input layer (B); atype=0 selects *point*; btype=0 selects *point*; operator=0 selects *overlap*; relate=value — Intersection Matrix Pattern used for 'relate' operator; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.split — v.split

**group:** Vector (v.\*)

Split lines to shorter segments by length.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input lines layer                                                                                       |
| length                                 | number            | no       | —       | Maximum segment length                                                                                  |
| units                                  | enum              | no       | 0       | Length units — options: 0=map, 1=meters, 2=kilometers, 3=feet, 4=surveyfeet, 5=miles, 6=nautmiles       |
| vertices                               | number            | no       | —       | Maximum number of vertices in segment                                                                   |
| -n                                     | boolean           | no       | False   | Add new vertices, but do not split                                                                      |
| -f                                     | boolean           | no       | False   | Force segments to be exactly of given length, except for last one                                       |
| output                                 | vectorDestination | yes      | —       | Split by length                                                                                         |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.split input=input.gpkg length=10 units=0 vertices=10 output=output.gpkg
```

This calls **v.split**: input=input.gpkg — Input lines layer; length=10 — Maximum segment length; units=0 selects *map*; vertices=10 — Maximum number of vertices in segment; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.surf.bspline — v.surf.bspline

**group:** Vector (v.\*)

Bicubic or bilinear spline interpolation with Tykhonov regularization.

| Parameter    | Type   | Required | Default | Description                                                                                        |
|--------------|--------|----------|---------|----------------------------------------------------------------------------------------------------|
| input        | source | yes      | —       | Input points layer                                                                                 |
| column       | field  | no       | —       | Attribute table column with values to interpolate                                                  |
| sparse_input | source | no       | —       | Sparse points layer                                                                                |
| ew_step      | number | no       | 4.0     | Length of each spline step in the east-west direction                                              |
| ns_step      | number | no       | 4.0     | Length of each spline step in the north-south direction                                            |
| method       | enum   | no       | 0       | Spline interpolation algorithm — options: 0=bilinear, 1=bicubic                                    |
| lambda_i     | number | no       | 0.01    | Tykhonov regularization parameter (affects smoothing)                                              |
| solver       | enum   | no       | 0       | Type of solver which should solve the symmetric linear equation system — options: 0=cholesky, 1=cg |
| maxit        | number | no       | 10000.0 | Maximum number of iteration used to solve the linear equation system                               |
| error        | number | no       | 1e-06   | Error break criteria for iterative solver                                                          |
| memory       | number | no       | 300.0   | Maximum memory to be used (in MB)                                                                  |

| Parameter                                | Type              | Required | Default | Description                                                                                             |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| output                                   | vectorDestination | no       | —       | Output vector                                                                                           |
| raster_output                            | rasterDestination | no       | —       | Interpolated spline                                                                                     |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                                               |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                                                        |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*                | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector), raster\_output (outputRaster)

### Example usage

```
run grass:v.surf.bspline input=input.gpkg column=field1 sparse_input=input.gpkg ew_step=4
↳ ns_step=4 method=0 lambda_i=0.01 output=output.gpkg
```

This calls **v.surf.bspline**: input=input.gpkg — Input points layer; column=field1 — Attribute table column with values to interpolate; sparse\_input=input.gpkg — Sparse points layer; ew\_step=4 — Length of each spline step in the east-west direction; ns\_step=4 — Length of each spline step in the north-south direction; method=0 selects *bilinear*; lambda\_i=0.01 — Tykhonov regularization parameter (affects smoothing); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.surf.idw — v.surf.idw

**group:** Vector (v.\*)

Surface interpolation from vector point data by Inverse Distance Squared Weighting.

| Parameter                                | Type              | Required | Default | Description                                                               |
|------------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------|
| input                                    | source            | yes      | —       | Input vector layer                                                        |
| npoints                                  | number            | no       | 12.0    | Number of interpolation points                                            |
| power                                    | number            | no       | 2.0     | Power parameter; greater values assign greater influence to closer points |
| column                                   | field             | yes      | —       | Attribute table column with values to interpolate                         |
| -n                                       | boolean           | no       | False   | Don't index points by raster cell                                         |
| output                                   | rasterDestination | yes      | —       | Interpolated IDW                                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —       | GRASS GIS region extent                                                   |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                           |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —       | Output Rasters format options (createopt)                                 |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —       | Output Rasters format metadata options (metaopt)                          |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                    |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001  | v.in.ogr min area                                                         |

**Outputs:** output (outputRaster)

## Example usage

```
run grass:v.surf.idw input=input.gpkg column=field1 npoints=12 power=2 output=output.tif
```

This calls **v.surf.idw**: input=input.gpkg — Input vector layer; column=field1 — Attribute table column with values to interpolate; npoints=12 — Number of interpolation points; power=2 — Power parameter; greater values assign greater influence to closer points; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.surf.rst — v.surf.rst

**group:** Vector (v.\*)

Performs surface interpolation from vector points map by splines.

| Parameter                               | Type              | Required | Default | Description                                                               |
|-----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------|
| input                                   | source            | yes      | —       | Input points layer                                                        |
| zcolumn                                 | field             | no       | —       | Name of the attribute column with values to be used for approximation     |
| where                                   | string            | no       | —       | WHERE conditions of SQL statement without 'where' keyword                 |
| mask                                    | raster            | no       | —       | Name of the raster map used as mask                                       |
| tension                                 | number            | no       | 40.0    | Tension parameter                                                         |
| smooth                                  | number            | no       | —       | Smoothing parameter                                                       |
| smooth_column                           | field             | no       | —       | Name of the attribute column with smoothing parameters                    |
| segmax                                  | number            | no       | 40.0    | Maximum number of points in a segment                                     |
| npmin                                   | number            | no       | 300.0   | Minimum number of points for approximation in a segment (>segmax)         |
| dmin                                    | number            | no       | —       | Minimum distance between points (to remove almost identical points)       |
| dmax                                    | number            | no       | —       | Maximum distance between points on iso-line (to insert additional points) |
| zscale                                  | number            | no       | 1.0     | Conversion factor for values used for approximation                       |
| theta                                   | number            | no       | —       | Anisotropy angle (in degrees counterclockwise from East)                  |
| scalex                                  | number            | no       | —       | Anisotropy scaling factor                                                 |
| -t                                      | boolean           | no       | False   | Use scale dependent tension                                               |
| -d                                      | boolean           | no       | False   | Output partial derivatives instead of topographic parameters              |
| elevation                               | rasterDestination | no       | —       | Interpolated RST                                                          |
| slope                                   | rasterDestination | no       | —       | Slope                                                                     |
| aspect                                  | rasterDestination | no       | —       | Aspect                                                                    |
| pcurvature                              | rasterDestination | no       | —       | Profile curvature                                                         |
| tcurvature                              | rasterDestination | no       | —       | Tangential curvature                                                      |
| mcurvature                              | rasterDestination | no       | —       | Mean curvature                                                            |
| deviations                              | vectorDestination | no       | —       | Deviations                                                                |
| treeseg                                 | vectorDestination | no       | —       | Quadtree Segmentation                                                     |
| overwin                                 | vectorDestination | no       | —       | Overlapping Windows                                                       |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                                   |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                           |
| GRASS_RASTER_FORMAT_OPT *(adv)*         | string            | no       | —       | Output Rasters format options (createopt)                                 |
| GRASS_RASTER_FORMAT_META *(adv)*        | string            | no       | —       | Output Rasters format metadata options (metaopt)                          |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*  | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                    |
| GRASS_MIN_AREA_PARAMETER *(adv)*        | number            | no       | 0.0001  | v.in.ogr min area                                                         |

| Parameter                           | Type    | Required | Default | Description                                                                                             |
|-------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)* | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*           | string  | no       | —       | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*            | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*   | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** elevation (outputRaster), slope (outputRaster), aspect (outputRaster), pcurvature (outputRaster), tcurvature (outputRaster), mcurvature (outputRaster), deviations (outputVector), treeseg (outputVector), overwin (outputVector)

### Example usage

```
run grass:v.surf.rst input=input.gpkg zcolumn=field1 where=value mask=dem.tif tension=40
↳ smooth=10 smooth_column=field1 elevation=elevation.tif
```

This calls **v.surf.rst**: input=input.gpkg — Input points layer; zcolumn=field1 — Name of the attribute column with values to be used for approximation; where=value — WHERE conditions of SQL statement without 'where' keyword; mask=dem.tif — Name of the raster map used as mask; tension=40 — Tension parameter; smooth=10 — Smoothing parameter; smooth\_column=field1 — Name of the attribute column with smoothing parameters; elevation=elevation.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.surf.rst.cvdev — v.surf.rst.cvdev

**group:** Vector (v.\*)

v.surf.rst.cvdev - Performs surface interpolation from vector points map by splines.

| Parameter     | Type              | Required | Default | Description                                                                                 |
|---------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------|
| input         | source            | yes      | —       | Input points layer                                                                          |
| zcolumn       | field             | no       | —       | Name of the attribute column with values to be used for approximation                       |
| where         | string            | no       | —       | WHERE conditions of SQL statement without 'where' keyword                                   |
| mask          | raster            | no       | —       | Name of the raster map used as mask                                                         |
| tension       | number            | no       | 40.0    | Tension parameter                                                                           |
| smooth        | number            | no       | —       | Smoothing parameter                                                                         |
| smooth_column | field             | no       | —       | Name of the attribute column with smoothing parameters                                      |
| segmax        | number            | no       | 40.0    | Maximum number of points in a segment                                                       |
| npmin         | number            | no       | 300.0   | Minimum number of points for approximation in a segment (>segmax)                           |
| dmin          | number            | no       | —       | Minimum distance between points (to remove almost identical points)                         |
| dmax          | number            | no       | —       | Maximum distance between points on iso-line (to insert additional points)                   |
| zscale        | number            | no       | 1.0     | Conversion factor for values used for approximation                                         |
| theta         | number            | no       | —       | Anisotropy angle (in degrees counterclockwise from East)                                    |
| scalex        | number            | no       | —       | Anisotropy scaling factor                                                                   |
| -t            | boolean           | no       | False   | Use scale dependent tension                                                                 |
| -c            | boolean           | no       | True    | Perform cross-validation procedure without raster approximation [leave this option as True] |
| cvdev         | vectorDestination | no       | —       | Cross Validation Errors                                                                     |

| Parameter                                | Type    | Required | Default | Description                                                                                             |
|------------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*           | extent  | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number  | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                                                         |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number  | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*      | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*                | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*                 | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*        | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** cvdev (outputVector)

### Example usage

```
run grass:v.surf.rst.cvdev input=input.gpkg zcolumn=field1 where=value mask=dem.tif
↳ tension=40 smooth=10 smooth_column=field1 cvdev=cvdev.gpkg
```

This calls **v.surf.rst.cvdev**: input=input.gpkg — Input points layer; zcolumn=field1 — Name of the attribute column with values to be used for approximation; where=value — WHERE conditions of SQL statement without 'where' keyword; mask=dem.tif — Name of the raster map used as mask; tension=40 — Tension parameter; smooth=10 — Smoothing parameter; smooth\_column=field1 — Name of the attribute column with smoothing parameters; cvdev=cvdev.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.to.3d — v.to.3d

**group:** Vector (v.\*)

Performs transformation of 2D vector features to 3D.

| Parameter                              | Type              | Required | Default      | Description                                                                                             |
|----------------------------------------|-------------------|----------|--------------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —            | Name of input vector map                                                                                |
| type                                   | enum              | no       | [0, 1, 2, 3] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid                                   |
| height                                 | number            | no       | —            | Fixed height for 3D vector features                                                                     |
| column                                 | field             | no       | —            | Name of attribute column used for height                                                                |
| -r *(adv)*                             | boolean           | no       | False        | Reverse transformation; 3D vector features to 2D                                                        |
| -t *(adv)*                             | boolean           | no       | False        | Do not copy attribute table                                                                             |
| output                                 | vectorDestination | yes      | —            | 3D                                                                                                      |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —            | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0         | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001       | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0            | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —            | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —            | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False        | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.to.3d input=input.gpkg type=0 height=10 column=field1 output=output.gpkg
```

This calls **v.to.3d**: input=input.gpkg — Name of input vector map; type=0 selects *point*; height=10 — Fixed height for 3D vector features; column=field1 — Name of attribute column used for height; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.to.lines — v.to.lines

**group:** Vector (v.\*)

Converts vector polygons or points to lines.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Name of input vector map                                                                                |
| method                                 | enum              | no       | 0       | Method used for point interpolation — options: 0=delanay                                                |
| output                                 | vectorDestination | yes      | —       | Lines                                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.to.lines input=input.gpkg method=0 output=output.gpkg
```

This calls **v.to.lines**: input=input.gpkg — Name of input vector map; method=0 selects *delanay*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.to.points — v.to.points

**group:** Vector (v.\*)

Create points along input lines

| Parameter | Type              | Required | Default         | Description                                                                                     |
|-----------|-------------------|----------|-----------------|-------------------------------------------------------------------------------------------------|
| input     | source            | yes      | —               | Input lines layer                                                                               |
| type      | enum              | no       | [0, 1, 2, 3, 5] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area, 5=face, 6=kernel |
| use       | enum              | no       | 0               | Use line nodes or vertices only — options: 0=node, 1=vertex                                     |
| dmax      | number            | no       | 100.0           | Maximum distance between points in map units                                                    |
| -i        | boolean           | no       | False           | Interpolate points between line vertices                                                        |
| -t        | boolean           | no       | False           | Do not create attribute table                                                                   |
| output    | vectorDestination | yes      | —               | Points along lines                                                                              |

| Parameter                              | Type    | Required | Default | Description                                                                                             |
|----------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_REGION_PARAMETER *(adv)*         | extent  | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number  | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number  | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum    | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string  | no       | —       | v.out.ogr output data source options (dsko)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.to.points input=input.gpkg type=0 use=0 dmax=100 output=output.gpkg
```

This calls **v.to.points**: input=input.gpkg — Input lines layer; type=0 selects *point*; use=0 selects *node*; dmax=100 — Maximum distance between points in map units; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.to.rast — v.to.rast

**group:** Vector (v.\*)

Converts (rasterize) a vector layer into a raster layer.

| Parameter                                | Type              | Required | Default   | Description                                                         |
|------------------------------------------|-------------------|----------|-----------|---------------------------------------------------------------------|
| input                                    | source            | yes      | —         | Input vector layer                                                  |
| type                                     | enum              | no       | [0, 1, 3] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=area   |
| where                                    | string            | no       | —         | WHERE conditions of SQL statement without 'where' keyword           |
| use                                      | enum              | no       | 0         | Source of raster values — options: 0=attr, 1=cat, 2=val, 3=z, 4=dir |
| attribute_column                         | field             | no       | —         | Name of column for 'attr' parameter (data type must be numeric)     |
| rgb_column                               | field             | no       | —         | Name of color definition column (with RRR:GGG:BBB entries)          |
| label_column                             | field             | no       | —         | Name of column used as raster category labels                       |
| value                                    | number            | no       | 1.0       | Raster value (for use=val)                                          |
| memory                                   | number            | no       | 300.0     | Maximum memory to be used (in MB)                                   |
| output                                   | rasterDestination | yes      | —         | Rasterized                                                          |
| GRASS_REGION_PARAMETER *(adv)*           | extent            | no       | —         | GRASS GIS region extent                                             |
| GRASS_REGION_CELL_SIZE_PARAMETER *(adv)* | number            | no       | 0.0       | GRASS GIS region cellsize (leave 0 for default)                     |
| GRASS_RASTER_FORMAT_OPT *(adv)*          | string            | no       | —         | Output Rasters format options (createopt)                           |
| GRASS_RASTER_FORMAT_META *(adv)*         | string            | no       | —         | Output Rasters format metadata options (metaopt)                    |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*   | number            | no       | -1.0      | v.in.ogr snap tolerance (-1 = no snap)                              |
| GRASS_MIN_AREA_PARAMETER *(adv)*         | number            | no       | 0.0001    | v.in.ogr min area                                                   |



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**Outputs:** output (outputRaster)

### Example usage

```
run grass:v.to.rast input=input.gpkg type=0 where=value use=0 attribute_column=field1  
↳ rgb_column=field1 label_column=field1 output=output.tif
```

This calls **v.to.rast**: input=input.gpkg — Input vector layer; type=0 selects *point*; where=value — WHERE conditions of SQL statement without 'where' keyword; use=0 selects *attr*; attribute\_column=field1 — Name of column for 'attr' parameter (data type must be numeric); rgb\_column=field1 — Name of color definition column (with RRR:GGG:BBB entries); label\_column=field1 — Name of column used as raster category labels; output=output.tif is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.transform — v.transform

**group:** Vector (v.\*)

Performs an affine transformation on a vector layer.

| Parameter                              | Type              | Required | Default | Description                                                                                                                |
|----------------------------------------|-------------------|----------|---------|----------------------------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input vector layer                                                                                                         |
| xshift                                 | number            | no       | 0.0     | X shift                                                                                                                    |
| yshift                                 | number            | no       | 0.0     | Y shift                                                                                                                    |
| zshift                                 | number            | no       | 0.0     | Z shift                                                                                                                    |
| xscale                                 | number            | no       | 1.0     | X scale                                                                                                                    |
| yscale                                 | number            | no       | 1.0     | Y scale                                                                                                                    |
| zscale                                 | number            | no       | 1.0     | Z scale                                                                                                                    |
| zrotation                              | number            | no       | 0.0     | Rotation around z axis in degrees counter-clockwise                                                                        |
| columns                                | string            | no       | —       | Name of attribute column(s) used as transformation parameters (Format: parameter:column, e.g. xshift:xs,yshift:ys,zrot:zr) |
| -t *(adv)*                             | boolean           | no       | False   | Shift all z values to bottom=0                                                                                             |
| -w *(adv)*                             | boolean           | no       | False   | Swap coordinates x, y and then apply other parameters                                                                      |
| -x *(adv)*                             | boolean           | no       | False   | Swap coordinates x, z and then apply other parameters                                                                      |
| -y *(adv)*                             | boolean           | no       | False   | Swap coordinates y, z and then apply other parameters                                                                      |
| -a *(adv)*                             | boolean           | no       | False   | Swap coordinates after the other transformations                                                                           |
| -b *(adv)*                             | boolean           | no       | False   | Do not build topology                                                                                                      |
| output                                 | vectorDestination | yes      | —       | Transformed                                                                                                                |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                                    |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                                     |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                                          |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                           |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsko)                                                                                |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                                       |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported                    |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.transform input=input.gpkg xshift=0 yshift=0 zshift=0 xscale=1 yscale=1  
↳ zscale=1 output=output.gpkg
```

This calls **v.transform**: input=input.gpkg — Input vector layer; xshift=0 — X shift; yshift=0 — Y shift; zshift=0 — Z shift; xscale=1 — X scale; yscale=1 — Y scale; zscale=1 — Z scale; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.type — v.type

**group:** Vector (v.\*)

Change the type of geometry elements.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Name of existing vector map                                                                             |
| from_type                              | enum              | no       | 1       | Feature type to convert from — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=face, 5=kernel       |
| to_type                                | enum              | no       | 2       | Feature type to convert to — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=face, 5=kernel         |
| output                                 | vectorDestination | yes      | —       | Typed                                                                                                   |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

## Example usage

```
run grass:v.type input=input.gpkg from_type=1 to_type=2 output=output.gpkg
```

This calls **v.type**: input=input.gpkg — Name of existing vector map; from\_type=1 selects *line*; to\_type=2 selects *boundary*; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.univar — v.univar

**group:** Vector (v.\*)

Calculates univariate statistics for attribute. Variance and standard deviation is calculated only for points if specified.

| Parameter | Type   | Required | Default   | Description                                                                   |
|-----------|--------|----------|-----------|-------------------------------------------------------------------------------|
| map       | source | yes      | —         | Name of input vector map                                                      |
| type      | enum   | no       | [0, 1, 4] | Input feature type — options: 0=point, 1=line, 2=boundary, 3=centroid, 4=area |
| where     | string | no       | —         | WHERE conditions of SQL statement without 'where' keyword                     |

| Parameter                              | Type            | Required | Default     | Description                                                   |
|----------------------------------------|-----------------|----------|-------------|---------------------------------------------------------------|
| column                                 | field           | yes      | —           | Column name                                                   |
| percentile                             | number          | no       | 90.0        | Percentile to calculate                                       |
| -g                                     | boolean         | no       | True        | Print the stats in shell script style                         |
| -e                                     | boolean         | no       | False       | Calculate extended statistics                                 |
| -w                                     | boolean         | no       | False       | Weigh by line length or area size                             |
| -d                                     | boolean         | no       | False       | Calculate geometric distances instead of attribute statistics |
| html                                   | fileDestination | no       | report.html | Statistics                                                    |
| GRASS_REGION_PARAMETER *(adv)*         | extent          | no       | —           | GRASS GIS region extent                                       |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number          | no       | -1.0        | v.in.ogr snap tolerance (-1 = no snap)                        |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number          | no       | 0.0001      | v.in.ogr min area                                             |

**Outputs:** html (outputHtml)

### Example usage

```
run grass:v.univar map=input.gpkg column=field1 type=0 where=value percentile=90
↳ html=html.txt
```

This calls **v.univar**: map=input.gpkg — Name of input vector map; column=field1 — Column name; type=0 selects *point*; where=value — WHERE conditions of SQL statement without 'where' keyword; percentile=90 — Percentile to calculate; html=html.txt is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.vect.stats — v.vect.stats

**group:** Vector (v.\*)

Count points in areas and calculate statistics.

| Parameter                              | Type              | Required | Default | Description                                                                                                                                                             |
|----------------------------------------|-------------------|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| points                                 | source            | yes      | —       | Name of existing vector map with points                                                                                                                                 |
| areas                                  | source            | yes      | —       | Name of existing vector map with areas                                                                                                                                  |
| type                                   | enum              | no       | [0]     | Input feature type — options: 0=point, 1=centroid                                                                                                                       |
| method                                 | enum              | no       | 0       | Method for aggregate statistics — options: 0=sum, 1=average, 2=median, 3=mode, 4=minimum, 5=min_cat, 6=maximum, 7=max_cat, 8=range, 9=stddev, 10=variance, 11=diversity |
| points_column                          | field             | yes      | —       | Column name of points map to use for statistics                                                                                                                         |
| count_column                           | string            | yes      | —       | Column name to upload points count (integer, created if doesn't exists)                                                                                                 |
| stats_column                           | string            | yes      | —       | Column name to upload statistics (double, created if doesn't exists)                                                                                                    |
| output                                 | vectorDestination | yes      | —       | Updated                                                                                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                                                                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                                                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                                                                                    |

| Parameter                            | Type    | Required | Default | Description                                                                                             |
|--------------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_VECTOR_EXPORT_NOCAT<br>*(adv)* | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.vect.stats points=input.gpkg areas=input.gpkg points_column=field1
↳ count_column=value stats_column=value type=0 method=0 output=output.gpkg
```

This calls **v.vect.stats**: `points=input.gpkg` — Name of existing vector map with points; `areas=input.gpkg` — Name of existing vector map with areas; `points_column=field1` — Column name of points map to use for statistics; `count_column=value` — Column name to upload points count (integer, created if doesn't exists); `stats_column=value` — Column name to upload statistics (double, created if doesn't exists); `type=0` selects *point*; `method=0` selects *sum*; `output=output.gpkg` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.voronoi — v.voronoi

**group:** Vector (v.\*)

**v.voronoi** - Creates a Voronoi diagram from an input vector layer containing points.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input points layer                                                                                      |
| -l *(adv)*                             | boolean           | no       | False   | Output tessellation as a graph (lines), not areas                                                       |
| -t *(adv)*                             | boolean           | no       | False   | Do not create attribute table                                                                           |
| output                                 | vectorDestination | yes      | —       | Voronoi                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DS_CO *(adv)*             | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.voronoi input=input.gpkg output=output.gpkg
```

This calls **v.voronoi**: `input=input.gpkg` — Input points layer; `output=output.gpkg` is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.voronoi.skeleton — v.voronoi.skeleton

**group:** Vector (v.\*)

**v.voronoi.skeleton** - Creates a Voronoi diagram for polygons or compute the center line/skeleton of polygons.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| input                                  | source            | yes      | —       | Input polygons layer                                                                                    |
| smoothness                             | number            | no       | 0.25    | Factor for output smoothness                                                                            |
| thin                                   | number            | no       | -1.0    | Maximum dangle length of skeletons (-1 will extract the center line)                                    |
| -a *(adv)*                             | boolean           | no       | False   | Create Voronoi diagram for input areas                                                                  |
| -s *(adv)*                             | boolean           | no       | True    | Extract skeletons for input areas                                                                       |
| -l *(adv)*                             | boolean           | no       | False   | Output tessellation as a graph (lines), not areas                                                       |
| -t *(adv)*                             | boolean           | no       | False   | Do not create attribute table                                                                           |
| output                                 | vectorDestination | yes      | —       | Voronoi                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.voronoi.skeleton input=input.gpkg smoothness=0.25 thin=-1 output=output.gpkg
```

This calls **v.voronoi.skeleton**: input=input.gpkg — Input polygons layer; smoothness=0.25 — Factor for output smoothness; thin=-1 — Maximum dangle length of skeletons (-1 will extract the center line); output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.what.rast — v.what.rast

**group:** Vector (v.\*)

Uploads raster values at positions of vector centroids to the table.

| Parameter                               | Type              | Required | Default | Description                                                      |
|-----------------------------------------|-------------------|----------|---------|------------------------------------------------------------------|
| map                                     | source            | yes      | —       | Name of vector points map for which to edit attributes           |
| raster                                  | raster            | yes      | —       | Raster map to be sampled                                         |
| type                                    | enum              | no       | 0       | Input feature type — options: 0=point, 1=centroid                |
| column                                  | field             | yes      | —       | Name of attribute column to be updated with the query result     |
| where                                   | string            | no       | —       | WHERE conditions of SQL statement without 'where' keyword        |
| -i *(adv)*                              | boolean           | no       | False   | Interpolate values from the nearest four cells                   |
| output                                  | vectorDestination | yes      | —       | Sampled                                                          |
| GRASS_REGION_PARAMETER *(adv)*          | extent            | no       | —       | GRASS GIS region extent                                          |
| GRASS_REGION_CELLSIZE_PARAMETER *(adv)* | number            | no       | 0.0     | GRASS GIS region cellsize (leave 0 for default)                  |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)*  | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                           |
| GRASS_MIN_AREA_PARAMETER *(adv)*        | number            | no       | 0.0001  | v.in.ogr min area                                                |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*     | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area |

| Parameter                         | Type    | Required | Default | Description                                                                                             |
|-----------------------------------|---------|----------|---------|---------------------------------------------------------------------------------------------------------|
| GRASS_VECTOR_DSCO *(adv)*         | string  | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*          | string  | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)* | boolean | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.what.rast map=input.gpkg raster=dem.tif column=field1 type=0 where=value
↳ output=output.gpkg
```

This calls **v.what.rast**: map=input.gpkg — Name of vector points map for which to edit attributes; raster=dem.tif — Raster map to be sampled; column=field1 — Name of attribute column to be updated with the query result; type=0 selects *point*; where=value — WHERE conditions of SQL statement without 'where' keyword; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.

## grass:v.what.vect — v.what.vect

**group:** Vector (v.\*)

Uploads vector values at positions of vector points to the table.

| Parameter                              | Type              | Required | Default | Description                                                                                             |
|----------------------------------------|-------------------|----------|---------|---------------------------------------------------------------------------------------------------------|
| map                                    | source            | yes      | —       | Name of vector points map for which to edit attributes                                                  |
| column                                 | field             | yes      | —       | Column to be updated with the query result                                                              |
| query_map                              | source            | yes      | —       | Vector map to be queried                                                                                |
| query_column                           | field             | yes      | —       | Column to be queried                                                                                    |
| dmax                                   | number            | no       | 0.0     | Maximum query distance in map units                                                                     |
| output                                 | vectorDestination | yes      | —       | Updated                                                                                                 |
| GRASS_REGION_PARAMETER *(adv)*         | extent            | no       | —       | GRASS GIS region extent                                                                                 |
| GRASS_SNAP_TOLERANCE_PARAMETER *(adv)* | number            | no       | -1.0    | v.in.ogr snap tolerance (-1 = no snap)                                                                  |
| GRASS_MIN_AREA_PARAMETER *(adv)*       | number            | no       | 0.0001  | v.in.ogr min area                                                                                       |
| GRASS_OUTPUT_TYPE_PARAMETER *(adv)*    | enum              | no       | 0       | v.out.ogr output type — options: 0=auto, 1=point, 2=line, 3=area                                        |
| GRASS_VECTOR_DSCO *(adv)*              | string            | no       | —       | v.out.ogr output data source options (dsco)                                                             |
| GRASS_VECTOR_LCO *(adv)*               | string            | no       | —       | v.out.ogr output layer options (lco)                                                                    |
| GRASS_VECTOR_EXPORT_NOCAT *(adv)*      | boolean           | no       | False   | Also export features without category (not labeled). Otherwise only features with category are exported |

**Outputs:** output (outputVector)

### Example usage

```
run grass:v.what.vect map=input.gpkg column=field1 query_map=input.gpkg
↳ query_column=field1 dmax=0 output=output.gpkg
```

This calls **v.what.vect**: map=input.gpkg — Name of vector points map for which to edit attributes; column=field1 — Column to be updated with the query result; query\_map=input.gpkg — Vector map to be queried; query\_column=field1 — Column to be queried; dmax=0 — Maximum query distance in map units; output=output.gpkg is where the result is written. Enum values are integer indices; the paths and values here are illustrative — substitute your own.